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Classification and Tabulation of 2-String Tangles: the Astronomy of Subtangle Decompositions

by

Nicholas Connolly

A thesis submitted in partial fulfillment of the
requirements for the Doctor of Philosophy
degree in Mathematics
in the Graduate College of
The University of Iowa

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Abstract

Tangles are a type of knot theoretic structure originally introduced by Conway to tabulate knots. Intuitively, tangles can be thought of as the building blocks of mathematical knots. The two principal goals of this dissertation are classification and tabulation of 2-string tangle diagrams, and we approach both endeavors through the careful study of diagram notations. Broadly speaking, tangles can be divided into two types: algebraic and non-algebraic. Many special cases of algebraic tangle are well understood and have been fully classified. We begin by reviewing the structure of these special types of tangles, each of which may be uniquely represented using a canonical diagram.

Unlike these special cases, generic algebraic and non-algebraic tangle diagrams are not fully classified. To better understand these types of tangles, we propose a graph-based notation for tangle diagrams, referred to as *subtangle planar diagram* (SPD) code. SPD code can be used to represent generic tangle diagrams using decompositions into subtangles. Based on this description, we show how algebraic and non-algebraic tangles can be distinguished using planar graphs. This description enables any algebraic diagram to be converted into an equivalent semi-canonical form. Furthermore, we show how certain types of planar graphs known as *constellations* arise in connection with subtangle decompositions for non-algebraic diagrams. All 2-string tangle diagrams with at most 9 crossings can be represented using a combination of tangle algebra and the first ten constellations.

In addition to this theoretical work, we also discuss a number of related computational results. In particular, we give an overview of an algorithm to compute several variations of the SPD code, which has been successfully implemented as a program in C/C++. This algorithm has been used in combination with tangle tabulation programs to classify tangle diagrams. We conclude by discussing the progress on the prototype of a web-accessible database of 2-string tangles currently being developed to catalog these results.

Public Abstract

Knots, the principal objects of study in the mathematical field of knot theory, can be visualized as an entwined piece of string with the two ends glued together. This dissertation is concerned with studying the parts of a knot obtained by chopping up this string into pieces while preserving the shape. These parts, known as *tangles*, can be thought of as the building blocks of mathematical knots; they can be pieced back together to build more complicated types of structures. There are many different types of tangles that can be identified based on how these pieces are put together.

Understanding, classifying, and tabulating these different types of tangles is the principal goal of this research. Some types of tangles are well understood in the literature, and we begin by reviewing these known results. Other types of tangles are not yet uniquely classified. In this dissertation, we propose a method for identifying these different types of tangles based on how they decompose into smaller pieces known as *subtangles*. In addition to this theoretical work, we also discuss progress on a web-accessible database of tangles currently under development.

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1 Introduction

Pattern recognition is the heart of mathematics. Patterns are everywhere, and perhaps the most recognizable is the notion of *shape*. Topology is the field of mathematics concerned with the study of shape, but how does one formalize something so inherently intuitive? A pattern is not a particular type of object, rather, it is a relationship between the parts of a whole, in the same spirit that a musical melody is independent from the instrument it is played on.

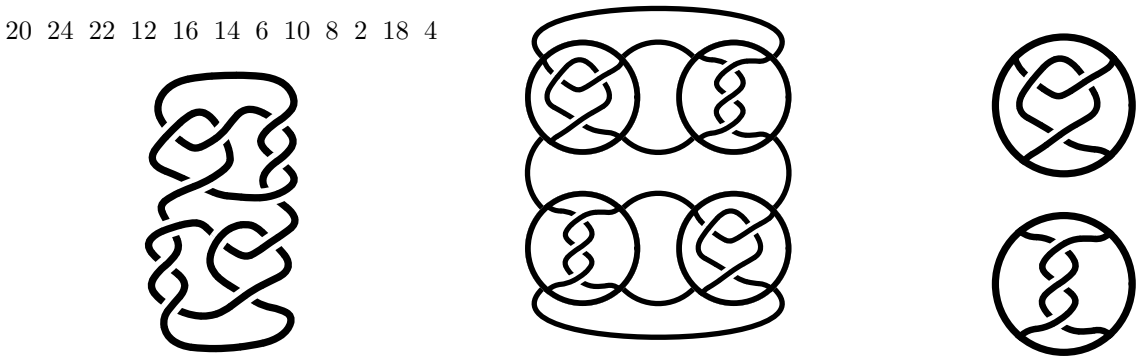
The job of the mathematician is to decide what axioms govern a pattern of interest. He declares: *This is a mathematical object! It is defined by these rules and behaves in this way.* He then dubs his discovery an appropriately scientific name and proceeds to catalog its properties. He invents the notation to enumerate what he finds. Eventually, his labor may lay the groundwork for proving theorems. My objective in this dissertation is to lay such a groundwork in a particular area of knot theory.

1.1 A Friendly Intuition for Knots and Knot Tabulation

A shoelace is the first image conjured in my mind by the word *knot*. I picture the two ends of a string tied together in the shape of a bow. Stepping back from the abstractions, mathematical knots are merely the formalization of this mental image, with the added rule that the two string ends are joined together to form a closed curve. A tied shoelace consists of a string and crossings: loops passing over each other in a certain order. When we tie a shoe, we trace out each of these components in sequence. Expressing this sequence of steps symbolically is one of many ways to interpret a knot using mathematics.

Any outdoorsman knows a plethora of ways to tie a rope, but how does he distinguish a square knot from a half hitch or a carrick bend? Simply put, different knots are the result of crossing the ropes in a different combination of over and under passes. Mathematically, this is equivalent to a different sequence of symbols. Such a list of symbols describing the strings and crossings in a knot is formally known as a *diagram notation*. While knots are three dimensional objects, mathematicians often study them using *knot diagrams*: the projected shadow of a knot onto a two dimensional surface. A diagram notation encodes the important information from the inherent pattern of a knot (Figure 1.1a).

Just as our tied shoelaces come apart when pulled taut, the shape of a mathematical knot can also wiggle and change. However, unlike physical knots, a mathematical knot is still considered the same object no matter how it folds and wiggles. Thus, a diagram notation only



(a) Knot diagram and notation **(b)** Knot building blocks **(c)** Parts of a knot

Figure 1.1: Some friendly pictures of mathematical knots

reflects one of many possible diagrams corresponding to a single mathematical object. How to distinguish between different mathematical knots is a surprisingly challenging question which has prompted many researchers to tabulate lists of known knots [37, 22, 15, 11, 7]. The enumeration of knots has been the quintessential goal of knot theory since the field's inception.

The major difficulty of this endeavor stems from the fact that a single knot can be realized by many different diagrams. Diagram notations help mitigate this difficulty by providing a formulaic lexicon for cataloging diagrams. Every knot can be represented using a knot diagram, albeit non-uniquely. It is far simpler to tabulate knot diagrams than to tabulate knots, but this is a significant first step. For a fixed number of crossings, there are only finitely many knot diagrams and hence only finitely many diagram notations.

The *Dowker code*, developed by the Dowker and Thistlewaite [11] and shown in Figure 1.1a, is an example of one such diagram notation. By systematically enumerating Dowker codes, Dowker and Thistlewaite were able generate computationally a list of possible knot diagrams up to a fixed crossing number. While this method is known to create redundancies, it is also exhaustive. Every knot admitting a diagram with this many or fewer crossings must show up on this list. Other mathematical techniques can then be leveraged to detect and eliminate repeated knots.

Another example of a knot diagram notation was developed by the knot theorist J. H. Conway [7]. Conway's notation described how to build up a knot diagram piece by piece using what he referred to as *tangles*. Much of my doctoral research is a generalization of Conway's work.

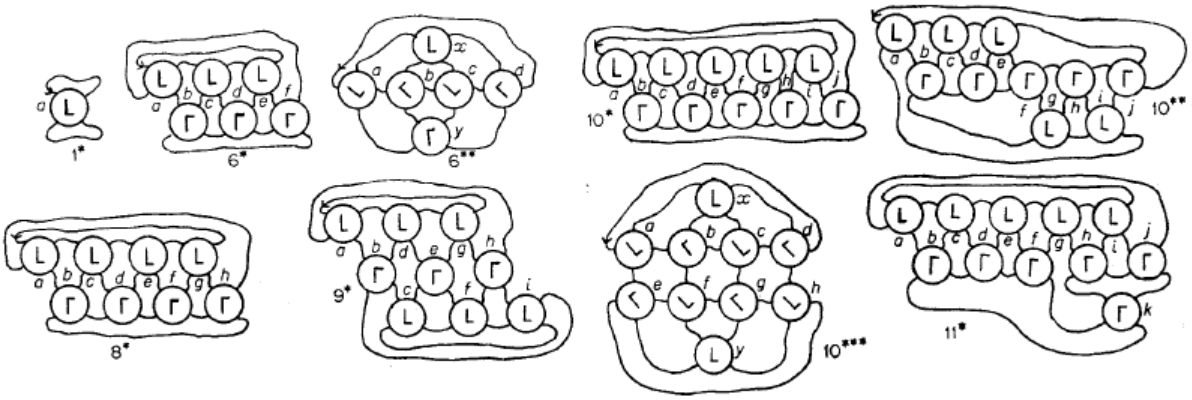


Figure 1.2: Conway's basic graphs, reprinted from [7] with permission from Elsevier

1.2 Tangles and the Missing Infrastructure Project

Tangles are the principal objects of study in this dissertation. They can be thought of most intuitively as the building blocks of mathematical knots in the sense that any knot diagram can be decomposed into a configuration of tangles (Figure 1.1b). Conway introduced tangles as a way to tabulate knots [7]. He showed that it was possible to describe any knot with up to 11 crossings using one of the planar graphs shown in Figure 1.2, work which was later improved upon by Caudron [4]. Conway's notation involved replacing the vertices in one of these planar graphs with certain kinds of tangles, thus building up to a knot diagram.

As mathematical objects, tangles can be studied independently from knots, although they share many features in common. The principal difference is that, whereas a knot consists of a single unbroken string without ends, a tangle is made up of multiple entwined string-segments. Figure 1.1c shows examples of tangles with 2 strings. The endpoints of these string-segments are embedded on the surface of a ball. A tangle can have any number of strings, but we will only be concerned with 2-string tangles in this manuscript. Much like knots, tangles can be studied using tangle diagrams and diagram notations. Most notations for tangle diagrams are direct generalizations from a corresponding knot diagram notation, with additional information describing the endpoints.

The tabulation of tangles has been less thoroughly researched than knot tabulation, although progress has been made in special cases and under certain assumptions [7, 20, 31, 38, 17]. Even so, a thorough understanding of tangles is necessary for many practical applications involving knots. For example, computing knot invariants is one of the techniques commonly used to distinguish between types knots, a process which could be simplified if

a tangle decomposition were known. This deficiency has led topologist Dror Bar-Natan to describe tangle tabulation as "the most important missing infrastructure project" in the field of knot theory [3], a deficiency which this dissertation seeks to address.

My approach to this objective is similar in spirit to Conway's. By generalizing a diagram notation introduced by Ewing and Millett [12], I have developed my own notation using planar graphs to decompose a diagram into tangles. Tangles can be divided into different families based on their structure, and while some families are well understood, others are quite challenging to describe. One of the difficulties of working with tangles is the lack of a suitable notation for distinguishing between these tangle families. My notation readily identifies the family of a tangle diagram and is the first step towards classifying the tangle.

1.3 Mathematical Notation and Nomenclature

They are all like that, the eggheads. The most important thing for them is to come up with a name... when he finds a label like "graviconcentrate," he thinks he's figured it all out.

Arkady and Boris Strugatsky,
Roadside Picnic

Ironically, mathematical notation can be major a barrier to understanding mathematics. A good notation captures the intuition of the pattern while excluding extraneous information, but the balance between these competing tensions changes depending on the context. In relation to knot tabulation, the Dowker code's simplicity makes it very convenient to iterate through possible diagram notations [11]. However, it is difficult to reconstruct the mental image of the diagram without reverse engineering the notation by hand.

By contrast, the knot diagram notation introduced by Ewing and Millett provides a graph-like description of a knot [12]. As diagramatic objects, knots and graphs share many features in common; by replacing the crossings in a knot diagram with vertices, one recovers a 4-valent planar graph. One of the common ways of describing a graph is with an *adjacency list*: a list of adjacent vertices for each vertex. Ewing and Millett's notation is similar to an adjacency list organized by crossing, with additional information concerning orientation and crossing sign. While far less compact than the Dowker code, it is more easily read both by human and machine.

The notation I propose, the *subtangle planar diagram* (SPD) code, also visualizes a diagram as a planar graph, but vertices represent subtangles rather than mere crossings. This is the same idea employed by Conway, but the SPD code also encodes the underlying graph. Conway's notation used only certain types of subtangles and a handful of planar graphs, but his technique can be applied more generally. Like Ewing and Millett's notation, the SPD code describes the adjacency list for a graph with additional information concerning the structure of each subtangle. Unlike other notations, the SPD code is only unique up to certain equivalencies of subtangles. We will see how leveraging this property enables tangle diagrams to be easily classified.

Computing the SPD code of a diagram is equivalent to finding a subtangle decomposition, which also makes it easy to reconstruct a diagram from this notation. While a generic subtangle decomposition for a given diagram is non-unique, it will be when restricting to certain types of subtangles. The same diagram can be examined using different types of subtangle decompositions, leading to several variants of SPD code. These alternate notations are particularly helpful for verifying whether or not a given diagram satisfies a preferred form. The SPD code and its variations strike a balance between compactness and intuition and are readily implemented in a computer. Furthermore, the SPD code provides a partial solution to the challenge of identifying redundant diagrams when using Dowker and Thistlewaite's method.

While there are many ways to describe a diagram using planar graphs, the particular graphs that Conway obtained (Figure 1.2) have a special structure. By studying similar graph-like descriptions for tangle diagrams, I have recovered Conway's graphs in my own research. Conway's list of graphs was sufficient for representing knot diagrams with up to 11 crossings, but these are just the first few examples of a class of graphs that have a special relationship with knot and tangle diagrams. Since planar graphs are reminiscent of astrological images, I refer to graphs satisfying these special properties as *constellations*.

The study of constellations and their connection to tangle diagrams is the central focus of one of the final sections. We will see that these graphs arise in the subtangle decomposition of what are known as *non-algebraic* tangles, which are particularly difficult to study. All of the tangle families which have been well studied are special cases of *algebraic* tangles, but the proportion of tangles which are non-algebraic grows exponentially as crossing number increases. Constellations are a promising tool for grappling with this less-well-understood class of tangles. Towards this goal, I propose a *constellation notation* for describing these types of tangle diagrams.

1.4 Overview

The main objective of this dissertation is tangle classification, and I approach this endeavor through the careful study of diagram notations. Tangles can be divided into two types: algebraic and non-algebraic. Many special cases of algebraic tangle are well understood and have been studied in detail. Much of this thesis is devoted to reviewing these important results. Generic algebraic and non-algebraic tangles are less well understood, and I propose a graph-based approach for studying these types of diagrams. In particular, I show how each algebraic tangle can be represented in a semi-canonical form using *algebraic tangle trees*, and how each non-algebraic tangle can be represented using *constellations*. These descriptions are based on generalizing the work of Caudron [4] and Conway [7], respectively, to tangle diagrams.

In addition to these theoretical descriptions, computation plays a central role in this research. I have written programs in C/C++ to implement much of my work concurrently with developing the theory [5]. The major achievement is my classification algorithm based on computing the SPD code. Expanding on software that was developed in previous student research projects [25, 9], I have combined my classification algorithm with a tabulation program that generates an exhaustive list of tangles using Dowker codes. The culmination of this computational research is a web-accessible database of 2-string tangles [6]: `http://www.nick-connolly.com/tangles/tangles.php`. The curious reader is encouraged to consult and play around with this resource while reviewing this manuscript!

My thesis is organized into seven chapters covering both the theoretical and computational aspects of my research, as well as a review of many of the important known results from the literature. The following is a brief outline of each chapter.

1. **Introduction:** Tabulation of knots is a longstanding goal of knot theory. The introduction provides some intuition for motivating this goal in the context of tangles.

2. **Background:** The second chapter is devoted to preliminaries and formal definitions. It also provides a detailed explanation of some existing diagram notations that will be important in later sections, including the Dower Code [11], the Gauss Code, and the Ewing-Millett notation [12].

3. **Rational Tangles:** Among 2-string tangles in general, the class of *rational* tangles are the most well-understood and have been classified completely by Conway [7]. Thanks to their rich and well-behaved structure, rational tangles are particularly useful for describing

subtangle decompositions. This section is devoted to a survey of these tangles' most important properties and a discussion of some of their major results. Much of this material constitutes a review of the existing literature based on the work of Kauffman, Goldman, and Lambropoulou [14, 20].

4. **Subtangle Decompositions:** Subtangle decompositions are a powerful tool for studying tangle diagrams. The structure of a tangle diagram can be identified based on how it decomposes into maximal subtangles of a certain type. This chapter has two principal goals: to introduce several special types of subtangle decomposition, and then to show how these decompositions can be computed in practice. Towards these goals, I formally define the *subtangle planar diagram* (SPD) code, a notation of my own invention based on generalizing the Ewing-Millett notation [12]. I conclude by walking through the computational algorithms I have developed to calculate several special variations of SPD code [5].

5. **Tangle Classification:** Broadly speaking, there are two types of 2-string tangle diagram: algebraic and non-algebraic. Algebraic tangles can be divided into a hierarchy of increasingly general classes based on their structure: integer, rational, Montesinos, generalized Montesinos, and algebraic. The first four classes in this hierarchy are special cases of algebraic tangle which can be uniquely characterized using a *canonical* diagram. The first half of this chapter, based on the work of Montesinos, Moon, and Darcy [29, 30, 31], focuses on the study of Montesinos and generalized Montesinos tangles. In the second half, I generalize this work to propose a *semi-canonical* form that applies to all algebraic tangle diagrams, albeit non-uniquely. Furthermore, I show how the construction of a semi-canonical diagram can be visualized using *algebraic tangle trees*, which are based on a method introduced by Caudron to represent algebraic knots using acyclic graphs [4].

6. **Constellations:** Every decomposition of a tangle diagram using algebraic subtangles can be described using a 4-valent planar graph. The collection of these graphs for a fixed diagram forms a category, where there exists a morphism between any graphs related by a sequence of collapsing bigons. The Maximal Algebraic Subtangle Graph (MASG) of a tangle diagram arises as a terminal object in this category, and can be used to distinguish between algebraic and non-algebraic tangle diagrams. *Constellations* are a special type of 4-valent planar graph which arise in connection with non-algebraic MASGs, and these graphs are closely related to Conway's basic polyhedra (Figure 1.2). All non-algebraic

2-string tangle diagrams with a most 9 crossings can be described using one of the first 10 constellations, making these graphs a promising tool for tangle tabulation. The first half of this chapter is devoted to formalizing the categorical description of the MASG, and the second half is devoted to classifying constellations. Using these special graphs, I propose a *constellation notation* for non-algebraic diagrams that parallels Conway notation [7].

7. **Database Development and Future Work:** The final chapter of this dissertation addresses the future directions of this research. In particular, I discuss my contributions to the broader project of tangle tabulation as well as my progress on designing a web-accessible database of 2-string tangles. Building on the computational work of previous students [25, 9], I have written my own programs in C/C++ capable of identifying sub-tangle decompositions by computing SPD codes [5]. In the spirit of such resources as *The Knot Atlas* and *KnotInfo* [2, 23], I have created *The Tanglenomicon* [6] to serve as repository of cataloged tangles. I conclude with an overview of my database's existing capabilities and my plans for future functionality.

2 Background

This dissertation is devoted to the study of tangles, but various differing notions of tangle arise in different mathematical contexts. The goal of this preliminary section is to formalize our definition of tangle along with cataloging some of their important characteristics. A general familiarity in topology such as what is covered in Munkres [32] will be assumed. More general information can also be found in standard introductory knot theory texts such Rolfsen, Cromwell, or Adams [34, 8, 1].

We begin with a review of tangles' basic properties. Tangles have a nice modular structure that allows them to be transformed and combined to build up more complicated knot theoretic objects. We include a discussion of those operations on tangles that will frequently be used in later sections. Finally, this chapter concludes with a review of some of the commonly used diagram notations. Since these are usually generalizations of notations for knot diagrams, we introduce them first in the context of knots and then show how the notations can be modified to accommodate tangles.

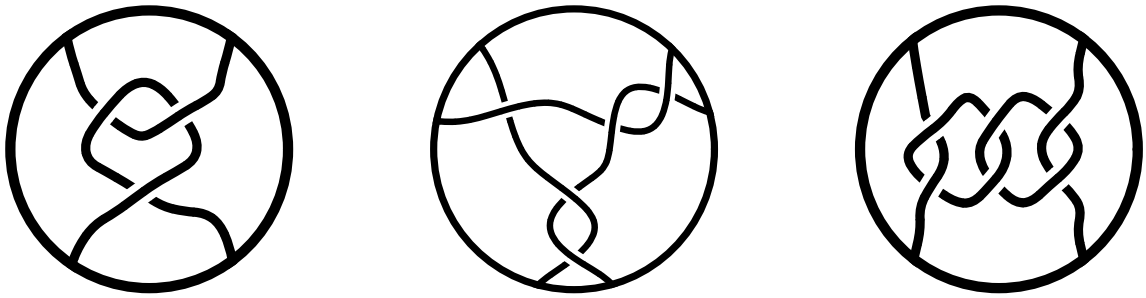
2.1 Properties of 2-String Tangles

Recall that the formal definition of a *knot* is a homeomorphic embedding of the 1-dimensional circle S^1 into 3-dimensional space $\mathbb{R}^3$. Tangles are also defined via embeddings into 3-space. Many familiar properties of knots and links such as orientations and isotopies readily extend to tangles. However, tangles have additional structure with no counterpart among knots.

2.1.1 Tangles and Tangle Diagrams

An *n-string tangle* is an embedding of n disjoint string-segments into the interior of a 3-ball. The endpoints of the string-segments are embedded on the surface of the ball. Figures 2.1a and 2.1b show examples of diagrams for a 2-string tangle and a 3-string tangle, respectively. Some definitions of tangle allow for the existence of closed loops in the interior of the ball (Figure 2.1c). We exclude the possibility of closed loops in this paper; for our purposes, a tangle contains only string-segments.

Other definitions of tangle also allow the endpoints of the strings to move on the boundary of the ball, an assumption which may allow some crossings to unwind. We will on occasion be interested in this property when examining tangle families, but not in general. Unless



(a) 2-string tangle **(b)** 3-string tangle **(c)** Non-tangle with closed loops

Figure 2.1: Examples of diagrams for a tangle and some common variations

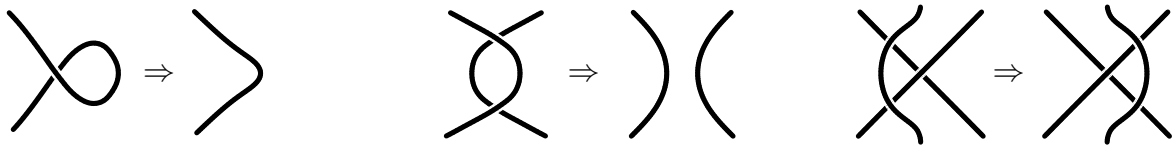
otherwise stated, the endpoints of the strands are assumed to be fixed on the boundary of the sphere. While a tangle may contain any number of strings, we will only be concerned with the study of 2-string tangles and will often drop the number. Henceforth, it should be assumed that *tangle* refers to a 2-string tangle unless otherwise stated.

All of the images in Figure 2.1 are examples of diagrams generated using KnotPlot [35]. Tangles, like knots, are primarily visualized using diagrams. Formally, a *tangle diagram* is the projected shadow of a tangle onto a two dimensional surface. For each crossing, the under-strand is distinguished by a break in the string as it passes beneath the over-strand. We may assume that a diagram has finitely many crossings and that these do not overlap. The boundary of the 3-ball is projected onto a circle enclosing the entwined strings. The endpoints of the strings intersect this circle.

For 2-string tangle diagrams, the four endpoints of the strings are arranged around the

(a) Tangle diagram with endpoints labeled **(b)** Non-equivalent tangle diagram after rotation

Figure 2.2: Examples of compass labels for tangle diagram endpoints: NW, NE, SE, SW



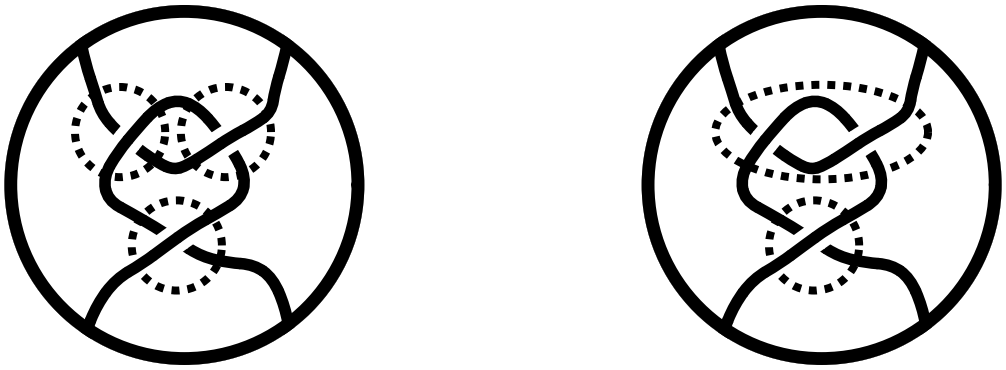
(a) RI move **(b)** RII move **(c)** RIII move

Figure 2.3: The three Reidemeister moves: RI, RII, and RIII

four corners of the circle: NW, NE, SE, SW (Figure 2.2). We will refer to the endpoints by these labels throughout this text. The positions of these endpoints are fixed; in general, rotating the endpoints yields a different tangle. For example, Figure 2.2a and Figure 2.2b are considered different tangles even though the second diagram is only a 90 degree rotation of the first. Accounting for the behavior of the endpoints is an important part of understanding the combinatorics of tangle diagrams.

Another property of tangles familiar from knots is the notion of equivalence. Two tangles are said to be *equivalent* if there exists an ambient isotopy between them which leaves the boundary of the 3-ball fixed. Two diagrams describe equivalent tangles if and only if they are related by a sequence of *Reidemeister moves* (Figure 2.3) and planar isotopies, keeping the endpoints of the strings fixed. We will use the notation $T_1 \cong T_2$ to denote when T_1 and T_2 are equivalent tangles.

A *subtangle* is any tangle contained inside of a tangle, knot, or link (Figure 2.4). Any



(a) 1-crossing subtangle decomposition **(b)** Another possible subtangle decomposition

Figure 2.4: Examples of subtangles and subtangle decompositions

sphere which intersects the strings in exactly four places contains a subtangle, although we will continue to exclude closed loops in the case of links. These are known as *separating spheres* and can be visualized in diagrams using *separating circles* (Figure 2.4). A *subtangle decomposition* is a partition of the tangle, knot, or link into subtangles. It is always possible to isolate each crossing with a separating sphere, which defines the *1-crossing subtangle decomposition* (Figure 2.4a). In general, there exist many possible subtangle decompositions for a given knot theoretic structure.

2.1.2 Orientation and Parity

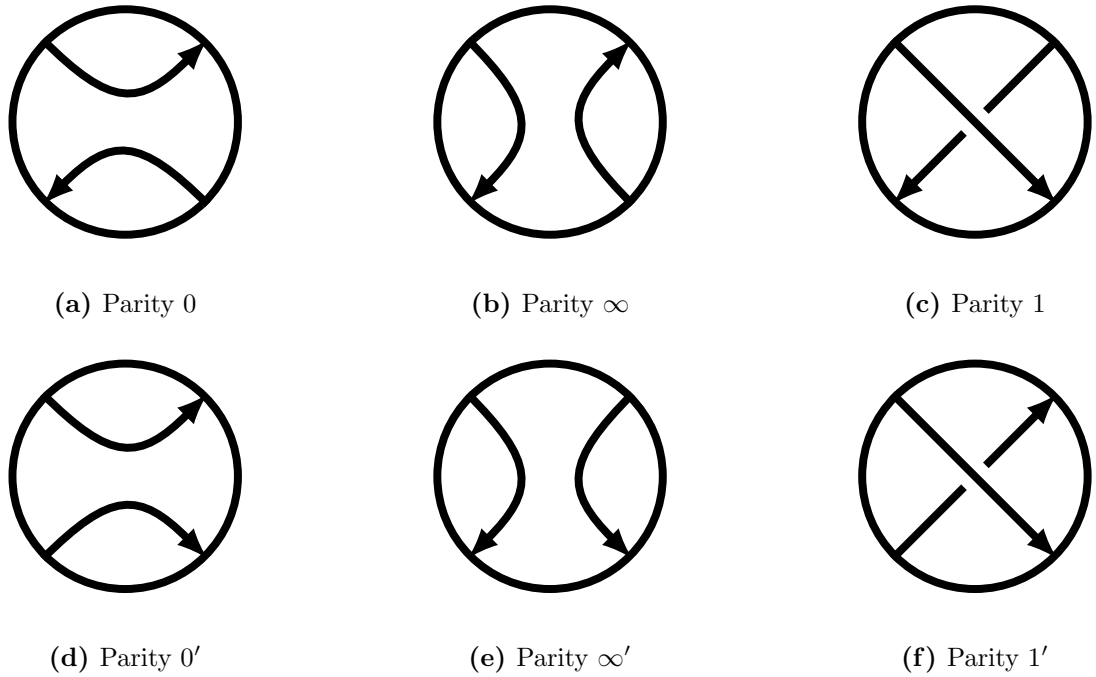
Much like knots, tangles can be oriented. An *orientation* on a tangle is a choice of direction for each strand. A choice of orientation on the strings also determines a *crossing sign* for each crossing in the diagram. A crossing is said to be either *positive* or *negative* depending on the orientations of the over-strand and the under-strand (Figure 2.5). Orientation and crossing sign are often used in diagram notations, so it will be helpful to choose some default conventions for working with generic tangle diagrams.

Each string inside a tangle connects two endpoints. For 2-string tangle diagrams, there are three ways for the entwined strands to connect the four corners of the bounding circle, as illustrated in the first row of Figure 2.6. This feature is known as the *parity* of a tangle. Ignoring orientation, a 2-string tangle is either parity 0, parity ∞, or parity 1, depending on which pairs of corners are connected by the interior strings.

We make an additional assumption for oriented 2-string tangle diagrams. Treating the NW corner as the natural starting place, we adopt the convention that the orientation always points inward from the NW corner. The string connected to the NW corner is considered

(a) Positive crossing **(b)** Negative crossing

Figure 2.5: Possible crossing signs in oriented diagrams



(a) Parity 0 **(b)** Parity ∞ **(c)** Parity 1

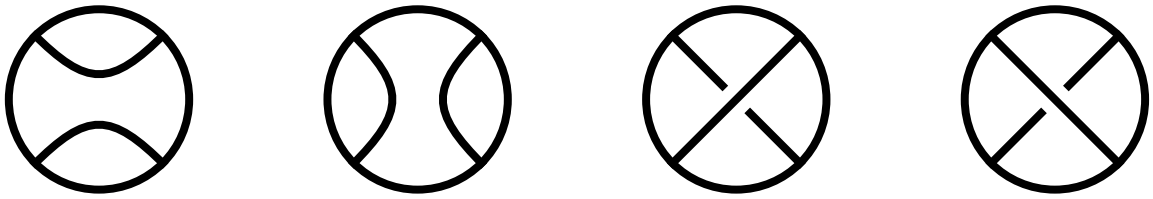
(d) Parity $0'$ **(e)** Parity ∞' **(f)** Parity $1'$

Figure 2.6: The six possible parities of an oriented 2-string tangle diagram

to be the first strand; there are three possible positions for the second strand depending on parity. When considering a tangle diagram in isolation, the default orientation will match that shown in the first row of Figure 2.6.

In some contexts, it will be helpful to choose the reverse direction on the second strand, as shown in the second row of Figure 2.6. To distinguish these cases, we expand the definition of parity to account for the orientation on the second strand. There are six possible parities for *oriented* 2-string tangles matching the figure: 0, ∞, 1, $0'$, ∞', and $1'$. These additional parities allow for consistent orientations on subtangles when finding a subtangle decomposition of an oriented diagram.

Ignoring orientations again, the two possible zero crossing tangles are referred to as *trivial tangles* (Figures 2.7a and 2.7b). These are the simplest examples of a tangle having parity 0 or parity ∞, and they are often denoted simply by 0 and ∞, respectively. There are two possible 1-crossing tangles, which differ from each other by a crossing change (Figures 2.7c and 2.7d). These are denoted $+1$ and -1, respectively, matching the slope of the over-strand. Also notice that assigning these tangles the default parity 1 orientation yields a positive crossing for the $+1$ tangle and a negative crossing for the -1 tangle. The four simple tangles shown in Figure 2.7 will often arise in proofs involving diagram manipulations.



(a) 0 **(b)** ∞ **(c)** $+1$ **(d)** -1

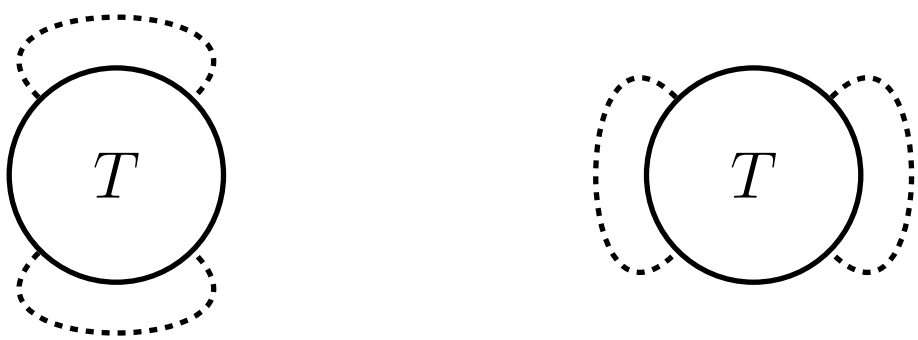
Figure 2.7: The four simplest 2-string tangles: 0, ∞, $+1$, and -1

2.2 Operations on 2-String Tangles

If tangles are the building blocks of knots, then tangle operations are the mortar. These describe the many ways to manipulate, transform, and combine tangles to build more complicated structures, actions which are easily visualized using tangle diagrams. Reverse engineering a diagram to identify such constructions is the essential goal of diagram classification.

2.2.1 Closure

A knot or link can be recovered from a tangle by joining the endpoints together. For 2-string tangles, the two most common ways to do this are by capping off the north and south endpoints, known as *numerator closure* (Figure 2.8a) or by capping off the east and west endpoints, known as *denominator closure* (Figure 2.8b).



(a) Numerator Closure: $N(T)$ **(b)** Denominator Closure: $D(T)$

Figure 2.8: Tangle closures obtained by connecting endpoints

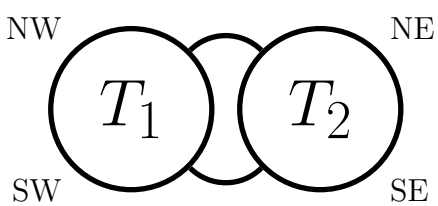
While these two closures are the most obvious, any choice of strings joining the endpoints defines a valid closure, even if those strings are knotted. Whether these yield a knot or a 2-component link depends on the parity of the tangle. For oriented tangles, note that certain choices of closures may be inconsistent with the direction on the strings. Closures offer a simple way to extend properties of knots, such as invariants, to tangles.

2.2.2 Sum and Product

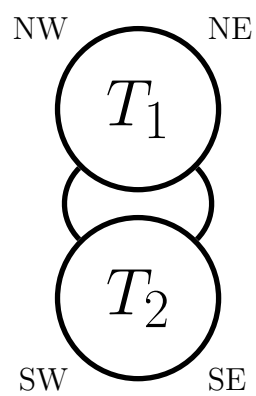
Two tangles can be combined to form a single, larger tangle by gluing together their endpoints. Attaching two tangles horizontally is known as *tangle sum*, denoted $T_1 + T_2$ (Figure 2.9a). Attaching two tangles vertically is known as *tangle product*, denoted $T_1 * T_2$ (Figure 2.9b). Observe that T_1 and T_2 become subtangles when combined in this way. In general, these operations are not commutative.

The parity of the combined tangle will depend on the parities of T_1 and T_2. Table 2.1 summarizes all of the possible combinations of tangle sum and product and the effect these have on parity. It is possible that these operations will yield a closed loop, such as a product of two parity 0 tangles. These types of combinations do not arise in our search for subtangle decompositions and we exclude them from consideration. We define the operations of sum and product on unoriented tangles, but these extend naturally to oriented diagrams provided the directions of the strings are consistent.

Since tangle sum and product always yield a new 2-string tangle, these operations can be



(a) Tangle Sum: $T_1 + T_2$



(b) Tangle Product: $T_1 * T_2$

Figure 2.9: Algebraic operations for combining tangles

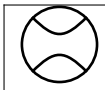
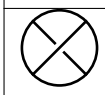
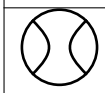
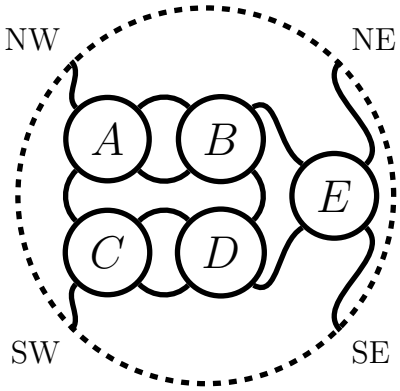
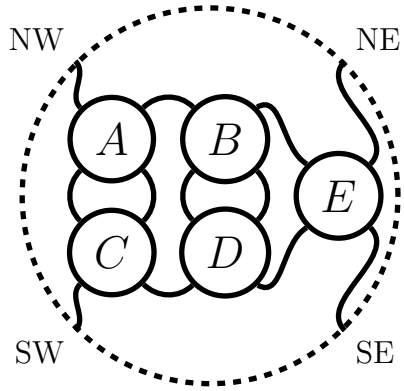
T_1		T_2		$T_1 + T_2$		$T_1 * T_2$	
tangle	parity	tangle	parity	tangle	parity	tangle	parity
	0		0	$0+0$	0	$0*0$	invalid
			1	$0+1$	1	$0*1$	0
			∞	$0+\infty$	∞	$0*\infty$	0
	1		0	$1+0$	1	$1*0$	0
			1	$1+1$	0	$1*1$	∞
			∞	$1+\infty$	∞	$1*\infty$	1
	∞		0	$\infty+0$	∞	$\infty*0$	0
			1	$\infty+1$	∞	$\infty*1$	1
			∞	$\infty+\infty$	invalid	$\infty*\infty$	∞

Table 2.1: Effect of tangle sum and product on parity

applied successively to build up more intricate tangles. An *algebraic construction formula* is an expression involving tangle sums and products which describes how to build up a tangle diagram (Figure 2.10). Tangle sum and product are associative operations, but parentheses should be used to determine the order of operations in formulas involving both.



(a) $((A+B)*(C+D))+E$



(b) $((A*C)+(B*D))+E$

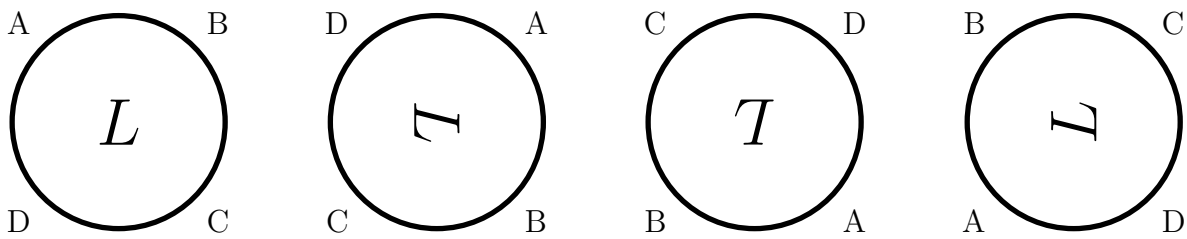
Figure 2.10: Examples of algebraic construction formulas

Construction formulas give a natural subtangle decomposition and are a particularly compact way to denote certain types of diagrams. Tangle algebra makes it very convenient to describe these types of diagrams using a formula, although strictly speaking this is not a proper diagram notation; many diagrams cannot be described using simple tangle algebra. Nevertheless, tangle sum and product are very useful tools for visualizing diagram constructions and will be leveraged in later proofs.

2.2.3 Transformations

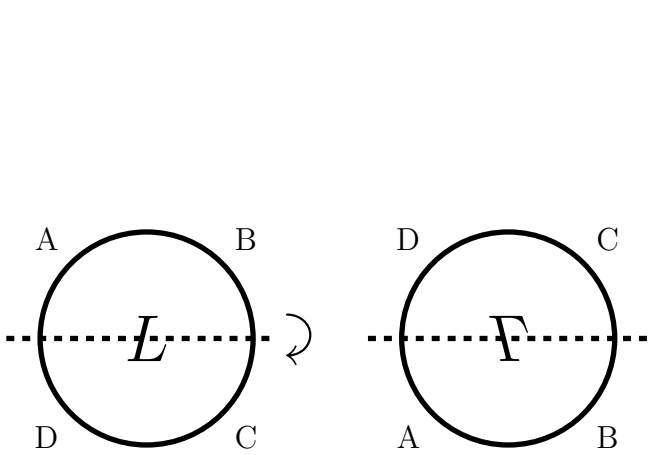
Recall that our definition of tangle requires that the endpoints of the strings be fixed on the surface of the ball. Except for some special cases, operations which move the ball change the equivalence class of the tangle. To study these special cases in more detail, we will catalog some of the commonly used transformations. These transformations are are also very useful for describing local operations on subtangles which preserve the equivalence class of a knot or tangle.

For a given tangle diagram L, let the notation $R_\theta(L)$ denote the clockwise rotation of L by θ degrees in the plane. Since we are working with 2-string tangles, we require that the angle θ be a multiple of 90 degrees. Figure 2.11 shows the rotations' effects on a tangle's endpoints. While we visualize a rotation using diagrams, this transformation is equivalent to permuting the end points of the original tangle. This will have a corresponding effect on the parity of the tangle diagram. Under a 90 degree rotation, 0 switches to ∞, ∞ switches to 0, and a parity 1 diagram remains parity 1.

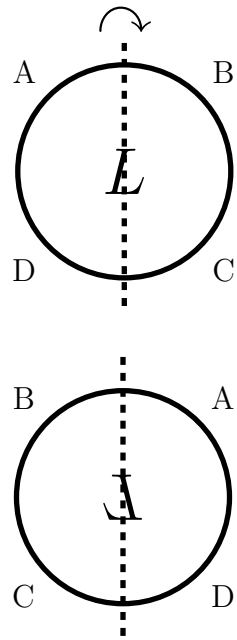


(a) $R_0(L)$ **(b)** $R_{90}(L)$ **(c)** $R_{180}(L)$ **(d)** $R_{270}(L)$

Figure 2.11: Clockwise rotations of a tangle L by θ degrees: $R_\theta(L)$



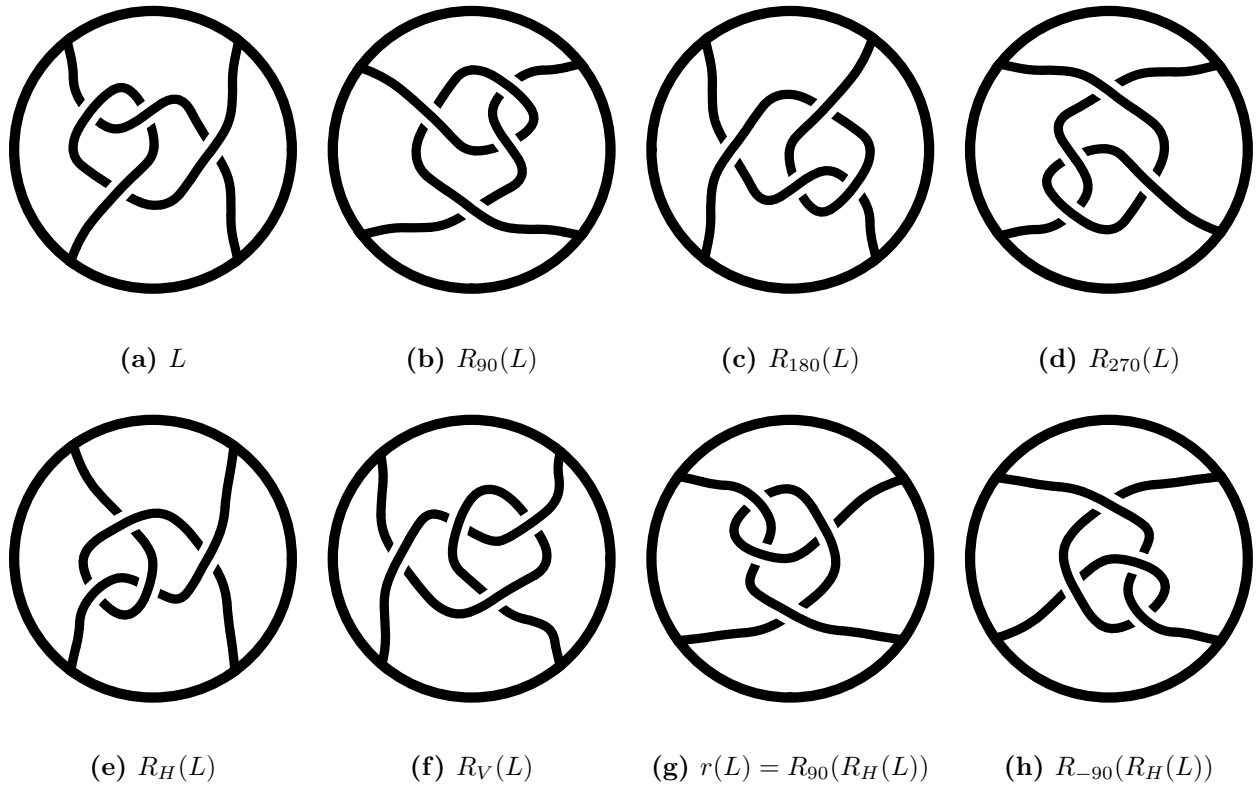
(a) Flip over horizontal axis: $R_H(L)$



(b) Flip over vertical axis: $R_V(L)$

Figure 2.12: 180 degree rotations of a tangle L over the horizontal and vertical axes

In addition to rotating a tangle diagram in the plane, we can also flip it across an axis. In terms of the three dimensional tangle, this corresponds to a 180 degree rotation along the chosen axis. To denote this transformation, we introduce the notation $R_H(L)$ and $R_V(L)$ to denote a rotation across either the horizontal or vertical axis, respectively (Figure 2.12). The final position will be the same for either a clockwise or counterclockwise rotation, so



(a) L **(b)** $R_{90}(L)$ **(c)** $R_{180}(L)$ **(d)** $R_{270}(L)$

(e) $R_H(L)$ **(f)** $R_V(L)$ **(g)** $r(L) = R_{90}(R_H(L))$ **(h)** $R_{-90}(R_H(L))$

Figure 2.13: Examples of tangle transformations corresponding to symmetries of a square

we make no distinction between these two directions. This transformation will permute the positions of the endpoints in the tangle diagram as shown in the figure.

We can also imagine flipping a tangle diagram across one of the diagonal axes. This transformation can also be achieved by composing flips and rotations in the plane. For example, $R_{90}(R_H(L))$ is equivalent to rotating a tangle L 180 degrees across the NW-SE diagonal axis; $R_{-90}(R_H(L))$ is equivalent to a rotation over the NE-SW diagonal axis. The first of these two operations is used frequently enough to warrant its own shorthand. We will use $r(L) = R_{90}(R_H(L))$ to denote flipping L across the NW-SE diagonal.

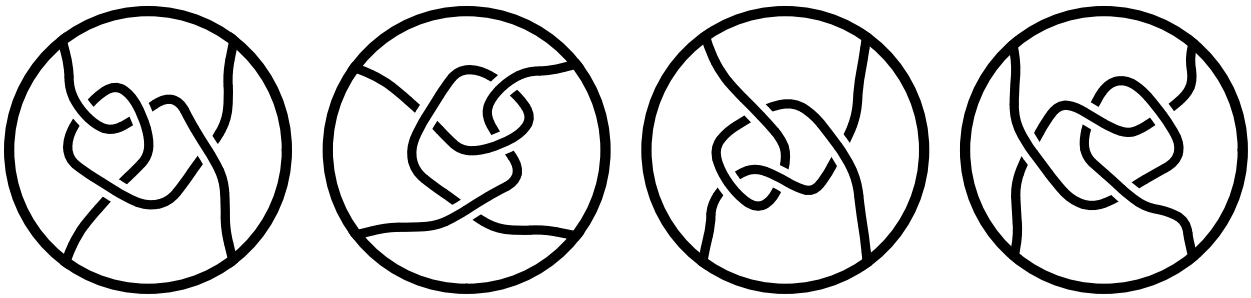
In fact, each of the transformations introduced so far corresponds to one of the symmetries of a square. Recall that these symmetries form a group of order 8. Hence, there exist a total of eight distinct positions for a tangle diagram that can be obtained by rotating the boundary sphere of the tangle. Figure 2.13 shows an example of each of these positions for a simple tangle. If the tangle endpoints are allowed to move on the boundary of the ball, then each of these diagrams would describe equivalent tangles.

(a) L **(b)** $-L$

Figure 2.14: Reversing the over- and under-strands: mirror image

Consider the effect of these transformations on individual crossings. Recall that the two 1-crossing tangles, $+1$ (Figure 2.7c) and -1 (Figure 2.7d), were distinguished by the slope of the over-strand. A 90 degree rotation in the plane reverses this slope, so that $R_{90}(\pm 1) \cong \mp 1$. However, a flip across the horizontal or vertical axis effectively switches the over- and under-strands in addition to reversing the slope. In other words, $R_H(\pm 1) \cong \pm 1 \cong R_V(\pm 1)$. This interesting behavior will enable us to describe transformations of more complicated tangle diagrams in later sections.

Another way to transform a tangle diagram is by reversing the over- and under-strands at each crossing, an operation known as the *mirror image* (Figure 2.14). The mirror image of a tangle L will be denoted as $-L$. Notice that $- \pm 1 = \mp 1$, and we will see later that a similar relationship holds for rational tangles in general. Figure 3.3 shows the effect of combining the mirror image with some of the transformations of Figure 2.13.



(a) $-L$ **(b)** $R_{90}(-L)$ **(c)** $R_H(-L)$ **(d)** $R_V(-L)$

Figure 2.15: More examples of tangle transformations involving mirror images

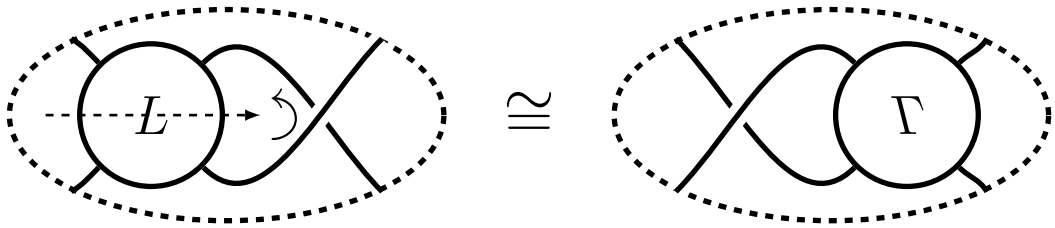


Figure 2.16: Example of a horizontal flype: $L + 1 \cong 1 + R_H(L)$

We conclude our discussion of tangle operations with some theorems. When performing various moves in a localized part of a diagram, transformations are useful for describing the effect of these moves on subtangles. An example of one such move first introduced by Tait [37] is a *flype* (Figure 2.16).

Lemma 2.1 (Flyping [15])**:**
For any 2-string tangle L, we have the following tangle equivalencies.

$$\pm 1 + L \cong R_H(L) + \pm 1 \qquad\qquad \pm 1 * L \cong R_V(L) * \pm 1$$

The equivalent tangles are said to be related by a horizontal or vertical *flype*.

Proof:

For this proof, we refer to figure 2.16, which shows a horizontal flype through a positive twist. A similar proof holds for a negative twist or vertical flype.

We imagine the isotopic motion of sliding the subtangle L through this twist in a corkscrew motion. As long as the endpoints of the strands remain fixed on the dashed circle, all interior movements involving rotations, flips, and wiggles of the subtangle L preserve the equivalence class of the tangle as a whole. By flipping L upside down while sliding, the figure shows how this motion unwinds the twist on the right side of the subtangle, but introduces a new twist on the left side. Observe that the slope of the over-strand in the new twist must match that of the old twist. Using the tangle notations previously introduced, this figure shows that $L + 1 \cong 1 + R_H(L)$.

□

Many of the diagrams we are interested in studying have constructions which are most easily described using tangle sums and products. This motivates the next theorem, which

enumerates the effects of applying various transformations to a sum or product of subtangles. Figure 2.17 gives a visual summary of these transformations. The formulas are also useful for writing programs to compute subtangle decompositions.

Theorem 2.2 (Transformations with tangle sum and product)**:**
Let L_1 and L_2 be two diagrams for 2-string tangles. Consider the effect of applying transformations on $L_1 + L_2$ and $L_1 * L_2$. We have the following list of tangle equivalences.

a. $R_H(L_1 + L_2) \cong R_H(L_1) + R_H(L_2)$

b. $R_V(L_1 + L_2) \cong R_V(L_2) + R_V(L_1)$

c. $R_H(L_1 * L_2) \cong R_H(L_2) * R_H(L_1)$

d. $R_V(L_1 * L_2) \cong R_V(L_1) * R_V(L_2)$

e. $R_{90}(L_1 + L_2) \cong R_{90}(L_1) * R_{90}(L_2)$

f. $R_{180}(L_1 + L_2) \cong R_{180}(L_2) + R_{180}(L_1)$

g. $R_{270}(L_1 + L_2) \cong R_{270}(L_2) * R_{270}(L_1)$

h. $R_{90}(L_1 * L_2) \cong R_{90}(L_2) + R_{90}(L_1)$

i. $R_{180}(L_1 * L_2) \cong R_{180}(L_2) * R_{180}(L_1)$

j. $R_{270}(L_1 * L_2) \cong R_{270}(L_2) + R_{270}(L_2)$

k. $-(L_1 + L_2) \cong (-L_1) + (-L_2)$

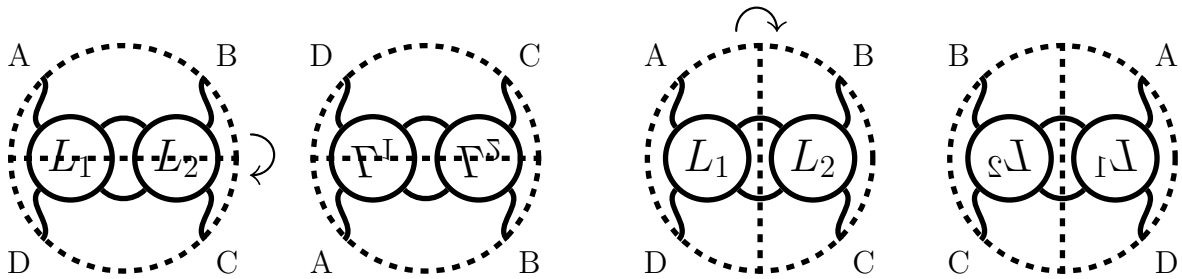
l. $-(L_1 * L_2) \cong (-L_1) * (-L_2)$

m. $r(L_1 + L_2) = R_{90}(R_H(L_1 + L_2)) \cong R_{90}(R_H(L_1)) * R_{90}(R_H(L_2)) = r(L_1) * r(L_2)$

n. $r(L_1 * L_2) = R_{90}(R_H(L_1 * L_2)) \cong R_{90}(R_H(L_1)) + R_{90}(R_H(L_2)) = r(L_1) + r(L_2)$

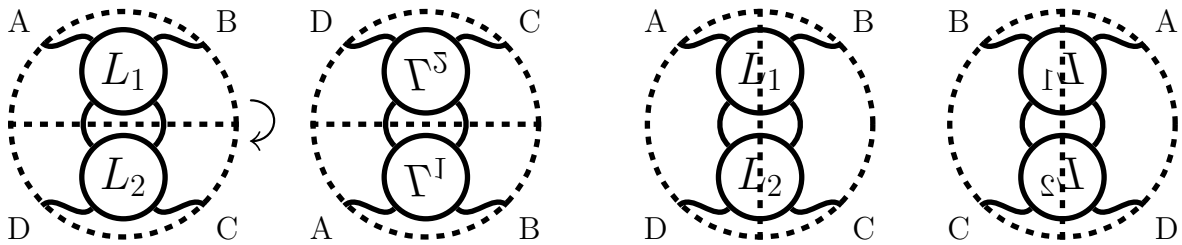
Proof:

The first ten equivalences are proven visually in Figure 2.17. The two equivalences involving mirror images follow immediately from switching the over-strands and under-strands in a tangle diagram. The final two equivalences involving diagonal flips are computed using a composition of R_{90} with R_H. □



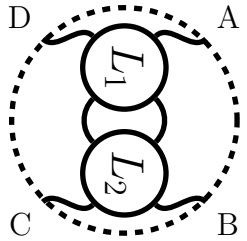
(a) $R_H(L_1 + L_2) \cong R_H(L_1) + R_H(L_2)$

(b) $R_V(L_1 + L_2) \cong R_V(L_2) + R_V(L_1)$

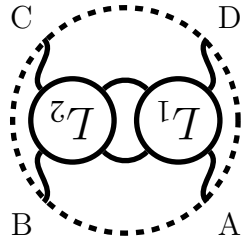


(c) $R_H(L_1 * L_2) \cong R_H(L_2) * R_H(L_1)$

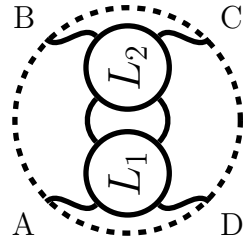
(d) $R_V(L_1 * L_2) \cong R_V(L_1) * R_V(L_2)$



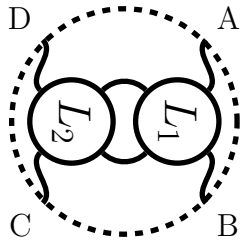
(e) $R_{90}(L_1 + L_2) \cong R_{90}(L_1) * R_{90}(L_2)$



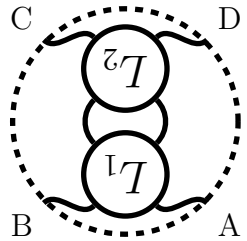
(f) $R_{180}(L_1 + L_2) \cong R_{180}(L_2) + R_{180}(L_1)$



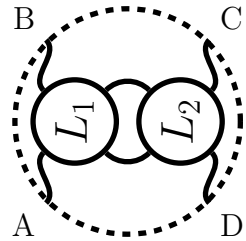
(g) $R_{270}(L_1 + L_2) \cong R_{270}(L_2) * R_{270}(L_1)$



(h) $R_{90}(L_1 * L_2) \cong R_{90}(L_2) + R_{90}(L_1)$



(i) $R_{180}(L_1 * L_2) \cong R_{180}(L_2) * R_{180}(L_1)$



(j) $R_{270}(L_1 * L_2) \cong R_{270}(L_1) + R_{270}(L_2)$

Figure 2.17: Effect of applying transformations to tangle sum and product

$$-4 \mid -6 \quad -2$$

(a) Dowker code

$$|)a1{+}b2{+}(|)a3{+}b1{+}a2{+}b3{+}(|$$

(b) Gauss code

$$X_{5,2,6,1}X_{2,7,3,6}X_{7,5,8,4}$$

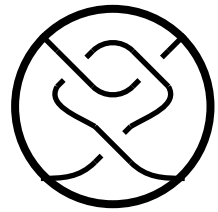
(c) Planar diagram code

$$\begin{bmatrix} 1 & -2 & 0 & 1 & 0 \\ -2 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & -2 & 1 \end{bmatrix}$$

(d) Coloring matrix

$$\begin{array}{ccccc} + & 2b & 3a & 0\ \mathrm{NW} & 2c \\ + & 3b & 1a & 1d & 0\ \mathrm{NE} \\ + & 1b & 2a & 0\ \mathrm{SE} & 0\ \mathrm{SW} \end{array}$$

(e) Crossing planar diagram code



(f) Tangle diagram

Figure 2.18: Examples of notations for the tangle diagram shown in 2.18f

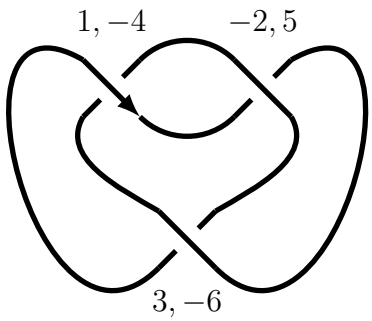
2.3 Diagram Notations

The last part of this section is devoted to explaining some of the most commonly used diagram notations. Figure 2.18 shows an example of five different notations for the tangle diagram of 2.18f. Each of the notations listed above was originally introduced to describe knot diagrams, but these can be extended to tangle diagrams using closures. Notations are most easily understood through examples, and so we will briefly summarize how to compute each of the notations shown in Figure 2.18, starting with the corresponding knot diagram notation. Generalizing these to tangle diagrams requires modifying the notation to account for the endpoints of the strands.

2.3.1 Dowker Code

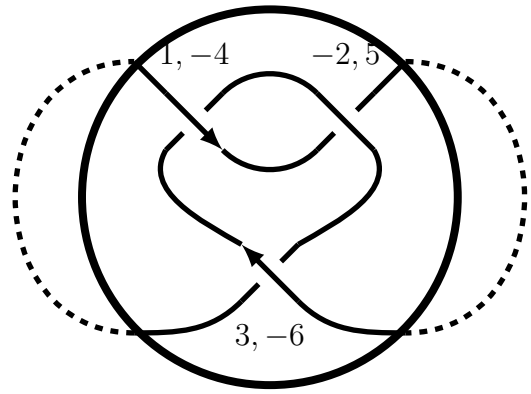
One of the simplest diagram notations was developed by Dowker and Thistlewaite [11]. It is properly known as the *Dowker-Thistlewaite code*, but often shortened to just Dowker code. A detailed method for computing the Dowker code by hand is outlined in Adams, as well as the method for reconstructing a diagram from a given Dowker code [1]. The method described here is based on Adams' with a few adjustments for consistency.

To calculate the Dowker code of a knot diagram, we begin by fixing an orientation and choosing a starting point somewhere along the string. Traveling along the strand in the direction of the orientation, we label each crossing sequentially with an integer until returning to the starting position. Note that each crossing is encountered twice, once on the



1	3	5
−4	−6	−2

(a) Knot Dowker code: $-4\ -6\ -2$



1	3	5
−4	−6	−2

(b) Tangle Dowker code: $-4 \mid -6\ -2$

Figure 2.19: Example of diagram annotations for computing the Dowker code

over-strand and once on the under-strand, and hence will be labeled with two integers: one even and one odd. We include the extra rule that, if passing along the under-strand on an even integer, then the label is made negative.

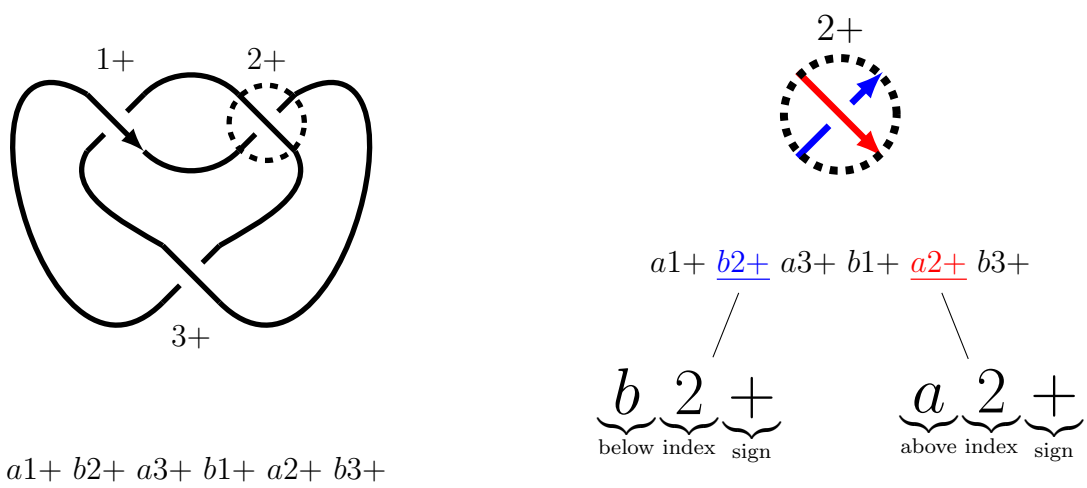
Figure 2.19a shows an example of this labeling for the knot diagram obtained from the denominator closure of the tangle diagram in Figure 2.18. A choice of orientation is made and the NW corner of the diagram is selected as the starting place. After the crossings are labeled, the pairs of labels are ordered so that the odd integer labels are increasing. The Dowker code of this diagram is the corresponding sequence of even integers. In general, a single diagram can have more than one valid Dowker code since we can choose a different starting location on the string.

With some additional rules, we use a similar method to compute the Dowker code for a tangle diagram. We adopt the convention that the NW corner is always chosen as the starting place and the strands are assigned the default orientation matching the tangle parity. For the annotated tangle diagram shown in Figure 2.19b, the orientation on the strands matches the default direction for a parity 0 tangle, described in Figure 2.6a. The two oriented strands are traced out in order and each crossing is labeled with two integers, just as before. When ordering the pairs of labels, a bar is placed to indicate the break between the first and second strands. The position of this bar precedes the odd integer label of the first crossing encountered on the second strand.

The tangle Dowker code is the sequence of even integers including the bar. In general, the parity of the tangle diagram must also be specified. It is possible that two tangle diagrams of different parities have the same Dowker code. There is another potential for ambiguity, and so some additional comments should be made concerning the Dowker code in connection with the larger goal of tangle tabulation. We've commented that a given knot diagram can have multiple valid Dowker codes. In general, non-equivalent diagrams can also have identical Dowker codes in the case of composite knots, although these can be avoided with certain additional restrictions. Tangles containing connect sums with knots are said to be locally knotted, and we choose to exclude these from the tabulation project.

2.3.2 Gauss Code

Another notation for describing diagrams is known as the *Gauss code*. This method for encoding a diagram is originally attributed to Gauss and there are numerous variations on the notation. These are all based on the idea of representing a diagram as a string of symbols, sometimes referred to as a *Gauss word*, which encodes information about each crossing. We walk through how to compute one such Gauss notation for the diagram shown in Figure 2.20. While all Gauss codes enumerate a sequence of crossing indices, the variation we will prefer also incorporates crossing sign and strand position. These features generally make the Gauss code easier to read than the Dowker code.



(a) Gauss code for a knot diagram

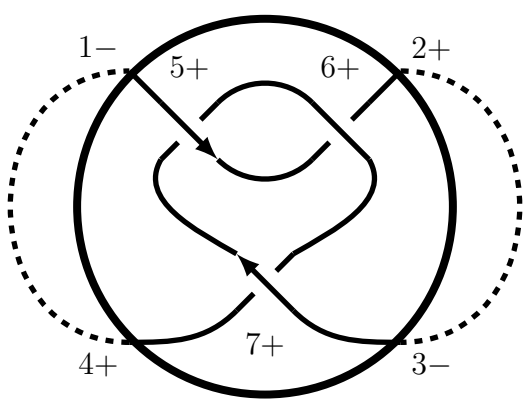
(b) Detailed Guass crossing labels

Figure 2.20: Visual explanation of the Gauss code labeling conventions

The first step is to enumerate each crossing in a diagram. We choose an orientation and a starting point somewhere along the strand, and then index each crossing with an integer until returning to the starting position. The indices are not required to increase sequentially, but we will follow the convention that the crossing index matches the order in which we first encountered the crossing while traveling along the strand. Unlike the Dowker code, each crossing is only indexed once, af the first pass. We also mark the crossing with $+$ or $-$, matching with the crossing sign. Figure 2.20a shows this labeling.

After each crossing is indexed in this way, we return to the starting position and trace out the string one more time. Each time we pass through a crossing, we record three pieces of information: whether we pass above (a) or below (b) the crossing, the index of the crossing, and the crossing sign. Figure 2.20b shows this information in detail for the second crossing in the diagram. Upon returning to the starting point, we obtain a string of symbols describing each crossing in the order in which we passed through them. This notation is the Gauss code for the diagram. Note that each crossing is listed twice in the code: once on the over-strand and once on the under-strand.

The Gauss code can be extended to tangle diagrams by accounting for the behavior of the endpoints. Figure 2.21 shows two examples of tangle Gauss codes which also incorporate some of my own modifications. We will always follow the conventions of starting at the NW corner of the tangle and assigning the two strands the default orientation based on parity. For the full Gauss code, we also treat the boundary circle as though it's a separate



(a) Full Gauss code: $a1{-}a2{+}a3{-}a4{+}|b1{-})\mathbf{a5}{+}\mathbf{b6}{+}(b2{+}|b3{-})\mathbf{a7}{+}\mathbf{b5}{+}\mathbf{a6}{+}\mathbf{b7}{+}(b4{+}|$

(b) Simplified Gauss code: $|)a1{+}b2{+}(|)a3{+}b1{+}a2{+}b3{+}(|$

Figure 2.21: Gauss code for a tangle diagram

unlinked component sitting on top of the strands' endpoints. Using a clockwise orientation for this circle, we imagine four additional crossings at the endpoints and index these before the interior crossings.

To compute the full Gauss code, we compute the Gauss word for this diagram just as in Figure 2.20. We then insert bars and parentheses as shown in 2.21a. Bars are used to separate each of the three components: the exterior circle and the two interior strands. Parentheses are placed to separate interior crossings from crossings on the circle. These additional symbols also make reading the notation more fluid since often we are only interested in examining the interior crossings. Note that the four crossings along the circle make it possible to distinguish the diagram's parity, something lacking from the Dowker code.

However, this also means that every parity 0 tangle will include the expressions

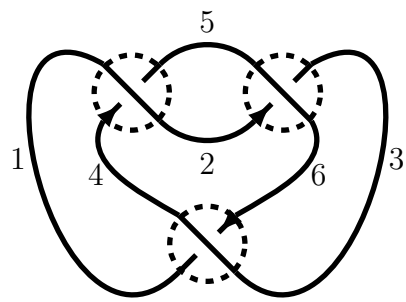
$$a1{-}a2{+}a3{-}a4{+}|b1{-})\cdots(b2{+}|b3{-})\cdots(b4{+}|$$

in its full Gauss code. Similar facts hold for parity 1 and parity ∞ tangles. Hence, as long as one keeps track of the tangle's parity, these additional crossings are unnecessary. This is the logic behind the simplified Gauss code for tangles, which only uses the internal crossings. Notice that the Gauss code shown in 2.21b matches the knot Gauss code of Figure 2.20a, but with the breaks between strands. In general, we will prefer the simplified Gauss code and this notion is commonly used in the computational algorithms discussed later.

2.3.3 Planar Diagram Code

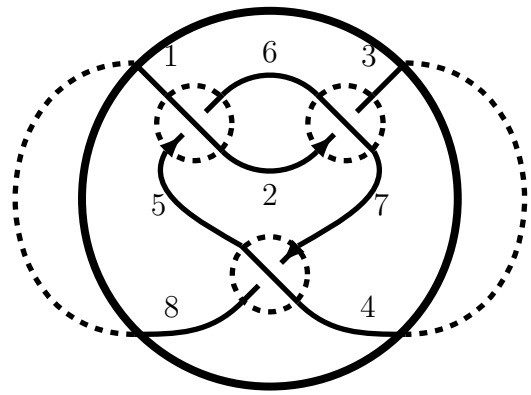
The next notation we consider is known as the *planar diagram* (PD) code. This notation for knots and links was developed by Dror Bar-Natan and his students as part of the KnotTheory Mathematica package, which is described in detail on the the Knot Atlas [2]. Additional properties and results concerning this notation are also discussed in [24]. As before, we will walk through how to compute this code for knot and tangle diagrams with an example (Figure 2.22).

Instead of indexing crossings, the planar diagram code indexes string segments. Choosing an orientation and a starting location on the string, we travel along the knot and label each arc between crossings sequentially with an integer. Note that we move to a new string segment when passing along an over-strand even through the arc looks unbroken in the diagram. Figure 2.22a shows an example of these labels for the three crossing knot of the previous examples. An n-crossing knot will have $2n$ string segments.



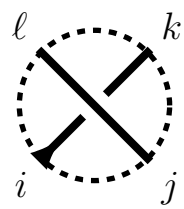
$X_{4,2,5,1}X_{2,6,3,5}X_{6,4,1,3}$

(a) Planar diagram code for a knot diagram



$X_{5,2,6,1}X_{2,7,3,6}X_{7,5,8,4}$

(b) Planar diagram code for a tangle diagram



(c) Planar diagram crossing labels: $X_{i,j,k,\ell}$

Figure 2.22: Example of diagram annotations for the planar diagram code

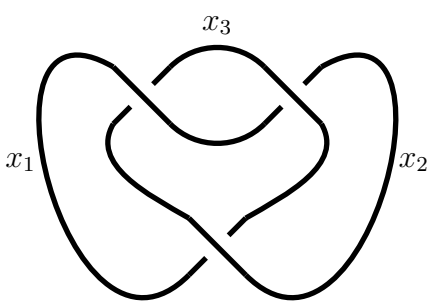
Next, we assign each crossing a label $X_{i,j,k,\ell}$ as shown in Figure 2.22c. Each crossing has four endpoints. The first endpoint i matches the index of the arc at the inward-pointing under-strand. The remaining endpoints are listed moving counterclockwise from the first. Thus, each crossing encodes the indices of the adjacent string segments. The corresponding PD code for this diagram is the list of these crossing labels, where the crossings are ordered by when we pass through them while traveling along the strand. Since each arc connects two crossings, each index on a string segment is used in exactly two crossing labels.

The planar diagram notation readily extends to tangle diagrams. The only difference is that the arcs connecting to endpoints will only be used in one crossing label. We follow the usual convention of starting at the NW corner and using the default orientation based on parity. For a 2-string tangle with n crossings, there will be $2n + 2$ string segments (Figure 2.22b). The previous ambiguities concerning parity are also possible with the planar diagram code, and so parity should be specified when using this notation for tangle diagrams.

2.3.4 Coloring Matrix

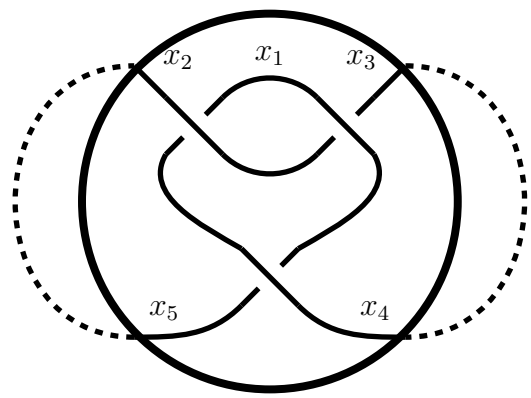
The *coloring matrix* is a method for describing a diagram using a system of equations. Each string segment is indexed with a variable and each crossing defines an equation. The coloring matrix was originally used to compute a knot invariant known as Fox coloring [13]. Darcy and Navarra-Madsen showed how to extend this invariant to n-string tangles [10, 9]. In addition to its use for computing invariants, the coloring matrix is itself a diagram notation well suited for computer implementation. We continue our discussion of notations by walking through an example of how to compute this matrix for a diagram (Figure 2.23).

Beginning with the 3-crossing knot diagram considered earlier, each string segment in the diagram is labeled with a variable x_i (Figure 2.23a). Unlike the planar diagram notation,



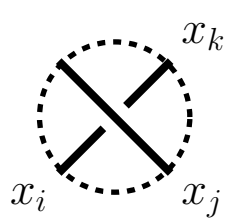
$$\begin{bmatrix} -2 & 1 & 1 \\ 1 & 1 & -2 \\ 1 & -2 & 1 \end{bmatrix}$$

(a) Coloring matrix for a knot diagram



$$\begin{bmatrix} 1 & -2 & 0 & 1 & 0 \\ -2 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & -2 & 1 \end{bmatrix}$$

(b) Coloring matrix for a tangle diagram



(c) Crossing equation: $x_i - 2x_j + x_k$

Figure 2.23: Example of the system of equations defining the coloring matrix

passing through a crossing along the over-strand does not split an arc into two different string segments. Hence, the number of string segments in a knot diagram will match the number of crossings. The diagram does not need to be oriented nor do the arcs need to be ordered, although we will choose to index the arcs in the order we pass through them while traveling along the string.

Each crossing defines an equation of the form $x_i - 2x_j + x_k$, where x_j denotes the variable of the over-strand and x_i and x_k correspond to the two under-strands (Figure 2.23c). The coloring matrix for this diagram comes from representing this system of equations in matrix form. The columns represent the arcs and the rows can be thought of as representing crossings. Since the number of arcs will be the same as the number of crossings, the coloring matrix will always be square for a knot. Any row equivalent matrix corresponds to a re-labeling of the arcs and is also a valid coloring matrix for the diagram.

We use a similar approach to find the coloring matrix for a tangle diagram, although we include some additional conventions concerning the endpoints that are important for computing tangle coloring invariants. In general, a diagram for an n-string tangle with m crossings will have $m+n$ string segments. Since the number of equations matches the number of crossings, the corresponding system of equations defines a non-square $m \times (m+n)$ matrix. The endpoint arcs behave somewhat differently from the interior arcs, and so the convention is to assign variables to the string segments such that these endpoint arcs come last. This way, the last $2n$ columns of the matrix always describe the endpoints.

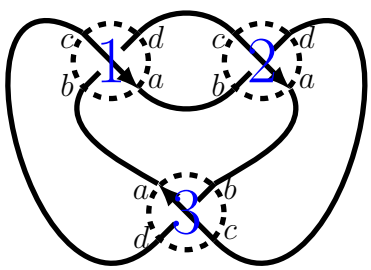
Consider this rule in the context of the 2-string tangle with three crossings shown in Figure 2.23b. The coloring matrix will define a system of three equations in five variables, corresponding to the three crossings and five string segments. Four of these five arcs connect to the tangle endpoints, and so we reserve the last four variables for these arcs. The one interior arc is labeled x_1 and the remaining four endpoints arcs are labeled clockwise, starting from the NW corner. This yields the 3×5 coloring matrix shown in the figure. In the special case where a single arc connects two endpoints, such as in the $+1$ and -1 tangles, this arc will be labeled twice: once for each endpoint.

As long as this convention for endpoints is preserved, the interior arcs can be labeled in any order and the diagram need not be oriented. As with knot diagrams, any row equivalent matrix is also a valid coloring matrix for the knot diagram. We will comment on how this notation can be used to compute an invariant in a later section.

2.3.5 Ewing-Millett Notation and Crossing Planar Diagram Code

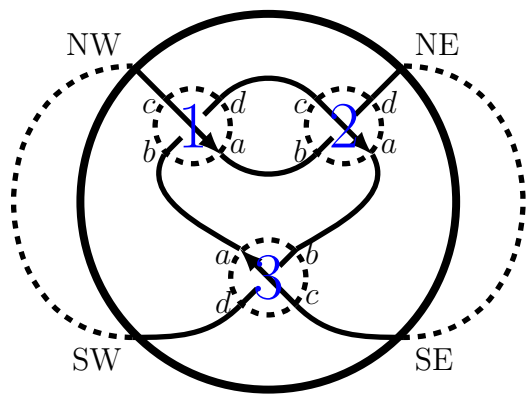
The next notation we consider was introduced in a paper by Ewing and Millett [12], which we will refer to as *Ewing-Millett notation*. Although it predates the planar diagram code, Ewing and Millett's method for computing their notation is quite similar to the method for the PD code. The principal difference is that the Ewing-Millett notation indexes the crossings rather than the string segments. Hence, we will refer to Ewing-Millett notation as the *crossing planar diagram* (CPD) code when we generalize it to tangles. This name also parallels my own subtangle planar diagram (SPD) code, which is based on Ewing-Millett notation, albeit with heavy modifications.

We will refer to Figure 2.24 to illustrate how to compute this notation. We begin by fixing



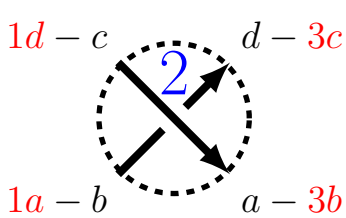
1	+	$2b$	$3a$	$3d$	$2c$
2	+	$3b$	$1a$	$1d$	$3c$
3	+	$1b$	$2a$	$2d$	$1c$

(a) Ewing-Millett code for a knot diagram



1	+	$2b$	$3a$	0 NW	$2c$
2	+	$3b$	$1a$	$1d$	0 NE
3	+	$1b$	$2a$	0 SE	0 SW

(b) Crossing planar diagram code for a tangle diagram



$$\underbrace{+}_{\text{sign}}\ \underbrace{3b}_{a}\ \underbrace{1a}_{b}\ \underbrace{1d}_{c}\ \underbrace{3c}_{d}$$

(c) Ewing-Millett crossing labels and corresponding row

Figure 2.24: Example of diagram annotations for the EM and CPD codes

an orientation for the diagram and enumerating each of the crossings. Strictly speaking, the crossings need not have a particular order, but we will choose to enumerate them in the order that we first pass through them while traveling along the strand. Each crossing considered in isolation has four endpoints. For each crossing, the corner matching the outward-pointing over-strand is labeled a. The remaining three endpoints are labeled b, c, and d, moving clockwise. Figure 2.24a shows an example of these crossing labels for a knot diagram.

For each crossing, we record five pieces of information in a row: the crossing sign, followed by a list of connections for each of the four corners starting at a. Figure 2.24c demonstrates this connection information visually: the position in the row represents a certain corner of the crossing; the value in the row describes the connected crossing. The Ewing-Milllett code for the knot diagram is the list of these rows, one for each crossing. Note that the blue numbers listed in Figure 2.24 next to the rows are merely a visual aid and are not part of the notation.

As with the other notations, extending this code to a tangle diagram requires accounting for endpoints. This is illustrated in Figure 2.24b. We will choose the standard tangle orientation based on parity and index the crossings in order while traveling along the strand, starting from the NW endpoint. We will also treat the exterior circle as though it were a crossing with index 0, and label the four endpoints using the compass directions. The crossing planar diagram code for this tangle diagram is then obtained in the same way as the Ewing-Millett code for a knot diagram, except that four entries in the code now refer to endpoint connections.

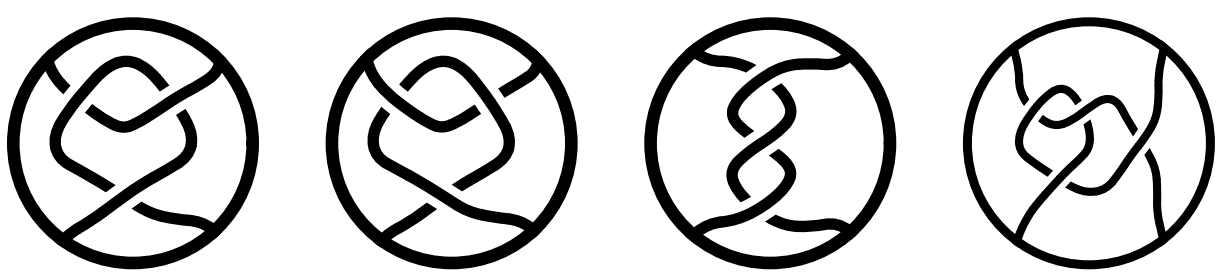
Some additional remarks should be made about the crossing planar diagram code. This is a very graph-like method for describing a diagram, where we imagine crossings as vertices and string segments as edges. The CPD is a list of connections for each crossings and is essentially equivalent to the adjacency list of the corresponding planar graph. With additional rules for subtangles, this method from Ewing and Millett for organizing a diagram will become the foundation of the subtangle planar diagram code.

3 Rational Tangles

In general, 2-string tangles can be divided into several increasingly complex classes based on their internal structure. The most well-studied of these is the class of rational tangles, which have a particularly rich and well-behaved form. The name *rational* is due to Conway, who first formalized these tangles and showed that they are in bijection with the extended rational numbers [7]. For this reason, we often denote rational tangles by a corresponding fraction $\frac{p}{q}$. Figure 3.1 shows some examples of rational tangles we have already encountered together with their corresponding fractions. The four simple tangles we previously denoted 0, ∞, $+1$, and -1 (Figure 2.7) are also special cases of rational tangles.

One of the beautiful properties of rational tangles is their formulaic construction. Any rational tangle diagram can be described in terms of a sequence of twists on two parallel strands, an idea which we will make rigorous in the coming section. This description also makes it easy to describe the effect of various transformations on rational tangles, such as flipping and rotating. This makes them particularly well-suited for describing subtangle decompositions, and generic 2-string tangle diagrams can be classified based on how they are constructed from rational subtangles.

Our objective in this section is to build up to Conway's major result establishing the bijection between rational tangles and the extended rational numbers. In addition to Conway's original paper, many of the results introduced here are adapted from an excellent introductory paper on rational tangles due to Goldman and Kauffman and a series of papers due to Kauffman and Lambropoulou [7, 14, 18, 19, 20]. Rational tangles are also the primary topic of Chapter 8 in the introductory text on knot theory by Cromwell [8].



(a) $\frac{2}{3}$ **(b)** $-\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{3}$

Figure 3.1: Examples of some rational tangles and their corresponding fractions

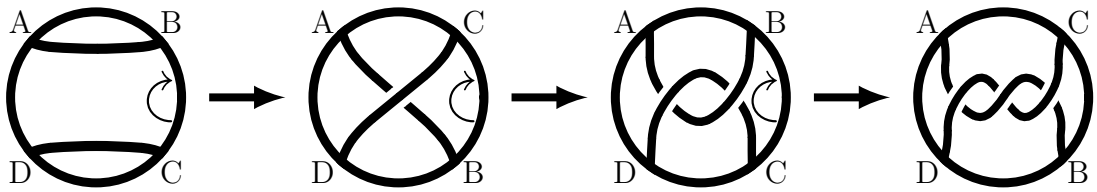


Figure 3.2: Integer tangle created from three horizontal twists

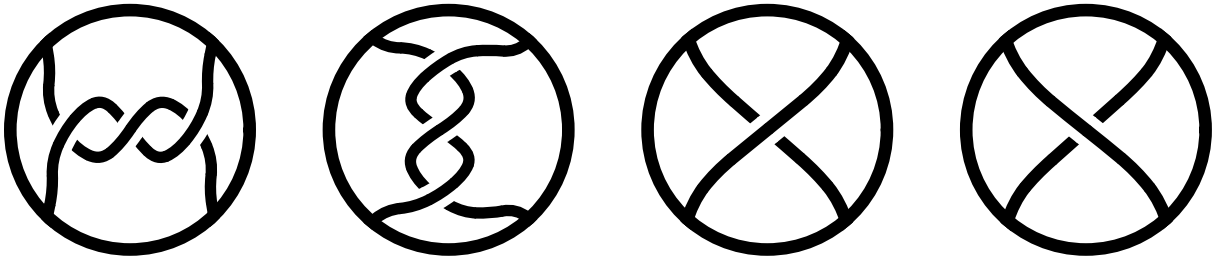
3.1 Constructing Rational Tangles

There are several equivalent definitions of rational tangle, but these tangles are most commonly characterized by a certain inductive construction. Intuitively, a 2-string tangle is said to be *rational* if it can be obtained from the 0 or ∞ tangle by applying finitely many twists to pairs of adjacent endpoints. Figure 3.2 shows an example of one such tangle constructed from three horizontal twists on the east endpoints. We can use the language of tangle operations to formalize this construction into a proper definition. Before this, we consider a special case of rational tangle with a very simple structure.

3.1.1 Integer Tangles

Integer tangles are the simplest type of rational tangle. They consist purely of a sequence of horizontal or vertical twists. The tangle shown in Figure 3.2 is a horizontal integer tangle with three twists. Twisting is considered horizontal (Figure 3.3a) or vertical (Figure 3.3b) relative to the tangle's NW corner. These are known as *horizontal* or *vertical* integer tangles, respectively. Moreover, twists may be either positive (Figure 3.3c) or negative (Figure 3.3d) depending on the slope of the over-strand.

Twisting the endpoints of a tangle diagram is equivalent to a tangle sum or product with either the $+1$ or -1 tangle. These correspond to postive twisting or negative twisting, respectively. Note that twist sign is different from crossing sign; twist sign only depends on the slope of the over-strand, it need not match the crossing sign nor must the diagram be oriented. Based on this idea of twisting, integer tangles may also be understood as a combination of 1-crossing subtangles. Horizontal tangles represent a tangle sum, while vertical tangles represent a tangle product. Note that twists of opposite sign will cancel each other through a Reidemeister II move (Figure 3.4). Hence, an integer tangle will always reduce to some number of twists of the same sign, and this determines whether the integer tangle is positive or negative.



(a) Horizontal Tangle $\frac{3}{1}$ **(b)** Vertical Tangle $\frac{1}{3}$ **(c)** Positive Twist $+1$ **(d)** Negative Twist -1

Figure 3.3: Examples of integer tangles

Integer tangles are a special case of rational tangle and are represented by a fraction. Horizontal integer tangles are denoted by $\pm\frac{n}{1}$ and vertical integer tangles are denoted by $\pm\frac{1}{n}$, where n is the number of crossings and the sign matches the sign of the twists. Horizontal integer tangles can be combined using tangle sum and vertical integer tangles can be combined using tangle product. Since twists of like sign are additive and twists of opposite sign cancel, this behavior for integer tangles follows the rule that

$$\frac{n}{1} + \frac{m}{1} = \frac{n+m}{1} \qquad \text{and} \qquad \frac{1}{n} * \frac{1}{m} = \frac{1}{n+m}$$

for any integers n and m. Note that $\frac{0}{1}$ and $\frac{1}{0}$ denote the parity 0 and parity ∞ tangles with no crossings, respectively.

Given their simple structure, it is easy to describe the effects of transformations on integer tangles. We summarize these here with a lemma and refer to Figure 3.5 for a visual explanation. Although less obvious, similar rules can also be obtained for the more complicated rational tangle diagrams we study later.

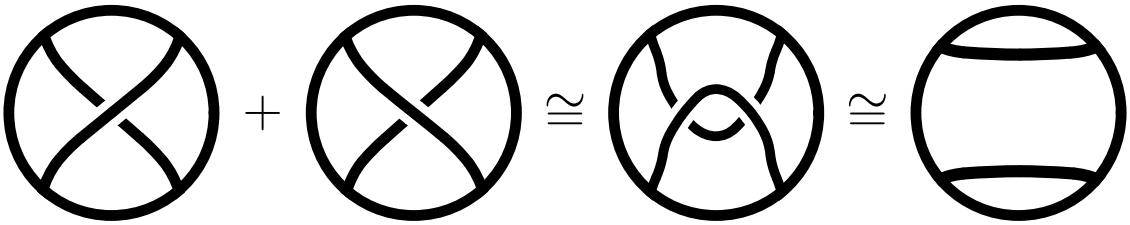
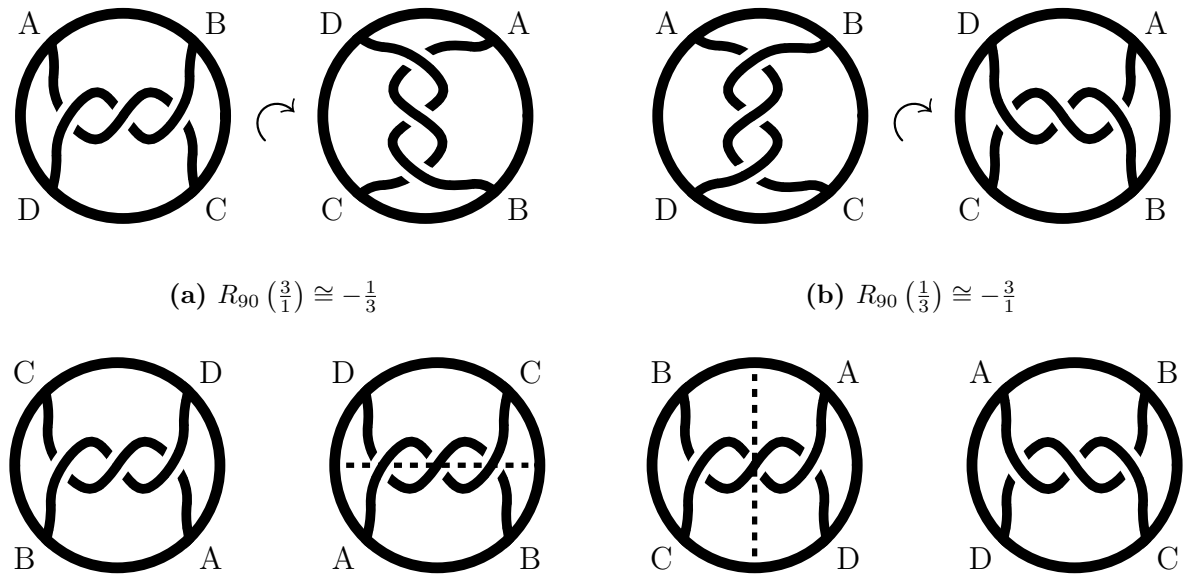


Figure 3.4: Cancellation of twists with opposite signs: $\pm 1 + \mp 1 \cong 0$



(a) $R_{90}\left(\frac{3}{1}\right) \cong -\frac{1}{3}$ **(b)** $R_{90}\left(\frac{1}{3}\right) \cong -\frac{3}{1}$

(c) $R_{180}\left(\frac{3}{1}\right) \cong \frac{3}{1}$ **(d)** $R_H\left(\frac{3}{1}\right) \cong \frac{3}{1}$ **(e)** $R_V\left(\frac{3}{1}\right) \cong \frac{3}{1}$ **(f)** $-\left(\frac{3}{1}\right) \cong \frac{-3}{1}$

Figure 3.5: Transformations of integer tangles

Lemma 3.1 (Transformations of integer tangles)**:**
For any integer $n \in \mathbb{Z}$, let $\frac{n}{1}$ and $\frac{1}{n}$ denote the corresponding horizontal and vertical integer tangles, respectively. We have the following equivalences under tangle transformations.

a. Rotations by 90 **Degrees:** $R_{90}\left(\frac{n}{1}\right) \cong \frac{1}{-n}$ and $R_{90}\left(\frac{1}{n}\right) \cong \frac{-n}{1}$.

b. Rotations by 180 **Degrees:** $R_{180}(T) \cong T$ for any integer tangle T.

c. Flips: $R_H(T) \cong T \cong R_V(T)$ for any integer tangle T.

d. Mirror Image: $-\left(\frac{n}{1}\right) \cong \frac{-n}{1}$ and $-\left(\frac{1}{n}\right) \cong \frac{1}{-n}$.

Proof:

For each equivalence in this proof, we refer to Figure 3.5. Observe that a 90 degree rotation of an integer tangle diagram converts its type from horizontal to vertical (3.5a), or from vertical to horizontal (3.5b). The rotated tangle has the same number of crossings. However, as shown, the slope of the over-stand changes sign after this rotation. Hence, rotating $\frac{n}{1}$ yields $\frac{1}{-n}$, and rotating $\frac{1}{n}$ yields $\frac{-n}{1}$, proving the first two equivalences in part **a**.

It is immediately apparent by symmetry that every integer tangle diagram is equivalent to its 180 degree rotation in the plane, as shown in 3.5c. This may also be seen by applying a 90 degree rotation twice. For a horizontal integer tangle $\frac{n}{1}$, this gives

$$R_{180}\left(\tfrac{n}{1}\right) \cong R_{90}\left(R_{90}\left(\tfrac{n}{1}\right)\right) \cong R_{90}\left(\tfrac{1}{-n}\right) \cong \tfrac{n}{1}.$$

A similar result holds for a vertical integer tangle $\frac{1}{n}$, proving **b**.

For a 180 degree rotation across either the horizontal (3.5d) or vertical (3.5e) axis, a horizontal integer tangle remains horizontal and likewise for a vertical integer tangle. On individual crossings, a flip across either axis has the effect of switching the over-strand and under-strand while simultaneously reversing the slope of each. The operations effectively cancel each other so that the twist sign remains constant. Depending on the axis, the crossings may be turned upside down or have their order reversed, but the diagram will have the same form by symmetry. Therefore, for any integer tangle T, we have that $R_H(T) \cong T \cong R_V(T)$, establishing **c**.

Finally, the mirror image of a tangle diagram switches the over-strand and under-strand at each crossing (3.5f). Since the diagram for an integer tangle always reduces to a sequence of horizontal or vertical twists of constant sign, taking the mirror image yields the same number of horizontal or vertical twists but with sign reversed, thus proving **d**. Note that the minus sign in the notation $-T$ for the mirror image of a tangle is consistent with the sign of an integer $\pm n$ used to indicate positive or negative twisting in the notation $\frac{\pm n}{1}$ or $\frac{1}{\pm n}$ for integer tangles.

□

3.1.2 Building Rational Tangles from Integer Tangles

Whereas integer tangles are constructed from a sequence of twists in either the horizontal or vertical direction, *rational* tangles can be constructed using both horizontal and vertical twists. For example, Figure 3.6 shows an example of a rational tangle similar to the integer tangle shown in Figure 3.2, except that the last twist was made vertical instead of horizontal. Furthermore, twisting for a rational tangle may occur between any of the four pairs of adjacent endpoints: north, south, east, or west, and twists need not have constant sign. We will work to show that there exist preferred constructions for rational tangle diagrams, but a priori any of type of twisting starting from the 0 or ∞ tangle defines a valid rational tangle.

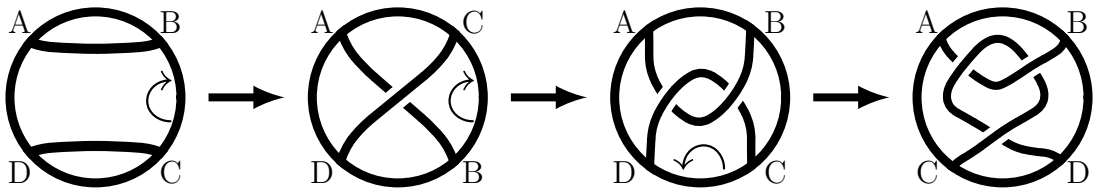


Figure 3.6: Rational tangle created from two horizontal twists and a third vertical twist

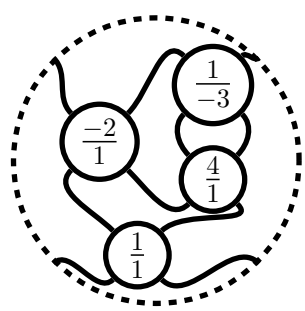
As an example of this, consider the more complicated rational tangle diagram shown in Figure 3.7, which uses a combination of horizontal and vertical twists of non-constant twist sign. Since each twist corresponds to a sum or product with the $+1$ or -1 tangle, the construction of the diagram shown in 3.7a can be described by the expression

$$(-1 + -1 + (-1 * -1 * -1 * (1 + 1 + 1 + 1))) * 1.$$

This suggests an equivalent definition of rational tangle: any combination of tangle sums and products using the $+1$ and -1 tangles yields a rational tangle. This expression is unwieldy, but observe that adjacent horizontal twists and vertical twists of the same sign can be grouped together. These groups of twists correspond exactly to horizontal or vertical integer subtangles. Figure 3.7b shows a subtangle decomposition for the rational tangle diagram of 3.7a which corresponds to these integer subtangles. This decomposition implies an equivalent formula to describe the construction of this tangle:

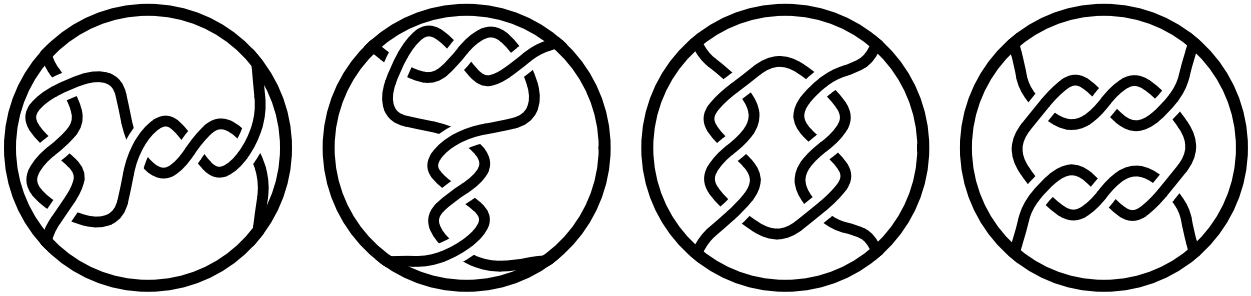
$$\left(\tfrac{-2}{1} + \left(\tfrac{1}{-3} * \tfrac{4}{1}\right)\right) * \tfrac{1}{-1}.$$

(a) Rational tangle diagram



(b) $\left(\frac{-2}{1} + \left(\frac{1}{-3} * \frac{4}{1}\right)\right) * \frac{1}{-1}$

Figure 3.7: Example of a rational tangle and decomposition into integer subtangles



(a) $\frac{1}{3} + \frac{3}{1}$ **(b)** $\frac{3}{1} * \frac{1}{3}$ **(c)** $\frac{1}{3} + \frac{1}{3}$ **(d)** $\frac{3}{1} * \frac{3}{1}$

Figure 3.8: Examples of rational (3.8a and 3.8b) and non-rational (3.8c and 3.8d) tangles

This second expression suggests an alternative way to characterize rational tangles using integer subtangles rather than merely the $+1$ and -1 tangles. However, arbitrary sums and products of integer tangles need not be rational; consider the examples shown in Figure 3.8. The first two diagrams (3.8a and 3.8b) are rational since they can be obtained from the 0 or ∞ tangle merely by twisting endpoints, but this is not possible for the second two diagrams (3.8c and 3.8d). When constructing a tangle diagram from integer subtangles, the resulting tangle will be rational if and only if each attachment is parallel with the direction of twisting. Accounting for this behavior, we may assert that a tangle is rational if it can be constructed through any combination of *sums with horizontal integer tangles* and *products with vertical integer tangles*. Such a rational diagram is in *twist form*.

Based on this this idea, any rational tangle diagram can be described using an algebraic construction formula which explains how to build up the tangle from integer subtangles. The order in which subtangles are joined is determined by the placement of parentheses in this formula. While every rational tangle can be uniquely identified with a fraction $\frac{p}{q}$, unfortunately this fraction is not obtained by simplifying a construction formula. This formula is merely a symbolic expression summarizing how to construct a diagram, and different diagrams for equivalent rational tangles will have different constructions formulas.

While many diagrams are possible for the same rational tangle, there does exist a preferred diagram that we refer to as the *canonical form* of the rational tangle. We will say that a rational tangle diagram is *canonical* if it is constructed using only right twisting and bottom twisting, and if all twists have the same sign. The tangle diagram shown in Figure 3.7a is non-canonical because it includes twisting of the top and left endpoints as well as

(a) $\left(\left(\frac{4}{1} * \frac{1}{-3}\right) + \frac{-2}{1}\right) * \frac{1}{-1}$

(b) $(((\frac{3}{1} * \frac{1}{1}) + \frac{2}{1}) * \frac{1}{1}) + \frac{1}{1}$

Figure 3.9: More examples of rational tangle diagrams from integer construction formulas

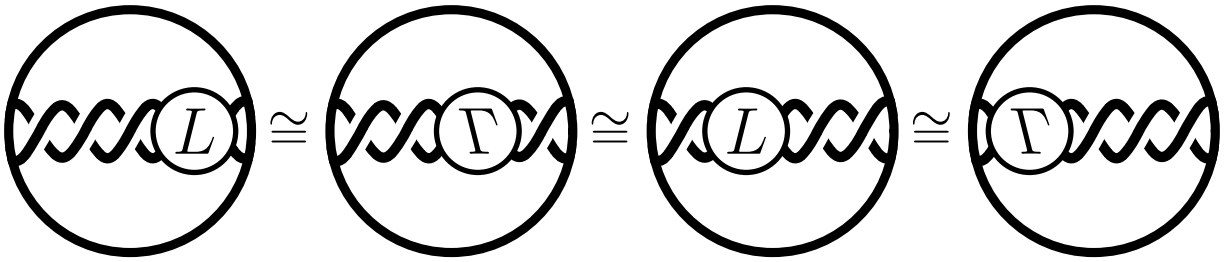
using both positive and negative twists. Figure 3.9 shows two more diagrams which describe rational tangles equivalent to the one from 3.7a. Observe that the rightmost diagram (3.9b) is in canonical form. The canonical form can be readily identified from the construction formula using integer subtangles: the fractions must have constant sign, and tangle sum and product must always be applied on the right side of the parentheses in the formula.

3.2 Twist Vectors and Canonical Form

Every rational tangle diagram is equivalent to a unique diagram in canonical form. It will take quite some effort to prove this fact, which is the goal of this subsection. Canonicity for rational tangle diagrams has two main components: a preferred direction for twisting and a constant twist sign. We will set aside the second component for the time being and start by showing that every rational tangle can be described with a diagram using only right and bottom twists. This fact will enable us to encode the construction information for a rational tangle as a sequence of twists known as a *twist vector*.

3.2.1 Right and Bottom Twisting for Rational Tangles

As we originally defined, any diagram obtained from the 0 or ∞ tangle by twisting pairs of adjacent endpoints corresponds to a rational tangle. Twisting can be performed on all four sides of the tangle: top, right, bottom, or left. However, all of these twists can be shifted to the right and bottom sides of the diagram through a sequence of transformations on subtangles which preserves the equivalence class of the tangle. We will build up to this result through a series of lemmas, beginning with a more generalized notion of flyping.



$$\frac{3}{1}+L \quad\cong\quad \frac{2}{1}+R_H(L)+\frac{1}{1} \quad\cong\quad \frac{1}{1}+L+\frac{2}{1} \quad\cong\quad R_H(L)+\frac{3}{1}$$

Figure 3.10: Flyping a subtangle through an integer tangle

Corollary 3.2 (Shifting exterior twists by flyping)**:**
Given any tangle L and integer tangles $\frac{n}{1}$ and $\frac{1}{n}$, we have the following equivalences:

$$\frac{n}{1}+L \cong \begin{cases} R_H(L)+\frac{n}{1} & , \quad \text{if } n \text{ is odd} \\ L+\frac{n}{1} & , \quad \text{if } n \text{ is even} \end{cases} \qquad \frac{1}{n} * L \cong \begin{cases} R_V(L) * \frac{1}{n} & , \quad \text{if } n \text{ is odd} \\ L * \frac{1}{n} & , \quad \text{if } n \text{ is even} \end{cases}.$$

Proof:

Recall that a *flype* (Lemma 2.1) refers to the isotopic motion of sliding a subtangle through a twist. For a 2-string tangle L, this motion is described by the formulas

$$\pm 1 + L \cong R_H(L) + \pm 1 \quad \text{and} \quad \pm 1 * L \cong R_V(L) * \pm 1.$$

By rotating the subtangle 180 degrees over the horizontal or vertical axis, this operation removes a twist on one side of L while simultaneously introducing a twist on the other side. If there are two twists, this action can be performed twice.

$$\frac{2}{1}+L \quad\cong\quad \frac{1}{1}+R_H(L)+\frac{1}{1} \quad\cong\quad R_H(R_H(L))+\frac{2}{1} \quad\cong\quad L+\frac{2}{1}$$

More generally, flyping can be be used to slide a subtangle through any number of twists in a corkscrew motion. If the number of twists is odd, the subtangle will be flipped, but if the number of twists is even, the subtangle will return to its original orientation. Figure 3.10 shows the effect of flyping a subtangle through three horizontal twists in sequence.

When an integer tangle $\frac{n}{1}$ or $\frac{1}{n}$ is attached to L parallel to the direction of twisting (either using a tangle sum for $\frac{n}{1}$ or a tangle product for $\frac{1}{n}$), then L can be slid through each

twist in sequence. Effectively, this motion shifts the integer tangle to the opposite side of L, with L possibly flipped depending on the number of twists. Since these operations on subtangles preserve the equivalence class of the original tangle, we obtain the two formulas in the corollary statement.

□

The formulas in Corollary 3.2 apply for any 2-string tangle L, but these can be simplified even further if L is rational. In general, transformations change the equivalence class of a tangle, but we will prove in the next Lemma that flipping a rational tangle yields an equivalent rational tangle.

Lemma 3.3 (Equivalence of rational tangles under horizontal and vertical flips [20])**:** If T is a rational tangle, then $R_H(T) \cong T$ and $R_V(T) \cong T$.

Proof:

Let T be a rational tangle. By our previous definition, this means that T is constructed through some finite combination of *sums with horizontal integer tangles* and *products with vertical integer tangles*. We will proceed by induction on the number r of integer subtangles used to construct T. If $r = 1$, then T is itself an integer tangle. Hence, $R_H(T) \cong T \cong R_V(T)$ by Lemma 3.1, which proves the base case.

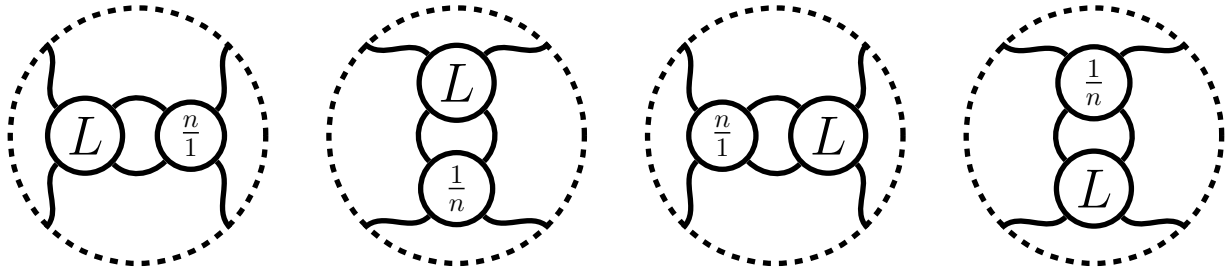
Now suppose that every rational tangle constructed from $r \geq 1$ or fewer integer tangles is equivalent to its flip across either the horizontal or vertical axis. Suppose that T is a rational tangle constructed from $r + 1$ integer subtangles. Thus, there exists an integer n and a rational subtangle L constructed from r integer subtangles such that T satisfies one of the following four constructions:

$$T \cong L + \frac{n}{1} \qquad T \cong L * \frac{1}{n} \qquad T \cong \frac{n}{1} + L \qquad T \cong \frac{1}{n} * L.$$

These four possibilities are displayed in Figure 3.11; each corresponds to the last direction of endpoint twisting in the construction of T: right, bottom, left, or top.

Consider the first case, where $T \cong L + \frac{n}{1}$. By the inductive hypothesis, we have that $R_H(L) \cong L \cong R_V(L)$. Recall that Theorem 2.2 summarizes the effects of transforming a sum or product of tangles. By the first equivalence from this theorem, we have that

$$R_H(T) \quad \cong \quad R_H(L + \tfrac{n}{1}) \quad \cong \quad R_H(L) + R_H(\tfrac{n}{1}) \quad \cong \quad L + \tfrac{n}{1} \quad \cong \quad T.$$



(a) $T \cong L + \frac{n}{1}$ **(b)** $T \cong L * \frac{1}{n}$ **(c)** $T \cong \frac{n}{1} + L$ **(d)** $T \cong \frac{1}{n} * L$

Figure 3.11: Possible constructions for a rational tangle T in the proof of Lemma 3.3

This shows that T is equivalent under horizontal flipping. To show equivalence under vertical flipping, we will also invoke Corollary 3.2 to flype the subtangle L through the integer subtangle.

$$\begin{aligned} R_V(T) &\cong R_V(L + \tfrac{n}{1}) \\ &\cong R_V(\tfrac{n}{1}) + R_V(L) \\ &\cong \tfrac{n}{1} + L \\ &\cong \begin{cases} R_H(L) + \frac{n}{1} & , \quad \text{if } n \text{ is odd} \\ L + \frac{n}{1} & , \quad \text{if } n \text{ is even} \end{cases} \\ &\cong L + \tfrac{n}{1} \\ &\cong T \end{aligned}$$

Therefore, if $T \cong L + \frac{n}{1}$, then $R_H(T) \cong T \cong R_V(T)$.

The remaining three cases for the construction of T displayed in 3.11 follow parallel arguments with similar algebra, and so we choose not to compute these explicitly. This competes the proof by induction that a rational tangle T constructed from any number of integer subtangles will be equivalent under horizontal and vertical flips. This result is also known as the *Flipping Lemma* in the paper by Kauffman and Lambropoulou [20].

□

Together, Corollary 3.2 and Lemma 3.3 imply that any number of exterior twists on a rational tangle can be moved to the other side without changing the tangle's equivalence class. We state this formally in a Corollary.

Corollary 3.4 (Shifting exterior twists through rational tangles)**:**
Given a rational tangle T and integer tangles $\frac{n}{1}$ and $\frac{1}{n}$, we have the two equivalences

$$\frac{n}{1} + T \cong T + \frac{n}{1} \qquad \text{and} \qquad \frac{1}{n} * T \cong T * \frac{1}{n}.$$

Proof:

If T is a rational tangle, then $R_H(T) \cong T \cong R_V(T)$ by Lemma 3.3. Replacing these expression in the formula from Corollary 3.2 completes the proof.

□

It remains to show that all of the horizontal twists in a rational tangle diagram can be shifted to the right and that all of the vertical twists can be shifted to the bottom. This will follow from the preceding corollary by induction.

Lemma 3.5 (Rational tangles from right and bottom twists)**:**
Every rational tangle diagram is equivalent to a diagram constructed using only right and bottom twists.

Proof:

Let T be a rational tangle with a diagram constructed from some combination of sums with horizontal integer tangles and products with vertical integer tangles. Proceed by induction the number r of integer subtangles used in this construction. If $r = 1$, then T is an integer tangle, either $\frac{n}{1}$ or $\frac{1}{n}$ for some integer n.

If $T \cong \frac{n}{1}$ is horizontal, then it is constructed from $|n|$ horizontal twists starting from the diagram for the 0 tangle. Without loss of generality, we may assume that all twists are performed on the right side, as was shown in Figure 3.2. If $T \cong \frac{1}{n}$ is vertical, then it is constructed from $|n|$ vertical twists starting from the diagram for the ∞ tangle. By the same logic, we may assume without loss of generality that all twists are performed on the bottom side. These establish the base case.

Suppose that every diagram for a rational tangle constructed from $r \geq 1$ or fewer integer tangles is equivalent to a diagram constructed using only right and bottom twists. Let T be a diagram for a rational tangle constructed from $r+1$ integer subtangles, so that there exists an integer n and a rational subtangle L such that T satisfies one of the four constructions of Figure 3.11:

$$T \cong L + \frac{n}{1} \qquad T \cong L * \frac{1}{n} \qquad T \cong \frac{n}{1} + L \qquad T \cong \frac{1}{n} * L,$$

where L is constructed from r integer subtangles. Thus, we may assume by our inductive hypothesis that L has a diagram constructed using only right and bottom twists.

The first possible construction $T \cong L + \frac{n}{1}$ corresponds to right twisting (Figure 3.11a) and the second possibility $T \cong L * \frac{1}{n}$ corresponds to bottom twisting (Figure 3.11b). In these two cases, the diagram for T satisfies the statement of the theorem. The third possible construction $T \cong \frac{n}{1} + L$ corresponds to left twisting (Figure 3.11c) and the fourth possibility $T \cong \frac{1}{n} * L$ corresponds to top twisting (Figure 3.11d). In both cases, we shift these twists to the other side of L by Corollary 3.4. While this will flip the diagram for L when n is odd, this can be replaced with the diagram for L using only right and bottom twists since these define equivalent rational tangles.

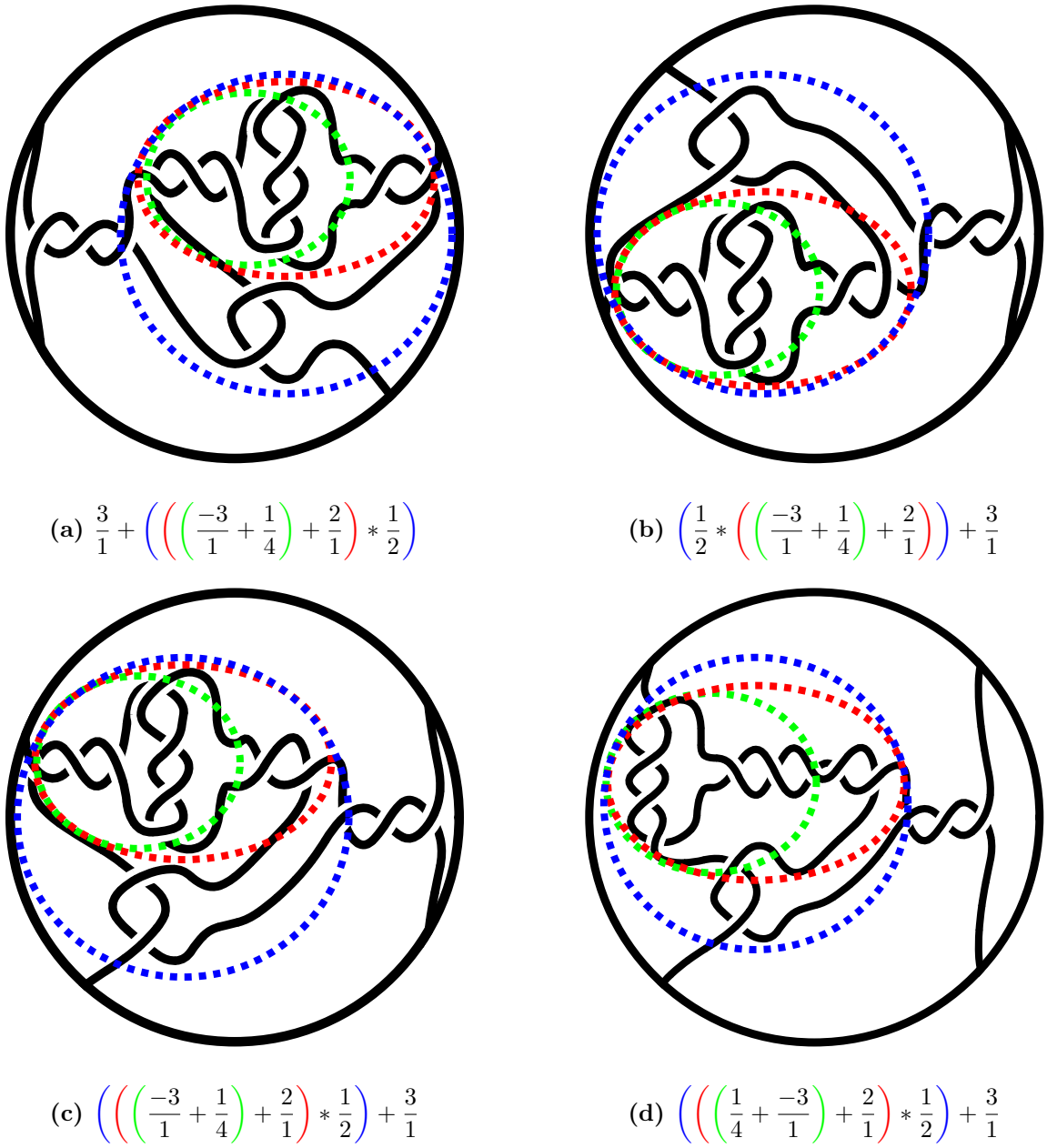
In all four cases, we are able to obtain a rational tangle diagram that is equivalent to T but which is constructed using only right and bottom twists. This completes the proof by induction.

□

If working with a rational tangle diagram which contains left and top twists, the inductive argument in the preceding proof suggests a way to shuffle subtangles through integer tangles one at a time until the diagram satisfies this condition of right and bottom twisting. Each of these shuffles matches a corresponding change in this diagram's construction formula from integer subtanlges, allowing each isotopy to be described algebraically. We will illustrate how these transformations apply in practice using an example.

Consider Figure 3.12, which shows four diagrams for equivalent rational tangles. The colored parentheses in the integer construction formulas correspond to the colored subtangles in the diagrams. Moving from 3.12a to 3.12b, the integer subtangle $\frac{3}{1}$ is moved from the left side to the right side of the blue subtangle. However, because $\frac{3}{1}$ indicates an odd number of twists, the blue subtangle is flipped upside down in the diagram of 3.12b. As a consequence of this, the blue subtangle of 3.12b now has two top twists that must be shifted. When shifting a rational tangle diagram into the preferred position using only right and bottom twists, working from the outermost twists towards the innermost guarantees that this process will finish in finitely many steps.

Moving from Figure 3.12b to 3.12c, the red subtangle is moved through these two top twists. Moving from Figure 3.12c to 3.12d, the left twists in the green subtangle are moved to the right. The fourth tangle shown in 3.12d satisfies the condition of only right and bottom twisting, although now some additional simplifications are possible.



(a) $\frac{3}{1} + \left(\left(\left(\frac{-3}{1} + \frac{1}{4}\right) + \frac{2}{1}\right) * \frac{1}{2}\right)$

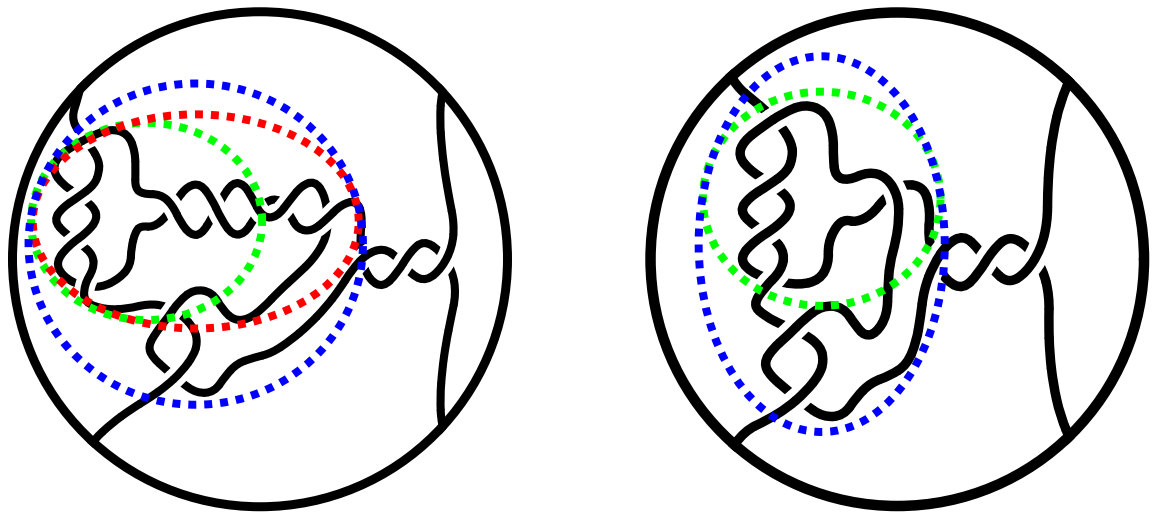
(b) $\left(\frac{1}{2} * \left(\left(\frac{-3}{1} + \frac{1}{4}\right) + \frac{2}{1}\right)\right) + \frac{3}{1}$

(c) $\left(\left(\left(\frac{-3}{1} + \frac{1}{4}\right) + \frac{2}{1}\right) * \frac{1}{2}\right) + \frac{3}{1}$

(d) $\left(\left(\left(\frac{1}{4} + \frac{-3}{1}\right) + \frac{2}{1}\right) * \frac{1}{2}\right) + \frac{3}{1}$

Figure 3.12: Sequence of diagrams showing how to shift twists in a rational tangle

Notice that some of the twists in the red subtangle of the diagram in 3.12d can now be combined and canceled by RII moves. In general, after shifting all twists to the right and bottom sides, adjacent horizontal or vertical twists of the same type can be grouped together. This corresponds to a diagram which alternates between horizontal and vertical integer subtangles in the construction formula for the diagram. For the preceding example,



(a) $\left(\left(\left(\frac{1}{4}+\frac{-3}{1}\right)+\frac{2}{1}\right)*\frac{1}{2}\right)+\frac{3}{1}$ **(b)** $\left(\left(\frac{1}{4}+\frac{-1}{1}\right)*\frac{1}{2}\right)+\frac{3}{1}$

Figure 3.13: Combining adjacent twists of the same type

this alternating behavior is illustrated in Figure 3.13. Moving from 3.13a to 3.13b, the positive and negative twists in the red subtangle cancel each other, whereby $\frac{-3}{1}+\frac{2}{1}\cong\frac{-1}{1}$ is replaced in the construction formula.

Figures 3.12 and 3.13 show several intermediate diagrams, but the exact sequence of isotopies to transform diagram 3.12a into diagram 3.13b can also be described algebraically.

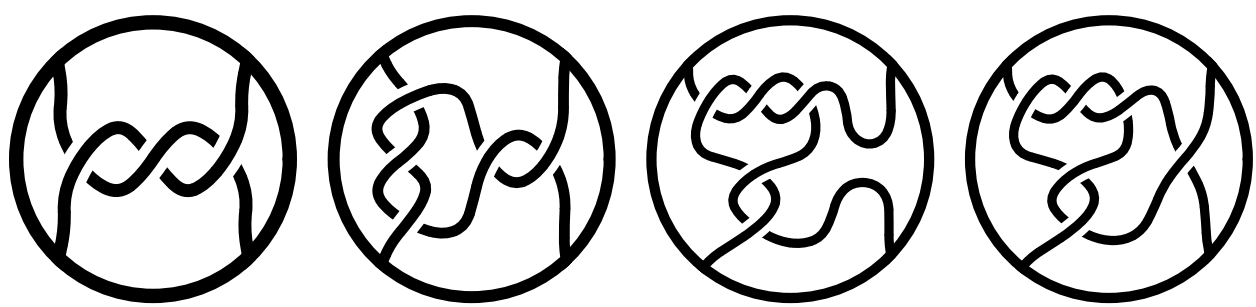
$$
\begin{aligned}
\tfrac{3}{1}+\left(\left(\left(\tfrac{-3}{1}+\tfrac{1}{4}\right)+\tfrac{2}{1}\right)*\tfrac{1}{2}\right) \quad (3.12a) \quad &\cong R_H\left(\left(\left(\tfrac{-3}{1}+\tfrac{1}{4}\right)+\tfrac{2}{1}\right)*\tfrac{1}{2}\right)+\tfrac{3}{1}\\
&\cong \left(R_H\left(\tfrac{1}{2}\right)*R_H\left(\left(\tfrac{-3}{1}+\tfrac{1}{4}\right)+\tfrac{2}{1}\right)\right)+\tfrac{3}{1}\\
&\cong \left(\tfrac{1}{2}*\left(R_H\left(\tfrac{-3}{1}+\tfrac{1}{4}\right)+R_H\left(\tfrac{2}{1}\right)\right)\right)+\tfrac{3}{1}\\
&\cong \left(\tfrac{1}{2}*\left(\left(R_H\left(\tfrac{-3}{1}\right)+R_H\left(\tfrac{1}{4}\right)\right)+\tfrac{2}{1}\right)\right)+\tfrac{3}{1}\\
&\cong \left(\tfrac{1}{2}*\left(\left(\tfrac{-3}{1}+\tfrac{1}{4}\right)+\tfrac{2}{1}\right)\right)+\tfrac{3}{1} \quad (3.12b)\\
&\cong \left(\left(\left(\tfrac{-3}{1}+\tfrac{1}{4}\right)+\tfrac{2}{1}\right)*\tfrac{1}{2}\right)+\tfrac{3}{1} \quad (3.12c)\\
&\cong \left(\left(\left(R_H\left(\tfrac{1}{4}\right)+\tfrac{-3}{1}\right)+\tfrac{2}{1}\right)*\tfrac{1}{2}\right)+\tfrac{3}{1}\\
&\cong \left(\left(\left(\tfrac{1}{4}+\tfrac{-3}{1}\right)+\tfrac{2}{1}\right)*\tfrac{1}{2}\right)+\tfrac{3}{1} \quad (3.12d)\\
&\cong \left(\left(\tfrac{1}{4}+\tfrac{-3+2}{1}\right)*\tfrac{1}{2}\right)+\tfrac{3}{1}\\
&\cong \left(\left(\tfrac{1}{4}+\tfrac{-1}{1}\right)*\tfrac{1}{2}\right)+\tfrac{3}{1} \quad (3.13b)
\end{aligned}
$$

Lemma 3.5 guarantees that any rational tangle diagram is equivalent to a diagram constructed using only right and bottom twists. However, as the preceding example shows, it can be confusing to describe the operations which move all of the twists into the desired positions. Fortunately, with the aid of Theorem 2.2 and Corollary 3.2, finding this sequence of isotopies reduces to pure algebra.

Observe that the final formula in the preceding algebraic derivation alternates between tangle sum and tangle product, and that all operations are applied on the right side of the parentheses. In general, given any integer construction formula for a rational tangle diagram, we can use the algebraic shortcut of shifting all operations to the right and then combining twists of the same type. By doing this, we obtain the integer construction formula for the corresponding diagram with only right and bottom twists. The fact that any rational tangle diagram can be described in this way enables us to introduce the more compact notation of a twist vector.

3.2.2 Twist Vector Construction for Rational Tangles

For a rational tangle diagram T, define the *twist vector* of T to be the sequence of integers $(n_1, n_2, \cdots, n_r)$ denoting alternating right and bottom twists used to construct the diagram. Each entry n_i encodes both the number and sign of the twists in the ith integer subtangle used in the construction, and we allow for $n_i = 0$. We include the rule that the last entry n_r of the twist vector always denotes horizontal twisting, possibly by using $n_r = 0$ if the last non-zero integer subtangle of T is vertical (3.14c). Since the twists must alternate between horizontal and vertical, this condition on n_r also determines the twist type for each n_i. Figure 3.14 shows some examples of twist vectors for rational tangle diagrams.



(a) $(3) \cong \frac{3}{1}$ **(b)** $(3,2) \cong \frac{1}{3} + \frac{2}{1}$ **(c)** $(3,2,0) \cong \left(\frac{3}{1} * \frac{1}{2}\right) + \frac{0}{1}$ **(d)** $(3,2,1) \cong \left(\frac{3}{1} * \frac{1}{2}\right) + \frac{1}{1}$

Figure 3.14: Examples of rational twist vectors and corresponding diagrams

We will use the notation $T \cong (n_1, \cdots, n_r)$ to denote the rational tangle diagram obtained from a twist vector. A twist vector is not defined for a diagram which includes top or left twists, but our previous work shows that horizontal twists in a diagram can always be shuffled to the right and vertical twists can always be shuffled to the bottom. As in the example of Figure 3.13b, it may also be necessary to group together adjacent twists of the same type to guarantee that twisting alternates between horizontal and vertical. Hence, while not every diagram for a rational tangle admits a twist vector description, it will always be possible to find an equivalent diagram which does. We state this formally as a corollary.

Corollary 3.6 (Twist vector construction for rational tangle diagrams)**:**
Every rational tangle T has a diagram which admits an integer construction formula defined by some twist vector $(n_1, \cdots, n_r)$, whereby $T \cong \left(\left(\cdots + \frac{n_{r-2}}{1}\right) * \frac{1}{n_{r-1}}\right) + \frac{n_r}{1}$.

Proof:

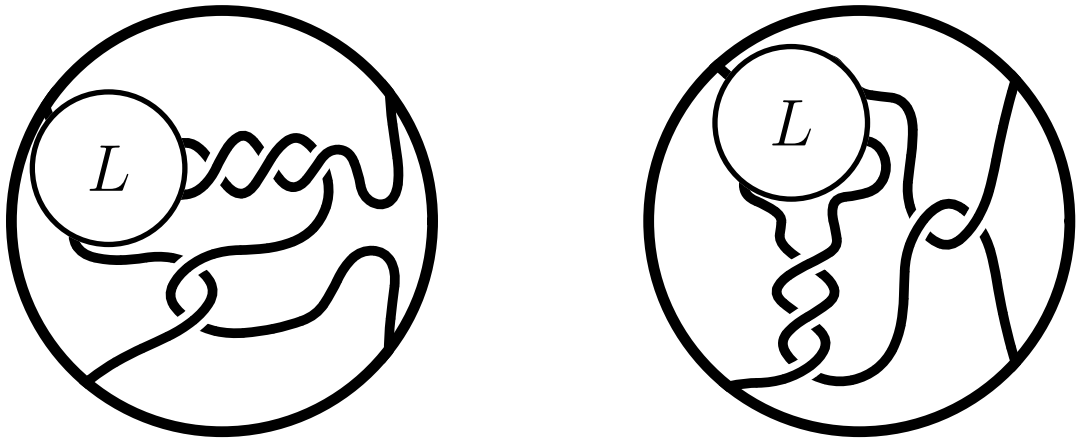
If T is a rational tangle, then it can be built from the 0 or ∞ tangle through some combination of twisting pairs of adjacent endpoints. By Lemma 3.5, there exists a diagram for T constructed using only right and bottom twists. Algebraically, this diagram corresponds to an integer construction formula of the form

$$T \cong \left(\left(\cdots + \frac{n_{r-2}}{1}\right) * \frac{1}{n_{r-1}}\right) + \frac{n_r}{1},$$

where we use $n_r = 0$ if the last nonzero twist in T is vertical. By definition, this construction formula corresponds to the twist vector $(n_1, \cdots, n_r)$, proving the claim.

□

While our study of twist vectors is initially motivated by rational tangles, twist vectors are also useful for describing endpoint twisting on tangles more generally. For a 2-string tangle L and a twist vector $(n_1, \cdots, n_r)$, let $L \circ (n_1, \cdots, n_r)$ denote the tangle obtained by alternately twisting the right and bottom endpoints of L (Figure 3.15). This endpoint twisting can also be described using integer tangles. The exact formula depends on whether r is even or odd and is stated in the next lemma. Since rational tangles are obtained from twisting the endpoints of the 0 or ∞ tangle, they can also be denoted using either $0 \circ (n_1, \cdots, n_r)$ or $\infty \circ (n_1, \cdots, n_r)$, although it is sufficient to use only the twist vector.



(a) $L \circ (3,2,0) \cong \left(\left(L + \frac{3}{1}\right) * \frac{1}{2}\right) + \frac{0}{1}$ **(b)** $L \circ (3,2) \cong \left(L * \frac{1}{3}\right) + \frac{2}{1}$

Figure 3.15: Effect of applying a twist vector to a tangle (Lemma 3.7)

Lemma 3.7 (Effect of twist vectors on generic 2-string tangles)**:**
If L is a tangle and $(n_1, \cdots, n_r)$ a twist vector, then the effect of twisting the endpoints of L according to $(n_1, \cdots, n_r)$ satisfies the following formula:

$$L \circ (n_1, \cdots, n_r) \cong \begin{cases} \left(\cdots\left(\left(L + \frac{n_1}{1}\right) * \frac{1}{n_2}\right) + \cdots\right) + \frac{n_r}{1} & , \quad \text{if } r \text{ is odd} \\ \left(\cdots\left(\left(L * \frac{1}{n_1}\right) + \frac{n_2}{1}\right) * \cdots\right) + \frac{n_r}{1} & , \quad \text{if } r \text{ is even} \end{cases}.$$

Proof:

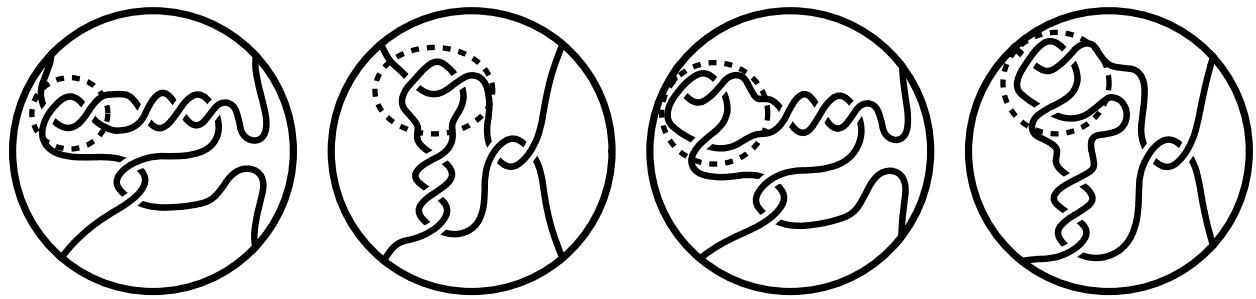
By definition, a twist vector $(n_1, \cdots, n_r)$ denotes a sequence of alternately horizontal and vertical twists, and the last entry n_r always denotes a horizontal twist. Since twists are always applied on the right and bottom sides of the diagram, the effect of applying these twists to L can be described algebraically using integer tangles. If r is odd, it follows that n_1 also denotes horizontal twisting, in which case the twisted diagram satisfies

$$L \circ (n_1, \cdots, n_r) \cong \left(\cdots\left(\left(L + \frac{n_1}{1}\right) * \frac{1}{n_2}\right) + \cdots\right) + \frac{n_r}{1}.$$

If r is even, then n_1 must denote vertical twisting, and hence the diagram is described by the formula

$$L \circ (n_1, \cdots, n_r) \cong \left(\cdots\left(\left(L * \frac{1}{n_1}\right) + \frac{n_2}{1}\right) * \cdots\right) + \frac{n_r}{1}.$$

□



(a) $(2) \circ (3,2,0) \cong (5,2,0)$ **(b)** $(2) \circ (3,2) \cong (2,3,2)$ **(c)** $(2,1,0) \circ (3,2,0) \cong (2,1,3,2,0)$ **(d)** $(2,1,0) \circ (3,2) \cong (2,1,0,3,2)$

Figure 3.16: Composing twist vectors (Lemma 3.8)

When applying a twist vector $(n_1, \cdots, n_r)$ to a generic tangle L, Lemma 3.7 describes the effect of performing only right and bottom twists on the diagram. However, one could also consider the effect of twisting L on the top or left. In this case, Corollary 3.2 can be leveraged to shift all horizontal twists to the right and all vertical twists to the bottom, possibly by flipping L in the process, so that this twisting behavior is described by some twist vector. This idea of characterizing external twisting will play a role in distinguishing certain families of tangles, but for the present we will focus only on rational tangles. If L is known to be rational, then there exists a twist vector so that $L \cong (m_1, \cdots, m_s)$. In this case, $L \circ (n_1, \cdots, n_r)$ can be described using a composition of twist vectors (Figure 3.16). The exact formula again depends on whether r is even or odd and is stated in the next theorem.

Lemma 3.8 (Composition of twist vectors)**:**
If $(m_1, \cdots, m_s)$ and $(n_1, \cdots, n_r)$ are two twist vectors, then their composition satisfies the following formula:

$$(m_1 \cdots, m_s) \circ (n_1, \cdots, n_r) \cong \begin{cases} (m_1, \cdots, m_s + n_1, \cdots, n_r) & , \quad \text{if } r \text{ is odd} \\ (m_1, \cdots, m_s, n_1, \cdots, n_r) & , \quad \text{if } r \text{ is even} \end{cases}.$$

Proof:

Let $v_1 = (m_1, \cdots, m_s)$ and $v_2 = (n_1, \cdots, n_r)$ be two twist vectors. Note that their last entries m_s and n_r both correspond to horizontal twists. The composition $v_1 \circ v_2$ can be interpreted as applying v_2 to the rational tangle diagram obtained from v_1. By using Corollary 3.6 and Lemma 3.7, we may compute an explicit formula.

While it is generally best practice to avoid ambiguity in a tangle construction formula by using parentheses to indicate the order of operations, we will deviate from this convention to make reading easier in the following derivation. It should be understood that the operations of tangle sum and tangle product are applied reading from left to right.

$$
\begin{aligned}
v_1 \circ v_2 &\cong (m_1 \cdots, m_s) \circ (n_1, \cdots, n_r) \\
&\cong \left(\cdots + \frac{m_{s-2}}{1} * \frac{1}{m_{s-1}} + \frac{m_s}{1} \right) \circ (n_1 \cdots n_r) \\
&\cong \begin{cases} \left(\cdots + \frac{m_{s-2}}{1} * \frac{1}{m_{s-1}} + \frac{m_s}{1} \right) + \frac{n_1}{1} * \frac{1}{n_2} + \cdots + \frac{n_r}{1} & , \quad \text{if } r \text{ is odd} \\ \left(\cdots + \frac{m_{s-2}}{1} * \frac{1}{m_{s-1}} + \frac{m_s}{1} \right) * \frac{1}{n_1} + \frac{n_2}{1} * \cdots + \frac{n_r}{1} & , \quad \text{if } r \text{ is even} \end{cases} \\
&\cong \begin{cases} \cdots + \frac{m_{s-2}}{1} * \frac{1}{m_{s-1}} + \frac{m_s + n_1}{1} + \frac{n_1}{1} * \frac{1}{n_2} + \cdots + \frac{n_r}{1} & , \quad \text{if } r \text{ is odd} \\ \cdots + \frac{m_{s-2}}{1} * \frac{1}{m_{s-1}} + \frac{m_s}{1} * \frac{1}{n_1} + \frac{n_2}{1} * \cdots + \frac{n_r}{1} & , \quad \text{if } r \text{ is even} \end{cases} \\
&\cong \begin{cases} (m_1, \cdots, m_s + n_1, \cdots, n_r) & , \quad \text{if } r \text{ is odd} \\ (m_1, \cdots, m_s, n_1, \cdots, n_r) & , \quad \text{if } r \text{ is even} \end{cases}
\end{aligned}
$$

Since m_s always denotes horizontal twisting regardless of whether s is even or odd, the formula only depends on r. This completes the proof.

□

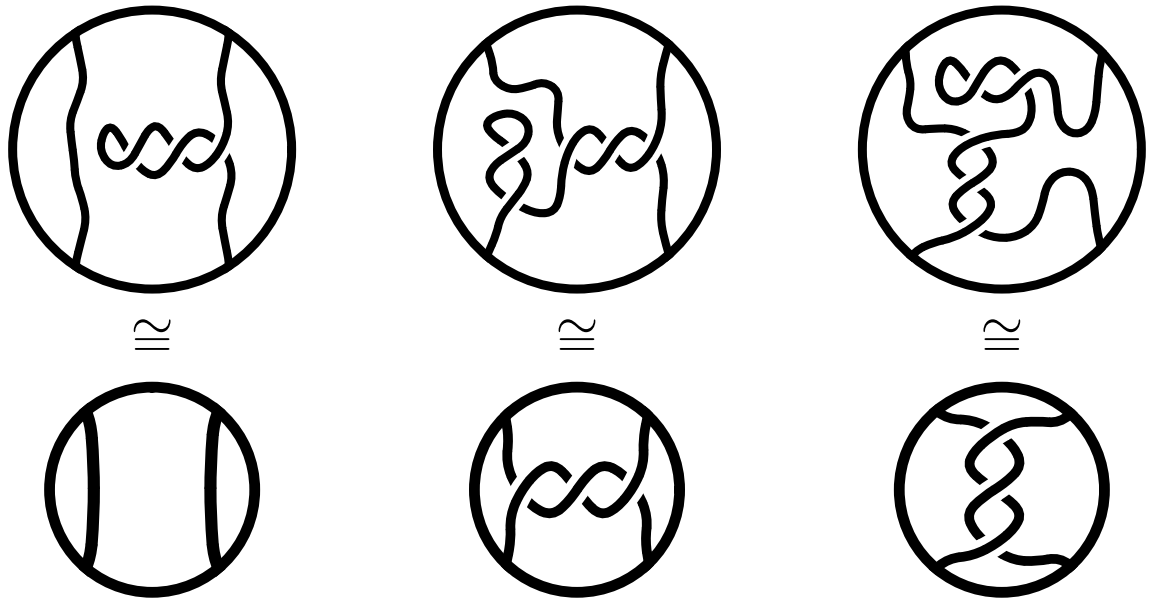
Figure 3.16 shows some examples of twist vector compositions. Notice that the twist vector from the fourth image (3.16d) contains a 0 entry, although it is clear from the diagram that this could be simplified. Entries denoting zero twists correspond to either the 0 or the ∞ tangle in the algebraic construction formula. Except when these occur as the final entry, zeros in the twist vector can be collapsed. We consider the cases of leading and interior twist vector zeros separately.

Lemma 3.9 (Collapsing leading twist vector zeros)**:**
If T is a rational tangle with twist vector construction $(n_1, \cdots, n_r)$ where $n_1 = 0$, then T is equivalent to the rational tangle with twist vector given by one of the following:

a. if $r = 1$, then $T \cong (0) \cong 0$;

b. if $r = 2$, then $T \cong (0, 0) \cong \infty$;

c. if $r \geq 3$, then $T \cong (n_3, \cdots, n_r)$.



(a) $(0,3) \cong (0,0)$ **(b)** $(0,2,3) \cong (3)$ **(c)** $(0,2,3,0) \cong (3,0)$

Figure 3.17: Collapsing the leading zero of a twist vector (Lemma 3.9)

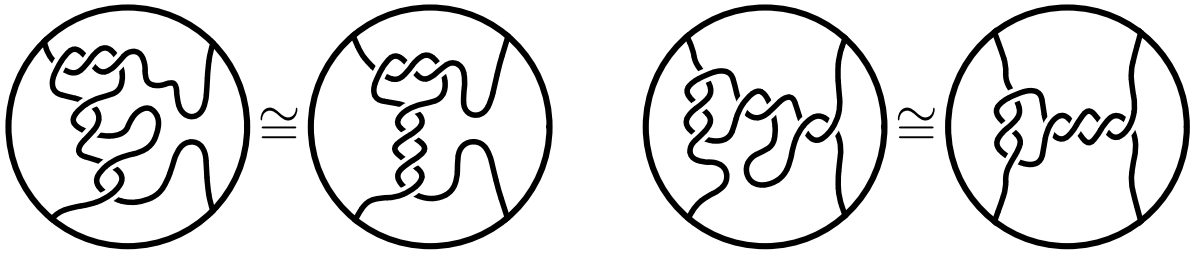
Proof:

If $r = 1$, then $(n_1) = (0)$ and thus $T \cong \frac{0}{1}$, which is the 0 tangle. If $r = 2$, then $(n_1, n_2) = (0, n_2)$ and thus $T \cong \frac{1}{0} + \frac{n_2}{1}$. In this case, T has a diagram of the form shown in Figure 3.17a, whereby all of the horizontal twists can be removed using RI moves. After unwinding these twists, we obtain the diagram for a parity ∞ tangle with no crossings, which is to say that $T \cong \infty$. We use the twist vectors (0) and $(0,0)$ to denote the two 0-crossing tangles 0 and ∞, respectively.

Now suppose that $r \geq 3$. Whether the first twist $n_1 = 0$ is considered horizontal or vertical depends on whether r is odd (3.17b) or even (3.17c). Figure 3.17 shows examples of these two possibilities. In either case, the twists corresponding to n_2 can again be unwound using RI moves. Removing these twists corresponds to deleting the first two entries of the twist vector, establishing the equivalence $T \cong (n_3, \cdots, n_r)$.

□

We originally introduced twist vectors starting from a rational tangle diagram, but any sequence of integers $(n_1, \cdots, n_r)$ defines a valid twist vector for some rational tangle. The preceding lemma shows our first example of different twist vectors which define diagrams for equivalent rational tangles. We continue with a similar result concerning interior zeros.



(a) $(3,2,0,2,0) \cong (3,4,0)$ **(b)** $(3,2,0,2) \cong (3,4)$

Figure 3.18: Collapsing interior zeros in a twist vector (Lemma 3.10)

Lemma 3.10 (Collapsing interior twist vector zeros)**:**
If T is a rational tangle with twist vector construction $(n_1, \cdots, n_r)$, where $n_i = 0$ for some $1 < i < r$, then equivalently $T \cong (n_1, \cdots, n_{i-1} + n_{i+1}, \cdots, n_r)$.

Proof:

Let $T \cong (n_1, \cdots, n_r)$ be a rational tangle and suppose that $n_i = 0$ for some $1 < i < r$. Recall that each twist vector entry denotes either horizontal or vertical twisting in alternation. For any tangle L, we know that $L + \frac{0}{1} \cong L$ and $L * \frac{1}{0} \cong L$. Thus, if n_i denotes horizontal twisting, then the corresponding integer subtangle may be collapsed in the construction formula as follows.

$$\begin{aligned}
T &\cong (n_1, \cdots, n_{i-1}, n_i, n_{i+1}, \cdots, n_r) \\
&\cong \cdots * \frac{1}{n_{i-1}} + \frac{0}{1} * \frac{1}{n_{i+1}} + \cdots + \frac{n_r}{1} \\
&\cong \cdots * \frac{1}{n_{i-1}} * \frac{1}{n_{i+1}} + \cdots + \frac{n_r}{1} \\
&\cong \cdots * \frac{1}{n_{i-1} + n_{i+1}} + \cdots + \frac{n_r}{1} \\
&\cong (n_1, \cdots, n_{i-1} + n_{i+1}, \cdots, n_r)
\end{aligned}$$

If n_i denotes vertical twisting, a similar collapse is possible using $\frac{1}{0}$. In both cases, because twist vector entries must alternate between horizontal and vertical twists, the entries n_{i-1} and n_{i+1} will denote twists of the same type. Hence, after collapsing n_i, the remaining adjacent twists corresponding to n_{i-1} and n_{i+1} can be combined into a single integer subtangle. Figure 3.18 shows examples of this collapsing.

□

Together, the preceding two lemmas imply that all zeros in a twist vector can be collapsed, with the exception of the last entry that determines the twist type. We formalize this conclusion in the following corollary.

Corollary 3.11 (Nonzero twist vector entries)**:**
Every nontrivial rational tangle T admits a twist vector construction $(n_1, \cdots, n_r)$ such that $n_i \neq 0$ for $1 \leq i < r$.

Proof:

Recall that *trivial tangle* refers to the two possible 0-crossing tangles, 0 and ∞, which are denoted by the twist vectors (0) and $(0, 0)$, respectively. Suppose that T is a nontrivial rational tangle with twist vector $(n_1, \cdots, n_r)$. By Lemma 3.9, we may assume without loss of generality that $n_1 \neq 0$, else the leading zero could be collapsed. If $r = 1$ or $r = 2$, then the claim follows immediately, so it is sufficient to consider $r \geq 3$. We may apply Lemma 3.10 repeatedly until all interior zeros are collapsed, possibly applying Lemma 3.9 if a leading zero is introduced. Since T is assumed to be nontrivial, this process will eventually terminate with a twist vector containing all nonzero entries, except possibly the last.

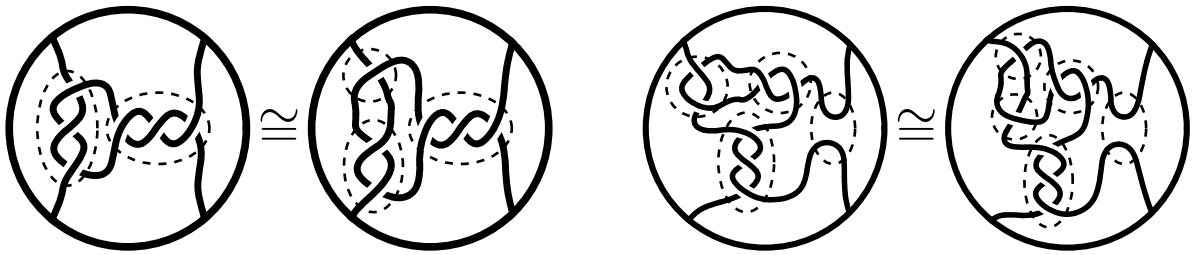
□

Many of the preceding results have slightly different formulas depending on whether the length of the twist vector is even or odd. This length corresponds to the number of integer tangles in a subtangle decomposition, but the next lemma will show that it is always possible to split the first nonzero twist into two integer subtangles (Figure 3.19). Effectively, this adds another entry to the twist vector, allowing the length of a twist vector to be changed from even to odd and vice versa. These modifications may introduce zero entries, but Corollary 3.11 guarantees that these zeros may always be collapsed. This equivalence justifies our adoption of a default *odd length convention* for generic twist vectors.

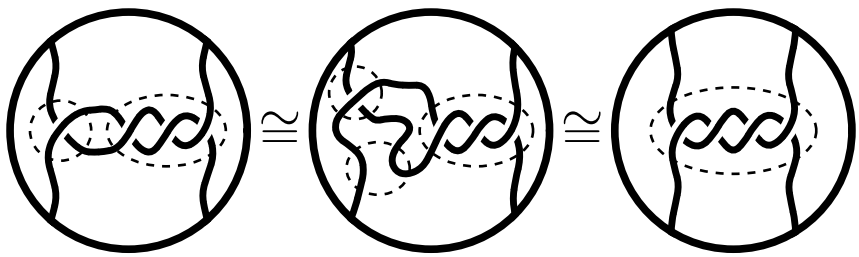
Lemma 3.12 (Incrementing twist vector length)**:**
If T is a nontrivial rational tangle diagram defined by a twist vector $(n_1, n_2, \cdots, n_r)$ of length r, where $n_1 \neq 0$, then T is equivalent to a rational tangle constructed from one of the following two twist vectors of length $r + 1$:

a. if $n_1 > 0$, then $T \cong (1, n_1 - 1, n_2, \cdots, n_r)$;

b. if $n_1 < 0$, then $T \cong (-1, n_1 + 1, n_2, \cdots, n_r)$.



(a) $(3,3) \cong (1,2,3)$ **(b)** $(-2,-2,-3,0) \cong (-1,-1,-2,-3,0)$



(c) $(1,3) \cong (1,0,3) \cong (4)$

Figure 3.19: Converting the length of a twist vector from even to odd (Lemma 3.12)

Proof:

Let T be a diagram for a nontrivial rational tangle with twist vector $(n_1, \cdots, n_r)$. By Lemma 3.9, we may assume without loss of generality that $n_1 \neq 0$. Suppose that r is even. Since the last entry n_r must denote horizontal twisting, the first twist n_1 must denote vertical twisting. Thus, the twist vector defines the integer construction formula

$$T \cong \frac{1}{n_1} + \frac{n_2}{1} * \cdots + \frac{n_r}{1}$$

for this diagram. The vertical integer tangle corresponding to n_1 can be decomposed as a tangle product: $\frac{1}{n_1} \cong \frac{1}{\pm 1} * \frac{1}{n_1 \mp 1}$. The sign of the twist $\frac{1}{\pm 1}$ matches the sign of n_1. This gives the equivalent integer construction formula

$$T \cong \frac{1}{\pm 1} * \frac{1}{n_1 \mp 1} + \frac{n_2}{1} * \cdots + \frac{n_r}{1}$$

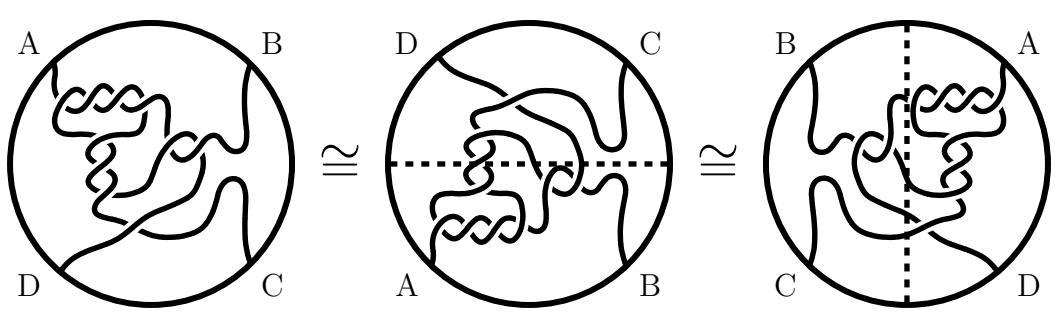
for the rational tangle diagram. A *leading* 1-crossing integer subtangle can be regarded as either a vertical twist or a horizontal twist: $\frac{1}{\pm 1} \cong \pm 1 \cong \frac{\pm 1}{1}$. Thus, the preceding integer construction formula corresponds to the twist vector $(\pm 1, n_1 \mp 1, n_2, \cdots, n_r)$, proving the desired formula. A similar derivation holds if r is odd. $\square$

Lemma 3.12 shows that any nontrivial rational tangle diagram can be described using a twist vector of odd length. In the special case where $n_1 = \pm 1$, as in Figure 3.19c, observe that this conversion will introduce an interior zero. However, this zero can then be collapsed using Lemma 3.10. Since this collapse reduces the total number of twist vector entries by two, it preserves the odd length convention.

Taken together, Corollary 3.11 and Lemma 3.12 show that any rational tangle obtained from a twist vector construction is equivalent to some twist vector of odd length with all nonzero entries, except possibly the last. Thus, we may assume without loss of generality that the twist vector for a rational tangle diagram satisfies these conditions. In general, a twist can include both positive and negative twists. Our next goal is to show that any twist vector can be modified so that all twists have the same sign. To achieve this, it will be helpful to detail the effects of various tangle transformations on the twist vector.

3.2.3 Transformations on Twist Vectors

We have already seen that rational tangles are equivalent under horizontal and vertical flips (Lemma 3.3). However, flipping a diagram over the horizontal axis converts bottom twists into top twists, while flipping over the vertical axis converts right twists into left twists (Figure 3.17). By definition, twist vectors only account for right and bottom twisting, but Lemma 3.5 ensures that these twists can always be shuffled back to the right and bottom sides without changing the equivalence class of the rational tangle. With this in mind, it should be understood that tangle transformations applied to twist vector constructions must also incorporate this extra step of shuffling twists back into the preferred position. In the case of flipping, the corresponding twist vector constructions will be identical after this shuffle.



$$(4,3,2,1,0) \cong R_H(4,3,2,1,0) \cong R_V(4,3,2,1,0)$$

Figure 3.20: Twist vector equivalence under flips (Lemma 3.13), before shuffling twists

Lemma 3.13 (Effect of flips on twist vector)**:**
The rational tangle diagram defined by the twist vector $(n_1, \cdots, n_r)$ is equivalent under horizontal and vertical flips: $R_H(n_1, \cdots, n_r) \cong (n_1, \cdots, n_r) \cong R_V(n_1, \cdots, n_r)$.

Proof:

Let $(n_1, \cdots, n_r)$ be a twist vector defining the construction for some rational tangle diagram T. By Lemma 3.3, we already know that $R_H(T) \cong T \cong R_V(T)$. The objective of this proof is to show that we obtain identical twist vectors for the flipped diagrams after shuffling all twists back into the preferred right and bottom positions.

We may suppose without loss of generality that this twist vector satisfies the odd length convention, so that n_1 denotes horizontal twisting (Lemma **??**). As shown in Figure 3.20, flipping this diagram over the horizontal axis converts all bottom twists into top twists, although it preserves the number, sign, and type. Algebraically, this is equivalent to reversing the order of all products with vertical integer subtangles in the construction formula for this diagram. We may apply Corollary 3.4 on rational subtangles to shift vertical integer subtangles back into the preferred position.

$$\begin{aligned}
R_H(n_1, \cdots, n_r) &\cong R_H\left(\tfrac{n_1}{1} * \tfrac{1}{n_2} + \tfrac{n_3}{1} * \tfrac{1}{n_4} + \cdots * \tfrac{1}{n_{r-1}} + \tfrac{n_r}{1}\right) \\
&\cong \left(\tfrac{1}{n_{r-1}} * \left(\cdots\left(\tfrac{1}{n_4} * \left(\left(\tfrac{1}{n_2} * \tfrac{n_1}{1}\right) + \tfrac{n_3}{1}\right)\right) + \cdots\right)\right) + \tfrac{n_r}{1} \\
&\cong \left(\left(\cdots\left(\left(\left(\tfrac{n_1}{1} * \tfrac{1}{n_2}\right) + \tfrac{n_3}{1}\right) * \tfrac{1}{n_4}\right) + \cdots\right) * \tfrac{1}{n_{r-1}}\right) + \tfrac{n_r}{1} \\
&\cong (n_1, \cdots, n_r)
\end{aligned}$$

The above derivation also follows from Lemma 3.5, which ensures that each top twist in the flipped diagram can be shuffled back to the bottom while preserving the equivalence class of the rational tangle. A similar derivation holds for vertical flips. As twist vectors, the resulting sequence of integers is identical, completing the proof.

□

Unlike flipping, rotating a rational tangle 90 degrees in the plane changes the type of each twist: horizontal twists become vertical and vertical twists become horizontal (Figure 3.21). Furthermore, the sign of each twist changes along with the slope of each over-strand. We previously characterized how rotations affect integer tangles in Lemma 3.1; rotating a rational tangle diagram has the effect of converting each integer subtangle to its negative reciprocal. The next lemma describes this behavior in terms of the twist vector.

(a) $(4,3,2,1,0)$

(b) $R_{90}(4,3,2,1,0)$
$\cong (-4,-3,-2,-1,0,0)$
$\cong (-1,-3,-3,-2,-1)$

(c) $R_{180}(4,3,2,1,0)$
$\cong (4,3,2,1,0,0,0)$
$\cong (4,3,2,1,0)$

Figure 3.21: Planar rotations on twist vectors (Lemma 3.14 and 3.15), before shuffling twists

Lemma 3.14 (Effect of 90 degree clockwise rotation on twist vector)**:**
If T is a rational tangle with twist vector construction $(n_1, \cdots, n_r)$, then its 90 degree clockwise rotation is a rational tangle with twist vector construction $R_{90}(T) \cong (-n_1, \cdots, -n_r, 0)$ (uncorrected for the the odd length convention and collapsing zeros).

Proof:

Let T be a rational tangle diagram defined by the twist vector $(n_1, \cdots, n_r)$, which we may assume satisfies the odd length convention. As shown in Figure 3.21b, rotating T clockwise by 90 degrees converts right twists to bottom twists and converts bottom twists to left twists. It also reverses the sign of each twist. In terms of the integer construction formula for this rotate diagram, we may again apply Corollary 3.4 to shift integer subtangles through rational subtangles.

$$\begin{aligned}
R_{90}(n_1, \cdots, n_r) &\cong R_H\left(\tfrac{n_1}{1} * \tfrac{1}{n_2} + \tfrac{n_3}{1} * \tfrac{1}{n_4} + \cdots * \tfrac{1}{n_{r-1}} + \tfrac{n_r}{1}\right) \\
&\cong \left(\tfrac{-n_{r-1}}{1} + \left(\cdots\left(\tfrac{-n_4}{1} + \left(\left(\tfrac{-n_2}{1} + \tfrac{1}{-n_1}\right) * \tfrac{1}{-n_3}\right)\right) * \cdots\right)\right) * \tfrac{1}{-n_r} \\
&\cong \left(\left(\cdots\left(\left(\left(\tfrac{1}{-n_1} + \tfrac{-n_2}{1}\right) * \tfrac{1}{-n_3}\right) + \tfrac{-n_4}{1}\right) * \cdots\right) + \tfrac{-n_{r-1}}{1}\right) * \tfrac{1}{-n_r} \\
&\cong \left(\tfrac{1}{-n_1} + \tfrac{-n_2}{1} * \tfrac{1}{-n_3} + \tfrac{-n_4}{1} * \cdots + \tfrac{-n_{r-1}}{1} * \tfrac{1}{-n_r}\right) + \tfrac{0}{1} \\
&\cong (-n_1, -n_2, \cdots, -n_{r-1}, -n_r, 0)
\end{aligned}$$

Since the last entry n_r becomes a vertical twist after rotation, we must introduce a zero twist so that the last twist vector entry is horizontal. Note that doing so increases the length of the twist vector by one. □

Since Lemma 3.14 changes the length of the twist vector and introduces a zero entry, the modified twist vector will have even length and possibly an interior zero. Figure 3.21b shows one such example. In this case, the rotated twist vector can be modified again to collapse zeros and satisfy the odd length convention. Applying Lemma 3.14 twice yields a rational tangle diagram equivalent under a 180 degree rotation in the plane (Figure 3.21c).

Corollary 3.15 (180 degree clockwise rotational equivalence of rational tangles)**:**
If T is a rational tangle, then T is equivalent to its 180 degree clockwise rotation $R_{180}(T)$.

Proof:

If T is a rational tangle diagram with twist vector $(n_1, \cdots, n_r)$, then the 180 degree rotation of this diagram can be computed using twist vectors by applying Lemma 3.14 twice. Doing so will introduce two zero entries, which can be collapsed by Lemma 3.10.

$$\begin{aligned} R_{180}(n_1, \cdots, n_r) &\cong R_{90}(R_{90}(n_1, \cdots, n_r)) \\ &\cong R_{90}(-n_1, \cdots, -n_r, 0) \\ &\cong (n_1, \cdots, n_r, 0, 0) \\ &\cong (n_1, \cdots, n_r + 0) \\ &\cong (n_1, \cdots, n_r) \end{aligned}$$

Hence, both T and its 180 degree rotation have the same twist vector construction, which proves that $T \cong R_{180}(T)$.

□

Mirror images are the next type of transformation we consider. Since the mirror image switches the over-strand and under-strand at each crossing, it reverses the sign of each twist but preserves the twist type and the order of the twists. This makes it simple to describe the mirror image of a rational tangle diagram using twist vectors. As shown in Figure 3.22, this transformation corresponds to reversing the sign on each twist vector entry. We formalize this result in the next corollary.

Corollary 3.16 (The mirror image of a twist vector)**:**
If T is a rational tangle diagram with twist vector construction $(n_1, \cdots, n_r)$, then its mirror image is $-T \cong (-n_1, \cdots, -n_r)$.

(a) $(4,3,2,1,0)$

(b) $-(4,3,2,1,0)$
$\cong (-4,-3,-2,-1,0)$

(c) $-R_{90}(4,3,2,1,0)$
$\cong (4,3,2,1,0,0)$
$\cong (1,3,3,2,1)$

Figure 3.22: Effect of mirror image on twist vectors (Corollary 3.16 and 3.17)

Proof:

Let $T \cong (n_1, \cdots, n_r)$ be a rational tangle defined by a twist vector, which we may assume satisfies the odd length convention. Since mirror image commutes with tangle sum and product (Theorem 2.2) and reverses the twist sign on integer tangles (Lemma 3.1), the effect on the algebraic construction formula for a rational tangle diagram is computed as follows.

$$\begin{aligned} -(n_1, \cdots, n_r) &\cong -\left(\tfrac{n_1}{1} * \tfrac{1}{n_2} + \cdots * \tfrac{1}{n_{r-1}} + \tfrac{n_r}{1}\right) \\ &\cong \tfrac{-n_1}{1} * \tfrac{1}{-n_2} + \cdots * \tfrac{1}{-n_{r-1}} + \tfrac{-n_r}{1} \\ &\cong (-n_1, \cdots, -n_r) \end{aligned}$$

Therefore, the mirror image of T satisfies $-T \cong (-n_1, \cdots, -n_r)$, completing the proof.

□

So far, we have listed the effects of basic transformations on rational tangle diagrams using the twist vector. These transformations can also be combined. In the next corollary, we will characterize the composition of mirror image with a 90 degree rotation, a transformation which will be needed in an upcoming proof. Figure 3.22c shows the effect of such a composition on the twist vector, where it merely adds an additional zero entry. After shuffling twists back to the preferred position, the resulting diagram can be interpreted as a reflection across the NW-SE diagonal.

Corollary 3.17 (The diagonal reflection of a rational tangle)**:**
If T is a rational tangle with twist vector construction $(n_1, \cdots, n_r)$, then $-R_{90}(T)$ is a rational tangle with twist vector construction $(n_1, \cdots, n_r, 0)$. In particular, these diagrams are reflections of each other across the NW-SE diagonal.

Proof:

Let T be a rational tangle diagram defined by the twist vector $(n_1, \cdots, n_r)$. We may combine Corollary 3.16 and Lemma 3.14 to show the following.

$$\begin{aligned} -R_{90}(T) &\cong -R_{90}(n_1, \cdots, n_r) \\ &\cong -(-n_1, \cdots, -n_r, 0) \\ &\cong (n_1, \cdots, n_r, 0) \end{aligned}$$

As shown in Figures 3.22a and 3.22c, the diagrams for T and $-R_{90}(T)$ are reflections across the NW-SE diagonal. Note that this formula does not depend on the length of the twist vector, although it can be modified to collapse zeros and satisfy the odd length convention. □

Corollary 3.17 shows that a rational tangle diagram and its diagonal reflection have very similar twist vectors. They both have the same number, sign, and order of twists, but the type of each twist is reversed: horizontal becomes vertical and vertical becomes horizontal. This observation plays a role in identifying rational subtangles within a twist vector.

Lemma 3.8 showed how to compose twist vectors, but the same idea can also be applied in reverse. Any twist vector can be divided into a rational tangle with endpoint twisting of the form introduced in Lemma 3.7, but as before, there are two possibilities depending on the length of the twist vector. The next lemma shows that these possibilities are related by the diagonal reflection.

Lemma 3.18 (Twist vector rational subtangle decomposition)**:**
If L is a rational tangle with a diagram defined by a twist vector $(m_1, \cdots, m_s)$, and if T is a rational tangle with a diagram defined by $(m_1, \cdots, m_s, n_1, \cdots, n_r)$, then

$$T \cong \begin{cases} -R_{90}(L) \circ (n_1, \cdots, n_r) & , \quad \text{if } r \text{ is odd} \\ L \circ (n_1, \cdots, n_r) & , \quad \text{if } r \text{ is even} \end{cases}.$$

Proof:

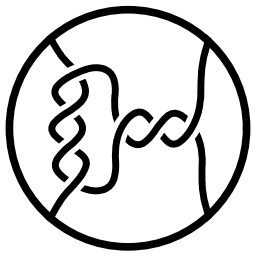
Let $L \cong (m_1, \cdots, m_s)$ and $T \cong (m_1, \cdots, m_s, n_1, \cdots, n_r)$ be two rational tangle diagrams defined by twist vectors. Note that m_s always denotes a horizontal twist in L since it is the last twist vector entry. However, m_s could denote either vertical or hoizontal twisting in T, depending on whether r is even or odd. Consequently, there are two possibilities for the construction formula for T: either m_s corresponds to the vertical integer subtangle $\frac{1}{m_s}$ or it corresponds to horizontal integer subtangle $\frac{m_s}{1}$. In the former case, we may insert a 0-crossing subtangle into the construction formula after $\frac{1}{m_s}$ to obtain a rational subtangle ending on a horizontal twist. We show this algebraically as follows.

$$
\begin{aligned}
T &\cong (m_1, \cdots, m_s, n_1, \cdots, n_r) \\
&\cong \begin{cases} \cdots + \frac{m_{s-1}}{1} * \frac{1}{m_s} + \frac{n_1}{1} * \frac{1}{n_2} + \cdots * \frac{1}{n_{r-1}} + \frac{n_r}{1} & , \text{ if } r \text{ is odd} \\ \cdots * \frac{1}{m_{s-1}} + \frac{m_s}{1} * \frac{1}{n_1} + \frac{n_2}{1} * \cdots * \frac{1}{n_{r-1}} + \frac{n_r}{1} & , \text{ if } r \text{ is even} \end{cases} \\
&\cong \begin{cases} \left(\cdots + \frac{m_{s-1}}{1} * \frac{1}{m_s} + \frac{0}{1}\right) + \frac{n_1}{1} * \frac{1}{n_2} + \cdots * \frac{1}{n_{r-1}} + \frac{n_r}{1} & , \text{ if } r \text{ is odd} \\ \left(\cdots * \frac{1}{m_{s-1}} + \frac{m_s}{1}\right) * \frac{1}{n_1} + \frac{n_2}{1} * \cdots * \frac{1}{n_{r-1}} + \frac{n_r}{1} & , \text{ if } r \text{ is even} \end{cases} \\
&\cong \begin{cases} (m_1, \cdots, m_s, 0) \circ (n_1, \cdots, n_r) & , \text{ if } r \text{ is odd} \\ (m_1, \cdots, m_s) \circ (n_1, \cdots, n_r) & , \text{ if } r \text{ is even} \end{cases} \\
&\cong \begin{cases} -R_{90}(L) \circ (n_1, \cdots, n_r) & , \text{ if } r \text{ is odd} \\ L \circ (n_1, \cdots, n_r) & , \text{ if } r \text{ is even} \end{cases}
\end{aligned}
$$

This proves the formula in the claim. Note that inserting the extra zero twist when r is odd also makes the above expression consistent with the formula for twist vector composition introduced in Lemma 3.8.

□

Figure 3.23 shows two examples of rational subtangles identified from the twist vector for the rational tangle diagram $T \cong (4, 3, 2, 1, 0)$. The motivation for examining rational subtangles in this way is connected to the question of sign changes in the twist vector. Most of our examples have used positive twists, but twist vectors in general can use a combination of positive and negative twists. However, we will see that the corresponding rational tangle diagram can be modified so that all twists have the same sign. This will be achieved through operations on rational subtangles, which are most easily described using twist vectors.



(a) $L_1 \cong (4,3)$

(b) $-R_{90}(L_1) \cong (4,3,0)$

(c) $L_2 \cong (4,3,2)$

(d) $-R_{90}(L_2) \cong (4,3,2,0)$

(e) $T \cong -R_{90}(L_1) \circ (2,1,0) \cong (4,3,2,1,0)$

(f) $T \cong L_2 \circ (1,0) \cong (4,3,2,1,0)$

Figure 3.23: Rational subtangle decomposition within a twist vector (Lemma 3.18)

We have seen before that two adjacent twists of the same type (horizontal or vertical) but opposite sign cancel each other through a Reidemeister II move (Figure 3.4). Perhaps less obvious is the fact that two adjacent twists of opposite sign and different type will simplify to a single twist (Figure 3.24). If we imagine these as exterior twists on a subtangle, the resulting isotopy is reminiscent of flyping. We will refer to this operation as a *flupe*.

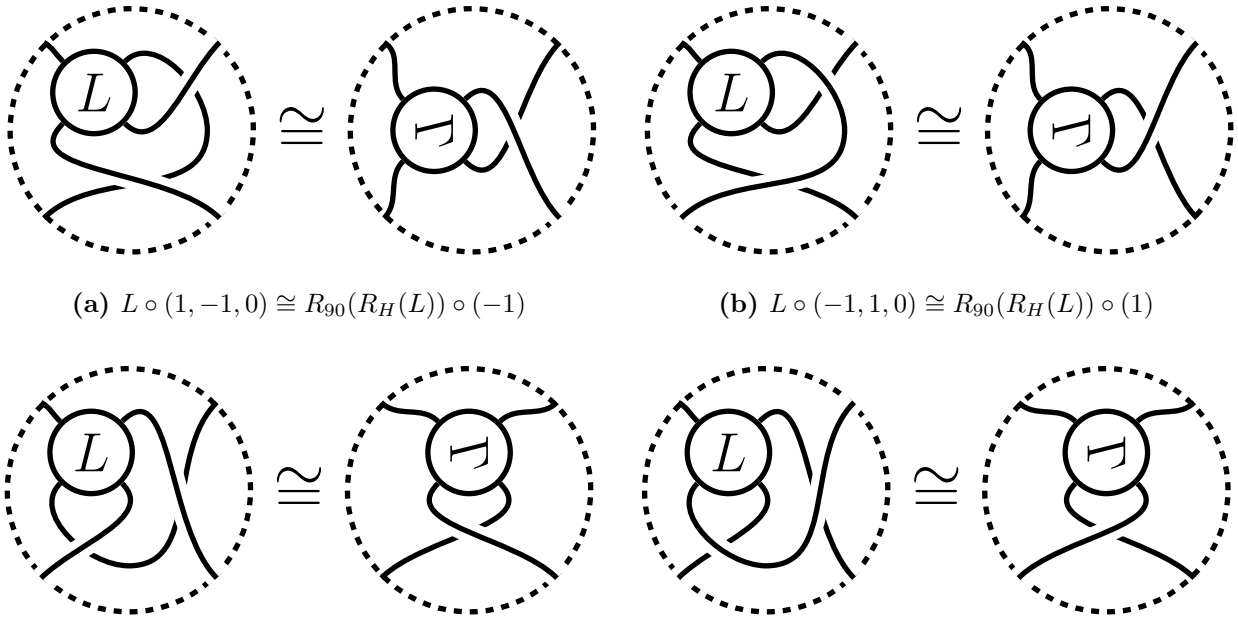
Lemma 3.19 (Fluping subtangles)**:**
Any tangle L with an exterior right twist and bottom twist of opposite signs is equivalent to a tangle with one exterior twist. The equivalent tangles are said to be related by a *flupe*.

a. $L \circ (1,-1,0) \cong R_{90}(R_H(L)) \circ (-1)$

b. $L \circ (-1,1,0) \cong R_{90}(R_H(L)) \circ (1)$

c. $L \circ (1,-1) \cong R_{90}(R_H(L)) \circ (-1,0)$

d. $L \circ (-1,1) \cong R_{90}(R_H(L)) \circ (1,0)$



(a) $L \circ (1, -1, 0) \cong R_{90}(R_H(L)) \circ (-1)$ **(b)** $L \circ (-1, 1, 0) \cong R_{90}(R_H(L)) \circ (1)$

(c) $L \circ (1, -1) \cong R_{90}(R_H(L)) \circ (-1, 0)$ **(d)** $L \circ (-1, 1) \cong R_{90}(R_H(L)) \circ (1, 0)$

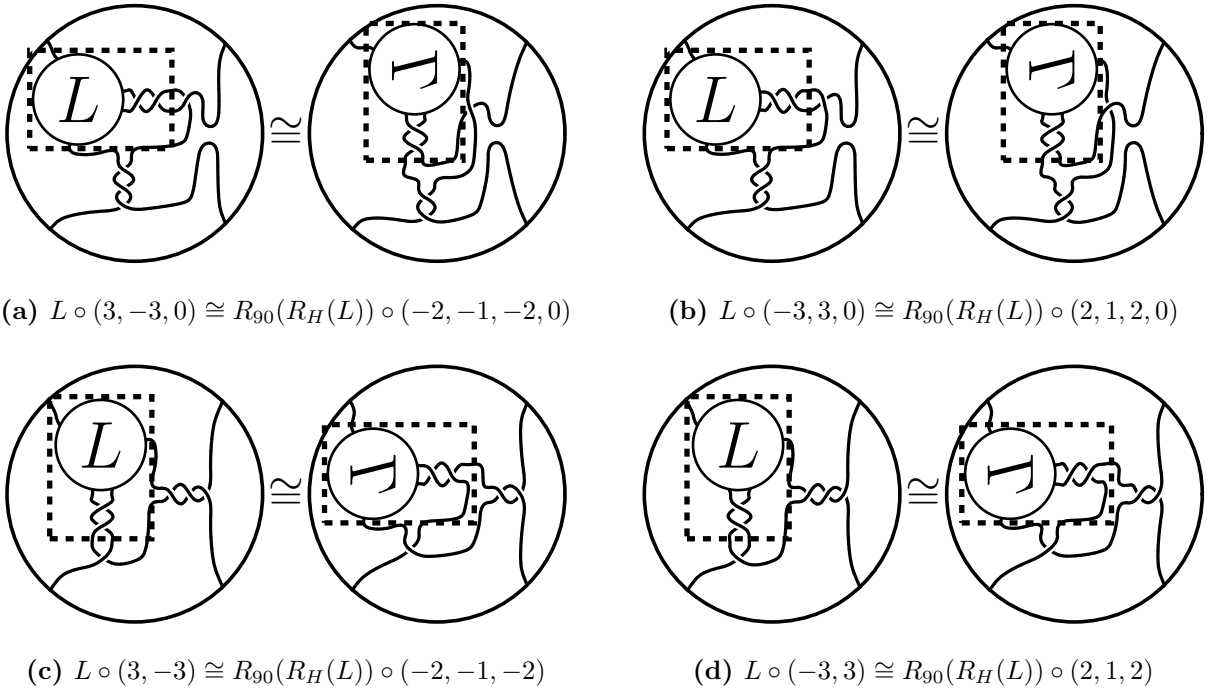
Figure 3.24: The four possible ways to flupe a subtangle (Lemma 3.19)

Proof:

There are four possible ways for a tangle L to a have an exterior right twist and bottom twist of opposite sign. These depend on the type and order of the twists and each can be described using a twist vector. Figure 3.24 shows these four possibilities and the resulting isotopy. In all four cases, there exists a strand that passes either in front of or behind both twists. If we imagine pulling this strand taught, this will flip L across the diagonal, removing both twists but introducing a new one. Recall that flipping a tangle across the the NW-SE diagonal can be obtained from the composition of a horizontal flip and a 90 degree rotation (Figure 2.13g). Hence, after fluping L, the expression for the modified tangle uses $R_{90}(R_H(L))$ in each case. The corresponding twist vectors match the figure above.

□

In the same way that flyping can be generalized to multiple twists (Corollary 3.2), a similar generalization is possible using flupes. We would like to characterize this idea using the language of twist vectors. Since twist vectors define a construction formula using integer subtangles, the effect of multiple right and bottom twists of opposite sign can be described using tangle algebra. These can always be simplified to twisting of constant sign.



(a) $L \circ (3, -3, 0) \cong R_{90}(R_H(L)) \circ (-2, -1, -2, 0)$

(b) $L \circ (-3, 3, 0) \cong R_{90}(R_H(L)) \circ (2, 1, 2, 0)$

(c) $L \circ (3, -3) \cong R_{90}(R_H(L)) \circ (-2, -1, -2)$

(d) $L \circ (-3, 3) \cong R_{90}(R_H(L)) \circ (2, 1, 2)$

Figure 3.25: Fluping exterior twists of opposite sign to twists of constant sign (Lemma 3.20)

Lemma 3.20 (Converting adjacent exterior twists of opposite sign)**:**
Any tangle L with exterior right and bottom twisting of opposite signs is equivalent to a tangle with up to three exterior twists of the same sign.

a. $L \circ (n, -m, 0) \cong R_{90}(R_H(L)) \circ (-n + 1, -1, -m + 1, 0)$

b. $L \circ (-n, m, 0) \cong R_{90}(R_H(L)) \circ (n - 1, 1, m - 1, 0)$

c. $L \circ (n, -m) \cong R_{90}(R_H(L)) \circ (-n + 1, -1, -m + 1)$

d. $L \circ (-n, m) \cong R_{90}(R_H(L)) \circ (n - 1, 1, m - 1)$

The above formulas hold for any 2-string tangle L and any positive integers n and m. The sign of the resulting three exterior twists matches the sign of the outermost twist, $\pm m$.

Proof:

Figure 3.25 shows an example of the four possible cases, and these can be interpreted as a generalization of the fluping shown in Figure 3.24. In each case, we imagine isolating the adjacent horizontal and vertical twists so that Lemma 3.19 can be applied.

Let L be a 2-string tangle and let $n, m \in \mathbb{Z}$ be positive integers. Each of the four formulas in the claim can be derived using tangle algebra. We will only show this computation explicitly for the first equivalence, but the remaining three cases are similar.

$$\begin{aligned}
L \circ (n, -m, 0) &\cong L + \tfrac{n}{1} * \tfrac{1}{-m} + \tfrac{0}{1} \\
&\cong L + \left(\tfrac{n-1}{1} + \tfrac{1}{1}\right) * \left(\tfrac{1}{-1} * \tfrac{1}{-m+1}\right) + \tfrac{0}{1} \\
&\cong \left[\left(L + \tfrac{n-1}{1}\right) + \tfrac{1}{1} * \tfrac{1}{-1} + \tfrac{0}{1}\right] * \tfrac{1}{-m+1} + \tfrac{0}{1} \\
&\cong \left[\left(L + \tfrac{n-1}{1}\right) \circ (1, -1, 0)\right] \circ (-m+1, 0) \\
&\cong R_{90}\left(R_H\left(L + \tfrac{n-1}{1}\right)\right) \circ (-1) \circ (-m+1, 0) \\
&\cong \left[R_{90}\left(R_H(L) + \tfrac{n-1}{1}\right)\right] \circ (-1, -m+1, 0) \\
&\cong \left[R_{90}(R_H(L)) * R_{90}\left(\tfrac{n-1}{1}\right)\right] \circ (-1, -m+1, 0) \\
&\cong \left[R_{90}(R_H(L)) * \tfrac{1}{-n+1}\right] \circ (-1, -m+1, 0) \\
&\cong R_{90}(R_H(L)) \circ (-n+1, 0) \circ (-1, -m+1, 0) \\
&\cong R_{90}(R_H(L)) \circ (-n+1, -1, -m+1, 0)
\end{aligned}$$

The twist vector compositions above are computed using Lemma 3.8. If n and m are greater than 1, then each entry in the twist vector $(-n+1, -1, -m+1, 0)$ denotes negative twisting. Otherwise, this operation will introduce an interior zero entry that could then be collapsed. In either case, the remaining exterior twists have the same sign after performing this isotopy. The other three formulas follow from a similar derivation.

□

The formulas from Lemma 3.20 apply to any 2-string tangle L. In the special case where L is rational, we can use the corresponding twist vector to compute $R_{90}(R_H(L))$. Combining this with Lemma 3.18 will enable us to convert adjacent twist vector entries of opposite sign to the same sign, although it is challenging to derive this equivalence elegantly. Formulating this result using twist vectors is the goal of the next lemma.

Lemma 3.21 (Converting adjacent twist vector entries of opposite sign)**:**
If $T \cong (n_1, \cdots, n_r)$ is a rational tangle defined by a twist vector for which there exist adjacent entries $\pm n_i$ and $\mp n_{i+1}$ of opposite sign, then T satisfies the following equivalence:

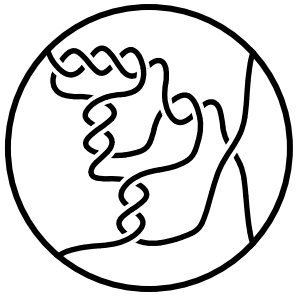
$$T \cong (n_1, \cdots, n_{i-1}, \underbrace{\pm n_i, \mp n_{i+1}}_{\text{opposite sign}}, \cdots, n_r) \cong (-n_1, \cdots, -n_{i-1}, \underbrace{\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1}_{\text{same sign}}, \cdots, n_r).$$

Proof:

Let $T \cong (n_1, \cdots, n_r)$ be a rational tangle defined by a twist vector with some pair of adjacent entries $\pm n_i$ and $\mp n_{i+1}$ of opposite sign. Let $L \cong (n_1, \cdots, n_{i-1})$ be a rational tangle defined by the first $i-1$ twist vector entries. By Lemma 3.18, either L or $-R_{90}(L)$ is a rational subtangle of T. There will be two cases to consider depending on whether $r-i$ is even or odd, but these reduce to the same formula leveraging previous results.

$$
\begin{aligned}
T &\cong (n_1, \cdots, n_{i-1}, \underbrace{\pm n_i, \mp n_{i+1}}_{\text{opposite sign}}, \cdots, n_r) \\
&\cong \begin{cases} (n_1, \cdots, n_{i-1}, \underbrace{\pm n_i}_{\text{horiz.}}, \underbrace{\mp n_{i+1}}_{\text{vert.}}, n_{i+2}, \cdots, n_r) & , \quad \text{if } r-i \text{ is even} \\ (n_1, \cdots, n_{i-1}, \underbrace{\pm n_i}_{\text{vert.}}, \underbrace{\mp n_{i+1}}_{\text{horiz.}}, n_{i+2}, \cdots, n_r) & , \quad \text{if } r-i \text{ is odd} \end{cases} \\
&\cong \begin{cases} (n_1, \cdots, n_{i-1}, 0) \circ (\pm n_i, \mp n_{i+1}, 0) \circ (n_{i+2}, \cdots, n_r) & , \text{even} \\ (n_1, \cdots, n_{i-1}) \circ (\pm n_i, \mp n_{i+1}) \circ (n_{i+2}, \cdots, n_r) & , \text{odd} \end{cases} \quad \text{(Lemma 3.8)} \\
&\cong \begin{cases} [-R_{90}(L) \circ (\pm n_i, \mp n_{i+1}, 0)] \circ (n_{i+2}, \cdots, n_r) & , \text{even} \\ [L \circ (\pm n_i, \mp n_{i+1})] \circ (n_{i+2}, \cdots, n_r) & , \text{odd} \end{cases} \quad \text{(Lemma 3.18)} \\
&\cong \begin{cases} [R_{90}(R_H(-R_{90}(L))) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, 0)] \circ (n_{i+2}, \cdots, n_r) & , \text{even} \\ [R_{90}(R_H(L)) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1)] \circ (n_{i+2}, \cdots, n_r) & , \text{odd} \end{cases} \quad \text{(Lemma 3.20)} \\
&\cong \begin{cases} -R_{180}(L) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & , \text{even} \\ R_{90}(L) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & , \text{odd} \end{cases} \quad \text{(Lemma 3.13)} \\
&\cong \begin{cases} -L \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, \cdots, n_r) & , \text{even} \quad \text{(Corollary 3.15)} \\ R_{90}(L) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, \cdots, n_r) & , \text{odd} \end{cases} \\
&\cong \begin{cases} (-n_1, \cdots, -n_{i-1}) \circ \underbrace{(\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, \cdots, n_r)}_{\text{odd length}} & , \text{even} \quad \text{(Corollary 3.16)} \\ (-n_1, \cdots, -n_{i-1}, 0) \circ \underbrace{(\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, \cdots, n_r)}_{\text{even length}} & , \text{odd} \quad \text{(Lemma 3.14)} \end{cases} \\
&\cong (-n_1, \cdots, -n_{i-1}, \underbrace{\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1}_{\text{same sign}}, \cdots, n_r) \quad \text{(Lemma 3.8)}
\end{aligned}
$$

This derivation proves the formula in the statement of the lemma. □



(a) $T \cong (-4, 3, -2, 3, 1)$

(b) $T \cong (-4, 3, -2, 3, 1)$
$\cong (4, -3, 2-1, 1, 3-1, 1)$
$\cong (4, -3, 1, 1, 2, 1)$

(c) $T \cong (4, -3, 1, 1, 2, 1)$
$\cong (-4, 3-1, 1, 1-1, 1, 2, 1)$
$\cong (-4, 2, 1, 0, 1, 2, 1)$

(d) $T \cong (-4, 2, 1, 0, 1, 2, 1)$
$\cong (4-1, 1, 2-1, 1, 0, 1, 2, 1)$
$\cong (3, 1, 1, 1, 0, 1, 2, 1)$

(e) $T \cong (3, 1, 1, 1, 0, 1, 2, 1)$
$\cong (3, 1, 1, 1+1, 2, 1)$
$\cong (3, 1, 1, 2, 2, 1)$

(f) $T \cong (3, 1, 1, 2, 2, 1)$
$\cong (1, 3-1, 1, 1, 2, 2, 1)$
$\cong (1, 2, 1, 1, 2, 2, 1)$

Figure 3.26: Converting a twist vector to canonical form

Note that the formula from Lemma 3.21 increases the length of the twist vector by one, but the number of crossings in the equivalent diagram is reduced by one. Furthermore, because the corresponding isotopy involves fluping a rational subtangle, the leading twist vector entries matching this subtangle all have their sign reversed. On the other hand, the outermost twist vector entries are unaffected. This means that Lemma 3.21 can be applied multiple times, working from the outside-in, until all twist vector entries have the same sign, with the possible exception of the last entry.

Figure 3.26 shows an example of this for the rational tangle defined by the twist vector $T \cong (-4, 3, -2, 3, 1)$. The diagram can be modified in stages based on the twist vector until all entries have the same sign. As in part 3.26c, these operations may introduce a zero entry in the twist vector. When this happens, the interior zero can be collapsed using Lemma 3.10, which occurs in part 3.26e of the figure. Finally, Lemma 3.12 is used to increment the length of the twist vector in part 3.26f so that it satisfies the odd length convention.

The sequence of transformations illustrated in Figure 3.26 shows how to convert a rational tangle diagram into canonical form based on its twist vector. The most challenging step involves converting the twist vector entries to constant sign, and this is achieved through multiple applications of Lemma 3.21. However, there is a subtlety to keep track of when introducing interior zero entries since these can separate entries of opposite sign. In this case, the zeros must be collapsed to apply the preceding lemma, as outlined in the next proof.

Lemma 3.22 (Twist vectors of constant sign)**:**
Every nontrivial rational tangle T admits a twist vector construction $(n_1, \cdots, n_r)$ such that either $n_i \geq 0$ or $n_i \leq 0$ for all $1 \leq i \leq r$.

Proof:

Let T be a nontrivial rational tangle. By Corollary 3.6, it has a diagram defined by some twist vector $v = (n_1, \cdots, n_r)$. Recall that leading and interior zeros can be collapsed by Lemmas 3.9 and 3.10, so we may assume without loss of generality that all twist vector entries are nonzero, with the possible exception of n_r. Furthermore, the claim holds automatically if $r = 1$, so we may assume that $r \geq 2$.

If the entries in v do not have constant sign, then there exists at least one pair of adjacent entries of opposite sign. Let $1 \leq i < r$ be the largest index for which n_i and n_{i+1} have opposite sign. By Lemma 3.21, there exists an equivalent twist vector $v' \cong v$ of length $r+1$ of the form $v' = (n'_1, \cdots, n'_{i-1}, n'_i, \mp 1, n'_{i+1}, \cdots, n'_r) = (-n_1, \cdots, -n_{i-1}, \mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, \cdots, n_r)$, where n'_i and n'_{i+1} do not have opposite signs, allowing for the special case where either $n'_i = 0$ or $n'_{i+1} = 0$. Since i was chosen to be the largest index corresponding to a pair of adjacent entries of opposite sign in v, it follows that all entries in v' between n'_i and n'_r have the same sign, possibly with some zero entries. Hence, if there exist any remaining pairs of adjacent entries of opposite sign in v', these must occur in the first $i-1$ indices.

If the entries in v' have constant sign, then the claim is proven. If not, then let $1 \leq i' < i$ be the largest index for which $n'_{i'}$ and $n'_{i'+1}$ have opposite sign. By the same argument as above, Lemma 3.21 guarantees that there exists a twist vector $v'' \cong v' \cong v$ of length $r+2$ for which all but the first $i'-1 < i-1$ entries are known to have non-opposite sign. Since the number of leading twist vector entries decreases after each iteration, we may repeat this process as many times as needed until we obtain a twist vector $v_2 \cong v$ where no pairs of adjacent entries have opposite signs, possibly with some interior zeros.

It is still possible that the twist vector v_2 has non-constant sign at this stage if an interior zero separates a positive and negative entry, such as $v_2 = (1, 0, -1, 0, 1)$ for example. In this

case, we may collapse all leading and interior zeros to obtain a twist vector $v_2' \cong v_2$ containing all nonzero entries except possibly the last. After canceling any combined twists of opposite sign using Reidemeister II moves, we may repeat the algorithm from the beginning again to obtain an equivalent twist vector with no adjacent entries of opposite sign.

Observe that each application of Lemma 3.21 and each Reidemeister II simplification reduces the total number of crossings in the diagram. Hence, this algorithm must eventually terminate, either when all crossings are eliminated or when we obtain a twist vector where each nonzero entry has the same sign. Since each twist vector transformation used in this proof preserves the equivalence class of the rational tangle, we eventually obtain some twist vector of constant sign v_c which defines an equivalent diagram for the original rational tangle, $T \cong v_c$. This twist vector will satisfy the claim, completing the proof.

□

3.2.4 Rational Tangles in Canonical Form

The preceding series of lemmas describe a number of twist vector modifications that can be used to transform a rational tangle diagram. Those transformations which preserve the equivalence class of the rational tangle provide a set of rules for performing isotopies on rational tangle diagrams. We consolidate these results into the following theorem.

Theorem 3.23 (Summary of rules for equivalent twist vectors)**:**
If two twist vectors are related by the operations of:

a. collapsing a leading zero (Lemma 3.9): $(0, n_2, n_3 \cdots, n_r) \cong (n_3, \cdots, n_r)$;

b. collapsing an interior zero (Lemma 3.10):
$(n_1, \cdots, n_{i-1}, 0, n_{i+1}, \cdots, n_r) \cong (n_1, \cdots, n_{i-1} + n_{i+1}, \cdots, n_r)$;

c. incrementing length by 1 (Lemma 3.12): $(n_1, n_2, \cdots, n_r) \cong (\pm 1, n_1 \mp 1, n_2, \cdots, n_r)$;

d. or converting two adjacent entries of opposite sign to three adjacent entries of the same sign, with a consequent sign reversal on leading twists (Lemma 3.21):
$(n_1, \cdots, n_{i-1}, \underbrace{\pm n_i, \mp n_{i+1}}_{\text{opposite sign}}, \cdots, n_r) \cong (-n_1, \cdots, -n_{i-1}, \underbrace{\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1}_{\text{same sign}}, \cdots, n_r)$;

then these two twist vectors define equivalent rational tangle diagrams.

The four operations outlined in Theorem 3.23 will always yield equivalent twist vectors, which is to say twist vectors that define equivalent rational tangle diagrams. However, this does not imply that all equivalent twist vectors are related by only these operations. Even so, these will be sufficient to obtain a canonical diagram for any rational tangle. Our interest in characterizing canonical rational tangle diagrams motivates our definition of a *canonical twist vector* in the next lemma.

Lemma 3.24 (Canonical twist vectors)**:**
Every nontrivial twist vector is equivalent to a *canonical twist vector* $(n_1, \cdots, n_r)$ of odd length, where each entry is nonzero and has the same sign, except possibly $n_r = 0$.

Proof:

Let $v = (n_1, \cdots, n_r)$ be any nontrivial twist vector, meaning that v is not equivalent to a twist vector with only zero twists. By definition, v defines the construction for some nontrivial rational tangle diagram $T \cong (n_1, \cdots, n_r)$. By Lemma 3.22, we know that T admits some twist vector construction of constant sign which is equivalent to v. Thus, we may assume without loss of generality that $v = (n_1, \cdots, n_r)$ satisfies this condition, so either $n_i \geq 0$ or $n_i \leq 0$ for all $1 \leq i \leq r$. Again without loss of generality, suppose that $n_i \geq 0$.

No twist vector entries in v are negative, but some entries may be zero. By applying Lemmas 3.9 and 3.10 as many times as needed, we may collapse all leading and interior zero twists until we obtain an equivalent twist vector $v' = (m_1, \cdots, m_s) \cong v$, where $m_i \neq 0$ for all $1 \leq i < s$ (Corollay 3.11). Since all nonzero entries in v were assumed to be positive, it follows that all entries in v' obtained through this process will be positive, with the possible exception of $m_s = 0$.

It remains to find an equivalent twist vector of odd length. If s is odd, then the twist vector v' satisfies the claim. If s is even, then we may apply Lemma 3.12 to convert to an equivalent twist vector $v'' = (1, m_1 - 1, m_2, \cdots, m_s) \cong v' \cong v$ of odd length $s+1$. If $m_1 > 1$, then all leading entries in v'' are positive and hence this twist vector satisfies the claim. If $m_1 = 1$, then the second entry of v'' is an interior zero which may be collapsed using Lemma 3.10. This yields an equivalent twist vector $v''' = (1 + m_2, \cdots, m_s) \cong v'' \cong v' \cong v$ of odd length $s - 1$ and for which all entries are positive, except possibly $m_s = 0$.

In all cases, we are able to obtain a twist vector of odd length which is equivalent to v and for which all entries have the same sign, with the possible exception of a final zero entry. This is the desired canonical twist vector, completing the proof.

□

Recall that a diagram for a rational tangle is said to be *canonical* if it is constructed using only right and bottom twists of constant sign. All twist vectors denote right and bottom twisting, and we introduced canonical twist vectors to satisfy the remaining conditions on canonicity for a rational tangle diagram. Requiring that these also satisfy the odd length convention is a convenience to ensure that the first and last twist vector entries denote horizontal twisting. This convention simplifies tangle algebra involving rational tangles since we need not consider a separate even case. Proving that every rational tangle admits a canonical diagram is now simple in light of our previous work.

Theorem 3.25 (Existence of canonical diagrams for rational tangles)**:**
Every nontrivial rational tangle admits a *canonical diagram* constructed using only right and bottom twists of constant sign, and which is defined by a canonical twist vector.

Proof:

If T is a rational tangle, then it has a diagram with construction defined by some twist vector (Corollary 3.6). If T is nontrivial, then by Lemma 3.24, this twist vector is equivalent to a canonical twist vector $T \cong (n_1, \cdots, n_r)$. By definition, this canonical twist vector defines an equivalent diagram for T constructed using only right and bottom twists of constant sign.

□

While Lemma 3.24 guarantees that any twist vector is equivalent to some canonical twist vector, it does not imply that this canonical twist vector is unique. This deficiency extends to Theorem 3.25, which establishes the existence but not uniqueness of canonical diagrams for rational tangles. We would also like to prove that different canonical diagrams define non-equivalent rational tangles. To accomplish this, we can leverage a variation of the well known Tait Flyping Conjecture, which was originally posed by Tait [37] and later proven by Menasco and Thistlethwaite [27, 26].

Theorem 3.26 (The Tait Flyping Conjecture for knots [37, 27, 26])**:**
Any two reduced, alternating diagrams of the same knot are equivalent on the sphere if and only if those diagrams are related by a finite sequence of flypes.

To extend this result to rational tangles, we will follow the approach used in a paper by Kauffman and Lambropoulou [20]. In particular, they reference an earlier work due to Sundberg and Thistlethwaite which argues that the solution to the Tait Flyping Conjecture

(a) 2-string tangle T **(b)** Vertex closure $V(T)$

Figure 3.27: Vertex closure for a 2-string tangle

also applies to vertex closures of alternating 2-string tangle diagrams [36]. A *vertex closure* of a 2-string tangle diagram T refers to the diagram $V(T)$ obtained by collapsing the boundary ball to a point (Figure 3.27). Visualizing the tangle diagram as sitting on the surface of a sphere, we can imagine this vertex as the point ∞ on the opposite side of the sphere. Note that this vertex is marked: the edges connected to it are labeled NW, NE, SE, SW, as shown in the figure. With this in mind, we may state the following variation on the Tait Flyping Conjecture for vertex closures of tangles.

Corollary 3.27 (The Tait Flyping Conjecture for vertex closures of tangles [36])**:**
Two vertex closures of reduced, alternating 2-string tangle diagrams are equivalent on the sphere if and only if they are related by a finite sequence of flypes.

We will assume that Corollary 3.27 holds, but note that this result is only stated for vertex closures of tangle diagrams; it does not immediately apply to tangle diagrams alone. In particular, two types of flypes are possible in a vertex closure: those which involve subtangles (Figure 2.16) and those which involve the boundary vertex (Figure 3.28). Recall that our

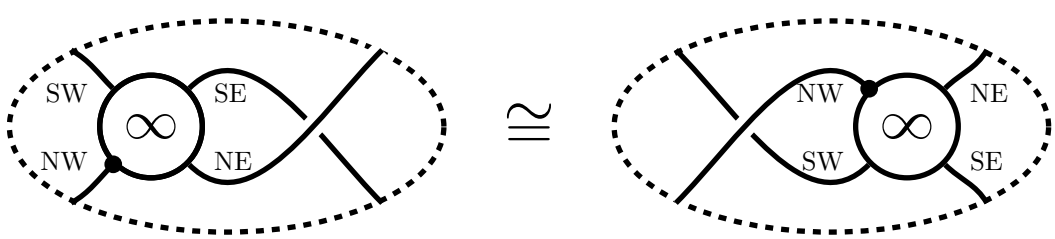


Figure 3.28: Horizontal vertex flype, with NW corner marked

definition of tangle isotopy requires that the boundary sphere be fixed, and hence flypes which involve the boundary vertex do not correspond to a tangle isotopy in general. However, in the special case of a rational tangle, we will show that a vertex flype is equivalent to a sequence of subtangle flypes and isotopies on the sphere which preserve the equivalence class of the tangle diagram. Proving this will require the following lemma concerning rational subtangles.

Lemma 3.28 (Subtangles of rational tangles [20])**:**
Every 2-string subtangle of a rational tangle diagram in twist form is rational.

Proof:

Let T be a rational tangle diagram in twist form, that is, defined by some finite combination of vertical and horizontal twists on two strings. We will proceed by induction on the number of twists n in this diagram, starting with $n = 1$ for the base case. If T has a single twist, then it is either the $+1$ or -1 tangle. Any 2-string subtangle of T will either contain the twist or will contain two untwisted string segments. Figure 3.29 shows all of the possible 2-string subtangles of the $+1$ tangle. The 1-crossing subtangle is again the $+1$ tangle and hence rational, and the 0-crossing subtangles are either the 0 or ∞ tangle, both of which are rational. In all cases, a 2-string subtangle of T is rational, proving the base case.

Now suppose that every 2-string subtangle of a rational tangle diagram with $n \geq 1$ twists is rational. Let T be a rational tangle with $n + 1$ twists, and let L be the subtangle defined by the first n of these twists. Hence, T is obtained by adding the $n + 1^{\text{st}}$ twist to L, which is to say that T is the either the sum or product of L with the $+1$ or -1 tangle. Suppose without loss of generality that $T \cong L + 1$. Note that L is a rational tangle with n twists by construction, and hence all 2-string subtangles of L are rational by the inductive hypothesis.

Let U be a 2-string subtangle of T. If U is entirely contained in L, then it is rational. If U is completely disjoint from L, then it contains either one or zero twists, in which case it

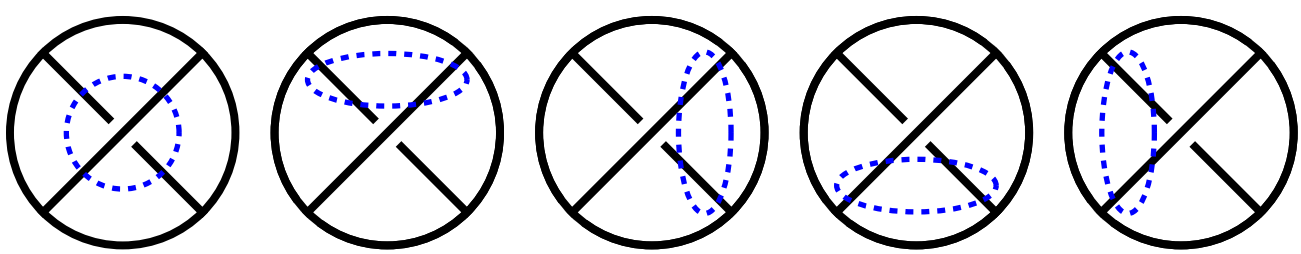
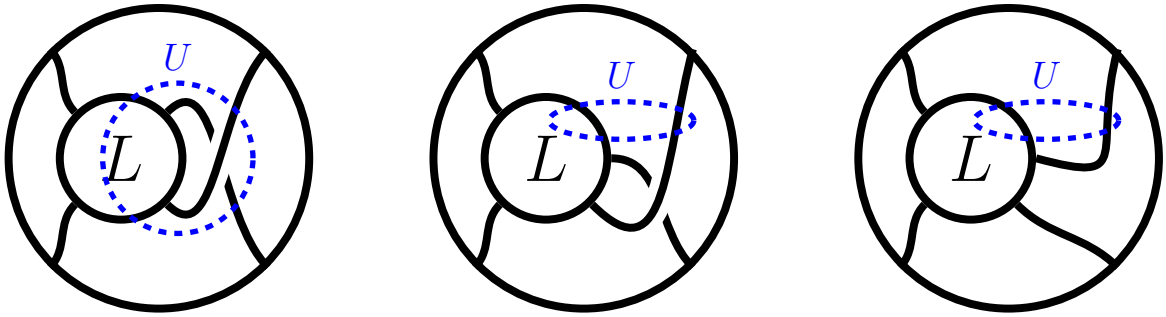


Figure 3.29: All possible 2-string subtangles of the $+1$ tangle



(a) Subtangle containing twist **(b)** Subtangle not containing twist **(c)** Subtangle with twist removed

Figure 3.30: Possible subtangles U of a rational tangle $T \cong L + 1$ which intersect L

will be rational by the same argument used in the base case. Consider what happens if U overlaps partially with the diagram contained in L and partially with the diagram exterior to L (Figure 3.30). There are two cases depending on whether or not U contains the $n + 1^{\text{st}}$ twist, each illustrated in the figure above.

If U does not contain this twist (3.30b), then observe that U is equivalent to a subtangle of a rational tangle diagram with n twists after removing the $n + 1^{\text{st}}$ twist from T (3.30c). In this case, U is rational. Suppose instead that U does contain the $n + 1^{\text{st}}$ twist (3.30a). Since U overlaps with L, there exists a subtangle L' of L for which $U \cong L' + 1$. Note that L' must be rational by the inductive hypothesis. Hence, U is obtained by adding a twist to a rational tangle, in which case U must also be rational. In all cases, we see that a 2-string subtangle U of T must be rational, completing the proof by induction.

□

To perform the vertex flype introduced in Figure 3.28, we must be able to isolate the boundary vertex and an adjacent crossing from the rest of the diagram. For the vertex closure $V(T)$ of a rational tangle diagram T in twist form, Lemma 3.28 guarantees that the portion of the diagram away from the region of the vertex flype is a rational subtangle. Furthermore, any subtangle flypes in $V(T)$ will involve only rational subtangles.

Observe that performing a vertex flype in $V(T)$ flips the marked vertex upside down, but it is not obvious how to visualize this in the corresponding tangle diagram T without the vertex closure. To keep the boundary circle of T fixed with the NW corner at the top left, we may instead interpret flipping the vertex as flipping the entire tangle diagram. Since rational tangles are equivalent under flipping via a sequence of flypes (Lemma 3.3), we can leverage this idea to extend the Tait Flyping Conjecture to rational tangles.

Theorem 3.29 (The Tait Flyping Conjecture for rational tangles [20])**:**
Two reduced, alternating rational tangle diagrams in twist form are equivalent on the sphere if and only if they are related by a finite sequence of flypes of rational subtangles.

Proof:

Let T_1 and T_2 be two reduced, alternating tangle diagrams in twist form for the same rational tangle. Hence, there exists an isotopy $T_1 \cong T_2$. We imagine these diagrams as sitting on the surface of a sphere rather than in the plane, which will allow us to slide the strings around the back of the sphere. Let $V(T_1)$ and $V(T_2)$ be the respective vertex closures of these diagrams, which will also be reduced and alternating. The tangle isotopy $T_1 \cong T_2$ induces a corresponding isotopy of vertex closures $V(T_1) \cong V(T_2)$. Thus, by the Tait Flyping Conjecture, $V(T_1)$ and $V(T_2)$ are related by a finite sequence of flypes (Corollary 3.27).

Earlier, we made the observation that flypes in a vertex closure will either involve a subtangle or they will involve the special marked vertex denoting the endpoints. A subtangle flype in a vertex closure $V(T)$ induces a corresponding subtangle flype in the tangle diagram T. If T is assumed to be a rational tangle diagram in twist form, then these must be flypes of rational subtangles by Lemma 3.28. Our goal is to show that a vertex flype in $V(T)$ is equivalent to a sequence of subtangle flypes in the special case where T is a reduced, alternating rational tangle diagram in twist form. Showing this will imply that an isotopic sequence of flypes in $V(T)$ induces an isotopic sequence of flypes of rational subtangles in T. We consider a horizontal vertex flype, but a similar argument holds for the vertical case.

Suppose without loss of generality that T is a rational tangle diagram in twist form which is alternating and reduced. Consider the sequence of operations on the vertex closure $V(T)$ shown in Figure 3.31. If it is possible to perform a vertex flype on $V(T)$ as introduced in Figure 3.28, then we must be able to isolate the vertex and an adjacent crossing from T in a separating circle that intersects the diagram in exactly four places. An example of this is shown in the dotted region of the figure, where the endpoint vertex is denoted by ∞ with a black dot used to mark the NW corner. The remainder of the diagram outside of this circle is a subtangle L of T, where we note that L is rational by Lemma 3.28.

Following Figure 3.31 counterclockwise from the top left corner, we obtain a sequence of operations that can be used to achieve the same effect as a vertex flype. We allow for the strings to slide along the surface of the sphere as long as these do not change the crossings. In particular, the final *sphere slide* operation is obtained by sliding the subtangle L around the back of the sphere to place it below the dashed region. A similar effect could also be

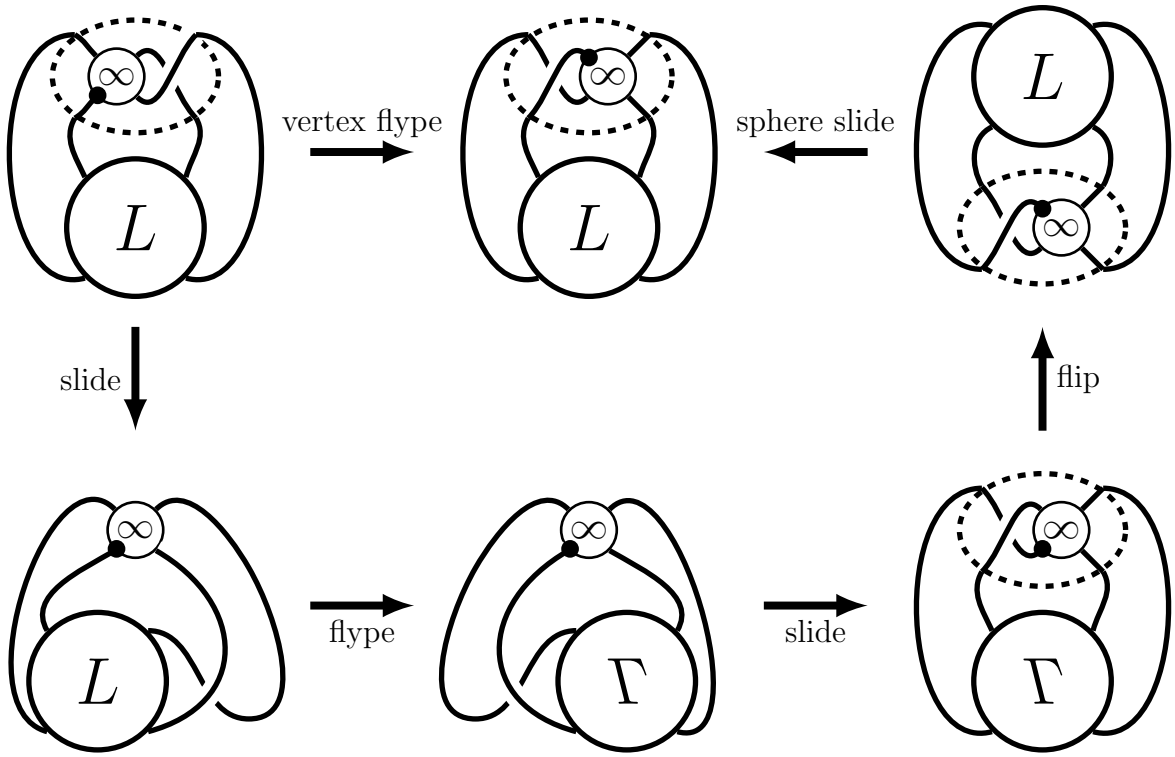


Figure 3.31: Vertex flype reconfigured using subtangle flypes and isotopies on the sphere

achieved by sliding L beneath the dotted region using Reidemeister moves, but the Tait Flyping Conjecture only allows for flypes; using a sphere slide avoids any crossing changes.

The only operation in Figure 3.31 which requires that T be rational is the flip. The entire diagram for $V(T)$ is flipped upside down so that the endpoint vertex matches the orientation following a vertex flype. Observe that the rational subtangle L is also flipped. In general, flipping is not a tangle isotopy since it moves the boundary of the tangle, but because L is rational, it is equivalent under flipping via a sequence of flypes (Lemma 3.3). Thus, the sequence of operations shown in Figure 3.31 reduces to a series of subtangle flypes.

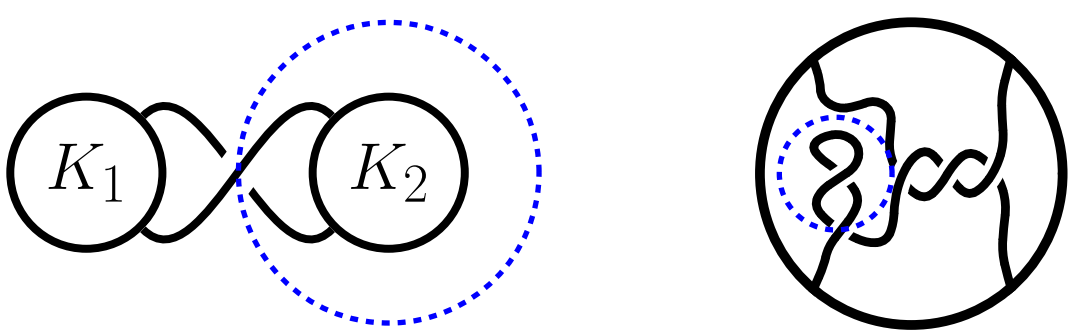
Returning to the original premise, we have two reduced, alternating tangle diagrams in twist form T_1 and T_2 for the same rational tangle. Their vertex closures $V(T_1)$ and $V(T_2)$ are isotopic via a sequence of vertex flypes and subtangle flypes. Our preceding argument shows that these induce an isotopic sequence of rational subtangle flypes from T_1 to T_2. This completes one direction of the proof. The other direction is immediate since the existence of a sequence of flypes between two rational tangle diagrams proves they are isotopic.

□

While we did not leverage it explicitly in the proof of Theorem 3.29, applying the Tait Flyping Conjecture requires that the diagrams be reduced and alternating. We will show in the next lemma that rational tangle diagrams in canonical form satisfy both conditions, but first we pause to define these terms. A diagram is called *alternating* if it alternates between over-crossings and under-crossings while traveling along the strand. Part 3.26a from Figure 3.26 shows a non-alternating rational tangle diagram, while part 3.26f shows an equivalent diagram which is alternating. In fact, this alternating behavior for rational tangle diagrams does not require the canonical form, it is sufficient that the diagram be constructed using twists of constant sign.

A diagram is called *reduced* if it contains no nugatory crossings, and a crossing is called *nugatory* if there exists a circle intersecting the diagram only at this crossing and nowhere else (Figure 3.32). Figure 3.32a is adapted from [15] and shows an example of a nugatory crossing in a knot diagram, where we imagine separating the rest of the diagram into two disjoint spheres K_1 and K_2. Observe that this crossing could be removed by flipping the portion of the knot in either of these two spheres, indicating that the diagram can be further reduced. Figure 3.32b shows an example of a non-reduced rational tangle diagram with nugatory crossings that are clearly removable through Reidemeister I moves. We will see that a canonical diagram for a rational tangle does not contain nugatory crossings.

Lastly, Theorem 3.29 also required that the rational tangle diagrams be in *twist form*, meaning constructed from a combination of horizontal and vertical twists. We took this as the definition of rational tangle (Figure 3.6), but one could conceivably introduce crossings into a diagram using Reidemeister moves which do not correspond to twists. We are only interested in diagrams for rational tangles in twist form.



(a) Separating circle for a nugatory crossing [15] **(b)** Rational tangle diagram with nugatory crossings

Figure 3.32: Examples of reducible knot and tangle diagrams with nugatory crossings

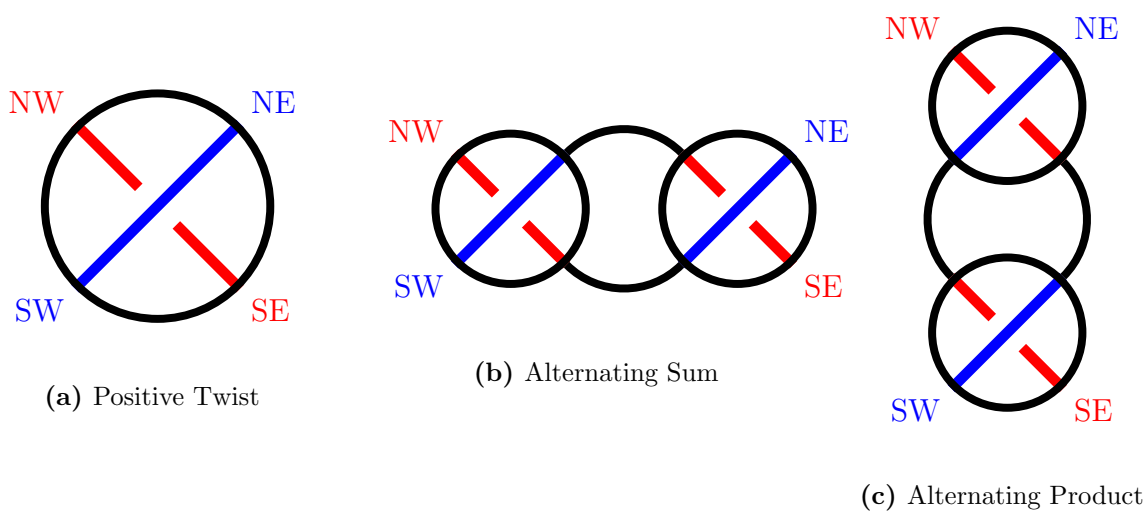
Lemma 3.30 (Reduced and alternating canonical diagrams)**:**
Rational tangle diagrams in canonical form are reduced, alternating, and in twist form.

Proof:

Note that the claim holds vacuously for 0-crossing rational tangles. Let T be a non-trivial rational tangle diagram in canonical form defined by a canonical twist vector $T \cong (n_1, \cdots, n_r)$ (Lemma 3.25). Recall that a twist vector denotes the sequence of alternately right and bottom twists used to construct the diagram, and hence T is in twist form by definition. We will first show that T is alternating and then that T is reduced.

Since a canonical twist vector has odd length, we know that the first twist is horizontal. Since the entries in a canonical twist vector must have constant sign, suppose without loss of generality that T is constructed from only positive twists. Hence, T is a constructed from the 0 tangle through some finite combination of sums and products with the $+1$ tangle. Figure 3.33 shows how to characterize the alternating behavior of the crossings in T based on its construction from 1-crossing subtangles.

Since the $+1$ tangle contains only a single crossing, its diagram is automatically alternating. Observe that the NW and SE endpoints of the $+1$ tangle connect to an under-strand while the NE and SW endpoints connect to an over-strand (3.33a). A sum of two tangle diagrams connects the east endpoints of one summand with the west endpoints of the other. Hence, a sum of two alternating tangle diagrams will continue to be alternating if and only if both connections join an under-strand with an over-strand.



(a) Positive Twist

(b) Alternating Sum

(c) Alternating Product

Figure 3.33: Alternating behavior for a diagram constructed from positive twists

Thus, a sum of the +1 tangle with the +1 tangle yields an alternating diagram (3.33b). Furthermore, this sum preserves the alternating behavior of the endpoints: under-strands at the NW and SE endpoints; over-strands at the NE and SW endpoints. Analogous facts hold using tangle products (3.33b). Therefore, in building up the tangle diagram for T from 1-crossing subtangles, we see that each sum and product with the +1 tangle preserves this alternating behavior. Hence, we may conclude that a rational tangle diagram in canonical form is alternating.

It remains to prove that a canonical diagram for a rational tangle is reduced. To show that no crossing in T is nugatory, we must prove that there does not exist a separating circle for any crossing in the diagram. Note that any separating circles which exist must intersect the diagram only at the nugatory crossing and be entirely contained in the boundary ball of the tangle, as in Figure 3.32b. We will proceed by induction on the number of crossings n in the diagram. The result is vacuously true using $n = 0$ as the base case since a trivial tangle contains no crossings and hence no nugatory crossings.

Suppose that every canonical diagram for a rational tangle with at most $n \geq 0$ crossings is reduced. Let T be a canonical diagram for a rational tangle with $n+1$ crossings. As before, we will assume that T is constructed using only positive twists. Let L be the subtangle of T constructed from the first n twists. Hence, L is a rational tangle in canonical form with n crossings, and T is obtained from either the sum or product of L with the +1 tangle. Suppose without loss of generality that $T \cong L + 1$.

Since L has a canonical rational tangle diagram with n crossings, it is reduced by the inductive hypothesis and thus contains no nugatory crossings. Hence, if T contains a nugatory crossing, then it cannot be contained in L. In other words, the only possible nugatory crossing in T corresponds to the $n + 1^{\text{st}}$ twist. Suppose by way of contradiction that T is not reduced, and so this twist is a nugatory crossing. Hence, there exists a separating circle through this crossing, as shown in Figure 3.34.

Observe that this circle must pass through the interior of the subtangle L, but cannot intersect any portion of the diagram. Hence, the separating circle splits L into two disjoint parts, as shown in Figure 3.34. If the diagram for $T \cong L + 1$ splits apart in this way, then the two strings in T never cross. However, this contradicts the fact that the very first twist in a nontrivial canonical rational tangle diagram introduces a crossing between these two strings. Thus, we may conclude that T cannot contain a nugatory crossing, completing the proof by induction. Therefore, a canonical diagram for a rational tangle must be reduced.

□

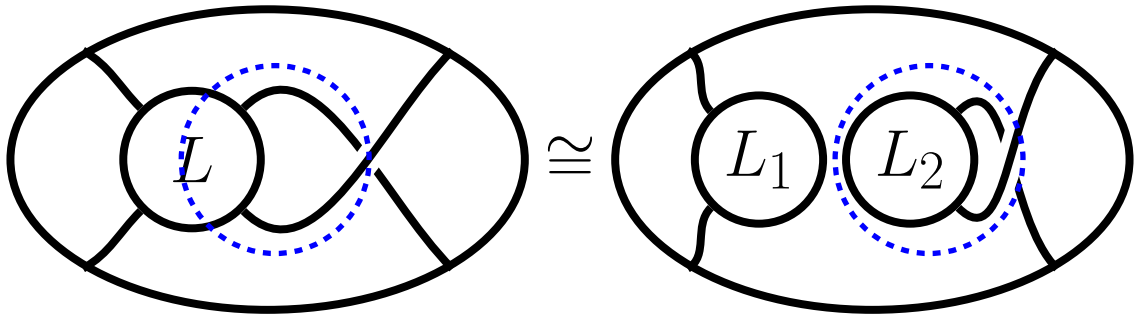


Figure 3.34: Contradictory nugatory crossing in a canonical rational tangle diagram

Lemma 3.30 ensures that rational tangle diagrams in canonical form satisfy the conditions of the Tait Flyping Conjecture, which is the basis for proving that canonical diagrams are unique. We have already seen how flypes can be used to shuffle twists in a rational tangle diagram to the preferred right and bottom sides. Uniqueness will follow from the fact that these are the only moves possible between equivalent canonical diagrams.

Theorem 3.31 (Uniqueness of canonical diagrams for rational tangles)**:**
Every nontrivial rational tangle admits a unique canonical diagram.

Proof:

Let T be a nontrivial rational tangle. By Theorem 3.25, we know that T has a canonical diagram defined by a canonical twist vector. Suppose by way of contradiction that this canonical diagram is not unique. Thus, there exist two different canonical twist vectors v_1 and v_2 for which $T \cong v_1 \cong v_2$, but $v_1 \neq v_2$. Note that v_1 and v_2 are equivalent by definition since they define diagrams for equivalent rational tangles, but this alone does not imply they are identical twist vectors.

Let D_1 and D_2 be the two different canonical diagrams for T defined by v_1 and v_2, respectively. By Lemma 3.30, we know that D_1 and D_2 are both reduced, alternating, and in twist form. Since $D_1 \cong D_2$ as rational tangles, Theorem 3.29 implies that these two diagrams are related by a finite sequence of flypes of rational subtangles. By definition, a flype moves a twist from one side of a subtangle to the other side and flips the subtangle in the process (Lemma 2.1). By Lemma 3.28, all subtangles of a rational tangle diagram are rational and hence isotopic under flipping (Lemma 3.3), where we note that this isotopy reduces to a finite sequence of flypes. Therefore, a flype in a rational tangle diagram has the effect of shifting a twist from the left side to the right side of a subtangle and vice versa, or from the top side to the bottom side and vice versa.

Since we assumed that the rational tangle diagrams D_1 and D_2 are in canonical form, all twists have already been shuffled to the right and bottom sides of both diagrams by definition. If D_1 and D_2 are not already identical diagrams, then D_1 cannot be converted into D_2 only using flypes. However, this violates the Tait Flyping Conjecture for rational tangles, yielding our desired contradiction. Therefore, we may conclude that the rational tangle T must have a unique canonical diagram.

□

3.3 Rational Tangles and Rational Numbers

Our next goal is to build up to the proof that there exists a bijection between rational tangles and the extended rational numbers [7]. We start by exploring how to assign fractions to rational tangle diagrams, an idea which revolves around the interesting combinatorial properties of twist vectors. Every twist vector defines a continued fraction that can be used to compute a rational number $\frac{p}{q}$ for the associated diagram. We will see that continued fractions from equivalent twist vectors simplify to the same rational number. The desired bijection will follow from the uniqueness of canonical diagrams.

In contrast to the preceding subsection's focus on topological results using diagrams, this section will include primarily algebraic results which rely on manipulating continued fractions. A detailed treatment of continued fractions is given in Section XIII of a text on number theory by Roberts [33]. We will be particularly interested in leveraging Roberts' results concerning rational numbers. A number of similar of results relating rational tangles to continued fractions are also proven in the paper by Kauffman and Lambropoulou [20].

3.3.1 The Continued Fraction from a Twist Vector

Any finite sequence of integers $(n_1, \cdots, n_r)$ defines a *twist vector*, and this twist vector defines a diagram for some rational tangle (Corollary 3.6). Define a map from twist vectors to the extended rational numbers $\mathrm{CF} : \{\text{twist vectors}\} \to \mathbb{Q} \cup \{\infty\}$ by the formula

$$\mathrm{CF}(n_1, \cdots, n_r) = n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} = \frac{p}{q},$$

where $\frac{p}{q}$ is obtained by rationalizing the *continued fraction* corresponding to a given twist vector. Notice that the order of the twist vector entries is reversed in the continued fraction.

$$(-3,-3,3) \cong (3,2,1,2) \cong (1,2,2,1,2)$$

$$\mathrm{CF}(-3,-3,3) = \mathrm{CF}(3,2,1,2) = \mathrm{CF}(1,2,2,1,2) = \frac{27}{10}$$

Figure 3.35: Equivalent twist vectors which define identical fractions

The extended rational number $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$ is computed through a repeated application of the Euclidean algorithm, where we allow that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$. Figure 3.35 shows examples of rational tangle diagrams obtained from equivalent twist vectors. One can verify that the corresponding continued fractions simplify to the same rational number. We will prove that equivalent twist vectors always define identical fractions, but first we state some basic algebraic results concerning continued fractions.

Lemma 3.32 (The negative of a continued fraction)**:**
For any twist vector $(n_1, \cdots, n_r)$, the negative of its continued fraction satisfies:

$$-\mathrm{CF}(n_1, \cdots, n_r) = \mathrm{CF}(-n_1, \cdots, -n_r).$$

Proof:

This result is proven through direct compoutation.

$$\begin{aligned} -\mathrm{CF}(n_1, \cdots, n_r) &= -\left(n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} \right) \\ &= -n_r + \frac{1}{-n_{r-1} + \cdots + \frac{1}{-n_2 + \frac{1}{-n_1}}} \\ &= \mathrm{CF}(-n_1, \cdots, -n_r) \end{aligned}$$

□

Lemma 3.33 (Composing continued fractions)**:**
The continued fraction for any twist vector $(n_1, \cdots, n_r)$ can be rewritten using a composition:

$$\mathrm{CF}(n_1, \cdots, n_{i-1}, n_i, \cdots, n_r) = \mathrm{CF}(\mathrm{CF}(n_1, \cdots, n_{i-1}), n_i, \cdots, n_r).$$

Proof:

Note that $(\mathrm{CF}(n_1, \cdots, n_{i-1}), n_i, \cdots, n_r)$ does not represent a twist vector when the expression $\mathrm{CF}(n_1, \cdots, n_{i-1})$ in the first entry is not an integer. The formula for a continued fraction still makes sense with this slight abuse of notation, but this trick only works for the first entry. The claim again follows from direct computation.

$$\begin{aligned}
\mathrm{CF}(n_1, \cdots, n_{i-1}, n_i, \cdots, n_r) &= n_r + \cfrac{1}{n_{r-1} + \cdots + \frac{1}{n_i + \frac{1}{n_{i-1} + \cdots + \frac{1}{n_1}}}} \\
&= n_r + \cfrac{1}{n_{r-1} + \cdots + \frac{1}{n_i + \frac{1}{\mathrm{CF}(n_1, \cdots, n_{i-1})}}} \\
&= \mathrm{CF}(\mathrm{CF}(n_1, \cdots, n_{i-1}), n_i, \cdots, n_r)
\end{aligned}$$

□

Recall that Theorem 3.23 gives a list of rules for equivalent twist vectors, each of which was proven topologically using rational tangles. Now we can use the algebra of continued fractions to prove that any twist vectors related by these rules define the same fraction under CF. We begin with the following lemma that parallels Theorem 3.23.

Lemma 3.34 (Continued fraction identities)**:**
If $\mathrm{CF}(n_1, \cdots, n_r) = n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}}$ denotes the continued fraction associated to the twist vector $(n_1, \cdots, n_r)$, then the following identities are satisfied:

a. $\mathrm{CF}(0, n_2, n_3, \cdots, n_r) = \mathrm{CF}(n_3, \cdots, n_r)$;

b. $\mathrm{CF}(n_1, \cdots, n_{i-1}, 0, n_{i+1}, \cdots, n_r) = \mathrm{CF}(n_1, \cdots, n_{i-1} + n_{i+1}, \cdots, n_r)$;

c. $\mathrm{CF}(n_1, n_2, \cdots, n_r) = \mathrm{CF}(\pm 1, n_1 \mp 1, n_2, \cdots, n_r)$; and

d. $\mathrm{CF}(n_1, \cdots, n_{i-1}, \underline{\pm n_i, \mp n_{i+1}}, \cdots, n_r)$
$= \mathrm{CF}(-n_1, \cdots, -n_{i-1}, \underline{\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1}, \cdots, n_r)$.

Proof:

The four identities in question reduce to straightforward albeit tedious computations involving continued fractions. Equivalent computations leveraging the Euler bracket function can be found in both Roberts [33] and Kohn [21], but a proof directly working with continued fractions is given here. A choice of sign is made in deriving the third and fourth identities, but the computations are similar for the other cases.

$$\begin{aligned}
\mathrm{CF}(0, n_2, n_3, \cdots, n_r) &= \cdots + \frac{1}{n_3 + \frac{1}{n_2 + \frac{1}{0}}} \\
&= \cdots + \frac{1}{n_3 + \frac{1}{n_2 + \infty}} \\
&= \cdots + \frac{1}{n_3 + 0} \\
&= \mathrm{CF}(n_3, \cdots, n_r)
\end{aligned}$$

$$\begin{aligned}
\mathrm{CF}(n_1, \cdots, n_{i-1}, 0, n_{i+1}, \cdots, n_r) &= \cdots + \frac{1}{n_{i-1} + \frac{1}{0 + \frac{1}{n_{i+1} + \cdots}}} \\
&= \cdots + \frac{1}{n_{i-1} + \frac{1}{\left(\frac{1}{n_{i+1} + \cdots}\right)}} \\
&= \cdots + \frac{1}{n_{i-1} + (n_{i+1} + \cdots)} \\
&= \mathrm{CF}(n_1, \cdots, n_{i-1} + n_{i+1}, \cdots, n_r)
\end{aligned}$$

$$\begin{aligned}
\mathrm{CF}(n_1, n_2, \cdots, n_r) &= \cdots + \frac{1}{n_2 + \frac{1}{n_1}} \\
&= \cdots + \frac{1}{n_2 + \frac{1}{n_1 - 1 + 1}} \\
&= \cdots + \frac{1}{n_2 + \frac{1}{n_1 - 1 + \frac{1}{1}}} \\
&= \mathrm{CF}(1, n_1 - 1, n_2, \cdots, n_r)
\end{aligned}$$

Before computing the fourth identity, we will prove the following intermediate result:

$$\mathrm{CF}(c, -n, m) = \mathrm{CF}(-c, n-1, 1, m-1).$$

In this three entry twist vector, n and m are integers while c is either an integer or another continued fraction that can be substituted using the formula from Lemma 3.33.

$$\begin{aligned}
\mathrm{CF}(c,-n,m) &= m+\frac{1}{-n+\frac{1}{c}} \\
&= m-1+1-\frac{1}{n-\frac{1}{c}} \\
&= m-1+\frac{n-\frac{1}{c}}{n-\frac{1}{c}}-\frac{1}{n-\frac{1}{c}} \\
&= m-1+\frac{n-1-\frac{1}{c}}{n-\frac{1}{c}} \\
&= m-1+\frac{1}{\frac{n-\frac{1}{c}}{n-1-\frac{1}{c}}} \\
&= m-1+\frac{1}{\frac{n-1-\frac{1}{c}}{n-1-\frac{1}{c}}+\frac{1}{n-1-\frac{1}{c}}} \\
&= m-1+\frac{1}{1+\frac{1}{n-1+\frac{1}{-c}}} \\
&= \mathrm{CF}(-c,n-1,1,m-1)
\end{aligned}$$

Now, by Lemmas 3.32 and 3.33, we may let $c=\mathrm{CF}(n_1,\cdots,n_{i-1})$ and note that $-c=\mathrm{CF}(-n_1,\cdots,-n_{i-1})$. Let $n=n_i$ and $m=n_{i+1}$. We finish the fourth identity as follows.

$$\begin{aligned}
\mathrm{CF}(n_1,\cdots,n_{i-1},-n_i,n_{i+1},\cdots,n_r) &= \mathrm{CF}(c,-n,m,\cdots,n_r) \\
&= \mathrm{CF}(\mathrm{CF}(c,-n,m),\cdots,n_r) \\
&= \mathrm{CF}(\mathrm{CF}(-c,n-1,1,m-1),\cdots,n_r) \\
&= \mathrm{CF}(-c,n-1,1,m-1,\cdots,n_r) \\
&= \mathrm{CF}(-n_1,\cdots,-n_{i-1},n_i-1,1,m_{i+1}-1,\cdots,n_r)
\end{aligned}$$

□

Notice that the intermediate result used to prove the fourth identity in the preceding proof is similar to how Lemma 3.20 was used to prove Lemma 3.21 in the corresponding rule for twist vectors. Lemma 3.34 implies that any twist vectors related by one of the four rules outlined in Theorem 3.23 define the same extended rational number. We can use this to prove that the map CF is invariant for equivalent twist vectors.

Lemma 3.35 (Continued fractions for equivalent twist vectors)**:**
Equivalent twist vectors define the same extended rational number under the map CF.

Proof:

Let v_1 and v_2 be two equivalent twist vectors, which is to say that they define equivalent diagrams for the same rational tangle. If we can show that v_1 and v_2 are related by a sequence of the four twist vector transformations from Theorem 3.23, then we may use Lemma 3.34 to conclude that these twist vectors define the same fraction under CF.

By Lemma 3.24, both v_1 and v_2 are equivalent to some canonical twist vectors using the rules of Theorem 3.23, say v_1' and v_2' respectively. We know that $v_1 \cong v_1' \cong v_2' \cong v_2$ are all equivalent as twist vectors, but this does not immediately imply that $v_1' = v_2'$. By Theorem 3.25, the canonical twist vectors v_1' and v_2' both define canonical diagrams for the same rational tangle. Since a canonical diagram is unique by Theorem 3.31, this implies that $v_1' = v_2'$ are identical twist vectors.

Therefore, v_1 and v_2 are related by a sequence of transformations of Theorem 3.23. By Lemma 3.34, the continued fraction of each of these twist vector transformations simplifies to the same extended rational number. Thus, $\mathrm{CF}(v_1) = \mathrm{CF}(v_2)$, which proves that the map CF is invariant for equivalent twist vectors.

□

Since canonical diagrams for rational tangles are uniquely defined by a canonical twist vector (Theorem 3.31), these can be used to associate a unique fraction to any rational tangle. However, Lemma 3.35 implies that this fraction can be obtained from any equivalent twist vector, whether or not it is canonical.

Corollary 3.36 (The fraction of a rational tangle)**:**
Every rational tangle T defines a unique extended rational number $\frac{p}{q} = \mathrm{CF}(v)$ obtained by simplifying the continued fraction of any twist vector v for T.

Proof:

Let T be a rational tangle. Hence, T admits a diagram defined by some twist vector, $T \cong v$ (Corollary 3.6). Let $\frac{p}{q} = \mathrm{CF}(v)$ be the fraction obtained by rationalizing the continued fraction defined by v. Since CF is invariant for equivalent twist vectors (Lemma 3.35), any choice of twist vector for T yields the same fraction. Therefore, $\frac{p}{q}$ is unique.

□

3.3.2 Conway's Bijection

Lemma 3.35 establishes that CF is a well-defined map from *equivalence classes of twist vectors* to the extended rational numbers. Since all rational tangles can be described using a twist vector, Corollary 3.36 allows us to broaden our definition of CF to mean a map from rational tangles to fractions $\frac{p}{q}$. With this in mind, our goal is to prove that CF defines a bijection from rational tangles to $\mathbb{Q} \cup \{\infty\}$. Before we begin, we pause to make some comments on the special case of trivial rational tangles.

Our notion of canonical diagram and canonical twist vector is only defined for nontrivial rational tangles (Theorem 3.25). Recall that a rational tangle is said to be *trivial* if it is equivalent to either the 0 or ∞ tangle, which have no crossings. By Lemma 3.9, these correspond to twist vectors (0) and $(0,0)$, respectively. If the algorithm outlined in the proof of Lemma 3.24 to find a canonical twist vector is applied to a twist vector for a trivial tangle, it will terminate with either (0) or $(0,0)$. In fact, (0) satisfies the original definition of a canonical twist vector, but $(0,0)$ does not. To accommodate the ∞ tangle, we will simply define $(0,0)$ to be canonical as a special case. All of our previous results concerning equivalent twist vectors still apply.

We will see that CF restricted to canonical twist vectors, including $(0,0)$, defines a bijection with $\mathbb{Q} \cup \{\infty\}$. Before we begin proving this, we will need the following result concerning rational numbers from Roberts [33], restated here in the language of twist vectors.

Lemma 3.37 (Continued fraction representation of a rational number [33])**:**
For each rational number $\frac{p}{q} \in \mathbb{Q}$, there exists a unique set of integers $\{a_1, \cdots, a_r\}$ for which $a_i > 0$ for $1 \leq i < r$ and such that

$$\frac{p}{q} = \mathrm{CF}(1, a_1, a_2, \cdots, a_r) = \mathrm{CF}(a_1 + 1, a_2, \cdots, a_r),$$

and hence no other continued fraction with a_i satisfying these conditions simplifies to $\frac{p}{q}$.

For a given rational number $\frac{p}{q}$, the unique set of integers that exist by Lemma 3.37 is closely related to a canonical twist vector. We have already seen that rational tangles correspond to a unique canonical twist vector; next we study how rational numbers uniquely correspond to canonical twist vectors. We will need the following intermediate result before we can make this connection explicit.

Lemma 3.38 (The sign of a continued fraction)**:**
If $v = (n_1, \cdots, n_r)$ is a twist vector for which $n_i > 0$ for $1 \leq i < r$, then $0 \leq \mathrm{CF}(v) < \infty$ if and only if $n_r \geq 0$.

Proof:

Let $v = (n_1, \cdots, n_r)$ be a twist vector and suppose that all but the final entry is guaranteed to be positive. Consider the corresponding continued fraction,

$$\mathrm{CF}(v) = n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} < \infty.$$

Since no term in a denominator can be zero, the expression above is guaranteed to simplify to a finite rational number. Since each twist vector entry is an integer, and all but the final entry are known to be positive, it follows that each denominator in the continued fraction is a positive integer plus a positive fraction. Therefore, each denominator is greater than or equal to 1, and hence each fractional part is less than or equal to 1. In particular, this means that

$$0 < \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} < 1.$$

Since n_r is an integer, either $n_r \geq 0$ or $n_r \leq -1$. If $n_r \geq 0$, then $\mathrm{CF}(v) \geq 0$, but if $n_r \leq -1$, then $\mathrm{CF}(v) < 0$. This establishes both directions of the claim.

□

Lemma 3.38 implies that the sign of a nonzero rational number $\frac{p}{q}$ matches the sign of the last integer in the unique set of integers $\{a_1, \cdots, a_r\}$ that exist by Lemma 3.37. We can leverage this to consider the positive and negative rational numbers separately.

Lemma 3.39 (Canonical twist vector for positive rational number)**:**
If $\frac{p}{q} \in \mathbb{Q}$ is positive, then there exists a unique canonical twist vector v such that $\mathrm{CF}(v) = \frac{p}{q}$.

Proof:

Let $\frac{p}{q}$ be a positive rational number. By Lemma 3.37, there exists a unique set of integers $\{a_1, \cdots, a_r\}$ such that $\frac{p}{q} = \mathrm{CF}(1, a_1, a_2, \cdots, a_r) = \mathrm{CF}(a_1 + 1, a_2, \cdots, a_r)$, where all elements but a_r are known to be positive. Since $\frac{p}{q} > 0$, it follows that $a_r > 0$ by Lemma 3.38. Let $v_1 = (1, a_1, a_2, \cdots, a_r)$ and $v_2 = (a_1 + 1, a_2, \cdots, a_r)$ denote these two corresponding twist vectors. All entries in v_1 and v_2 are positive, and hence they have constant sign.

Notice that the length of v_1 is exactly one longer than the length of v_2. This means that one of v_1 and v_2 has odd length and the other has even length. The twist vector of odd length is canonical by definition, and its continued fraction rationalizes to $\frac{p}{q}$. Uniqueness of this canonical twist vector follows from the uniqueness of the set integers $\{a_1, \cdots, a_r\}$. If a different canonical twist vector with this property existed, then it would be possible to obtain a different set of integers satisfying the conditions of Lemma 3.37, which would be a contradiction.

□

Lemma 3.40 (Canonical twist vector for negative rational number)**:**
If $\frac{p}{q} \in \mathbb{Q}$ is negative, then there exists a unique canonical twist vector v such that $\mathrm{CF}(v) = \frac{p}{q}$.

Proof:

Let $\frac{p}{q}$ be a negative rational number. Then $-\frac{p}{q}$ is a positive rational number. By Lemma 3.39, there exists a unique canonical twist vector $v = (n_1, \cdots, n_r)$ for which $\mathrm{CF}(v) = -\frac{p}{q}$. By Lemma 3.32, we can use the negative of the continued fraction for $-\frac{p}{q}$ to show that

$$\frac{p}{q} = -\mathrm{CF}(v) = -\mathrm{CF}(n_1, \cdots, n_r) = \mathrm{CF}(-n_1, \cdots, -n_r) = \mathrm{CF}(-v).$$

Since $v = (n_1, \cdots, n_r)$ is a canonical twist vector, it has odd length and constant sign. Hence, $-v = (-n_1, \cdots, -n_r)$ also has odd length and constant sign, and therefore must be a canonical twist vector. The uniqueness of $-v$ follows from the uniqueness of v.

□

It remains to consider the fractions $\frac{0}{1}, \frac{1}{0} \in \mathbb{Q} \cup \{\infty\}$, which will correspond to the trivial tangles 0 and ∞, respectively. Recall that (0) satisfies the definition of a canonical twist vector and denotes the construction of the 0 tangle. Furthermore, $(0, 0)$ denotes the construction of the ∞ tangle and we defined this to be canonical as a special case. We prove next that these are the unique canonical twist vectors for the two trivial rational tangles.

Lemma 3.41 (Canonical twist vector for 0 and ∞)**:**
If $\frac{p}{q} = \frac{0}{1} = 0$, then $v_0 = (0)$ is the unique canonical twist vector such that $\mathrm{CF}(v_0) = 0$.
If $\frac{p}{q} = \frac{1}{0} = \infty$, then $v_\infty = (0, 0)$ is the unique canonical twist vector such that $\mathrm{CF}(v_\infty) = \infty$.

Proof:

Let $v_0 = (0)$ and $v_\infty = (0, 0)$. Since v_0 has odd length, constant sign, and only the last entry is 0, it is a canonical twist vector. We note that v_∞ was defined to be canonical as a special case. Observe that $\mathrm{CF}(v_0) = 0$ and $\mathrm{CF}(v_\infty) = 0 + \frac{1}{0} = \infty$. Hence, we only need to prove uniqueness. We will consider v_0 first.

For the rational number $0 \in \mathbb{Q}$, observe that $\{a_1\} = \{-1\}$ is the unique set of integers satisfying the conditions of Lemma 3.37 for which

$$\mathrm{CF}(1, a_1) = \mathrm{CF}(1, -1) = 1 + \tfrac{1}{-1} = 1 - 1 = 0$$

and

$$\mathrm{CF}(a_1 + 1) = \mathrm{CF}(-1 + 1) = \mathrm{CF}(0) = 0.$$

Since the set of integers $\{a_1\}$ is unique for the rational number 0, it follows that $v_0 = (0) = (a_1 + 1)$ is the unique canonical twist vector for which $\mathrm{CF}(v_0) = 0$.

Now consider v_∞. Suppose by way of contradiction that there exists a canonical twist vector $v = (n_1, \cdots, n_r) \neq v_\infty$ for which $\mathrm{CF}(v) = \infty$. However, if such a v exists, then the leading entries $n_1, \cdots, n_{r-1}$ must all be nonzero and of the same sign. In particular, this means that no denominator in the continued fraction will be zero, and hence

$$\mathrm{CF}(v) = \mathrm{CF}(n_1, \cdots, n_r) = n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} < \infty.$$

This a contradiction, and so we may conclude that v_∞ is the unique canonical twist vector satisfying $\mathrm{CF}(v_\infty) = \infty$.

□

The preceding three lemmas account for all possible extended rational numbers. These will allow us to obtain a canonical twist vector corresponding to any $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$, which establishes surjectivity in the desired bijection. We state this as a corollary.

Corollary 3.42 (Canonical twist vector for any extended rational number)**:**
If $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$, then there exists a unique canonical twist vector v such that $\mathrm{CF}(v) = \frac{p}{q}$.

Proof:

For any $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$, the existence of the desired canonical twist vector follows immediately from Lemmas 3.39, 3.40, and 3.41. □

Injectivity in Conway's bijection will follow from the fact that these canonical twist vectors are unique, which we prove in the next lemma.

Lemma 3.43 (Equivalence of rational tangles with the same fraction)**:**
If $T_1 \cong v_1$ and $T_2 \cong v_2$ are rational tangles for which $\mathrm{CF}(v_1) = \mathrm{CF}(v_2)$, then $T_1 \cong T_2$.

Proof:

Let $T_1 \cong v_1$ and $T_2 \cong v_2$ be two rational tangles defined by twist vectors v_1 and v_2, respectively. Since every rational tangle admits a unique canonical diagram corresponding to a canonical twist vector (Theorem 3.31), we may suppose without loss of generality that v_1 and v_2 are canonical twist vectors. Suppose that $\mathrm{CF}(v_1) = \frac{p}{q} = \mathrm{CF}(v_2)$. For any $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$, we know by Lemmas 3.39, 3.40, and 3.41 that there exists a unique canonical twist vector v for which $\mathrm{CF}(v) = \frac{p}{q}$. Hence, if v_1 and v_2 are both canonical such that $\mathrm{CF}(v_1) = \frac{p}{q} = \mathrm{CF}(v_2)$, it follows that $v_1 = v_2$ by this uniqueness. Therefore, $T_1 \cong T_2$.

□

We are now ready to consolidate these facts into a proof for Conway's bijection between rational tangles and rational numbers. The proof is almost immediate thanks to our preceding work.

Theorem 3.44 (Conway's bijection [7])**:**
The map CF defines a bijection between rational tangles and the extended rational numbers.

Proof:

Recall that every rational tangle can be described using a twist vector (Corollary 3.6), and any twist vector defines the construction for some rational tangle. By Corollary 3.36, there exists a well-defined map $\mathrm{CF} : \{\text{equivalence classes of rational tangles}\} \to \mathbb{Q} \cup \{\infty\}$ defined by simplifying the continued fraction corresponding to any twist vector representative.

For any $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$, there exists a unique canonical twist vector v such that $\mathrm{CF}(v) = \frac{p}{q}$ (Corollary 3.43). Letting $T \cong v$ be the rational tangle constructed from this twist vector establishes surjectivity. If $T_1 \cong v_1$ and $T_2 \cong v_2$ are two rational tangles for which $\mathrm{CF}(v_1) = \mathrm{CF}(v_2)$, then it follows from Lemma 3.43 that $T_1 \cong T_2$ are equivalent rational tangles, establishing injectivity. This completes the proof of the bijection.

□

3.3.3 Properties of the Fraction of a Rational Tangle

Now that we know how to associate an extended rational number to any rational tangle, we may denote rational tangles unambiguously using the corresponding fraction $\frac{p}{q}$. We will use the notation $T \cong \frac{p}{q}$ to say when a 2-string tangle T is equivalent to the rational tangle corresponding to $\frac{p}{q}$, and we will assume that p and q are in lowest terms. Recall that we denote integer tangles by the fractions $\frac{n}{1}$ and $\frac{1}{n}$; these notations were introduced because they align with the corresponding rational numbers for integer tangles under Conway's bijection. This is also true for the two trivial tangles, 0 and ∞.

Corollary 3.45 (Special types of rational tangle)**:**
Let $T \cong \frac{p}{q}$ be a rational tangle corresponding to the extended rational number $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$.

a. $T \cong (0) \cong 0$ is the parity 0 trivial tangle if and only if $\frac{p}{q} = \frac{0}{1} = 0$.

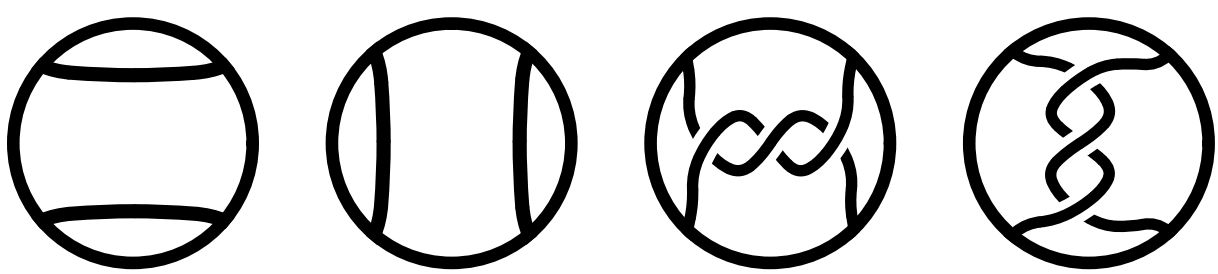
b. $T \cong (0,0) \cong \infty$ is the parity ∞ trivial tangle if and only if $\frac{p}{q} = \frac{1}{0} = \infty$.

c. $T \cong (n) \cong \frac{n}{1}$ is a horizontal integer tangle if and only if $\frac{p}{q} = \frac{n}{1} = n$.

d. $T \cong (0,n) \cong \frac{1}{n}$ is a vertical integer tangle if and only if $\frac{p}{q} = \frac{1}{n}$.

Proof:

The fraction $\frac{p}{q}$ for a rational tangle can be computed from any twist vector for that tangle using the map CF. Thanks to Conway's bijection (Theorem 3.44), we need only prove one direction of the claim. It suffices to calculate $\frac{p}{q}$ from a representative twist vector for each of these special cases. These are also summarized in Figure 3.36.



(a) $T \cong (0) \cong \frac{0}{1}$ **(b)** $T \cong (0,0) \cong \frac{1}{0}$ **(c)** $T \cong (3) \cong \frac{3}{1}$ **(d)** $T \cong (3,0) \cong \frac{1}{3}$

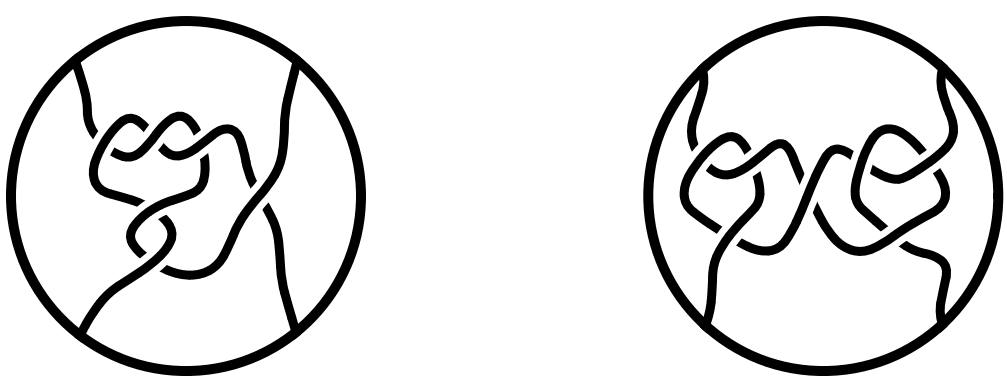
Figure 3.36: Fractions for special cases of rational tangles (Corollary 3.45)

Recall that the two trivial tangles were characterized in Lemma 3.41. These correspond to the twist vectors $0 \cong (0)$ and $\infty \cong (0,0)$, respectively, which satisfy $\mathrm{CF}(0) = 0$ and $\mathrm{CF}(0,0) = \infty$. Any horizontal integer tangle $\frac{n}{1}$ with $n \in \mathbb{Z}$ twists is described by the length one twist vector (n), which satisfies $\mathrm{CF}(n) = n$. Any vertical integer tangle $\frac{1}{n}$ with $n \in \mathbb{Z}$ twists is described by the length two twist vector $(n,0)$, which satisfies $\mathrm{CF}(n,0) = \frac{1}{n}$. This completes the proof.

□

We now have three ways of denoting rational tangle diagrams: integer construction formulas, twist vectors, and fractions (Figure 3.37a). Moving forward, the fraction $\frac{p}{q}$ will be our preferred way to denote rational tangles. Just as construction formulas using integer subtangles were used to describe rational tangle diagrams, we now have a convenient way to describe construction formulas for more complicated tangles using sums and products of rational subtangles. Denoting these subtangles with the corresponding rational number yields a very compact notation (Figure 3.37b). The next section will be devoted to classifying these more complicated tangle diagrams based on how they decompose into rational subtangles.

We conclude our brief adventure into the theory of rational tangles by cataloging a number of corollaries. Beyond its use as a convenient notation, the fraction $\frac{p}{q}$ conveys a surprising amount of information about the corresponding rational tangle diagram. Many interesting properties of a rational tangle can be inferred from its rational number. Generally, a rational tangle diagram has some property if and only if its corresponding fraction satisfies some condition. As in the preceding proof, Conway's bijection will allow us to conclude both directions of these claims only using a representative twist vector.



(a) $T \cong \left(\frac{3}{1} * \frac{1}{2}\right) + \frac{1}{1} \cong (3,2,1) \cong \frac{10}{7}$ **(b)** $T \cong (2,1,1) + (2,1,0) \cong \frac{5}{3} + \frac{2}{3}$

Figure 3.37: Examples of tangle construction formulas using subtangles

Corollary 3.46 (Negative and reciprocal rational tangles)**:**
Let $T \cong \frac{p}{q}$ be a rational tangle corresponding to the extended rational number $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$. Certain transformations of T can be characterized using its fraction as follows.

a. The mirror image corresponds to the negative fraction, $-T \cong -\frac{p}{q}$.

b. The diagonal reflection corresponds to the reciprocal fraction, $-R_{90}(T) \cong \frac{q}{p}$.

c. The 90 degree rotation corresponds to the negative reciprocal fraction, $R_{90}(T) \cong -\frac{q}{p}$.

Proof:

Let $T \cong (n_1, \cdots, n_r) \cong \frac{p}{q}$ be a rational tangle, where $(n_1, \cdots, n_r)$ is any twist vector for T. We compute each of the three fractions in the claim based on the corresponding twist vector transformation. By Corollary 3.16, the mirror image of T has twist vector construction $-T \cong (-n_1, \cdots, -n_r)$. By Lemma 3.32, we may compute the fraction of $-T$ as

$$\mathrm{CF}(-n_1, \cdots, -n_r) = -\mathrm{CF}(n_1, \cdots, n_r) = -\frac{p}{q}.$$

Thus, $-T \cong -\frac{p}{q}$. By Corollary 3.17, the diagonal reflection of T has twist vector construction $-R_{90}(T) \cong (n_1, \cdots, n_r, 0)$. By Lemma 3.33, we may compute the fraction of $-R_{90}(T)$ as

$$\mathrm{CF}(n_1, \cdots, n_r, 0) = \mathrm{CF}(\mathrm{CF}(n_1, \cdots, n_r), 0) = \mathrm{CF}(\tfrac{p}{q}, 0) = 0 + \frac{1}{\frac{p}{q}} = \frac{q}{p}.$$

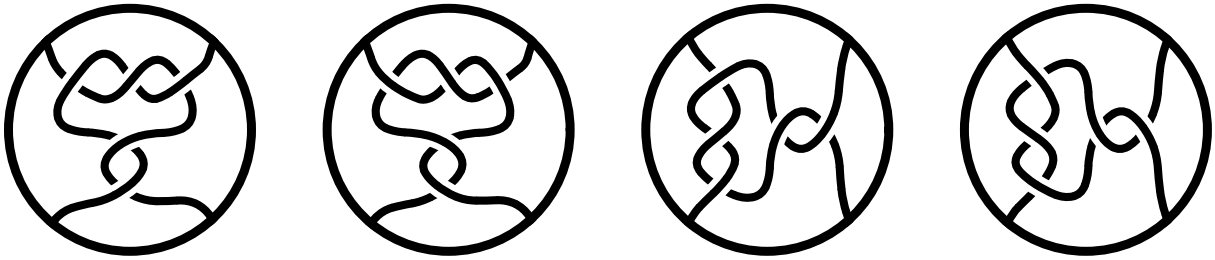
Thus, $-R_{90}(T) \cong \frac{q}{p}$. By Lemma 3.14, the 90 degree rotation of T has twist vector construction $R_{90}(T) \cong (-n_1, \cdots, -n_r, 0)$. We may compute the fraction of $R_{90}(T)$ by combining the preceding two results:

$$\mathrm{CF}(-n_1, \cdots, -n_r, 0) = -\mathrm{CF}(n_1, \cdots, n_r, 0) = -\frac{q}{p}.$$

Thus, $R_{90}(T) \cong -\frac{q}{p}$. This completes the proof of all three transformations.

□

Figure 3.38 shows an example of each of the three transformations of Corollary 3.46. In addition to describing tangle transformations, the fraction of a rational tangle also implies information about the corresponding canonical diagram. The behavior of the twists in a canonical diagram can be inferred from the sign and magnitude of $\frac{p}{q}$.



(a) $T \cong (3,2,0) \cong \frac{3}{7}$ **(b)** $-T \cong (-3,-2,0) \cong -\frac{3}{7}$ **(c)** $-R_{90}(T) \cong (1,2,2) \cong \frac{7}{3}$ **(d)** $R_{90}(T) \cong (-1,-2,-2) \cong -\frac{7}{3}$

Figure 3.38: Negative and reciprocal rational tangles (Corollary 3.46)

Corollary 3.47 (The sign of a rational tangle)**:**
Let $T \cong \frac{p}{q}$ be a nontrivial rational tangle corresponding to the rational number $\frac{p}{q} \in \mathbb{Q} \setminus \{0\}$.

a. The canonical diagram for T uses only positive twists if and only if $\frac{p}{q} > 0$.

b. The canonical diagram for T uses only negative twists if and only if $\frac{p}{q} < 0$.

Proof:

Let $T \cong (n_1, \cdots, n_r) \cong \frac{p}{q}$ be a nontrivial rational tangle, where $(n_1, \cdots, n_r)$ is a canonical twist vector. Since the entries in a canonical twist vector have constant sign by definition, each nonzero term n_i in the continued fraction has the same sign. Hence, computing the fraction of T as

$$\frac{p}{q} = \mathrm{CF}(n_1, \cdots, n_r) = n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}}$$

shows that the sign of $\frac{p}{q}$ must match the constant sign of the entries n_i. This establishes both parts of the claim.

□

Corollary 3.48 (The last nonzero twist in the canonical diagram)**:**
Let $T \cong \frac{p}{q}$ be a nontrivial rational tangle corresponding to the rational number $\frac{p}{q} \in \mathbb{Q} \setminus \{0\}$.

a. The last nonzero twist in the canonical diagram of T is horizontal if and only if $1 \leq \left|\frac{p}{q}\right|$.

b. The last nonzero twist in the canonical diagram of T is vertical if and only if $0 < \left|\frac{p}{q}\right| < 1$.

Proof:

Let $T \cong (n_1, \cdots, n_r) \cong \frac{p}{q}$ be a nontrivial rational tangle, where $(n_1, \cdots, n_r)$ is a canonical twist vector. Note that $0 < \left|\frac{p}{q}\right| < \infty$ since T is assumed to be non-trivial. By definition, the last entry of a twist vector n_r always denotes horizontal twisting, allowing that $n_r = 0$ if the last nonzero twist is vertical. Note that the leading entries $n_1, \cdots, n_{r-1}$ are all nonzero and of constant sign. Suppose without loss of generality that these are positive, and hence $n_r \geq 0$. This will also imply that $\frac{p}{q} > 0$ by Corollary 3.47.

Each denominator in the continued fraction $\mathrm{CF}(n_1, \cdots, n_r)$ is greater than or equal to 1, and thus each fractional part is less than or equal to 1. In particular, this means that

$$0 < \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} < 1.$$

Therefore, if $n_r = 0$, we have that

$$0 < \frac{p}{q} = \mathrm{CF}(n_1, \cdots, n_r) = n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} = \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} < 1.$$

If $n_r \neq 0$, then $n_r \geq 1$ since it must be an integer. In this case, it follows that

$$\frac{p}{q} = \mathrm{CF}(n_1, \cdots, n_r) = n_r + \frac{1}{n_{r-1} + \cdots + \frac{1}{n_2 + \frac{1}{n_1}}} \geq n_r \geq 1.$$

In the case where twists are negative, a similar result follows by reversing the direction of the inequalities above. This establishes both parts of the claim.

□

Figure 3.38 also illustrates the preceding two corollaries' characterization of twists in a canonical diagram. The behavior of these twists will play a role in understanding non-rational tangle diagrams. Consider the example of a non-rational tangle $T \cong \frac{5}{3} + \frac{2}{3}$ constructed from two rational subtangles shown back in Figure 3.37b. In this diagram, it is possible to shift a twist from the left subtangle to the right using a flype. Corollary 3.48 allows us to detect when operations such as this are possible from a decomposition into rational subtangles.

In our final corallary for this section, we prove how the parity of a rational tangle can be inferred from its fraction $\frac{p}{q}$. In combination with the table of rules describing the effect of tangle sums and products on parity (Table 2.1), this result allows us to determine the parity of any diagram constructed from algebraic combinations of rational subtangles.

Corollary 3.49 (The parity of a rational tangle)**:**
Let $T \cong \frac{p}{q}$ be a rational tangle corresponding to the extended rational number $\frac{p}{q} \in \mathbb{Q} \cup \{\infty\}$. Assume that p and q are in lowest terms and hence share no common factors.

a. $T \cong \frac{p}{q}$ is parity 0 if and only if p is even and q is odd.

b. $T \cong \frac{p}{q}$ is parity ∞ if and only if p is odd and q is even.

c. $T \cong \frac{p}{q}$ is parity 1 if and only if both p and q are odd.

Proof:

Observe that the claim is satisfied for the two trivial tangles $0 \cong \frac{0}{1}$ and $\infty \cong \frac{1}{0}$. Suppose that $T \cong (n_1, \cdots, n_r) \cong \frac{p}{q}$ is a nontrivial rational tangle defined by the twist vector $(n_1, \cdots, n_r)$. We do not assume that $(n_1, \cdots, n_r)$ is canonical nor that it satisfies the odd length convention. We will proceed by induction on the length r of the twist vector.

If $r = 1$, then $T \cong (n_1) \cong \frac{n_1}{1}$ is defined by a twist vector of length one, in which case T must be a horizontal integer tangle by Corollary 3.45. Hence, T is constructed from a tangle sum of $|n_1|$ copies of the $+1$ or -1 tangle, depending on the sign of n_1. These tangles both have parity 1. Referring back Table 2.1, recall that a sum of two parity 1 tangles has parity 0, and a sum of a parity 0 tangle with a parity 1 tangle has parity 1. By applying this rule multiple times, it follows that T must have parity 0 if n_1 is even and parity 1 if n_1 is odd. In either case, the claim is satisfied, proving the base case.

Now suppose that there exists $r \geq 1$ for which the claim holds for any rational tangle defined by a twist vector of length r or less. Let $T \cong (n_1, \cdots, n_r, n_{r+1}) \cong \frac{p}{q}$ be a rational tangle defined by a twist vector of length $r+1$, where $\frac{p}{q} = \mathrm{CF}(n_1, \cdots, n_r, n_{r+1})$. Let $L \cong (n_1, \cdots, n_r) \cong \frac{a}{b}$ be the related rational tangle defined by the twist vector of length r corresponding to the first r twists of T, noting that L satisfies the induction hypothesis. By Corollary 3.46, we know that the diagonal reflection of L corresponds to $-R_{90}(L) \cong \frac{b}{a}$. Furthermore, $-R_{90}(L)$ arises as a subtangle of T by Lemma 3.18:

$$T \cong (n_1, \cdots, n_r, n_{r+1}) \cong -R_{90}(L) \circ (n_{r+1}) \cong \frac{b}{a} + \frac{n_{r+1}}{1}.$$

Note that the expression above refers to a *sum of rational tangles* rather than a sum of rational numbers. The parity of T depends on the parity of the rational tangles $\frac{b}{a}$ and $\frac{n_{r+1}}{1}$, which both satisfy the induction hypothesis.

Since $L \cong (n_1, \cdots, n_r) \cong \frac{a}{b}$, its fraction is computed as $\frac{a}{b} = \mathrm{CF}(n_1, \cdots, n_r)$. Thus, we may use Lemma 3.33 to compute the fraction of $T \cong \frac{p}{q}$ as

$$\frac{p}{q} = \mathrm{CF}(n_1, \cdots, n_r, n_{r+1}) = \mathrm{CF}(\mathrm{CF}(n_1, \cdots, n_r), n_{r+1}) = \mathrm{CF}\left(\tfrac{a}{b}, n_{r+1}\right) = n_{r+1} + \frac{1}{\frac{a}{b}} = \frac{an_{r+1} + b}{a}.$$

Thus, we have an explicit expression for the fraction of T as $\frac{p}{q} = \frac{an_{r+1}+b}{a}$. Determining whether the numerator and denominator of this fraction are even or odd breaks into six cases depending on the values of a, b, and n_{r+1}, noting that a and b cannot be simultaneously even. Since $T \cong \frac{b}{a} + \frac{n_r}{1}$ as a rational tangle, we can use the induction hypothesis and Table 2.1 to determine the parity of T in each of these six cases. We will compute one of these six cases explicitly, and then list the remaining cases in a table for convenience.

Depending on which of the integers a, b, and n_{r+1} are even and which are odd, there exist integers s, t, and ℓ for which: either $a = 2s$ or $a = 2s + 1$; either $b = 2t$ or $b = 2t + 1$; and either $n_{r+1} = 2\ell$ or $n_{r+1} = 2\ell + 1$. In the case where a and n_{r+1} are even but b is odd, we have that

$$\frac{p}{q} = \frac{an_{r+1} + b}{a} = \frac{2s2\ell + 2t + 1}{2s} = \frac{2(2s\ell + t) + 1}{2s} = \frac{\text{odd}}{\text{even}}.$$

Since b is odd and a is even, $-R_{90}(L) \cong \frac{b}{a}$ has parity ∞. Since n_{r+1} is even, $\frac{n_{r+1}}{1}$ has parity 0. Thus, $T \cong -R_{90}(L) + \frac{n_{r+1}}{1}$ must have parity ∞, which is consistent with the claim. Referring to Table 3.1 for the remaining five cases completes the proof by induction.

□

Fraction				**Tangle Parity**		
a	b	n_{r+1}	$\frac{an_{r+1}+b}{a} = \frac{p}{q}$	$-R_{90}(L) \cong \frac{b}{a}$	$\frac{n_{r+1}}{1}$	$T \cong -R_{90}(L) + \frac{n_{r+1}}{1}$
$2s$	$2t+1$	2ℓ	$\frac{2(2s\ell+t)+1}{2s} = \frac{\text{odd}}{\text{even}}$	∞	0	∞
		$2\ell+1$	$\frac{2(2s\ell+s+t)+1}{2s} = \frac{\text{odd}}{\text{even}}$		1	∞
$2s+1$	$2t$	2ℓ	$\frac{2(2s\ell+\ell+t)}{2s+1} = \frac{\text{even}}{\text{odd}}$	0	0	0
		$2\ell+1$	$\frac{2(2s\ell+s+\ell+t)+1}{2s+1} = \frac{\text{odd}}{\text{odd}}$		1	1
$2s+1$	$2t+1$	2ℓ	$\frac{2(2s\ell+\ell+t)+1}{2s+1} = \frac{\text{odd}}{\text{odd}}$	1	0	1
		$2\ell+1$	$\frac{2(2s\ell+s+\ell+t+1)}{2s+1} = \frac{\text{even}}{\text{odd}}$		1	0

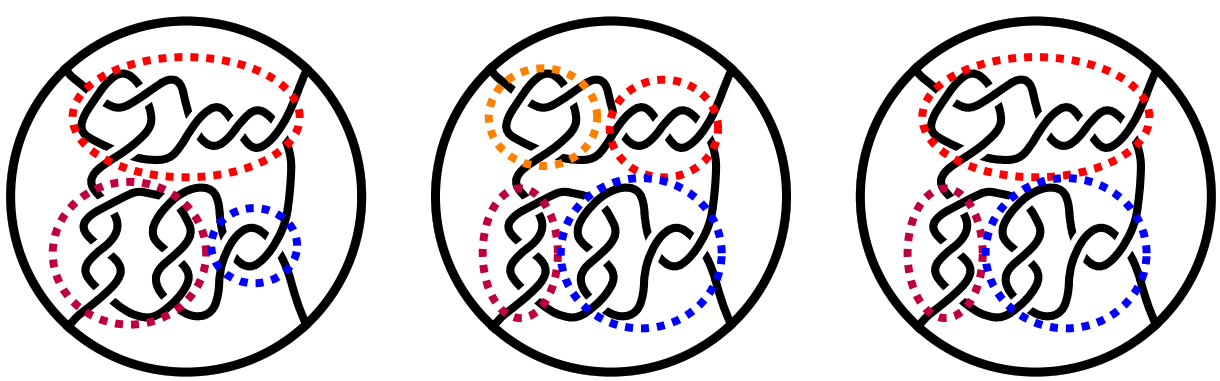
Table 3.1: The six cases determining the tangle parity in the proof of Corollary 3.49

4 Subtangle Decompositions

Recall that a *subtangle* refers to any tangle properly contained inside of a knot, link, or another tangle. Subtangles can be isolated from the rest of the structure using separating spheres, which can be identified with the boundary sphere of the subtangle. These separating spheres project to separating circles in the corresponding diagram (Figure 4.1). We will only be interested in studying 2-string subtangles, and so the separating circle of a subtangle intersects the strings of the diagram in exactly four places.

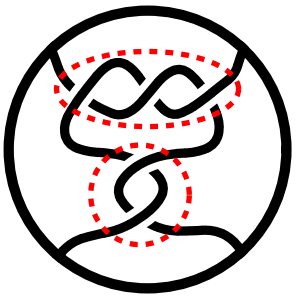
Recall that a *subtangle decomposition* of a knot, link, or tangle refers to a partition of the structure into subtangles. Algebraic construction formulas like those shown in Figure 4.1 give a natural subtangle decomposition into rational subtangles, but we can see from the figure that such decompositions are not unique in general. However, in many cases, it is possible to find a unique decomposition of a given tangle diagram into subtangles which meet certain conditions. Our first goal in this section is to define these special subtangle decompositions and show how they can be used to characterize tangle diagrams.

Our second goal is to compute these decompositions in practice. The decompositions shown in Figure 4.1 are immediate from the corresponding algebraic construction formulas, but formulas such as these only apply to algebraic diagrams. To capture this idea for tangle diagrams more generally, we propose a new diagram notation, referred to as the *subtangle planar diagram* (SPD) code (Figure 4.2). The SPD code is a generalization of the CPD code introduced in Section 2, only using subtangles rather than crossings. Computing an SPD code for a diagram is equivalent to identifying a subtangle decomposition.



(a) $T \cong \frac{11}{3} * \left(\left(\frac{1}{3} + \frac{1}{3}\right) + \frac{2}{1}\right)$ **(b)** $T \cong \left(\frac{2}{3} + \frac{3}{1}\right) * \left(\frac{1}{3} + \frac{7}{3}\right)$ **(c)** $T \cong \frac{11}{3} * \left(\frac{1}{3} + \frac{7}{3}\right)$

Figure 4.1: Examples of equivalent subtangle decompositions for the same algebraic tangle



(a) $T \cong \frac{3}{1} * \frac{1}{2} \cong \frac{3}{7}$

$\binom{3}{1}$	0 NW	0 NE	2 NE	2 NW	(P : 1)
$\binom{1}{2}$	1 SW	1 SE	0 SE	0 SW	(P : ∞)

(b) SPD Code

+	$4b$	0 NW	$2d$	$2c$
+	$3b$	$3a$	$1d$	$1c$
+	$2b$	$2a$	0 NE	$4c$
+	$5b$	$1a$	$3d$	$5c$
+	0 SW	$4a$	$4d$	0 SE

(c) CPD Code

Figure 4.2: Example of a subtangle decomposition identified by an SPD code

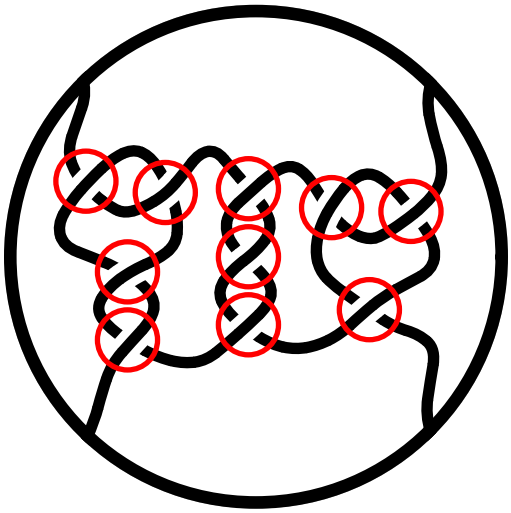
4.1 Subtangles in a Tangle Diagram

In this section, we always work with fixed tangle diagrams. In this case, a *subtangle* in a diagram T refers to a subset of the crossings in T that can be isolated from the rest of the diagram by a separating circle. This circle intersects the strings of T in exactly four places, as illustrated by the separating circles from Figure 4.1. We are free to wiggle and slide this circle along the strings as long as we do not pass over a crossing or intersect the string in a new location. This circle is smooth and consists of a single connected component.

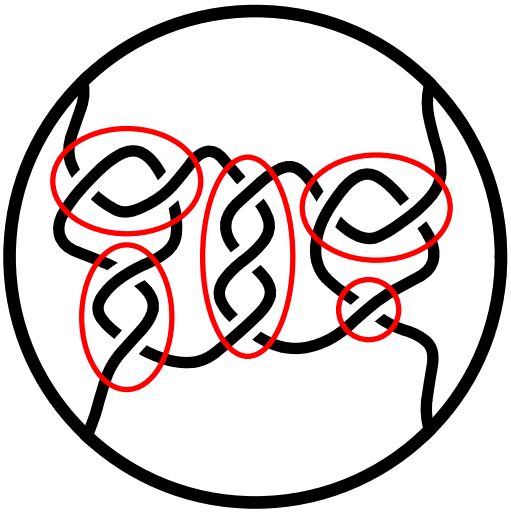
Two separating circles represent the *same* subtangle if they contain the same subset of crossings. Hence, whenever we speak of the separating circle of a subtangle, it should be understood that we are really referring to the equivalence class of separating circles that represent the same subtangle. Two separating circles define *overlapping* subtangles if they share one or more crossings in common. Hence, a *subtangle decomposition* of a diagram is a partition of the crossings into non-overlapping subtangles. Many of the results in this section involve analyzing overlapping subtangles based on comparing their separating circles.

4.1.1 Choosing Subtangles

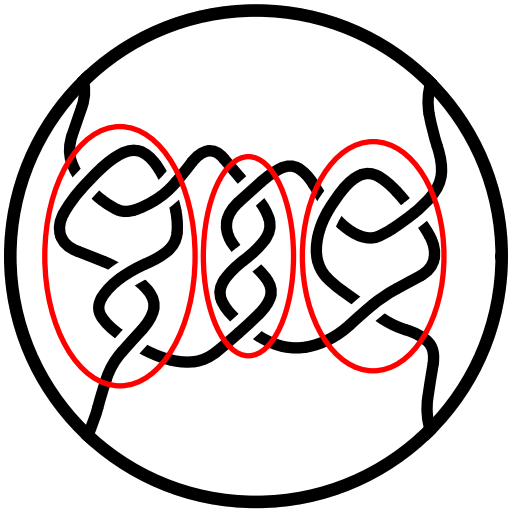
There are many equivalent ways to decompose a given tangle diagram into subtangles, but certain decompositions can be used to identify the structure of the tangle. By placing restrictions on how subtangles chosen, we may define several special types of subtangle decomposition. For example, every tangle diagram can be decomposed into 1-crossing subtangles by identifying each crossing with the $+1$ or -1 tangle (Figure 4.3a). We can expand on this idea by choosing subtangles of a certain type that are as large as possible in the sense that they are not properly contained in another subtangle of the same type.



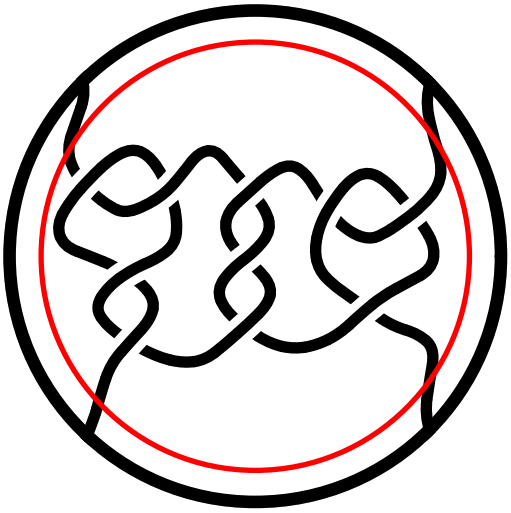
(a) 1-crossing subtangle decomposition



(b) Maximal integer subtangle decomposition



(c) Maximal rational subtangle decomposition



(d) Maximal algebraic subtangle decomposition

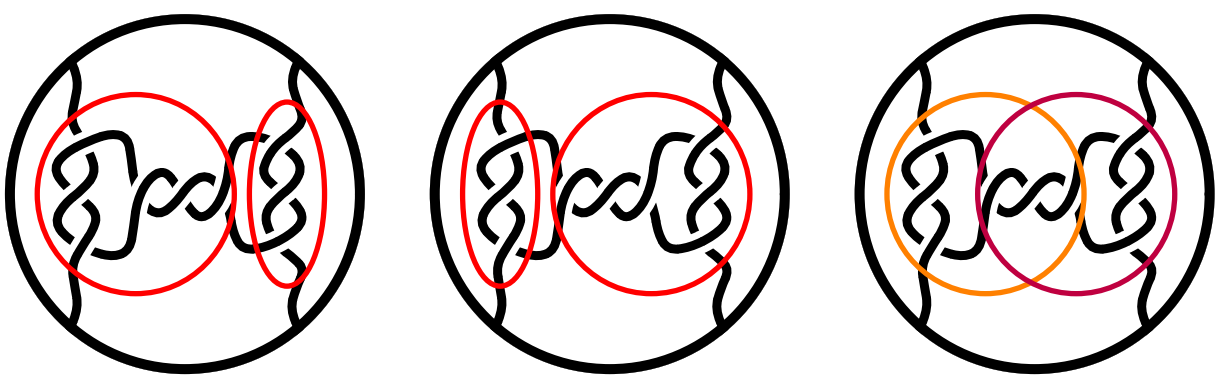
Figure 4.3: Examples of subtangle decompositions for the algebraic tangle $T \cong \frac{2}{5} + \frac{1}{3} + \frac{2}{3}$

We can formalize this idea into a definition. We will say that an integer subtangle is *maximal* in a given diagram if it is not a proper subtangle of another integer tangle in the diagram. We introduce analogous definitions for maximal rational and algebraic subtangles. For a given diagram, a *maximal subtangle decomposition* is a choice of maximal subtangles of a given type. Figure 4.3 shows examples of the same algebraic tangle diagram decomposed into maximal integer, rational, and algebraic subtangles. Since the tangle is algebraic, its decomposition into maximal algebraic subtangles consists of a single subtangle.

Furthermore, there is natural relationship between these maximal subtangle decompositions. Given two subtangle decompositions D_1 and D_2 for the same diagram, we will say that D_1 is a *refinement* of D_2, written $D_1 \leq D_2$, if each subtangle of D_1 is contained in a subtangle of D_2. Note that this containment need not be proper, the same subtangle can exist in both decompositions. We can see that the four subtangle decompositions of Figure 4.3 are refinements of each other. If $D_1 \leq D_2$ and $D_2 \leq D_1$, then they define identical subtangle decompositions and we write $D_1 = D_2$.

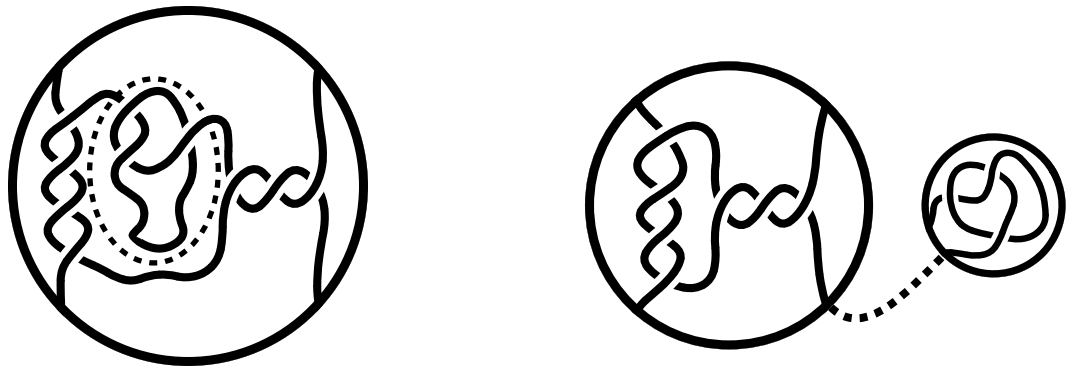
For any tangle diagram, we will show that decompositions using 1-crossing subtangles, maximal integer subtangles, and maximal algebraic subtangles always exist. Unfortunately, a decomposition into maximal rational subtangles does not always exist; Figure 4.4 shows one such counterexample. In this example, the three horizontal twists in the middle can be grouped into two possible overlapping maximal rational subtangles (Figure 4.4c). Consequently, the tangle diagram cannot be divided into disjoint rational subtangles which are maximal; the subtangle $\frac{1}{3}$ is non-maximal in both 4.4a and 4.4b.

However, such a decomposition will exist provided the rational subtangles meet certain conditions. Observe that the diagram for the rational subtangle $\frac{10}{3}$ in Figure 4.4b is non-canonical since it includes left twisting. If the three horizontal twists in the middle were shuffled to the right, we would obtain an equivalent tangle diagram for which a unique decomposition into maximal rational subtangles does exist. We will address these special conditions later when we discuss Montesinos tangles, but for now we can state some general results about subtangle decompositions which hold for any diagram.



(a) $T \cong \frac{10}{3} + \frac{1}{3}$ **(b)** $T \cong \frac{1}{3} + \frac{10}{3}$ **(c)** Overlapping rational subtangles

Figure 4.4: Example of a diagram with no decomposition into maximal rational subtangles



(a) Tangle with a local knot

(b) Sliding a local knot outside of a tangle

Figure 4.5: Example of how to isolate a local knot from the rest of a tangle

We originally defined a 2-string tangle as an embedding of two disjoint string segments into the interior of a 3-ball. This broad definition allows for the existence of *local knots* within a tangle, which we may identify as 1-string subtangles (Figure 4.5a). Observe that a local knot can be moved around the tangle by sliding it along the string. Hence, any local knot can be isolated from the rest of the tangle just as composite knots can be divided into parts using connect sums. We would like to exclude local knots from our study of tangles.

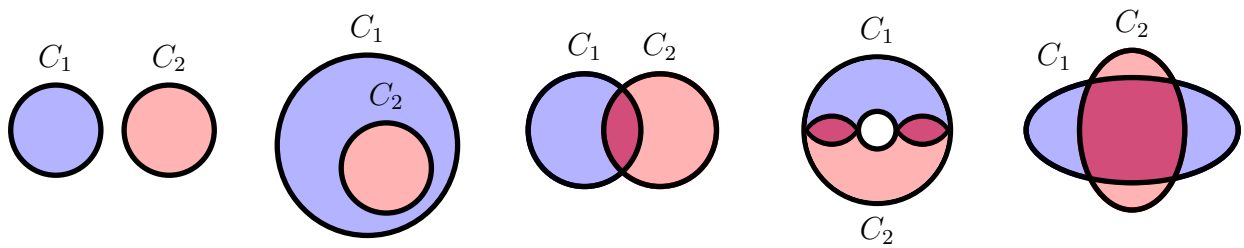
Lemma 4.1 (Excluding local knots from tangles)**:**
If a tangle contains a local knot, then we can slide this knot outside of the rest of the tangle. Thus, we are justified in excluding local knots from our study of tangles.

Proof:

We refer to Figure 4.5 for a visual proof. By shrinking the 1-string subtangle containing the local knot, we are free to slide it along the strand. Hence, we can pull a local knot outside of the boundary sphere while preserving the rest of the tangle.

□

For the same reason that knot tabulators often restrict themselves to the study of prime knots, we will make a similar exclusion concerning the existence of local knots. We will only be interested in classifying and tabulating 2-string tangles that do not contain local knots, and so we will henceforth include the blanket assumption that local knots do not exist. The reason to state this assumption now is because local knots make it more challenging to understand the structure and behavior of 2-string subtangles. If we know that a tangle contains no local knots, subtangle decompositions become much easier to describe.



(a) Disjoint **(b)** Fully contained **(c)** Overlapping **(d)** Mult. component **(e)** Cross through

Figure 4.6: Possible ways for two topological circles C_1 and C_2 to overlap

Any separating circle that intersects the strings in a tangle diagram in exactly four places defines a 2-string subtangle. By examining how these separating circles overlap, we can state a handful of results concerning how the subtangles in a given diagram interact. Figure 4.6 summarizes some of the possible ways that two topological circles can overlap. In general, two overlapping circles could intersect in multiple places or cross through each other, but these possibilities can be simplified for subtangle separating circles.

Given a separating circle for a 2-string subtangle, the subtangle's four endpoints match the four locations where the separating circle intersects the strings in the diagram. If separating circles for two subtangles overlap, then we can divide the diagram into different regions like those shown in Figure 4.6. Each of these regions can be thought of as a separating circle for a smaller portion of the diagram. Since a string can only pass into or out of a region at one of the original subtangle endpoints, we can describe overlapping subtangles by enumerating the possible locations of endpoints on their separating circles.

Lemma 4.2 (Overlapping subtangle separating circles)**:**
If S_1 and S_2 are two 2-string subtangles of a tangle diagram T, then it is possible to choose separating circles C_1 and C_2 for these subtangles which are either completely disjoint (4.6a), one fully contained in the other (4.6b), or which overlap in a single bigon (4.6c).

Proof:

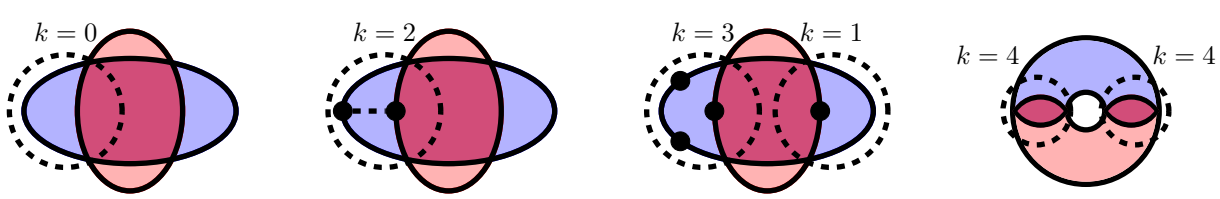
Let T be a tangle diagram and let S_1 and S_2 be two 2-string subtangles of T. If S_1 and S_2 do not share any crossings in common, then we can choose disjoint separating circles as in Figure 4.6a. If all of the crossings in S_2 are also crossings in S_1, then we can choose separating circles such that $C_2 \subseteq C_1$, as in Figure 4.6b. A similar fact holds if all of the crossings in S_1 are contained in S_2.

Suppose instead that S_1 and S_2 share at least one crossing in common, but that neither is fully contained in the other. As 2-string subtangles, there exist separating circles C_1 and C_2 for S_1 and S_2, respectively. Since they share a crossing in the diagram, C_1 and C_2 must overlap, although they could overlap in multiple places (Figure 4.6d). More generally, $C_1 \cap C_2$ consists of a number of connected components. We may assume without loss of generality that each of these connected components contains a shared crossing, otherwise the component could be removed by pushing the circles off of each other.

When C_1 and C_2 overlap, $C_1 \cup C_2$ can be divided into three possibly disconnected regions: $C_1 \setminus C_2$, $C_1 \cap C_2$, and $C_2 \setminus C_1$, as colored in Figure 4.6. If C_1 and C_2 pass through each other as in Figure 4.6e, observe that $C_1 \setminus C_2$ and $C_2 \setminus C_1$ will not be connected. However, some of these components may be simplified based on the behavior of the diagram in these regions.

As separating circles for subtangles, C_1 and C_2 each have four endpoints located along their edges. A string can only enter or exit one the regions in $C_1 \cup C_2$ through these endpoints. If a component in $C_1 \setminus C_2$ or $C_2 \setminus C_1$ contains no adjacent endpoints, then the string never passes into this portion of the region (Figure 4.7a). Hence, we can slide the separating circles to eliminate this component. If a component in $C_1 \setminus C_2$ or $C_2 \setminus C_1$ has exactly 2 adjacent endpoints, then either this component contains no crossings or it contains a local knot (Figure 4.7b). Since, we assumed that local knots do not exist (Lemma 4.1), it follows that this component contains no crossings and can be eliminated by sliding C_1 and C_2.

Any component of any region in $C_1 \cup C_2$ cannot be adjacent to an odd number of endpoints. Otherwise, it would be impossible for the strings to pass through all of these endpoints without becoming trapped inside of the region (Figure 4.7c). Thus, we may infer that each component of $C_1 \cup C_2$ is adjacent to an even number of endpoints. Since components with 0 or 2 endpoints can be removed, each component of each region in $C_1 \cup C_2$ must be bordered by a minimum of 4 endpoints.



(a) Removable region **(b)** Remove or local knot **(c)** Trapped string **(d)** Disconnected strings

Figure 4.7: Problematic regions from overlapping separating circles containing k endpoints

Since there are only 8 endpoints in total, 4 from C_1 and 4 from C_2, there are not enough endpoints available for more than two connected components in $C_1 \cap C_2$. However, with two connected components, these would exhaust the available endpoints (Figure 4.7d). If this happens, the strings in $C_1 \cup C_2$ would form a closed loop and be disconnected from the endpoints of the original tangle diagram. Our definition of tangle does not allow for closed loops in the interior, and so it follows that $C_1 \cap C_2$ contains only a single connected component.

This still allows for C_1 and C_2 to pass through each other as in Figure 4.6e. However, in this case, the regions $C_1 \backslash C_2$ and $C_2 \backslash C_1$ each split into two disjoint components, each of which must be bordered by a minimum of 4 endpoints. Since none of these four components shares an edge with any other, the 8 available endpoints are not enough to satisfy this condition, and so this cannot happen. Therefore, we may conclude that C_1 and C_2 overlap in a single connected component without passing through each other, as shown in Figure 4.6c. Note that this connected component forms a bigon with one edge from C_1 and the other from C_2. □

We can take the analysis of the preceding lemma a step further. Given two 2-string subtangles with shared crossings, we may decompose these into an algebraic combination of subtangles corresponding to the regions of their overlapping separating circles.

Lemma 4.3 (Intersections, unions, and differences of 2-string subtangles)**:**
If S_1 and S_2 are two 2-string subtangles of a tangle diagram T which share at least one crossing in common, then either one subtangle is contained in the other, or the union of these two subtangles is a 2-string subtangle which can be decomposed in the following way:

a. $S_1 \cap S_2$ is a 2-string subtangle of S_1, S_2, and T;

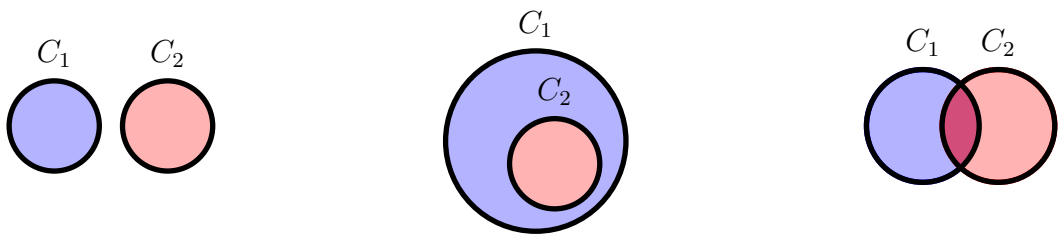
b. $S_1 \setminus S_2$ is a 2-string subtangle of S_1 and T;

c. $S_2 \setminus S_1$ is a 2-string subtangle of S_2 and T;

d. $S_1 \cup S_2$ is a 2-string subtangle of T which contains S_1 and S_2 as subtangles;

e. $S_1 \cup S_2$ decomposes as an algebraic combination of subtangles, either sum or product:

$$\begin{aligned} S_1 \cup S_2 &\cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1) \text{ or} \\ S_1 \cup S_2 &\cong (S_1 \setminus S_2) * (S_1 \cap S_2) * (S_2 \setminus S_1). \end{aligned}$$



(a) Disjoint, $C_1 \cap C_2 = \emptyset$ **(b)** Fully contained, $C_2 \subseteq C_1$ **(c)** Overlapping, $C_1 \cap C_2$ is a bigon

Figure 4.8: Possible intersections of subtangle separating circles, up to locating endpoints

Proof:

Let T be a tangle diagram and let S_1 and S_2 be two 2-string subtangles of T which share at least one crossing in common. By Lemma 4.2, we can choose separating circles C_1 and C_2 for these subtangles which are either completely disjoint (Figure 4.8a), one fully contained in the other (Figure 4.8b), or which overlap in a single bigon (Figure 4.8c). Note that disjoint separating circles are not possible since S_1 and S_2 share a crossing. If $C_1 \subseteq C_2$ or $C_2 \subseteq C_1$, then all of the crossings of one subtangle are contained in the other. In this case, either S_1 is a subtangle of S_2 or S_2 is a subtangle of S_1.

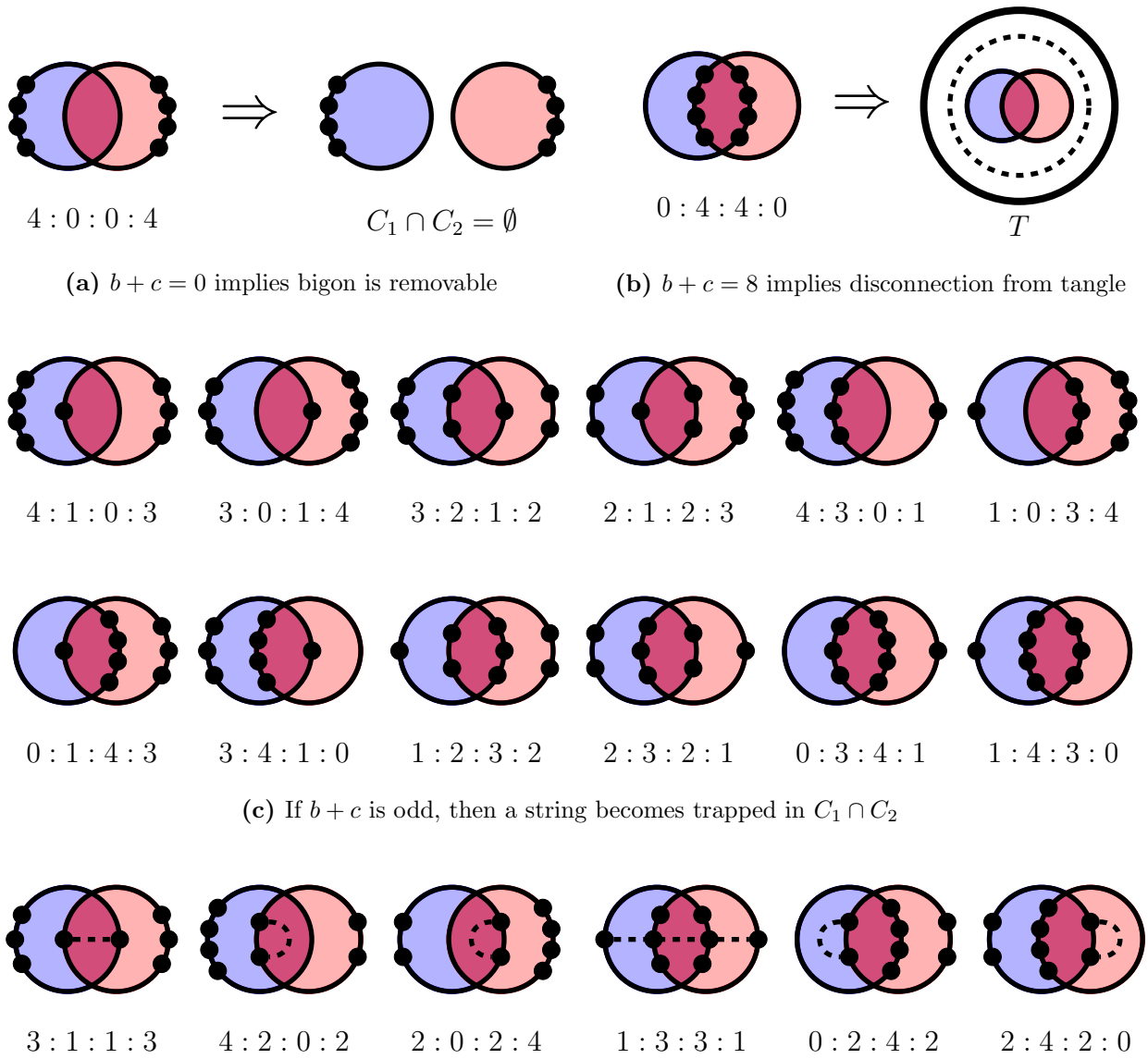
Suppose instead that neither subtangle is fully contained in the other, and so $C_1 \cap C_2$ consists of a single bigon (Figure 4.8c). Thus, the union of these two circles can be divided into three regions: $C_1 \setminus C_2$, $C_1 \cap C_2$, and $C_2 \setminus C_1$. Observe that each of these three regions is a bigon with one edge defined by a portion of C_1 and the other defined by C_2. Let $S_1 \setminus S_2$, $S_1 \cap S_2$, and $S_2 \setminus S_1$ denote the portion of the tangle diagram contained in each of these regions, respectively. Let $S_1 \cup S_2$ denote the portion of the diagram contained in $C_1 \cup C_2$.

It is clear that $S_1 \setminus S_2$, $S_1 \cap S_2$, $S_2 \setminus S_1$, and $S_1 \cup S_2$ are all subtangles of T with separating circles defined by $C_1 \setminus C_2$, $C_1 \cap C_2$, $C_2 \setminus C_1$, and $C_1 \cup C_2$, respectively. However, it is not obvious that each is a 2-string subtangle. Recall that 2-string tangles have exactly four endpoints, and these correspond to the locations in the diagram where the strings intersect the separating circle. Thus, for each of the four regions $C_1 \setminus C_2$, $C_1 \cap C_2$, $C_2 \setminus C_1$, and $C_1 \cup C_2$, we must show that the strings intersect the edges of this region in exactly four places.

Since S_1 and S_2 are 2-string subtangles with separating circles C_1 and C_2, respectively, there exist 4 endpoints along the border of C_1 and 4 endpoints along the border of C_2. Depending on how these endpoints are arranged, they may or may not lie along the edges of the $C_1 \cap C_2$ bigon. Each edge of the $C_1 \cap C_2$ bigon may contain between 0 and 4 endpoints, for a total of 25 possible arrangements. In fact, there is only one possible arrangement of

the endpoints which does not contradict one of our preceding assumptions. Figures 4.9 and 4.10 illustrate an exhaustive list of these possibilities and the resulting contradictions.

To enumerate these possible arrangements concisely, we can denote the number of endpoints along each edge using $a : b : c : d$, as in Figure 4.9. Note that $a + b + c + d = 8$ is the total number of endpoints, $a + c = 4$ and $b + d = 4$ are the endpoints on C_1 and C_2, respectively, and that $b + c$ is the number of endpoints on the $C_1 \cap C_2$ bigon.



(a) $b + c = 0$ implies bigon is removable

(b) $b + c = 8$ implies disconnection from tangle

(c) If $b + c$ is odd, then a string becomes trapped in $C_1 \cap C_2$

(d) $b + c = 2$ implies local knot inside $C_1 \cap C_2$

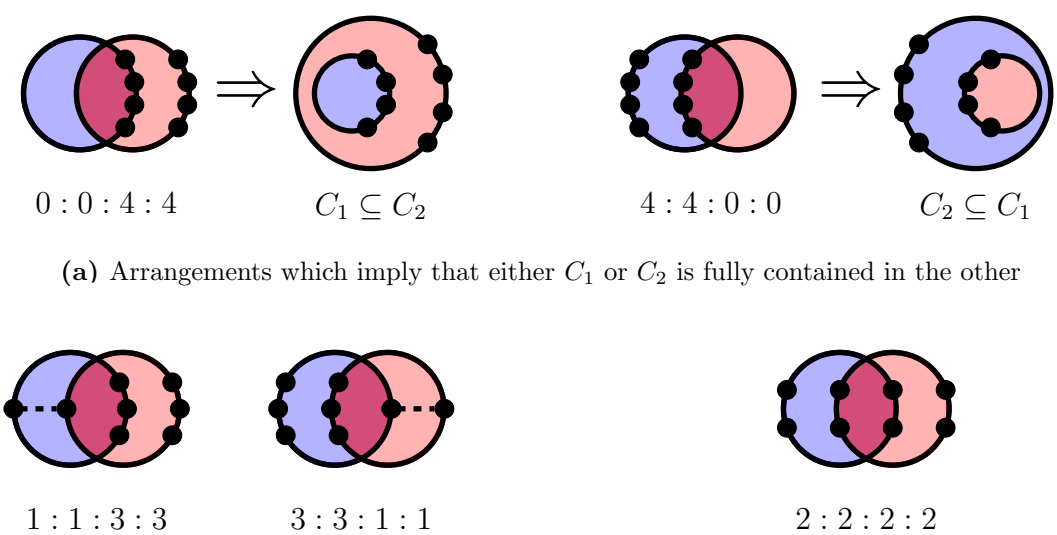
(e) $b + c = 6$ implies local knot inside $C_1 \cup C_2$

Figure 4.9: List of endpoint arrangements $a : b : c : d$ in $C_1 \cup C_2$ which give contradictions

The cases can be divided into groups based on the value of $b + c$. If $b + c = 0$, then a string never passes into the $C_1 \cap C_2$ bigon and the region can be removed (Figure 4.9a). However, this contradicts the assumption that $C_1 \cap C_2$ contains at least one shared crossing. If $b+c = 8$, then all of the endpoints lie within the overlapping region of $C_1 \cup C_2$ (Figure 4.9b). However, if this happens, the strings in $S_1 \cup S_2$ form a closed loop which is disconnected from the endpoints of the tangle T. This is a contradiction since our definition of tangle does not allow for the existence of closed loops in the interior.

If $b + c$ is an odd number, then the strings cannot pass through all of the endpoints on $C_1 \cap C_2$ without becoming trapped inside this region (Figure 4.9c). The strings can only pass into and out of these regions at an endpoint. Since all of the strings must terminate on the original endpoints of the tangle T, we have a contradiction when $b + c$ is odd. Since there are a maximum of 8 endpoints in total, it follows that $b + c$ is either 2, 4, or 6.

In the case where $b+c = 2$, then either $C_1 \cap C_2$ contains no crossings or it contains a local knot (Figure 4.9d). The former cannot occur since we assumed that $C_1 \cap C_2$ contains at least one crossing and the latter cannot occur since we assumed that tangles do not contain local knots (Lemma 4.1). If $b + c = 6$, then we run into precisely the same dilemma elsewhere in $C_1 \cup C_2$ (Figure 4.9e). Hence, both $b + c = 2$ and $b + c = 6$ yield contradictions. The only remaining possibility is that $b + c = 4$. This could happen in several ways, all of which are summarized in Figure 4.10, but some of these lead to contradictions.



(a) Arrangements which imply that either C_1 or C_2 is fully contained in the other

(b) Arrangements implying region with a local knot

(c) $2 : 2 : 2 : 2$ arrangement is the only valid option

Figure 4.10: Possible arrangements of 4 endpoints on the $C_1 \cap C_2$ bigon; only 4.10c is valid

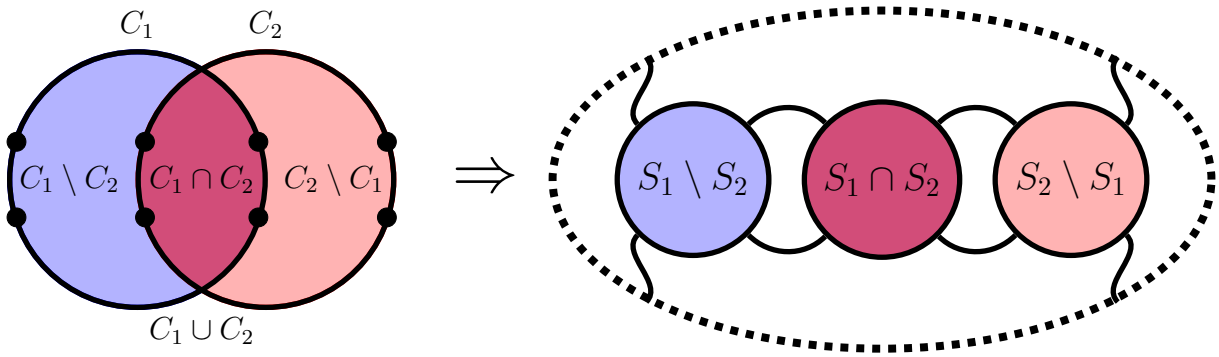


Figure 4.11: Decomposition of overlapping subtangles: $S_1 \cup S_2 \cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1)$

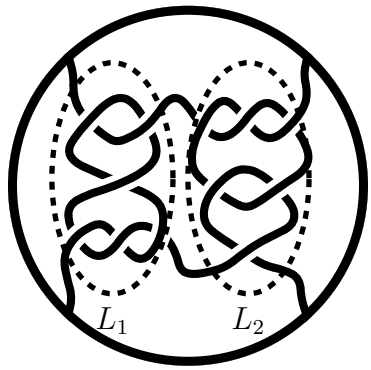
Using either the $0:0:4:4$ or $4:4:0:0$ arrangement implies that one of the separating circles can be fully contained within the other (Figure 4.10a). However, this is a contradiction since we assumed this does not happen. Using either the $1:1:3:3$ or $3:3:1:1$ arrangement implies that either $C_1 \setminus C_2$ or $C_2 \setminus C_1$ contains no crossings or contains a local knot (4.10b). We have already ruled out the existence of local knots, but if the region contains no crossings, then one separating circle can be fully contained within the other. This is not possible either, and so both of these arrangements lead to contradictions.

Therefore, $2:2:2:2$ is the only arrangement which does not cause a contradiction (Figure 4.10c). Furthermore, this particular arrangement of endpoints implies that the three regions comprising $C_1 \cup C_2$ describe an algebraic combination of subtangles like that shown in Figure 4.11. By algebraic combination, we mean that two adjacent subtangles are connected by exactly two parallel strings. These connections can be described using either tangle sum or tangle product, depending on a choice of NW corner for the subtangles. This proves the desired decomposition from the claim.

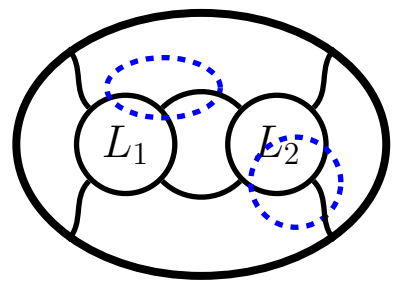
□

Note that we made no additional assumptions about the structure of the overlapping 2-string subtangles; S_1 and S_2 need not be algebraic subtangles to find such a decomposition, although $S_1 \cup S_2$ will also be algebraic if they are. The preceding result will help us prove the existence of certain maximal subtangle decompositions. Before we do this, it will be helpful to establish the following lemma concerning subtangles in algebraic diagrams.

Lemma 4.4 (2-string subtangles of an algebraic tangle are algebraic)**:**
If T is an algebraic tangle diagram, then all 2-string subtangles of T are also algebraic.



(a) $T \cong (\frac{2}{3} * \frac{3}{1}) + (\frac{3}{1} * \frac{2}{3}) \cong L_1 + L_2$



(b) Examples of subtangles which retract to L_1 or L_2

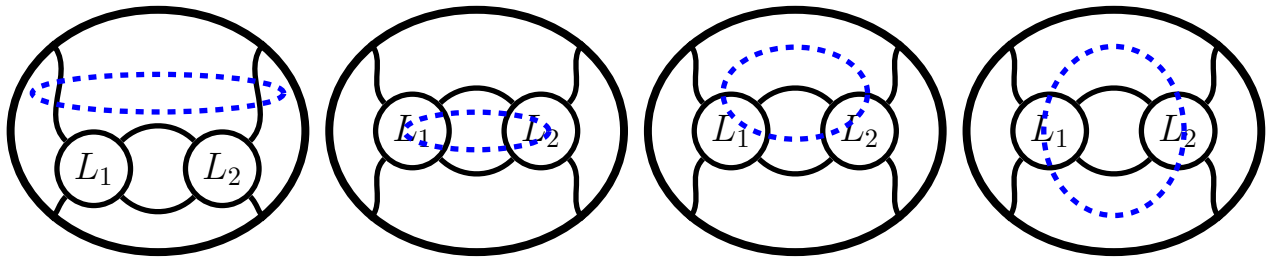
Figure 4.12: Example of dividing an algebraic construction formula into parts

Proof:

If T is an alebraic tangle diagram, then it is defined by some algebraic construction formula using sums and products of rational subtangles. The algebraic tangle $T \cong (\frac{2}{3} * \frac{3}{1}) + (\frac{3}{1} * \frac{2}{3})$ shown in Figure 4.12a is one such example. We will proceed by induction on the number k of rational subtangles used in some construction formula for an algebraic tangle diagram. While the same diagram may have many equivalent construction formulas using different choices of subtangles, the exact decomposition is not important in this proof.

Suppose for the base case that T is an algebraic tangle diagram constructed from $k = 1$ rational subtangle. In this case, T is a rational tangle diagram, and we may suppose without loss of generality that it is in twist form. By Lemma 3.28, every 2-string subtangle of T is rational, and hence algebraic. This proves the base case.

Now suppose by way of induction that there exists $k \geq 1$ such that the claim holds for any algebraic tangle diagram defined by a construction formula using k or fewer rational subtangles. Let T be an algebraic tangle diagram defined by a construction formula using $k + 1$ rational subtangles. Recall that the order of adjacent tangle sums and products used in this formula is determined by the placement of parentheses, as in the example of Figure 4.12a. Hence, there exist one or more outermost operations in this formula. In Figure 4.12a, this operation is a tangle sum. Thus, for an algebraic construction formula using two or more rational subtangles, it will always be possible to divide this formula into two parts joined by a sum or product. These two parts define the formulas for two disjoint algebraic subtangles in the diagram, as in Figure 4.12.



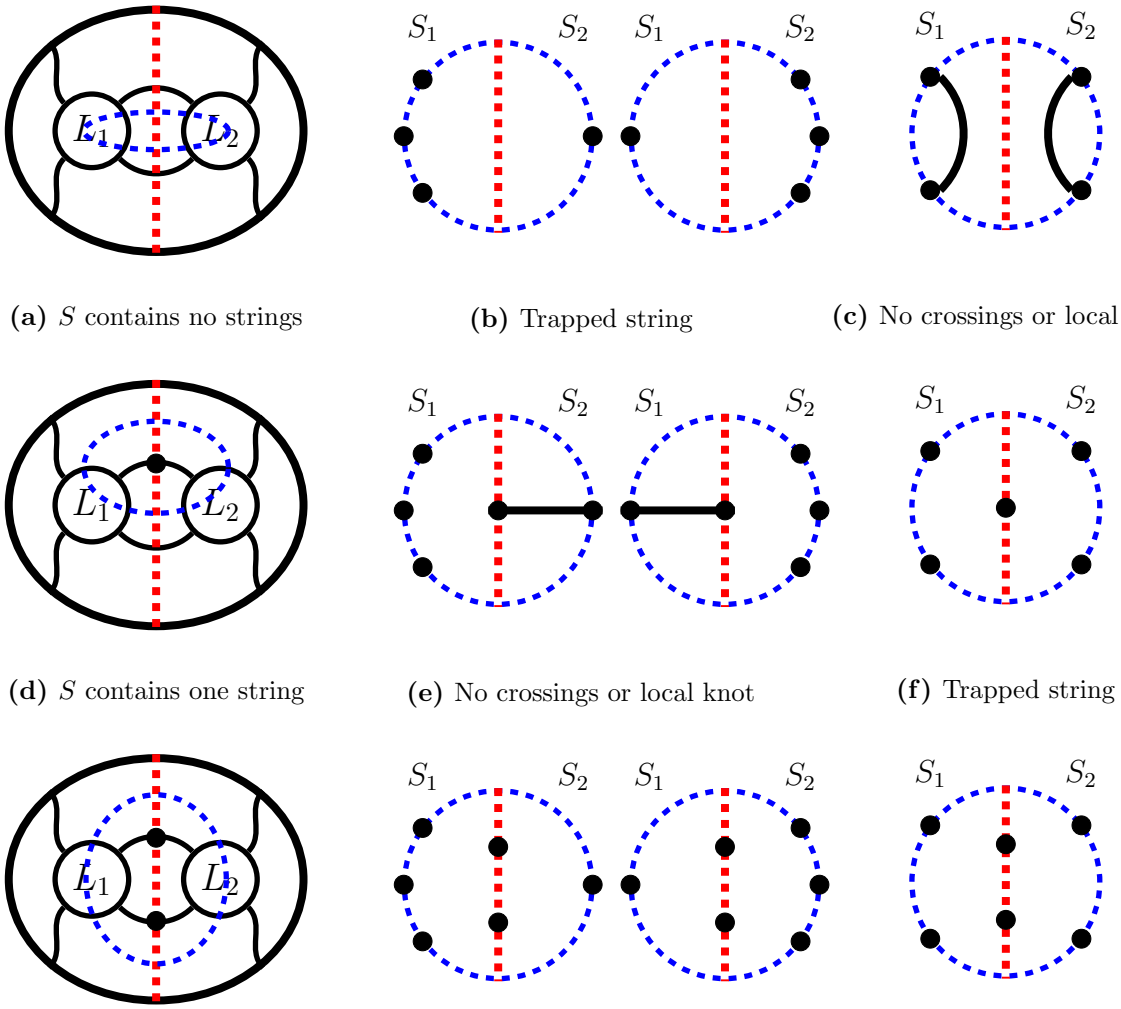
(a) Trivial subtangle **(b)** Containing no strings **(c)** Containing one string **(d)** Containing two strings

Figure 4.13: Examples of subtangles in $T \cong L_1 + L_2$ which do not retract to L_1 or L_2

Suppose without loss of generality that the outermost operation in T is a tangle sum, and hence the algebraic construction formula can be rewritten as $T \cong L_1 + L_2$. Hence, L_1 and L_2 are two disjoint algebraic subtangles of T. In particular, the formulas for L_1 and L_2 both contain k or fewer rational subtangles and thus satisfy the induction hypothesis. Let S be a 2-string subtangle of T. If S is fully contained in L_1 or L_2, then it must be algebraic. This is also true if the separating circle for S can be retracted into either L_1 or L_2 without passing over any crossings (Figure 4.12b).

Suppose that S cannot be fully retracted into only L_1 or only L_2. There are several possibilities for the separating circle of S (Figure 4.13). If S does not overlap with L_1 or L_2, then it contains no crossings and must be a trivial tangle, which is algebraic (Figure 4.13a). More generally, since all of the crossings are located within L_1 and L_2, we may slide the separating circle for S without passing over any crossings so that it does not intersect the strings outside of L_1 or L_2. Since S cannot be fully retracted into either subtangle, S must contain at least one crossing from L_1 and at least one crossing from L_2. This means that the separating circle for S must intersect both L_1 and L_2.

This leads to the three remaining cases illustrated in Figure 4.13, which depend on whether S contains zero (4.13b), one (4.13c), or both (4.13d) string segments connecting L_1 and L_2. We will show that only this last option does not lead to a contradiction. In all cases, S must have at least one of its four endpoints in L_1 and at least one endpoint in L_2, otherwise it could be fully retracted into one of these subtangles. Observe that we can cut the separating circle for S down the middle so that it splits into a part overlapping with L_1 and a part overlapping with L_2 (Figure 4.14). Denote these two halves as S_1 and S_2 and note that they become separating circles for a subtangle of L_1 and L_2, respectively.



(a) S contains no strings **(b)** Trapped string **(c)** No crossings or local knot

(d) S contains one string **(e)** No crossings or local knot **(f)** Trapped string

(g) S contains two strings **(h)** Trapped string **(i)** $S \cong S_1 + S_2$

Figure 4.14: Possible arrangements of the endpoints on the subtangle S in $T \cong L_1 + L_2$

In each of the three cases for S, we can analyze the possible locations of the four endpoints from S relative to S_1 and S_2. If one or both of the strings connecting L_1 and L_2 is contained in S, then there is also a corresponding endpoint along the cut where the string passes from S_1 into S_2. Since S has at least one crossing and at least one endpoint in L_1 and L_2, it follows that so do S_1 and S_2, respectively. Recall that a separating circle bordered by an odd number of endpoints is not possible since this would result in a trapped string. A separating circle bordered by exactly two endpoints either contains no crossings or a local knot. Hence, we may eliminate any arrangements that result in one of these scenarios for S_1 or S_2.

For all three cases, the possible arrangements of endpoints on the separating circles are summarized in Figure 4.14. If S contains no strings (4.14a), then either S_1 and S_2 contain a trapped string (4.14b) or they both contain a local knot (4.14c). Both possibilities are contradictions, and so we may eliminate this case. If S contains a single string (4.14d), then we have exactly the same dilemma that either S_1 or S_2 contains a local knot (4.14e) or a trapped string (4.14f). Hence, it follows that S must contain both strings (4.14g). Furthermore, S_1 and S_2 will contain a trapped string (4.14h), unless S has exactly two endpoints in L_1 and two endpoints in L_2 (4.14i). This is the only possibility that does not cause a contradiction.

Therefore, S decomposes as a sum of two 2-string subtangles, $S \cong S_1 + S_2$. Since S_1 is a 2-string subtangle of the algebraic tangle L_1, it follows that S_1 is also algebraic. A similar fact holds for S_2. Hence, S is a sum of two algebraic tangles, and must therefore be algebraic. Thus, we may conclude that any 2-string subtangle of an algebraic tangle diagram is also algebraic.

□

In Lemma 3.28, we proved that all 2-string subtangles of an rational tangle diagram are are rational, and now Lemma 4.4 gives us an analogous result concerning algebraic tangles more generally. We can also state a similar result for integer tangles, although this will be immediate in light of our previous work.

Corollary 4.5 (2-string subtangles of an integer tangle are integer)**:**
If T is an integer tangle diagram, then all 2-string subtangles of T are also integer.

Proof:

Let T be a diagram for an integer tangle. Hence, T consists of a finite sequence of horizontal or vertical twists. It follows that T is automatically a diagram for a rational tangle in twist form. Let S be a 2-string subtangle of T. By Lemma 3.28, S is also a rational tangle. If S is not an integer tangle, then it includes both horizontal and vertical twisting. However, this would imply that T also contains both horizontal and vertical twisting, which cannot happen in an integer tangle. Thus, S must be an integer subtangle of T.

□

With these results in hand, we have enough to prove the existence and uniqueness of certain types of subtangle decompositions as well as the relationships between them.

4.1.2 Refinements of Maximal Subtangle Decompositions

Figure 4.3 showed several examples of maximal subtangle decompositions for the same algebraic tangle diagram, each of which was a refinement of the next. If we exclude trivial subtangles, then it is clear that the 1-crossing subtangle decomposition is as refined as possible (Figure 4.3a). Similarly, if we use the entire diagram as a decomposition with one maximal subtangle, as in Figure 4.3d, then this is the least refined decomposition possible. For any tangle diagram, we may prove that these two subtangle decompositions exist, are unique, and will bound any other non-trivial subtangle decompositions for this diagram.

Lemma 4.6 (The 1-crossing and maximal subtangle decompositions)**:**
Every 2-string tangle diagram has a unique decomposition into 1-crossing subtangles and a unique decomposition into a single maximal subtangle, denoted $D_{\min}$ and $D_{\max}$, respectively.

Proof:

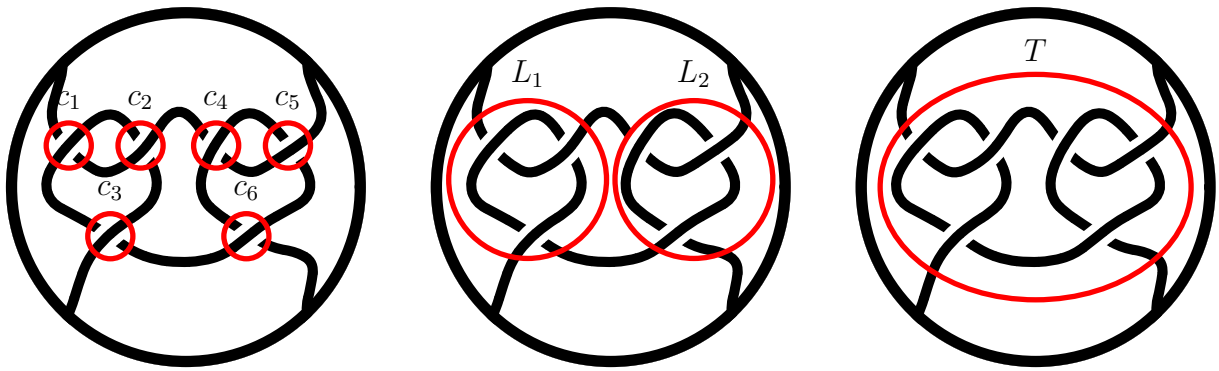
Let T be a diagram for a 2-string tangle. By isolating each crossing in T with a separating circle, we obtain a decomposition of T into 1-crossing subtangles, proving existence. Denote this decomposition $D_{\min}$. If $D'_{\min}$ is any other decomposition of T into 1-crossing subtangles, then the crossings define a bijection between the subtangles in $D_{\min}$ and $D'_{\min}$. Based on this bijection, we may establish that each subtangle of one decomposition is obtained in a subtangle from the other, which is to say that they are both refinements of each other: $D_{\min} \leq D'_{\min}$ and $D'_{\min} \leq D_{\min}$. In other words, $D_{\min} = D'_{\min}$, proving uniqueness.

Now let $D_{\max}$ denote the subtangle decomposition of T obtained by using all of T as a single subtangle. By definition, this subtangle is maximal in T since it is not properly contained in any other subtangle of T. Uniqueness is also immediate by maximality. Hence, we may conclude that both of the subtangle decompositions stated in the claim exist and are unique.

□

For any 2-string tangle diagram, $D_{\min}$ and $D_{\max}$ are, respectively, the most and least refined subtangle decompositions possible. Any other subtangle decomposition that does not contain trivial tangles must lie between these two.

Corollary 4.7 (Bounds on subtangle decomposition refinements)**:**
If D is any subtangle decomposition for a 2-string tangle diagram, then $D_{\min} \leq D \leq D_{\max}$.



(a) 1-crossing decomposition: $D_{\min}$ **(b)** Subtangle decomposition: D **(c)** Maximal decomposition: $D_{\max}$

Figure 4.15: Subtangle decomposition refinements for $T \cong \frac{2}{3} + \frac{2}{3}$: $D_{\min} \leq D \leq D_{\max}$

Proof:

Let T be a diagram for a 2-string tangle and let D be any decomposition of T into subtangles. Let $D_{\min}$ and $D_{\max}$ be the 1-crossing and maximal subtangle decompositions for T, which we know exist by Lemma 4.6. Since the subtangles in D partition the crossings in T, it follows that each 1-crossing subtangle of $D_{\min}$ is contained in some subtangle from D, and so $D_{\min} \leq D$. Since each subtangle in D is by definition contained in T, follows that $D \leq D_{\max}$. Thus, $D_{\min} \leq D \leq D_{\max}$, proving the claim.

□

Figure 4.15 shows an example of these bounds on refinements for the subtangle decomposition $D = \{L_1, L_2\}$ of $T \cong \frac{2}{3} + \frac{2}{3}$. In this figure, the 1-crossing subtangle decomposition is $D_{\min} = \{c_1, c_2, c_3, c_4, c_5, c_6\}$ and the maximal subtangle decomposition is $D_{\max} = \{T\}$. Since $c_1, c_2, c_3 \subseteq L_1$ and $c_4, c_5, c_6 \subseteq L_2$, it follows that $D_{\min} \leq D$; since $L_1, L_2 \subseteq T$, it follows that $D \leq D_{\max}$. Corollary 4.7 establishes that these bounds will hold for any choice of subtangle decomposition D.

Any collection of disjoint subtangles which partitions the crossings in a tangle diagram defines a valid subtangle decomposition. We will be particularly interested in decompositions into maximal subtangles of a certain type, especially using integer, rational, and algebraic subtangles. These will allow us to classify the structure of the tangle diagram. While a maximal decomposition using rational subtangles may not always exist (Figure 4.4), we can prove that maximal decompositions using integer or algebraic subtangles do exist and are unique.

Lemma 4.8 (Maximal integer subtangle decomposition)**:**
Every tangle diagram has a unique decomposition into maximal integer subtangles, D_{int}.

Proof:

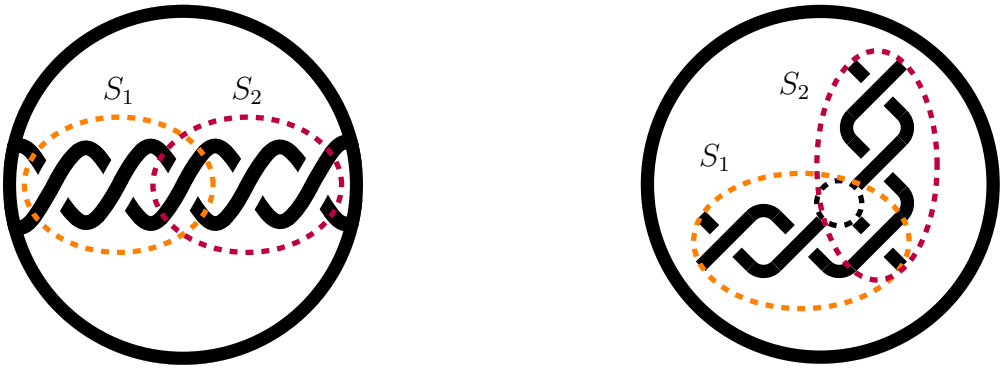
Let T be a 2-string tangle diagram. To prove the claim, we will show that each crossing in the diagram is contained in a unique maximal integer subtangle. By Lemma 4.6, there exists a decomposition $D_{\min}$ of T into 1-crossing subtangles, either the $+1$ or -1 tangle, both of which are integer tangles. If any of these 1-crossing subtangles is not already a maximal integer subtangle, then it is properly contained in some larger integer subtangle. Since there are only finitely many crossings in the diagram, it follows that there exists a maximal integer subtangle containing each crossing.

Thus, each crossing in T is contained in some maximal integer subtangle. We must show that these maximal integer subtangles are unique. Suppose by way of contradiction that there exists a crossing which is contained in two non-identical maximal integer subtangles S_1 and S_2. Hence, S_1 and S_2 overlap, but neither is fully contained within the other since they are both maximal integer subtangles.

By Lemma 4.3, $S_1 \cup S_2$ is also a subtangle of T which contains both S_1 and S_2 as subtangles. Note that $S_1 \cup S_2$ cannot be an integer tangle, or this would contradict the maximality of S_1 and S_2. Again by Lemma 4.3, we may decompose $S_1 \cup S_2$ as an algebraic combination of subtangles using sum or product. Suppose without loss of generality that this decomposition uses tangle sum, and hence $S_1 \cup S_2 \cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1)$.

Recall that all 2-string subtangles of an integer tangle are also integer (Corollary 4.5). Hence, $S_1 \setminus S_2$ is integer since it is a subtangle of S_1 and $S_2 \setminus S_1$ is integer since it is a subtangle of S_2. Likewise, $S_1 \cap S_2$ is also integer. Observe that $S_1 \cong (S_1 \setminus S_2) + (S_1 \cap S_2)$ and $S_2 \cong (S_1 \cap S_2) + (S_2 \setminus S_1)$. Since integer tangles use only one kind of algebraic connection, either tangle sum for horizontal integer tangles or tangle product for vertical integer tangles, we may infer that both S_1 and S_2 are horizontal integer tangles, as are all of their subtangles. Thus, $S_1 \cup S_2 \cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1)$ is a sum of three horizontal integer tangles and must therefore be a horizontal integer tangle itself.

Thus, $S_1 \cup S_2$ is an integer tangle which properly contains both S_1 and S_2, contradicting their assumed maximality. Hence, we may conclude that each crossing in T is contained in a unique maximal integer subtangle of T. In other words, the crossings in T may be uniquely partitioned into a collection of disjoint maximal integer subtangles; denote this collection D_{int}. This completes the proof. □



(a) $S_1 \cup S_2 \cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1)$
$\cong (\frac{2}{1}) + (\frac{1}{1}) + (\frac{2}{1}) \cong \frac{5}{1}$

(b) Horizontal and vertical subtangles cannot overlap

Figure 4.16: Integer subtangles which share a crossing must have the same type

Figure 4.16 shows an example of the contradiction that would be obtained in the preceding proof using non-maximal horizontal integer subtangles. The union of two overlapping integer subtangles is an integer subtangle containing both (Figure 4.16a). A similar contradiction is obtained using non-maximal vertical integer subtangles if $S_1 \cup S_2$ decomposes using tangle products. The decomposition of $S_1 \cup S_2$ obtained from Lemma 4.3 forces both integer subtangles to have the same type, but this fact can be inferred even without leveraging this result. Figure 4.16b illustrates the impossibility of a shared crossing between horizontal and vertical integer tangles; there is no way to connect the strings that allows for both horizontal and vertical twisting on the same crossing.

We can leverage Lemma 4.3 in a similar way to prove the existence and uniqueness of a decomposition into maximal algebraic subtangles. In fact, the proof is even easier since we need not distinguish between horizontal and vertical as we must with integer subtangles.

Lemma 4.9 (Maximal algebraic subtangle decomposition)**:**
Every tangle diagram has a unique decomposition into maximal algebraic subtangles, D_{alg}.

Proof:

Let T be a 2-string tangle diagram. As in the preceding lemma, we may prove the claim by showing that each crossing in the diagram is contained in a unique maximal algebraic subtangle. The proof is nearly identical to that of Lemma 4.8, but using algebraic subtangles instead of integer subtangles.

By Lemma 4.6, there exists a decomposition $D_{\min}$ of T into 1-crossing subtangles, either the $+1$ or -1 tangle, both of which are algebraic tangles. If any of these 1-crossing subtangles is not already a maximal algebraic subtangle, then it is properly contained in some larger algebraic subtangle. Since there are only finitely many crossings in the diagram, it follows that there exists a maximal algebraic subtangle containing each crossing.

Thus, each crossing in T is contained in some maximal algebraic subtangle. We must show that these maximal algebraic subtangles are unique. Suppose by way of contradiction that there exists a crossing which is contained in two non-identical maximal algebraic subtangles S_1 and S_2. Hence, S_1 and S_2 overlap, but neither is fully contained within the other since they are both maximal algebraic subtangles. By Lemma 4.3, $S_1 \cup S_2$ is also a 2-string subtangle of T which contains both S_1 and S_2 as subtangles. This union decomposes as an algebraic combination of the subtangles $S_1 \setminus S_2$, $S_1 \cap S_2$, and $S_2 \setminus S_1$, either using tangle sum or tangle product. Note that these are all 2-string subtangles of algebraic tangles. Hence, by Lemma 4.4, each of these must also be algebraic.

Suppose without loss of generality that $S_1 \cup S_2 \cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1)$. Therefore, $S_1 \cup S_2$ is a sum of three algebraic tangles by Lemma 4.4, and hence must itself be algebraic. Since S_1 and S_2 are both proper subtangles of the algebraic tangle $S_1 \cup S_2$, they cannot be maximal algebraic subtangles, which yields the desired contradiction. Thus, each crossing in T is contained in a unique maximal algebraic subtangle. In other words, T may be uniquely partitioned into a collection of disjoint maximal algebraic subtangles, which we will denote D_{alg}. This completes the proof.

□

The four subtangle decompsositions we have just enumerated exist for any 2-string tangle diagram. Furthermore, they are all refinements of each other.

Corollary 4.10 (Refinements of maximal subtangle decompositions)**:**
For any tangle diagram, the subtangle decompositions $D_{\min}$, D_{int}, D_{alg}, and $D_{\max}$ all exist, are unique, and satisfy $D_{\min} \leq D_{\text{int}} \leq D_{\text{alg}} \leq D_{\max}$.

Proof:

Let T be a 2-string tangle diagram. By Lemmas 4.6, 4.8, and 4.9, we know that the subtangle decompositions $D_{\min}$, D_{int}, D_{alg}, and $D_{\max}$ all exist and that each is the unique decomposition of T into maximal subtangles of a certain type. We will verify that each of these decompositions is a refinement of the next.

By Corollary 4.7, we know that $D_{\text{min}} \leq D_{\text{int}}$ and that $D_{\text{alg}} \leq D_{\text{max}}$, and so we need only compare D_{int} and D_{alg}. Recall that integer tangles are a special case of algebraic tangle. Hence, each subtangle in D_{int} is algebraic. If a subtangle from D_{int} is a maximal algebraic tangle, then it is also contained in D_{alg} by uniqueness. If a subtangle from D_{int} is not a maximal algebraic tangle, then it is properly contained in some maximal algebraic subtangle, which again must be contained in D_{alg} by uniqueness. Since each subtangle from D_{int} is equal to or contained within a subtangle from D_{alg}, we have that $D_{\text{int}} \leq D_{\text{alg}}$ by definition. Therefore, $D_{\text{min}} \leq D_{\text{int}} \leq D_{\text{alg}} \leq D_{\text{max}}$, completing the proof.

□

While a decomposition into maximal rational subtangles may not always be possible, as we saw previously in Figure 4.4, the decomposition will be unique when it does exist. Moreover, it will lie between D_{int} and D_{alg} in the list of subtangle decomposition refinements from the preceding corollary.

Lemma 4.11 (Maximal rational subtangle decomposition)**:**
If a decomposition into maximal rational subtangles D_{rat} does exist for a given 2-string tangle diagram, then it is unique and satisfies $D_{\text{min}} \leq D_{\text{int}} \leq D_{\text{rat}} \leq D_{\text{alg}} \leq D_{\text{max}}$.

Proof:

Let T be a 2-string tangle diagram. Suppose that there exists a decomposition of T into maximal rational subtangles, which we will denote D_{rat}. Suppose by way of contradiction that this decomposition is not unique, and hence there exists another decomposition of T into maximal rational subtangles D'_{rat}. We may assume without loss of generality that each of the rational subtangles from D_{rat} and D'_{rat} is in twist form, and hence Lemma 3.28 implies that any subtangles of these subtangles will also be rational.

Since D_{rat} and D'_{rat} are not identical, there exists at least one crossing in T which is contained in $S_1 \in D_{\text{rat}}$ and contained in $S_2 \in D'_{\text{rat}}$, where $S_1 \neq S_2$. That is to say, S_1 and S_2 are two overlapping maximal rational subtangles taken from these two different subtangle decompositions. Since each subtangle decomposition partitions the crossings of T into disjoint subtangles, we know that $S_1 \notin D'_{\text{rat}}$ and $S_2 \notin D_{\text{rat}}$ since they overlap. Furthermore, we know that neither is fully contained within the other since each is a maximal rational subtangle. Thus, by Lemma 4.3, $S_1 \cup S_2$ is also a 2-string subtangle of T which contains both S_1 and S_2 as subtangles. Suppose without loss of generality that $S_1 \cup S_2$ decomposes as a sum of subtangles: $S_1 \cup S_2 \cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1)$.

(a) Rational subtangles with shared twists **(b)** Rational subtangles with a shared core

Figure 4.17: Examples of the two possibilities for overlapping rational subtangles

Observe that we also have the decompositions $S_1 \cong (S_1 \setminus S_2) + (S_1 \cap S_2)$ and $S_2 \cong (S_1 \cap S_2) + (S_2 \setminus S_1)$. Moreover, the subtangles $S_1 \setminus S_2$, $S_1 \cap S_2$, and $S_2 \setminus S_1$ must all be rational by Lemma 3.28. Since S_1 and S_2 are rational tangles in twist form, they are constructed using some combination of sums with horizontal integer tangles and products with vertical integer tangles. Thus, at least one of the subtangle summands in $S_1 \cong (S_1 \setminus S_2) + (S_1 \cap S_2)$ must be a horizontal integer tangle, as must one of the subtangle summands in $S_2 \cong (S_1 \cap S_2) + (S_2 \setminus S_1)$. Since both S_1 and S_2 share the subtangle $S_1 \cap S_2$, this leads to two the two possibilities shown in Figure 4.17.

Either $S_1 \cap S_2$ is a horizontal integer tangle (Figure 4.17a), or both $S_1 \setminus S_2$ and $S_2 \setminus S_1$ are horizontal integer tangles (Figure 4.17b). However, if this second case is true, then $S_1 \cup S_2 \cong (S_1 \setminus S_2) + (S_1 \cap S_2) + (S_2 \setminus S_1)$ is a rational subtangle of T. Since $S_1 \cup S_2$ properly contains S_1 and S_2 as subtangles, this cannot happen because these were both assumed to be maximal rational subtangles. Thus, the only remaining possibility is that $S_1 \cap S_2$ is a horizontal integer subtangle.

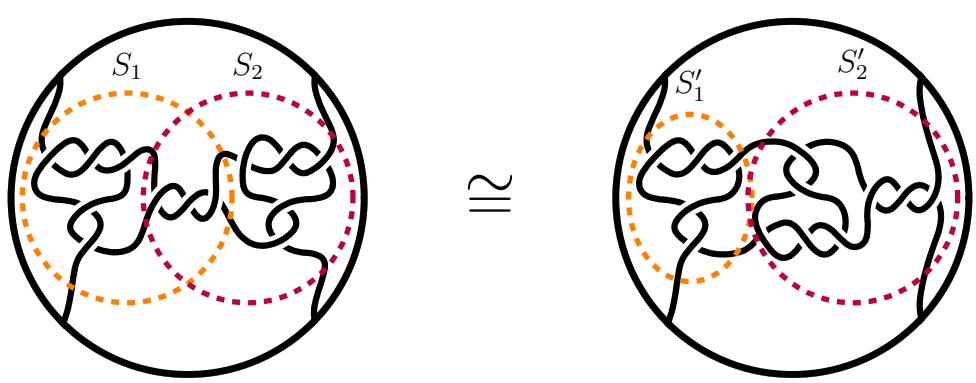
Since $S_2 \setminus S_1$ is contained in S_2, it is a non-maximal rational subtangle in T. Since $S_2 \notin D_{\text{rat}}$, there must exist some maximal rational subtangle $S_3 \in D_{\text{rat}}$ which contains $S_2 \setminus S_1$. Since the subtangles in D_{rat} are disjoint, we know that S_1 and S_3 do not overlap. In particular, this means that $S_1 \cap S_2$ is not contained in S_3, although they must be connected by a tangle sum since $S_2 \setminus S_1$ is contained in S_3. This means that $(S_1 \cap S_2) + S_3$ is a subtangle of T. Since $S_1 \cap S_2$ is a horizontal integer tangle and S_3 is a rational tangle, it follows that $(S_1 \cap S_2) + S_3$ is a rational tangle which properly contains S_3. However, this contradicts the fact that S_3 is maximal.

To avoid this contradiction, we must exclude the existence of a crossing which is contained in two overlapping but non-identical maximal rational subtangles. In other words, either every crossing in T is contained in a unique maximal rational subtangle, or there does not exist a decomposition of T into maximal rational subtangles. Since we assumed the existence of D_{rat}, we may conclude that this is the unique maximal rational subtangle decomposition of T. It remains to consider how D_{rat} compares to the other maximal subtangle decompositions.

By Corollary 4.10, D_{min}, D_{int}, D_{alg}, and D_{max} all exist for T. Since each maximal integer subtangle in D_{int} is also a rational subtangle of T, it follows that each of these is either properly contained in a maximal rational subtangle from D_{rat} or is itself a maximal rational subtangle, in which case it is contained in D_{rat} by uniqueness. Thus, $D_{\text{int}} \leq D_{\text{rat}}$. By exactly the same logic, $D_{\text{rat}} \leq D_{\text{alg}}$. Therefore, by Corollary 4.10, we may conclude that $D_{\text{min}} \leq D_{\text{int}} \leq D_{\text{rat}} \leq D_{\text{alg}} \leq D_{\text{max}}$, completing the proof.

□

The preceding proof shows that a decomposition into maximal rational subtangles fails to exist precisely when there are adjacent maximal rational subtangles which share some twists in common (Figure 4.18a). This is only possible for algebraic tangles with two or more rational subtangles, but it will be possible to identify such diagrams from an algebraic construction formula. Fortunately, it will always be possible to shuffle these shared twists to one side using flypes without changing the equivalence class of the diagram (Figure 4.18b). While a decomposition into maximal rational subtangles may not exist for a given diagram, this ensures that such a decomposition will exist for an equivalent diagram. We will hold off on addressing the details of this until we are ready to discuss Montesinos tangles.



(a) Overlapping maximal rational subtangles **(b)** Non-overlapping maximal rational subtangles

Figure 4.18: Shared twists between rational subtangles can be moved apart using flypes

Our study of subtangle decompositions is motivated by the goal of classifying the structure of a given tangle diagram. Setting aside the more challenging question of equivalent *tangles*, we may determine the class of a *tangle diagram* based on how it decomposes into maximal subtangles of a certain type. Since $D_{\max}$ exists for any 2-string tangle diagram and consists of a single subtangle comprising the entire diagram, this classification is determined by comparison of $D_{\max}$ with its various refinements.

Theorem 4.12 (Tangle diagram classification by subtangle decomposition)**:**
If T is a tangle *diagram*, then it can be classified based on its decomposition into subtangles:

a. if $D_{\text{int}} = D_{\max}$, then T is an integer diagram;

b. if D_{rat} exists and $D_{\text{rat}} = D_{\max}$, then T is a rational diagram;

c. if $D_{\text{alg}} = D_{\max}$, then T is an algebraic diagram;

d. if $D_{\text{alg}} < D_{\max}$, then T is a non-algebraic diagram.

Proof:

Let T be a 2-string tangle diagram. By Corollary 4.11, we know that $D_{\min}$, D_{int}, D_{alg} and $D_{\max}$ all exist. Recall that $D_{\max}$ consists of a single subtangle corresponding to all of T. If $D_{\text{int}} = D_{\max}$, then T consists of a single integer tangle. If D_{rat} exists and $D_{\text{rat}} = D_{\max}$, then T consists of a single rational tangle. If $D_{\text{alg}} = D_{\max}$, then T consists of a single algebraic tangle. In each of these cases, this implies that T is integer, rational, or algebraic, respectively.

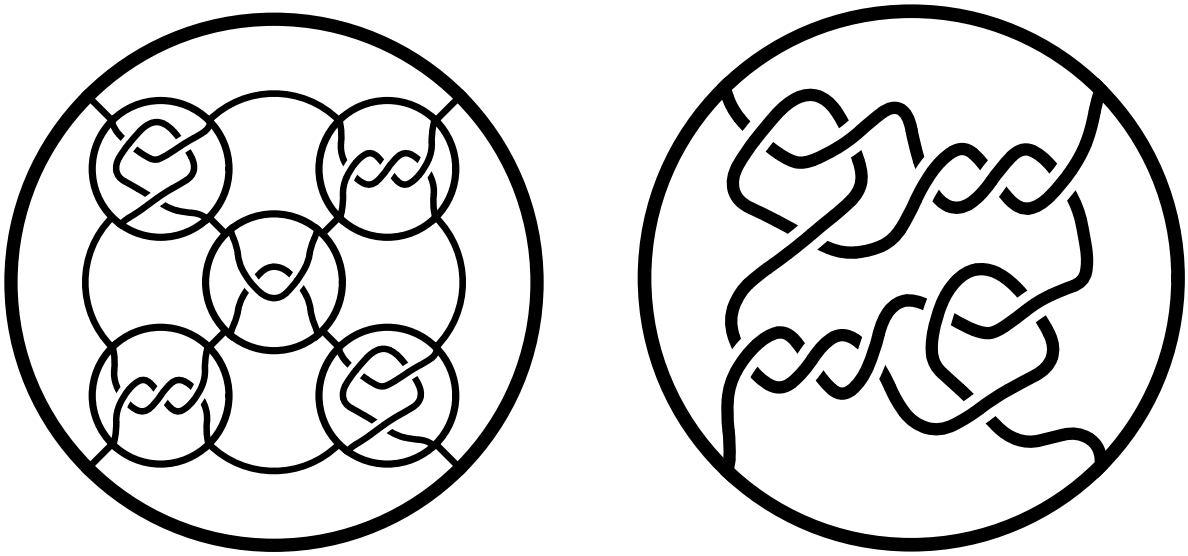
If $D_{\text{alg}} < D_{\max}$, then T contains multiple maximal algebraic subtangles. Since sums and products of algebraic tangles are also algebraic, none of these maximal algebraic subtangles can share an algebraic connection with any other. Hence, this diagram cannot be constructed only using sums and products of rational subtangles, which implies that T is non-algebraic. □

Note that the possibilities in Theorem 4.12 are not mutually exclusive. For example, an integer tangle diagram is also rational and algebraic, and it satisfies $D_{\text{int}} = D_{\text{rat}} = D_{\text{alg}} = D_{\max}$. On the other hand, a rational tangle diagram which is not integer will satisfy $D_{\text{int}} < D_{\text{rat}} = D_{\text{alg}} = D_{\max}$, and so maximal subtangle decompositions provide a convenient way of distinguishing between classes of tangle diagrams.

As for the cases of Montesinos and generalized Montesinos tangles that we have not yet discussed, while it would be possible to define corresponding subtangle decompositions, these are more readily identified from an algebraic construction formula. Finding D_{alg} will be sufficient for determining whether a given diagram falls into one of these cases.

We must be careful about the distinction between *tangles* and *tangle diagrams*; Theorem 4.12 only classifies the latter. In general, a tangle of a certain class might permit a diagram of a different class. Figure 4.19 shows an example of a non-algebraic diagram (4.19a) and an algebraic diagram (4.19b) which describe equivalent tangles. The leftmost diagram is non-algebraic; it is impossible to describe the shown configuration of subtangles using only tangle sums and products. However, the crossings in the center subtangle can be eliminated using a Reidemeister II simplification, which results in the algebraic diagram on the right. The underlying tangle is algebraic since it can be constructed using sums and products of rational subtangles, even though it admits a non-algebraic diagram.

The existence of a certain class of tangle diagram is a necessary but not sufficient condition for classifying the underlying tangle. While the example from Figure 4.19 is overly simplistic, it is an open question whether a given non-algebraic tangle diagram is equivalent to an algebraic tangle diagram. Similarly, it also an open question whether a given algebraic diagram can be simplified to one of the special cases.



(a) Non-algebraic tangle diagram

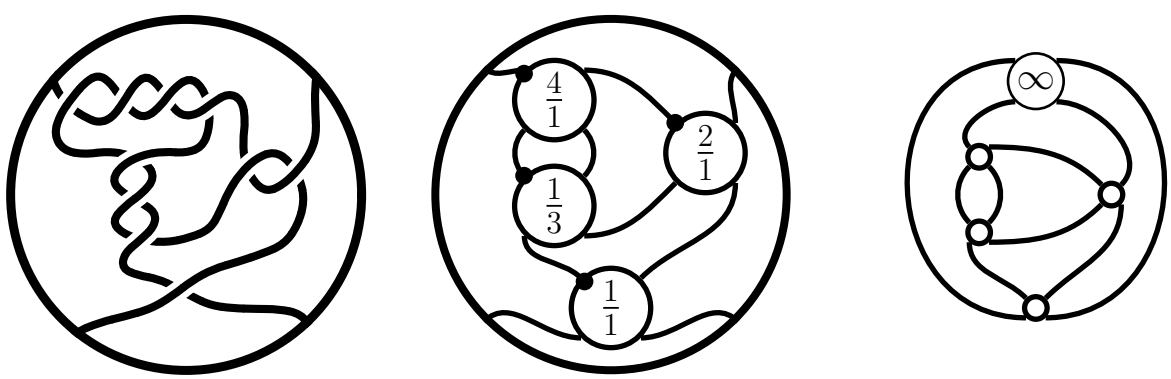
(b) Equivalent algebraic tangle diagram

Figure 4.19: Examples of equivalent algebraic and non-algebraic diagrams for $T \cong \frac{11}{3} * \frac{11}{3}$

4.1.3 Subtangle Graphs and Conway Notation

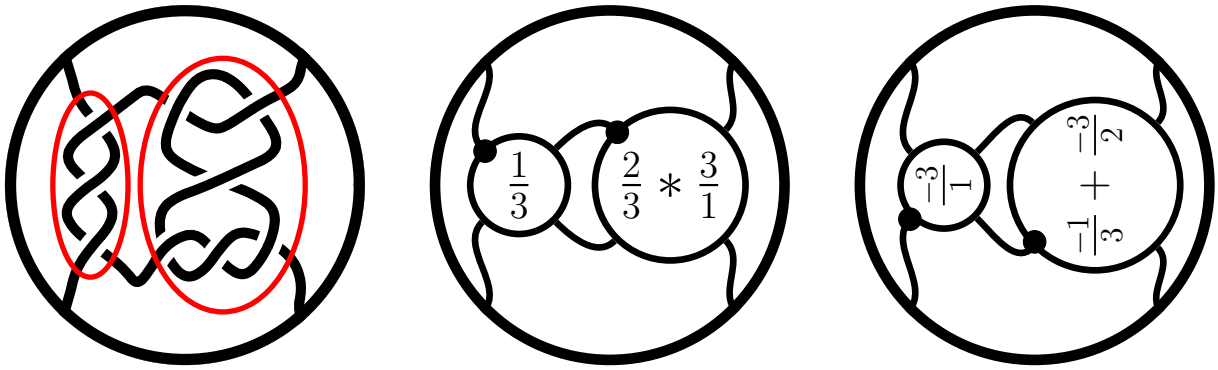
Using subtangle decompositions to classify tangle diagrams is a first step towards studying these more complicated questions. One convenient way to work with decompositions is using *subtangle graphs* (Figure 4.20). By partitioning the crossings in a diagram into disjoint subtangle separating circles, we can interpret a subtangle decomposition using a graph where each vertex represents a subtangle and edges represent the strings between these subtangles. If we imagine this diagram as sitting on the surface of a sphere, we can collapse the boundary circle of the tangle to a point on the back of the sphere and treat this as a special vertex. The boundary vertex is denoted by ∞ in the graph illustrated in Figure 4.20c. This idea is similar to the vertex closure of a tangle diagram used in the generalization of the Tait Flyping Conjecture (Figure 3.27).

Since we have restricted ourselves to the study of 2-string tangles, each subtangle has exactly 4 endpoints where the separating circle connects to the strings, as does the boundary vertex. Hence, the subtangle graph is guaranteed to be a 4-valent planar graph. A subtangle decomposition thus specifies such a graph and how to slot in subtangles at each vertex. Algebraic construction formulas define this graph inherently, as for example the construction $\frac{30}{43} \cong ((\frac{4}{1} * \frac{1}{3}) + \frac{2}{1}) * \frac{1}{1}$ in Figure 4.20b. When we group the crossings in a tangle diagram together into subtangles, it is important to keep track of which subtangle endpoint denotes the NW corner. This corner is marked using a bold black dot in the subtangle decomposition shown in Figure 4.20b.



(a) Rational Tangle $\frac{30}{43}$ **(b)** Subtangle Decomposition **(c)** Subtangle Graph

Figure 4.20: Rational tangle $\frac{30}{43}$ decomposed into maximal integer subtangles



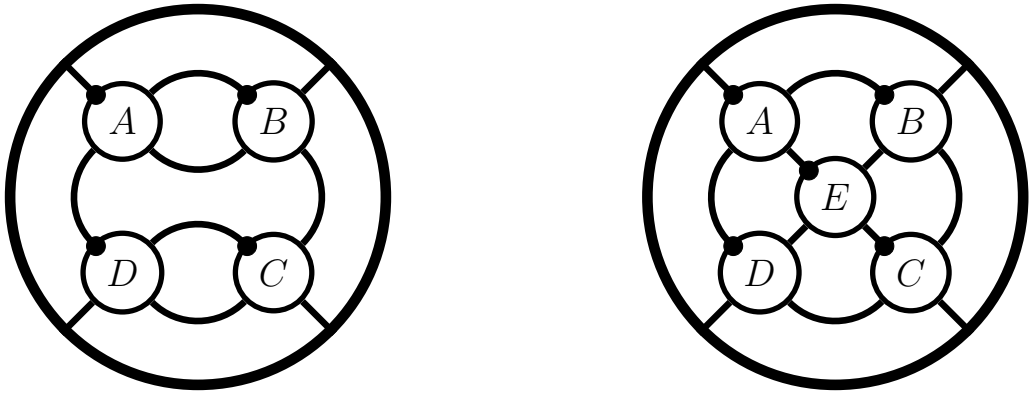
(a) $T \cong \frac{1}{3} + (\frac{2}{3} * \frac{3}{1})$ **(b)** Standard choice of NW corner **(c)** Different choice of NW corner

Figure 4.21: Subtangle decomposition using different choices of NW corner for subtangles

Since subtangles in an algebraic diagram are stacked together like blocks, using the top left corner of each subtangle as the NW endpoint is the natural choice (4.21b). However, we could have made a different choice. Had we chosen the bottom left corner as the NW endpoint in the subtangle decomposition of Figure 4.21, then locally the subtangles would appear rotated (Figure 4.21c). If the subtangle is algebraic, a different choice of NW corner can change the construction formula. There is no particular advantage in making a different choice when presented with an obviously algebraic tangle diagram, but this becomes a real concern when choosing subtangles in a generic diagram that may or may not be algebraic.

There may not always be an obvious choice of NW corner for a non-algebraic subtangle decomposition, and a choice which looks reasonable in one presentation of the diagram may look unreasonable in a different presentation. For computing subtangle decompositions in practice, we will address this dilemma in two ways. The first method is to make a choice of NW corner for each subtangle in a given decomposition. In this case, a local algebraic construction formula for the subtangle may not match the original construction formula, as we see in Figure 4.21c. While this choice looks unintuitive for an algebraic diagram, this information is easy to track with a computer and will play an important role in developing a algorithm to compute subtangle decompositions.

The second option for addressing this dilemma is choosing a presentation for every subtangle decomposition in which there is an obvious choice of the NW corner for each subtangle. Algebraic diagrams constructed by attaching together subtangles horizontally and vertically are an example of this since we may unambiguously choose the top left corner (Figure 4.22a).



(a) Choice of NW corner for an algebraic diagram **(b)** Choice of NW corner for a non-algebraic diagram

Figure 4.22: Examples of standard choices of NW corner for subtangle decompositions

We may attempt to extend this to non-algebraic diagrams by choosing a preferred corner for each subtangle in a given decomposition (Figure 4.22b). While there are infinitely many non-algebraic diagrams, we can reasonably do this for sufficiently small tangle diagrams. Essentially, this choice is built into the subtangle decompositions used in *Conway notation*, which Conway developed along with his classification of rational tangles [7].

The corresponding subtangle graph is independent from this choice of NW corner, but this is additional data necessary for reconstructing a subtangle decomposition from a subtangle graph. We will hold off on a detailed examination of subtangle graphs for non-algebraic tangle diagrams until a later section, but this discussion of subtangle graphs is a good precursor to understanding Conway notation. This notation encodes a knot diagram using a decomposition into maximal algebraic subtangles, an idea which will also be central to our study of tangle diagrams. Before expanding on this concept, we will pause to give a brief overview of how Conway accomplishes this in his notation.

A key observation concerning subtangle decompositions is that tangle sums and products always involve exactly two connected strings between the subtangles. These correspond to bigons in the subtangle decomposition and subtangle graph. By collapsing bigons into a single vertex, we can reduce the number of vertices in a subtangle graph, which is equivalent to choosing larger subtangles in the original diagram (Figure 4.23). If we repeat this process until there are no remaining bigons between subtangles, the resulting graph either consists of a single subtangle or some non-algebraic combination of subtangles.

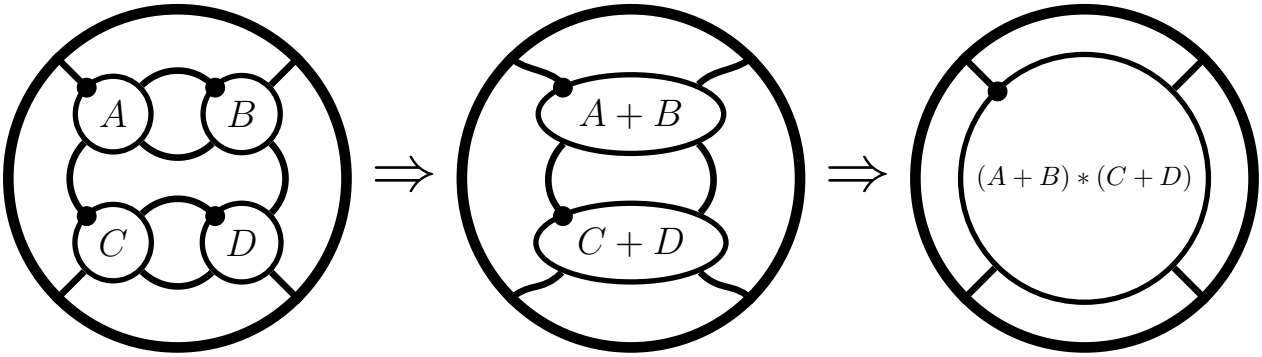


Figure 4.23: Collapsing bigons in a subtangle decomposition

This idea of collapsing bigons will be made more precise in Section 6, but for now we can take this as the motivation behind the choice of graphs used in Conway notation. Conway showed that it was possible to decompose all knots with up to 11 crossings using one of the *basic polyhedra* illustrated in Figure 4.24, where the vertices represent algebraic subtangles [7]. Observe that these graphs contain no bigons. Note that we would need to introduce the special boundary vertex to accommodate tangle diagrams, but otherwise the idea of using a graph to represent a decomposition of a diagram into subtangles is essentially the same for knots, links, and tangles.

In Conway's construction, any tangle diagram can be constructed from the 0, +1, and −1 tangles using three operations: *sum*, *product*, and *ramification* (Figure 4.25). Unfortunately for us, our choice of tangle operations is inconsistent with that of Conway. We use the same definition of tangle sum, but what Conway refers to as a product of tangles would be defined

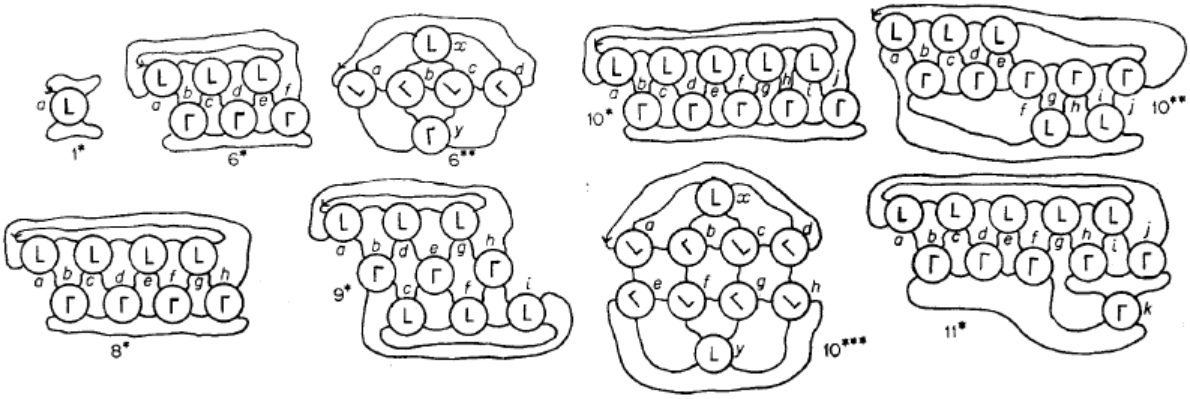


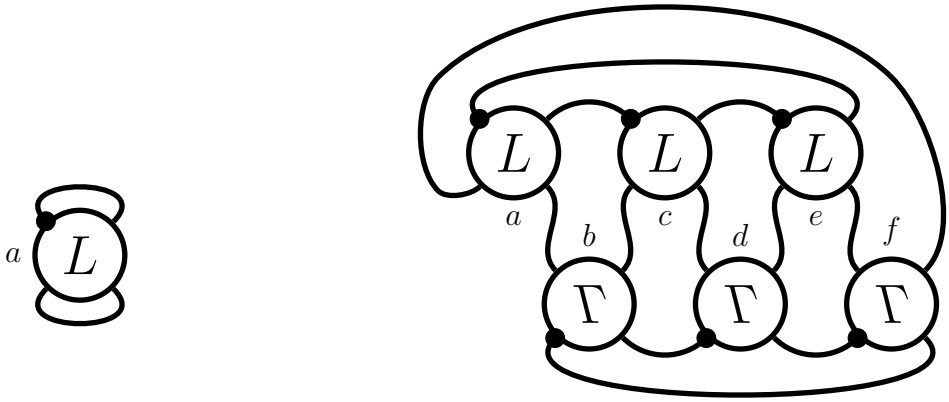
Figure 4.24: Conway's basic polyhedra [7], reprinted with permission from Elsevier

(a) Conway tangle sum: $L_1 + L_2$ **(b)** Conway tangle product: $L_1 L_2 = R_{90}(R_H(L_1)) + L_2$ **(c)** Conway tangle ramification: $L_1, L_2 = R_{90}(R_H(L_1)) + R_{90}(R_H(L_2))$

Figure 4.25: Conway's three operations on tangles

as a sum using the diagonal reflection for the first summand in our notation. Ramification is a sum using the diagonal reflection of both summands. Even more unfortunately, Conway uses the expression $-T$ to denote the diagonal reflection of a tangle T, whereas we have used this to denote the mirror image.

Thanks to these different choices, Conway's notation for algebraic tangles is very different from ours. We will exclude a complete explanation of how Conway denotes algebraic tangles to avoid unnecessary confusion, but a proper introduction to this topic can be found in the literature [7, 4, 34, 1, 16]. Setting aside the specific details of algebraic subtangles, we will examine how Conway leverages graphs to denote knot diagrams. The brief discussion of Conway notation which follows is based in part on the description outlined in the text *LinKnot* by Jablan and Sazdanovic [16]. Figure 4.26 shows a closeup of the first two Conway graphs, which are used as templates for slotting in subtangles.



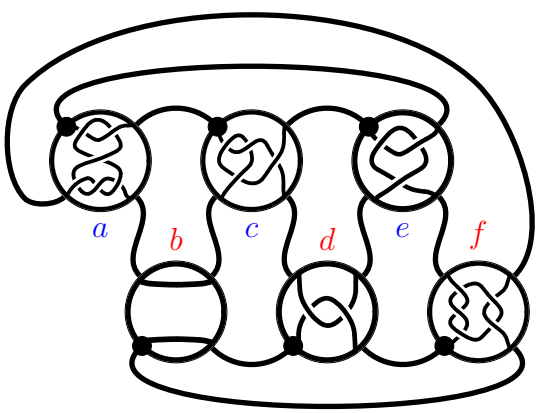
(a) $K \cong 1^* a.$ **(b)** $K \cong 6^* a.b.c.d.e.f.$

Figure 4.26: Conway notation template for decomposing a knot diagram using subtangles

The Conway notation for a knot diagram has the form: $k^a t_1.t_2.\cdots t_k.$, as in the two templates of Figure 4.26. The first symbol k^a denotes one of Conway's basic polyhdera, where k is the number of vertices and a is a number of asterisks used to distinguish between the graphs with k vertices. The symbols $t_1, \cdots, t_k$ denote the k algebraic subtangles that slot into these vertices. In the expression $k^a t_1.t_2.\cdots t_k.$, each tangle term is followed by a period separating it from the next subtangle. In this expression, the subtangles are listed alphabetically according to the letters next to each vertex in the Conway graph. The orientation of the letter L in each vertex indicates the position of the NW corner for the corresponding subtangle; these are indicated with a bold black dot in the template of Figure 4.26.

Figure 4.27 shows a full example of a subtangle decomposition for a knot diagram denoted using Conway notation. The first symbol, 6^*, denotes the the basic polyhderon from the list of those shown in Figure 4.24. The remaining string of symbols denotes the algebraic subtangles that slot into the vertices, each separated by a period. In this case, there are six subtangles, one for each vertex, and they are ordered alphabetically as indicated on the nodes in part 4.27a. These are color coded in the example for greater visibility. Part 4.27b shows the Conway notation for each algebraic subtangle, along with the translation into sums and products of rational tangles that we have become familiar with.

More generally, any of Conway's basic polyhedra can be used to denote a knot diagram in a similar way, although Conway introduces a few more notational shorthands that we will skip. Reconstructing a diagram from its Conway notation reduces to slotting subtangles into



(a) $K \cong 6^*(2\ 1\ 0)(3), 0.0.2\ 1\ 1.2.2\ 1\ 0.3, 2.$

a	$\cong$	$(2\ 1\ 0)(3), 0$	$\cong$	$\frac{2}{3} * \frac{3}{1}$
b	$\cong$	0	$\cong$	0
c	$\cong$	$2\ 1\ 1$	$\cong$	$\frac{5}{3}$
d	$\cong$	2	$\cong$	$\frac{2}{1}$
e	$\cong$	$2\ 1\ 0$	$\cong$	$\frac{2}{3}$
f	$\cong$	$3, 2$	$\cong$	$\frac{1}{3} + \frac{1}{2}$

(b) Conway notation for subtangles

Figure 4.27: Example of a knot diagram described using Conway notation

the appropriate graph. One must pay attention to the orientation of the subtangle vertex in the graph when reconstructing the diagram; a rotated or flipped subtangle can change the equivalence class of the knot diagram. The polyhedron consisting of a single vertex is only used to describe algebraic diagrams (Figure 4.26a).

Conway notation is defined for a choice of subtangle decomposition, but it may not be unique even for a given diagram. For example, the second subtangle used in the example from Figure 4.27 was chosen to be a trivial tangle. Since they contain no crossings, trivial subtangles are unnecessary in a subtangle decomposition. If we remove this trivial subtangle, then the remaining subtangles can be merged together by collapsing bigons (Figure 4.28). Hence, The knot diagram from Figure 4.27 is actually algebraic and could also be denoted using the Conway graph with a single vertex. This also illustrates why we excluded trivial subtangles from our earlier discussion of subtangle decompositions.

While Conway introduced this notation in the context of knot diagrams, it is clear that we can take a similar approach to denoting tangle diagrams. Since algebraic diagrams can always be collapsed down to a single vertex, this really only becomes interesting for non-algebraic diagrams. The major takeaway from our discussion of Conway notation is the utility of graphs for encoding subtangle decompositions, although we must be precise when we make choices such as how to orient the subtangle vertices.

The second half of this dissertation is merely a generalization of how Conway studied knot diagrams. In combination with algebraic subtangles, subtangle graphs offer a very compact way of denoting a diagram in such a way that can be easily reconstructed. The remainder of this section focuses on the subtangle planar diagram code, which is a notation for tangle diagrams based on subtangle graphs. We will return to these ideas again in Chapter 6 to discuss subtangle decompositions of non-algebraic diagrams.

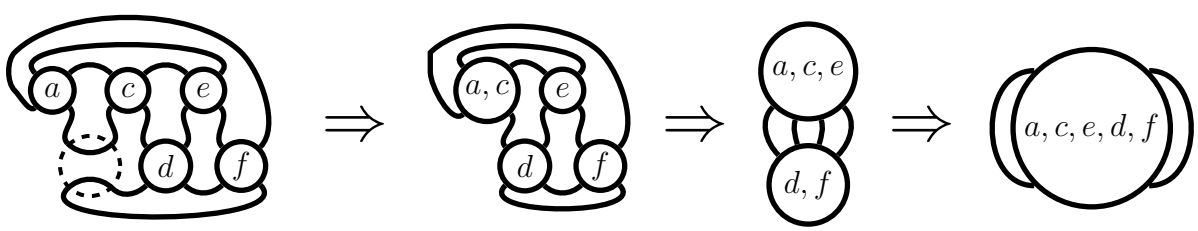


Figure 4.28: The decomposition from Figure 4.27 collapsed into a single algebraic subtangle

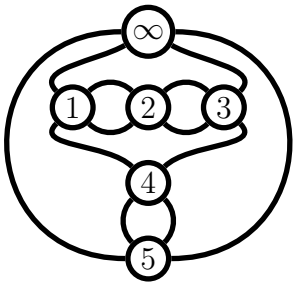
4.2 Subtangle Planar Diagram Code

Recall that the crossing planar diagram (CPD) code for tangle diagrams, which is based on the Ewing-Millett notation for knot diagrams [12], is a list of information recording the connections between each crossing (Figure 4.29). This code is essentially the adjaceny list for a graph, with some additional information based on oriented crossings. The graph hiding in the CPD code is exactly the subtangle graph for a diagram using 1-crossing subtangles. Hence, the CPD code provides a template for how a diagram could be described using subtangle graphs more generally: a list of connection information describing the edges between subtangles, and then some additional information to describe these subtangle vertices.

Using the CPD code as a model, we propose how a diagram can be encoded using a similar *subtangle planar diagram* (SPD) code. Unlike the CPD code, the SPD code is not unique for a fixed diagram. Rather, it is defined relative to a choice of subtangle decomposition for the given diagram. Hence, we will speak more generally of *an* SPD code rather than *the* SPD code, although we will introduce a few variations of SPD code which overcome this by using maximal subtangle decompositions. Our first objective is to show how a choice of subtangle decomposition can be encoded.

+	$4b$	0 NW	$2d$	$2c$
+	$3b$	$3a$	$1d$	$1c$
+	$2b$	$2a$	0 NE	$4c$
+	$5b$	$1a$	$3d$	$5c$
+	0 SW	$4a$	$4d$	0 SE

(a) Tangle diagram and CPD code



$$\begin{aligned} \{0 &: \{1,3,5,5\}, \\ 1 &: \{4,0,2,2\}, \\ 2 &: \{3,3,1,1\}, \\ 3 &: \{2,2,0,4\}, \\ 4 &: \{5,1,3,5\}, \\ 5 &: \{0,4,4,0\}\} \end{aligned}$$

(b) Subtangle graph and adjacency list

Figure 4.29: Example of CPD code matching a subtangle graph

4.2.1 Generalizing the CPD Code to Subtangles

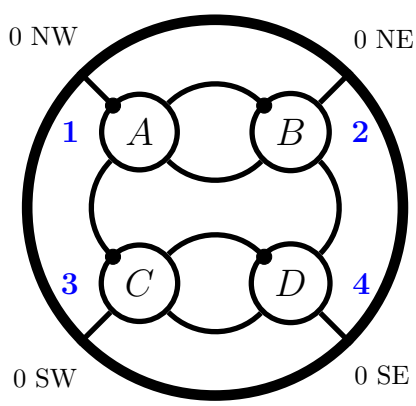
Suppose that we have a fixed tangle diagram T and a choice of subtangle decomposition D. The standard CPD code consists of a list of rows, one for each crossing, and it assumes that the crossings in the diagram are *ordered* and *oriented*. Hence, to obtain a similar description of T using D, we will assume that the subtangles in this decomposition are ordered in some way, and that each is oriented by a choice of NW corner in the diagram.

For each subtangle $S \in D$, we write out a row including the label, connection information, and parity of S. The connection information specifies the index and corner of each subtangle connected to an endpoint of S, treating the boundary as index 0.

$$\underbrace{S}_{\text{label}} \qquad \underbrace{\underbrace{\quad}_{\text{NW}} \quad \underbrace{\quad}_{\text{NE}} \quad \underbrace{\quad}_{\text{SE}} \quad \underbrace{\quad}_{\text{SW}}}_{\text{connection info.}} \qquad \underbrace{(\text{P} : ___)}_{\text{parity}}$$

In this way, we may write out a list of information for each subtangle in D. As in Figure 4.30b, these become the rows in a subtangle planar diagram code for the diagram T, indexed according to how they are ordered in D. Although the index is included in the figure, we do not write this down in general since it matches the order of the rows in the SPD code.

If the subtangles in D are algebraic, then we label these using the corresponding algebraic construction formula. Whereas the CPD code includes crossing sign, the SPD code a labeled



(a) $T \cong (A + B) * (C + D)$

index	label	connection information				parity
1	A	0 NW	2 NW	2 SW	3 NW	$(\text{P} : p_A)$
2	B	1 NE	0 NE	4 NE	1 SE	$(\text{P} : p_B)$
3	C	1 SW	4 NW	4 SW	0 SW	$(\text{P} : p_C)$
4	D	3 NE	2 SE	0 SE	3 SE	$(\text{P} : p_D)$

(b) Subtangle planar diagram code

Figure 4.30: Example of a subtangle decomposition converted into an SPD code

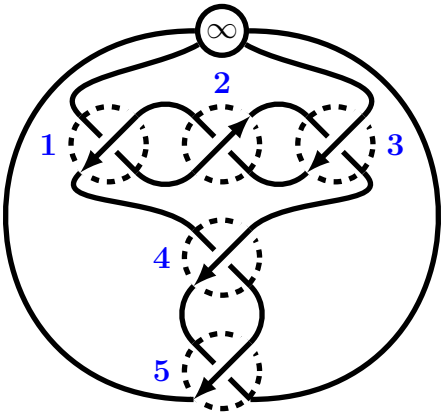
subtangle. The parity of each subtangle is relative to the choice of NW corner, although there are some subtleties to be aware of. Setting these aside momentarily, the process we have just described shows how an SPD code can be computed for a diagram by hand. This is a straightforward method to write down a subtangle decomposition in practice, although it assumes that we can identify the connection information visually.

Often times we are limited to working with only a diagram notation, and we may not know whether the subtangles in a given decomposition can be joined algebraically. In light of this, we will impose some systematic rules about how subtangles are ordered and oriented that will make it easier for a computer to track and identify the connection information. These choices are not a requirement for generic SPD codes, but we will adopt these rules for the special variations of code that we consider shortly.

Recall how similar choices are made in the CPD code. The crossings are ordered and oriented based on the standard choice of orientation for the diagram. Starting from the NW corner of the tangle diagram, this order is determined by following the string in the direction of the orientation and indexing each new crossing we pass through (Figure 4.31a). Recall that the four endpoints of each crossing are labeled alphabetically moving clockwise around the crossing, taking the outward-pointing over-strand as a.

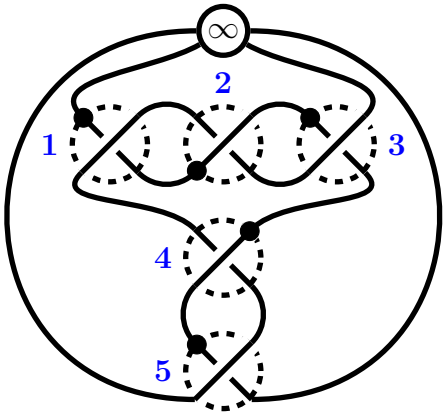
We can make similar choices to order and orient the subtangles in a given subtangle decomposition for a diagram. Tracing through the strings in the tangle diagram, we pass through each subtangle twice. Using the standard orientation and traveling along the string from the NW corner, we may index these subtangles based on when we encounter them for the first time. Unlike oriented crossings, an outward-pointing over-strand does not make sense for generic subtangles. Instead, we can choose to mark the first corner of the subtangle we pass through while traveling along the the oriented strings. These are marked using a bold black dot in the example of Figure 4.31b. By default, we will take this to be the NW corner of the subtangle.

Figure 4.31 shows a side-by-side comparison of annotated tangle diagrams using the CPD labeling conventions (4.31a) and the SPD labeling conventions (4.31b), in this case using the 1-crossing subtangle decomposition. Observe that the choice of NW corner obtained in this way for some subtangles does not match the obvious choice of the top left corner. Hence, relative to these choices, some of these subtangles are labeled -1, even though this particular rational tangle diagram is constructed using only positive twists. Although this may seem unintuitive at first glance, this information is more easily tracked by computational algorithms in the absence of a nice diagram.



1	+	$4b$	0 NW	$2d$	$2c$
2	+	$3b$	$3a$	$1d$	$1c$
3	+	$2b$	$2a$	0 NE	$4c$
4	+	$5b$	$1a$	$3d$	$5c$
5	+	0 SW	$4a$	$4d$	0 SE

(a) CPD Code



1	$+1$	0 NW	2 NE	2 NW	4 SW	(P : 1)
2	-1	1 SE	1 NE	3 NW	3 SW	(P : 1′)
3	$+1$	2 SE	0 NE	4 NW	2 SW	(P : 1)
4	-1	3 SE	5 NE	5 NW	1 SW	(P : 1′)
5	$+1$	4 SE	4 NE	0 SE	0 SW	(P : 1)

(b) 1CPD Code

Figure 4.31: Comparison of the CPD code and the SPD code using 1-crossing subtangles

This choice of orientation on subtangles also determines the *oriented parity* of each subtangle, which we originally introduced in Figure 2.6. The two 1-crossing tangles $+1$ and -1 both have parity 1, but observe that some of the rows in Figure 4.31b have oriented parity $1'$. This is to account for the fact that the standard orientation on the strands in a subtangle considered in isolation may be different from the orientation these strands inherit from the larger tangle diagram. Our choice of NW corner ensures that the first string of each subtangle always points inward from the NW corner. However, the inherited orientation on the second string may point in the opposite direction from the standard orientation.

This reversed direction on the second string may occur in subtangle decompositions more generally. In these cases, we mark the parity as $0'$, $1'$, or ∞' in accordance with Figure 2.6. Note that there does not exist an analogue of this behavior in the CPD code; the orientation of the crossing in the diagram matches the orientation recorded in the CPD code.

4.2.2 SPD Codes for Maximal Subtangle Decompositions

The choice of subtangles illustrated in Figure 4.31 matches the 1-crossing subtangle decomposition, $D_{\min}$. We will refer to the corresponding SPD code as the *1-crossing subtangle planar diagram* (1CPD) code. This code carries equivalent information to the standard CPD code, except that we treat the crossings as subtangles. The 1CPD code is the first of several variations on the SPD code arising from the maximal subtangle decompositions summarized in Theorem 4.12.

The crossing planar diagram code can be converted into the 1-crossing subtangle planar diagram code by mapping the alphabetical CPD labels to the corresponding compass 1CPD labels, and then updating the connection information. As summarized in Table 4.1, the label conversion can be determined independently from the connection information. There are four

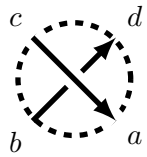
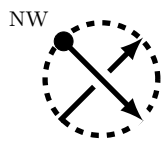
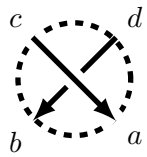
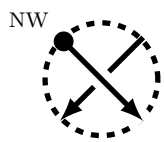
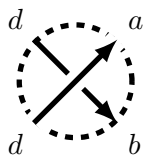
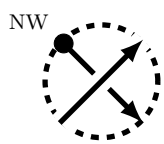
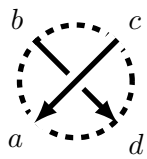
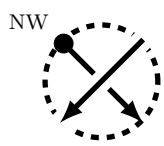
Cross. Sign	CPD Labels	1CPD Labels	Subtangle	Parity	Label Conversion
$+$			-1	$1'$	$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mapsto \begin{pmatrix} \text{SE} \\ \text{SW} \\ \text{NW} \\ \text{NE} \end{pmatrix}$
$-$			-1	1	$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mapsto \begin{pmatrix} \text{SE} \\ \text{SW} \\ \text{NW} \\ \text{NE} \end{pmatrix}$
$-$			$+1$	$1'$	$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mapsto \begin{pmatrix} \text{NE} \\ \text{SE} \\ \text{SW} \\ \text{NW} \end{pmatrix}$
$+$			$+1$	1	$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} \mapsto \begin{pmatrix} \text{SW} \\ \text{NW} \\ \text{NE} \\ \text{SE} \end{pmatrix}$

Table 4.1: Conversion from CPD labels to 1CPD labels

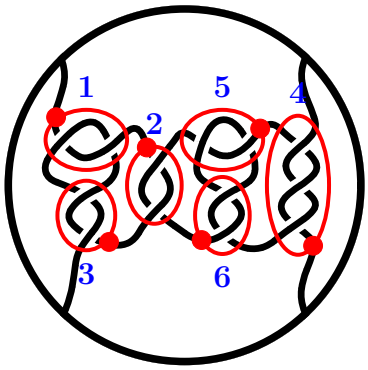
possibilities which depend on on the crossing sign and whether we first pass along the over-strand or the under-strand in the oriented diagram. As in the second and third columns from the table, we can examine these four possibilities using diagrams for 1-crossing subtangles, where the local picture is rotated so that the NW corner indicates the inward-pointing strand encountered first.

In the 1CPD code, the NW corner matches the first encountered inward-pointing strand, and so the local subtangle diagrams used in Table 4.1 are rotated to match this. Based on this presentation, the 1-crossing subtangle is identified as $+1$ or -1 based on the slope of the over-strand. There are two possibilities for this slope, and two possibilities for the orientation on second strand, leading to the four local pictures illustrated in the table. The oriented parity of the 1-crossing subtangle is either 1 or $1'$, as determined by the string directions described in Figure 2.6.

The directions on the strands in these local pictures also determine whether the oriented crossing sign is positive or negative. The alphabetical CPD labels always assign a to the outward pointing over-strand, and the remaining labels are chosen moving clockwise around the circle. Hence, the corresponding conversion from CPD labels to 1CPD labels is determined by the four cases illustrated in Table 4.1. The connection information is identical in the CPD and 1CPD codes, except that the labels are changed and the order of these labels is permuted cyclically to start at the NW corner. This cyclic permutation can be seen in the example from Figure 4.31; each row of the two codes contains exactly the same indices, but listed in a different order.

The 1CPD code is an example of a subtangle planar diagram code matching one of our previously studied subtangle decompositions, in this case D_{min}. Generalizing this to the other subtangle decompositions from Theorem 4.12, we will give analogous definitions for the *integer planar diagram* (IPD) code, the *rational planar diagram* (RPD) code, and *algebraic planar diagram* (APD) code. These codes match the maximal subtangle decompositions D_{int}, D_{rat}, and D_{alg}, respectively. We skip defining a code for D_{max} since this always consists of a single subtangle.

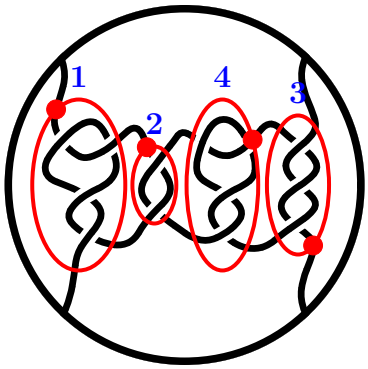
Figure 4.32 shows an example of each of these codes for the algebraic tangle diagram $T \cong \frac{2}{5} + \frac{1}{2} + \frac{2}{5} + \frac{1}{3}$. Since this diagram is algebraic, its APD code contains a single term corresponding to the single maximal algebraic subtangle. Each SPD code in this list can be computed from the preceding code through a process of refinements, starting from the 1CPD code. A decomposition into maximal rational subtangles may not always exist for a given diagram, and so we will make some exceptions when computing the RPD code.



(a) Maximal Integer Subtangles

1	$\left(\frac{2}{1}\right)$	0 NW	2 NW	3 SW	3 SE	(P:0)
2	$\left(\frac{1}{2}\right)$	1 NE	5 SW	6 NW	3 NW	(P:∞')
3	$\left(\frac{1}{2}\right)$	2 SW	0 SW	1 SW	1 SE	(P:∞)
4	$\left(\frac{1}{3}\right)$	0 SE	6 SW	5 NW	0 NE	(P:1)
5	$\left(\frac{1}{-2}\right)$	4 SE	6 SE	6 NE	2 NE	(P:∞)
6	$\left(\frac{2}{-1}\right)$	2 SE	5 SE	5 NE	4 NE	(P:0)

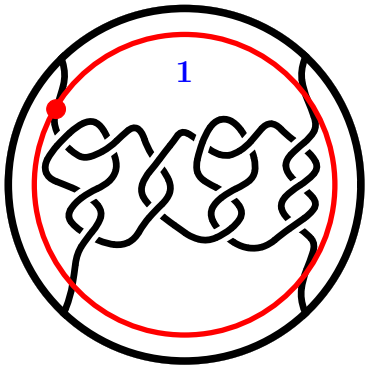
(b) Integer Planar Diagram (IPD) Code



(c) Maximal Rational Subtangles

1	$\left(\frac{2}{5}\right)$	0 NW	2 NW	2 SW	0 SW	(P:0)
2	$\left(\frac{1}{2}\right)$	1 NE	4 SW	4 SE	1 SE	(P:∞')
3	$\left(\frac{1}{3}\right)$	0 SE	4 NE	4 NW	0 NE	(P:1)
4	$\left(\frac{5}{-2}\right)$	3 SE	3 NE	2 SE	2 NE	(P:∞)

(d) Rational Planar Diagram (RPD) Code



(e) Maximal Algebraic Subtangles

1	$\frac{2}{5}+\frac{1}{2}+\frac{2}{5}+\frac{1}{3}$	0 NW	0 NE	0 SE	0 SW	(P:∞)

(f) Algebraic Planar Diagram (APD) Code

Figure 4.32: Example of special SPD codes using maximal subtangle decompositions

4.3 Computing Maximal SPD Codes

We previously discussed how the 1CPD code for a diagram can be computed from the CPD code based on the label conversions summarized in Table 4.1. This algorithm has successfully been implemented as a program written in C/C++, as have algorithms to compute the IPD, RPD, and APD codes [5]. In each case, the algorithm uses the connection information in an SPD code to identify algebraic connections between subtangles, and then collapses these into a single subtangle. As can be seen from Figure 4.32, an algebraic connection between subtangles exists whenever a row of the SPD code includes the same subtangle index twice.

The final part of this section is devoted to explaining these algorithms. Although the combinatorics is different in each case, each of these SPD codes can be computed from the previous code without an explicit diagram. However, because of how we choose to orient the subtangles, often we must rotate adjacent subtangles before they can be combined. The most challenging aspect of these algorithms involves keeping track of subtangle transformations. Fortunately, rotations of rational subtangles are easy to describe using Corollary 3.46. In computing each of these codes, we give up some information about the diagram in favor of a more compact description, and hence diagrams which have equivalent but non-identical subtangles may have the same SPD code.

4.3.1 Converting 1CPD Code to IPD Code

The IPD code if a diagram is obtained from the 1CPD code by joining together neighboring 1-crossing subtangles into integer subtangles. Integer tangles are considered either horizontal or vertical depending on the direction of twisting relative to the NW corner. Recall

Type	**Horizontal Integer**		**Vertical Integer**	
Tangle				
Fraction	$\frac{n}{1}$	$-\frac{n}{1}$	$\frac{1}{n}$	$-\frac{1}{n}$

Table 4.2: Generic integer tangles and their fractions

that horizontal integer tangles are denoted by the fraction $\pm\frac{n}{1}$ and vertical integer tangles are denoted by the fraction $\pm\frac{1}{n}$. In both cases, n denotes the number of crossings in the tangle and the sign of the fraction matches the sign of the twisting. This leads to the four generic types of integer tangle summarized in Table 4.2.

Relative to the NW corner, observe that horizontal integer tangles twist from east to west whereas vertical integer tangles twist from north to south. We refer to this behavior as the *twist axis* of an integer tangle. Distinguishing between horizontal and vertical integer tangles without a diagram depends on identifying the twist axis relative to a chosen NW corner. When an integer subtangle shares an algebraic connection with a 1-crossing subtangle, these can be joined into a larger subtangle. However, this larger subtangle is integer if and only if the algebraic connection is aligned with the twist axis, as described in Table 4.3.

A 1-crossing subtangle is considered both horizontal and vertical, but tracking the twist axis becomes important when considering how two or more 1-crossing subtangles can be combined into larger integer subtangles. When two 1-crossing subtangles share an algebraic connection, they can be joined into an integer subtangle provided their twist signs are compatible. These twists are considered incompatible if they could be simplified by a Reidemeister II move. This way, the algorithm is still applicable to non-simplified diagrams without changing the number of crossings in a fixed diagram.

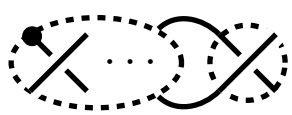
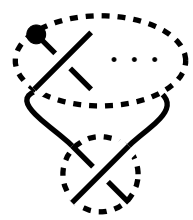
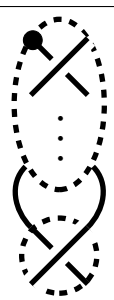
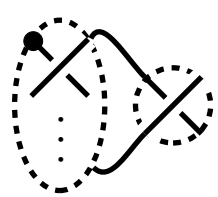
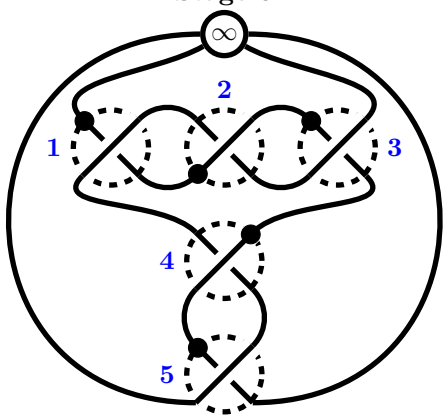
Type	**Twist Axis Alignment**	
	aligned	unaligned
Horizontal		
Vertical		

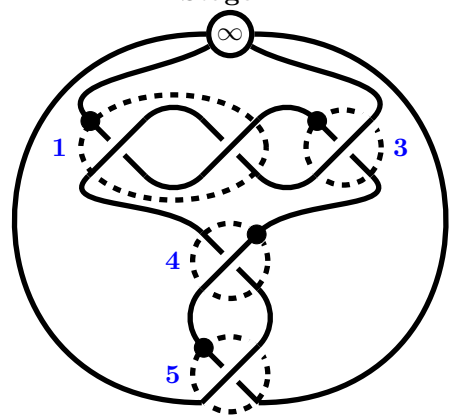
Table 4.3: Integer subtangle twist axis alignment

Stage 0



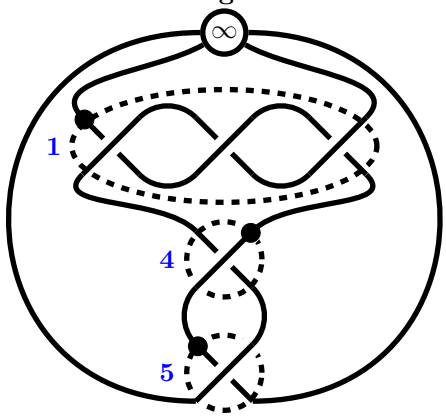
1	$+1$	0 NW	2 NE	2 NW	4 SW	$(P:1)$
2	-1	1 SE	1 NE	3 NW	3 SW	$(P:1')$
3	$+1$	2 SE	0 NE	4 NW	2 SW	$(P:1)$
4	-1	3 SE	5 NE	5 NW	1 SW	$(P:1')$
5	$+1$	4 SE	4 NE	0 SE	0 SW	$(P:1)$

Stage 1



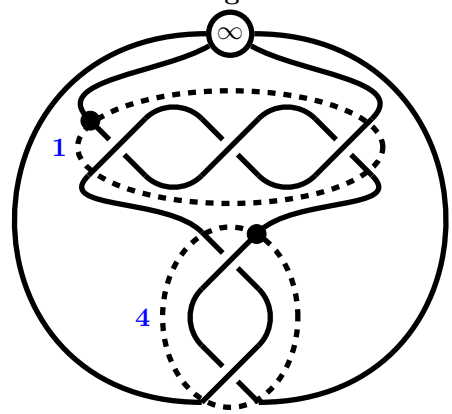
1	$\underline{\left(\frac{2}{1}\right)}$	0 NW	3 NW	3 SW	4 SW	$(\underline{P:0})$
~~**2**~~						
3	$+1$	1 NE	0 NE	4 NW	1 SE	$(P:1)$
4	-1	3 SE	5 NE	5 NW	1 SW	$(P:1')$
5	$+1$	4 SE	4 NE	0 SE	0 SW	$(P:1)$

Stage 2



1	$\underline{\left(\frac{3}{1}\right)}$	0 NW	0 NE	4 NW	4 SW	$(\underline{P:1})$
~~**2**~~						
~~**3**~~						
4	-1	1 SE	5 NE	5 NW	1 SW	$(P:1')$
5	$+1$	4 SE	4 NE	0 SE	0 SW	$(P:1)$

Stage 3



1	$\left(\frac{3}{1}\right)$	0 NW	0 NE	4 NW	4 SW	$(P:1)$
~~**2**~~						
~~**3**~~						
4	$\underline{\left(-\frac{2}{1}\right)}$	1 SE	$\underline{0\ \text{SE}}$	$\underline{0\ \text{SW}}$	1 SW	$(\underline{P:0'})$
~~**5**~~						

The IPD code is obtained from the 1CPD code by computing a sequence of intermediate SPD codes. Each stage in the sequence of panels above represents one of these intermediate SPD codes obtained by joining a 1-crossing subtangle into a larger integer subtangle. The portions of the SPD code highlighted in red indicate an algebraic connection between subtangles that can be joined together. The next intermediate SPD code is obtained by combining these two subtangles into a single row, and then deleting the row of higher index. When no more valid joins are possible, the IPD code is obtained by collapsing missing subtangle indices.

Figure 4.33: Example of converting 1CPD code to IPD code for the rational tangle $\frac{3}{7}$

We refer to Figure 4.33 to illustrate an explicit example of how the IPD code is computed from the 1CPD code. This is accomplished through computing a sequence of intermediate SPD codes, each of which is obtained by absorbing a 1-crossing subtangle into an integer subtangle. An algebraic connection between subtangles can be identified from an SPD code whenever the same index occurs twice in one row. The connections highlighted in red in Figure 4.33 indicate when two subtangles can be combined into a larger integer subtangle. However, when computing the IPD code, we only join integer subtangles that have the same twist sign and aligned twist axis.

Starting from the 1CPD code, the algorithm iterates through each row in the notation. If there exists an algebraic connection between this subtangle and an adjacent 1-crossing subtangle, then these two subtangles may be combined, provided the twists are compatible and the connection aligns with the direction of the twist axis. When a valid join is possible, an intermediate SPD code is computed joining these two subtangles. By default, this combined subtangle uses the lower index of the two joined subtangles; the row corresponding to the higher index is deleted. The connection information in the intermediate SPD code is updated to reflect this.

This process repeats for each intermediate SPD code until there do not exist any more algebraic connections that can be combined into a larger integer subtangle. When this search terminates, the final IPD code is obtained by collapsing missing indices. Hence, the IPD code obtained from the example of Figure 4.33 consists of the following two rows.

$$\begin{array}{lllll} \left(\frac{3}{1}\right) & 0\text{ NW} & 0\text{ NE} & 2\text{ NW} & 2\text{ SW} \quad (P:1) \\ \left(-\frac{2}{1}\right) & 1\text{ SE} & 0\text{ SE} & 0\text{ SW} & 1\text{ SW} \quad (P:0') \end{array}$$

Since the 1-crossing subtangles are ordered by the standard orientation on the strings, with the NW corner matching the first pass of the string, this subtangle matches the first crossing in the corresponding integer subtangle obtained in the IPD code. In accordance with Table 4.2, the fraction of this tangle is determined by the number of crossings and whether the twisting occurs to the east or the south. The sign of these twists matches the sign of the first 1-crossing subtangle. The unoriented parity of the tangle can be determined from the fraction. The oriented parity depends on the direction of the second string in the subtangle, and this will also match the first 1-crossing subtangle.

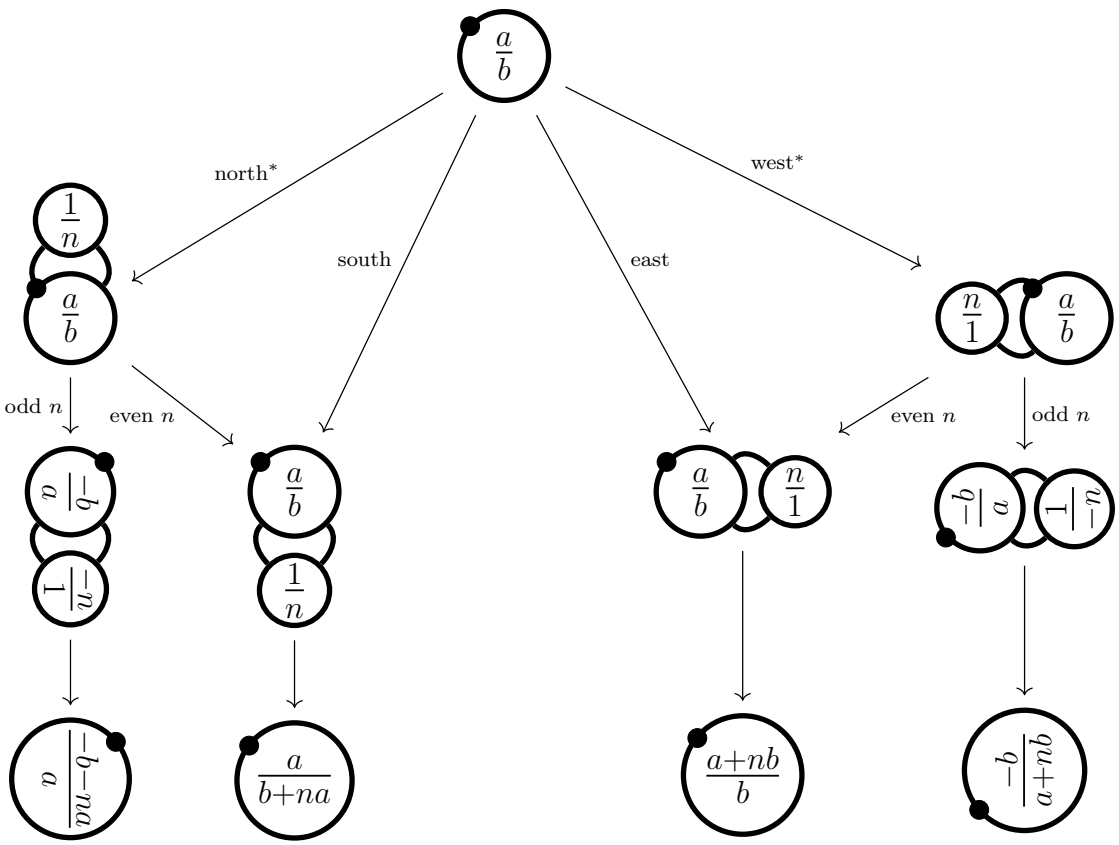
4.3.2 Converting IPD Code to RPD Code

Although we must keep track of a large amount of information, the preceding example illustrates how this method can be applied more generally. A similar approach is used to convert the IPD code into the RPD through a sequence of intermediate SPD codes. Recall that a decomposition into maximal rational subtangles is not possible when there exist distinct maximal rational subtangles that overlap. In practice, we may still compute an SPD code for a decomposition into rational subtangles which aligns with the maximal rational subtangle decomposition when it does exist. We will still refer to this as the RPD code and follow the convention that, whenever there exist overlapping maximal rational subtangles, we use the one encountered first along the oriented strings.

When computing the IPD code, we iterated through integer subtangles and searched for adjacent twists, initially treating each 1-crossing subtangle as an integer subtangle. To compute the RPD code, we will iterate through each rational subtangle and search for adjacent twists, initially treating each integer subtangle as a rational subtangle. If there exists an algebraic connection between a rational subtangle and a sequence of twists, then these can be combined into a larger rational subtangle provided the connection aligns with the twist axis. A rational subtangle identified in this way may not be in canonical form, but we may compute the corresponding fraction based on the flowchart shown in Figure 4.34.

When combining an integer subtangle with a rational subtangle in this way, the fraction of the combined rational subtangle is relative to the choice of NW corner. By default, this corner is determined by the first pass of the oriented strings in an SPD code. Hence, when combining two subtangles into a larger subtangle, it is possible that the NW endpoint changes. This may occur in the RPD algorithm whenever the algebraic connection involves the NW endpoint of the rational subtangle. In the decision tree of Figure 4.34, this happens when an algebraic connection occurs on the north or west side of this rational subtangle. If the connection occurs on the south or east side, then the NW endpoint does not change.

A connection involving the north or west side of the rational subtangle corresponds to non-canonical twisting. However, we know that such a subtangle is equivalent to a canonical diagram via flyping. Hence, we may compute the fraction of the equivalent rational subtangle that would be obtained by moving these twists to the preferred side of the subtangle. Recall that flyping involves flipping a subtangle; hence, there are two cases depending on whether the total number of flyped twists is even or odd. These two possibilities determine whether we rotate our choice of NW corner in the subtangle and are summarized in the decision tree.



When a rational subtangle shares an algebraic connection with an integer subtangle that aligns with the twist axis, these can be combined into a larger rational subtangle. The fraction of the combined rational subtangle depends on the location of this connection relative the original NW corner, possibly with a rotation*.

Figure 4.34: Decision tree to determine rational subtangle fraction in the RPD algorithm

After determining the local choice of NW corner for the combined rational subtangle, we may compute its extended rational number based on the cases shown in Figure 4.34. The joined integer tangle represents either horizontal or vertical twisting, which can be encoded as a twist vector. The new fraction is determined by rationalizing the corresponding continued fraction, where we may leverage the composition of Lemma 3.33 to compute this.

$$\frac{a}{b} + \frac{n}{1} \quad \cong \quad \mathrm{CF}\big(\tfrac{a}{b}, 0, n\big) \quad = \quad n + \frac{1}{0 + \frac{1}{\frac{a}{b}}} \quad = \quad \frac{a+nb}{b}$$

$$\frac{a}{b} * \frac{1}{n} \quad \cong \quad \mathrm{CF}\big(\tfrac{a}{b}, n, 0\big) \quad = \quad 0 + \frac{1}{n + \frac{1}{\frac{a}{b}}} \quad = \quad \frac{a}{b+na}$$

Original Parity	**East Join** n		**South Join** n		**West Join*** n		**North Join*** n	
	even	odd	even	odd	even	odd	even	odd
0	0	1	0	$0'$	0	$1'$	0	∞'
$0'$	$0'$	$1'$	$0'$	0	$0'$	1	$0'$	∞
1	1	0	1	∞'	1	∞	1	$0'$
$1'$	$1'$	$0'$	$1'$	∞	$1'$	∞'	$1'$	0
∞	∞	∞'	∞	$1'$	∞	$0'$	∞	1
∞'	∞'	∞	∞'	1	∞'	0	∞'	$1'$

*West and north joins must incorporate possible rotations of the NW corner.

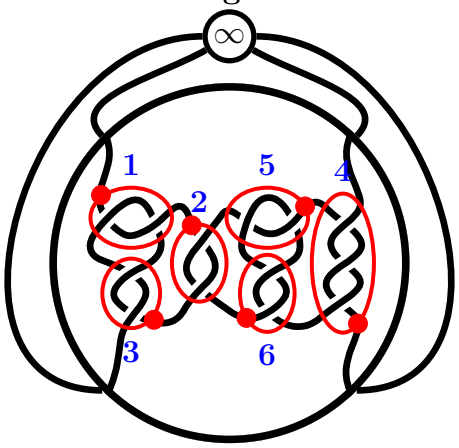
Table 4.4: Cases to determine modified oriented parity in the RPD algorithm

In addition to computing a new fraction, we must also determine how the oriented parity of the combined rational subtangle is modified by these changes. This will depend on the parity of the original rational subtangle before joining, the number and location of the twists, and whether or not this choice involves a rotation of the NW corner. Table 4.4 summarizes how the new parity can be determined based on these factors. Although the *unoriented* parity can be determined directly from the fraction using Corollary 3.49, the *oriented* parity is affected by the choices made in the decision tree of Figure 4.34. Each time a larger rational subtangle is obtained during the intermediate steps of the RPD algorithm, the corresponding oriented parity of this new subtangle follows from the cases in Table 4.4.

Similar to the algorithm to compute the IPD code, the algorithm to compute the RPD code generates a sequence of intermediate SPD codes. Each of these intermediate codes is obtained from the preceding code by joining together a rational subtangle and an integer subtangle which share an algebraic connection, provided this connection aligns with the integer twist axis described in Table 4.3. This process starts from the IPD code an iterates through each row, treating these as rational subtangles.

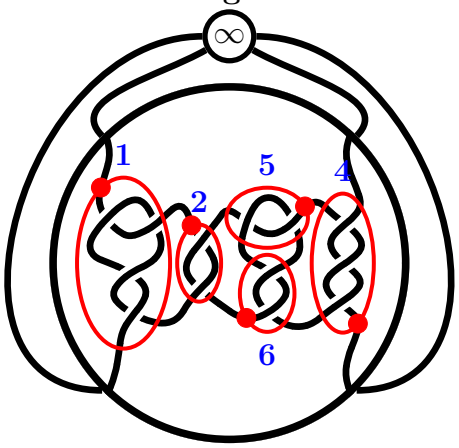
We refer to Figure 4.35 to show an example of these intermediate stages in practice. Each time two subtangles are combined into a larger rational subtangle, the new fraction is determined based on the Figure 4.34 decision tree, the connection information is updated, and the new oriented parity is determined. When no more subtangles can be combined in this way, the final RPD code is obtained by collapsing missing indices. If a decomposition of the diagram into maximal rational subtangles exists, then this matches the RPD code.

Stage 0



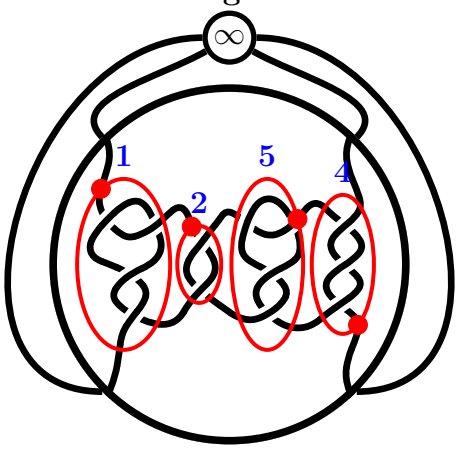
1	$\frac{2}{1}$	0 NW	2 NW	3 SW	3 SE	$(P:0)$
2	$\frac{1}{2}$	1 NE	5 SW	6 NW	3 NW	$(P:\infty')$
3	$\frac{1}{2}$	2 SW	0 SW	1 SW	1 SE	$(P:\infty)$
4	$\frac{1}{3}$	0 SE	6 SW	5 NW	0 NE	$(P:1)$
5	$-\frac{1}{2}$	4 SE	6 SE	6 NE	2 NE	$(P:\infty)$
6	$-\frac{2}{1}$	2 SE	5 SE	5 NE	4 NE	$(P:0)$

Stage 1



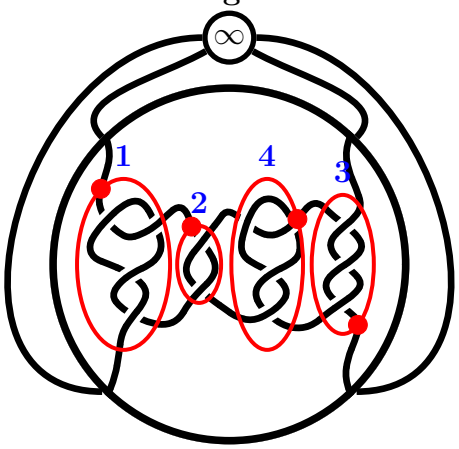
1	$\frac{2}{5}$	0 NW	2 NW	2 SW	0 SW	$(P:0)$
2	$\frac{1}{2}$	1 NE	5 SW	6 NW	1 SE	$(P:\infty')$
~~**3**~~						
4	$\frac{1}{3}$	0 SE	6 SW	5 NW	0 NE	$(P:1)$
5	$-\frac{1}{2}$	4 SE	6 SE	6 NE	2 NE	$(P:\infty)$
6	$-\frac{2}{1}$	2 SE	5 SE	5 NE	4 NE	$(P:0)$

Stage 2



1	$\frac{2}{5}$	0 NW	2 NW	2 SW	0 SW	$(P:0)$
2	$\frac{1}{2}$	1 NE	5 SW	5 SE	1 SE	$(P:\infty')$
~~**3**~~						
4	$\frac{1}{3}$	0 SE	5 NE	5 NW	0 NE	$(P:1)$
5	$-\frac{5}{2}$	4 SE	4 NE	2 SE	2 NE	$(P:\infty)$
~~**6**~~						

Stage 3



1	$\frac{2}{5}$	0 NW	2 NW	2 SW	0 SW	$(P:0)$
2	$\frac{1}{2}$	1 NE	4 SW	4 SE	1 SE	$(P:\infty')$
3	$\frac{1}{3}$	0 SE	4 NE	4 NW	0 NE	$(P:1)$
4	$-\frac{5}{2}$	3 SE	3 NE	2 SE	2 NE	$(P:\infty)$

As illustrated in the panels above, the RPD code is obtained from the IPD code through a sequence of intermediate SPD codes using larger rational subtangles, and then collapsing missing subtangle indices.

Figure 4.35: Example of converting IPD code to RPD code for the tangle $T \cong \frac{2}{5} + \frac{1}{2} + \frac{2}{5} + \frac{1}{3}$

4.3.3 Converting RPD Code to APD Code

Finally, the APD code of a diagram can be determined from the RPD code through a similar sequence of intermediate SPD codes. Each intermediate code corresponds to choosing a larger algebraic subtangle. Instead of computing a new fraction, we must compute a new algebraic construction formula for the combined subtangle. As in the RPD algorithm, the construction formula obtained in the APD algorithm is relative to a choice of NW corner. However, as the preceding examples have shown, the local choice of NW corner in a subtangle from an SPD code may appear rotated relative to another subtangle. If there exists an algebraic connection between two such subtangles, then these orientations must be made to match before the two subtangles can be combined.

When an algebraic connection between subtangles is identified while iterating through the rows in an SPD code, we adopt the convention that the subtangle of higher index is rotated to match the subtangle of lower index before combining. This choice ensures that the algebraic connection occurs to the east or south of the subtangle of lower index, and hence the NW corner of this lower index subtangle matches the NW corner of the combined

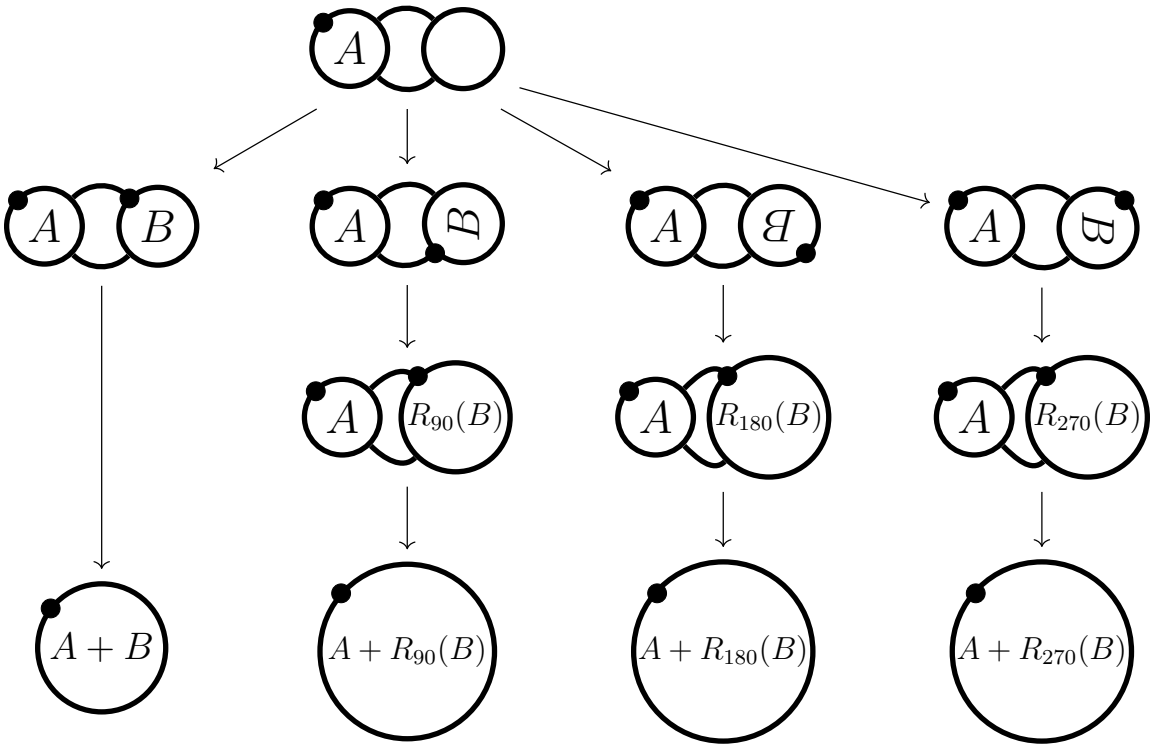


Figure 4.36: Decision tree to determine an algebraic sum in the APD algorithm

algebraic subtangle. Furthermore, the new construction formula is obtained using a sum or product of the joined subtangles' construction formulas, possibly with a rotation on the one of higher index. A sum is used when the algebraic connection occurs to the east (Figure 4.36), and a product is used when the algebraic connection occurs to the south (Figure 4.37).

Figures 4.36 and 4.37 show the decision trees used to determine these new construction formulas. If it is necessary to rotate a rational subtangle before joining, then the fraction of this rotated subtangle is determined according to Corollary 3.46, possibly using the negative reciprocal. Since the APD algorithm is applied iteratively, a rotated subtangle may itself be algebraic. In this case, the modified algebraic construction formula of the rotated subtangle is determined using the rules summarized in Theorem 2.2.

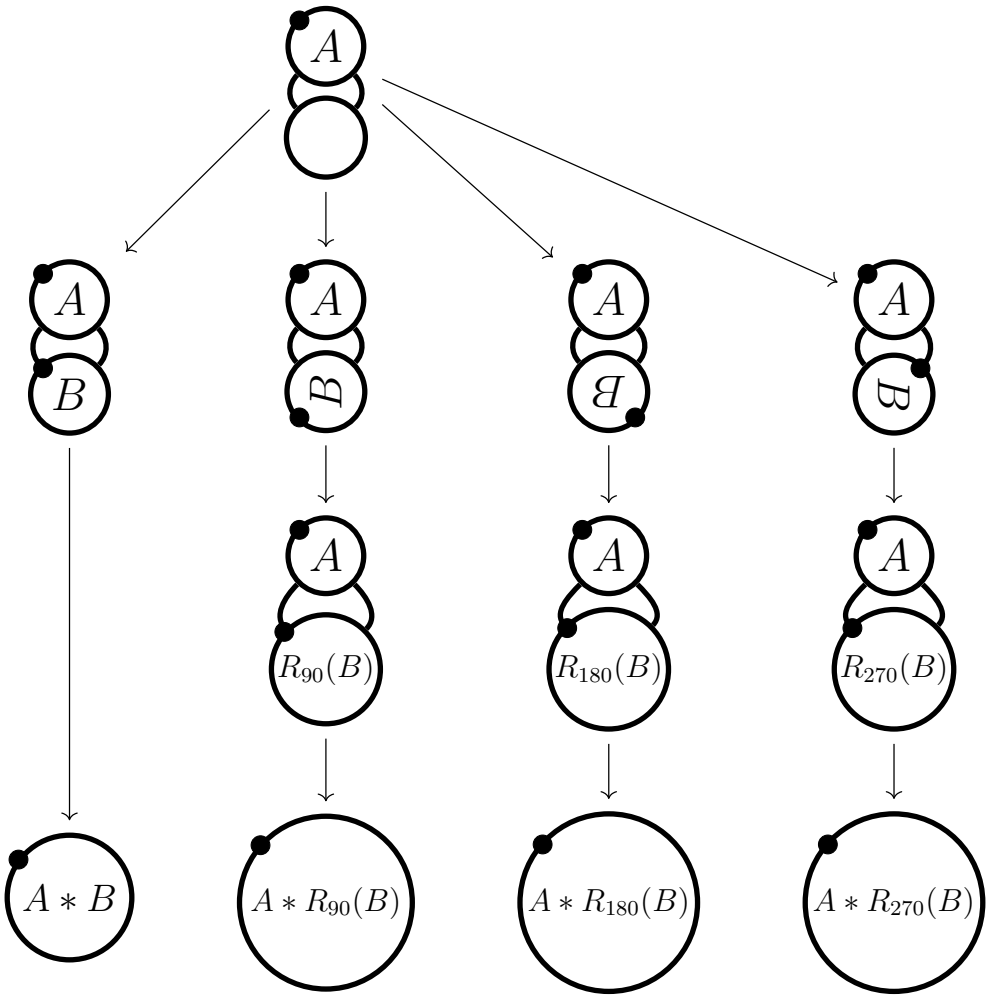


Figure 4.37: Decision tree to determine an algebraic product in the APD algorithm

In this way, the construction formulas for the algebraic subtangles can be built up recursively in the APD algorithm. Each time a larger algebraic subtangle is chosen, we must also determine its parity. Since this algebraic subtangle is the result of a sum or product, its unoriented parity can be determined using the cases from Table 2.1. However, if the higher index subtangle is rotated, we must also account for how this effects the combined parity. Table 4.5 summarizes the possibilities for the unoriented parity of the combined algebraic subtangle based these considerations for both sum and product.

Recall that a sum of two parity ∞ tangles results in a closed loop, as does a product of two parity 0 tangles. We exclude closed loops from our study of tangles, and so these combinations are not allowed; these entries are marked with $\times$ in Table 4.5. If the original tangle diagram does not contain closed loops, then these invalid possibilities will never arise in the APD algorithm and can be safely ignored. This table shows the possibilities for unoriented parity; the oriented parity is slightly more challenging since it may not be defined for a rotated subtangle.

A	B	$R_{90}(B)$	$R_{180}(B)$	$R_{270}(B)$	$A+B$	$A+R_{90}(B)$	$A+R_{180}(B)$	$A+R_{270}(B)$
0	0	∞	0	∞	0	∞	0	∞
	1	1	1	1	1	1	1	1
	∞	0	∞	0	∞	0	∞	0
1	0	∞	0	∞	1	∞	1	∞
	1	1	1	1	0	0	0	0
	∞	0	∞	0	∞	1	∞	1
∞	0	∞	0	∞	∞	$\times$	∞	$\times$
	1	1	1	1	∞	∞	∞	∞
	∞	0	∞	0	$\times$	∞	$\times$	∞

A	B	$R_{90}(B)$	$R_{180}(B)$	$R_{270}(B)$	$A*B$	$A*R_{90}(B)$	$A*R_{180}(B)$	$A*R_{270}(B)$
0	0	∞	0	∞	$\times$	0	$\times$	0
	1	1	1	1	0	0	0	0
	∞	0	∞	0	0	$\times$	0	$\times$
1	0	∞	0	∞	0	1	0	1
	1	1	1	1	∞	∞	∞	∞
	∞	0	∞	0	1	0	1	0
∞	0	∞	0	∞	0	∞	0	∞
	1	1	1	1	1	1	1	1
	∞	0	∞	0	∞	0	∞	0

Table 4.5: Table of unoriented parities for tangle sums and products in the APD algorithm

The six possible oriented parities were originally introduced in Figure 2.6. By default, the orientation is assumed to point inward from the NW corner. A tangle of unoriented parity p is said to have oriented parity p or p' depending on the direction of the second string. Given a subtangle decomposition of an oriented tangle diagram, we chose the NW corners in the subtangles to match the first pass of the string. This ensures that the directions on the strings in a subtangle always point inward from the NW corner, and hence the orientations on the subtangles induced by the oriented diagram always fall into one of these six cases. If the NW corner of the subtangle of higher index is rotated, as shown in the decision trees of Figures 4.36 and 4.37, then the oriented parity of this rotated subtangle may not be defined. Despite this, the combined algebraic subtangle will have an oriented parity since its NW corner matches that of the subtangle of lower index, which is not rotated.

Suppose that T represent a combined algebraic subtangle obtained in the APD algorithm by joining the subtangles A and B, where A has the lower index and B has the higher index.

A oriented parity	B' unoriented parity	$A + B'$ oriented parity	$A * B'$ oriented parity
0	0	0	$\times$
$0'$	0	$0'$	$\times$
0	1	1	$0'$
$0'$	1	$1'$	0
0	∞	∞ or ∞'	0
$0'$	∞	$\times$	$0'$
1	0	1	$\times$
$1'$	0	$1'$	0 or $0'$
1	1	0	∞'
$1'$	1	$0'$	∞
1	∞	∞ or ∞'	1
$1'$	∞	$\times$	$1'$
∞	0	∞	0 or $0'$
∞'	0	∞'	$\times$
∞	1	∞'	$1'$
∞'	1	∞	1
∞	∞	$\times$	∞
∞'	∞	$\times$	∞'

Let A denote the lower index subtangle, let B' denotes the rotated subtangle of higher index, and let $A+B'$ or $A * B'$ denote the combined algebraic subtangle obtained in the APD algorithm. In most cases, the oriented parity of the combined algebraic subtangle can be determined from the *oriented parity of* A and the *unoriented parity of* B', with a few exceptions that are highlighted in blue above.

Table 4.6: Table to determine the oriented parity in the APD algorithm

Let B' denote the necessary rotation of B, so that either $T \cong A + B'$ or $T \cong A * B'$. In most cases, the oriented parity of T can be determined from the *oriented parity of* A and the *unoriented parity of* B', noting the oriented parity of B' may not be defined. These cases are summarized in Table 4.6.

However, there are a few exceptions where the oriented parity of T cannot be fully identified in this way. These correspond to the entries of Table 4.6 which are highlighted in blue. In these cases, the second string of T is fully contained in B', and hence its direction cannot be inferred from the directions on the strings in A alone. However, if B' is rotated, it may be necessary to refer back to the oriented parity of B to identify this direction. This leads to many possibilities, but fortunately we need not consider every combination of cases.

In the case of $T \cong A + B'$, these exceptions only occur when B' has unoriented parity ∞ and A has oriented parity 0 or 1. Similarly, in the case of $T \cong A * B'$, these exceptions only occur when B' has unoriented parity 0 and A has oriented parity $1'$ or ∞. Table 4.7 lists only those cases which correspond to these scenarios, as well as the resulting parity of T. The entries marked with $\times$ denote cases where the directions on the strings are incompatible, and hence are not possible in an oriented diagram.

We conclude this section with one final example illustrating how the APD code of a diagram can be obtained from the RPD code (Figure 4.38). As in the preceding examples, this figure shows the sequence of intermediate SPD codes built up by identifying joined

cases where B' has unoriented parity ∞		$A + B'$ oriented parity	
B or. par.	B' rotation	If $p_A = 0$	If $p_A = 1$
∞	B	∞	$\times$
∞'	B	∞'	$\times$
0	$R_{90}(B)$	$\times$	∞'
$0'$	$R_{90}(B)$	$\times$	∞
∞	$R_{180}(B)$	∞	$\times$
∞'	$R_{180}(B)$	$\times$	∞
0	$R_{270}(B)$	$\times$	∞'
$0'$	$R_{270}(B)$	∞'	$\times$

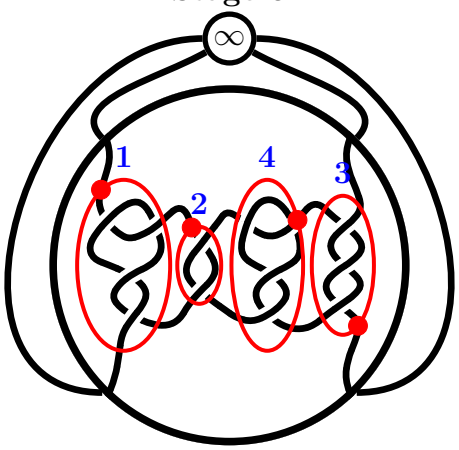
cases where B' has unoriented parity 0		$A * B'$ oriented parity	
B or. par.	B' rotation	If $p_A = 1'$	If $p_A = \infty$
∞	B	$\times$	0
∞'	B	$\times$	$0'$
0	$R_{90}(B)$	$0'$	$\times$
$0'$	$R_{90}(B)$	$\times$	$0'$
∞	$R_{180}(B)$	$\times$	0
∞'	$R_{180}(B)$	0	$\times$
0	$R_{270}(B)$	$0'$	$\times$
$0'$	$R_{270}(B)$	0	$\times$

In the special cases from Table 4.6 marked in blue, the oriented parity of $A + B'$ or $A * B'$ cannot be fully identified. The two exceptions for $A + B'$ arise when B' has unoriented parity ∞; the two exceptions for $A * B'$ arise when B' has unoriented parity 0. This table exhausts the possibilities for B which could lead to one of these scenarios, although some combinations are not possible in a diagram. The final oriented parity of $A + B'$ or $A * B'$ then depends on the oriented parity of A, denoted as p_A in the table above.

Table 4.7: Table to determine special cases of oriented parity in the APD algorithm

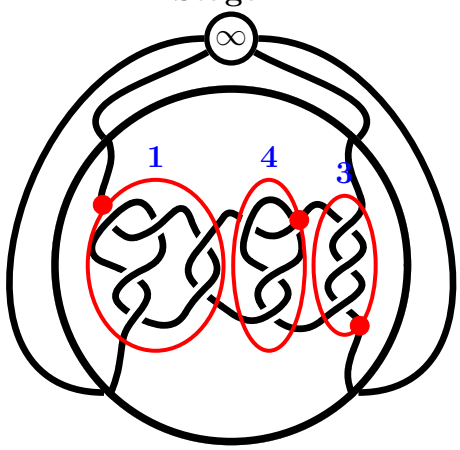
subtangles from the connection information in the code. When a larger algebraic subtangle is identified in this way, these are combined into a single row and the connection information is updated. In the absence of a diagram, the subtleties involved with identifying the modified construction formula and parity are best left to a computer.

Stage 0



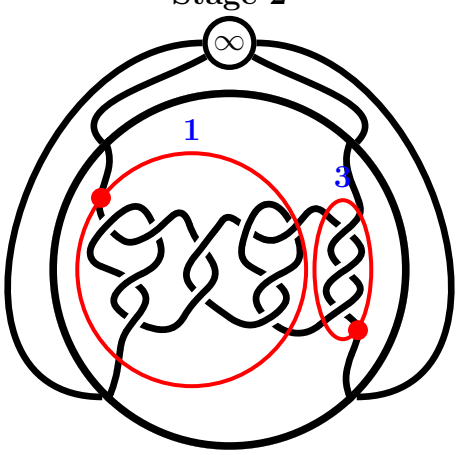
1	$\frac{2}{5}$	0 NW	2 NW	2 SW	0 SW	$(P:0)$
2	$\frac{1}{2}$	1 NE	4 SW	4 SE	1 SE	$(P:\infty')$
3	$\frac{1}{3}$	0 SE	4 NE	4 NW	0 NE	$(P:1)$
4	$-\frac{5}{2}$	3 SE	3 NE	2 SE	2 NE	$(P:\infty)$

Stage 1



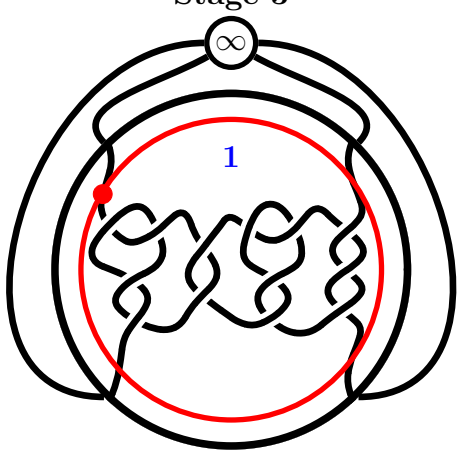
1	$\frac{2}{5}+\frac{1}{2}$	0 NW	4 SW	4 SE	0 SW	$(P:\infty')$
~~**2**~~						
3	$\frac{1}{3}$	0 SE	4 NE	4 NW	0 NE	$(P:1)$
4	$-\frac{5}{2}$	3 SE	3 NE	1 SE	1 NE	$(P:\infty)$

Stage 2



1	$\frac{2}{5}+\frac{1}{2}+\frac{2}{5}$	0 NW	3 SE	3 NE	0 SW	$(P:\infty')$
~~**2**~~						
3	$\frac{1}{3}$	0 SE	1 SE	1 NE	0 NE	$(P:1)$
~~**4**~~						

Stage 3



1	$\frac{2}{5}+\frac{1}{2}+\frac{2}{5}+\frac{1}{3}$	0 NW	0 NE	0 SE	0 SW	$(P:\infty)$
~~**2**~~						
~~**3**~~						
~~**4**~~						

The APD code is obtained from the RPD code through a sequence of intermediate SPD codes which combine rational subtangles sharing algebraic connections, rotating to match the subtangle of lower index.

Figure 4.38: Example of converting RPD code to APD code for the tangle $T \cong \frac{2}{5}+\frac{1}{2}+\frac{2}{5}+\frac{1}{3}$

5 Tangle Classification

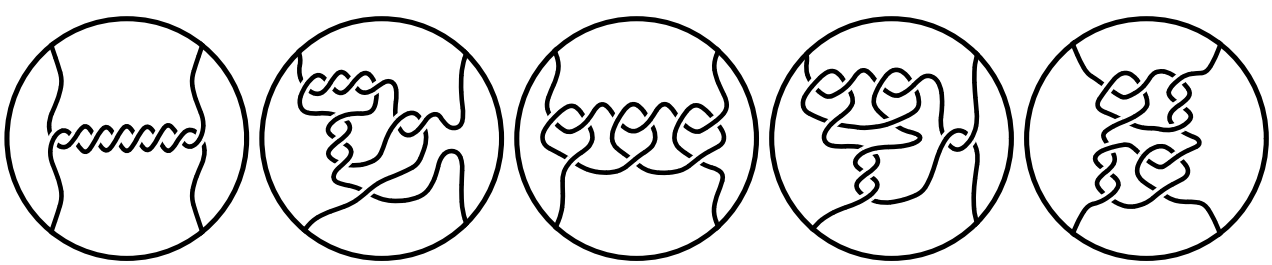
Thus far, our study has centered around rational tangles, but these represent just one class of 2-string tangle. We have encountered examples of non-rational tangles obtained by joining together rational subtangles using tangle algebra. The class of *algebraic tangles* refers to any 2-string tangle that can be obtained through some finite combination of tangle sums and products of rational subtangles. This means that algebraic tangle diagrams can be compactly described using *algebraic construction formulas*, like those shown in Figure 5.1.

Broadly speaking, 2-string tangles can be divided into *algebraic* and *non-algebraic* tangles. Algebraic tangles can be organized into a hierarchy of increasingly general types of tangle:

$$\{\text{Integer}\} \subseteq \{\text{Rational}\} \subseteq \{\text{Montesinos}\} \subseteq \{\text{Generalized Montesinos}\} \subseteq \{\text{Algebraic}\}.$$

Just as integer tangles are a special case of rational tangle, each class of tangles in this hierarchy is a special case of the next class. Figure 5.1 shows examples of tangles from each of these five classes. Unlike the canonical form for rational tangles, it is not yet known whether a unique preferred diagram exists for generic algebraic tangles. However, we will show that such diagrams do exist in special cases.

Tangles which cannot be constructed using only sums and products of rational subtangles are known as *non-algebraic tangles*. These are more challenging to describe in general, and it is an open question whether a given non-algebraic diagram is equivalent to an algebraic diagram. Despite this difficulty, both algebraic and non-algebraic tangle diagrams can be decomposed into integer, rational, and algebraic subtangles. Subtangle decompositions will be our principal tool for classifying tangle diagrams.



$\frac{8}{1}$ | $\frac{30}{43}$ | $\frac{2}{3}+\frac{2}{3}+\frac{2}{3}$ | $(\frac{2}{3}+\frac{2}{3})\circ(3,2)$ | $(\frac{2}{3}+\frac{1}{3})*(\frac{1}{3}+\frac{2}{3})$

(a) Integer **(b)** Rational **(c)** Montesinos **(d)** Gen. Montesinos **(e)** Algebraic

Figure 5.1: Examples of types of tangles in the hierarchy of algebraic tangles

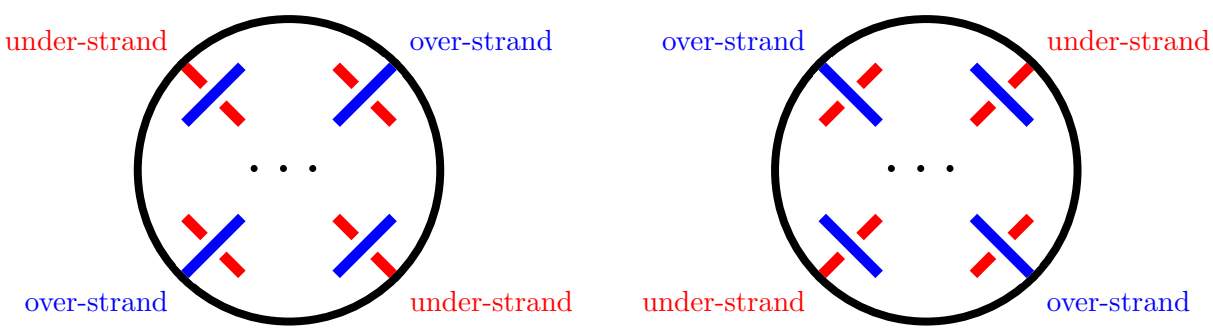
5.1 Alternating Algebraic Tangle Diagrams

We have seen how tangle diagrams can be identified as integer, rational, or algebraic based on a decomposition into maximal subtangles of a certain type. A decomposition can also be used to determine when a non-rational algebraic tangle diagram is Montesinos or generalized Montesinos, which are two more special types of algebraic tangle that we will study in detail. Although it is still an open question for generic algebraic tangles, the special cases of *integer*, *rational*, *Montesinos*, and *generalized Montesinos* all admit a preferred diagram.

One of the interesting features of these preferred diagrams is their alternating behavior, which can be determined from a construction formula. For alternating tangle diagrams more generally, there are two possibilities based on what happens at the endpoints. Before exploring the special types of algebraic tangle, we begin by studying some results concerning alternating diagrams.

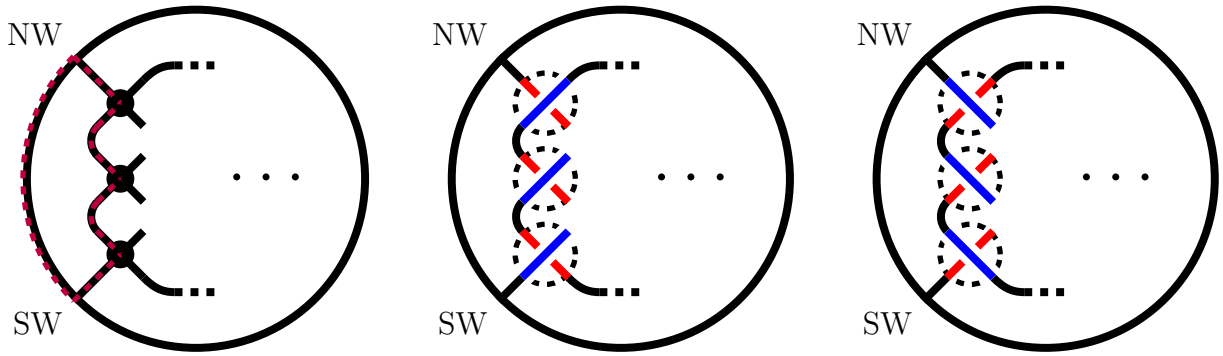
5.1.1 Under-alternating and Over-alternating

In the proof of Lemma 3.30, when we showed that rational tangles in canonical form are alternating, we relied on generalizing the fact that the strings at the endpoints in a 1-crossing tangle alternate between over-strands or under-strands (Figure 3.33). In fact, this property of the endpoints holds for any alternating tangle diagram, even those which are non-algebraic. This leads to two types of alternating diagram for tangles based on what kind of strand connects to each endpoint (Figure 5.2).



(a) Under-alternating diagram **(b)** Over-alternating diagram

Figure 5.2: The two possible endpoint behaviors for an alternating tangle diagram



(a) Polygonal region **(b)** Under-alternating **(c)** Over-alternating

Figure 5.3: Crossings along the polygonal path from NW to SW in an alternating diagram

Lemma 5.1 (Endpoint behavior in an alternating 2-string tangle diagram)**:**
If T is an *alternating diagram* for a 2-string tangle, then the strings connected to the endpoints of T must alternate between under-strands and over-strands while traveling around the boundary circle. There are two possibilities, as illustrated in Figure 5.2.

Proof:

Let T be an alternating 2-string tangle diagram. The claim holds vacuously if there are no crossings, so suppose that T has at least one crossing. By replacing the crossings in this diagram with vertices, observe that we create a polygonal region between NW and SW endpoints of the tangle (Figure 5.3a). Consider what happens as we travel along the polygonal path from the NW endpoint to the SW endpoint.

At each vertex, we can mark the locations of the over-strands and under-strands as shown in Figure 5.3. Suppose that the NW endpoint connects to an under-strand. Since T is alternating, this forces the position of the over-strands and under-strand at each vertex along this polygonal path. No matter how many vertices we pass through, when we eventually reach the SW corner, the connected string must correspond to an over-strand (Figure 5.3b). If the NW endpoint had connected to an over-strand instead, then this would force the string connected to the SW corner to be an under-strand (Figure 5.3c).

In either case, the strings connected to the NW and SW endpoints must correspond to opposite types of strand. By a parallel argument, we can draw similar conclusions for the NE and SE endpoints by traveling along polygonal paths in the same way. Thus, the strings connected to the endpoints must alternate between over-strands and under-strands while moving along the boundary circle of T. □

If a tangle diagram is alternating, then the behavior of the string at the NW endpoint forces the behavior of the strings at the other endpoints. As shown in Figure 5.2, this can happen in two ways since the string connected to the NW endpoint could either be an under-strand or an over-strand. To distinguish these two types of alternating tangle diagrams, we will say that a diagram is *under-alternating* if the string at the NW corner is an under-strand (5.2a) and *over-alternating* if the string at the NW corner is an over-strand (5.2b).

It is interesting to note that the argument used in the proof of Lemma 5.1 also applies to tangles with more than two strings, and so these two definitions can be applied to tangle diagrams more generally. In the context of 2-string tangle diagrams specifically, this behavior at the endpoints in an alternating diagram is preserved under tangle sums and products.

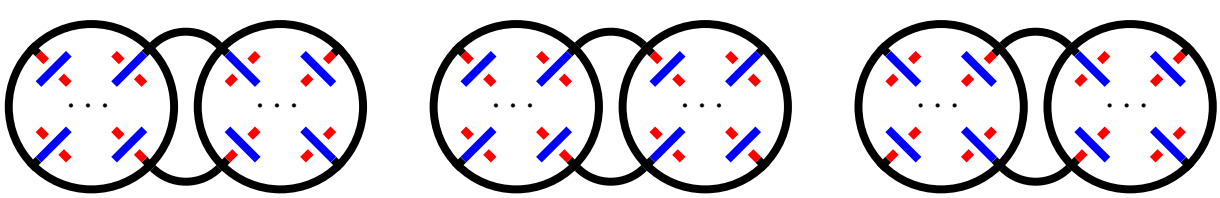
Corollary 5.2 (Algebraic combinations of alternating tangle diagrams)**:**
If L_1 and L_2 are alternating tangle diagrams, then:

a. $L_1 + L_2$ and $L_1 * L_2$ are under-alternating if and only if L_1 and L_2 are under-alternating;

b. $L_1 + L_2$ and $L_1 * L_2$ are over-alternating if and only if L_1 and L_2 are over-alternating.

Proof:

Let L_1 and L_2 be alternating tangle diagrams. We refer to Figure 5.4 for a visual proof using tangle sums. The claim for tangle products follows by a parallel argument. In general, a sum or product of alternating tangle diagrams will continue to be alternating if and only if under-strands are connected to over-strands. Hence, an algebraic combination of an under-alternating diagram with an over-alternating diagram will fail to be alternating (Figure 5.4a). On the other hand, a sum of two under-alternating 2-string tangle diagrams is also under-alternating (Figure 5.4b), and likewise for a sum of two over-alternating diagrams (Figure 5.4c). Similar facts hold for tangle products. □



(a) Non-alternating sum **(b)** Under-alternating sum **(c)** Over-alternating sum

Figure 5.4: Tangle sums using alternating tangle diagrams

In other words, Corollary 5.2 establishes that the type of alternating tangle diagram, under-alternating or over-alternating, is preserved under tangle algebra as long as the two connected subtangle diagrams have the same type. Furthermore, since algebraic tangles are defined using sums and products, we can extend this result more generally to arbitrary algebraic constructions. Since these two types of alternating tangle diagram depend on the choice of NW corner, this fact only applies when using the standard choice of NW corner on each subtangle in an algebraic subtangle decomposition.

Theorem 5.3 (Under-alternating and over-alternating algebraic tangle diagrams)**:**
If T is an algebraic tangle diagram, then relative to the standard choice of NW corner,

a. T is under-alternating if and only if every subtangle of T is under-alternating, and

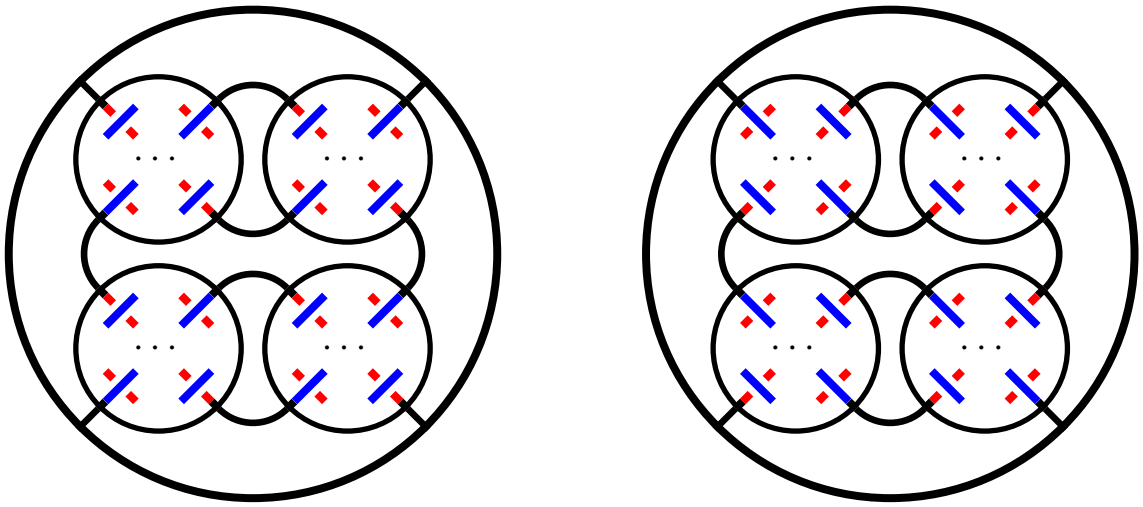
b. T is over-alternating if and only if every subtangle of T is over-alternating.

Proof:

Let T be an algebraic tangle diagram. The claim holds vacuously if the diagram contains no crossings, so suppose that T has at least one crossing. Since T is a subtangle of itself, the backwards direction of the claim is immediate. For the forward direction, note that T cannot be alternating if it contains a non-alternating subtangle. Hence, if T is alternating, then every subtangle of T must either be under-alternating or over-alternating. We must show that each subtangle in T matches the alternating behavior of T.

Let T be an alternating algebraic tangle diagram and suppose that L is an under-alternating subtangle of T. We will show that T must also be an under-alternating diagram. Let D be some decomposition of T into subtangles which contains L. Note that such a decomposition exists since we could choose 1-crossing subtangles for every crossing not contained in L. Every subtangle in D must be alternating, but we do not yet know whether they all have the same type of alternating behavior.

Since T is algebraic, every subtangle of T is also algebraic (Lemma 4.4). Hence, there exists some algebraic construction formula for T using the subtangles in D. Since the subtangle L is included in this construction formula, every subtangle connected to L by a sum or product must also be under-alternating by Corollary 5.2. By choosing increasingly long portions of this construction formula which contain L, we may reapply Corollary 5.2 to identify increasingly large under-alternating subtangles in T constructed from a subset of the subtangles in D (Figure 5.5). Eventually, this will exhaust all of the subtangles in D, and we establish that T is itself under-alternating.



(a) Under-alternating algebraic construction **(b)** Over-alternating algebraic construction

Figure 5.5: Subtangles in an alternating algebraic tangle diagram must be consistent

Therefore, if T is an alternating algebraic tangle diagram which contains an under-alternating subtangle, we have shown that T must be under-alternating (Figure 5.5a). A similar fact holds if T contains an over-alternating subtangle (Figure 5.5b). Since the subtangle L was taken arbitrarily, it follows that we would have a contradiction if T contains both an under-alternating and an over-alternating subtangle. Thus, if T is under-alternating, then every subtangle of T must be under-alternating, and a similar fact holds if T is over-alternating. This completes the proof.

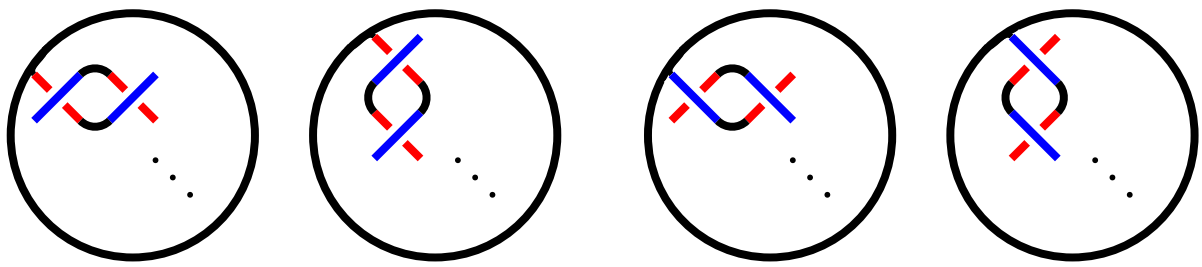
□

When we previously examined alternating canonical diagrams for rational tangles, we did not make a distinction between under-alternating and over-alternating. However, the type of alternating diagram can be inferred from the sign of the rational tangle's corresponding fraction (Figure 5.6).

Corollary 5.4 (Under-alternating and over-alternating rational tangle diagrams)**:**
If $T \cong \frac{p}{q}$ is a rational tangle diagram in canonical form, then T is alternating, where

a. T is under-alternating if and only if $\frac{p}{q} \geq 0$, and

b. T is over-alternating if and only if $\frac{p}{q} \leq 0$.



(a) Under-alternating if $\frac{p}{q} > 0$ **(b)** Over-alternating if $\frac{p}{q} < 0$

Figure 5.6: Alternating canonical rational tangle diagrams determined by fraction sign

Proof:

Let $T \cong \frac{p}{q}$ be a rational tangle diagram in canonical form. Recall that canonical diagrams for rational tangles are alternating by Lemma 3.30, so we just need to consider when such a diagram is under-alternating or over-alternating. If $T \cong \frac{0}{1}$ or $T \cong \frac{1}{0}$ is a diagram for a trivial tangle, then T is both under-alternating and over-alternating and hence satisfies the claim vacuously.

Suppose that $T \cong \frac{p}{q}$ is a canonical rational tangle diagram with at least one crossing, so $0 < \left|\frac{p}{q}\right| < \infty$. By Corollary 3.47, if $\frac{p}{q} > 0$, then the canonical diagram for T uses only positive twists. Hence, the slope of the over-strand on the very first twist in T is positive, which means that the under-strand connects to the NW corner (Figure 5.6a). Therefore, T is under-alternating. Also by Corollary 3.47, if $\frac{p}{q} < 0$, then the canonical diagram for T uses only negative crossings. Hence, T is over-alternating by a parallel argument (Figure 5.6b). This establishes one direction of the claim, and the other direction follows immediately by the contrapositive, completing the proof.

□

Theorem 5.3 and Corollary 5.4 can be combined to draw some conclusions about the rational subtangles in an alternating algebraic tangle diagram. If the rational subtangles in some decomposition of an algebraic tangle diagram are assumed to be in canonical form, then the diagram will only be alternating if the rational subtangles have the same sign, allowing for possible trivial tangles. This sign also determines whether the diagram is under-alternating or over-alternating. Unlike rational tangles, generic algebraic tangles need not admit an alternating diagram. However, when they do, we may leverage the generalized Tait Flyping Conjecture to work with them.

Corollary 5.5 (Rational subtangles in an alternating algebraic tangle diagram)**:**
If an algebraic tangle diagram T is decomposed into rational subtangles $\frac{p_1}{q_1}, \cdots, \frac{p_n}{q_n}$ in canonical form, then the alternating behavior of T depends on the sign of these fractions.

a. T is under-alternating if and only if $\frac{p_i}{q_i} \geq 0$ for each rational subtangle.

b. T is over-alternating if and only if $\frac{p_i}{q_i} \leq 0$ for each rational subtangle.

c. T is non-alternating if and only if there exist subtangles with $\frac{p_i}{q_i} < 0$ and $\frac{p_j}{q_j} > 0$.

Proof:

Let T be an algebraic tangle diagram. Hence, T is defined by some algebraic construction formula from rational subtangles $\frac{p_1}{q_1}, \cdots, \frac{p_n}{q_n}$. Suppose without loss of generality that each of these subtangles is in canonical form. T is under-alternating if and only if each subtangle $\frac{p_i}{q_i}$ is under-alternating (Theorem 5.3). A rational subtangle $\frac{p_i}{q_i}$ is under-alternating if and only if $\frac{p_i}{q_i} \geq 0$ (Corollary 5.4). Therefore, T is under-alternating if and only if $\frac{p_i}{q_i} \geq 0$ for each of these n rational subtangles. A similar fact holds if T is over-alternating. The third part of the claim follows immediately from these two facts.

□

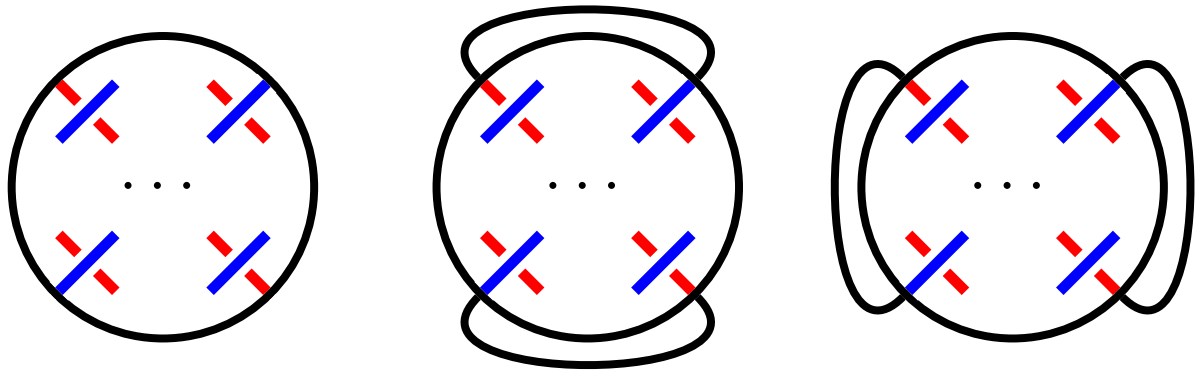
Finally, we conclude with an observation connecting alternating tangle diagrams to alternating diagrams for knots and links. Regardless of whether a tangle diagram is under-alternating or over-alternating, the behavior of the strings at the endpoints guarantees that the numerator or denominator closure is a diagram for an alternating knot or link.

Corollary 5.6 (Closure of an alternating tangle diagram is alternating)**:**
If T is an alternating tangle diagram, then $N(T)$ and $D(T)$ are also alternating diagrams.

Proof:

If T is an alternating 2-string tangle diagram, then the strings connected to the endpoints alternate between over-strands and under-strands as described in Lemma 5.1. Recall that the numerator closure $N(T)$ is defined by capping off the north and south endpoints of the diagram whereas the denominator closure $D(T)$ is defined by capping off the east and west endpoints. Whether these yield a diagram for a knot or a link depends on T, but we can see immediately that the connections introduced in these closures attach an over-strand with an under-strand (Figure 5.7). Therefore, the resulting diagram is still alternating.

□

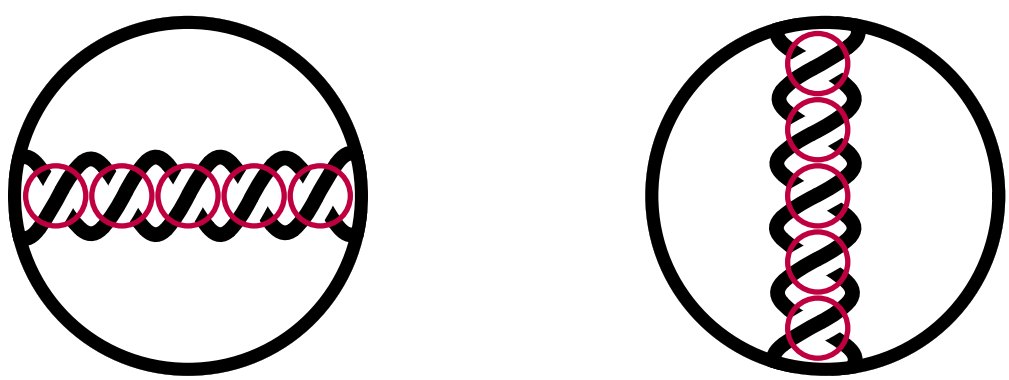


(a) Under-alternating diagram T **(b)** Numerator closure $N(T)$ **(c)** Denominator closure $D(T)$

Figure 5.7: Closures of an under-alternating tangle diagram which are alternating

5.1.2 Integer and Rational Tangles

Now we return to the examination of those special cases of algebraic tangle which are fully classified. We have already studied integer and rational tangles at length in Section 3, but we mention them here again for completion. *Integer tangles*, the simplest class of algebraic tangle, refer to any sequence of horizontal or vertical twists. These are distinguished as *horizontal* and *vertical* integer tangles, respectively. In terms of subtangle decompositions, integer tangles can be decomposed as a sum or product of 1-crossing subtangles, either $+1$ or -1 (Figure 5.8).



(a) $\frac{5}{1} \cong 1+1+1+1+1$ **(b)** $\frac{1}{5} \cong 1*1*1*1*1$

Figure 5.8: Horizontal and vertical integer tangles decomposed into 1-crossing subtangles

Theorem 5.7 (Integer decomposition into 1-crossing subtangles)**:**
A reduced integer tangle diagram T with n crossings can be uniquely decomposed into n 1-crossing subtangles of the same sign satisfying one of the following.

a. T is a *horizontal integer tangle* if and only if $T \cong \frac{\pm n}{1} \cong 0 + \pm 1 + \cdots + \pm 1$.

b. T is a *vertical integer tangle* if and only if $T \cong \frac{1}{\pm n} \cong \infty * \pm 1 * \cdots * \pm 1$.

Proof:

An integer tangle is constructed from two untwisted strings by applying finitely many horizontal twists or finitely many vertical twists. Recall that a horizontal twist is equivalent to a sum with the $+1$ or -1 tangle and that these denote positive or negative twists, respectively. Likewise, a vertical twist is equivalent to a product with the $+1$ or -1 tangle. Since adjacent twists of opposite sign undo each other by a Reidemeister II simplification, an integer tangle always reduces to some number of twists of the same sign. In the horizontal case, we start twisting from the parity 0 trivial tangle, and in the vertical case we start from the parity ∞ trivial tangle.

As rational tangles, recall that the fraction of a horizontal integer tangle is $T \cong \frac{n}{1}$ and the fraction of a vertical integer tangle is $T \cong \frac{1}{n}$ (Corollary 3.45), where $|n|$ matches the number of crossings and the sign of n matches the sign of the twists. Therefore, if T is a horizontal integer tangle, then $T \cong 0 + \pm 1 + \cdots + \pm 1$, where there are $|n|$ summands, and if T is a vertical integer tangle, then $T \cong \infty * \pm 1 * \cdots * \pm 1$, where there are $|n|$ products. This establishes the forward direction of the two possibilities in the claim; the reverse direction is immediate from the definition of an integer tangle.

Thus, choosing to isolate each of these $|n|$ twists with a subtangle defines the 1-crossing subtangle decomposition $D_{\min}$ for this diagram (Figure 5.8). By Lemma 4.6, this decomposition is unique. This completes the proof.

□

The two trivial tangles, 0 and ∞, satisfy Theorem 5.7 using an empty sum or an empty product. Hence, these are considered horizontal and vertical integer tangles, respectively. In this case, the 1-crossing subtangle decomposition is taken to be empty: $D_{\min} = \emptyset$. Furthermore, the 1-crossing tangles $+1$ and -1 are considered both horizontal and vertical integer tangles since $0 + \pm 1 \cong \pm 1 \cong \infty * \pm 1$. As a special case of rational tangle, recall that the fraction of an integer tangle is obtained from Conway's bijection (Theorem 3.44).

We previously considered a few equivalent definitions for rational tangle, and the one which lends itself most readily to subtangle decompositions is the construction from integer tangles. Recall that a 2-string tangle is *rational* if can be constructed through any combination of *sums with horizontal integer tangles* and *products with vertical integer tangles*. Hence, an integer tangle is automatically rational with a single subtangle term.

Lemma 5.8 (Rational decomposition into canonical integer subtangles)**:**
A canonical diagram for a rational tangle $T \cong (n_1, \cdots, n_r)$ can be uniquely decomposed into an odd number of integer subtangles $\frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}$ of constant sign such that

$$T \cong \frac{n_1}{1} * \frac{1}{n_2} + \frac{n_3}{1} * \cdots + \frac{n_r}{1}.$$

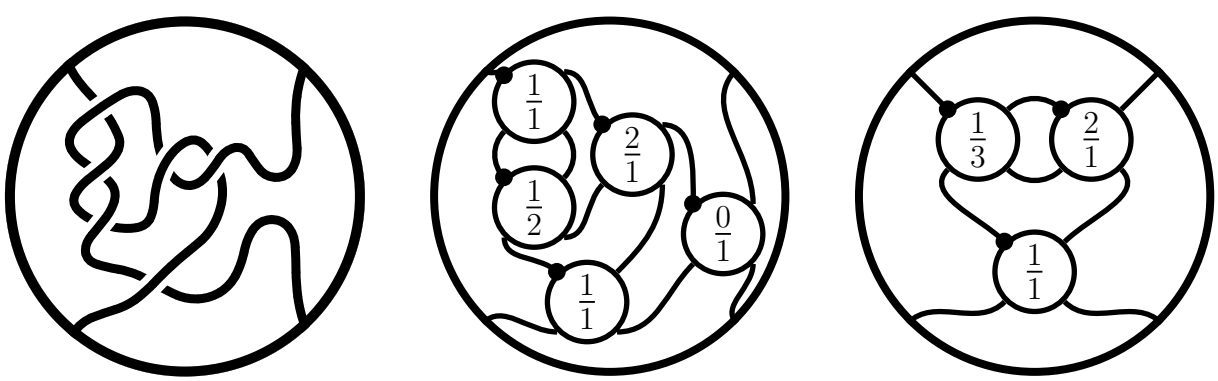
Proof:

If T is a rational tangle diagram in canonical form, then its construction is defined by a canonical twist vector of odd length $T \cong (n_1, \cdots, n_r)$, where each entry has constant sign except possibly $n_r = 0$ (Lemma 3.24). By Corollary 3.6, the twist vector construction formula is

$$T \cong \frac{n_1}{1} * \frac{1}{n_2} + \frac{n_3}{1} * \cdots + \frac{n_r}{1}.$$

The canonical diagram is unique by Theorem 3.31, which completes the proof.

□



(a) Canon. rational tangle diagram $T \cong (1, 2, 2, 1, 0) \cong \frac{7}{10}$ **(b)** Canon. integer decomposition $T \cong \frac{1}{1} * \frac{1}{2} + \frac{2}{1} * \frac{1}{1} + \frac{0}{1}$ **(c)** Maximal integer decomposition $T \cong \frac{1}{3} + \frac{2}{1} * \frac{1}{1}$

Figure 5.9: Integer subtangle decompositions for a canonical rational tangle diagram

Often times, the subtangles obtained by Lemma 5.8 will define a decomposition of a canonical rational tangle diagram into maximal integer subtangles, but not always. Since the last entry in a twist vector always denotes horizontal twisting, $n_r = 0$ when the last nonzero twist is vertical. In this case, $\frac{n_r}{1}$ corresponds to a trivial tangle, which we generally exclude from subtangle decompositions. Furthermore, $\frac{n_1}{1}$ may not be maximal when $|n_1| = 1$ in a canonical twist vector with multiple twists. Figure 5.9 illustrates these two exceptions. When these do not occur, the *canonical integer subtangle decomposition* obtained from a canonical twist vector will correspond to the decomposition into maximal integer subtangles.

Theorem 5.9 (Rational decomposition into maximal integer subtangles)**:**
A canonical diagram for a nontrivial rational tangle $T \cong (n_1, \cdots, n_r)$ admits a unique decomposition into maximal integer subtangles $D_{\text{int}} = \left\{\frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}\right\}$ with the following two exceptions.

a. If $|n_1| = 1$ for $r > 1$, then $\frac{n_1}{1} * \frac{1}{n_2}$ is a maximal integer subtangle replacing $\frac{n_1}{1}$ and $\frac{1}{n_2}$.

b. If $n_r = 0$, then $\frac{n_r}{1} \cong \frac{0}{1}$ is excluded from D_{int}.

Proof:

Let $T = (n_1, \cdots, n_r)$ be a diagram for a rational tangle in canonical form defined by a canonical twist vector. We know by Lemma 5.8 that $T \cong \frac{n_1}{1} * \frac{1}{n_2} + \frac{n_3}{1} * \cdots + \frac{n_r}{1}$. Hence, there exists a decomposition of T into integer subtangles $D = \left\{\frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}\right\}$, provided that $n_r \neq 0$. By Lemma 4.8, there exists a unique decomposition of T into maximal integer subtangles D_{int}. By maximality, it follows that $D \leq D_{\text{int}}$. To prove the claim, we must show that $D = D_{\text{int}}$, with possible corrections for the two exceptions.

We will address these exceptions first. Consider what happens if $|n_1| = 1$. If $r = 1$, then $T \cong \pm 1$ is a 1-crossing subtangle, and hence $D = \{\frac{\pm 1}{1}\}$ is the only subtangle decomposition possible. If $r > 1$, then there exists some non-trivial subtangle $\frac{1}{n_2}$. Observe that $\frac{n_1}{1} * \frac{1}{n_2} \cong \frac{1}{1+n_2}$ is a vertical integer tangle containing both $\frac{n_1}{1}$ and $\frac{1}{n_2}$ as subtangles, as in Figure 5.9b. In fact, this corresponds to the equivalent twist vector construction $T \cong (1+n_2, \cdots, n_r)$, which is considered non-canonical since it has even length. However, as an integer subtangle, we can show that $\frac{n_1}{1} * \frac{1}{n_2}$ is maximal. If this happens, redefine $D = \left\{\frac{n_1}{1} * \frac{1}{n_2}, \cdots, \frac{n_r}{1}\right\}$.

Now consider what happens in the second exception. If $n_r = 0$, then $\frac{n_r}{1} \cong 0$ is a trivial tangle and hence extraneous to the subtangle decomposition. If we remove this subtangle, then $T \cong \frac{n_1}{1} * \frac{1}{n_2} + \frac{n_3}{1} * \cdots * \frac{1}{n_{r-1}}$ is still a decomposition of T into integer subtangles. If

the first exception is not also true, then redefine $D = \left\{\frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{1}{n_{r-1}}\right\}$. If both exceptions occur, then we may include both modifications to redfine $D = \left\{\frac{n_1}{1} * \frac{1}{n_2}, \cdots, \frac{1}{n_{r-1}}\right\}$. Possibly with these corrections, let D be the decomposition of T into integer subtangles defined in this way. We must show that each subtangle in D is maximal.

Let $S_i \in D$ be any integer subtangle, horizontal or vertical, corresponding to the twist vector entry n_i. If S_i is not maximal, then it is properly contained in some larger integer subtangle. Thanks to the alternately horizontal and vertical twisting used to construct T, note that no two adjacent integer subtangles from D can be grouped into a single integer subtangle, with the possible exception of $\frac{n_1}{1}$ that we have already accounted for. Thus, the only way that S_i can fail to be maximal is if $n_{i-1} = 0$ or $n_{i+1} = 0$, in which case it could be combined with the next integer subtangle of the same type. However, no entry besides n_r can be zero in a canonical twist vector (Lemma 3.24). Thus, S_i must be a maximal integer subtangle, and we may therefore conclude that $D = D_{\text{int}}$ is the unique decomposition of T into maximal integer subtangles.

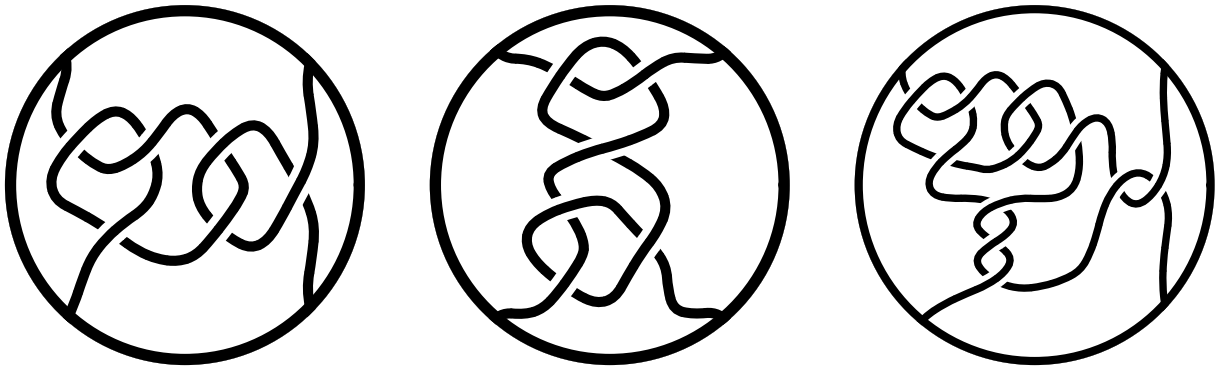
□

5.2 Montesinos Tangles

The next special class of tangle we study is that of the Montesinos tangles, named for their connection with Montesinos knots and links [29, 39, 40]. A *Montesinos tangle* refers to any tangle constructed as a finite sum of rational subtangles, excluding the parity ∞ trivial tangle. Tangles constructed using a finite product of rational subtangles rather than a sum behave similarly, and so we will be slightly more general and refer to both possibilities as Montesinos tangles (Figure 5.10). Unlike generic algebraic tangles, only one type of operation is allowed in a Montesinos construction, either sum or product. A numerator or denominator closure of a Montesinos tangle yields a Montesinos knot or link.

5.2.1 Horizontal and Vertical Montesinos Constructions

The construction of a Montesinos tangle is similar to how we decomposed integer tangles, except using rational subtangles rather than 1-crossing subtangles. We will distinguish these as *horizontal* (5.10a) and *vertical* (5.10b), respectively, although we will see that these constructions are equivalent as generalized Montesinos tangles. A *generalized Montesinos tangle* is a Montesinos tangle which also includes endpoint twisting (5.10c).

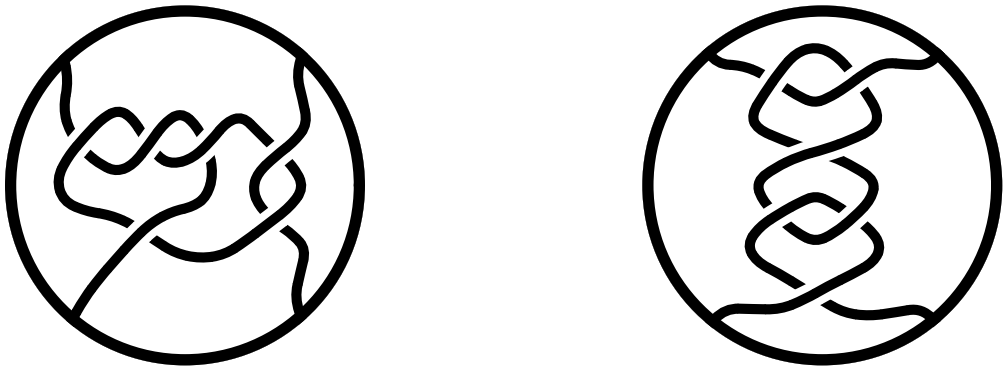


(a) Horizontal: $T \cong \frac{2}{3} + \frac{3}{2}$ **(b)** Vertical: $T \cong \frac{2}{3} * \frac{3}{2}$ **(c)** Generalized: $T \cong (\frac{2}{3} * \frac{3}{2}) \circ (3, 2)$

Figure 5.10: Examples of horizontal, vertical, and generalized Montesinos tangle diagrams

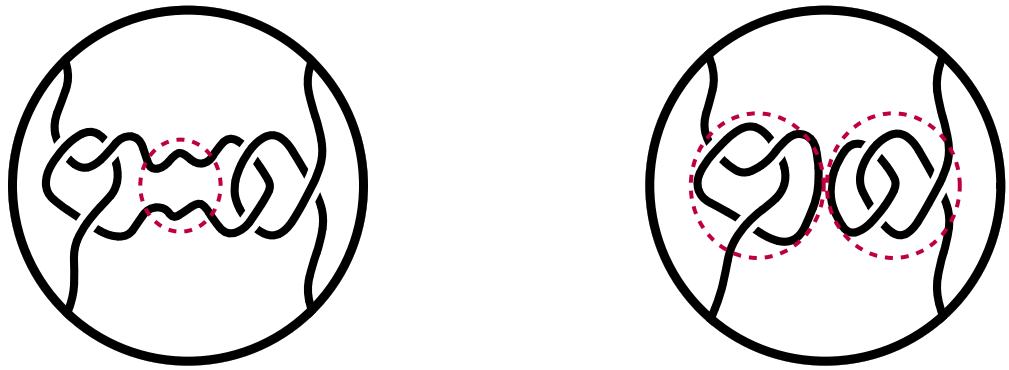
When denoting the construction for these types of tangles, we will usually write $T \cong L_1 + \cdots + L_m$ or $T \cong L_1 * \cdots * L_m$ for a Montesinos tangle, where each L_i is a rational subtangle. For a generalized Montesinos tangle, we use a twist vector to describe the endpoint twisting, which has the form $T \cong M \circ (n_1, \cdots, n_r)$, where M is a Montesinos tangle, either horizontal or vertical. While endpoint twisting in a generalized Montesinos tangle can occur on any side, these twists can always be shuffled to the right and bottom using flypes, just as we do with rational tangles (Lemma 3.5).

As a sum or product of tangles, note that the restrictions on parity mentioned in Table 2.1 still apply. In particular, there can be at most one parity ∞ summand in a horizontal Montesinos tangle and at most one parity 0 product in a vertical Montesinos tangle. Oth-



(a) Two parity ∞ summands: $T \cong \frac{3}{4} + \frac{1}{2}$ **(b)** Two parity 0 products: $T \cong \frac{2}{3} * \frac{2}{3}$

Figure 5.11: Examples of Montesinos non-tangles with closed loops



(a) Removable subtangle: $T \cong \frac{2}{3} + \frac{0}{1} + \frac{3}{2} \cong \frac{2}{3} + \frac{3}{2}$

(b) Local knots: $T \cong \frac{2}{3} + \frac{1}{0} + \frac{3}{2}$

Figure 5.12: Examples of trivial tangles in a Montesinos construction

erwise, the resulting structure may contain a closed loop and hence fail to be a 2-string tangle (Figure 5.11). Furthermore, while trivial tangles can be used to define a Montesinos tangle, they are either unnecessary or contradictory and can thus be omitted without loss of generality from the construction of any nontrivial Montesinos tangle (Figure 5.12).

Lemma 5.10 (Omitting trivial tangles from a Montesinos construction)**:**
If T is a nontrivial Montesinos tangle constructed from a sum or product of rational subtangles $L_1, \cdots, L_m$, then we may assume that each L_i is also nontrivial.

Proof:

Let $T \cong L_1 + \cdots + L_m$ be a nontrivial horizontal Montesinos tangle. Figure 5.12 illustrates what happens if any of these summands is a trivial tangle. If there exists a rational subtangle $L_i \cong \frac{0}{1}$ which is equivalent to the parity 0 trivial tangle, then this tangle can be excluded from the construction: $T \cong L_1 + \cdots + L_{i-1} + L_{i+1} + \cdots + L_m$ (5.12a). Thus, we may assume without loss of generality that no rational subtangle L_i is equivalent to the parity 0 trivial tangle. If there exists a rational subtangle $L_i \cong \frac{1}{0}$ which is equivalent to the parity ∞ trivial tangle, then $T \cong (L_1 + \cdots + L_{i-1}) + L_i + (L_{i+1} + \cdots + L_m)$ splits into two local knots (5.12b). Since we assumed that local knots do not exist (Lemma 4.1), it follows that no rational subtangle L_i is equivalent to the parity ∞ trivial tangle either.

Thus, we may conclude that each rational subtangle used to construct a horizontal Montesinos tangle $T \cong L_1 + \cdots + L_m$ is nontrivial. By switching the roles of the parity 0 and parity ∞ trivial tangles, a similar result follows for a vertical Montesinos tangle $T \cong L_1 * \cdots * L_m$ using a parallel argument. $\square$

Much like the canonical diagram for a rational tangle, there also exist preferred diagrams for Montesinos and generalized Montesinos tangles. This form is slightly different for horizontal and vertical tangles, but in the special case of a Montesinos tangle which is also rational, one of these preferred diagrams will match the canonical form of a rational tangle. Thus, we will use *canonical diagram* to refer to the preferred diagrams in general for rational, Montesinos, and generalized Montesinos tangles.

Moon and Darcy proved that generalized Montesinos tangles are uniquely classified by this canonical form [31], and our goal in this subsection is to establish those results concerning Montesinos tangles that will be necessary to state this theorem.

Lemma 5.11 (Rational sum and product of rational tangles)**:**
If L_1 and L_2 are two rational tangles, then the following are satisfied.

a. $L_1 + L_2$ is a rational tangle if and only if L_1 or L_2 is a horizontal integer tangle.

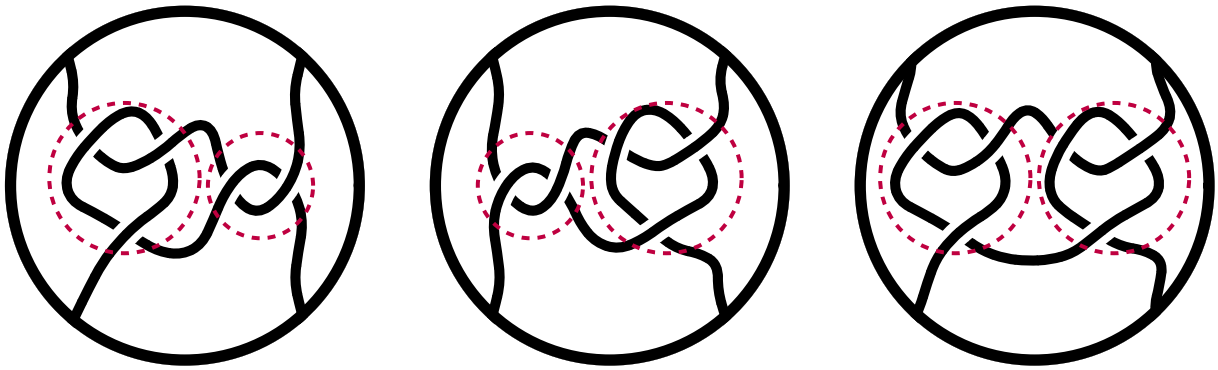
b. $L_1 * L_2$ is a rational tangle if and only if L_1 or L_2 is a vertical integer tangle.

Proof:

Let L_1 and L_2 be two rational tangles. We may assume without loss of generality that these are in canonical form. Recall that a tangle is rational if it is constructed using any combination of sums with horizontal integer tangles and products with vertical integer tangles. Hence, if neither summand in $L_1 + L_2$ is a horizontal integer tangle, then the resulting tangle $L_1 + L_2$ is not rational. Conversely, if either summand is a horizontal integer tangle, then $L_1 + L_2$ is a sum of a horizontal integer tangle with a rational tangle and thus still rational. This proves the first part of the claim, and the second part follows by a parallel argument using products and vertical integer tangles.

□

Figure 5.13 illustrates the preceding result with some examples of sums of rational tangles. This shows that a tangle defined by a Montesinos construction may also be rational. Furthermore, Lemma 5.11 readily extends to Montesinos tangles constructed from more than two rational subtangles using induction.



(a) Rational: $T \cong \frac{2}{3} + \frac{2}{1} \cong \frac{8}{3}$ **(b)** Rational: $T \cong \frac{2}{1} + \frac{2}{3} \cong \frac{8}{3}$ **(c)** Non-rational $T \cong \frac{2}{3} + \frac{2}{3}$

Figure 5.13: Examples of Montesinos constructions which are also rational

Corollary 5.12 (Montesinos tangles which are rational)**:**
If T is a Montesinos tangle constructed from a sum or product of rational subtangles $L_1, \cdots, L_m$, then T is rational if and only if at least one of the following is satisfied.

a. If $T \cong L_1 + \cdots + L_m$ is a horizontal Montesinos tangle, then T is rational if and only if all but at most one rational subtangle L_i is a horizontal integer tangle.

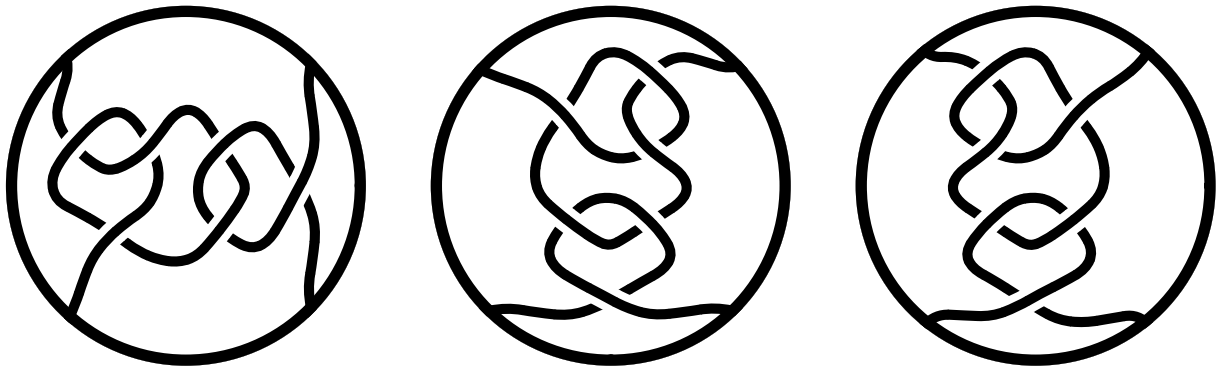
b. If $T \cong L_1 * \cdots * L_m$ is a vertical Montesinos tangle, then T is rational if and only if all but at most one rational subtangle L_i is a vertical integer tangle.

Proof:

If T is a horizontal or vertical Montesinos tangle defined by a sum or product of rational subtangles $L_1, \cdots, L_m$, then the claim follows immediately from Lemma 5.11 by induction on the number m of rational subtangles.

□

When $T \cong L_1$ is a Montesinos tangle constructed from a single rational tangle, it is both horizontal and vertical Montesinos. Both conditions from Corollary 5.12 are satisfied since the construction consists of a single summand and a single product. The converse is also true: a Montesinos tangle which is both horizontal and vertical must be rational. We omit the result here, but this fact can be proven using Montesinos links by taking a closure of the tangle [30, 31].



(a) Montesinos tangle $T \cong \frac{2}{3} + \frac{3}{2}$ **(b)** $R_{90}(T) \cong \left(-\frac{3}{2}\right) * \left(-\frac{2}{3}\right)$ **(c)** $r(T) \cong r\left(\frac{2}{3}\right) * \left(\frac{3}{2}\right) \cong \frac{3}{2} * \frac{2}{3}$

Figure 5.14: Examples of some transformations of Montesinos tangle diagrams

5.2.2 Transformations of Montesinos Tangles

Consider some of the transformations of Montesinos tangles shown in Figure 5.14. Notice that rotating a Montesinos tangle 90 degrees in the plane transforms a sum of subtangles into a product and a product of subtangles into a sum. Recall that Theorem 2.2 summarizes how various tangle transformations interact with sums and products. These results naturally generalize to Montesinos tangles by induction, in some cases requiring that the order of the subtangles be reversed in the construction. The *diagonal flip* across the NW-SE axis is a particularly useful transformation for Montesinos tangles (Figure 5.14c). This operation is equivalent to a composition of a 90 degree clockwise rotation with a horizontal flip, and we originally introduced the notation $r(L) = R_{90}(R_H(L))$ to denote this.

Lemma 5.13 (Flipping a Montesinos tangle across the diagonal)**:**
If T is a Montesinos tangle constructed from a sum or product of rational subtangles $L_1, \cdots, L_m$, then the diagonal flip $r(T) = R_{90}(R_H(T))$ is also a Montesinos tangle.

a. If $T \cong L_1 + \cdots + L_m$ is horizontal, then $r(T) \cong r(L_1) * \cdots * r(L_m)$ is vertical.

b. If $T \cong L_1 * \cdots * L_m$ is vertical, then $r(T) \cong r(L_1) + \cdots + r(L_m)$ is horizontal.

Proof:

Let $T \cong L_1 + \cdots + L_m$ be a horizontal Montesinos tangle. Consider transforming the diagram of T by flipping it across the NW-SE diagonal axis; Figure 5.14c shows one such

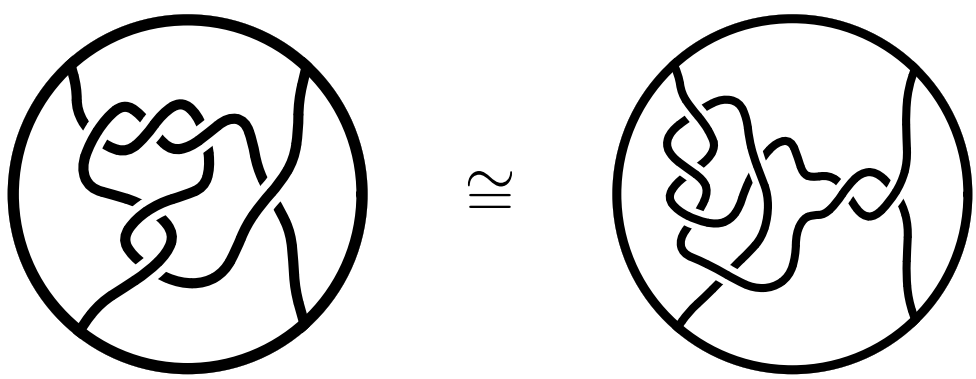
example. We use Theorem 2.2 to compute the following.

$$\begin{aligned} r(T) &\cong r(L_1 + \cdots + L_m) \\ &\cong R_{90}(R_H(L_1 + \cdots + L_m)) \\ &\cong R_{90}(R_H(L_1) + \cdots + R_H(L_m)) \\ &\cong R_{90}(R_H(L_1)) * \cdots * R_{90}(R_H(L_m)) \\ &\cong r(L_1) * \cdots * r(L_m) \end{aligned}$$

This establishes the first part of the claim for horizontal Montesinos tangles. The corresponding result for vertical Montesinos tangles follows by a parallel argument.

□

Lemma 5.13 will enable us to extend results for horizontal Montesinos tangles to vertical Montesinos tangles using the diagonal flip, which will streamline some proofs. Rather than flipping an entire Montesinos tangle, we may also consider the effect of flipping individual rational subtangles in a Montesinos construction. Since this operation involves a 90 degree rotation, one of the effects is reversing the sign of the twists in the flipped subtangle. By manipulating the twist vector, this operation can be exploited to change the sign of a rational tangle by shuffling out a certain number of twists (Figure 5.15). The next lemma makes this idea precise.



$$\frac{10}{7} \cong (3,2,1) \cong (3,2,-1) \circ (2) \cong (-3,-1,-1,0) \circ (2) \cong -\frac{4}{7} + \frac{2}{1}$$

Figure 5.15: Example of shuffling twists to reverse the sign of a rational tangle

Lemma 5.14 (Converting the sign of a rational tangle with horizontal twists)**:**
If $L \cong \frac{p}{q}$ is a nontrivial rational tangle, then there exists a rational tangle $L' \cong \frac{p'}{q'}$ of opposite sign and a horizontal integer tangle $\frac{k}{1}$ of the same sign such that $L \cong L' + \frac{k}{1}$.

Proof:

Let $L \cong \frac{p}{q}$ be a nontrivial rational tangle and suppose without loss of generality that $\frac{p}{q} > 0$. Suppose that the canonical diagram for L has a construction defined by the canonical twist vector $(n_1, \cdots, n_r)$ of odd length and constant sign (Theorem 3.25). Since $\frac{p}{q} = \mathrm{CF}(n_1, \cdots, n_r) > 0$, the canonical diagram for L is constructed using only positive twists (Corollary 3.47). This means that all twist vector entries n_i are positive, with the possible exception of $n_r = 0$. To prove the claim, we will show how to reverse the sign of the twist vector by shuffling out twists. Figure 5.15 shows one example of this in practice.

Let $k = n_r + 1$. By leveraging the rules for manipulating twist vectors, observe that we can modify the construction for L in the following way.

$$\begin{aligned}
L &\cong (n_1, \cdots, n_{r-1}, n_r) \\
&\cong (n_1, \cdots, \underbrace{n_{r-1}, -1}_{\text{opposite sign}}) \circ (n_r + 1) \qquad \text{(Lemma 3.8)} \\
&\cong (-n_1, \cdots, \underbrace{-n_{r-1} + 1, -1, -1 + 1}_{\text{same sign}}) \circ (k) \qquad \text{(Lemma 3.21)} \\
&\cong (-n_1, \cdots, -n_{r-1} + 1, -1, 0) + \frac{k}{1} \qquad \text{(Lemma 3.7)} \\
&\cong L' + \frac{k}{1}
\end{aligned}$$

In the expression above, let L' denote the rational tangle defined by the twist vector $v = (-n_1, \cdots, -n_{r-1} + 1, -1, 0)$, and let $\frac{p'}{q'} = \mathrm{CF}(v)$. We must verify that $L' \cong \frac{p'}{q'}$ and $\frac{k}{1}$ defined in this way satisfy the claim. Since $k = n_r + 1 > 0$, it follows that $\frac{k}{1}$ is a horizontal integer tangle of the same sign as $L \cong \frac{p}{q}$.

It remains to check that $\frac{p'}{q'} < 0$. Note that v is not a canonical twist vector for L' since it has even length and possibly contains an interior zero entry. However, all nonzero entries in v are negative. Hence, by applying some combination of the first three rules in Theorem 3.23, we may convert v into an equivalent canonical twist vector v' (Lemma 3.24). Since these three rules only involve rewriting the notation and do not modify the diagram for L', it follows that each entry in the canonical twist vector v' is negative, except possibly the

last. Thus, each twist in the canonical diagram for L' is negative, which means that $\frac{p'}{q'} < 0$ (Corollary 3.47).

Therefore, we have shown that, for any rational tangle $L \cong \frac{p}{q}$ with a positive fraction, there exists a rational tangle $L' \cong \frac{p'}{q'}$ with a negative fraction and a positive horizontal integer tangle $\frac{k}{1}$ such that we have the equivalence $L \cong L' + \frac{k}{1}$. If $L \cong \frac{p}{q}$ has a negative fraction, a similar argument applies by reversing the signs in the preceding derivations. This shows that any rational tangle can have its sign reversed by extracting a certain number of horizontal twists.

□

There was nothing special about the twists used in Lemma 5.14 being horizontal, the same effect can be accomplished using vertical twists. Rather than manipulate twist vectors, we may prove this fact by combining the preceding result with a flip across the diagonal.

Lemma 5.15 (Converting the sign of a rational tangle with vertical twists)**:**
If $L \cong \frac{p}{q}$ is a nontrivial rational tangle, then there exists a rational tangle $L' \cong \frac{p'}{q'}$ of opposite sign and a vertical integer tangle $\frac{1}{k}$ of the same sign such that $L \cong L' * \frac{1}{k}$.

Proof:

Let $L \cong \frac{p}{q}$ be a nontrivial rational tangle and suppose without loss of generality that $\frac{p}{q} > 0$. Consider the related tangle $r(L) = R_{90}(R_H(L))$, which is obtained by flipping L across the NW-SE diagonal axis. Note that $r(L)$ is also a rational tangle. Since rational tangles are equivalent under horizontal and vertical flips (Lemma 3.3), we have that $r(L) = R_{90}(R_H(L)) \cong R_{90}(L)$. By Corollary 3.46, the corresponding fraction of $r(L)$ is $-\frac{q}{p} < 0$.

By Lemma 5.14, there exists a positive rational tangle L_r and a negative horizontal integer tangle $\frac{k_r}{1}$ such that $r(L) \cong L_r + \frac{k_r}{1}$. Thus, we may flip $r(L)$ back across the diagonal to compute the following expression for L using the rules summarized in Theorem 2.2.

$$\begin{aligned}
L &\cong r(r(L)) \\
&= R_{90}(R_H(L_r + \tfrac{k_r}{1})) \\
&= R_{90}(R_H(L_r) + R_H(\tfrac{k_r}{1})) \\
&= R_{90}(L_r + \tfrac{k_r}{1}) \qquad \text{(Lemma 3.3)} \\
&= R_{90}(L_r) * R_{90}(\tfrac{k_r}{1}) \\
&= R_{90}(L_r) * \frac{1}{-k_r} \qquad \text{(Lemma 3.1)}
\end{aligned}$$

Since L_r is a rational tangle with a positive fraction, $R_{90}(L_r)$ is a rational tangle with a negative fraction by Corollary 3.46. Hence, we may take $L' \cong R_{90}(L_r)$ as a negative rational tangle. Since $k_r < 0$, taking $k = -k_r > 0$ gives a positive vertical integer tangle $\frac{1}{k}$. Thus, $L \cong L' * \frac{1}{k}$ defined in this way yields the desired expression for L.

This establishes the claim when $L \cong \frac{p}{q}$ has a positive fraction. If $\frac{p}{q}$ is negative, we obtain a similar result by reversing the signs in the preceding argument. Therefore, we may conclude that any rational tangle can be expressed as a product of a rational tangle of opposite sign and a vertical integer tangle of the same sign.

□

In connection with Montesinos tangles, Lemmas 5.14 and 5.15 can be applied to rational subtangles at the cost of introducing some additional twists. In general, a horizontal Montesinos tangle $T \cong L_1 + \cdots + L_m$ can be constructed as a sum using both positive and negative rational tangles. Using Lemma 5.14, we may convert each rational subtangle L_i to the same sign, and then flype the additional horizontal twists to one side (Figure 5.16). A similar construction is possible for vertical Montesinos tangles using Lemma 5.15. This idea leads to the following construction for Montesinos tangles, wherein the leading rational subtangles have constant sign.

$$T \quad \cong \quad -\tfrac{10}{7} + \tfrac{2}{3} \quad \cong \quad \left(\tfrac{4}{7} + \tfrac{-2}{1}\right) + \tfrac{2}{3} \quad \cong \quad \tfrac{4}{7} + \tfrac{2}{3} + \tfrac{-2}{1}$$

Figure 5.16: Montesinos conversion for leading rational subtangles of constant sign

Lemma 5.16 (Montesinos tangles with rational subtangles of constant sign)**:**
If T is a Montesinos tangle constructed from a sum or product of rational subtangles $L_1, \cdots, L_m$, then there exists an integer k and rational tangles $L'_1, \cdots, L'_m$ with fractions of the same sign such that at least one of the following constructions is satisfied.

a. If $T \cong L_1 + \cdots + L_m$ is a horizontal Montesinos tangle, then $T \cong L'_1 + \cdots + L'_m + \frac{k}{1}$.

b. If $T \cong L_1 * \cdots * L_m$ is a vertical Montesinos tangle, then $T \cong L'_1 * \cdots * L'_m * \frac{1}{k}$.

Proof:

Let $T \cong L_1 + \cdots + L_m$ be a horizontal Montesinos tangle. By Lemma 5.10, we may assume without loss of generality that each L_i is nontrivial. Suppose that each rational subtangle corresponds to the fraction $L_i \cong \frac{p_i}{q_i}$. Our first goal is to rewrite each of these subtangles as $L_i \cong L'_i + \frac{k_i}{1}$, where all of the L'_i subtangles are rational and have the same sign. We can choose to convert these to positive rational subtangles or negative rational subtangles; we choose positive in this proof, but the negative choice follows by a parallel argument.

For each subtangle $L_i \cong \frac{p_i}{q_i}$, there are two possibilities depending on if the fraction is positive or negative. If $\frac{p_i}{q_i} > 0$, then define $L'_i \cong L_i$ and take $k_i = 0$. Thus, $L_i \cong L'_i + \frac{k_i}{1}$, where the fraction of L'_i is positive. If $\frac{p_i}{q_i} < 0$, then by Lemma 5.14 there exists a rational tangle L'_i with a positive fraction and an integer k_i such that $L_i \cong L'_i + \frac{k_i}{1}$. Hence, each L'_i subtangle chosen in this way is a positive rational tangle.

Now consider replacing each L_i in the construction for T with the expressions $L'_i + \frac{k_i}{1}$ chosen in this way. By Corollary 3.4, we may flype all of the horizontal integer tangles to the right without changing the equivalence class of T or any of the new L'_i rational subtangles.

$$\begin{aligned}
T &\cong L_1 + \cdots + L_m \\
&\cong (L'_1 + \tfrac{k_1}{1}) + \cdots + (L'_m + \tfrac{k_m}{1}) \\
&\cong L'_1 + \cdots + L'_m + \frac{k_1}{1} + \cdots + \frac{k_m}{1} \\
&\cong L'_1 + \cdots + L'_m + \frac{k_1 + \cdots + k_m}{1} \\
&\cong L'_1 + \cdots + L'_m + \frac{k}{1}
\end{aligned}$$

Taking $k = k_1 + \cdots + k_m$, we obtain the desired construction for a horizontal Montesinos tangle using the rational subtangles L'_i which all have the same sign. We chose to find positive rational subtangles, but we could also have found negative subtangles in a similar

way. If $T \cong L_1 * \cdots * L_m$ is a vertical Montesinos tangle, then the claim follows by a parallel argument using Lemma 5.15.

□

Since integer tangles are a special case of rational tangle, the construction obtained through Lemma 5.16 still defines a Montesinos diagram. Note that the sign of this integer tangle at the end may not match the sign of the other rational subtangles, although it could. These rational subtangles can be taken as either positive or negative, the only difference is the number and sign of twists that are pushed into the final integer tangle. We can take this idea of pushing twists to the end a step further to place some additional constraints on the structure of the leading rational subtangles.

In addition to converting the sign of a rational subtangle's fraction, a similar technique can be used to bound its magnitude. Corollary 3.48 states that the magnitude of a rational tangle's fraction is greater than one if the last nonzero twist is horizontal and less than one if vertical. By shuffling any flypable twists to the end of the Montesinos construction, we may force each rational subtangle to meet one of these conditions. The removed twists are grouped into an integer subtangle at the end.

Lemma 5.17 (Montesinos tangles with bounded rational subtangles)**:**
If T is a Montesinos tangle constructed from a sum or product of rational subtangles $L_1, \cdots, L_m$, then there exists an integer k and rational tangles $L'_1, \cdots, L'_m$ such that at least one of the following constructions is satisfied.

a. If $T \cong L_1 + \cdots + L_m$ is a horizontal Montesinos tangle, then $T \cong L'_1 + \cdots + L'_m + \frac{k}{1}$, where the magnitude of each corresponding fraction $L'_i \cong \frac{p'_i}{q'_i}$ satisfies $0 < \left|\frac{p'_i}{q'_i}\right| < 1$.

b. If $T \cong L_1 * \cdots * L_m$ is a vertical Montesinos tangle, then $T \cong L'_1 * \cdots * L'_m * \frac{1}{k}$, where the magnitude of each corresponding fraction $L'_i \cong \frac{p'_i}{q'_i}$ satisfies $1 < \left|\frac{p'_i}{q'_i}\right| < \infty$.

Proof:

Let $T \cong L_1 + \cdots + L_m$ be a horizontal Montesinos tangle. By Lemma 5.10, we may assume without loss of generality that each L_i is nontrivial. Suppose that each rational subtangle corresponds to the fraction $L_i \cong \frac{p_i}{q_i}$. As in the proof of Lemma 5.16, our first goal is to rewrite each of these subtangles as $L_i \cong L'_i + \frac{k_i}{1}$, but this time where each L'_i is a rational tangle whose fraction has a magnitude bounded between 0 and 1.

For each subtangle $L_i \cong \frac{p_i}{q_i}$, let $v_i = (n^i_1, \cdots, n^i_{k_i})$ denote the corresponding canonical twist vector for L_i. Let $v'_i = (n^i_1, \cdots, n^i_{k_{i-1}}, 0)$ denote the twist vector obtained by setting the last entry to zero, and let $L'_i \cong v'_i$ be the rational tangle defined by this modified twist vector. Note that v'_i is still canonical, but the last nonzero twist is guaranteed to be vertical. If $\frac{p'_i}{q'_i} = \mathrm{CF}(v'_i)$ is the fraction corresponding to L'_i, then $0 < \left|\frac{p'_i}{q'_i}\right| < 1$ by Corollary 3.48. Thus, we may use Lemma 3.8 to rewrite the expression of L_i using twist vector compositions as follows.

$$L_i \cong (n^i_1, \cdots, n^i_{k_{i-1}}, n^i_{k_i}) \cong (n^i_1, \cdots, n^i_{k_{i-1}}, 0) \circ (n^i_{k_i}) \cong L'_i + \frac{n^i_{k_i}}{1}$$

Suppose that we obtain such an expression for each rational subtangle L_i in the original construction for T. Note that the sign of each of these rational subtangles is unimportant and that the corresponding fraction always satisfies the bound $0 < \left|\frac{p'_i}{q'_i}\right| < 1$.

Now we replace each L_i in the construction for T with the expressions $L'_i + \frac{n^i_{k_i}}{1}$ chosen in this way. As in the proof of Lemma 5.16, we may leverage Corollary 3.4 to flype all of the horizontal integer tangles to the right without changing the equivalence class of T or any of the new L'_i rational subtangles.

$$\begin{aligned}
T &\cong L_1 + \cdots + L_m \\
&\cong (L'_1 + \tfrac{n^1_{k_1}}{1}) + \cdots + (L'_m + \tfrac{n^m_{k_m}}{1}) \\
&\cong L'_1 + \cdots + L'_m + \frac{n^1_{k_1}}{1} + \cdots + \frac{n^m_{k_m}}{1} \\
&\cong L'_1 + \cdots + L'_m + \frac{n^1_{k_1} + \cdots + n^m_{k_m}}{1} \\
&\cong L'_1 + \cdots + L'_m + \frac{k}{1}
\end{aligned}$$

Taking $k = n^1_{k_1} + \cdots + n^m_{k_m}$, we obtain the desired construction for a horizontal Montesinos tangle using rational subtangles L'_i, all which have fractions that are bounded between 0 and 1. This proves the first part of the claim.

For the second part, let $T \cong L_1 * \cdots * L_m$ be a vertical Montesinos tangle. Hence, $r(T) \cong r(L_1) + \cdots + r(L_m)$ is a horizontal Montesinos tangle (Lemma 5.13). By the first part of the claim, there exists an integer k_r and rational tangles $L^r_1, \cdots, L^r_m$ such that $r(T) \cong L^r_1 + \cdots + L^r_m + \frac{k_r}{1}$, where the fraction of each $L^r_i \cong \frac{p^r_i}{q^r_i}$ satisfies the bound $0 < \left|\frac{p^r_i}{q^r_i}\right| < 1$. By reapplying Lemma 5.13 to flip $r(T)$ back across the diagonal, we obtain the following

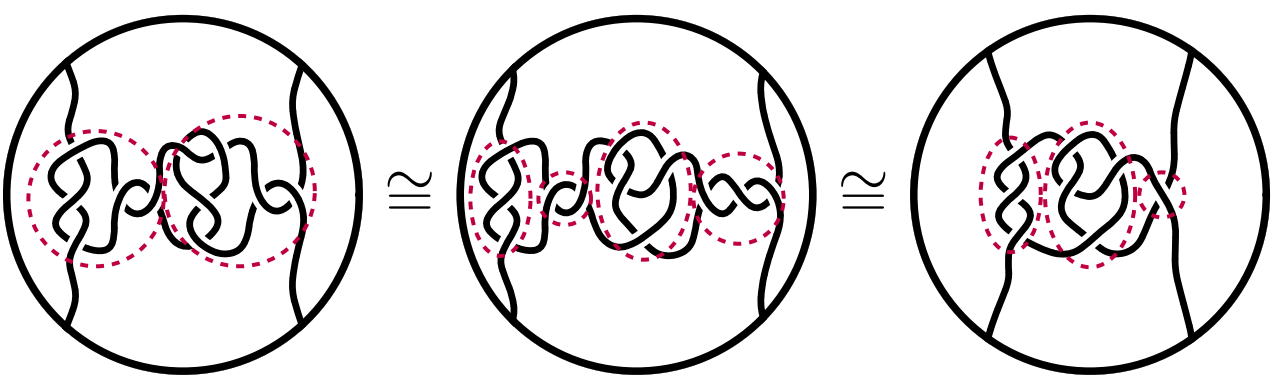
construction for T.

$$\begin{aligned}
T &\cong r(r(T)) \\
&\cong r(L_1^r + \cdots + L_m^r + \tfrac{k_r}{1}) \\
&\cong r(L_1^r) * \cdots * r(L_m^r) * r(\tfrac{k_r}{1}) \qquad \text{(Lemma 5.13)} \\
&\cong R_{90}(R_H(\tfrac{p_1^r}{q_1^r})) * \cdots * R_{90}(R_H(\tfrac{p_m^r}{q_m^r})) * R_{90}(R_H(\tfrac{k_r}{1})) \\
&\cong R_{90}(\tfrac{p_1^r}{q_1^r}) * \cdots * R_{90}(\tfrac{p_m^r}{q_m^r}) * R_{90}(\tfrac{k_r}{1}) \qquad \text{(Lemma 3.3)} \\
&\cong (-\tfrac{q_1^r}{p_1^r}) * \cdots * (-\tfrac{q_m^r}{p_m^r}) * (\tfrac{1}{-k_r}) \qquad \text{(Corollary 3.46)} \\
&\cong L_1' * \cdots * L_m' * \frac{1}{k}
\end{aligned}$$

In the final expression above, each L_i' is taken to be the rational tangle defined by the fraction $\frac{p_i'}{q_i'} = -\frac{q_i^r}{p_i^r}$. We take $k = -k_r$ to obtain the vertical integer tangle $\frac{1}{k}$ at the end. Since $0 < \left|\frac{p_i^r}{q_i^r}\right| < 1$, we have that $1 < \left|\frac{q_i^r}{p_i^r}\right| = \left|-\frac{q_i^r}{p_i^r}\right| = \left|\frac{p_i'}{q_i'}\right| < \infty$. Therefore, $T \cong L_1' * \cdots * L_m' * \frac{1}{k}$ defined in this way satisfies the claim.

□

Although the details in the preceding proofs are rather technical, the essential idea is quite intuitive. The last twists of every rational subtangle in the construction are merely shuffled to the end using flypes, and these are consolidated into a single integer tangle. When a Montesinos tangle diagram is converted into this form, the fractions of its rational subtangles satisfy the bounds stated in Corollary 3.48. If the leading rational subtangles



$$T \cong \tfrac{7}{3} + \tfrac{-12}{5} \cong \left(\tfrac{1}{3} + \tfrac{2}{1}\right) + \left(\tfrac{3}{5} + \tfrac{-3}{1}\right) \cong \tfrac{1}{3} + \tfrac{3}{5} + \tfrac{-1}{1}$$

Figure 5.17: Canonical Montesinos conversion for bounded, positive rational subtangles

are also converted to constant sign in addition to satisfying this bound, then we obtain the preferred diagram for a Montesinos tangle (Figure 5.17). Based on this idea, we combine Lemmas 5.16 and 5.17 to define the canonical form for Montesinos tangles. The rational subtangles used in this diagram construction are also assumed to be in canonical form.

Theorem 5.18 (Existence of canonical diagrams for Montesinos tangles [29])**:**
Every non-rational Montesinos tangle T admits a *canonical diagram* satisfying one of the following constructions for horizontal and vertical Montesinos tangles, respectively.

a. $T \cong L_1 + \cdots + L_m + \frac{k}{1}$, where each $L_i \cong \frac{p_i}{q_i}$ is a rational subtangle in canonical form with fraction satisfying $0 < \frac{p_i}{q_i} < 1$, and $\frac{k}{1}$ is a horizontal integer subtangle.

b. $T \cong L_1 * \cdots * L_m * \frac{1}{k}$, where each $L_i \cong \frac{p_i}{q_i}$ is a rational subtangle in canonical form with fraction satisfying $-\infty < \frac{p_i}{q_i} < -1$, and $\frac{1}{k}$ is a vertical integer subtangle.

Proof:

Let T be a horizontal Montesinos tangle which is non rational. By Lemma 5.16, we know that there exist positive rational tangles $L'_1, \cdots, L'_m$ and a horizontal integer tangle $\frac{k'}{1}$ such that $T \cong L'_1 + \cdots + L'_m + \frac{k'}{1}$. Let $T' \cong L'_1 + \cdots + L'_m$ be the Montesinos subtangle obtained by removing $\frac{k'}{1}$ from the end. By Lemma 5.17, there exist rational subtangles $L''_1, \cdots, L''_m$ and a horizontal integer tangle $\frac{k''}{1}$ such that $T' \cong L''_1 + \cdots + L''_m + \frac{k''}{1}$, where the magnitude of the fraction corresponding to each L''_i is bounded between 0 and 1.

Note that each rational tangle L''_i from this construction is the subtangle of L'_i obtained by removing the rightmost horizontal twists. In particular, this means that each L''_i has the same sign as the twists in L'_i. Since each L'_i has a positive fraction, it follows that each L''_i has a positive fraction between 0 and 1. Taking $L_i = L''_i$ and $k = k'' + k'$, we obtain the following construction for T.

$$\begin{aligned}
T &\cong L'_1 + \cdots + L'_m + \frac{k'}{1} \\
&\cong T' + \frac{k'}{1} \\
&\cong L''_1 + \cdots + L''_m + \frac{k''}{1} + \frac{k'}{1} \\
&\cong L_1 + \cdots + L_m + \frac{k'' + k'}{1} \\
&\cong L_1 + \cdots + L_m + \frac{k}{1}
\end{aligned}$$

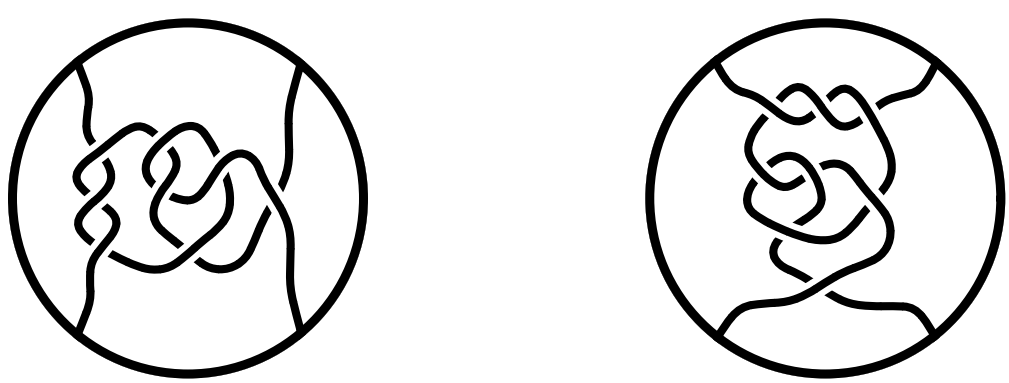
Each of the rational subtangles $L_i \cong \frac{p_i}{q_i}$ has been chosen so that $0 < \frac{p_i}{q_i} < 1$. The canonical diagram for T is thus obtained by stacking together the canonical rational tangle diagrams for each of these rational subtangles. This proves the claim for horizontal Montesinos tangles, and the proof for vertical Montesinos tangles follows by a parallel argument.

□

5.2.3 Montesinos Tangles in Canonical Form

In Theorem 5.18, we make a choice in the canonical diagram to use positive rational subtangles for horizontal Montesinos tangles and negative rational subtangles for vertical Montesinos tangles, although we could have made a different choice instead. In the context of tangle tabulation, any choice is sufficient for enumerating Montesinos tangles since every such tangle admits a diagram of this form. The reason to choose opposite signs for the rational subtangles in horizontal and vertical Montesinos tangles is so that flipping a canonical Montesinos diagram across the diagonal yields a canonical diagram for a Montesinos tangle of the opposite type (Figure 5.18). This property will be very helpful when working with generalized Montesinos tangles.

Corollary 5.19 (Diagonal flip of a canonical Montesinos tangle diagram)**:**
If T is a Montesinos tangle diagram in canonical form, then the diagonal flip $r(T)$ is a Montesinos tangle diagram of opposite type in canonical form.



(a) Horizontal: $T \cong \frac{1}{3} + \frac{3}{5} + \frac{-1}{1}$ **(b)** Vertical: $r(T) \cong \frac{-3}{1} * \frac{-5}{3} * \frac{1}{1}$

Figure 5.18: Example of canonical Montesinos tangle diagrams related by a diagonal flip

Proof:

Let T be a Montesinos tangle diagram in canonical form. Suppose without loss of generality that T is horizontal, and hence has the form $T \cong L_1 + \cdots + L_m + \frac{k}{1}$, where each rational subtangle $L_i \cong \frac{p_i}{q_i}$ has a fraction satisfying $0 < \frac{p_i}{q_i} < 1$ (Theorem 5.18). By Lemma 5.13, the diagonal flip of T is a vertical Montesinos tangle with construction $r(T) \cong r(L_1) * \cdots * r(L_m) * r(\frac{k}{1})$. By Corollary 3.46, we may compute the fraction of each rational subtangle $r(L_i)$ from $r(T)$ in the following way.

$$r(L_i) \cong r(\tfrac{p_i}{q_i}) \cong R_{90}(R_H(\tfrac{p_i}{q_i})) \cong R_{90}(\tfrac{p_i}{q_i}) \cong -\frac{q_i}{p_i}$$

This also applies to the horizontal integer tangle $\frac{k}{1}$ from M. We have that $r(\frac{k}{1}) = \frac{1}{-k}$ is a vertical integer tangle in $r(M)$. For the other rational subtangles, since $0 < \frac{p_i}{q_i} < 1$, it follows that $-\infty < -\frac{q_i}{p_i} < -1$. Hence, $r(M) \cong r(L_1) * \cdots * r(L_m) * \frac{1}{-k}$ is a vertical Montesinos tangle in canonical form.

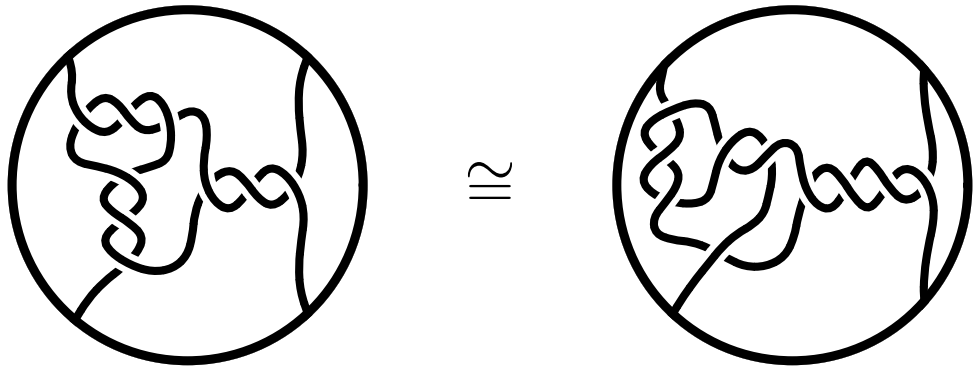
This shows that the diagonal flip of a horizontal Montesinos tangle diagram in canonical form yields a vertical Montesinos tangle diagram in canonical form. The proof using the canonical diagram of a vertical Montesinos tangle follows by a parallel argument.

□

As with rational tangle diagrams, the canonical form of a Montesinos tangle is unique. Hence, there will be no redundancies when tabulating only these types of Montesinos diagrams. However, also like rational tangle diagrams, proving the uniqueness of the canonical form takes significantly more work. Since this fact is a special case of Moon and Darcy's classification of generalized Montesinos tangles [31], we will state this result here without proof.

Theorem 5.20 (Uniqueness of canonical diagrams for Montesinos tangles [31])**:**
The canonical diagram of a non-rational Montesinos tangle is unique.

A non-rational Montesinos tangle $T \cong L_1 + \cdots + L_m + \frac{k}{1}$ or $T \cong L_1 * \cdots * L_m * \frac{1}{k}$ in canonical form must satisfy $m \geq 2$. This restriction of the canonical diagram to non-rational Montesinos tangles is to avoid a conflict with canonical diagrams for rational tangles. For example, if representing a negative rational tangle with a canonical diagram for a horizontal Montesinos tangle, then this diagram would use a positive rational subtangle with negative twists pushed into the integer tangle at the end, which is unintuitive (Figure 5.19). Restricting canonical Montesinos diagrams to non-rational tangles avoids this.



(a) Canonical rational diagram: $T \cong -\frac{33}{10}$ **(b)** Canonical Montesinos diagram: $T \cong \frac{7}{10} + \frac{-4}{1}$

Figure 5.19: Negative rational tangle represented with a horizontal Montesinos diagram

Another difference between the canonical diagrams for rational and Montesinos tangles is their alternating behavior. We saw previously that rational tangle diagrams in canonical form are always alternating (Lemma 3.30). However, since the sign of the final integer tangle in a canonical Montesinos diagram need not agree with the sign of the other rational subtangles, a canonical diagram for a Montesinos tangle may not be alternating (Figure 5.20).

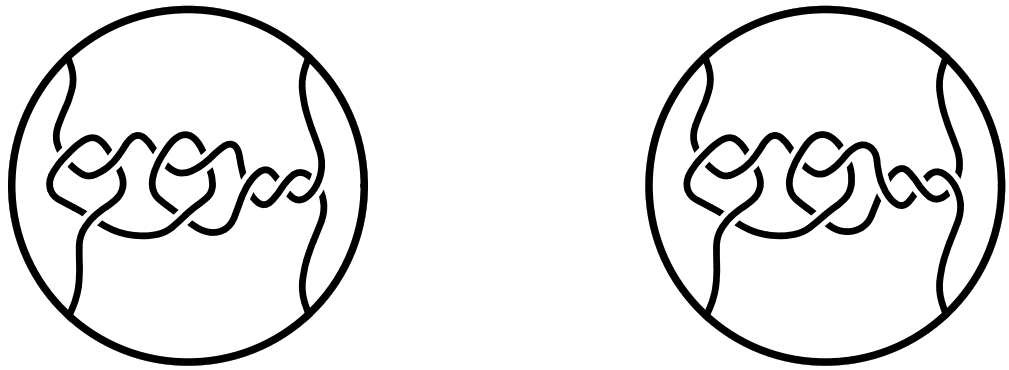
Corollary 5.21 (Alternating canonical Montesinos diagrams)**:**
If T is a Montesinos tangle diagram in canonical form, then the alternating behavior of T is determined by the final integer subtangle based on the following conditions.

a. If $T \cong L_1 + \cdots + L_m + \frac{k}{1}$ is horizontal, then T is alternating if and only if $k \geq 0$.

b. If $T \cong L_1 * \cdots * L_m * \frac{1}{k}$ is vertical, then T is alternating if and only if $k \leq 0$.

Proof:

Let T be a non-rational Montesinos tangle diagram in canonical form. Suppose without loss of generality that T is horizontal, and hence has the form $T \cong L_1 + \cdots + L_m + \frac{k}{1}$ for some integer k and some collection of rational subtangles $L_i \cong \frac{p_i}{q_i}$ satisfying $0 < \frac{p_i}{q_i} < 1$ (Theorem 5.18). Since each of these rational subtangles has a non-negative fraction, the canonical diagram for each L_i is under-alternating (Corollary 5.4). Hence, $L_1 + \cdots + L_m$ is also under-alternating (Corollary 5.2). Thus, whether or not T is alternating will depend on whether $\frac{k}{1}$ is under-alternating.



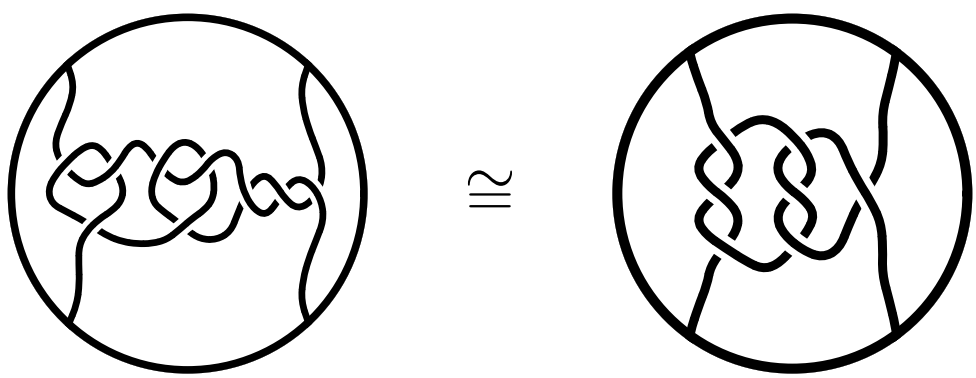
(a) Alternating: $T_1 \cong \frac{2}{3} + \frac{2}{3} + \frac{3}{1}$ **(b)** Non-alternating: $T_2 \cong \frac{2}{3} + \frac{2}{3} + \frac{-3}{1}$

Figure 5.20: Alternating and non-alternating canonical Montesinos diagrams

Since $T \cong L_1 + \cdots + L_m + \frac{k}{1}$ is algebraic, we may use the characterization given in Corollary 5.5 to determine its alternating behavior. If $k \geq 0$, then each fraction in the construction for T is non-negative and therefore T is under-alternating. If $k < 0$, then L_1 and $\frac{k}{1}$ are rational subtangles of T with fractions of opposite sign. In this case, T is non-alternating. Therefore, the canonical diagram for T is alternating if and only if $n \geq 0$. The proof when T is a vertical Montesinos tangle follows by a parallel argument.

□

While a canonical Montesinos diagram may not be alternating, in some cases it may be



(a) Canonical but non-alternating: $T \cong \frac{2}{3} + \frac{2}{3} + \frac{-3}{1}$ **(b)** Non-canonical but alternating: $T \cong \frac{-1}{3} + \frac{-1}{3} + \frac{-1}{1}$

Figure 5.21: Example of converting a non-alternating canonical Montesinos diagram

possible to obtain an equivalent Montesinos diagram which is alternating by converting the sign of each of the leading rational subtangles in the canonical form (Figure 5.21). Changing the sign of the rational subtangles is always possible by Lemma 5.16. However, since this conversion requires introducing positive or negative twists via Lemmas 5.14 and 5.15, this will only yield an alternating diagram when $|k|$ is sufficiently large.

Lemma 5.22 (Alternating negative Montesinos diagrams)**:**
If T is a non-rational Montesinos tangle diagram in canonical form which is non-alternating, then there exists an equivalent alternating Montesinos diagram in the following cases.

a. If $T \cong L_1 + \cdots + L_m + \frac{k}{1}$ is a non-alternating canonical horizontal Montesinos tangle diagram, where $k \leq -m$, then there exists an equivalent non-canonical horizontal Montesinos tangle diagram $T \cong L'_1 + \cdots + L'_m + \frac{k+m}{1}$ which is alternating.

b. If $T \cong L_1 * \cdots * L_m * \frac{1}{k}$ is a non-alternating canonical vertical Montesinos tangle diagram, where $k \geq m$, then there exists an equivalent non-canonical vertical Montesinos tangle diagram $T \cong L'_1 * \cdots * L'_m * \frac{1}{k-m}$ which is alternating.

Proof:

Let T be a non-rational Montesinos tangle diagram in canonical form which is non-alternating. Suppose without loss of generality that T is horizontal, so that $T \cong L_1 + \cdots + L_m + \frac{k}{1}$. Hence, by Corollary 5.21, we know that $k < 0$. Suppose that $k \leq -m$ and that each rational subtangle $L_i \cong \frac{p_i}{q_i}$ is in canonical form and defined by a canonical twist vector $v_i = (n^i_1, \cdots, n^i_{k_i})$. Since each fraction in the canonical diagram satisfies $0 < \frac{p_i}{q_i} < 1$, note that the last nonzero twist in each L_i is vertical by Corollary 3.48. Since the final twist vector entry always denotes horizontal twisting, this means that $n^i_{k_i} = 0$ for each twist vector v_i. The other twist vector entries are strictly positive.

By Lemma 5.14, for each positive rational tangle $L_i \cong \frac{p_i}{q_i}$, there exists a negative rational tangle $L'_i \cong \frac{p'_i}{q'_i}$ and a positive horizontal integer tangle $\frac{\ell_i}{1}$ such that $L_i \cong L'_i + \frac{\ell_i}{1}$. In the proof of Lemma 5.14, we computed the integer ℓ_i to be one greater than the final twist vector entry of v_i. In other words, $\ell_i = n^i_{k_i} + 1 = 1$, and this holds for each ℓ_i. Knowing this fact,

we may obtain the following equivalent construction for T.

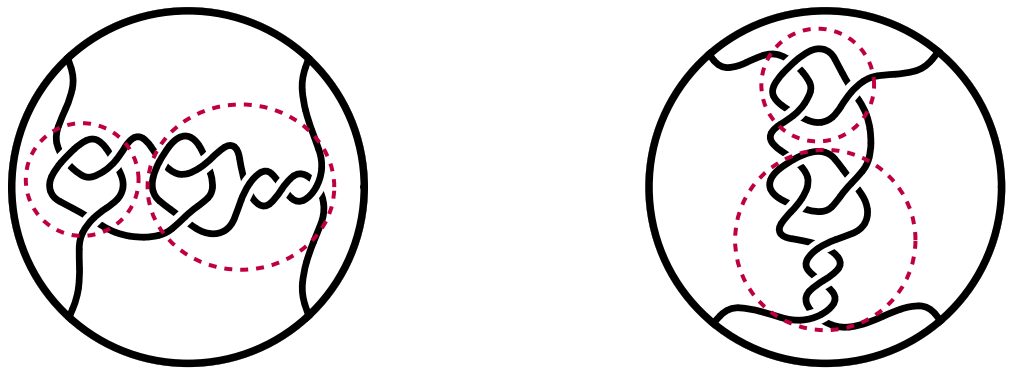
$$\begin{aligned} T &\cong L_1 + \cdots + L_m + \frac{k}{1} \\ &\cong (L_1' + \tfrac{\ell_1}{1}) + \cdots + (L_m' + \tfrac{\ell_m}{1}) + \frac{k}{1} \\ &\cong (L_1' + \tfrac{1}{1}) + \cdots + (L_m' + \frac{1}{1}) + \frac{k}{1} \\ &\cong L_1' + \cdots + L_m' + \frac{k}{1} + \frac{1}{1} + \cdots + \frac{1}{1} \\ &\cong L_1' + \cdots + L_m' + \frac{k+m}{1} \end{aligned}$$

Since $k \leq -m$, we have that $k + m \leq 0$, which means that $\frac{k+m}{1}$ is either a negative rational subtangle or a trivial subtangle if $k + m = 0$. Since each L_i' is also a negative rational subtangle, the above expression is an algebraic construction formula for T in which each rational subtangle has a non-positive fraction. Taking each of these subtangles to be in canonical form, this defines an over-alternating diagram for T by Corollary 5.5. Therefore, T admits the proposed alternating diagram, which proves the claim in the horizontal case. The claim for a vertical Montesinos tangle follows by a parallel argument.

□

Regardless of whether or not it is alternating, a Montesinos tangle diagram in canonical form admits a unique decomposition into maximal rational subtangles. We have seen previously that such a decomposition may not exist when adjacent rational subtangles share twists. However, in a canonical diagram of the form $T \cong L_1 + \cdots + L_m + \frac{k}{1}$, all of these shared twists have been flyped into the final integer subtangle $\frac{k}{1}$. If these twists at the end are grouped into the final rational subtangle as $L_m + \frac{k}{1}$, then the canonical Montesinos construction formula can be used to obtain the maximal rational subtangle decomposition (Figure 5.22). Note that $L_m + \frac{k}{1}$ may not be canonical if L_m and $\frac{k}{1}$ have opposite signs, but this still defines a rational subtangle. A similar decomposition exists for a canonical vertical Montesinos tangle $T \cong L_1 + \cdots + L_m * \frac{1}{k}$ by taking $L_m * \frac{1}{k}$ as a single rational subtangle.

Theorem 5.23 (Montesinos decomposition into maximal rational subtangles)**:**
A canonical diagram for a horizontal Montesinos tangle $T \cong L_1 + \cdots + L_m + \frac{k}{1}$ or a vertical Montesinos tangle $T \cong L_1 * \cdots * L_m * \frac{1}{k}$ admits a unique decomposition into maximal rational subtangles $D_{\text{rat}} = \{L_1, \cdots, L_m + \frac{k}{1}\}$ or $D_{\text{rat}} = \{L_1, \cdots, L_m * \frac{1}{k}\}$, respectively.



(a) $T_1 \cong \frac{2}{3} + \frac{2}{3} + \frac{3}{1}$ with $D_{\text{rat}} = \left\{\frac{2}{3}, \frac{11}{3}\right\}$ **(b)** $T_2 \cong \frac{-3}{2} * \frac{-3}{2} * \frac{-1}{3}$ with $D_{\text{rat}} = \left\{\frac{-3}{2}, \frac{-3}{11}\right\}$

Figure 5.22: Maximal rational subtangle decompositions for canonical Montesinos diagrams

Proof:

Let $T \cong L_1 + \cdots + L_m + \frac{k}{1}$ be a horizontal Montesinos tangle diagram in canonical form. Let $D = \{L_1, \cdots, L_m + \frac{k}{1}\}$ be the decomposition of T into rational subtangles defined by this construction formula. Note that we take the combined $L_m + \frac{k}{1}$ as the final subtangle in D, and this is still rational by Lemma 5.11. While a decomposition into maximal rational subtangles may not exist in general, Lemma 4.11 guarantees that such a decomposition is unique if it does exist. Thus, we must show that each of the rational subtangles from D is maximal.

Suppose by way of contradiction that there exists a non-maximal $L_i \in D$. Hence, there exists some larger rational subtangle S of T which properly contains L_i. Since S contains more crossings than L_i, it must overlap with one of the adjacent subtangles from D, either L_{i-1} or L_{i+1}. The proof for both cases uses a similar argument, so suppose without loss of generality that S overlaps with L_{i+1}. Since neither L_i nor L_{i+1} is a horizontal integer tangle, note that $L_i + L_{i+1}$ is non-rational by Lemma 5.11. Thus, L_{i+1} cannot be fully contained in S, because this would imply that a rational tangle S contains a non-rational subtangle $L_i + L_{i+1}$, which cannot happen by Lemma 3.28.

Thus, we have that $S \cap L_{i+1} \neq \emptyset$, but that neither subtangle is fully contained in the other. Therefore, by Lemma 4.3 the union of these two overlapping subtangles decomposes as $S \cup L_{i+1} \cong (S \setminus L_{i+1}) + (S \cap L_{i+1}) + (L_{i+1} \setminus S)$. Note that, since T is horizontal Montesinos, this decomposition must be a tangle sum rather than a tangle product. This also gives a decomposition $L_{i+1} \cong (S \cap L_{i+1}) + (L_{i+1} \setminus S)$. Since L_{i+1} is rational, one of these summands must be a horizontal integer tangle (Lemma 5.11).

Recall that a sum with a horizontal integer tangle denotes horizontal twisting. Since L_{i+1} is in canonical form as a rational subtangle of T, all of its twists occur to the bottom and the right (Theorem 3.25). Hence, of the two summands in the decomposition $L_{i+1} \cong (S \cap L_{i+1}) + (L_{i+1} \setminus S)$, it follows that $L_{i+1} \setminus S$ must be a horizontal integer tangle, and these denote the final nonzero twists in L_{i+1}. However, since T is in canonical form, the fraction of L_{i+1} is between 0 and 1 (Theorem 5.18). Hence, the final nonzero twists in L_{i+1} must be horizontal (Corollary 3.48), and this contradictions the existence of the right twisting implied by a sum with the horizontal integer tangle $L_{i+1} \setminus S$.

Therefore, there does not exist a rational subtangle S of T which properly contains L_i, which means that L_i is a maximal rational subtangle of T. This same arguemnts holds for the subtangles $L_1, \cdots, L_{m-1} \in D$. In the special case of $L_m + \frac{k}{1}$, there does not exist an adjacent subtangle on the right and hence the non-existence of an overlapping subtangle S holds vacuously. A similar argument applies if we had assumed that S overlaps with L_{i-1}. Thus, we may conclude that each rational subtangle in D is maximal.

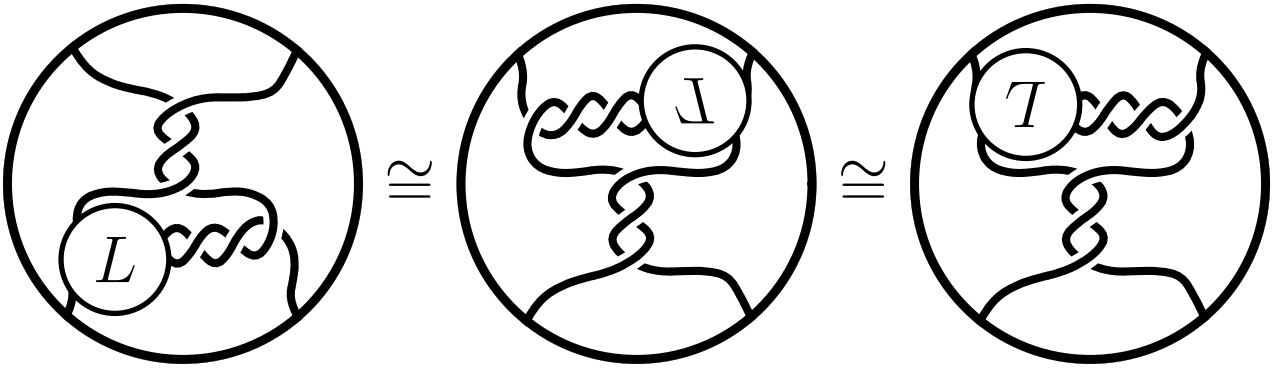
Therefore, $D = \{L_1, \cdots, L_m + \frac{k}{1}\} = D_{\text{rat}}$ is the unique decomposition of T into maximal rational subtangles, which proves the claim for a horizontal Montesinos tangle in canonical form. The claim follows by a parallel argument for a vertical Montesinos tangle $T \cong L_1 * \cdots * L_m * \frac{1}{k}$ in canonical form using $D = \{L_1, \cdots, L_m * \frac{1}{k}\}$.

□

5.3 Generalized Montesinos Tangles

We touched on generalized Montesinos tangles briefly in the preceding subsection, but these types of tangles have enough additional structure to warrant considering them separately in more detail. Recall that a generalized Montesinos tangle is a Montesinos tangle which also includes endpoint twisting. These are denoted $T \cong (L_1 + \cdots + L_m) \circ (n_1, \cdots, n_r)$ or $T \cong (L_1 * \cdots * L_m) \circ (n_1, \cdots, n_r)$ depending on whether we start from a horizontal or vertical Montesinos tangle, respectively.

Recall that a composition of a subtangle with a twist vector is defined by an alternating sequence of right sums and bottom products with horizontal and vertical integer subtangles, respectively (Lemma 3.7). However, the idea of endpoint twisting is not limited to the right and bottom sides. To verify that this notation makes sense, we must show that endpoint twists can always be shuffled to the right and bottom (Figure 5.23).



$$\tfrac{1}{3} * (L + \tfrac{3}{1}) \quad \cong \quad (\tfrac{3}{1} + R_V(L)) * \tfrac{1}{3} \quad \cong \quad (R_H(R_V(L)) + \tfrac{3}{1}) * \tfrac{1}{3} \quad \cong \quad R_{180}(L) \circ (3, 3, 0)$$

Figure 5.23: Example of endpoint twists on a subtangle flyped to the right and bottom

5.3.1 GM Composition Twist Vector Construction

In general, endpoint twisting on a subtangle can occur on any of the four sides of the tangle: top, right, bottom, or left. Horizontal twists are equivalent to a tangle sum on the right or left with a horizontal integer tangle, and vertical twists are equivalent to a tangle product on the top or bottom with a vertical integer tangle. For rational tangles, we saw these twists can always be flyped to the right and bottom sides without changing the equivalence class of the tangle (Corollary 3.6). This equivalence is what determines the twist vector of a rational tangle.

A similar equivalence is also possible for endpoint twisting on a non-rational tangle, but the corresponding subtangle may be flipped after pushing all of the twists to one side (Figure 5.23). Since a flyped subtangle is flipped across the horizontal or vertical axis, there are different cases depending on whether the number of shuffled twists is even or odd, and whether these represent horizontal or vertical twisting.

Lemma 5.24 (Shifting subtangle endpoint twisting to the right and bottom)**:**
If T is a 2-string tangle defined by some combination of top, right, bottom, and left endpoint twisting on a subtangle L, then equivalently $T \cong L' \circ (n_1, \cdots, n_r)$ for some twist vector $(n_1, \cdots, n_r)$ and a subtangle L' which is either equivalent to L, $R_H(L)$, $R_V(L)$, or $R_{180}(L)$.

Proof:

Suppose that T is a 2-string tangle constructed via endpoint twisting on a subtangle L. Since endpoint twisting can be described via sums with horizontal integer tangles and

products with vertical integer tangles, then there exists some construction formula for T that has a form such as

$$T \cong \frac{n_k}{1} + \frac{1}{n_{k-1}} * \cdots + \frac{1}{n_2} * \frac{n_1}{1} + L * \frac{1}{n_{k+1}} + \frac{n_{k+2}}{1} * \cdots * \frac{1}{n_{r-1}} + \frac{n_r}{1}.$$

We make a number of simplifying assumptions in this formula which require some justification, particularly regarding the order of operations. Since tangle sum and tangle product are neither commutative nor associative, parentheses are required for a construction such as this to be well defined, but we exclude these to avoid unnecessary clutter.

Assume without loss of generality that the operations are ordered by the index on the integer subtangles, and note that the construction above is just one possible example. Furthermore, the construction need not alternate between horizontal and vertical twisting, but we make this assumption since adjacent twists on the same side of the tangle can be grouped together into a single integer tangle. We also assume for convenience that this construction starts and ends with horizontal twisting, but these could be taken as zero twists in the vertical case.

To prove the claim, we must show how each of the operations on the left side of the above construction formula for T can be moved to the right, possibly by transforming L. This is achieved through successive applications of Corollary 3.2 to shift each set of left and top twists to the right and bottom, respectively, via flyping. Starting from the highest-index integer subtangles on the left side of the the formula, each can be be flyped to the other side of the subtangle one at a time, possibly with a flip across the horizontal or vertical axis when n_i is odd.

Since there are k integer subtangles on the left side of the construction formula for T, at most k applications of Corollary 3.2 are required to move all the twists into the desired position. Note that a flip across the horizontal axis switches the order of products in the flipped subtangle, and a flip across the vertical axis switches the order of sums in the flipped subtangle. Hence, if working from the outside in, this modification may be possible with fewer than k operations, but k is an upper bound.

Let L_i denote the transformation of L obtained after each application of Corollary 3.2. Assuming exactly k iterations are required, this sequence of flypes can be described in the

following way.

$$\begin{aligned}
T &\cong \tfrac{n_k}{1} + \tfrac{1}{n_{k-1}} * \cdots + \tfrac{1}{n_2} * \tfrac{n_1}{1} + L * \tfrac{1}{n_{k+1}} + \cdots + \tfrac{n_r}{1} \\
&\cong \tfrac{1}{n_{k-1}} * \cdots + \tfrac{1}{n_2} * \tfrac{n_1}{1} + L_1 + \tfrac{n_k}{1} * \tfrac{1}{n_{k+1}} + \cdots + \tfrac{n_r}{1} \\
&\cong \tfrac{n_{k-2}}{1} + \cdots + \tfrac{1}{n_2} * \tfrac{n_1}{1} + L_2 * \tfrac{1}{n_{k-1}} + \tfrac{n_k}{1} * \tfrac{1}{n_{k+1}} + \cdots + \tfrac{n_r}{1} \\
&\vdots \\
&\cong \tfrac{n_1}{1} + L_{k-1} * \tfrac{1}{n_2} + \cdots + \tfrac{n_k}{1} * \tfrac{1}{n_{k+1}} + \cdots + \tfrac{n_r}{1} \\
&\cong L_k + \tfrac{n_1}{1} * \tfrac{1}{n_2} + \cdots + \tfrac{n_k}{1} * \tfrac{1}{n_{k+1}} + \cdots + \tfrac{n_r}{1} \\
&\cong L' \circ (n_1, n_2, \cdots, n_k, n_{k+1}, \cdots, n_r)
\end{aligned}$$

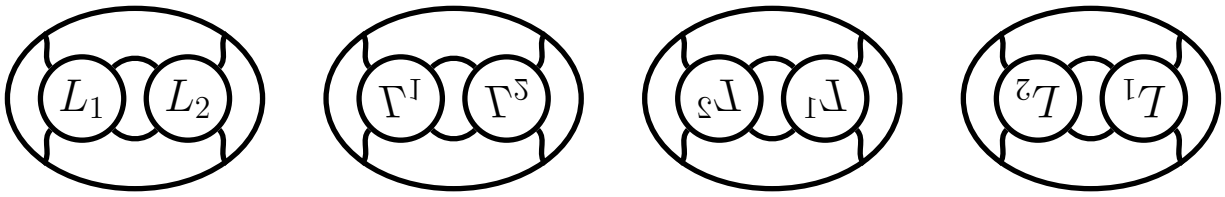
Let $L' = L_k$ denote the final transformation of L obtained through this process. Hence, L' corresponds to a composition of some combination of zero or more horizontal and vertical flips of L. By Lemma 3.7, the alternating sequence of sums with horizontal integer tangles and products with vertical integer tangles can be encoded as a composition of L' with the twist vector $(n_1, \cdots, n_r)$.

Since horizontal and vertical flips commute, and two flips of the same type undo each other, L' reduces to four possible transformations of L: either $L' \cong L$, $L' \cong R_H(L)$, $L' \cong R_V(L)$, or $L' \cong R_H(R_V(L))$. As a tangle transformation, this last possibility is equivalent to a 180 degree rotation in the plane, and hence corresponds to $L' \cong R_{180}(L)$. These give the four possibilities asserted in the claim, which completes the proof.

□

In the case of rational tangles, we always begin from a trivial tangle, either 0 or ∞, which are both invariant under the subtangle transformations used in Lemma 5.24. If L is taken to be a rational subtangle, it is also equivalent under these three possible transformations (Lemma 3.13 and Corollary 3.15). If L is non-rational, then it is possible that flyping the endpoint twists to the right and bottom results in a non-equivalent transformation of L. However, if L is Montesinos, then the transformed subtangle L' is also Montesinos, although L and L' may not be equivalent tangles.

Corollary 5.25 (Generalized Montesinos twist vector construction)**:**
If T is a generalized Montesinos tangle defined by some combination of top, right, bottom, and left endpoint twisting on a Montesinos subtangle M, then there exists a Montesinos tangle M' of the same type and a twist vector $(n_1, \cdots, n_r)$ such that $T \cong M' \circ (n_1, \cdots, n_r)$.



(a) $M \cong L_1 + L_2$ **(b)** $R_H(M) \cong R_H(L_1) + R_H(L_2)$ **(c)** $R_V(M) \cong R_V(L_2) + R_V(L_1)$ **(d)** $R_H(R_V(M)) \cong R_{180}(L_2) + R_{180}(L_1)$

Figure 5.24: Horizontal and vertical flips of Montesinos tangles are also Montesinos

Proof:

Let T be a generalized Montesinos tangle obtained from the Montesinos tangle M with some combination of endpoint twisting. By Lemma 5.24, there exists a twist vector $(n_1, \cdots, n_r)$ and a possibly transformed tangle M' such that $T \cong M' \circ (n_1, \cdots, n_r)$. To show that this equivalent diagram is also generalized Montesinos, we will verify that M' is also a Montesinos tangle. There are four possibilities for M': either $M' \cong M$, $M' \cong R_H(M)$, $M' \cong R_V(M)$, or $M' \cong R_{180}(M)$. Each of these cases is illustrated in Figure 5.24 for a Montesinos tangle constructed from two rational subtangles.

Clearly, M' is Montesinos if $M' \cong M$ (5.24a). If $M' \cong R_H(M)$, note that flipping M across the horizontal axis still defines a Montesinos tangle, although the order of the rational subtangles is reversed if M is vertical (5.24b). A similar fact holds when flipping M across the vertical axis if $M' \cong R_V(M)$ (5.24c). In both of these cases, M' is also Montesinos. If $M' \cong R_{180}(M)$, then each rational subtangle in M is turned upside down and their order is reversed, but the resulting diagram is still Montesinos (5.24d).

Therefore, if M is Montesinos, then the transformed tangle M' is also guaranteed to be Montesinos. Furthermore, if M is horizontal or vertical, then M' is also horizontal or vertical, respectively, although possibly with a reversal on the order of the rational subtangles. Therefore, $T \cong M' \circ (n_1, \cdots, n_r)$ is an equivalent generalized Montesinos diagram for T which is described explicitly as a composition of a Montesinos subtangle with a twist vector.

□

Corollary 5.25 justifies our use of twist vectors to describe the construction of a generalized Montesinos tangle. Hence, we may assume without loss of generality that every generalized Montesinos tangle diagram has the form $T \cong M \circ (n_1, \cdots, n_r)$, for some Montesinos tangle M.

5.3.2 Transformations of Generalized Montesinos Tangles

When working with Montesinos tangles, we made the observation that transformations of Montesinos diagrams such as flips and rotations also result in a Montesinos diagram. A similar fact holds for generalized Montesinos tangles. As before, the diagonal flip of a generalized Montesinos tangle is a particularly useful transformation. In terms of the twist vector, flipping a generalized Montesinos tangle across the NW-SE axis switches the right and bottom twists and changes the slope of the over-strand. Hence, this diagram can be described by a related twist vector of opposite sign (Figure 5.25).

Lemma 5.26 (Flipping a generalized Montesinos tangle across the diagonal)**:**
If $T \cong M \circ (n_1, \cdots, n_r)$ is a generalized Montesinos tangle defined by a Montesinos tangle M and a twist vector $(n_1, \cdots, n_r)$, then flipping T across the NW-SE diagonal axis yields a generalized Montesinos tangle defined by the construction $r(T) \cong r(M) \circ (-n_1, \cdots, -n_r, 0)$.

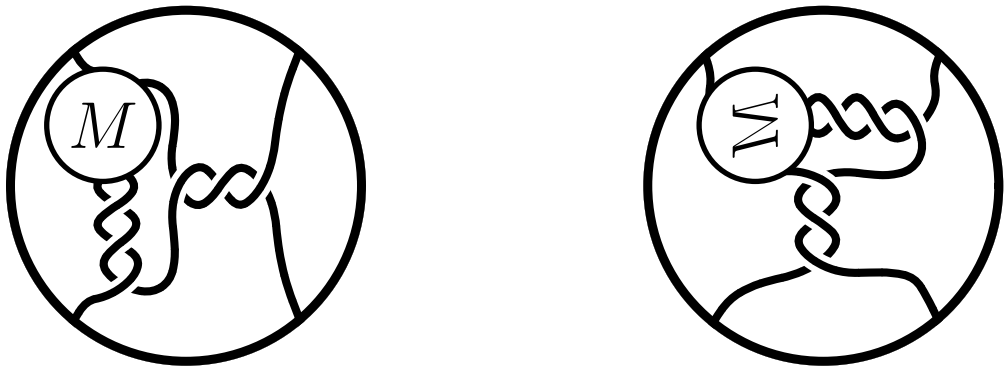
Proof:

Let $T \cong M \circ (n_1, \cdots, n_r)$ be a generalized Montesinos tangle, where M is a Montesinos tangle, either vertical or horizontal. Suppose without loss of generality that r is odd so that n_1 denotes horizontal twisting, possibly by allowing $n_1 = 0$. By Lemma 3.7, we have the construction formula $T \cong M + \frac{n_1}{1} * \cdots + \frac{n_r}{1}$. We may compute $r(T)$ by applying the rules summarized in Theorem 2.2.

Note that, in the special case of a horizontal integer tangle $\frac{n}{1}$, we have by Lemma 3.1 that $r(\frac{n}{1}) = R_{90}(R_H(\frac{n}{1})) \cong R_{90}(\frac{n}{1}) \cong \frac{1}{-n}$. Similarly, $r(\frac{1}{n}) \cong \frac{-n}{1}$ for a vertical integer tangle. Leveraging these facts, we may compute the following transformation.

$$\begin{aligned}
r(T) &\cong r(M \circ (n_1, \cdots, n_r)) \\
&\cong r(M + \tfrac{n_1}{1} * \cdots + \tfrac{n_r}{1}) \\
&\cong r(M + \tfrac{n_1}{1} * \cdots * \tfrac{1}{n_{r-1}}) * r(\tfrac{n_r}{1}) \\
&\vdots \\
&\cong r(M) * r(\tfrac{n_1}{1}) + \cdots * r(\tfrac{n_r}{1}) \\
&\cong r(M) * \tfrac{1}{-n_r} + \cdots * \tfrac{1}{-n_r} + \tfrac{0}{1} \\
&\cong r(M) \circ (-n_1, \cdots, -n_r, 0)
\end{aligned}$$

Since M is a Montesinos tangle, we know by Lemma 5.13 that the diagonal flip $r(M)$ is also a Montesinos tangle. Therefore, the derivation above shows that the diagonal flip $r(T) \cong$



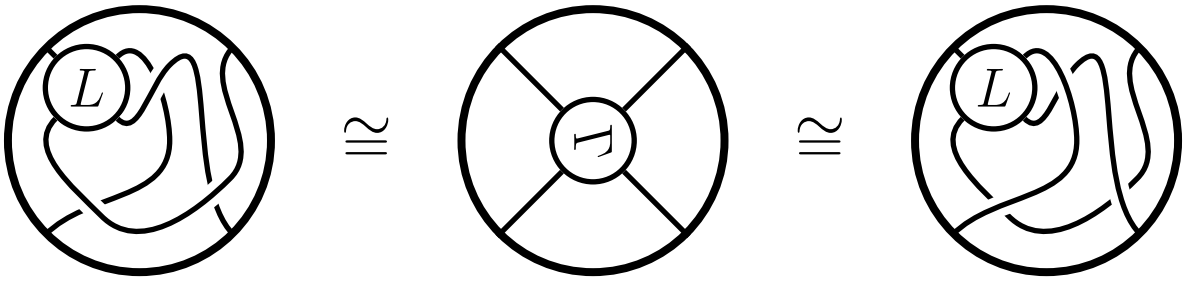
(a) $T \cong M \circ (3,3)$ **(b)** $r(T) \cong r(M) \circ (-3,-3,0)$

Figure 5.25: The diagonal flip of a GM tangle diagram is also a GM tangle diagram

$r(M) \circ (-n_1, \cdots, -n_r, 0)$ has the form of a generalized Montesinos tangle. In particular, the twist vectors for T and $r(T)$ satisfy the relationship stated in the claim.

□

The preceding result establishes that diagonal flips of generalized Montesinos tangle diagrams are still generalized Montesinos. Lemma 5.13 established a similar result for Montesinos tangle diagrams. This transformation does not define an equivalence of tangles in general because it moves the endpoints on the boundary sphere. However, if we instead flip a subtangle across the diagonal axis while holding the endpoints fixed, then the resulting diagram will be equivalent (Figure 5.26). To connect this idea to generalized Montesinos tangles, we must show that this operation can be described using endpoint twisting.



$$L \circ (1,-1,1) \quad \cong \quad r(L) \quad \cong \quad L \circ (-1,1,-1)$$

Figure 5.26: *Anti-fluping* a tangle defines an equivalence with the diagonal flip

Lemma 5.27 (Tangle anti-flupe equivalence with diagonal flip)**:**
If L is any 2-string tangle diagram, then the following two equivalences hold.

a. $L \circ (1, -1, 1) \cong r(L) \cong L \circ (-1, 1, -1)$

b. $r(L) \circ (1, -1, 1) \cong L \cong r(L) \circ (-1, 1, -1)$.

Proof:

Let L be a 2-string tangle diagram. We may prove the claim using a combination of subtangle flupes (Lemma 3.19) and twist vector compositions (Lemma 3.8).

$$\begin{aligned} L \circ (1, -1, 1) &\cong L \circ (1, -1, 0) \circ (1) && \text{(Lemma 3.8)} \\ &\cong r(L) \circ (-1) \circ (1) && \text{(Lemma 3.19)} \\ &\cong r(L) \circ (-1 + 1) && \text{(Lemma 3.8)} \\ &\cong r(L) \circ (0) \\ &\cong r(L) \end{aligned}$$

By a parallel argument, we also have that $L \circ (-1, 1, -1) \cong r(L)$, which establishes the first part of the claim. Since $L \cong r(r(L))$, the second part of the claim follows immediately from the first by switching the roles of L and $r(L)$.

□

The transformation of Lemma 5.27 invokes the operation known as *fluping*, which we originally introduced in Lemma 3.19 to describe how to modify adjacent twist vector entries of opposite sign. We can now imagine applying this idea in reverse by composing a tangle with $(\pm 1, \mp 1, \pm 1)$, an operation we will refer to as an *anti-flupe*.

Flipping a subtangle across the diagonal while holding the endpoints fixed introduces right and bottom twists of opposite sign which can be encoded using a twist vector. When a sum or product of tangles is rotated across the diagonal axis, it changes to a product or sum of tangles, respectively. Hence, if the anti-fluped subtangle is Montesinos, then the modified diagram will be generalized Montesinos of the opposite type.

Corollary 5.28 (Montesinos anti-flupe equivalence with generalized Montesinos)**:**
A vertical Montesinos tangle can be converted into a horizontal generalized Montesinos tangle, and a horizontal Montesinos tangle can be converted into a vertical generalized Montesinos tangle, via the following anti-flupe transformations, respectively.

a. $L_1 * \cdots * L_m \cong (r(L_1) + \cdots + r(L_m)) \circ (1, -1, 1) \cong (r(L_1) + \cdots + r(L_m)) \circ (-1, 1, -1)$.

b. $L_1 + \cdots + L_m \cong (r(L_1) * \cdots * r(L_m)) \circ (1, -1, 1) \cong (r(L_1) * \cdots * r(L_m)) \circ (-1, 1, -1)$.

Proof:

Let $T \cong L_1 * \cdots * L_m$ be a vertical Montesinos tangle. We may prove the claim by combining the anti-flupe equivalence of Lemma 5.27 with the diagonal flip of a Montesinos tangle established in Lemma 5.13.

$$\begin{aligned} T &\cong r(T) \circ (\pm 1, \mp 1, \pm 1) \\ &\cong r(L_1 * \cdots * L_m) \circ (\pm 1, \mp 1, \pm 1) \\ &\cong (r(L_1) + \cdots + r(L_m)) \circ (\pm 1, \mp 1, \pm 1) \end{aligned}$$

This shows the equivalence of a vertical Montesinos tangle with a horizontal generalized Montesnios tangle, establishing the first part of the claim. The second part follows by a parallel argument instead using a horizontal Montesinos tangle $T \cong L_1 + \cdots + L_m$.

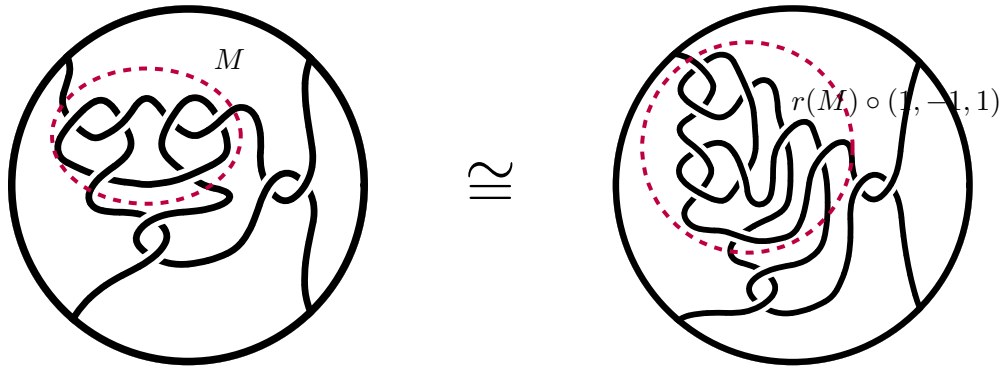
□

Since the transformation of Corollary 5.28 introduces three additional twists, it increases the crossing number of a diagram by three. Furthermore, these new twists define non-alternating crossings. In other words, an anti-fluped diagram is neither minimal nor alternating, although it can still be useful for writing proofs. The main utility of Corollary 5.28 is that it enables us to convert between horizontal and vertical generalized Montesinos tangles.

Lemma 5.29 (Converting the type of a generalized Montesinos tangle)**:**
If $T \cong M \circ v$ is a generalized Montesinos tangle, where M is either a horizontal or a vertical Montesinos subtangle, then there exists a Montesinos tangle M' of opposite type and a twist vector v' such that $T \cong M' \circ v'$, where the tangles M and M' are related by a diagonal flip.

a. If $M \cong L_1 * \cdots * L_m$ is vertical, then $M' \cong r(M) \cong r(L_1) + \cdots + r(L_m)$ is horizontal.

b. If $M \cong L_1 + \cdots + L_m$ is horizontal, then $M' \cong r(M) \cong r(L_1) * \cdots * r(L_m)$ is vertical.



(a) $T \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2, 2)$ **(b)** $T \cong \left(\frac{-3}{2} * \frac{-3}{2}\right) \circ (1, -1, 1, 2, 2)$

Figure 5.27: Example of equivalent horizontal and vertical generalized Montesinos tangles

Proof:

Let $T \cong M \circ v$ be a generalized Montesinos tangle and suppose without loss of generality that $M \cong L_1 * \cdots * L_m$ is a vertical Montesinos subtangle. By Corollary 5.28, we have that

$$M \cong L_1 * \cdots * L_m \cong (r(L_1) + \cdots + r(L_m)) \circ (1, -1, 1)$$

is a horizontal generalized Montesinos tangle. Let $M' \cong r(L_1) + \cdots + r(L_m)$ be the corresponding horizontal Montesinos tangle obtained by removing these endpoint twists. Hence, $M \cong M' \circ (1, -1, 1)$. Let $v' \cong (1, -1, 1) \circ v$ be the twist vector obtained by composition. We may compute the following equivalent diagram for T:

$$T \cong M \circ v \cong M' \circ (1, -1, 1) \circ v \cong M' \circ v'.$$

This proves that a vertical generalized Montesinos tangle can be converted into an equivalent horizontal generalized Montesinos tangle. A parallel argument shows that a horizontal generalized Montesinos tangle can be converted in a vertical generalized Montesinos tangle, which completes the proof.

□

Thus, every generalized Montesinos tangle is both horizontal and vertical. Figure 5.27 shows one such example of this equivalence. Unlike with Montesinos tangles, it is not necessary to treat horizontal and vertical Montesinos tangles separately. We may use either a horizontal or vertical diagram when working with generalized Montesinos tangles, whichever

construction is most convenient. Furthermore, since each of the rational subtangles in the two Montesinos subtangles of Lemma 5.29 are related by a flip across the diagonal, the corresponding fractions will be negative reciprocals of each other. Notice that the two core Montesinos subtangles of Figure 5.27 satisfy this relationship.

Corollary 5.30 (Equivalent GM tangles with negative reciprocal rational subtangles)**:** If $T \cong M \circ v \cong M' \circ v'$ is a generalized Montesinos tangle, where M and M' are Montesinos tangles of opposite type related by a diagonal flip $M' \cong r(M)$, then the fractions of the rational subtangles in M and M' are negative reciprocals.

Proof:

Suppose that $T \cong M \circ v$ is a generalized Montesinos tangle, and let $T \cong M' \circ v'$ be the equivalent generalized Montesinos tangle of opposite type which exists by Lemma 5.29. Hence, we know that the two subtangles M and M' are related by a flip across the NW-SE diagonal axis: $M' \cong r(M)$. Suppose without loss of generality that $M \cong L_1 + \cdots + L_m$ is horizontal so that $M' \cong r(L_1) * \cdots * r(L_m)$ is vertical. Suppose that each rational subtangle in M has the fraction $L_i \cong \frac{p_i}{q_i}$. Hence, by Corollary 3.46, the corresponding subtangle $r(L_i)$ in M' has the fraction

$$r(L_i) \cong r(\tfrac{p_i}{q_i}) \cong R_{90}(R_H(\tfrac{p_i}{q_i})) \cong R_{90}(\tfrac{p_i}{q_i}) \cong -\frac{q_i}{p_i}.$$

This relationship between the fractions holds for each pair of corresponding flipped rational subtangles L_i and $r(L_i)$ in the constructions of M and M', respectively, which proves the claim.

□

This relationship holds for any choice of Montesinos subtangle M in a generalized Montesinos tangle $T \cong M \circ v$. If we include additional restrictions on the fractions of the rational subtangles in M, such as when M is in canonical form, then this relationship imposes corresponding restrictions on the fractions of the rational subtangles in M'. One of the properties of Montesinos tangles in canonical form is that all of the shared twists between rational subtangles have been flyped into an integer subtangle at the end of the construction formula. Since this integer tangle corresponds to horizontal or vertical twisting, these twists can be subsumed into the twist vector of a generalized Montesinos tangle. This enables the following formulation.

Lemma 5.31 (GM tangles with a canonical Montesinos subtangle)**:**
If T is any generalized Montesinos tangle, then there exists a canonical horizontal Montesinos tangle $M' \cong L_1 + \cdots + L_m$, a canonical vertical Montesinos tangle $M'' \cong r(L_1) * \cdots * r(L_m) \cong r(M')$, and twist vectors v' and v'' such that T satisfies the following two constructions.

a. $T \cong M' \circ v'$, where each rational subtangle $L_i \cong \frac{p_i}{q_i}$ satisfies $0 < \frac{p_i}{q_i} < 1$.

b. $T \cong M'' \circ v''$, where each rational subtangle $r(L_i) \cong -\frac{q_i}{p_i}$ satisfies $-\infty < -\frac{q_i}{p_i} < -1$.

Proof:

Let $T \cong M \circ v$ be a generalized Montesinos tangle defined by a composition of a Montesinos subtangle M with a twist vector v. By Lemma 5.29, we may assume without loss of generality that M is horizontal Montesinos. By Theorem 5.18, we may assume without loss of generality that M is in canonical form with a construction $M \cong L_1 + \cdots + L_m + \frac{k}{1}$, where each rational subtangle $L_i \cong \frac{p_i}{q_i}$ has a fraction satisfying $0 < \frac{p_i}{q_i} < 1$. To prove the claim, we will modify this construction slightly.

Let $v = (n_1, \cdots, n_r)$ denote the twist vector of $T \cong M \circ v$. Suppose without loss of generality that r is odd so that n_1 denotes horizontal twisting, possibly by taking $n_1 = 0$. Since the horizontal integer tangle $\frac{k}{1}$ on the end of the construction formula for M also denotes horizontal twisting, it can be combined with v using a composition of twist vectors as follows.

$$\begin{aligned} T &\cong M \circ v \\ &\cong (L_1 + \cdots + L_m + \tfrac{k}{1}) \circ (n_1, \cdots, n_r) \\ &\cong (L_1 + \cdots + L_m) \circ (k) \circ (n_1, \cdots, n_r) \\ &\cong (L_1 + \cdots + L_m) \circ (k + n_1, \cdots, n_r) \\ &\cong M' \circ v' \end{aligned}$$

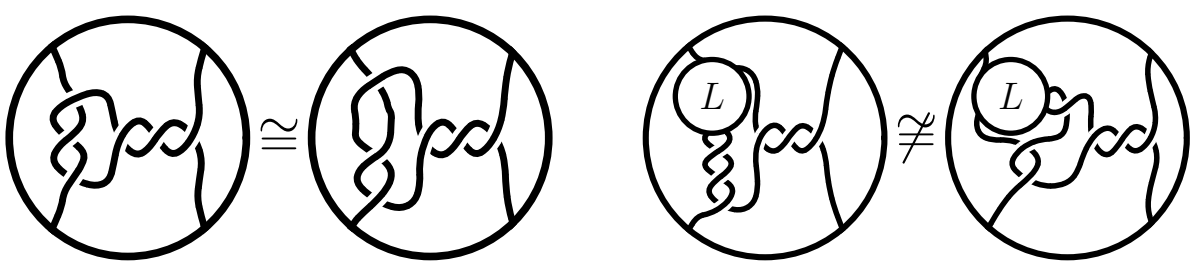
Let $M' \cong L_1 + \cdots + L_m$ denote the Montesinos subtangle of M obtained by removing the horizontal integer tangle $\frac{k}{1}$, and let $v' = (k + n_1, \cdots, n_r)$. Hence, $T \cong M' \circ v'$, where each rational subtangle $L_i \cong \frac{p_i}{q_i}$ of M' has a fraction satisfying $0 < \frac{p_i}{q_i} < 1$. This establishes the first part of the claim.

For the second part, we may leverage Lemma 5.29 to convert T from horizontal to vertical. Since $T \cong M' \circ v'$ is a horizontal generalized Montesinos tangle, there exists a twist vector v'' and a vertical Montesinos tangle $M'' \cong r(M') \cong r(L_1) * \cdots * r(L_m)$ such that $T \cong M'' \circ v''$.

Since M'' and M' are related by a diagonal flip, and M' was assumed to be in canonical form, Corollary 5.19 implies that M'' is also in canonical form. Furthermore, Corollary 5.30 implies that the fractions of their corresponding rational subtangles in M' and M'' are negative reciprocals of each other. Since each rational subtangle $L_i \cong \frac{p_i}{q_i}$ of M' has a fraction satisfying $0 < \frac{p_i}{q_i} < 1$, it follows that each rational subtangle $r(L_i) \cong -\frac{q_i}{p_i}$ of M'' has a fraction satisfying $-\infty < -\frac{q_i}{p_i} < -1$. This establishes the second part, completing the proof. □

Notice that the two equivalent generalized Montesinos tangles shown in Figure 5.27 satisfy the conditions of Lemma 5.31; the core Montesinos subtangle in each is in canonical form. Recall that we made a choice to use positive fractions in the canonical diagram for a horizontal Montesinos tangle and negative fractions in the canonical diagram for a vertical Montesinos tangle (Theorem 5.18). As a consequence of this choice, if M is in canonical form, then $r(M)$ is also in canonical form (Corollary 5.19). Since twist vector manipulations in a generalized Montesinos tangle diagram often involve a diagonal flip of the core Montesinos tangle, this property is a boon because canonicity is preserved. We may now build on this to extend the notion of canonical form to generalized Montesinos tangles.

To provide a classification of generalized Montesinos tangles, we must develop an understanding of the interactions between a subtangle and a composed twist vector. Much of this will be familiar from our work with rational tangles. However, the twist vector rules we summarized in Theorem 3.23 do not apply in the same way to a composition of a tangle with a twist vector, although there are similarities. In particular, for a composition of the form $L \circ (n_1, \cdots, n_r)$, the twist corresponding to the first entry n_1 is now attached to the subtangle



$(3,3) \cong (1,2,3)$

(a) Equivalent rational twist vectors

$L \circ (3,3) \ncong L \circ (1,2,3)$

(b) Non-equivalent composition twist vectors

Figure 5.28: Counterexample distinguishing rules for rational and composition twist vectors

L, and hence cannot be modified in the same way that we could for rational tangles (Figure 5.28). In all cases, we always assume that the final entry n_r denotes horizontal twisting.

We will make a distinction between the rules for *rational twist vectors* of Theorem 3.23 and the rules for *composition twist vectors* which we study now. If $v_1 \cong v_2$ are equivalent as rational twist vectors, this *does not imply* that $L \circ v_1 \cong L \circ v_2$. The counterexample of Figure 5.28 is one such example of this difference. Hence, before we proceed with our discussion of generalized Montesinos tangles specifically, we introduce some rules for composition twist vectors that hold in general. Although not identical, the rules for composition twist vectors parallel those for rational twist vectors, starting with collapsing zero entries.

Lemma 5.32 (Collapsing or inserting a leading composition twist vector zero)**:**
If $T \cong L \circ (n_1, n_2, \cdots, n_r)$ is a 2-string tangle, where $n_1 = 0$, then $T \cong L \circ (n_2, \cdots, n_r)$.

Proof:

Let $T \cong L \circ (n_1, n_2, \cdots, n_r)$ be a 2-string tangle defined by some subtangle L with endpoint twisting described by a twist vector. Suppose without loss of generality that r is odd so that n_1 denotes horizontal twisting. Suppose that $n_1 = 0$. We may use Lemma 3.7 to compute the following equivalent twist vector construction for T.

$$\begin{aligned} T &\cong L \circ (0, n_2, \cdots, n_r) \\ &\cong L + \tfrac{0}{1} * \tfrac{1}{n_2} + \cdots + \tfrac{n_r}{1} \\ &\cong L * \tfrac{1}{n_2} + \cdots + \tfrac{n_r}{1} \\ &\cong L \circ (n_2, \cdots, n_r) \end{aligned}$$

This establishes the equivalent formula in the claim. If r is even, the same formula follows by a parallel argument using $\frac{1}{0}$. This completes the proof.

□

The formula obtained above is different from the related result for rational twist vectors stated in Lemma 3.9. For rational twist vectors with a leading zero, the first two entries are both collapsed because the twists corresponding to n_2 denote nugatory crossings. This does not happen in a composition twist vector since these twists are now attached to the subtangle L. However, in the special case where L is a trivial tangle, then these twists become nugatory. In this case, the composition twist vector reduces to a rational twist vector. The next rule addresses interior twist vector zeros.

Lemma 5.33 (Collapsing or inserting interior composition twist vector zeros)**:**
If $T \cong L \circ (n_1, \cdots, n_r)$ is a 2-string tangle, where $n_i = 0$ for some $1 < i < r$, then equivalently $T \cong L \circ (n_1, \cdots, n_{i-1} + n_{i+1}, \cdots, n_r)$.

Proof:

Let $T \cong L \circ (n_1, \cdots, n_r)$ be a 2-string tangle defined by some subtangle L with endpoint twisting described by a twist vector. Suppose there exists an entry $n_i = 0$, where $1 < i < r$. Suppose without loss of generality that n_i corresponds to horizontal twisting. Similarly, suppose without loss of generality that r is odd so that n_1 also denotes horizontal twisting. Again using Lemma 3.7, we may expand the construction formula for T to compute an equivalent twist vector representation as follows.

$$\begin{aligned}
T &\cong L \circ (n_1, \cdots, n_{i-1}, 0, n_i, \cdots, n_r) \\
&\cong L + \tfrac{n_1}{1} * \cdots * \tfrac{1}{n_{i-1}} + \tfrac{0}{1} * \tfrac{1}{n_{i+1}} + \cdots + \tfrac{n_r}{1} \\
&\cong L + \tfrac{n_1}{1} * \cdots * \tfrac{1}{n_{i-1}} * \tfrac{1}{n_{i+1}} + \cdots + \tfrac{n_r}{1} \\
&\cong L + \tfrac{n_1}{1} * \cdots * \tfrac{1}{n_{i-1}+n_{i+1}} + \cdots + \tfrac{n_r}{1} \\
&\cong L \circ (n_1, \cdots, n_{i-1} + n_i, \cdots, n_r)
\end{aligned}$$

This establishes the formula in the claim. The choices we made concerning n_i and n_1 determined that we use horizontal integer tangles in the above construction formula, but a similar argument yields the same formula using vertical integer tangles in the opposite cases. This completes the proof.

□

The formula obtained above is consistent with the similar formula for rational twist vectors stated in Lemma 3.10. Unlike leading twist vector zeros, interior zeros can be collapsed in the same way for rational and composition twist vectors. The next rule we had for rational twist vectors involved incrementing the length by manipulating the first entry (Lemma 3.12). However, this only worked for rational twist vectors because the first twist in a rational tangle may be regarded as either horizontal or vertical (Figure 5.28a). This is not possible in a composition twist vector and so we do not have an analogous rule for this property.

The final and most unwieldy rule addresses adjacent twist vector entries of opposite sign. To make this proof more palatable, we will first establish the following auxiliary result to isolate a subtangle with endpoint twisting.

Lemma 5.34 (Identifying subtangles from a composition twist vector)**:**
If $T \cong L \circ (n_1, \cdots, n_r)$ is a 2-string tangle defined by a composition of a subtangle L with a twist vector of length r, then for each $0 \leq i \leq r$, the tangle M_i defined as

$$M_i \cong \begin{cases} L \circ (n_1, \cdots, n_i, 0) & , \quad \text{if } r-i \text{ is odd} \\ L \circ (n_1, \cdots, n_i) & , \quad \text{if } r-i \text{ is even} \end{cases}.$$

is a subtangle of T defined by the first i twists on L, and satisfies $T \cong M_i \circ (n_{i+1}, \cdots, n_r)$.

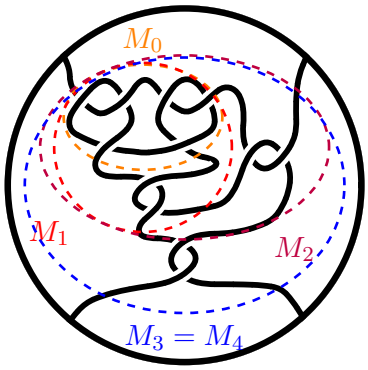
Proof:

Let $T \cong L \circ (n_1, \cdots, n_r)$ be a 2-string tangle defined by a composition of a subtangle L with a twist vector $v = (n_1, \cdots, n_r)$ of length r. Let $0 \leq i \leq r$ and define a tangle M_i by the formula in the claim. Note that $M_0 \cong L$ is clearly a subtangle of T when $i = 0$, so we may assume that $1 \leq i \leq r$. To prove the claim, we must show that M_i defined in this way arises as a subtangle in the construction formula for T.

Suppose without loss of generality that r is odd so that n_1 denotes horizontal twisting, which means that $\frac{n_1}{1}$ is used in the construction for T. The proof when r is even is identical except for starting with $\frac{1}{n_1}$ instead. There are two cases depending on whether $r-i$ is even or odd. If $r-i$ is odd, then n_i denotes vertical twisting in T; if $r-i$ is even, then n_i denotes horizontal twisting in T. By using Lemma 3.7 to expand the construction formula for T, we may consider these two cases simultaneously.

$$\begin{aligned} T &\cong L \circ (n_1, \cdots, n_r) \\ &\cong L + \tfrac{n_1}{1} * \cdots + \tfrac{n_r}{1} \\ &\cong \begin{cases} L + \frac{n_1}{1} * \cdots * \frac{1}{n_i} + \cdots + \frac{n_r}{1} & , \quad \text{if } r-i \text{ is odd} \\ L + \frac{n_1}{1} * \cdots + \frac{n_i}{1} * \cdots + \frac{n_r}{1} & , \quad \text{if } r-i \text{ is even} \end{cases} \\ &\cong \begin{cases} L \circ (n_1, \cdots, n_i, 0) \circ (n_{i+1}, \cdots, n_r) & , \quad \text{if } r-i \text{ is odd} \\ L \circ (n_1, \cdots, n_i) \circ (n_{i+1}, \cdots, n_r) & , \quad \text{if } r-i \text{ is even} \end{cases} \\ &\cong M_i \circ (n_{i+1}, \cdots, n_r) \end{aligned}$$

Hence, in both cases, M_i arises as the subtangle in T corresponding to the first i twists on L, which completes the proof. Figure 5.29 shows an example of a diagram with subtangles identified in this way. □



(a) $T \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2,2,2,0)$

$$\begin{array}{lcl} M_0 & \cong & \left(\frac{2}{3} + \frac{2}{3}\right) \\ M_1 & \cong & \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2,0) \\ M_2 & \cong & \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2,2) \\ M_3 & \cong & \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2,2,2,0) \\ M_4 & \cong & \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2,2,2,0) \end{array} \qquad \begin{array}{lcl} T & \cong & M_0 \circ (2,2,2,0) \\ T & \cong & M_1 \circ (2,2,0) \\ T & \cong & M_2 \circ (2,0) \\ T & \cong & M_3 \circ (0) \\ T & \cong & M_4 \end{array}$$

(b) Subtangles obtained from Lemma 5.34

Figure 5.29: Example of subtangles identified from a composition twist vector

This result can be leveraged to show how two adjacent entries of opposite sign in a composition twist vector can be converted into three adjacent entries of the same sign. By isolating the first portion of the diagram into a subtangle, this conversion is accomplished using a flupe via Lemma 3.20. The difficulty in stating this result elegantly is due to the fact that flupes can occur in four different ways depending on twist sign and type.

Lemma 5.35 (Converting adjacent composition twist vector entries of opposite sign)**:** If $T \cong L \circ (n_1, \cdots, n_r)$ is a 2-string tangle defined by a composition of a subtangle L with a twist vector $v = (n_1, \cdots, n_r)$, and there exist adjacent entries $\pm n_i$ and $\mp n_{i+1}$ of opposite sign in v, then T satisfies the following equivalence.

$$\begin{aligned} T &\cong L \circ (n_1, \cdots, n_{i-1}, \underbrace{\pm n_i, \mp n_{i+1}}_{\text{opposite sign}}, n_{i+2}, \cdots, n_r) \\ &\cong r(L) \circ (-n_1, \cdots, -n_{i-1}, \underbrace{\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1}_{\text{same sign}}, n_{i+2}, \cdots, n_r) \end{aligned}$$

Proof:

Let $T \cong L \circ (n_1, \cdots, n_r)$ be a 2-string tangle defined by a composition of a subtangle L with a twist vector $v = (n_1, \cdots, n_r)$ of length r. Suppose that there exists a pair of adjecent entries $\pm n_i$ and $\mp n_{i+1}$ in v which have opposite sign. Let M_{i-1} be the subtangle corresponding to the first $i-1$ twists obtained by Lemma 5.34. Hence, $T \cong M_{i-1} \circ (\pm n_i, \mp n_{i+1}, \cdots, n_r)$.

Starting from this expression, we may modify the construction of T as follows.

$$
\begin{aligned}
T &\cong L \circ (n_1, \cdots, n_{i-1}, \underbrace{\pm n_i, \mp n_{i+1}}_{\text{opposite sign}}, n_{i+2}, \cdots, n_r) \\
&\cong M_{i-1} \circ (\pm n_i, \mp n_{i+1}, n_{i+2}, \cdots, n_r) \qquad \text{(Lemma 5.34)} \\
&\cong \begin{cases} M_{i-1} \circ (\underbrace{\pm n_i}_{\text{horiz.}}, \underbrace{\mp n_{i+1}}_{\text{vert.}}, n_{i+2}, \cdots, n_r) & , \quad \text{if } r-i \text{ is even} \\ M_{i-1} \circ (\underbrace{\pm n_i}_{\text{vert.}}, \underbrace{\mp n_{i+1}}_{\text{horiz.}}, n_{i+2}, \cdots, n_r) & , \quad \text{if } r-i \text{ is odd} \end{cases} \\
&\cong \begin{cases} M_{i-1} \circ (\pm n_i, \mp n_{i+1}, 0) \circ (n_{i+2}, \cdots, n_r) & , \text{ even} \\ M_{i-1} \circ (\pm n_i, \mp n_{i+1}) \circ (n_{i+2}, \cdots, n_r) & , \text{ odd} \end{cases} \text{(Lemma 3.8)} \\
&\cong \begin{cases} r(M_{i-1}) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, 0) \circ (n_{i+2}, \cdots, n_r) & , \text{ even} \\ r(M_{i-1}) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1) \circ (n_{i+2}, \cdots, n_r) & , \text{ odd} \end{cases} \text{(Lemma 3.20)} \\
&\cong \begin{cases} r(M_{i-1}) \circ (\underbrace{\mp n_i \pm 1}_{\text{vert.}}, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & , \text{ even} \\ r(M_{i-1}) \circ (\underbrace{\mp n_i \pm 1}_{\text{horz.}}, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & , \text{ odd} \end{cases} \text{(Lemma 3.8)}
\end{aligned}
$$

The final two expressions above look the same, but recall that there there are two possibilities for M_{i-1} depending on whether $r-(i-1)$ is even or odd. These give slight variations in how we compute $r(M_{i-1})$ in terms of L.

$$
\begin{aligned}
r(M_{i-1}) &\cong \begin{cases} r(L \circ (n_1, \cdots, n_{i-1}, 0)) & , \quad \text{if } r-(i-1) \text{ is odd} \\ r(L \circ (n_1, \cdots, n_{i-1})) & , \quad \text{if } r-(i-1) \text{ is even} \end{cases} \qquad \text{(Lemma 5.34)} \\
&\cong \begin{cases} r(L) \circ (-n_1, \cdots, -n_{i-1}, 0, 0) & , \quad \text{if } r-(i-1) \text{ is odd} \\ r(L) \circ (-n_1, \cdots, -n_{i-1}, 0) & , \quad \text{if } r-(i-1) \text{ is even} \end{cases} \qquad \text{(Lemma 5.26)} \\
&\cong \begin{cases} r(L) \circ (-n_1, \cdots, -n_{i-1}) & , \quad \text{if } r-(i-1) \text{ is odd} \\ r(L) \circ (-n_1, \cdots, -n_{i-1}, 0) & , \quad \text{if } r-(i-1) \text{ is even} \end{cases} \qquad \text{(Lemma 5.33)}
\end{aligned}
$$

A composition of twist vectors changes depending on the length of the composed twist vector. However, the two possibilities for $r(M_{i-1})$ will result in these yielding the same

formula. Thus, we may finish our computation of T as follows.

$$
\begin{aligned}
T &\cong \begin{cases} r(M_{i-1}) \circ (\underbrace{\mp n_i \pm 1}_{\text{vert.}}, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & \text{, if } r-i \text{ is even} \\ r(M_{i-1}) \circ (\underbrace{\mp n_i \pm 1}_{\text{horz.}}, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & \text{, if } r-i \text{ is odd} \end{cases} \\
&\cong \begin{cases} (-n_1, \cdots, -n_{i-1}) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & \text{, even} \\ (-n_1, \cdots, -n_{i-1}, 0) \circ (\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1, n_{i+2}, \cdots, n_r) & \text{, odd} \end{cases} \\
&\cong r(L) \circ (-n_1, \cdots, -n_{i-1}, \underbrace{\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1}_{\text{same sign}}, n_{i+1}, \cdots, n_r) \qquad \text{(Lemma 3.8)}
\end{aligned}
$$

This establishes the formula in the claim, completing the proof.

□

This result parallels Lemma 3.21 for rational twist vectors, and the two proofs are very similar. In both cases, the twist vector length increases by one and the leading entries have their sign reversed. In the case of a composition twist vector, the subtangle L is also flipped across the diagonal. In both cases, the total number of crossings in the diagram decreases by one. As we did with rational twist vectors, we may apply this result multiple times to convert all of the entries in a composition twist vector to the same sign. Figure 5.30 shows an example of this for a generalized Montesinos tangle, preempting our definition of a canonical form for this class of tangle diagram.

(a) $T \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (3, -3, 3)$ **(b)** $T \cong r\left(\frac{2}{3} + \frac{2}{3}\right) \circ (-3, 2, 1, 2)$ **(c)** $T \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2, 1, 1, 1, 2)$

Figure 5.30: Converting a generalized Montesinos tangle to canonical form

Lemma 5.36 (Composition twist vectors of constant sign)**:**
If $T \cong L \circ v$ is any 2-string tangle defined by a composition of a subtangle L with a twist vector v, then there exists a tangle L' and a twist vector v' such that $T \cong L' \circ v'$, where either $L' \cong L$ or $L' \cong r(L)$, and the nonzero entries in v' all have constant sign.

Proof:

Let $T \cong L \circ v$ be any 2-string tangle defined by a composition of a subtangle L with a twist vector v. By Lemmas 5.32 and 5.33, we may assume without loss of generality that all of the leading and interior zeros in v have been collapsed. If the nonzero entries in v do not already have constant sign, then our goal is to convert v into a twist vector satisfying this property through successive applications of Lemma 5.35.

Note that, while Lemma 5.35 reverses the sign of the leading twists, the outermost twists are left unchanged. Hence, starting from the outermost twists and working our way in, we may check to see if every pair of adjacent twist vector entries has the same sign, and if not, convert the sign using the preceding lemma. We may repeat this process as many times as needed to convert all of the entries to the same sign, possibly pausing between iterations to collapse interior zeros that may have been introduced by Lemma 5.35. Since each application of the lemma reduces the total number of crossings by one, this algorithm will eventually terminate with a diagram of the form $T \cong L' \circ v'$, where v' is a twist vector where all of the nonzero entries have the same sign.

The exact details of the constructions involved in this algorithm are nearly identical to those in the proof of Lemma 3.22, and so we suppress those details here. However, because we are working with a composition twist vector rather than a rational twist vector, each iteration flips the subtangle corresponding to L across the diagonal. Since two diagonal flips cancel each other, it follows that the final subtangle L' obtained in this way is either $L' \cong L$ if the total number of iterations is even and $L' \cong r(L)$ if the total number of iterations is odd. This establishes the result stated in the claim.

□

Finally, to mirror the rational twist vector rules of Theorem 3.23, we may organize some of the preceding results into a theorem summarizing the rules for manipulating composition twist vectors. Any two 2-string tangle diagrams which are both defined by a composition of a subtangle with a twist vector, where these compositions satisfy one of these relationships, will define equivalent tangles, possibly with a diagonal flip on the subtangle in the last rule.

Theorem 5.37 (Summary of rules for equivalent composition twist vectors)**:**
If two *composition twist vectors* are related by the operations of:

a. collapsing or inserting a leading zero (Lemma 5.32):
$L \circ (0, n_2, \cdots, n_r) \cong L \circ (n_2, \cdots, n_r)$;

b. collapsing or inserting an interior zero (Lemma 5.33):
$L \circ (n_1, \cdots, n_{i-1}, 0, n_{i+1}, \cdots, n_r) \cong L \circ (n_1, \cdots, n_{i-1} + n_{i+1}, \cdots, n_r)$;

c. or converting two adjacent entries of opposite sign to three adjacent entries of the same sign, with a consequent sign reversal on leading twists and a subtangle flip across the diagonal (Lemma 5.35):

$$\begin{aligned} T &\cong L \circ (n_1, \cdots, n_{i-1}, \underbrace{\pm n_i, \mp n_{i+1}}_{\text{opposite sign}}, n_{i+2}, \cdots, n_r) \\ &\cong r(L) \circ (-n_1, \cdots, -n_{i-1}, \underbrace{\mp n_i \pm 1, \mp 1, \mp n_{i+1} \pm 1}_{\text{same sign}}, n_{i+1}, \cdots, n_r); \end{aligned}$$

then these two compositions define equivalent 2-string tangle diagrams.

Using these rules, it is possible to convert a generic composition twist vector into a preferred form, much like we did with rational twist vectors in Lemma 3.24. Note that the rules of Theorem 5.37 apply to any composition of a subtangle with a twist vector, not just to generalized Montesinos tangles, and so this concept can be applied more generally. With this idea in mind, we introduce the idea of a *canonical composition twist vector* which has constant sign, no interior zeros, and odd length. Since Lemma 5.36 already establishes the first of these three conditions, obtaining the remaining two is almost immediate.

Corollary 5.38 (Canonical composition twist vectors)**:**
If $T \cong L \circ v$ is any 2-string tangle defined using a twist vector composition, then there exists a tangle L' and a *canonical composition twist vector* v' such that $T \cong L' \circ v'$, where either $L' \cong L$ or $L' \cong r(L)$, and v' has constant sign, no interior zeros, and odd length.

Proof:

Let $T \cong L \circ v$ be a 2-string tangle defined by a composition of a subtangle L with a twist vector v. By Lemma 5.36, there exists a tangle L' and a twist vector of constant sign v' such that $T \cong L' \circ v'$, where either $L' \cong L$ or $L' \cong r(L)$. If v' has any leading or interior zeros, let

v'' be the twist vector obtained by collapsing these using the first two rules of Theorem 5.37, so that that $T \cong L' \circ v''$. If v'' has even length, then we may use the first rule to obtain an equivalent composition twist vector v''' of odd length by inserting a leading zero. Otherwise, take $v''' = v''$. Thus, the composition twist vector v''' obtained in this way has constant sign, no interior zeros, and odd length, which is to say that v''' is a canonical composition twist vector. Furthermore, this twist vector satisfies $T \cong L' \circ v'''$, proving the claim.

□

In the context of generalized Montesinos tangles, we may combine the canonical Montesinos subtangle from Lemma 5.31 with the canonical composition twist vector from Corollary 5.38. Taking these ideas together will allow us to prove the existence of a canonical diagram for generalized Montesinos tangles, as illustrated in Figure 5.30. The proof itself is quite simple thanks to previous work.

Theorem 5.39 (Existence of canonical diagrams for GM tangles [31])**:**
Every non-rational generalized Montesinos tangle T admits a *canonical diagram* defined by a canonical composition twist vector v and either a canonical horizontal Montesinos tangle M or a canonical vertical Montesinos tangle $r(M)$, satisfying one of the following constructions.

a. $T \cong M \circ v$, where $M \cong L_1 + \cdots + L_m$ is a horizontal Montesinos tangle diagram in canonical form where each rational subtangle $L_i \cong \frac{p_i}{q_i}$ satisfies $0 < \frac{p_i}{q_i} < 1$.

b. $T \cong r(M) \circ v$, where $r(M) \cong r(L_1) * \cdots * r(L_m)$ is a vertical Montesinos tangle diagram in canonical form where each rational subtangle $r(L_i) \cong -\frac{q_i}{p_i}$ satisfies $-\infty < -\frac{q_i}{p_i} < -1$.

Proof:

Let $T \cong M \circ v$ be a generalized Montesinos tangle defined by a composition of a Montesinos tangle M with a twist vector v. By Lemma 5.31, we may assume without loss of generality that $M \cong L_1 + \cdots + L_m$ is a horizontal Montesinos tangle in canonical form, where each rational subtangle $L_i \cong \frac{p_i}{q_i}$ has a fraction satisfying $0 < \frac{p_i}{q_i} < 1$. Furthermore, we know that the diagonal reflection of M is a vertical Montesinos tangle $r(M) \cong r(L_1) * \cdots * r(L_m)$ in canonical form (Corollary 5.19), and the fractions of the rational subtangles $r(L_i) \cong -\frac{q_i}{p_i}$ satisfy $-\infty < -\frac{q_i}{p_i} < -1$ (Corollary 5.30). By Corollary 5.38, there exists a tangle M' and a canonical composition twist vector v' such that $T \cong M' \circ v'$, where either $M' \cong M$ or $M' \cong r(M)$. Whichever possibility holds proves the claim.

□

5.3.3 Generalized Montesinos Tangles in Canonical Form

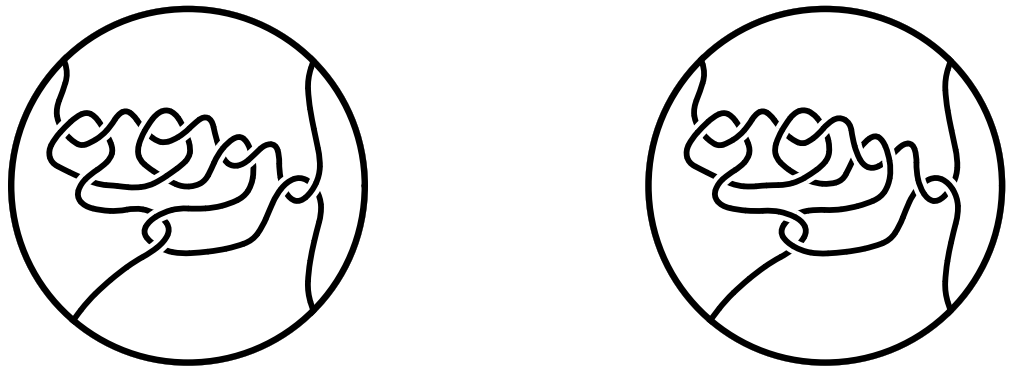
In the special case where T is a Montesinos tangle, observe that the canonical diagram of Theorem 5.39 aligns with the canonical diagram of Theorem 5.18, except that the twisting corresponding to the final integer tangle is now represented using a twist vector. For a horizontal Montesinos tangle in canonical form, these two representations are denoted as $L_1 + \cdots + L_m + \frac{k}{1} \cong (L_1 + \cdots + L_m) \circ (k)$, but these define exactly the same diagram. For a vertical Montesinos tangle in canonical form, the two representations take the form $L_1 * \cdots * L_m * \frac{1}{k} \cong (L_1 * \cdots * L_m) \circ (0, k, 0)$, where the twist vector is padded with zeros to ensure that $\frac{1}{k}$ properly represents vertical twisting while satisfying the odd length convention.

While every generalized Montesinos tangle can be described using both horizontal and vertical Montesinos tangles in canonical form with endpoint twisting (Lemma 5.31), only one of these constructions is possible using a canonical composition twist vector. Hence, it is still worthwhile to distinguish between horizontal and vertical generalized Montesinos tangles. As with the canonical form of a Montesinos tangle diagram, the canonical form for generalized Montesinos tangles is unique. The proof of this fact is due to Moon and Darcy [31] and will not be included here.

Theorem 5.40 (Uniqueness of canonical diagrams for GM tangles [31])**:**
The canonical diagram of a non-rational generalized Montesinos tangle is unique.

Thus, the formulation introduced in Theorem 5.39 provides a unique classification of all non-rational generalized Montesinos tangles. The restriction to non-rational tangles is again to avoid inconsistencies with the canonical form of a rational tangle diagram. However, if M is taken to be rational, then $T \cong M \circ v$ is also rational. If M is defined by the rational twist vector v_M, then $T \cong M \circ v \cong v_M \circ v$ defines a twist vector construction of T. Note that $v_M \circ v$ defined in this way is equivalent to a canonical rational twist vector, but it may not be canonical itself. In the special case where M is a trivial tangle, then $T \cong M \circ v \cong v$, and v may be regarded as both a composition twist vector and a rational twist vector.

We saw previously that the canonical diagram of a Montesinos tangle may be non-alternating (Corollary 5.21). A similar fact holds for the canonical diagram of a generalized Montesinos tangle whenever the twist vector and the Montesinos subtangle have opposite sign (Figure 5.31). Since the Montesinos subtangle is taken to be positive if horizontal and negative if vertical, this alternating behavior can be characterized in the following way.



(a) Alternating: $T_1 \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2, 2, 2)$ **(b)** Non-alternating: $T_2 \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (-2, -2, -2)$

Figure 5.31: Alternating and non-alternating canonical generalized Montesinos diagrams

Corollary 5.41 (Alternating canonical generalized Montesinos diagrams)**:**
If $T \cong M \circ v$ is a generalized Montesinos tangle diagram in canonical form, then the alternating behavior of T depends on whether the sign of M agrees with the sign of v.

a. If M is horizontal, then T is alternating if and only if v is non-negative.

b. If M is vertical, then T is alternating if and only if v is non-positive.

Proof:

Let $T \cong M \circ v$ be a non-rational generalized Montesinos tangle diagram in canonical form (Theorem 5.39). Hence, M is a Montesinos tangle in canonical form and v is a canonical composition twist vector. By construction, the rational subtangles used to define M all have the same sign; the integer subtangle at the end is taken to be trivial. Thus, M taken by itself is guaranteed to be alternating (Corollary 5.21). If v contains no non-zero twists, then it follows immediately that $T \cong M$ is alternating regardless of whether M is horizontal or vertical.

Suppose that $v = (n_1, \cdots, n_r)$ contains at least one nonzero twist. Since v is a canonical twist vector, its length r is odd, and its nonzero twists have constant sign. By Lemma 3.7, we have the construction $T \cong M \circ v \cong M + \frac{n_1}{1} * \cdots + \frac{n_r}{1}$. If v is non-negative, then the fraction of each of these integer subtangles, horizontal or vertical, satisfies $\frac{p}{q} \geq 0$. Likewise, if v is non-positive, then the fraction of each of these integer subtangles satisfies $\frac{p}{q} \leq 0$.

If M is a horizontal Montesinos tangle, then $M \cong L_1 + \cdots + L_m$, where each rational

subtangle $L_i \cong \frac{p_i}{q_i}$ satisfies $\frac{p_i}{q_i} \geq 0$. Furthermore, this implies that T has the algebraic construction formula $T \cong (L_1 + \cdots + L_m) + \frac{n_1}{1} * \cdots + \frac{n_r}{1}$. Therefore, by Corollary 5.5, T is under-alternating if and only if v is non-negative. If M is a vertical Montesinos tangle, then a similar argument using Corollary 5.5 shows that T is over-alternating if and only if v is non-positive. In both cases, the canonical diagram for T is alternating.

This establishes the forward direction of both parts of the claim. If M is horizontal with negative v, or if M is vertical with positive v, then the algebraic construction formula for T contains both a positive rational subtangle and a negative rational subtanlge. In this case, T is non-alternating by Corollary 5.5. This establishes the backward direction of both parts of the claim by the contrapositive.

□

Recall that Lemma 5.22 gives a condition for when a non-alternating canonical Montesinos tangle diagram is equivalent to an alternating diagram which is non-canonical. Since a canonical generalized Montesinos tangle contains a canonical Montesinos subtangle, we can extend this trick to generalized Montesinos tangles by borrowing crossings from the twist vector and grouping these into an integer tangle connected to the Montesinos subtangle (Figure 5.32).

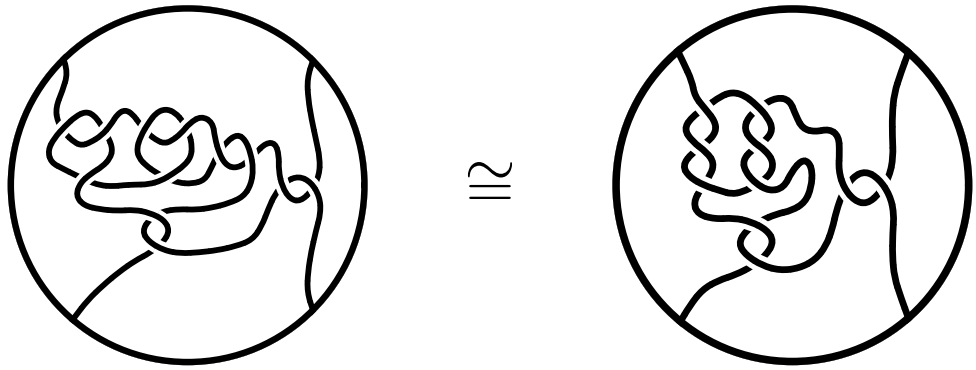
Lemma 5.42 (Alternating negative generalized Montesinos diagrams)**:**
If $T \cong M \circ v$ is a non-rational generalized Montesinos tangle diagram in canonical form, where $v = (n_1, \cdots, n_r)$ is a canonical composition twist vector of constant sign, then there exists an equivalent alternating generalized Montesinos tangle diagram in the following cases.

a. If $M \cong L_1 + \cdots + L_m$, where $n_1 \leq -m$, then there exists an equivalent alternating generalized Montesinos tangle diagram $T \cong (L'_1 + \cdots + L'_m) \circ (n_1 + m, n_2, \cdots, n_r)$.

b. If $M \cong L_1 * \cdots * L_m$, where $n_1 = 0$ and $n_2 \geq m$, then there exists an equivalent alternating generalized Montesinos tangle diagram $T \cong (L'_1 * \cdots * L'_m) \circ (0, n_2 - m, n_3, \cdots, n_r)$.

Proof:

Let $T \cong M \circ v$ be a non-rational generalized Montesinos tangle diagram in canonical form (Theorem 5.39). Let $v = (n_1, \cdots, n_r)$ denote the canonical composition twist vector of constant sign used in this construction. Note that M is a canonical Montesinos tangle which, by itself, has an alternating diagram. Suppose that T is non-alternating. There are two cases to consider depending on whether M is horizontal or vertical.



(a) Canonical but non-alternating: $T \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (-2, -2, -2)$

(b) Non-canonical but alternating: $T \cong \left(\frac{-1}{3} + \frac{-1}{3}\right) \circ (0, -2, -2)$

Figure 5.32: Example of converting a non-alternating canonical GM diagram

Suppose first that $M \cong L_1 + \cdots + L_m$ is horizontal Montesinos, and hence each rational subtangle $L_i \cong \frac{p_i}{q_i}$ has a fraction satisfying $0 < \frac{p_i}{q_i} < 1$. Since T is non-alternating, v must be non-positive by Corollary 5.41. By grouping the first twists from v into M, observe that we may obtain the following expression for T:

$$T \cong M \circ v \cong (L_1 + \cdots + L_m) \circ (n_1, \cdots, n_r) \cong (L_1 + \cdots + L_m + \tfrac{n_1}{1}) \circ (0, n_2, \cdots, n_r) \cong M' \circ v'.$$

Let $M' \cong L_1 + \cdots + L_m + \frac{n_1}{1}$ and $v' \cong (0, n_2, \cdots, n_r)$ denote the modified Montesinos subtangle and composition twist vector obtained in this way. Note that M' is in canonical form, but is non-alternating if $n_1 < 0$ (Corollary 5.21).

Since v is non-positive, this means that $n_1 \leq 0$. If $n_1 \leq -m$, then we may use Lemma 5.22 to obtain an equivalent construction $M' \cong L'_1 + \cdots + L'_m + \frac{n_1+m}{1}$, which is an alternating Montesinos diagram. Thus, we may use this expression for M' to rewrite the construction for T as follows:

$$T \cong M' \circ v' \cong (L'_1 + \cdots + L'_m + \tfrac{n_1+m}{1}) \circ (0, n_2, \cdots, n_r) \cong (L'_1 + \cdots + L'_m) \circ (n_1 + m, n_2, \cdots, n_r) \cong M'' \circ v''.$$

Let $M'' \cong L'_1 + \cdots + L'_m$ and $v'' \cong (n_1 + m, n_2, \cdots, n_r)$ denote this second modified Montesinos subtangle and composition twist vector obtained in this way. Since $n_1 \leq -m$, note that $n_1 + m \leq 0$. Hence, v'' is still non-positive. Furthermore, the transformed subtangles L'_i obtained through Lemma 5.22 are all negative. Hence, we have an algebraic construction

formula $T \cong (L'_1 + \cdots + L'_m) + \frac{n_1+m}{1} * \cdots + \frac{n_r}{1}$ in which every rational subtangle has a non-positive fraction. Therefore, this defines an over-alternating diagram for T by Corollary 5.5. This establishes the first part of the claim.

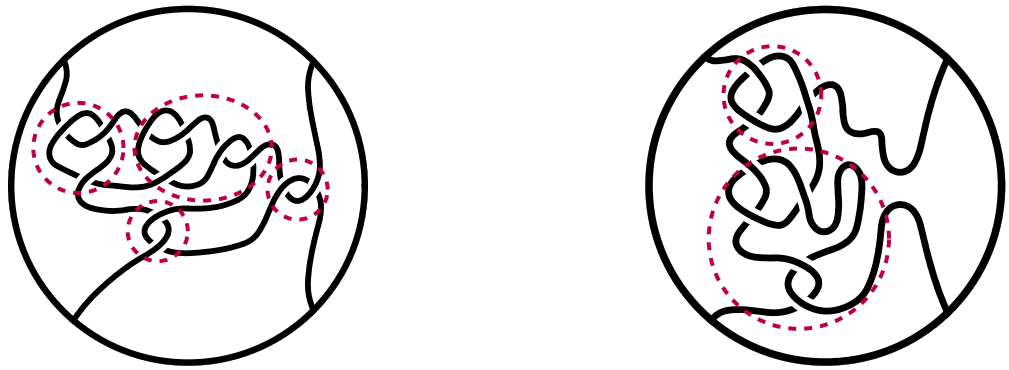
If we suppose instead that $M \cong L_1 * \cdots * L_m$ is vertical Montesinos, and that the first two entries of the twist vector $v = (n_1, \cdots, n_r)$ satisfy $n_1 = 0$ and $n_2 \geq m$, then the second part of the claim follows by a parallel argument. Since a canonical composition twist vector is assumed to have odd length, n_1 denotes horizontal twisting. Hence, the assumption that $n_1 = 0$ is to ensure that the first non-zero twist is vertical, in which case the twisting from n_2 can be grouped into M as a vertical integer tangle. Other than this detail, the argument for the second part of the claim is essentially identical to the first, but instead using the results for vertical Montesinos tangles.

□

On top of possibly being non-alternating, the canonical diagram for a generalized Montesinos tangle may also be non-reduced. In addition to their classification theorem for generalized Montesinos tangles (Theorem 5.39), Moon and Darcy also prove how every generalized Montesinos tangle which is not rational can be uniquely represented by a minimal diagram [31]. While we do not reproduce the proof here, we pause to make some comments about this result. This characterization also accounts for minimal diagrams for Montesinos tangles.

There are four possibilities for the unique minimal crossing diagram of a non-rational generalized Montesinos tangle. Two of these are identical to the form of a canonical diagram characterized in Theorem 5.39, but only when these canonical diagrams are alternating. Corollary 5.41 shows that there exist canonical generalized Montesinos diagrams which are non-alternating. These are non-minimal, but there are two ways in which such a diagram can be converted into a non-canonical diagram which is minimal. In these cases, the construction of a minimal diagram uses both positive and negative rational subtangles in the Montesinos subtangle, although the twist vector has constant sign.

While minimal diagrams are nice, they are not required for tabulation. Hence, we will prefer to keep with the slightly simpler canonical diagrams from Theorem 5.39. To conclude our discussion of generalized Montesinos tangles, we show how the canonical form can be used to obtain a unique decomposition into maximal rational subtangles (Figure 5.33).



(a) $T_1 \cong \left(\frac{2}{3} + \frac{2}{3}\right) \circ (2,2,2)$ with $D_{\text{rat}} = \left\{\frac{2}{3}, \frac{8}{3}, \frac{1}{2}, \frac{2}{1}\right\}$ **(b)** $T_2 \cong \left(\frac{-3}{2} * \frac{-3}{2}\right) \circ (0,-2,0)$ with $D_{\text{rat}} = \left\{\frac{-3}{2}, \frac{-3}{8}\right\}$

Figure 5.33: Maximal rational subtangle decompositions for canonical GM diagrams

Theorem 5.43 (GM decomposition into maximal rational subtangles)**:**
A canonical diagram for a generalized Montesinos tangle $T \cong M \circ v$ admits a unique decomposition D_{rat} into maximal rational subtangles based on its canonical construction formula.

a. If M is horizontal Montesinos so that $T \cong (L_1 + \cdots + L_m) \circ (n_1, \cdots, n_r)$, then $D_{\text{rat}} = \{L_1, \cdots, L_m + \frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}\}$.

b. If M is vertical Montesinos so that $T \cong (L_1 * \cdots * L_m) \circ (n_1, \cdots, n_r)$, then either $D_{\text{rat}} = \{L_1, \cdots, L_m, \frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}\}$, or $D_{\text{rat}} = \{L_1, \cdots, L_m * \frac{1}{n_2}, \frac{n_3}{1}, \cdots, \frac{n_r}{1}\}$ if $n_1 = 0$.

In the special case where $n_r = 0$, we exclude the trivial tangle $\frac{n_r}{1}$ from D_{rat}.

Proof:

Let $T \cong M \circ v$ be a generalized Montesinos tangle diagram in canonical form (Theorem 5.39). Let $v = (n_1, \cdots, n_r)$ denote the corresponding canonical composition twist vector. Recall that a decomposition D_{rat} into maximal rational subtangles may not exist for a given diagram, but that it is unique if it does exist (Lemma 4.11). Hence, to prove the claim, it is sufficient to show that each subtangle in the proposed decompositions is a maximal rational subtangle. In the special case where M is rational, then $T \cong M \circ v$ is also rational, and hence $D_{\text{rat}} = \{T\} = D_{\max}$ (Theorem 4.12). For generalized Montesinos tangles more generally, we may suppose that M is not rational, and hence it is constructed from a sum or product of two or more rational subtangles (Corollary 5.12).

There are two cases to consider depending on whether the Montesinos subtangle M is horizontal or vertical. In both cases, M is a canonical Montesinos subtangle defined using a sum or product of rational subtangles $L_1, \cdots, L_m$. As a consequence of $T \cong M \circ v$ being in canonical form, the integer tangle at the end of the canonical Montesinos construction for M is assumed to be trivial. Let $D_M = \{L_1, \cdots, L_m\}$ be the unique decomposition of M into maximal rational subtangles which exists by Theorem 5.23. Note that $m \geq 2$ since we assumed that M is constructed from two or more rational subtangles.

Suppose first that $M \cong L_1 + \cdots + L_m$ is horizontal Montesinos. Hence, the construction formula for T is

$$T \cong M \circ v \cong (L_1 + \cdots + L_m) \circ (n_1, \cdots, n_r) \cong (L_1 + \cdots + L_m) + \tfrac{n_1}{1} * \cdots + \tfrac{n_r}{1}.$$

Let $D = \{L_1, \cdots, L_m + \frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}\}$ be the decomposition of T into rational subtangles based on this formula. Each subtangle $L_i \in D_M$ is a maximal rational subtangle of M and hence automatically a maximal rational subtangle of T, with the exception of L_m, which is contained in the rational subtangle $L_m + \frac{n_1}{1}$. Hence, for $1 \leq i < m$, we take $L_i \in D$, but we replace L_m with $L_m + \frac{n_1}{1}$.

Although $L_m + \frac{n_1}{1}$ is properly contained in $L_j + \cdots + L_m + \frac{n_1}{1}$ for $1 \leq j < m$, none of these subtangles is rational by the canonicity of M. This could only happen if all but at most one of the rational subtangles $L_j, \cdots, L_m$ is a horizontal integer tangle (Corollary 5.12), but this would contradict the assumed maximality of these subtangles in M. Since $m \geq 2$, then by the same logic, no subtangle of the form $(L_1 + \cdots + L_m) + \frac{n_1}{1} * \cdots$ is rational. Since the only 2-string subtangles of T which properly contain $L_m + \frac{n_1}{1}$ have one of these two forms, we may conclude that $L_m + \frac{n_1}{1}$ is a maximal rational subtangle of T.

It remains to consider the maximality of the integer subtangles $\frac{1}{n_2}, \cdots, \frac{n_r}{1} \in D$. Each of these is a rational subtangle of T. However, for each such horizontal tangle $\frac{n_i}{1}$, the only 2-string subtangles of T which properly contain it have the form $M_j \cong M + \frac{n_1}{1} * \cdots + \frac{n_i}{1} \cdots$, where $i \leq j \leq r$ (Lemma 5.34). Each of these M_j subtangles is non-rational because it contains the non-rational subtangle M (Lemma 3.28). Similarly, the only 2-string subtanlges properly containing a vertical tangle $\frac{1}{n_i}$ have the form $M_j \cong M + \frac{n_1}{1} * \cdots * \frac{1}{n_i} \cdots$, where $i \leq j \leq r$, but these fail to be rational by the same logic. Therefore, each integer subtangle $\frac{1}{n_2}, \cdots, \frac{n_r}{1} \in D$ is a maximal rational subtangle of T.

Thus, we may conclude that $D = \{L_1, \cdots, L_m + \frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}\} = D_{\text{rat}}$ is the unique decomposition of T into maximal rational subtangles, which proves the claim when M is a

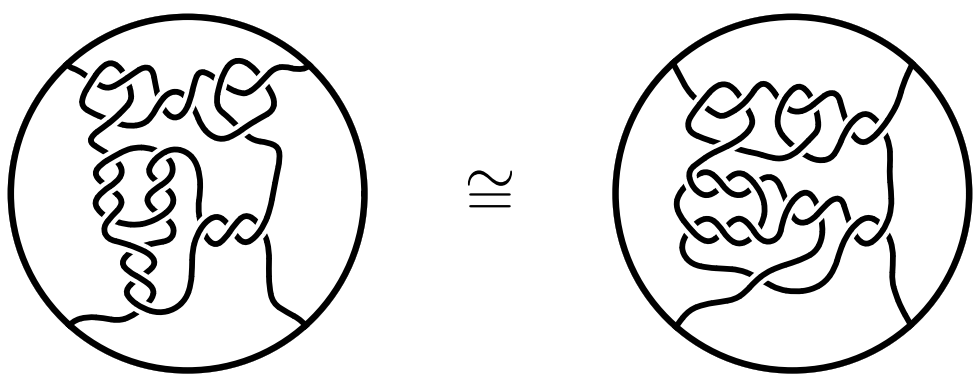
horizontal Montesinos subtangle. If $n_r = 0$, then $\frac{n_r}{1}$ is a trivial tangle and hence can be removed from the construction. In this case, we take $D_{\text{rat}} = \{L_1, \cdots, L_m + \frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{1}{n_{r-1}}\}$.

The case when $M = L_1 * \cdots * L_m$ is a vertical Montesinos tangle is similar with some slight modifications. In the special case when $n_1 = 0$, then $\frac{n_1}{1}$ is a trivial tangle and may be removed from the construction formula. When this happens, $L_m * \frac{1}{n_2}$ is a rational subtangle (Figure 5.33b). The proof that $D = \{L_1, \cdots, L_m * \frac{1}{n_2}, \frac{n_3}{1}, \cdots, \frac{n_r}{1}\} = D_{\text{rat}}$ then follows by an argument parallel to the one given above.

If $n_1 \neq 0$, then because $m \geq 2$, the subtangle L_m is not connected to one of the integer subtangles by a sum or product, which means that L_m is not properly contained in a larger rational subtangle. In this case, $D_{\text{rat}} = \{L_1, \cdots, L_m, \frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}\}$. As before, in both cases we exclude the trivial tangle $\frac{n_r}{1}$ from D_{rat} in the special case when $n_r = 0$.

□

This concludes our discussion of those known special classes of algebraic tangle which are completely classified via a unique canonical diagram. In the same way that the Montesinos canonical diagram built on the rational canonical diagram by requiring that rational subtangles be in canonical form, we may imagine doing something similar for generic algebraic tangle diagrams. Although this representation is not yet known to be unique, the generalized Montesinos subtangles in an algebraic tangle diagram can always be converted into canonical form while preserving the equivalence class of the tangle (Figure 5.34).



(a) $T \cong \left(\frac{8}{3} + \frac{2}{3}\right) * \left(\left(\frac{1}{3} + \frac{1}{3} \circ (-3, 3)\right)\right)$ **(b)** $T \cong \left(\left(\frac{2}{3} + \frac{2}{3}\right) \circ (2)\right) * \left(\left(\frac{-3}{1} * \frac{-3}{1} \circ (2, 1, 2)\right)\right)$

Figure 5.34: Equivalent algebraic diagrams with non-canonical and canonical GM subtangles

5.4 Algebraic Tangles

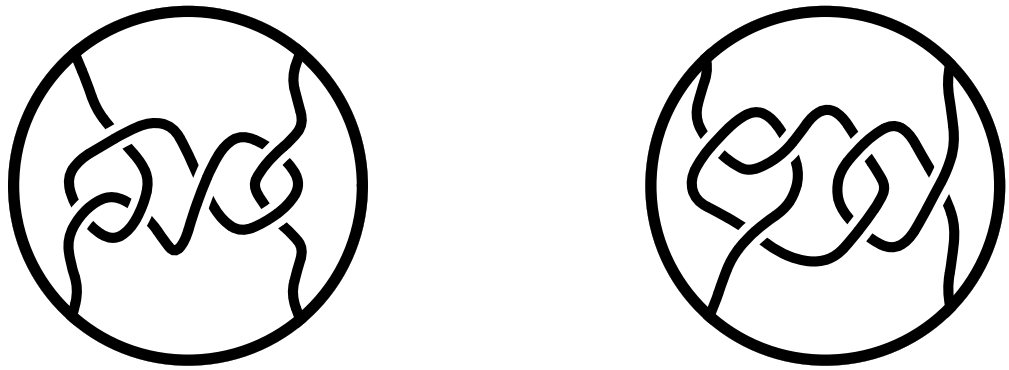
The final portion of this section is devoted to the study of algebraic tangle diagrams more generally. Ideally, we would like to discover a unique canonical form for generic algebraic tangles that arises as a natural generalization of the special class of tangle which we have already studied. As our previous work has shown, identifying such a construction uniquely is likely to be quite challenging, and this has not yet been accomplished in the existing theory. If we are willing to give up the requirement of uniqueness for the time being, identifying a preferred diagram for generic algebraic tangles is a more approachable goal. Ultimately, this will motivate our definition of a *semi-canonical form* for algebraic tangle diagrams.

5.4.1 Algebraic Tangle Trees

Before attempting to propose a formal definition of semi-canonical form, we will devote some time to another way of interpreting algebraic tangle diagrams. We have seen previously that, given an algebraic construction formula, we can construct a diagram by stacking together canonical diagrams for rational subtangles. However, this process alone does not always yield a preferred diagram; we encountered some issues for Montesinos diagrams whenever adjacent rational subtangles shared twists. In the Montesinos case, we overcame this by flyping all shared twists to one side, which guaranteed a unique decomposition of the diagram into maximal rational subtangles (Theorem 5.23). With some extra work, we obtained a similar result for generalized Montesinos tangles (Theorem 5.43).

Now we would like to extend this idea of flyping twists into a preferred position for algebraic diagrams more generally. This task becomes more difficult to describe without the added structure we had in the special cases of algebraic tangle, but is possible nevertheless. By decomposing an algebraic tangle diagram T using the 1-crossing subtangle decomposition $D_{\min}$ (Lemma 4.6), we may obtain an algebraic construction formula for T using only the tangles $+1$ and -1. Every flypable crossing in T exists somewhere in this formula, next to a sum or product (Figure 5.35). By flyping each sum or product that occurs on the left side of a formula, we can move all of these twists into a preferred position.

This process is identical to how we showed that rational tangles can be described using only right and bottom twists (Lemma 3.5), but the algorithm is not limited to rational tangles. As in the example of Figure 5.35, any algebraic construction formula using only 1-crossing subtangles can be modified to push all sums and products with ± 1 to the right side of the formula using flypes. However, it can be challenging to keep track of the details for



(a) $T \cong (1*(1+1)) + (1+(1*1))$ **(b)** $T \cong ((1+1)*1) + ((1*1)+1)$

Figure 5.35: Flyping twists in a non-canonical diagram for $T \cong \frac{2}{3} + \frac{3}{2}$ to the right and bottom

more complicated diagrams since flyping a subtangle reverses the order of sums or products, depending on the operation. While it is possible to describe this entire sequence of flypes using pure algebra, there is an elegant way to represent the combinatorics of this process visually using what we will refer to as an *algebraic tangle tree*.

This idea is based on a method introduced by Caudron to represent algebraic knots using acyclic graphs [4]. Caudron's technique involved encoding a diagram as a tree (Figure 5.36), where the leaves (vertices of degree 1) represent integer tangles, and interior vertices represent

(a) $K \cong N\left(\frac{1}{2} + \frac{-1}{3} + \left(\frac{-1}{2} * \frac{2}{1}\right)\right)$ **(b)** Caudron's corresponding acyclic graph

Figure 5.36: Example of a knot diagram encoded using Caudron's method [4]

either ramification with an integer tangle, or one of Conway's algebraic operations (Figure 4.25). These correspond to interior vertices of degree 2 or 3, respectively. Since Caudron's method was designed to encode knot diagrams, we will skip a full explanation, but the algebraic tangle trees we describe now are based on his work, with heavy modifications to adapt to our work with tangles.

Theorem 5.44 (Algebraic tangle diagrams encoded as algebraic tangle trees)**:**
If T is an algebraic tangle diagram, then this diagram can be encoded using an *algebraic tangle tree* $A(T)$, that is, a directed acyclic graph satisfying the following properties.

a. Each *leaf* (degree 1 vertex) in $A(T)$ matches a crossing in T.

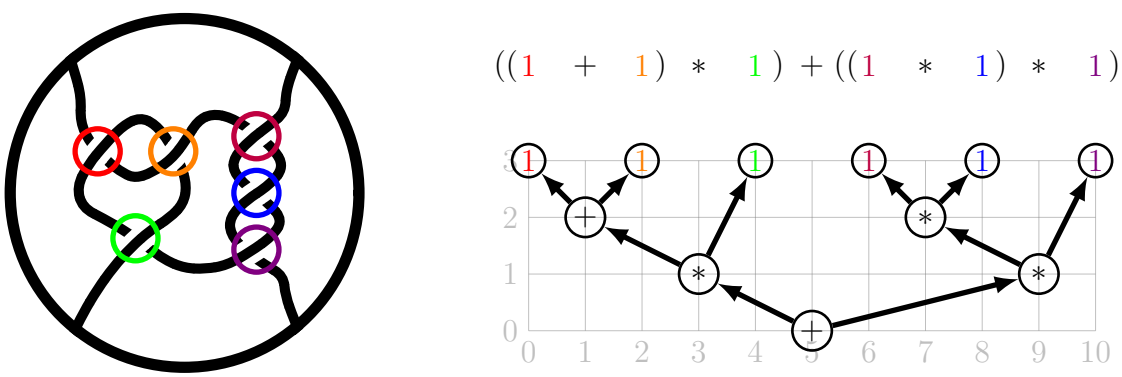
b. Each interior vertex in $A(T)$ corresponds to a sum or product of subtangles in T.

c. From each interior vertex, there exist two directed edges pointing towards the vertices matching the subtangles in the sum or product.

d. For T with more than one crossing, each interior vertex in $A(T)$ has *out-degree* 2, *in-degree* 1, and *total degree* 3, except for the *root* of the tree, which has in-degree 0.

e. If T has n crossings, then $A(T)$ has $2n-1$ vertices: n leaves and $n-1$ interior nodes.

The *standard embedding* of $A(T)$ has the form shown in Figure 5.37, where the order of the leaves matches the order of 1-crossing subtangles in the construction formula for T.



(a) Diagram: $T \cong ((1+1)*1)+((1*1)*1)$ **(b)** Algebraic tangle tree: $A(T)$

Figure 5.37: Example of an algebraic tangle tree for the algebraic diagram $T \cong \frac{2}{3} + \frac{1}{3}$

Proof:

Let T be a an algebraic tangle diagram. Hence, there exists a unique decomposition of T into 1-crossing subtangles $D_{\min}$ (Lemma 4.6). Based on this subtangle decomposition, T can be expressed (non-uniquely) using an algebraic construction formula defined by some combination of sums and products of the two 1-crossing subtangles, $+1$ and -1. Note that the order of operations in the construction formula is determined by the placement of nested parentheses, although we have often omitted these for convenience in previous examples.

Based on this construction formula for T, we visually construct a directed acyclic graph $A(T)$ on a grid in the following way, referring to Figure 5.37 as an example. If T has n crossings, then the grid is $2n-1$ units wide and $h \leq n$ units deep. The height h of the grid is 1 plus the maximum number of nested parentheses in the construction formula. On the top row of the grid, list out a horizontal sequence of n vertices, one at every other grid point, ordered to match the 1-crossing subtangles in the construction formula for T. These are the leaves in $A(T)$.

For each sum and product in the formula, define a vertex located in the following position: horizontally, this vertex lies in between the two vertices corresponding to the two adjacent 1-crossing subtangles from the construction formula; vertically, the height of the vertex matches the number of nested parentheses this operation is contained in, stopping just below the row of leaves on the top row. These are the interior vertices in $A(T)$. Note that each of the $2n-1$ columns in the grid has exactly one vertex, and these alternate between leaves and interior vertices.

Each interior vertex represents a tangle sum or product, and hence denotes a connection between two subtangles from T. For each interior vertex, define two directed edges pointing towards the vertices corresponding to the two joined subtangles. In the grid, each edge points up, one veering to the left and the other to the right, connecting to the lowest height vertex above it on the left or right side, respectively. Choosing directed edges in this way, each leaf has total degree 1, corresponding to a single incoming edge. Each interior vertex has out-degree 2 and in-degree 1, with the exception of the lowermost vertex of in-degree 0, which is the root of $A(T)$.

The graph $A(T)$ defined in this way is constructed to satisfy each of the properties listed in the claim. The particular diagram for $A(T)$ obtained through this construction and shown in Figure 5.37 is referred to as the *standard embedding* and will be our preferred way to visualize an algebraic tangle tree. In practice, we will drop the grid and the arrows for convenience, noting that each directed edge points upward. □

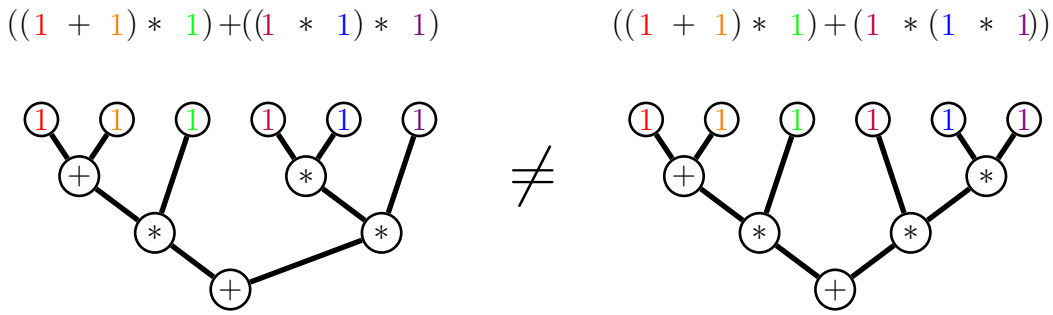


Figure 5.38: Distinct algebraic tangle trees for a fixed diagram $T \cong \frac{2}{3} + \frac{1}{3}$

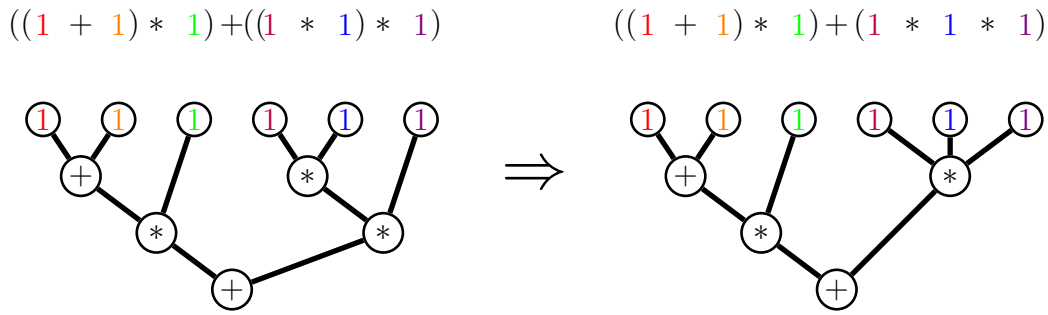
For any aglebraic tangle diagram T, the algebraic tangle tree $A(T)$ is defined in such a way that it can easily be sketched beneath an algebraic construction formula using 1-crossing subtangles. However, this graph is not unique in general. Recall that tangle sum and product, on their own, are associative operations: $A + (B + C) \cong (A + B) + C$. Hence, there is more than one valid way to place parentheses in a construction formula for T with repeated sums or repeated products. In the corresponding algebraic tangle tree, these will result in a different placement of interior vertices and edges for the same set of leaves.

For example, in the decomposition of the algebraic tangle diagram $T \cong \frac{2}{3} + \frac{1}{3}$ from Figure 5.37, we have equivalent constructions:

$$((1 + 1) * 1) + (\underline{(1 * 1) * 1}) \quad \cong \quad ((1 + 1) * 1) + (\underline{1 * (1 * 1)}).$$

While these constructions define the same tangle diagram, they yield slightly different algebraic tangle trees (Figure 5.38). Specifically, both trees use the same set of nodes, one for each crossing and one for each algebraic operation, but there is a different combination of edges between the vertices corresponding to the three twists on the right. Although the two trees from this example happen to be isomorphic as graphs, this need not hold in general. More importantly, these two trees represent distinct graphs on the same set of nodes because there is a different combination of edges.

This example shows that algebraic tangle trees are not unique, but we can make a slight modification to the construction from Theorem 5.44 to obtain a unique tree matching a given diagram (Figure 5.39). Any time there is a sequence of tangle sums or products in the construction formula, we may drop unnecessary parentheses without ambiguity. This is exactly what we do for Montesinos tangles when we write them as a sum $T \cong L_1 + \cdots + L_m$. In the tangle tree, we mirror this by combining a sequence of sums or products into a single



(a) Full algebraic tangle tree: $A(T)$ **(b)** Reduced algebraic tangle tree: $A^r(T)$

Figure 5.39: Converting $T \cong \frac{2}{3} + \frac{1}{3}$ from Figure 5.37b to a reduced algebraic tangle tree

interior vertex. Whenever there exists an edge between two interior vertices that denote the same operation, either sum or product, these can be collapsed into a single vertex. The out-degree of this new vertex is the number of subtangles in the sequence, which is 2 at minimum by default.

As with the subtangles in a construction formula, the order of the vertices matters. When multiple directed edges point outwards from the same interior vertex, the connected subtangles are ordered from left to right, matching the order from the construction formula. Given an algebraic tangle tree $A(T)$ for some tangle diagram T, the modified graph obtained by combining every sequence of sums or products into a single vertex will be referred to as the *reduced algebraic tangle tree* of T, denoted $A^r(T)$. As a consequence, there does not exist an edge between any two interior vertices denoting the same tangle operation. Figure 5.39 shows $A^r(T)$ for the diagram of $T \cong \frac{2}{3} + \frac{1}{3}$ we considered in Figure 5.37.

Note that $A^r(T)$ depends on the diagram for T. Equivalent algebraic tangle diagrams can have non-identical algebraic tangle trees, reduced or otherwise. However, a fixed diagram uniquely determines a reduced algebraic tangle tree.

Lemma 5.45 (Reduced algebraic tangle tree uniquely determined by diagram)**:**
If T is an algebraic tangle diagram, then this diagram can be uniquely encoded using a *reduced algebraic tangle tree* $A^r(T)$ which corresponds to the decomposition of T into 1-crossing subtangles defined by a construction formula with a minimal number of parentheses.

Proof:

Let T be an algebraic tangle diagram and let $D_{\min}$ be the unique decomposition of T into 1-crossing subtangles (Lemma 4.6). Hence, there exists an algebraic construction formula

for T using the subtangles from this decomposition. While $D_{\min}$ is unique, this construction formula may not be. In a proper construction formula, the order of operations is determined by the placement of parentheses. However, for any sequence of tangle sums in the decomposition, associativity of this operation implies that a different choice of parentheses will define an equivalent construction. A similar fact holds for any sequence of tangle products.

Define an equivalent construction formula for T by removing any unnecessary parentheses separating associative operations between subtangles. This formula uses a minimal number of parentheses; removing any more would make the construction ambiguous. Furthermore, this formula uniquely defines the diagram for T because changing the placement of any existing parentheses would yield a construction for a different diagram.

Let $A^r(T)$ be the reduced algebraic tangle tree obtained from this construction formula, as in the example of Figure 5.39b. Note that every sequence of repeated sums or products has been collapsed into a single node, and hence there does not exist an edge between any two interior vertices representing the same tangle operation. The uniqueness of $A^r(T)$ follows from the uniqueness of the modified construction formula.

□

Just as we refer to $A^r(T)$ as a *reduced* algebraic tangle tree, we will say an algebraic construction formula for T is *reduced* if all of the unnecessary parentheses between associative operations are removed. Since a reduced algebraic construction formula using 1-crossing subtangles uniquely defines a diagram, this leads to a one-to-one correspondence between algebraic tangle diagrams and reduced algebraic tangle trees. Hence, these two objects encode exactly the same information about tangles.

Lemma 5.46 (Bijection between algebraic tangles and reduced tangle trees)**:**
There exists a bijection between algebraic tangle diagrams T and reduced algebraic tangle trees $A^r(T)$ defined by a reduced construction formula for T using 1-crossing subtangles.

Proof:

Let T be an algebraic tangle diagram. By Lemma 5.45, T defines a unique reduced algebraic tangle tree $A^r(T)$. We will prove that this map defines a bijection by showing that it has an inverse. Let $A^r(T)$ be any reduced algebraic tangle tree for T. Our goal is to convert this tree into a construction formula by reversing the steps from Theorem 5.44, with some slight modifications for the associate operations represented by interior vertices with out-degree greater than 2.

If $A^r(T)$ is any reduced algebraic tangle tree for T, then the standard embedding of $A^r(T)$ introduced in Theorem 5.44 defines a construction formula for T obtained by writing out an ordered list of symbols which alternate between the 1-crossing subtangles and the algebraic operations from the leaves and interior nodes of $A^r(T)$, respectively. If we visualize the tree sitting on a grid such as in Figure 5.37b, then this ordered list of symbols is obtained by projecting each node to height zero, except that interior vertices with out-degree greater than 2 split up into multiple operations.

The edges from $A^r(T)$ determine the placement of parentheses in this list. Except for the root of the tree, for each interior vertex, we may insert a pair of parentheses into this formula which contains the leaves in the corresponding connected subtangles. These parentheses are nested according to their height in the tree, and the number of contained subtangles matches the out-degree of the vertex. This process yields a construction formula for T using 1-crossing subtangles.

Since there do not exist any edges between interior vertices denoting the same algebraic operation in the reduced tree $A^r(T)$, note that each pair of nested parentheses separates one or more sums from one or more products. In particular, since tangle sum and product are not associative with each other, it follows that this construction formula uses a minimal number of parentheses, which is to say that it is reduced. Hence, by Lemma 5.45, the unique reduced algebraic tangle tree defined by this formula for T must match $A^r(T)$.

Thus, we have establishes that any algebraic tangle diagram defines a reduced algebraic tangle tree, that any reduced algebraic tangle tree defines an algebraic tangle diagram, and that these maps are inverses of each other. Therefore, we may conclude that algebraic tangle diagrams are in bijection with reduced algebraic tangle trees, completing the proof.

□

The relationship of Lemma 5.46 is restricted to diagrams; equivalent but non-identical diagrams define different trees (Figure 5.40). However, we may now extend the notion of equivalence to include tangle trees as well. We will say that two algebraic tangle trees, reduced or otherwise, are *equivalent* if they define equivalent algebraic tangle diagrams. If $T_1 \cong T_2$ as algebraic tangles, then we will write $A(T_1) \cong A(T_2)$. In other words, transformations between tangle diagrams that preserve the equivalence class of the tangle can also be studied using tangle trees. For example, the two equivalent tangle diagrams shown in Figure 5.40 are related by a flype, and this induces a corresponding transformation in their reduced algebraic tangle trees.

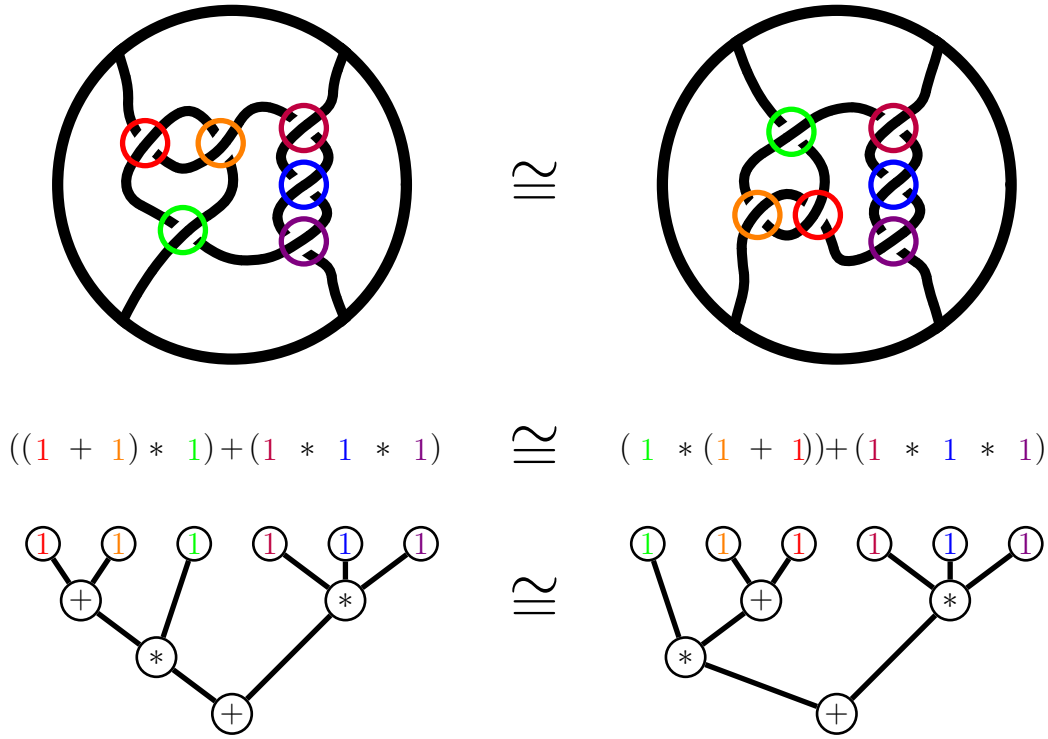
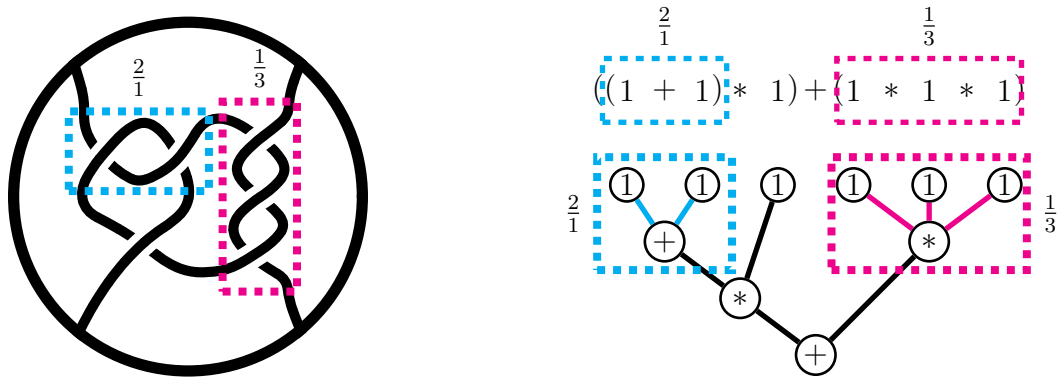


Figure 5.40: Equivalent diagrams for $T \cong \frac{2}{3} + \frac{1}{3}$ defining different but equivalent tangle trees

Such transformations are best described in terms of their effect on subtangles. Subtangles are easily identified in $A^r(T)$ thanks to the relationship between tangle trees and construction formulas. Since every subtangle S of an algebraic tangle diagram T is algebraic (Lemma 4.4), S consists of an algebraic combination of some subset of the crossings in T. In the construction formula for T using 1-crossing subtangles, this corresponds to a sub-segment of the formula. In $A^r(T)$, this corresponds to a *branch* of the tangle tree (Figure 5.41).

Formally, we define a *branch* in the reduced algebraic tangle tree to be a connected subtree of $A^r(T)$ terminating in one or more leaves. A branch may consist of a single leaf, but it cannot consist of interior nodes if these do not connect to any leaves. We also include the restriction that every interior vertex in a branch must have out-degree at least 2. Hence, a branch in $A^r(T)$ naturally defines the tangle tree for a subtangle of T. If the branch contains more than one leaf, then one of the interior nodes from $A^r(T)$ becomes the root in the subtree. In this way, every subtangle of T arises as a branch in $A^r(T)$.



(a) Subtangles in $T \cong ((1+1)*1)+(1*1*1)$ **(b)** Formula segment and branches in $A^r(T)$

Figure 5.41: Subtangles in $T \cong \frac{2}{3} + \frac{1}{3}$ corresponding to branches in $A^r(T)$

Lemma 5.47 (Bijection between subtangles in T and branches in $A^r(T)$)**:**
If T is an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$, then there exists a bijection between the *2-string subtangles of* T and the *branches in* $A^r(T)$.

Proof:

Let T be an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$. Let f_T denote the reduced algebraic construction formula using 1-crossing subtangles which relates T and $A^r(T)$ (Lemma 5.46). To prove the claim, we will show how every subtangle of T defines a branch in $A^r(T)$, that every branch in $A^r(T)$ defines a subtangle of T, and that these maps are inverses.

Let S be a subtangle of T, and note that S is algebraic (Lemma 4.4). Hence, there exists a reduced construction formula f_S for S using 1-crossing subtangles, and f_S defines a reduced tangle tree $A^r(S)$ (Lemma 5.46). Since S consists of a subset of the crossings from T, then f_S matches a sub-segment of the subtangles and operations in the construction formula f_T. The branch in $A^r(T)$ corresponding to this sub-segment is exactly $A^r(S)$ (Figure 5.41).

For the reverse direction of this map, let B denote any branch in $A^r(T)$. Let f_B denote the sub-segment of f_T corresponding to this branch. Hence, f_B defines the construction formula for a subtangle S of T. If $A^r(S)$ is the reduced algebraic tangle tree for this subtangle, then $A^r(S)$ exactly matches the branch B. Therefore, these identifications between subtangles with branches and branches with subtangles define inverse maps. Thus, we may conclude that the subtangles of T are in bijection with the branches in $A^r(T)$, completing the proof. □

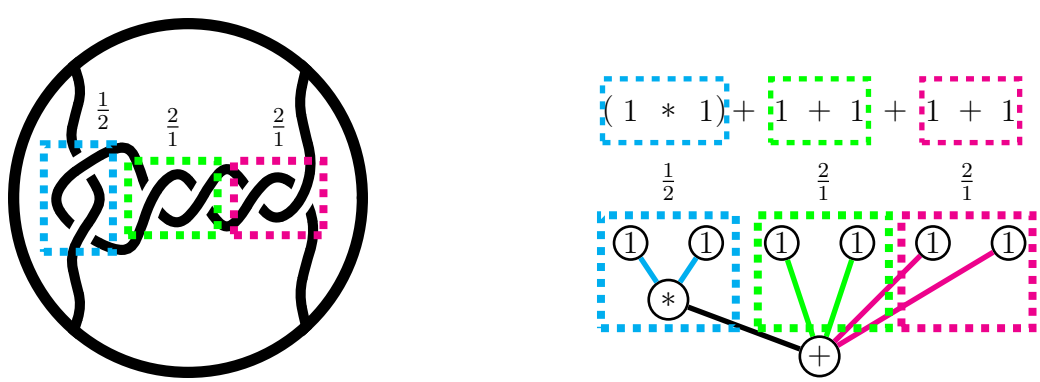
The preceding lemma leads to an interesting relationship between subtangle decompositions and tangle trees. Figure 5.42 illustrates how a decomposition of an algebraic tangle diagram T into disjoint subtangles induces a decomposition of $A^r(T)$ into disjoint branches. In a reduced tangle tree, it is possible that two different branches of $A^r(T)$ share the same root if the corresponding interior vertex has out-degree 3 or greater (Figure 5.42b). However, we still consider these branches to be disjoint if they do not share any leaves. We will refer to the set of branches in $A^r(T)$ induced by a subtangle decomposition D of T as B_D.

Corollary 5.48 (Subtangle decomposition induces branch partition)**:**
Every subtangle decomposition D of an algebraic tangle diagram T induces a unique partition of the leaves in the reduced algebraic tangle tree $A^r(T)$ into disjoint branches, denoted B_D.

Proof:

Let T be an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$. Let $D = \{S_1, \cdots, S_n\}$ be any subtangle decomposition of T. By Lemma 5.47, each subtangle $S_i \in D$ corresponds to a branch $A^r(S_i)$ of the tangle tree. Let $B_D = \{A^r(S_1), \cdots, A^r(S_n)\}$ denote this collection of branches. Since the subtangles in D define a partition of the crossings in T, it follows that the branches in B_D define a partition of the leaves in $A^r(T)$. Uniqueness of this partition follows from the bijection of Lemma 5.47. Furthermore, since the subtangles in D are disjoint, it follows that these branches are disjoint in $A^r(T)$, possibly allowing for a shared root, as in Figure 5.42.

□



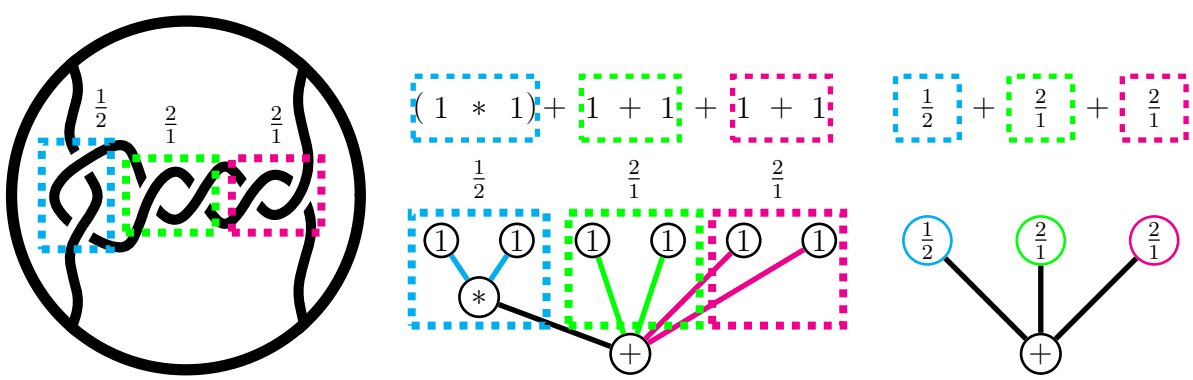
(a) Subtangle decomposition: $D = \{\frac{1}{2}, \frac{2}{1}, \frac{2}{1}\}$ **(b)** Branch partition: $B_D = \{A^r(\frac{1}{2}), A^r(\frac{2}{1}), A^r(\frac{2}{1})\}$

Figure 5.42: Subtangle decomposition of $T \cong \frac{9}{2}$ inducing a disjoint branch partition of $A^r(T)$

Given any branch of a tangle tree, we can imagine how to obtain a subgraph of the tree by chopping off this branch just above its root. With a few exceptions, if we do this for each branch in a branch partition of a tangle tree, then the resulting subgraph is referred to as a *pruned algebraic tangle tree*. Pruning a tangle tree yields a connected sub-tree rooted at the same vertex. Rather than represent crossings, the leaves in a pruned tangle tree represent algebraic subtangles, and hence are denoted using fractions or construction formulas.

If D is a subtangle decomposition of an algebraic tangle diagram T, then we can prune the reduced algebraic tangle tree $A^r(T)$ by each of the branches in the induced branch partition B_D (Figure 5.43). The resulting subgraph is referred to as the *D-pruning* of $A^r(T)$, and is denoted $A^r_D(T)$. Since this is a subgraph of a reduced tangle tree $A^r(T)$, it follows that $A^r_D(T)$ is also a reduced tangle tree, and hence we keep the r in the superscript. Although we will not study them, one may also work with non-reduced pruned tangle trees.

While pruning a branch in a tangle tree usually means removing everything above the root of the branch, we mentioned that there are a few exceptions. If pruning two disjoint branches which share a root, such as the two rightmost branches in the partition of Figure 5.43b, then each branch is collapsed to a single node while keeping the original root (Figure 5.43c). Similarly, when pruning a branch consisting of a single leaf, we do not actually remove the node, but now the vertex is denoted using $\pm\frac{1}{1}$ rather than ± 1. In both cases, the resulting graph is still a sub-tree.



(a) Subtangle decomposition D **(b)** Branch partition B_D of $A^r(T)$ **(c)** D-pruning $A^r_D(T)$

Part 5.43a above shows the subtangle decomposition $D = \left\{\frac{1}{2}, \frac{2}{1}, \frac{2}{1}\right\}$ of the rational tangle diagram $T \cong \frac{9}{2}$. Part 5.43b shows the corresponding induced branch partition B_D of the reduced algebraic tangle tree $A^r(T)$. Part 5.43c shows the D-pruned algebraic tangle tree $A^r_D(T)$ obtained by pruning the branches from B_D.

Figure 5.43: Pruning a tangle tree via the subtangle decomposition-induced branch partition

Just as there exists a bijection between algebraic diagrams and reduced tangle trees, a similar result exists for prunings. Starting from a fixed algebraic tangle diagram and reduced algebraic tangle tree, the possible subtangle decompositions are in bijection with the possible reduced prunings. Since we are not interested in working with non-reduced prunings, we will drop this descriptor and assume without loss of generality that all prunings are reduced.

Corollary 5.49 (Bijection between subtangle decompositions and prunings)**:**
If T is an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$, then there exists a bijection between subtangle decompositions D of T and the prunings $A^r_D(T)$ of $A^r(T)$.

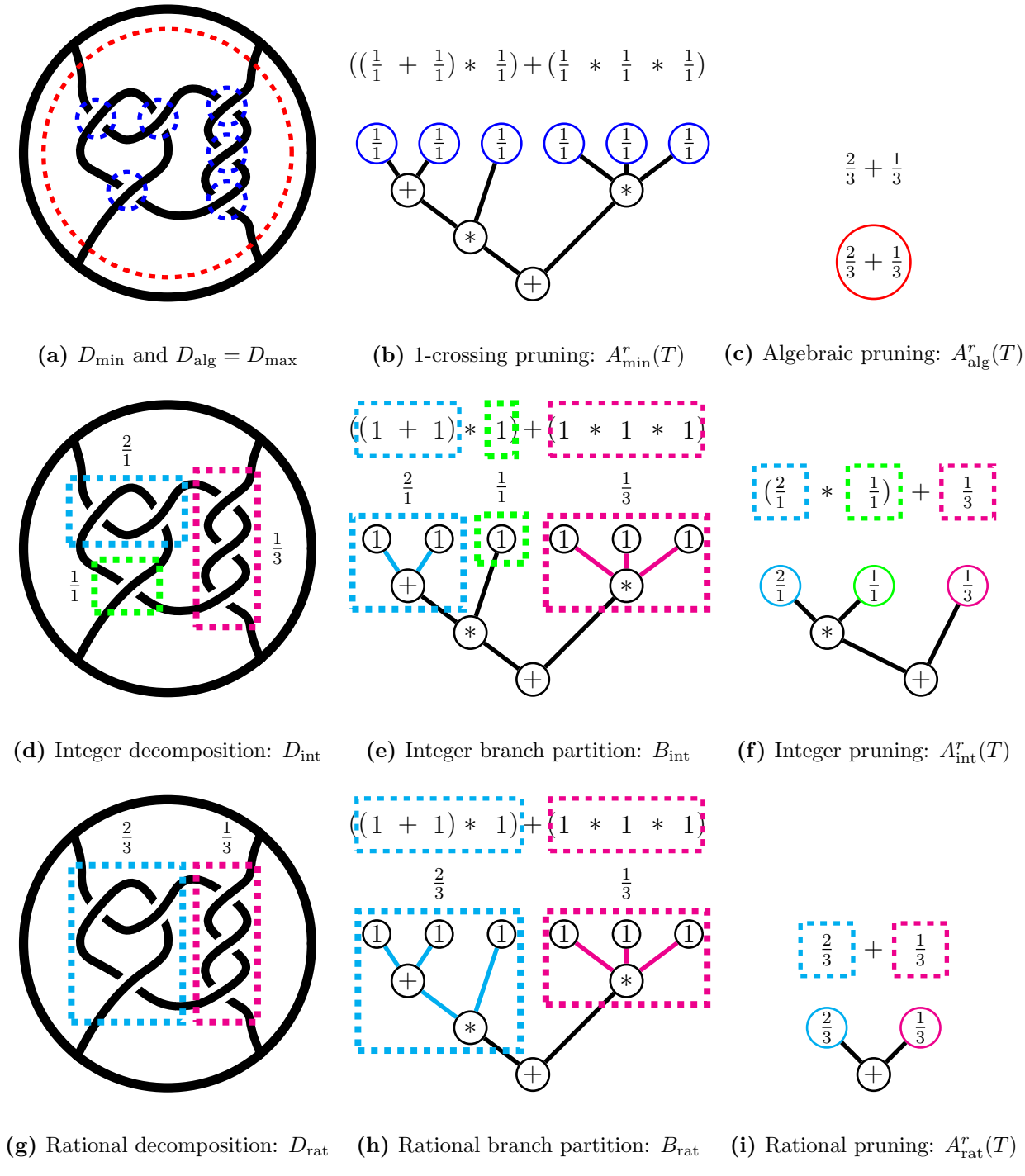
Proof:

Let T be an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$. By Lemma 5.47, the subtangles in T are in bijection with the branches in $A^r(T)$. Combining this fact with Corollary 5.48 implies that the subtangle decompositions of T are in bijection with the branch partitions of $A^r(T)$. By definition, any branch partition of $A^r(T)$ defines a corresponding pruned tangle tree. Therefore, this bijective correspondence extends to subtangle decompositions D of T with D-prunings $A^r_D(T)$ of $A^r(T)$, which proves the claim. □

In relation to tangle classification, recall that we are particularly interested in studying those subtangle decompositions which use maximal subtangles of a given a class. For a given algebraic tangle diagram, the bijection of Corollary 5.49 implies that each of these maximal subtangle decompositions defines a corresponding pruning of the reduced algebraic tangle tree. To refer to these special prunings, we will adopt the notation used in Table 5.1, which lists out the subtangle decompositions and corresponding pruned tangle trees. Although we list both in the table, note that we always have $A^r_{\max}(T) = A^r_{\text{alg}}(T)$ since algebraic tangle trees are only defined for algebraic tangle diagrams.

Class of Subtangle	Subtangle Decomposition of T	Pruning of $A^r(T)$
1-crossing	$D_{\min}$	$A^r_{\min}(T)$
Integer	D_{int}	$A^r_{\text{int}}(T)$
Rational	D_{rat}	$A^r_{\text{rat}}(T)$
Algebraic	D_{alg}	$A^r_{\text{alg}}(T)$
Maximal	$D_{\max}$	$A^r_{\max}(T)$

Table 5.1: Maximal subtangle decompositions of T and special prunings of $A^r(T)$



(a) D_{min} and $D_{\text{alg}} = D_{\text{max}}$ **(b)** 1-crossing pruning: $A^r_{\text{min}}(T)$ **(c)** Algebraic pruning: $A^r_{\text{alg}}(T)$

(d) Integer decomposition: D_{int} **(e)** Integer branch partition: B_{int} **(f)** Integer pruning: $A^r_{\text{int}}(T)$

(g) Rational decomposition: D_{rat} **(h)** Rational branch partition: B_{rat} **(i)** Rational pruning: $A^r_{\text{rat}}(T)$

Figure 5.44: Special prunings of $T \cong \frac{2}{3} + \frac{1}{3}$ using maximal subtangle decompositions

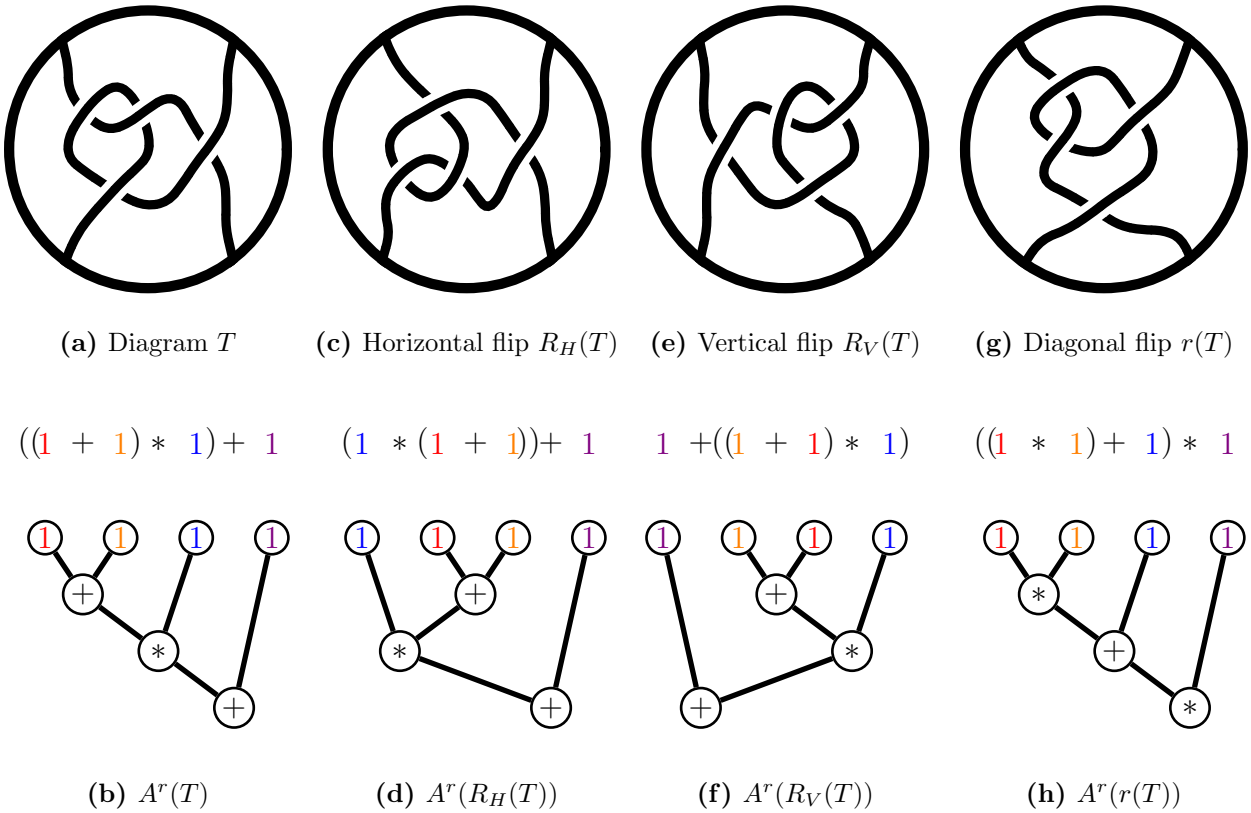
Figure 5.44 shows an example of each of these special prunings for the algebraic tangle diagram $T \cong \frac{2}{3} + \frac{1}{3}$, along with the corresponding subtangle decomposition and branch partition. In the case of the 1-crossing pruning (5.44b), note that $A^r_{\text{min}}(T)$ and $A^r(T)$ are identical as graphs, except that we identify the leaves in the pruning as 1-crossing subtangles rather than crossings. Since $D_{\text{alg}} = D_{\text{max}}$ for an algebraic tangle diagram, the corresponding pruning $A^r_{\text{alg}}(T) = A^r_{\text{max}}(T)$ consists of a single vertex (5.44c). Although not particularly interesting, this is a valid tangle tree. In general, the integer pruning $A^r_{\text{int}}(T)$ (5.44f) and rational pruning $A^r_{\text{rat}}(T)$ (5.44i) carry the most useful information about the structure of the tangle diagram, although $A^r_{\text{rat}}(T)$ does not exist if D_{rat} does not.

Each of the special tangle trees enumerated in Table 5.1 is a pruning of the previous tree. In Figure 5.44, observe that the height of each subsequent pruning decreases by 1. For a given algebraic tangle diagram T, if $D_{\text{min}} < D_{\text{int}}$, then height of $A^r_{\text{min}}(T)$ is always 1 greater than the height of $A^r_{\text{int}}(T)$. This is because each integer subtangle in T lives on the top two rows of the tangle tree $A^r_{\text{min}}(T)$, but these collapse to a single row in $A^r_{\text{int}}(T)$. Similar results hold for $A^r_{\text{rat}}(T)$ if $D_{\text{int}} < D_{\text{rat}}$ and for $A^r_{\text{alg}}(T)$ if $D_{\text{rat}} < D_{\text{alg}}$, although the height could decrease by more than 1.

5.4.2 Transformations of Algebraic Tangle Trees

Now that we have an elegant way of visualizing the construction for algebraic tangle diagrams, our next goal is to examine the effect of tangle transformations on the corresponding tangle tree. We will show that a semi-canonical form for algebraic diagrams can be achieved through some combination of flypes, flips, and flupes on subtangles. Since subtangles in an algebraic diagram correspond to branches in a tangle tree (Lemma 5.47), each of these operations on subtangles in the diagram is mirrored by an operation on a branch in the tree. Much like our previous work with twist vectors, we now seek to catalog the rules for equivalent tangle tree transformations.

Flipping a tangle involves involves a 180 degree rotation across a choice of axis. We have examined three ways to flip a tangle diagram T: across the horizontal axis, $R_H(T)$; across the vertical axis, $R_V(T)$; and across the NW-SE diagonal axis, $r(T) = R_{90}(R_H(T))$. These operations move the boundary sphere and hence do not define an equivalence of tangles in general, but flipping an algebraic diagram yields a different algebraic diagram. Since these diagrams are constructed using tangle sums and products, both the construction formula and the tangle tree of a flipped algebraic diagram can be computed using Theorem 2.2.



(a) Diagram T **(c)** Horizontal flip $R_H(T)$ **(e)** Vertical flip $R_V(T)$ **(g)** Diagonal flip $r(T)$

(b) $A^r(T)$ **(d)** $A^r(R_H(T))$ **(f)** $A^r(R_V(T))$ **(h)** $A^r(r(T))$

Figure 5.45: Effect of horizontal, vertical, and diagonal flips on algebraic tangle trees

Lemma 5.50 (Effect of flips on algebraic tangle trees)**:**
If T is an algebraic tangle diagram with algebraic tangle tree $A(T)$, then the tangle tree for the algebraic diagram obtained by flipping T across the horizontal, vertical, or diagonal axis corresponds to a transformation of $A(T)$ according to the following rules (Figure 5.45).

a. **Horizontal Flip:** $A(R_H(T))$ is obtained from $A(T)$ by reversing the order of branches at each interior vertex representing a product (Figure 5.45d).

b. **Vertical Flip:** $A(R_V(T))$ is obtained from $A(T)$ by reversing the order of branches at each interior vertex representing a sum (Figure 5.45f).

c. **Diagonal Flip:** $A(r(T))$ is obtained from $A(T)$ by switching the algebraic operation of each interior vertex, from sum to product and from product to sum (Figure 5.45h).

These rules extend to pruned tangle trees by also flipping the subtangle at each leaf.

Proof:

Let T be an algebraic tangle diagram with algebraic tangle tree $A(T)$. This tree need not be reduced, although it could be; the claim will hold in either case. Corresponding to $A(T)$, there exists a construction formula f_T for T using only 1-crossing subtangles. The statements in the claim are merely a visual representation using tangle trees of how an algebraic construction formula is modified when a corresponding diagram is flipped. Hence, these results are merely an extension of the rules from Theorem 2.2, generalized to generic algebraic construction formulas using induction.

If $R_H(T)$ is the diagram obtained by flipping T across the horizontal axis, then the corresponding construction formula $f_{R_H(T)}$ is determined from f_T by reversing the order of the subtangles in each product. Technically, each 1-crossing subtangle in the formula is also horizontally flipped, but we have that $R_H(\pm 1) \cong \pm 1$ by symmetry. In the tangle tree $A(R_H(T))$, this is equivalent to switching the order of the branches at each interior node representing a tangle product in $A(T)$.

If $R_V(T)$ is the diagram obtained by flipping T across the vertical axis, then a parallel argument shows that the corresponding construction formula $f_{R_V(T)}$ is obtained from f_T by reversing the order of the subtangles in each sum. Similarly, $A(R_V(T))$ is obtained from $A(T)$ by switching the order of the branches at each interior node representing a tangle sum.

In the case of a diagonal flip, where $r(T)$ is the diagram obtained by flipping T across the NW-SE diagonal axis, the order of the subtangles and operations is preserved in the formula, except that each tangle sum switches to a tangle product and each tangle product switches to a tangle sum. As with horizontal and vertical flips, 1-crossing subtangles are equivalent under diagonal flips by symmetry: $r(\pm 1) \cong \pm 1$. Hence, the construction formula $f_{r(T)}$ for $r(T)$ is obtained from f_T merely by switching the tangle operations. The corresponding tangle tree $A(r(T))$ is graph isomorphic to $A(T)$ with an identical standard embedding, but each interior vertex representing a tangle sum or product has its operation switched, matching the modified construction formula.

This establishes the claim for basic algebraic tangle trees, for which leaves represent crossings. For pruned algebraic tangle trees, each leaf represents a subtangle. The corresponding construction formulas, and therefore the corresponding tangle trees, are modified in exactly the same way, except that each subtangle is also flipped according to the rules from Theorem 2.2. Hence, pruned tangle trees require the additional step modifying the label of each leaf with a flipped subtangle.

□

Figure 5.45 illustrates each of the three transformations introduced in the preceding lemma for the rational tangle diagram $T \cong \frac{5}{3}$. It is interesting to observe that the construction formulas for $R_H(T)$ and $R_V(T)$ are anti-symmetric, and hence their corresponding tangle trees are mirror images of each other. This fact holds more generally, as does the fact that the tangle trees $A(T)$ and $A(r(T))$ have identical underlying graphs, if we ignore the vertex labels. Since rational tangle diagrams are equivalent under horizontal and vertical flips (Lemma 3.3), we need not flip the subtangles corresponding to each leaf in the integer and rational prunings. However, these flips must be accounted for in any pruning with non-rational subtangle leaves.

Lemma 5.50 establishes our first set of rules for transformations of tangle trees. Although flips in general do not preserve the equivalence class of an algebraic tangle diagram, they are useful for describing transformations which do, such as flypes. In the same way that we can swap out equivalent subtangles in a diagram while preserving the equivalence class of the tangle, we may swap out equivalent branches in an algebraic tangle tree. Now we may leverage this fact to characterize flyping in a tangle tree (Figure 5.46). Since a flyped subtangle is either horizontally or vertical flipped, the branch in the tangle tree corresponding to this subtangle is transformed according to the rules of Lemma 5.50.

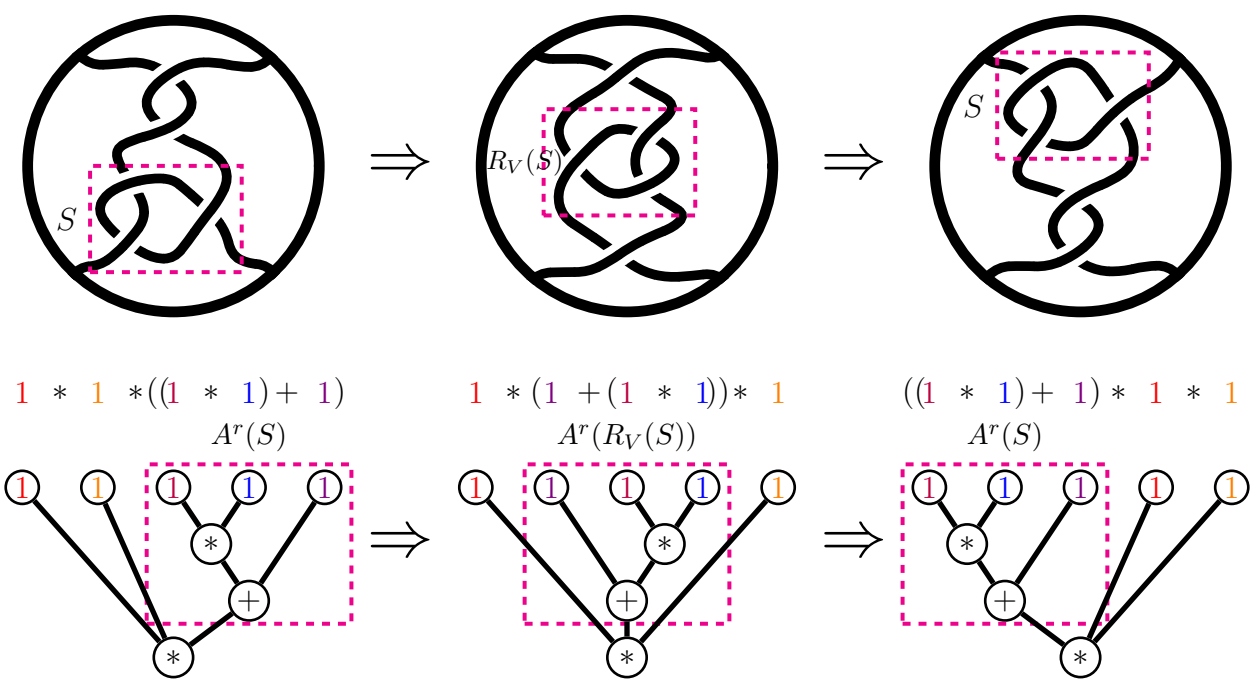


Figure 5.46: Flypes in a diagram for $T \cong \frac{3}{8}$ mirrored in the branches of $A^r(T)$

Lemma 5.51 (Flyping a subtangle branch in reduced algebraic tangle tree)**:**
If T is an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$, then it is possible to flype a subtangle S in T if and only if the branch $A^r(S)$ in $A^r(T)$ which corresponds to this subtangle shares a root with an adjacent leaf.

If $T' \cong T$ is the algebraic tangle diagram obtained by performing this flype, then the corresponding reduced algebraic tangle tree $A^r(T')$ is obtained from $A^r(T)$ as follows.

a. If the shared root denotes a tangle sum, then the order of the branch $A^r(S)$ and the adjacent leaf are switched in the tree, and the branch is replaced with $A^r(R_H(S))$.

b. If the shared root denotes a tangle product, then the order of the branch $A^r(S)$ and the adjacent leaf are switched in the tree, and the branch is replaced with $A^r(R_V(S))$.

Proof:

Let T be an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$. Let f_T be the reduced construction formula for T using 1-crossing subtangles which relates T and $A^r(T)$ (Lemma 5.46). Recall that the subtangles in T are in bijection with the branches in $A^r(T)$ (Lemma 5.47). Hence, if S is any subtangle of T, let $A^r(S)$ denote the corresponding branch of $A^r(T)$. Furthermore, recall that the sub-segment of f_T corresponding to this branch defines an algebraic construction formula for S.

Suppose that there exists subtangle S of T which can be flyped. In the diagram for T, this means there exists an adjacent 1-crossing subtangle which shares an algebraic connection with S. In the construction formula f_T, this means that the sub-segment corresponding to S is adjacent to a tangle sum or product with ± 1 on the left or right side. For example, in the first diagram from Figure 5.46, this formula looks like

$$f_T \quad = \quad 1 * \underbrace{1 * \overbrace{((1 * 1) + 1)}^{S}}_{\text{possible flype}}.$$

Since each interior vertex in $A^r(T)$ represents a sum or product of adjacent subtangles, the branch corresponding to S and the adjacent branch corresponding to this 1-crossing subtangle are rooted at the same node. This establishes the forward direction of the first part of the claim. The reverse direction follows from the bijective correspondence between subtangles and branches defined by the construction formula. A branch with this property indicates a subtangle with an adjacent twist, and hence a possible flype.

Now suppose that $T' \cong T$ is the equivalent algebraic tangle diagram obtained by performing a flype on a subtangle S. Let f_T and $f_{T'}$ denote their corresponding reduced construction formulas using 1-crossing subtangles. These formulas are identical, except for the sub-segment of each corresponding to the flyped subtangle and adjacent twist. For example, returning the diagram from Figure 5.46, these two construction formulas look like

$$f_T \quad = \quad 1 * \underbrace{1 * \overbrace{((1*1)+1)}^{S}}_{\text{possible flype}} \quad \cong \quad 1 * \underbrace{\overbrace{(1+(1*1))}^{R_V(S)} * 1}_{\text{possible flype}} \quad = \quad f_{T'}.$$

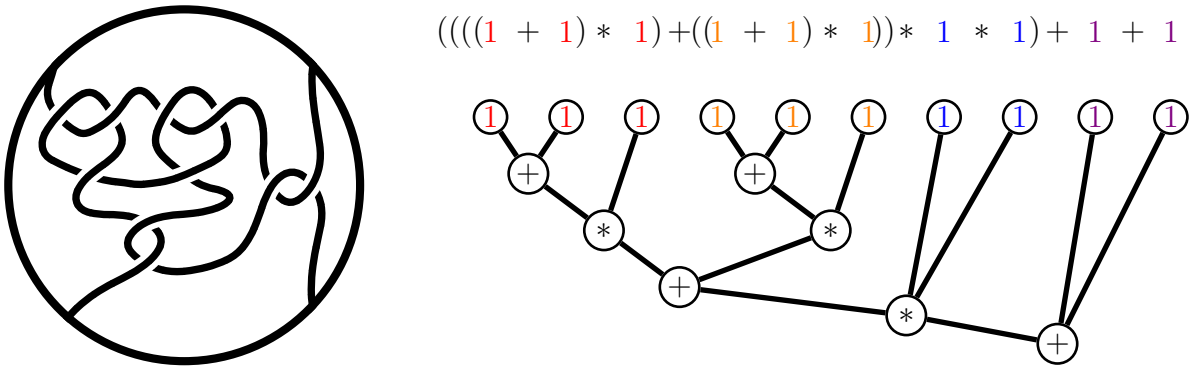
Hence, the subtangle and adjacent twist have their order reversed in the construction formula, and the subtangle S from T appears flipped in T' (Lemma 2.1). Whether we use a horizontal flip or a vertical flip depends on whether the adjacent twist is joined using a tangle sum or a tangle product, respectively.

Let $A^r(T)$ and $A^r(T')$ denote the two reduced algebraic tangle trees corresponding to these two equivalent diagrams. Since these trees are uniquely defined by f_T and $f_{T'}$, respectively, they are identical everywhere except at the branch corresponding to the flype. In both cases, the branch corresponding to the subtangle is rooted at the same node as the leaf corresponding to the adjacent twist, but their order is reversed in the two trees. Additionally, the subtangle S from T is flipped in T', which means that the corresponding branch in $A^r(T')$ is flipped based on the rules from Lemma 5.50.

There are two possibilities for $A^r(T')$ depending on whether T and T' are related by a horizontal or vertical flype. In the former case, the subtangle S is flipped horizontally, and hence the branch in $A^r(T')$ corresponding to this subtangle matches $A^r(R_H(S))$. In the case of a vertical flype, the subtangle S is flipped vertically, and hence the corresponding branch in $A^r(T')$ matches $A^r(R_V(S))$. This establishes both possibilities in the second part of the claim, completing the proof.

□

For each of the special cases of algebraic tangle, recall that the first step to obtaining a canonical diagram was to push twists into a preferred position by shuffling them to the right and bottom sides of the diagram using flypes. A flype in a tangle diagram is possible whenever a twist is adjacent to a subtangle. As we saw in the preceding proof, flypes can be identified from a construction formula wherever there exists a sum or product with ± 1, and from the reduced algebraic tangle tree whenever a leaf shares a root with a branch.



(a) $T \cong (\frac{2}{3} + \frac{2}{3}) \circ (0, 2, 2)$ **(b)** Right-veering reduced algebraic tangle tree $A^r(T)$

Figure 5.47: Reduced algebraic tangle tree for a canonical generalized Montesinos diagram

The reduced algebraic tangle tree corresponding to the canonical form of a special class of algebraic tangle diagram has a particular shape. The requirement that every flypable twist be pushed to the right and bottom results in a tangle tree in which the leaves tend to point to the right (Figure 5.47). To formalize this behavior with a definition, we will say that a tangle tree is *right-veering* if every leaf rooted at an interior vertex occurs to the right of every non-leaf branch rooted at that vertex. Hence, canonical diagrams for rational, Montesinos, and generalized Montesinos tangles are always right-veering.

Corollary 5.52 (Special algebraic tangles in canonical form are right-veering)**:**
If T is a rational, Montesinos, or generalized Montesinos tangle diagram in canonical form, then the reduced algebraic tangle tree $A^r(T)$ of this diagram is right-veering.

Proof:

Let T be an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$. Let f_T denote the corresponding reduced construction formula for T using 1-crossing subtangles. Suppose that T is one of the special classes of algebraic tangle, either rational, Montesinos, or generalized Montesinos, and suppose that T has the canonical form for this type of diagram.

By the definition of canonical form, all flyapable twists in T have been shuffled to the right and bottom sides of the diagram. Hence, for any tangle sum or product in the construction formula between ± 1 and a larger subtangle separated with parentheses, the 1-crossing subtangle occurs on the right side in the corresponding sub-segment of f_T. In the tangle tree, the interior vertex corresponding to this sum or product is the root of the branch matching

this subtangle and an adjacent leaf matching this twist. Since the twist occurs on the right in this sub-segment of f_T, the leaf is also to the right of the branch on their shared root. Since this property holds for every instance of ± 1 in f_T and hence for every shared root in the tangle tree, we may conclude that $A^r(T)$ is right-veering.

□

Nothing about the definition of right-veering is restricted to these special classes of algebraic tangle. Rather, this definition mirrors the fact that sums and products with ± 1 occur to the right in a construction formula. Hence, right-veering tangle trees allow us to generalize the idea of a preferred twisting direction for any algebraic tangle diagram. We will say that an algebraic tangle diagram is right-veering when its corresponding reduced algebraic tangle tree is right-veering. We will also say that prunings of right-veering tangle trees are right-veering. Notice that this behavior in the tangle tree is preserved under diagonal flips.

Corollary 5.53 (Diagonal flip of a right-veering algebraic tangle is right-veering)**:**
If T is an algebraic tangle diagram with right-veering reduced algebraic tangle tree $A^r(T)$, then the diagonal flip $r(T)$ also has a right-veering reduced algebraic tangle tree $A^r(r(T))$.

Proof:

Let T be an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$. Suppose that $A^r(T)$ is right-veering. Let $r(T)$ be the algebraic tangle diagram obtained by flipping T across the NW-SE diagonal axis. By Lemma 5.50, the corresponding tangle tree $A^r(r(T))$ for this flipped diagram is obtained from $A^r(T)$ by switching the algebraic operation of each interior vertex. Since this does not change the position of the leaves and branches in the graph, $A^r(r(T))$ still satisfies the condition of being right-veering.

□

This relationship involving diagonal flips can be seen in the symmetric tangles trees $A^r(T)$ and $A^r(r(T))$ shown in Figure 5.45, both of which are right-veering in that example. Corollary 5.53 can be viewed as a generalization of a similar result we stated for Montesinos tangles, wherein we showed that diagonal flips of canonical Montesinos diagrams are also canonical Montesinos (Lemma 5.19). Just as this lemma played a role in obtaining the canonical form for generalized Montesinos tangles, the preceding corollary will soon serve a similar role for algebraic diagrams more generally. Before approaching that question, we now prove that every algebraic tangle admits a right-veering diagram.

Lemma 5.54 (Equivalent right-veering algebraic tangle diagrams)**:**
If T is any algebraic tangle diagram, then there exists an equivalent algebraic diagram $T' \cong T$ which has a reduced algebraic tangle tree $A^r(T')$ which is right-veering.

Proof:

Let T be an algebraic tangle diagram with reduced algebraic tangle tree $A^r(T)$. Let f_T be the reduced construction formula for T using 1-crossing subtangles which relates T and $A^r(T)$ (Lemma 5.46). By definition, $A^r(T)$ is right-veering if and only if every instance of a tangle sum or product in f_T involving ± 1 is such that the corresponding leaf occurs on the right side of a sub-segment of the formula denoting a branch. Assuming that f_T is not already right-veering, we may convert it into the formula for an equivalent right-veering diagram through a sequence of branch flypes.

In f_T, these left sums or products with ± 1 correspond to flypable twists on the left or top side of a subtangle in T. For each such sub-segment of f_T, we can perform a flype in T to move this twist to the other side while preserving the equivalence class of the diagram (Lemma 2.1). By doing so, we move ± 1 to the right side of the sub-segment in f_T, and move a corresponding leaf to the right side of a branch in the tangle tree (Lemma 5.51).

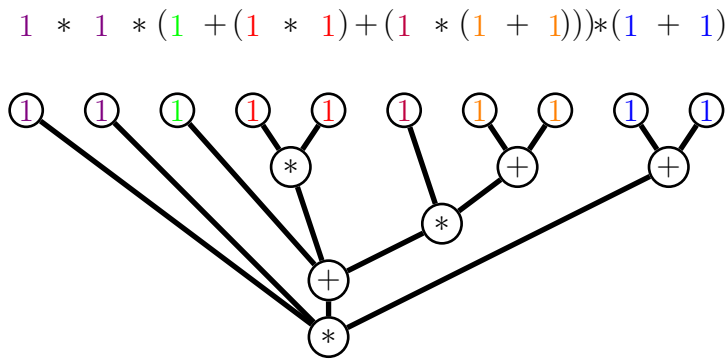
Note that the subtangle corresponding to this branch is flipped, which either reverses the order of all tangle sums in the branch or all tangle products, depending on the type of flype (Lemma 5.50). Consequently, it is possible that other twists in this branch may be moved away from the preferred position. Since our goal is to push all of the twists in T to the right and bottom sides, we must order these flypes such that we avoid undoing our work.

When using a flype to transform a tangle tree, only the branch corresponding to the flyped subtangle is modified, along with the adjacent leaf. Hence, if we start from the interior vertices at the bottom of the tree and work our way up along each branch, we can perform flypes as needed to push twists to the right side of the construction formula without affecting twists outside of this branch. Since there are only finitely many vertices, this process will eventually terminate with some right-veering reduced algebraic tangle tree. This tree defines the desired equivalent algebraic tangle diagram.

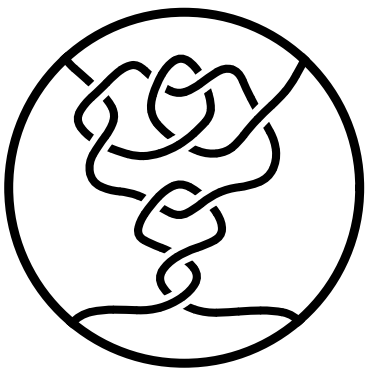
Thanks to the bijection of Lemma 5.46, each of these operations in the tree is mirrored by a subtangle flype in the diagram. Let $T' \cong T$ be the equivalent algebraic diagram obtained after pushing all flypable twists to the right and bottom in this way. The corresponding tangle tree $A^r(T')$ is thus constructed to be right-veering.

□

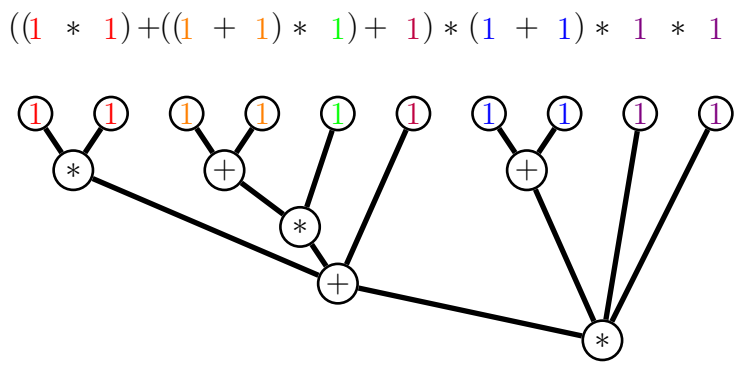
(a) $T \cong \frac{1}{2} * (\frac{1}{1} + \frac{1}{2} + (\frac{1}{1} * \frac{2}{1})) * \frac{2}{1}$



(b) Non-right-veering reduced algebraic tangle tree $A^r(T)$



(c) $T' \cong (\frac{1}{2} + \frac{5}{3}) * \frac{2}{5} \cong T$

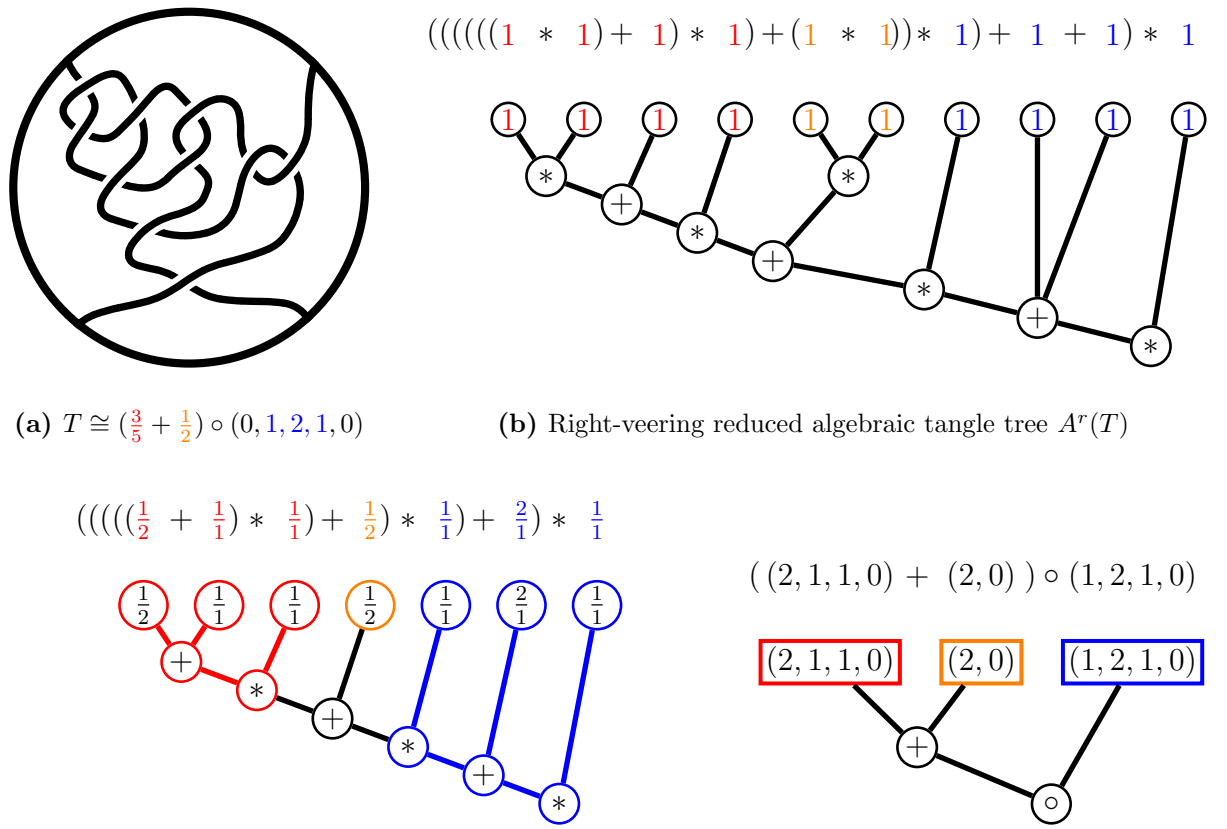


(d) Equivalent right-veering reduced algebraic tangle tree $A^r(T')$

Figure 5.48: Equivalent non-right veering and right-veering algebraic diagrams

Figure 5.48 shows an example of equivalent non-right-veering and right-veering diagrams for an algebraic tangle which does not fall into one of the special cases. By shuffling all twists into this preferred position, the equivalence of Lemma 5.54 allows us to ignore any algebraic diagrams which are not right-veering in favor of those which are. This generalized notion of a preferred direction for twisting gives us the first of two desired conditions for a semi-canonical algebraic diagram. The second condition relates to the sign of these twists.

For rational tangles, we are always able to convert to twists of constant sign in the canonical form. For the canonical form of Montesinos and generalized Montesinos tangles, we are always able to convert to a subtangle of constant sign along with external twists of constant sign, although these signs need not agree. For more complicated algebraic tangles, this relationship becomes difficult to characterize in general, but we may take a similar approach using twist vectors by emulating our work with generalized Montesinos tangles.



(a) $T \cong (\frac{3}{5} + \frac{1}{2}) \circ (0, 1, 2, 1, 0)$ **(b)** Right-veering reduced algebraic tangle tree $A^r(T)$

(c) Right-veering integer pruned tangle tree $A^r_{\text{int}}(T)$ **(d)** Twist vector pruned tangle tree $A^r_{\text{tv}}(T)$

Figure 5.49: Computing the twist vector pruning for a right-veering algebraic tangle tree

Consider the tangle tree corresponding to the canonical diagram for the generalized Montesinos tangle $T \cong (\frac{3}{2} + \frac{1}{2}) \circ (0, 1, 2, 1, 0)$ shown in Figure 5.49. Observe that T contains two rational subtangles within a Montesinos subtangle with endpoint twisting. In a right-veering diagram, each rational subtangle is in twist form, which is to say it can be encoded using a rational twist vector (Corollary 3.6). Similarly, endpoint twisting in non-rational subtangles can be encoded using composition twist vectors (Lemma 5.24).

In this way, we may imagine grouping together all of the twists in an algebraic diagram into twist vectors of one of these two types. This motivates us to define the *twist vector pruning* of a right-veering reduced algebraic tangle tree, which we will denote $A^r_{\text{tv}}(T)$ (Figure 5.49d). The twist vector pruning can be thought of as a modified version of the integer-pruned tangle tree for a right-veering diagram. Using the tangle trees of Figure 5.49 as an example, we now walk through how to compute the twist vector pruning.

Integer subtangles already represent a sequence of adjacent twists of the same type grouped together into a single tangle. Recall that twist vectors are merely a convenient way to encode an alternating sequence of sums with horizontal integer tangles and products with vertical integer tangles. In the integer pruning, leaves represent maximal integer subtangles. Hence, this alternating behavior is easily recognized in $A^r_{\text{int}}(T)$, but only when the twists are in the preferred position corresponding to a right-veering diagram. Twisting can be identified in the integer pruning by examining possible paths through those interior vertices just below the row leaves.

In a reduced tree, interior nodes always alternate between sums and products. If there exists a path in $A^r_{\text{int}}(T)$ through a subset of interior vertices just below the leaves, where each sum connects to a horizontal leaf and each product connects to a vertical leaf, then this path defines a twist vector. The number and sign of twists in each integer subtangle leaf determines the corresponding entry in the twist vector. The length of the twist vector is the number of leaves connected to this path, or one greater if the rightmost interior node denotes a product. In this case, we append a zero entry so that the twist vector ends on a horizontal twist. A path terminating just before a leaf defines a rational twist vector. Otherwise, a path terminating at the root of a non-leaf branch denotes a composition twist vector.

Figure 5.49c shows an example of these paths, and Figure 5.49d shows the corresponding twists vectors. Observe that the red path, which denotes a rational twist vector, encompasses an entire branch in the tangle tree corresponding to a rational subtangle. This will always occur for rational branches in the integer pruning. By contrast, the blue path terminates before reaching the end of the branch, which is characteristic for composition twist vectors. For the orange leaf between these two parts of the tree, there does not exist a corresponding path in $A^r_{\text{int}}(T)$. However, an isolated integer tangle such as this defines a rational twist vector with one nonzero entry.

Figure 5.49d shows the final twist vector pruned tangle tree, $A^r_{\text{tv}}(T)$. Each leaf represents a twist vector identified from the paths in the integer pruning, or else from an isolated integer subtangle for those not contained in a path. To account for the composition operation, we also introduce a new type of interior vertex in $A^r_{\text{tv}}(T)$ denoted by the $\circ$ symbol. Note that every node of this type must have out-degree 2: the left branch denotes a non-rational subtangle, and the right branch is a leaf denoting a composition twist vector. In this way, by identifying these twist vector paths in $A^r_{\text{int}}(T)$ which are as long as possible, we may construct the twist vector pruned algebraic tangle tree for any right veering algebraic tangle diagram.

5.4.3 Algebraic Tangle Diagrams in Semi-Canonical Form

Although the example from Figure 5.49 only uses positive twists, the twist vectors obtained through this construction of $A^r_{\text{tv}}(T)$ need not have constant sign. However, for individual twist vectors, we can transform the diagram so that the entries in a given twist vector do have constant sign. This is possible using canonical twist vectors of both types, which we previously characterized for rational twist vectors in Theorem 3.25 and for composition twist vectors in Corollary 5.38. This is the basis for our definition of a semi-canonical form for algebraic tangle diagrams.

We will say that an algebraic tangle diagram is in *semi-canonical form* if it is both right-veering and its twist vector pruned algebraic tangle tree uses only canonical twist vectors. Note that we require both conditions since the twist vector pruning is only defined for right-veering diagrams. We also include the restriction that any two rational twist vectors joined by a sum or product in $A^r_{\text{tv}}(T)$ have the same sign. This additional rule is to preserve a feature from the generalized Montesinos canonical form, where we showed how sums or products of rational subtangles could be converted to constant sign by modifying the composition twist vector (Theorem 5.39).

The diagram from Figure 5.49 is both canonical as a generalized Montesinos tangle and semi-canonical as an algebraic tangle. This fact holds true in general for the canonical diagrams of the special classes of algebraic tangle, of which the semi-canonical form is an extension. The additional restriction that adjacent rational twist vectors have constant sign ensures that the reverse implication is also true. This rule is equivalent to requiring that every maximal generalized Montesinos subtangle be in canonical form.

Lemma 5.55 (Special algebraic tangles in canonical form are also semi-canonical)**:**
If T is a rational, Montesinos, or generalized Montesinos tangle diagram, then T is in canonical form if and only if T is also in semi-canonical form as an algebraic tangle.

Proof:

Let T be an algebraic tangle diagram. Suppose that T is one of the special classes of algebraic tangle, either rational, Montesinos, or generalized Montesinos. If T has the canonical form for one of these types of diagram, then T is right-veering by Corollary 5.52, and hence defines a twist vector pruned algebraic tangle tree, $A^r_{\text{tv}}(T)$. We will prove both directions of the claim separately for each of these three special classes of algebraic tangle.

Starting with the forward direction, suppose first that T is in canonical form. The rational case is immediate since, if T is rational, then then $A^r_{\text{tv}}(T)$ consists of a single node denoting

a rational twist vector. Since T is in canonical form, this twist vector must also be canonical (Theorem 3.25), and the condition that adjacent rational twist vectors have constant sign holds vacuously. Hence, a canonical rational tangle diagram is also semi-canonical.

Now suppose that T is Montesinos. We will assume without loss of generality that T is horizontal, noting that the case for a vertical Montesinos tangle follows by a parallel argument. Hence, T has a canonical diagram of the form $T \cong L_1 + \cdots + L_m + \frac{k}{1}$, meeting the conditions of Theorem 5.18. In particular, each rational subtangle L_i is also in canonical form, and hence defined by a canonical rational twist vector v_i (Theorem 3.25). The horizontal integer subtangle at the end has twist vector representation $\frac{k}{1} \cong (k)$, which is canonical (Corollary 3.45).

By Theorem 5.18, each of the rational subtangles $L_1, \cdots, L_m$ has constant sign, and hence so do the twist vectors $v_1, \cdots, v_m$. The twist vector (k) may have opposite sign, but these twists can be grouped into a composition twist vector of length one. Thus, the twist vector pruning $A^r_{\text{int}}(T)$ is defined by a construction of the form $T \cong (v_1 + \cdots + v_1) \circ (k)$ consisting entirely of canonical twist vectors, and each pair of rational twist vectors joined by a sum have the same sign. Therefore, T is also in semi-canonical form.

Finally, suppose that T is generalized Montesinos. Hence, the canonical diagram has the form $T \cong M \circ v$, where M is a canonical Montesinos tangle and v is a canonical composition twist vector (Theorem 5.39). Note that all of the rational subtangles used to construct M have constant sign. Since the subtangles of T are in bijection with branches in the tangle tree (Lemma 5.47), the branch of $A^r_{\text{int}}(T)$ corresponding to M is semi-canonical by the case we considered above. Since the remainder of the twist vector pruned tangle tree away from this branch consists of a composition with v, which is canonical, we may conclude that T is semi-canonical.

Therefore, we have shown that each special class of algebraic tangle in canonical form is also in semi-canonical form. This proves the forward direction of the claim. To prove the backward direction, we will suppose that T has a semi-canonical diagram, and then show that T is also canonical if it falls into any of the special cases. The rational case is again immediate since T being rational implies that $A^r_{\text{tv}}(T)$ consists of a single node denoting a canonical rational twist vector. This twist vector defines a canonical rational diagram for T (Theorem 3.25). Hence, a semi-canonical rational diagram is also canonical.

Now suppose that T is Montesinos, again supposing without loss of generality that T is horizontal. The twist vector pruning $A^r_{\text{tv}}(T)$ is thus defined by a construction of the form $T \cong (v_1 + \cdots + v_m) \circ (k)$, where $v_1, \cdots, v_m$ are canonical rational twist vectors of constant

sign and (k) is a canonical composition twist vector. Reversing our steps from the forward direction, each of these rational twist vectors defines a rational subtangle $L_i \cong v_i$ in canonical form, all of which have the same sign. Since the composition twist vector has length one, it represents a sum with a horizontal integer tangle $\frac{k}{1}$. Thus, $T \cong L_1 + \cdots + L_m + \frac{k}{1}$, satisfying the canonical form of a Montesinos tangle defined in Theorem 5.18. This establishes the horizontal Montesinos case, and as before, the vertical case follows by a parallel argument. Hence, a semi-canonical Montesinos diagram is also canonical.

Lastly, suppose that T is generalized Montesinos, which means it has the form $T \cong M \circ v$. If we suppose that T is semi-canonical, then the branch of the twist vector pruning corresponding to the Montesinos subtangle M is also semi-canonical. This branch is defined by a sum or product of canonical composition twist vectors of constant sign, which corresponds to a Montesinos subtangle in canonical form by the preceding case. Furthermore, $T \cong M \circ v$ being semi-canonical implies that v is a canonical composition twist vector. Hence, $T \cong M \circ v$ satisfies the conditions of Theorem 5.39, showing it is also canonical.

Therefore, we have established that a rational, Montesinos, or generalized Montesinos tangle diagram in semi-canonical form also satisfies the conditions of a canonical diagram for that special type of algebraic tangle. This shows the backward direction of the claim, completing the proof.

□

The preceding lemma shows that the semi-canonical form is consistent with all of our previous work. Now that we have a preferred type of diagram which applies to algebraic tangles more generally, our next step is to show that every algebraic diagram can be converted into a semi-canonical form while preserving the equivalence class of the tangle. Figure 5.50 shows this conversion for one such example of a non-semi-canonical algebraic tangle diagram which does not fall into one of these special cases, but the equivalent semi-canonical diagram recovered at the end turns out be generalized Montesinos (Figure 5.50d). Finally, we will prove that every algebraic tangle admits a diagram in semi-canonical form.

Theorem 5.56 (Existence of semi-canonical form for algebraic tangle diagrams)**:**
If T is any algebraic tangle diagram, then there exists an equivalent right-veering *semi-canonical* algebraic tangle diagram $T' \cong T$ which is defined by a right-veering twist vector pruned algebraic tangle tree $A^r_{\text{tv}}(T')$ using only canonical twist vectors, where any two rational twist vectors joined by a sum or product have the same sign.

Proof:

Let T be an algebraic tangle diagram. By Lemma 5.54, we may suppose without loss of generality that T is right-veering, and hence the twist vector pruning $A^r_{\text{tv}}(T)$ exists. As illustrated in Figure 5.50a, the twist vector pruned tangle tree is defined by a construction using some combination of rational twist vectors and composition twist vectors. If T is not already in semi-canonical form, we will show how each twist vector can be modified until T satisfies the conditions of semi-canonicity.

Since each branch in the tangle tree corresponds to a subtangle in the diagram (Lemma 5.47), we may replace a given branch in the tree with a branch for an equivalent subtangle while preserving the overall equivalence class of the diagram. Hence, the semi-canonical conversion can be achieved in four stages. First, for each rational twist vector, we may replace any non-canonical twist vectors with an equivalent canonical rational twist vector (Lemma 3.24). Each such replacement involves modifying a rational twist vector leaf in $A^r_{\text{tv}}(T)$, and corresponds to replacing a rational subtangle in T with an equivalent canonical rational subtangle matching the canonical twist vector.

Let $T' \cong T$ denote the equivalent algebraic diagram obtained after completing all of these replacements. Note that the corresponding reduced tangle tree $A^r(T')$ is obtained from $A^r(T)$ by replacing branches for rational subtangles with equivalent branches for canonical rational subtangles, although these branches will have fewer leaves if the number of crossings decreases. By Corollary 5.52, these branches are right-veering, and hence T' is still right-veering, which means that the twist vector pruning $A^r_{\text{tv}}(T')$ still exists. At this stage, every rational twist vector in $A^r_{\text{tv}}(T')$ is in canonical form.

The second step is to convert any adjacent rational twist vectors to the same sign. In the twist vector pruned tangle tree, every collection of adjacent rational twist vector leaves rooted at the same interior node determines a Montesinos subtangle. This subtangle is horizontal if the interior vertex denotes a sum, and vertical if the interior vertex denotes a product. By examining adjacent rational twist vectors in this way, we may identify every maximal Montesinos subtangle containing more than one maximal rational subtangle. In a reduced tangle tree, the corresponding rational subtangle branches are all rooted at the same interior node, although it is possible that disjoint maximal Montesinos subtangles share the same root if they are separated by branch for a non-rational subtangle.

For each branch in $A^r_{\text{tv}}(T')$ corresponding to a maximal Montesinos subtangle identified in this way, the subtangle is semi-canonical if and only if it is canonical (Lemma 5.55). Since each rational twist vector in $A^r_{\text{tv}}(T')$ has already been made canonical, this means

that one of these maximal Montesinos branches can only fail to be canonical if it contains adjacent rational subtangles of opposite sign. If this happens, the non-canonical Montesinos subtangle in T' may be replaced with an equivalent canonical Montesinos (Theorem 5.18). In the tangle tree, this means replacing the corresponding branch.

Let $T'' \cong T'$ denote the equivalent algebraic diagram obtained after replacing each maximal Montesinos subtangle of T' identified above with an equivalent canonical Montesinos tangle. However, it is possible that the canonical Montesinos conversion from Theorem 5.18 also introduces some new horizontal twists, possibly of a different sign than the other rational subtangles. If this happens, it is possible that these twists make T'' a non-right-veering diagram. However, these new twists can always be flyped into a preferred position to obtain an equivalent right-veering diagram by Lemma 5.54. For the third step in this conversion, let $T''' \cong T''$ be the right-veering diagram obtained in this way.

At this stage, we now have an algebraic diagram T''' which is equivalent to our original diagram T, and which satisfies the conditions of being right-veering, using only canonical rational twist vectors, and adjacent rational twist vectors have the same sign. Let $A^r_{\text{tv}}(T''')$ be the corresponding twist vector pruned tangle tree. The fourth and final step is to convert any non-canonical composition twist vectors in this tree.

Recall that every interior vertex in $A^r_{\text{tv}}(T''')$ marked with by the symbol $\circ$ denotes a composition. This vertex has out-degree 2 and serves as the root of the branch for a non-rational subtangle on the left, and the root of a leaf for a composition twist vector on the right. In other words, every leaf denoting a composition twist vector v shares a root with a branch representing a non-rational subtangle L. Taken together, these denote a branch in $A^r_{\text{tv}}(T''')$ representing the subtangle $L \circ v$ in T'''. Every interior vertex denoting a composition in $A^r_{\text{tv}}(T''')$ defines a subtangle of this form. We will show that this tree can be transformed so that each composition twist vector is canonical.

If v is a non-canonical composition twist vector identified as above, then by Corollary 5.38, there exists an equivalent canonical composition twist vector v' and a tangle L' such that $L \circ v \cong L' \circ v'$, where either $L' \cong L$ or $L' \cong r(L)$. Since the branch of the tree corresponding to the subtangle L is right-veering, note that a branch corresponding to $r(L)$ is also right-veering (Corollary 5.53). Hence, for any non-canonical composition twist vector v in $A^r_{\text{tv}}(T''')$, we may replace the branch corresponding to $L \circ v$ with an equivalent branch $L' \circ v'$ which uses a canonical twist vector v'.

Each such change only modifies the leaf corresponding to the composition twist vector, possibly with a diagonal flip in the corresponding branch. While a diagonal flip reverses the

sign of each twist in the flipped branch and switches the operations of the nodes, it preserves the canonicity of any rational and Montesinos subtangles along this branch (Corollary 5.19). Hence, starting from the bottom of the tree $A^r_{\text{tv}}(T''')$ and working our way up along each branch, we may use this method to convert any non-canonical composition twist vectors into canonical form without undoing our previous work. In some cases,

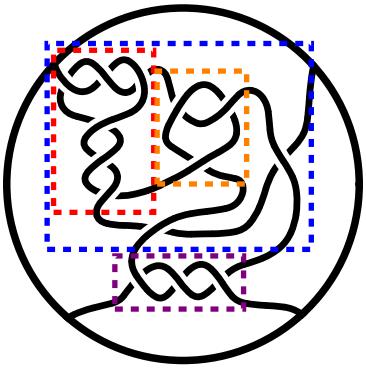
Let $T'''' \cong T'''$ denote the equivalent algebraic tangle diagram obtained by modifying each branch corresponding to a non-canonical composition twist vector in this way. Finally, we have obtained a right-veering diagram with an integer pruned tangle tree $A^r_{\text{tv}}(T)$ for which each rational and composition twist vector is canonical, and such that each pair of adjacent rational twist vectors joined by a sum or product have the same sign. In other words, T'''' is an algebraic diagram in semi-canonical form. Furthermore, each operation used to obtain this diagram preserved the equivalence class of our original algebraic diagram T, and so $T'''' \cong T$ is the equivalent semi-canonical diagram asserted in the claim. This completes the proof.

□

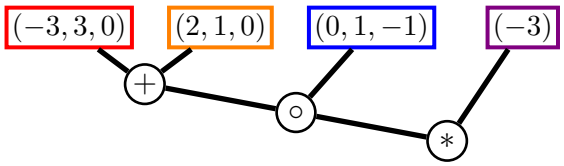
Figure 5.50 illustrates an example of this conversion into semi-canonical form. This includes converting a non-canonical rational twist vector (Part 5.50b) and a non-canonical composition twist vector (Part 5.50c). As this example shows, these conversions may lead to twists in the interior of the diagram that are not right-veering. The final step of this process involves flyping any such twists to the right and bottom, and updating each twist vector to its canonical form. In this case, the final diagram obtained turns out to be generalized Montesinos (Part 5.50d).

Theorem 5.56 proves that every algebraic tangle diagram can be represented in a semi-canonical form. The reason we call this *semi*-canonical is because this identification is not known to be unique. Even so, this result is extremely useful for tabulating algebraic tangles. Since every algebraic tangle admits a semi-canonical diagram, it is sufficient to tabulate only diagrams of this form; non-semi-canonical diagrams can be excluded as redundant. In other words, we may safely omit many extraneous diagrams, even allowing for the possibility of undetected repeats. An easy consequence of this theorem relates to the subtangles of a semi-canonical diagram.

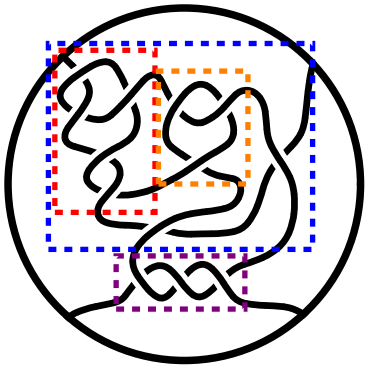
Corollary 5.57 (Subtangles of a semi-canonical diagram are semi-canonical)**:**
If T is an algebraic tangle diagram in semi-canonical form, then every subtangle of T is also in semi-canonical form.



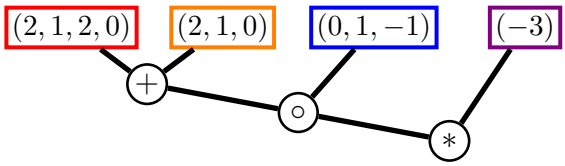
$T \cong [[(-3,3,0)+(2,1,0)] \circ (0,1,-1)] * (-3)$



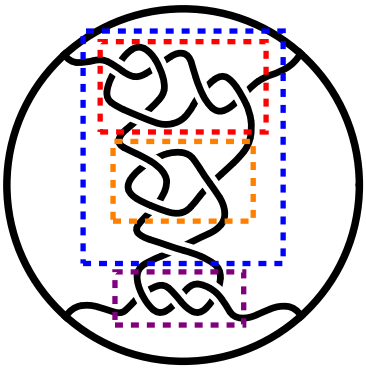
(a) Non-semi-canonical diagram T and tree $A^r_{\text{tv}}(T)$



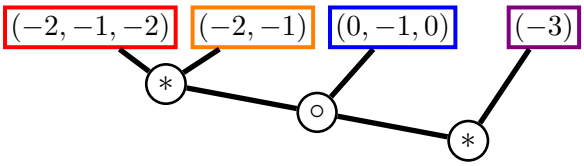
$T' \cong [[(2,1,2,0)+(2,1,0)] \circ (0,1,-1)] * (-3)$



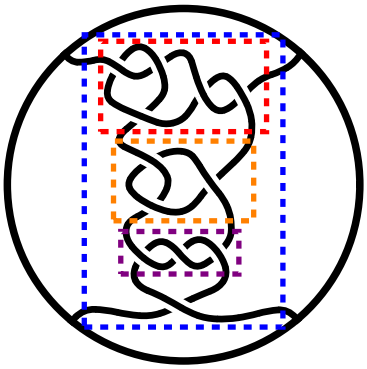
(b) Converting rational twist vectors



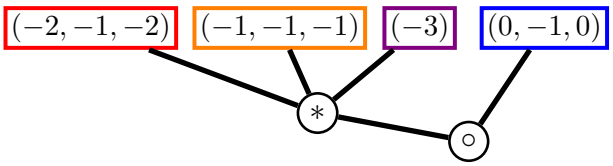
$T'' \cong [[(-2,-1,-2) * (-2,-1)] \circ (0,-1,0)] * (-3)$



(c) Converting composition twist vectors



$T''' \cong [(-2,-1,-2,) * (-1,-1,-1) * (-3)] \circ (0,-1,0)$



(d) Final semi-canonical construction

Figure 5.50: Converting an algebraic diagram into semi-canonical form via twist vectors

Proof:

Let T be an algebraic tangle diagram in semi-canonical form, let $A^r(T)$ be the corresponding reduced algebraic tangle tree, and let $A^r_{\text{tv}}(T)$ be the corresponding twist vector pruned tangle tree. Let S be any subtangle in T. Thus, by Lemma 5.47, there exist a branch $A^r(S)$ in the tangle tree which corresponds to this subtangle. Since $A^r(T)$ is a right-veering tree, note that this branch $A^r(S)$ is also right veering. Hence, this branch has a twist vector pruning $A^r_{\text{tv}}(S)$ which can be identified with a branch, leaf, or portion of a leaf in the twist vector pruned tangle tree $A^r_{\text{tv}}(T)$ for the original diagram.

If $A^r_{\text{tv}}(S)$ is not semi-canonical, then this would imply that the corresponding branch in $A^r_{\text{tv}}(T)$ violates one or more of the requirements of semi-canonicity stated in Theorem 5.56. However, this cannot happen since T is assumed to be in semi-canonical form. Therefore, we may conclude that the subtangle S is also in semi-canonical form.

□

Although we may not claim uniqueness in general, we may go slightly further for *alternating* semi-canonical diagrams. By Corollary 5.5, any algebraic diagram can be identified as alternating or not based on a construction formula using canonical rational subtangles. This construction is also inherent to the semi-canonical form.

Corollary 5.58 (Alternating semi-canonical algebraic tangle diagrams)**:**
If T is an algebraic tangle diagram in semi-canonical form, then T is alternating if and only if all nonzero twists in the twist vector pruned tangle tree $A^r_{\text{tv}}(T)$ have the same sign.

Proof:

Let T be an algebraic tangle diagram in semi-canonical form. Hence, the construction of T is defined by a twist vector pruned tangle tree $A^r_{\text{tv}}(T)$ satisfying the conditions from Theorem 5.56. Based on the construction of this tree, an algebraic construction formula for T can be recovered by replacing each rational twist vector with the corresponding rational subtangle, and each composition twist vector with an alternating sequence of sums and products with horizontal and vertical integer subtangles, respectively.

This construction formula defines a decomposition D of T into rational subtangles. Since T is semi-canonical, we may assume that each rational subtangle recovered from a canonical rational twist vector is in canonical form. Similarly, each of the integer subtangles obtained from the composition twist vectors is in canonical form. Hence, each of the subtangles in the decomposition D is a canonical rational subtangle of T. Note that the sign of these subtangles matches the sign of the corresponding twist vector from $A^r_{\text{tv}}(T)$.

The alternating behavior of T is characterized by Corollay 5.5. If T is alternating, then each rational subtangle in D has the same sign, and hence each twist vector in $A^r_{\text{tv}}(T)$ has the same sign. Conversely, if each twist vector in $A^r_{\text{tv}}(T)$ has the same sign, then each rational subtangle in D has the same sign. Therefore, T is alternating if and only if each nonzero twist used in the twist vector pruning $A^r_{\text{tv}}(T)$ has the same sign.

□

For alternating diagrams in particular, recall that the generalized version of the Tait Flyping Conjecture gives a condition relating equivalent alternating 2-string tangle diagrams using vertex closures (Corollary 3.27). This result only applies to *reduced* alternating diagrams. Since we have excluded local knots from our study of tangles, we may assume that every 2-string tangle diagram, alternating or otherwise, contains no nugatory crossings and hence is reduced (Lemma 4.1). Thus, the Tait Flyping Conjecture will enable us to give a general classification of alternating semi-canonical diagrams.

Recall that the central idea of the Tait Conjecture is that equivalent alternating diagrams can be isotoped into each other only using flypes. In the generalized version, this is extended to vertex closures of alternating 2-string tangle diagrams sitting on a sphere, which refers to the diagram obtained by collapsing the boundary circle to a special vertex on the back of the sphere. In this case, equivalent diagrams can be related either using subtangle flypes or flypes across the boundary vertex. Hence, we may classify the behavior of an alternating semi-canonical algebraic diagram based on these these two transformations in the following special case.

Theorem 5.59 (Classification of non-twisted semi-canonical algebraic diagrams)**:**
If T is an alternating algebraic tangle diagram in semi-canonical form, and there does not exist a proper subtangle L and a non-trivial twist vector v such that $T \cong L \circ v$, then the semi-canonical form of T is unique.

Proof:

Let T be an algebraic tangle diagram in semi-canonical form. Hence, the construction of T is defined by a twist vector pruned tangle tree $A^r_{\text{tv}}(T)$ satisfying the conditions from Theorem 5.56. Since T is alternating, we know by Corollary 5.58 that all nonzero twists used in this tree have the same sign. We will assume that there does not exist a proper subtangle L of T and a non-trivial twist vector v such that $T \cong L \circ v$.

Suppose by way of contradiction that there exists an equivalent alternating algebraic tangle $T' \cong T$ with a semi-canonical form that is different from T. Let $A^r_{\text{tv}}(T')$ denote the corresponding twist vector pruned tangle tree, and note that this tree must be distinct from $A^r_{\text{tv}}(T)$. Let $V(T)$ and $V(T')$ denote the vertex closures of these two algebraic diagrams. By the generalized Tait Flyping Conjecture (Corollary 3.27), $V(T)$ and $V(T')$ must be related by some combination of subtangle flypes and vertex flypes.

Since T and T' are in semi-canonical form, they both have right-veering diagrams. In particular, this means that every twist in both diagrams has already been shifted as far as possible to the right and bottom of the diagram. Flypes only have the effect of moving around twists in the diagram, and hence any subtangle flype in $V(T)$ would move a twist out of this preferred position. Thus, if T and T' are both right-veering but non-identical, then it is impossible to covert T into T' using only subtangle flypes. In other words, the conversion from $V(T)$ to $V(T')$ must involve at least one vertex flype.

A vertex flype in $V(T)$ is only possible if there exists a twist in the diagram adjacent to the vertex. In general, there could exist twists in the interior of $V(T)$ which are not adjacent to the vertex but which become adjacent after some combination of subtangle flypes. However, because T is right-veering, all such twists have already been pushed to the right and bottom sides of the diagram. Therefore, if there exists a possible vertex flype in $V(T)$, then there exists a twist in T at the far right or bottom of the diagram.

Suppose without loss of generality that the former case holds, meaning there exists a twist on the far right of the diagram. If this happens, then by isolating this twist from the rest of the crossings in T, we may decompose the diagram as $T \cong L + \frac{\pm 1}{1} \cong L \circ (\pm 1)$, where L is the proper subtangle of T obtained by removing this twist. However, this is not possible since it would violate our assumption that no such decomposition of T exists. A similar argument shows that no bottom twists are possible in T either.

This means that T has no endpoint twisting, nor does it contain any twists that could be pushed to the end of the diagram. Hence, it is not possible to perform a vertex flype in $V(T)$, even after performing some combination of subtangle flypes first. However, this contradicts the fact that the conversion from $V(T)$ to $V(T')$ guaranteed by the generalized Tait Flyping Conjecture must involve at least one vertex flype. Therefore, we may conclude that there does not exist an equivalent alternating algebraic diagram $T' \cong T$ that has a different semi-canonical form. In other words, the semi-canonical form of T is unique.

□

The assumption that algebraic diagrams in semi-canonical form be right-veering gives us an easy way to identify possible vertex flypes in the vertex closure. As in the preceding proof, these are only possible when the diagram can be described with endpoint twisting. In the absence of any endpoint twisting, Theorem 5.59 proves that this special case of algebraic tangle has a unique semi-canonical form.

For those diagrams which do include endpoint twisting, then the twists at the end can be used to perform vertex flypes. As we saw when studying rational tangle diagrams, the effect of a vertex flype can be described using a horizontal or vertical flip (Figure 3.31). This was fine for rational tangles, which define equivalent diagrams under these types of flips (Lemma 3.3), but this does not extend to more general types of diagrams. Hence, the uniqueness of the preceding characterization can only be extended to alternating algebraic diagrams in semi-canonical form more generally up to a possible horizontal or vertical flip.

Conjecture 5.60 (Semi-canonical twisted alternating algebraic diagrams)**:**
If T is an alternating algebraic tangle diagram in semi-canonical form, and there exists a proper subtangle L and a non-trivial twist vector v such that $T \cong L \circ v$, then the semi-canonical form of T is unique up to a possible horizontal or vertical flip of L.

As we did before for the special classes of algebraic tangle, we conclude this subsection with an observation concerning subtangle decompositions. If T is any algebraic tangle diagram in semi-canonical form, then the corresponding twist vector pruned tangle tree can be used to obtain a unique decomposition of this diagram into maximal rational subtangles. For the special classes of algebraic tangle, this decomposition agrees with our previous characterization thanks to Lemma 5.55.

Theorem 5.61 (Semi-canonical decomposition into maximal rational subtangles)**:**
If T is an algebraic tangle diagram in semi-canonical form with twist vector pruned tangle tree $A^r_{\text{tv}}(T)$, then this tree defines a unique decomposition of T into maximal rational subtangles, wherein each rational twist vector of $A^r_{\text{tv}}(T)$ defines a maximal rational subtangle, and each non-zero entry of each composition twist vector defines a maximal integer subtangle.

Proof:

Let T be an algebraic tangle diagram in semi-canonical form with right-veering reduced tangle tree $A^r(T)$. Let $A^r_{\text{int}}(T)$ denote the integer pruned tangle tree and let $A^r_{\text{tv}}(T)$ be the corresponding twist vector pruned tangle tree (Theorem 5.56). Each rational twist vector

leaf in $A^r_{\text{tv}}(T)$ corresponds to a branch in $A^r(T)$ and hence a rational subtangle in T (Lemma 5.47). Similarly, each composition twist vector leaf in $A^r_{\text{tv}}(T)$ denotes a sequence of sums with horizontal integer tangles and products with vertical integer tangles. Let D denote the set of rational and integer subtangles in T identified in this way from each twist vector leaf in $A^r_{\text{tv}}(T)$.

Since the leaves in $A^r_{\text{tv}}(T)$ exhaust the crossings in T, it follows that D defines a decomposition of T into rational subtangles. To prove the claim, we must show that each subtangle in D is a maximal rational subtangle. We will consider the subtangles obtained from each type of twist vector in $A^r_{\text{tv}}(T)$ separately, starting with those from the rational twist vectors. Let S be any such rational subtangle of T, with corresponding branch $A^r(S)$ in $A^r(T)$, and rational twist vector leaf v_S in $A^r_{\text{tv}}(T)$.

If S is not maximal, then it is properly contained in a some larger rational subtangle, and hence the branch $A^r(S)$ is rooted along some longer rational branch in the reduced algebraic tangle tree. Any twists in this larger subtangle with are not contained in S must occur further down the branch from where $A^r(S)$ is rooted. Passing to the integer pruned tangle tree, a similar fact must hold so that the corresponding branch $A^r_{\text{int}}(S)$ is rooted along some longer rational branch in $A^r_{\text{int}}(T)$. Since T is right-veering, each additional integer tangle along this longer rational branch must occur to the right of $A^r_{\text{int}}(S)$ in the tree, and these alternate between sums with horizontal integer tangles and products with vertical integer tangles.

By definition, each twist vector leaf in $A^r_{\text{tv}}(T)$ is determined by a path through the interior nodes in $A^r_{\text{int}}(T)$ which alternate between sums with horizontal integer tangles and products with vertical integer tangles, and we assumed that each of these paths is as long as possible. Hence, if there exists a longer rational branch which properly contains $A^r_{\text{int}}(S)$ in the integer pruning, then the path used to determine v_S could be extended along this branch, which would contradict the assumption that this path is as long as possible. Therefore, we may conclude that the branch corresponding to S is not properly contained in a larger rational branch of the tree, which is to say that S is a maximal rational subtangle.

Next, we must consider the composition twist vectors from $A^r_{\text{tv}}(T)$. Let $v = (n_1, \cdots, n_r)$ denote one such vector, noting that v is canonical. This vector defines r integer subtangles $\frac{n_1}{1}, \frac{1}{n_2}, \cdots, \frac{n_r}{1}$ in D, noting that any trivial tangles are excluded. We must verify that none of these is contained in a larger rational subtangle. In the twist vector pruning, the leaf corresponding to v shares a root with a branch for some subtangle L in T, and together these define the subtangle $L \circ v$. Note that L is not rational, otherwise v would have been

grouped into the corresponding rational twist vector.

Thanks to the alternating sums and products, the smallest subtangle of T which properly contains one of these horizontal integer subtangles $\frac{n_i}{1}$ is $L + \frac{n_1}{1} * \cdots + \frac{n_i}{1}$. In the case of a vertical integer subtangle $\frac{1}{n_i}$, this is $L + \frac{n_1}{1} * \cdots * \frac{1}{n_i}$. However, no subtangle of this form is rational since L is not rational. Hence, none of these integer tangles is properly contained in a larger rational subtangle, and so each is a maximal rational subtangle. This is essentially the same argument that we used for generalized Montesinos tangles in the proof of Theorem 5.43.

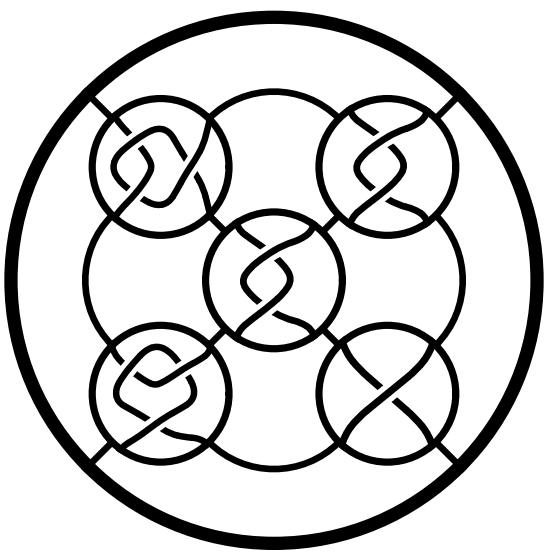
Thus, we may conclude that each rational subtangle of D identified in this way is maximal. Furthermore, this decomposition is unique by Lemma 4.11, which completes the proof.

□

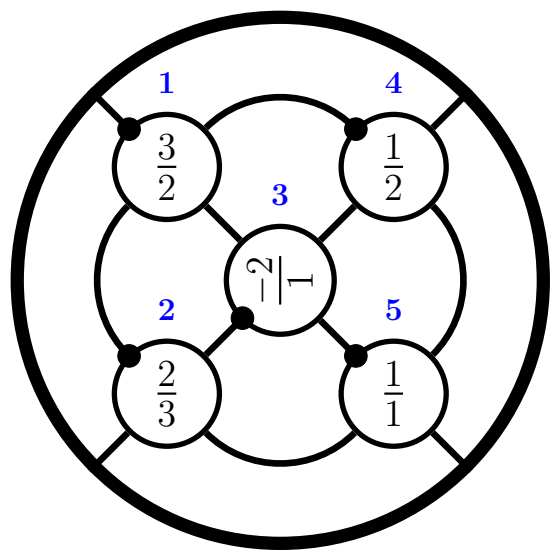
6 Constellations

The majority of our work up to this point has focused on algebraic tangles and their special cases. Non-algebraic tangles, when they have come up, have only been discussed in the context of how they fail to be algebraic. Whereas we have always been able to work from a concrete algebraic construction formula for algebraic tangle diagrams, we do not have an analogous construction method that applies to all non-algebraic tangle diagrams.

Subtangle decompositions are the solution to this (Figure 6.1). In particular, decompositions into maximal algebraic subtangles, which can be used to determine when a tangle diagram is non-algebraic. Furthermore, the algorithm to compute the APD code of a diagram provides a method for us to obtain this decomposition in practice. As a diagram notation, the APD code is "as algebraic as possible" in the sense that whatever subtangles remain cannot be joined together using simple tangle algebra.



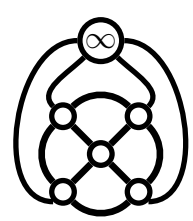
(a) Non-algebraic tangle diagram



(b) Maximal algebraic subtangle decomposition

1	$\left(\frac{3}{2}\right)$	0 NW	4 NW	3 NE	2 NW	(P: ∞)
2	$\left(\frac{2}{3}\right)$	1 SW	3 NW	5 SW	0 SW	(P: 0)
3	$\left(\frac{-2}{1}\right)$	2 NE	1 SE	4 SW	5 NW	(P: 0)
4	$\left(\frac{1}{2}\right)$	1 NE	0 NE	5 NE	3 SE	(P: ∞')
5	$\left(\frac{1}{1}\right)$	3 SW	4 SE	0 SE	2 SE	(P: 1)

(c) Algebraic planar diagram code



(d) Constellation

Figure 6.1: Example of a non-algebraic tangle diagram deconstructed using graphs

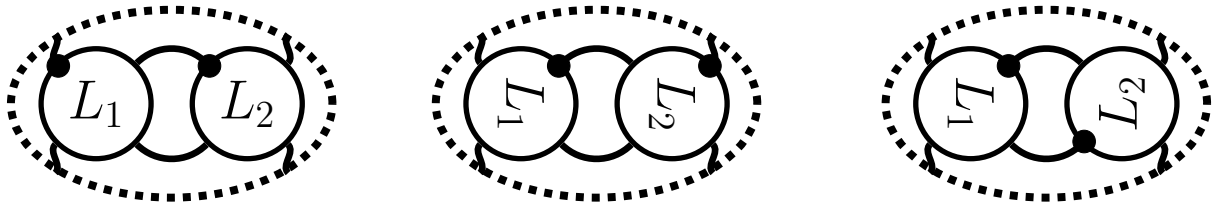
To understand non-algebraic tangle diagrams, we must seek to understand what non-algebraic configurations of subtangles are possible in the APD code. Certain configurations of subtangles necessarily arise in decompositions for non-algebraic tangle diagrams, and these can easily be described using planar graphs (Figure 6.1d). These graphs, which we will refer to as *constellations*, are closely related to the basic polyhedra used in Conway notation (Figure 4.24), but generalized to tangle diagrams. With the goal of building up to constellations, we will begin by formalizing the theory underlying subtangle graphs.

6.1 Algebraic Subtangle Graphs

How to identify non-algebraic subtangle decompositions is a surprisingly deep question. Abstracting away from familiar tangle diagrams with strings and crossings, the objects that we must really understand are subtangle graphs. Recall that any choice of subtangle decomposition for a diagram defines a corresponding *subtangle graph*, where the vertices represent subtangles, the edges represent the strings between these subtangles, and we collapse the boundary sphere to a special vertex marked ∞ (Figure 6.1d). Given a subtangle decomposition D of some tangle diagram, let $G(D)$ denote the subtangle graph obtained from D by this identification. Note that $G(D)$ is a connected 4-valent planar graph.

To characterize the structure of a non-algebraic tangle daigram, we will be most interested in studying its decomposition into maximal algebraic subtangles. We will refer to the graph which corresponds to this decomposition as the *maximal algebraic subtangle graph* (MASG) of a tangle diagram. Note that this graph is well-defined since the decomposition D_{alg} of a tangle diagram into maximal aglebraic subtangles always exists and is unique (Lemma 4.9). For a fixed tangle diagram, the MASG is exactly $G(D_{\text{alg}})$, and the adjacency list for this graph is built into the APD code.

Even without knowing the APD code, the MASG of a tangle diagram can be determined from any of its *algebraic subtangle graphs*, meaning those graphs which correspond to a decomposition into algebraic subtangles, maximal or otherwise. All of the examples of subtangle decompositions we examined when computing the IPD, RPD, and APD codes use algebraic subtangles. There is an interesting categorical relationship between the various algebraic subtangle graphs corresponding to a fixed tangle diagram. In particular, we can use the language of category theory to show how the MASG of a tangle diagram arises naturally from collapsing vertices together in these graphs.



(a) Local sum: $L_1 + L_2$ **(b)** Local product: $L_2 * L_1$ **(c)** Ambiguous algebraic connection

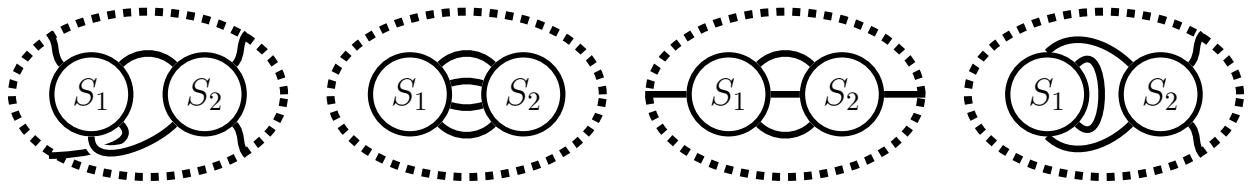
Figure 6.2: Examples of equivalent algebraic connections between subtangles

6.1.1 The Category of Algebraic Subtangle Graphs

When we discussed how to convert RPD code into APD code, the basic principle involved searching for adjacent subtangles that could be joined together using a tangle sum or product. When this is possible, whether we use sum or product depends on a choice of the NW corner in the subtangles. When examining connections between subtangles, sometimes it was necessary to choose a different NW corner on one of the subtangles as an intermediate step before these operations made sense. This happened whenever one subtangle appeared rotated relative to an adjacent subtangle, but it was always possible to make a different choice locally to identify a sum or product of subtangles (Figure 6.2).

When computing the various SPD codes, it is important to keep track of the NW corner for each subtangle, but these choices do not affect the corresponding subtangle graph. The three examples of Figure 6.2 look identical in the subtangle graph, even though the subtangles' orientations in the decomposition do not match in the third example. For the purpose of identifying when two adjacent subtangles can be joined together into a larger algebraic subtangle, it is not important to know whether these use a tangle sum or product. Rather, what matters is only that there are two pairs of connected endpoints between these neighboring subtangles. We will say that two adjacent subtangles in a subtangle decomposition share an *algebraic connection* if they are connected by exactly two strings. This will imply the existence of a tangle sum or product relative to some choice of NW corner.

Lemma 6.1 (Connections between subtangles)**:**
If D is a subtangle decomposition for a tangle diagram T, then every pair of subtangles in D is connected by at most two strings. If there exists a pair of subtangles in D with exactly two shared connections, then either their sum or product is a subtangle of T.



(a) Rogue crossing **(b)** Disconnected strings **(c)** Local knot **(d)** Trapped local knot

Figure 6.3: Invalid connections between subtangles in a subtangle decomposition

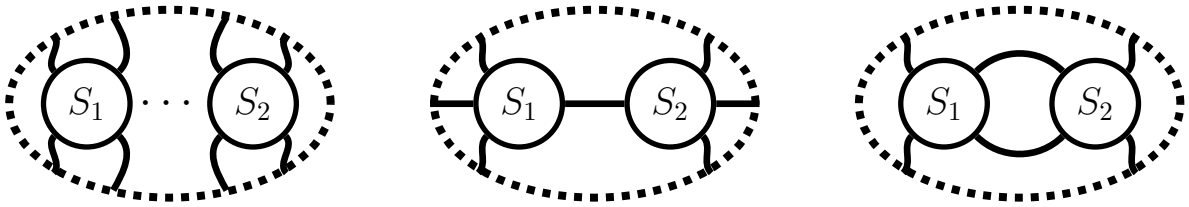
Proof:

Let T be a 2-string tangle diagram and let D be a subtangle decomposition for this diagram. Let S_1 and S_2 be two subtangles from D. Since we only consider 2-string tangles, the separating circles for S_1 and S_2 have exactly four endpoints each, and these must be non-overlapping in D. Hence, S_1 and S_2 can be joined by a maximum of four strings if each endpoint from S_1 is connected to an endpoint from S_2. We may prove the claim by examining the possible connections between these two subtangles.

Note that, in a subtangle decomposition, all of the crossings have been partitioned into disjoint subtangles. Hence, if two subtangles from a decomposition are connected by one or more strings, then there do not exist any *rogue crossings* along the strings in between these subtangles. Figure 6.3a illustrates this impossibility. If such a crossing did exist along the string between S_1 and S_2, then it would have to be contained in a subtangle, in which case this string is not a direct connection between S_1 and S_2. Thus, we may assume that all of the strings in the subtangle decomposition D are non-overlapping.

If all four endpoints from S_1 are connected to all four endpoints from S_2, then these would form a closed loop in the interior of T (Figure 6.3b). However, this would mean that S_1 and S_2 are completely disconnected from the rest of the diagram, which cannot happen under our definition of tangle. If S_1 and S_2 are connected by three strings along three pairs of endpoints, then these two subtangles form a local knot (Figure 6.3c). This cannot happen either since we have assumed that local knots do not exist (Lemma 4.1).

Thus, S_1 and S_2 can share at most two connections. Suppose that S_1 and S_2 are connected by exactly two strings. Hence, there exists a pair of endpoints from S_1 which connects to a pair of endpoints from S_2. Note that the two endpoints from S_1 must be directly next to each other on the separating circle. If another endpoint lies in between them, it would either have to make a third connection with S_2 (6.3c), or else overlap with another string (6.3a). These would result in a local knot or rogue crossing, respectively, and so neither is possible.



(a) No connections **(b)** One shared connection **(c)** Two shared connections

Figure 6.4: Valid connections between subtangles in a subtangle decomposition

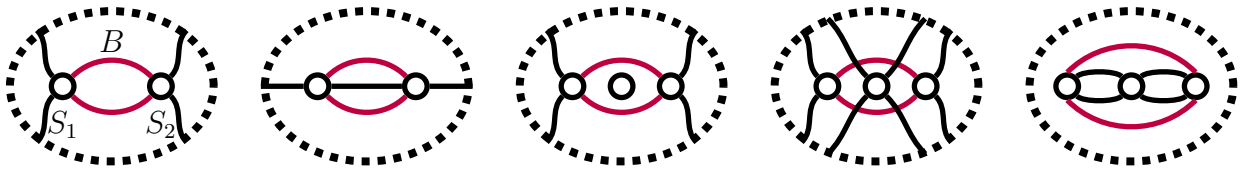
Similar restrictions apply to the locations of the connected endpoints on S_2. Furthermore, if there are two strings connecting S_1 and S_2, then the remaining non-connected endpoints cannot be trapped between these like the endpoints shown in Figure 6.3d. Since the strings from these trapped endpoints cannot connect to the other subtangle nor can they overlap with any other string, the only remaining option is that the trapped endpoints connect to each other. However, this would also create a local knot, and hence cannot happen.

Therefore, the only possibilities remaining for how S_1 and S_2 connect are summarized in Figure 6.4. Either these subtangles share no connections (6.4a), share a single connection (6.4b), or share two connections (6.4c). If they share two connections, then the connected endpoints must be directly next to each other on the separating circles as shown, with nothing else passing in between them. This has the form $S_1 + S_2$ in the figure, but note that these could also be expressed using a product or in a different order depending on how we choose the NW endpoints. In either case, this algebraic combination will be a subtangle of T and it can be expressed used a tangle sum or a tangle product.

□

Lemma 6.1 guarantees that subtangles connected by two strings can be joined together algebraically, and it justifies our definition of an algebraic connection between subtangles. In terms of the subtangle graph, an algebraic combination of subtangles in D defines a *bigon* in $G(D)$, meaning a planar face with exactly two edges (Figure 6.5a).

Lemma 6.2 (Bigons in a subtangle graph)**:**
If two subtangles in a subtangle decomposition D for a tangle diagram share an algebraic connection, then these define a bigon in the corresponding subtangle graph $G(D)$.



(a) Bigon B **(b)** Third connection **(c)** Trapped vertex **(d)** Non-planarity **(e)** Broken subgraph

Figure 6.5: Potential bigon conflicts that do not happen in a subtangle graph

Proof:

Let T be a 2-string tangle diagram, let D be a subtangle decomposition for this diagram, and let $G(D)$ be the subtangle graph corresponding to this decomposition. Suppose that S_1 and S_2 are subtangles in D which share an algebraic connection, meaning they are connected by exactly two strings. In $G(D)$, there exist vertices corresponding to S_1 and S_2 and two edges between them corresponding to the two connecting strings. Let B denote the cycle in $G(D)$ defined by these two edges (Figure 6.5a). We must verify that B is in fact a bigon.

Since $G(D)$ is a planar graph, B defines a bigon if and only if it contains no other vertices and edges in the interior. By Lemma 6.1, there cannot exist a third edge directly from S_1 to S_2, otherwise the corresponding subtangles would be connected by three strings. If there does exist a vertex in the interior of B, then there exists another subtangle S_3 lying in between S_1 and S_2. The last three images from Figure 6.5 illustrate why this cannot happen, but suppose by way of contradiction that such a vertex does exist.

Since $G(D)$ is 4-valent, the vertices corresponding to S_1, S_2, and S_3 each have four adjacent edges. Note that no edge in $G(D)$ is a loop since these would define a local knot in D. There exist two edges between S_1 and S_2 by assumption, and S_3 lies trapped in between these (Figure 6.5c). Since $G(D)$ is connected, there must exist an edge from S_3 to another vertex. If S_3 does not connect to S_1 nor to S_2, then the edges adjacent to S_3 must overlap with the edges of B to exit this region (Figure 6.5d). However, this cannot happen since $G(D)$ is planar; these would correspond to rogue crossings in D.

If the edges adjacent to S_3 only connect to S_1 and S_2, then these exhaust the adjacent edges. Since each vertex has degree four, there would not exist any other edges connected to S_1 and S_2. Hence, these three vertices would form a disconnected subgraph which is broken off from the rest of $G(D)$ (Figure 6.5e). However, this cannot happen since $G(D)$ consists of a single connected component; this would correspond to a disconnected subtangle in D.

Therefore, we may conclude that there do not exist any vertices or edges in $G(D)$ which lie in the interior of the cycle B. Hence, B defines a bigon in the subtangle graph. □

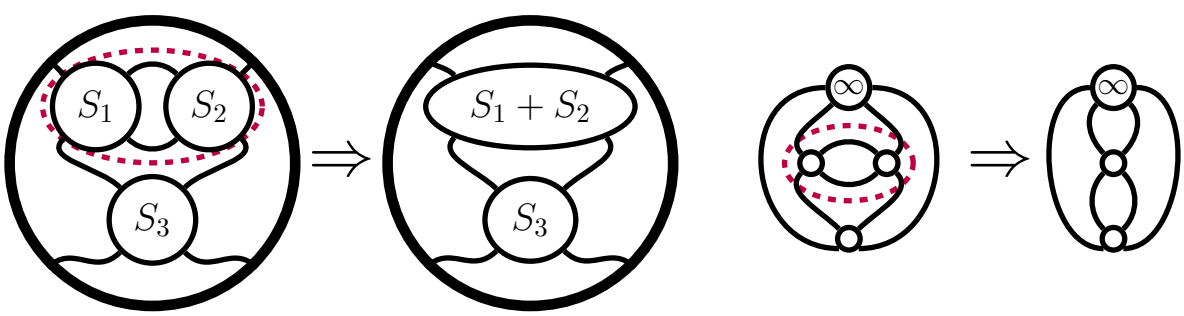
From a purely graph theoretic perspective, searching for bigons in a planar graph is relatively simple. Lemma 6.1 implies that subtangle vertices can share at most two edges, and this fact combined with planarity guarantees that every multi-edge defines a bigon. Hence, bigons can easily be identified even without a diagram. Since bigons in a subtangle graph are equivalent to algebraic combinations of subtangles in a decomposition, we can exploit this fact to choose increasingly large algebraic subtangles.

Lemma 6.3 (Bigon collapse in a subtangle graph)**:**
If two subtangles in a subtangle decomposition D for a tangle diagram which share an algebraic connection are combined into a single subtangle, then this is equivalent to collapsing a bigon to a single vertex in the corresponding subtangle graph $G(D)$.

Proof:

Let T be a 2-string tangle diagram, let D be a subtangle decomposition for this diagram, and let $G(D)$ be the subtangle graph corresponding to this decomposition. Suppose that S_1 and S_2 are two subtangles in D which share an algebraic connection (Figure 6.6a). Hence, by Lemma 6.2, there exists a corresponding bigon in $G(D)$ (Figure 6.6b). We refer to Figure 6.6 for a visual proof of the claim.

Since S_1 and S_2 share exactly two connections, either their sum or product is a subtangle of T (Lemma 6.1). Suppose without loss of generality that $S = S_1 + S_2$ denotes this combined subtangle, and note that a separating circle for S can be obtained by merging the separating circles of S_1 and S_2 (Figure 6.6a). Let D' be the subtangle decomposition of T obtained from D by replacing S_1 and S_2 with S. In the new subtangle graph $G(D')$, the bigon defined by S_1 and S_2 is collapsed to a single vertex corresponding to S, but the rest of the graph is left unchanged (Figure 6.6b). □



(a) Choosing a larger subtangle in D **(b)** Collapsing a bigon in $G(D)$

Figure 6.6: Bigon collapses in a subtangle graph correspond to a choice of larger subtangle

If we restrict ourselves to the study of *algebraic subtangle decompositions*, meaning only those decompositions in which every subtangle is algebraic, then we may leverage bigon collapses to identify when one subtangle decomposition is a refinement of another. Given two subtangle decompositions D_1 and D_2 for the same tangle diagram, recall that D_1 is a *refinement* of D_2, denoted $D_1 \leq D_2$, if each subtangle of D_1 is contained within a subtangle from D_2. Hence, an algebraic subtangle decomposition D satisfies $D \leq D_{\text{alg}}$.

Since subtangles of an algebraic tangle are also algebraic (Lemma 4.4), $D_1 \leq D_2 \leq D_{\text{alg}}$ implies that each subtangle in D_2 can be obtained by merging together some combination of subtangles in D_1 using algebraic connections. In terms of the corresponding subtangle graphs, this means that $G(D_1)$ and $G(D_2)$ are related by a sequence of collapsing bigons. This relationship can be formalized using categroy theory, wherein algebraic subtangle graphs become the objects with morphisms defined by any sequence of bigon collapses.

Theorem 6.4 (The category of algebraic subtangle graphs)**:**
If T is any fixed 2-string tangle diagram, then there exists a *category of algebraic subtangle graphs of* T, denoted $\mathscr{C}_{\text{ASG}}(T)$, where the objects and morphisms are defined as follows.

a. $\text{Ob}(\mathscr{C}_{\text{ASG}}(T)) = \{G(D)$, where $D \leq D_{\text{alg}}$ is a subtangle decomposition of $T\}$.

b. $\text{Hom}_{\mathscr{C}_{\text{ASG}}(T)}(G_1, G_2) = \{s : G_1 \to G_2$ is a sequence of zero or more bigon collapses$\}$.

Proof:

Let T be any fixed 2-string tangle diagram, and let $\mathscr{C}_{\text{ASG}}(T)$ be defined as in the theorem statement. Given two morphisms $s : G_1 \to G_2$ and $s' : G_2 \to G_3$, define a composition of these morphisms $s' \circ s : G_1 \to G_3$ to be the sequence of bigon collapses obtained by appending s and s' together. To verify that $\mathscr{C}_{\text{ASG}}(T)$ is a category, we must show that there exists an identity morphism for every object and that the composition rule is associative.

For any object $G \in \text{Ob}(\mathscr{C}_{\text{ASG}}(T))$, define $0_G : G \to G$ to be the empty sequence of bigon collapses in $\text{Hom}_{\mathscr{C}_{\text{ASG}}(T)}(G, G)$. For objects $G_1, G_2 \in \text{Ob}(\mathscr{C}_{\text{ASG}}(T))$, let $s : G_1 \to G_2$ be any morphism in $\text{Hom}_{\mathscr{C}_{\text{ASG}}(T)}(G_1, G_2)$. Since 0_{G_1} and 0_{G_2} are empty sequences, it follows immediately that appending them with s gives $s \circ 0_{G_1} = s = 0_{G_2} \circ s$. Hence, 0_G defined in this way is the identity morphism for $G \in \text{Ob}(\mathscr{C}_{\text{ASG}}(T))$.

Let G_1, G_2, G_3, G_4 be objects and let $s : G_1 \to G_2$, $s' : G_2 \to G_3$, and $s'' : G_3 \to G_4$ be morphisms in $\mathscr{C}_{\text{ASG}}(T)$. As a sequence of zero or more bigon collapses, we may write $s = (a_1, \cdots, a_n)$, $s' = (b_1, \cdots, b_m)$, and $s'' = (c_1, \cdots, c_k)$, where each a_i, b_i, and c_i term

represents a single bigon collapse in a subtangle graph. We may show that associativity is satisfied by writing out the composition of morphisms explicitly as a sequence of bigon collapses.

$$\begin{aligned}
s'' \circ (s' \circ s) &= (c_1 \cdots, c_k) \circ \big((b_1 \cdots, b_m) \circ (a_1, \cdots, a_n)\big) \\
&= (c_1 \cdots, c_k) \circ (a_1, \cdots, a_n, b_1, \cdots, b_m) \\
&= (a_1, \cdots, a_n, b_1, \cdots, b_m, c_1, \cdots, c_k) \\
&= (b_1 \cdots, b_m, c_1, \cdots, c_k) \circ (a_1, \cdots, a_n) \\
&= \big((c_1, \cdots, c_k) \circ (b_1, \cdots, b_m)\big) \circ (a_1, \cdots, a_n) \\
&= (s'' \circ s') \circ s
\end{aligned}$$

Therefore, we may conclude that $\mathscr{C}_{\mathrm{ASG}}(T)$ is indeed a category.

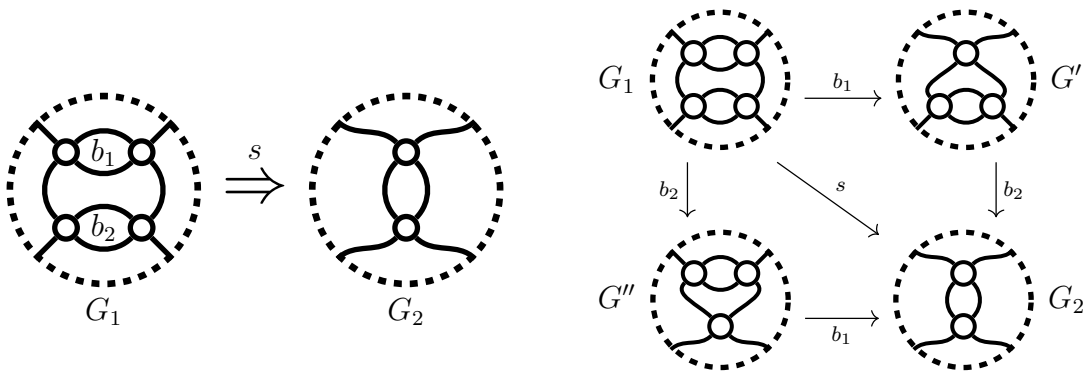
□

Note that bigon collapsing in the preceding theorem is restricted to *subtangle vertices* in an algebraic subtangle graph. There may exist bigons involving the special ∞ vertex marking the boundary sphere of the tangle diagram, but these do not correspond to algebraic connections between subtangles. Unless otherwise stated, it should be understood that *bigon collapse* in a subtangle graph only refers to bigons between subtangle vertices.

6.1.2 The Maximal Algebraic Subtangle Graph

In connection with finding the MASG of a tangle diagram, this graph can be obtained by collapsing as many bigons as possible. This is equivalent to choosing algebraic subtangles which are as large as possible, which is also the basis for how we compute the APD code. What may not be obvious is that we can obtain the MASG in this way starting from any algebraic subtangle graph, even when there are many different ways to collapse bigons. In general, when there exist multiple bigons in a subtangle graph, these can be collapsed in any order. In other words, when there exist two or more collapsible bigons in a given subtangle graph, these bigon collapses commute.

Lemma 6.5 (Commuting bigon collapses)**:**
If $G_1, G_2 \in \mathrm{Ob}(\mathscr{C}_{\mathrm{ASG}}(T))$ are algebraic subtangle graphs for a tangle diagram T, where G_2 is obtained by collapsing some combination of bigons which exist in G_1, then these bigon collapses can be performed in any order.



(a) $s : G_1 \to G_2$ collapses bigons b_1 and b_2

(b) These collapses commute: $(b_1, b_2) = s = (b_2, b_1)$

Figure 6.7: Example of commuting bigon collapses in a subtangle graph

Proof:

Let T be a 2-string tangle diagram and let $G_1, G_2 \in \mathrm{Ob}(\mathscr{C}_{\mathrm{ASG}}(T))$ be two algebraic subtangle graphs of T. Suppose that G_2 can be obtained from G_1 by collapsing some number of bigons which are initially present in G_1. We emphasize that each of these collapsible bigons must exist in G_1 and they cannot share edges; we do not consider any new bigons which arise after collapsing. We refer to Figure 6.7 for a visual proof.

With this restriction in mind, there exists a sequence of bigon collapses $s : G_1 \to G_2$ in $\mathrm{Hom}_{\mathscr{C}_{\mathrm{ASG}}(T)}(G_1, G_2)$. The example shown in Figure 6.7a illustrates the case when G_1 and G_2 are related by two bigon collapses; denote these bigons b_1 and b_2, respectively. Observe that these two collapses can be performed in either order, which is to say that the diagram in Figure 6.7b commutes: $(b_1, b_2) = s = (b_2, b_1)$. We slightly abuse notation since $b_1 : G_1 \to G'$ along the top row and $b_1 : G'' \to G_2$ along the bottom row, but context make this clear.

More generally, this argument extends to collapsing any number of bigons which are initially present in G_1. If G_1 and G_2 are related by only this type of bigon collapse, this means that there exists a set of bigons $\{b_1, \cdots, b_n\}$ in G_1 from which, when each is collapsed to a single vertex, we obtain G_2. If $s : G_1 \to G_2$ is a morphism in $\mathrm{Hom}_{\mathscr{C}_{\mathrm{ASG}}(T)}(G_1, G_2)$, then s is some ordered sequence of these n bigon collapses. Since these collapses commute, it follows that $s = \sigma(b_1, \cdots, b_n)$, where σ is any permutation of n terms. In other words, these bigon collapses can be performed in any order, which completes the proof.

$\square$

Since the preceding lemma only applies to the bigons which exist in a fixed algebraic subtangle graph, it is worth commenting that the order of two bigon collapses does matter when collapsing one bigon introduces a new one. For example, the final graph G_2 from Figure 6.7 contains a bigon that does not exist in G_1, which means that this bigon cannot be collapsed before b_1 and b_2. However, this does correspond to a *planar face* from G_1. More generally, a sequence of bigon collapses $s : G_1 \to G_2$ specifies a unique partially ordered subset of planar faces in G_1 which, when collapsed, yield G_2.

Lemma 6.6 (Sequence of bigon collapses induces a unique set of planar faces)**:**
If G and F are objects in $\mathscr{C}_{\text{ASG}}(T)$ for which $\text{Hom}_{\mathscr{C}_{\text{ASG}}(T)}(G, F) \neq \emptyset$, then there exists a unique partially ordered subset of planar faces in G which, when collapsed, yield F.

Proof:

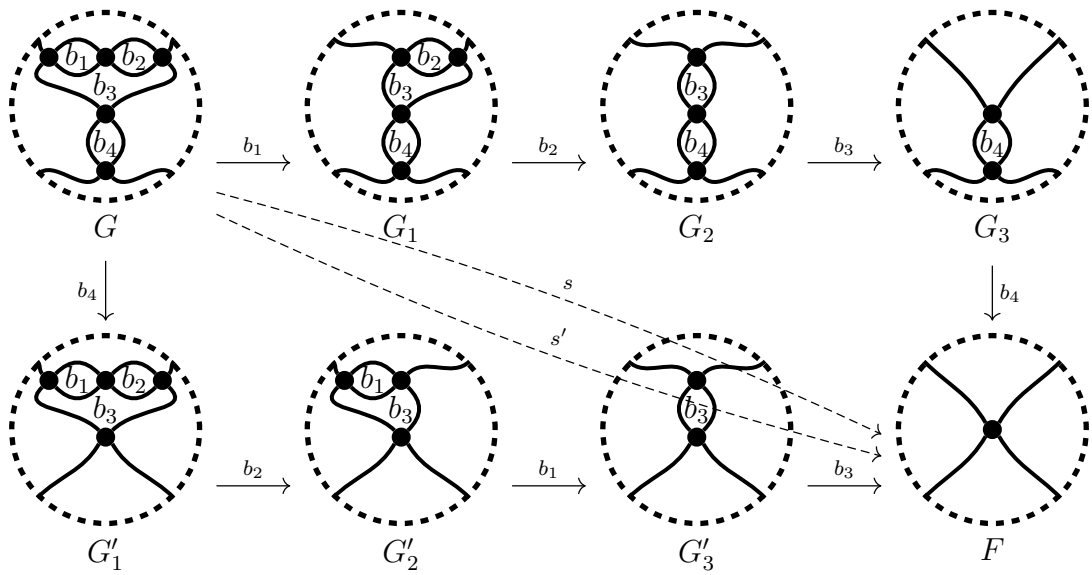
Let T be a 2-string tangle diagram, let G and F be objects in $\mathscr{C}_{\text{ASG}}(T)$, and suppose that there exists a morphism $s : G \to F$ in $\text{Hom}_{\mathscr{C}_{\text{ASG}}(T)}(G, F)$. We know that G and F are related by a sequence of bigon collapses, but the goal of this proof is to identify these explicitly with planar faces in G. Let $s = (b_1, \cdots, b_n)$, where each b_i term denotes a bigon collapse. Some of these collapsible bigons may not exist in G, but we can visualize these using a sequence of intermediate graphs (Figure 6.8).

Let $s_i = (b_1, \cdots, b_i)$ denote the partial sequence consisting of the first i bigon collapses in s, and let $G_i = s_i(G)$ denote the intermediate graph obtained after performing these. Hence $s : G \to F$ can be expressed term by term in the following way.

$$G = G_0 \xrightarrow{b_1} G_1 \xrightarrow{b_2} \cdots \xrightarrow{b_i} G_i \xrightarrow{b_{i+1}} \cdots \xrightarrow{b_n} G_n = F$$

Using these intermediate graphs, b_i denotes a bigon in G_{i-1} which is collapsed to obtain G_i. Every time we collapse a bigon in a planar graph, we remove one planar face, but the rest of the planar faces remain. Hence, we may keep track of the subset of planar faces from G which correspond to each b_i. After collapsing every bigon from s, the final graph $G_n = F$ is obtained by collapsing this specific subset of the planar faces from G.

The exact set is determined by G and F. It is unique in the sense that collapsing a different set of planar faces from G would yield a subtangle graph different from F, although they could be isomorphic as graphs if we ignore the additional structure imposed by the tangle diagram T. Furthermore, this set is partially ordered by the relation that, when two faces share an edge, the one with fewer edges must be collapsed first. □



In this diagram, $s, s' \in \mathrm{Hom}_{\mathscr{C}_{\mathrm{ASG}}(T)}(G, F)$, with $s = (b_1, b_2, b_3, b_4)$ and $s' = (b_4, b_2, b_1, b_3)$. Hence, $s' = \sigma(s)$, where $\sigma = (1, 3, 4) = (3, 4)(1, 4)$ is a permutation of bigon collapses which commute.

Figure 6.8: Example of morphisms related by a permutation of commuting bigon collapses

Although it is possible to specify a different sequence of bigon collapses between two subtangle graphs, Lemma 6.6 establishes that these correspond to a fixed set of planar faces. However, arbitrary planar faces in a subtangle graph do not indicate an algebraic connection between subtangles, and so the subtlety lies in ordering the faces so that only bigons are collapsed; hence the partial ordering. Non-commuting bigon collapses correspond to planar faces which share an edge. The face with fewer edges must be collapsed first.

Figure 6.8 illustrates this idea. The planar face labeled b_3 is not a bigon in G, and it only becomes a bigon after collapsing b_1 and b_2, with which it shares an edge. Hence, b_3 must be preceded by these two bigon collapses in any morphism which contains it. However, b_3 does not share an edge with b_4, and so the order of these may be reversed. The two morphisms $s, s' : G \to F$ shown in the figure collapse the same set of planar faces in G in a different order, and s' can be interpreted as a permutation of the collapses from s. When this permutation is decomposed as a product of 2-cycles as shown, each 2-cycle switches two commuting bigon collapses.

This fact holds more generally for morphisms in $\mathscr{C}_{\mathrm{ASG}}(T)$. The set of planar faces that must be collapsed is uniquely determined by the source and target graphs. The exact order of collapsing is unimportant for those faces which denote commuting bigon collapses.

Lemma 6.7 (Uniqueness up to permutations of commuting bigon collapses)**:**
If G and F are objects in $\mathscr{C}_{\text{ASG}}(T)$, and $\text{Hom}_{\mathscr{C}_{\text{ASG}}(T)}(G, F) \neq \emptyset$, then a morphism $s : G \to F$ is unique up to a permutation of commuting bigon collapses.

Proof:

Let T be a 2-string tangle diagram, let G and F be objects in $\mathscr{C}_{\text{ASG}}(T)$, and suppose that there exist morphisms $s, s' \in \text{Hom}_{\mathscr{C}_{\text{ASG}}(T)}(G, F)$. Let $s = (b_1, \cdots, b_n)$ and $s' = (b'_1, \cdots, b'_n)$ denote these two sequences of bigon collapses. Since each bigon collapse reduces the number of vertices in a graph by one, and s and s' both start from G and terminate at F, it follows that they must contain the same number of collapses.

By Lemma 6.6, there exists a unique set of n planar faces in G that must be collapsed to obtain F; these faces correspond exactly with the bigon collapses in s and s'. In particular, this set of faces can be labeled in order of how they are collapsed, either $\{b_1, \cdots, b_n\}$ or $\{b'_1, \cdots, b'_n\}$. Thus, this set of planar faces in G defines a permutation σ for which $\sigma(b_1, \cdots, b_n) = (b'_1, \cdots, b'_n)$, as in the example from Figure 6.8.

Certain planar faces in G only become bigons after collapsing other planar faces. These correspond to non-commuting bigon collapses, which only occur between planar faces which share edges. If σ is decomposed as a product of 2-cycles, each 2-cycle permutes two planar faces which do not share any edges. Therefore, σ reduces to a permutation of commuting bigon collapses, which completes the proof.

□

Lemma 6.7 establishes that the order of commuting bigon collapses is immaterial in $\mathscr{C}_{\text{ASG}}(T)$. These result in the same subtangle graph and thus define equivalent morphisms. This behavior is mirrored by the fact that there are generally multiple ways to combine subtangles which share algebraic connections in a subtangle decomposition. The exact order in which subtangles are combined only matters insofar as these correspond to algebraic connections, but the resulting descomposition with fewer subtangles does not change. Hence, we may prove that a refinement of algebraic subtanlge decompositions for a fixed diagram induces a unique morphism in $\mathscr{C}_{\text{ASG}}(T)$.

Theorem 6.8 (Refinement of subtangle decompositions induces unique morphism)**:**
If D and D' are two algebraic subtangle decompositions for a 2-string tangle diagram T, where D is a refinement of D', then there exists a morphism $s : G(D) \to G(D')$ in $\mathscr{C}_{\text{ASG}}(T)$ which is unique up to a permutation of commuting bigon collapses.

Proof:

Let T to be a 2-string tangle diagram and let D and D' be two algebraic subtangle decompositions for T. Suppose that $D \leq D'$ and consider the corresponding algebraic subtangle graphs $G(D)$ and $G(D)$ in $\mathscr{C}_{\mathrm{ASG}}(T)$. If $\mathrm{Hom}_{\mathscr{C}_{\mathrm{ASG}}(T)}(G(D), G(D')) \neq \emptyset$, then Lemma 6.7 implies that any morphism $s : G(D) \to G(D')$ is unique up to a permutation of commuting bigon collapses. Therefore, it will be sufficient to show that a morphism from $G(D)$ to $G(D')$ exists to prove the claim. Figure 6.9 illustrates one example of an induced morphism and we will show that such a construction is always possible.

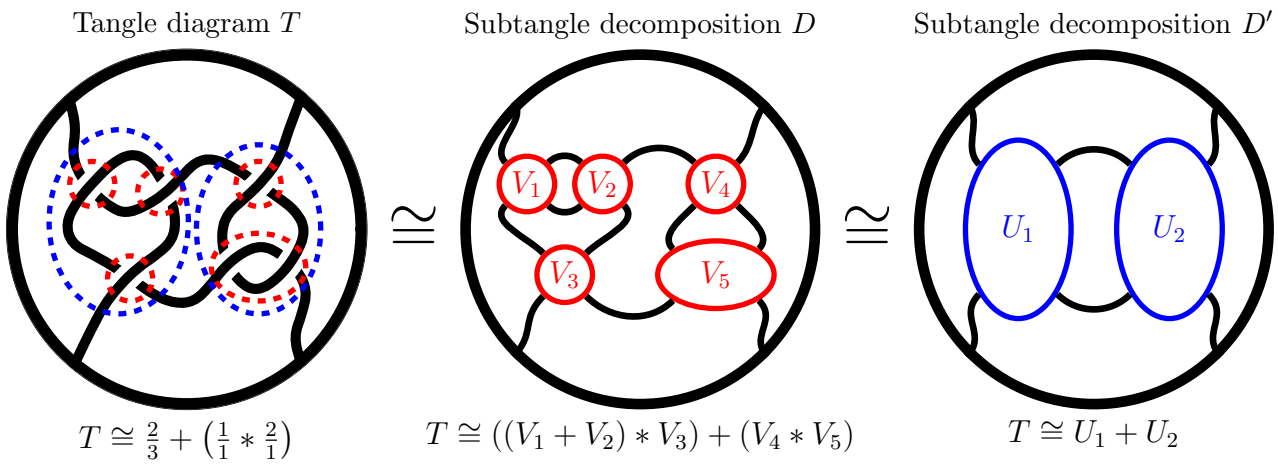
Recall that a subtangle decomposition partitions the crossings in a tangle diagram into disjoint subtangles. Hence, $D = \{V_1, \cdots, V_n\}$ and $D' = \{U_1, \cdots, U_m\}$ are two collections of subtangles which exhaust the crossings in T. Note that each of these subtangles is algebraic by assumption. Since D is a refinement of D', this means that, for each $V_i \in D$, there exists $U_j \in D'$ such that $V_i \subseteq U_j$. Since the subtangles in D are disjoint and account for all of the crossings from T, it follows that D can be partitioned into m disjoint subsets, each of which determines a decomposition of a subtangle $U_j \in D'$. In other words,

$$D = \{V_1, \cdots, V_n\} = \{V_1^1, \cdots, V_{k_1}^1\} \sqcup \cdots \sqcup \{V_1^m, \cdots, V_{k_m}^m\},$$

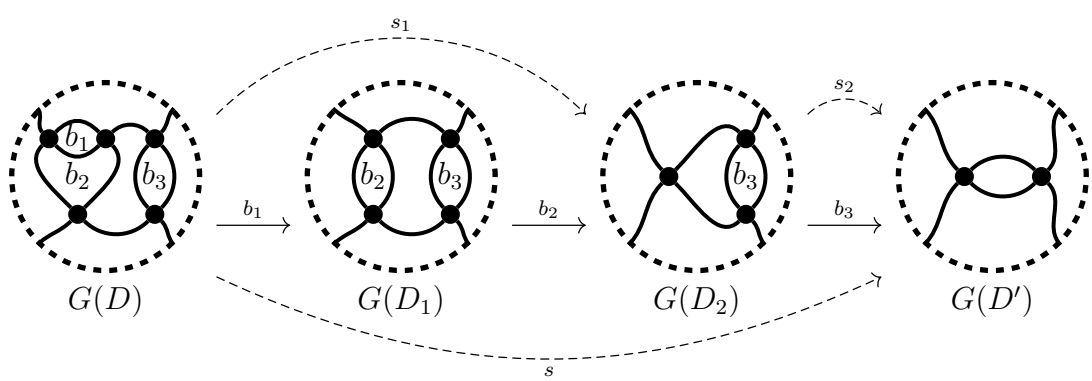
where each subset $\{V_1^j, \cdots, V_{k_j}^j\}$ determines a subtangle decomposition of $U_j \in D'$ (Figure 6.9a). For each of these subsets, we must show that the vertices in $G(D)$ corresponding to the subtangles from $\{V_1^j, \cdots, V_{k_j}^j\}$ can be combined into a single vertex corresponding to U_j in $G(D')$ by a sequence of bigon collapses.

Since each U_j is algebraic, it can be expressed in terms of an algebraic construction formula using the subtangles in $\{V_1^j, \cdots, V_{k_j}^j\}$ such as $U_j \cong ((V_1^j + V_2^j) * V_3^j) + \cdots$, for example. Each operation in such a formula denotes a tangle sum or product and hence an algebraic connection between subtangles. The placement of parentheses in the construction formula determines a partial ordering on these sums and products based on the standard order of operations, where we note that this exactly parallels the partial ordering on the planar faces in $G(D)$. Specifically, the innermost operations in this formula define an alagebraic connection between two subtangles from D, and hence a bigon in $G(D)$ (Lemma 6.2).

Starting from the innermost parentheses in the construction formula for U_j and working our way out, we may define a chain of refinements of intermediate subtangle decompositions $D_0^j \leq D_1^j \leq \cdots \leq D_{k_j}^j$, where each D_i^j is obtained from D_{i-1}^j by replacing two subtangles which share an algebraic connection with a single subtangle. In terms of the subtangle graphs,



(a) Refinement of algebraic subtangle decompositions $D \leq D'$ for the tangle diagram T



(b) Induced sequence of bigon collapses: $s = s_2 \circ s_1 : G(D) \to G(D')$

Figure 6.9: Example of a morphism induced by a refinement of subtangle decompositions

$G(D_i^j)$ is obtained from $G(D_{i-1}^j)$ by collapsing the bigon between the vertices corresponding to the two replaced subtangles (Lemma 6.3). Thus, this chain of refinements induces a sequence of bigon collapses $s_j : G(D_0^j) \to G(D_{k_j}^j)$.

We can repeat this process for each of the subtangles $U_j \in D'$ so that we obtain a sequence of bigon collapses s_j for each of them. Since any two subtangles $U_j, U_\ell \in D'$ are disjoint, note that the bigon collapses from s_j commute with the bigon collapses from s_ℓ (Lemma 6.5). Hence, these sequences can be appended together in any order so that all of the vertices in $G(D)$ corresponding to the subtangles from D can be collapsed down into vertices in $G(D')$ which correspond to the subtangles from D' (Figure 6.9b).

Defining $s = s_m \circ \cdots \circ s_1$, our construction guarantees that $s : G(D) \to G(D')$ is a sequence of bigon collapses. Therefore, $s \in \mathrm{Hom}_{\mathscr{C}_{\mathrm{ASG}}(T)}(G(D), G(D'))$ is a morphism between the subtangle graphs corresponding to this refinement of subtangle decompositions $D \leq D'$. We may conclude by Lemma 6.7 that s is unique up to permutations of commuting bigon collapses.

□

Theorem 6.8 applies to any refinement of algebraic subtangle decompositions for a fixed tangle diagram T. In connection with the MASG, this graph is $G(D_{\mathrm{alg}})$ by definition. Since every decomposition of T using algebraic subtangles is a refinement of D_{alg}, this means that we can obtain the MASG just by searching for bigon collapses in some algebraic subtangle graph. In particular, 1-crossing subtangles are algebraic, and whereas it may be difficult to identify some decompositions by eye, the 1-crossing subtangle decomposition $D_{\min}$ is immediate from the locations of the crossings in a tangle diagram.

Corollary 6.9 (Initial and terminal algebraic subtangle graphs)**:**
If T is a 2-string tangle diagram and $D_{\min}$ and D_{alg} denote the subtangle decompositions of T into 1-crossing subtangles and maximal algebraic subtangles, respectively, then $G(D_{\min})$ and $G(D_{\mathrm{alg}})$ are initial and terminal objects in $\mathscr{C}_{\mathrm{ASG}}(T)$, respectively.

Proof:

Let T be a 2-string tangle diagram. By Corollary 4.10, there exists a decomposition of T into 1-crossing subtangles, $D_{\min}$, and a decomposition of T into maximal algebraic subtangles, D_{alg}. Since these are both decompositions of T into algebraic subtangles, there exist corresponding objects $G(D_{\min})$ and $G(D_{\mathrm{alg}})$ in $\mathscr{C}_{\mathrm{ASG}}(T)$. To prove the claim, we must show that there exist unique morphisms from and to these objects, respectively, from any other objects in the category.

Let $G \in \mathrm{Ob}(\mathscr{C}_{\mathrm{ASG}}(T))$ be any object in this category. Hence, there exists an algebraic subtangle decomposition D of T for which $G = G(D)$. Since D is algebraic, it must satisfy $D \leq D_{\mathrm{alg}}$. Since $D_{\min}$ is by construction the most refined subtangle decomposition possible for T, it follows that $D_{\min} \leq D$ (Corollary 4.7). Thus, by Theorem 6.9, there exists in this category a unique morphism $s : G \to G(D_{\mathrm{alg}})$ and a unique morphism $s' : G(D_{\min}) \to G$. Since G was arbitrary, we may conclude that $G(D_{\min})$ and $G(D_{\mathrm{alg}})$ are, respectively, initial and terminal objects in $\mathscr{C}_{\mathrm{ASG}}(T)$.

□

This result is useful computationally, but it is also a boon for the pen-and-paper mathematician. Given a tangle diagram, one can obtain the corresponding MASG by hand merely by marking off the parts of the diagram which correspond to collapsing bigons in the subtangle graph. Similar to the preceding examples, this can be accomplished without reference to a diagram notation. Furthermore, this method provides a reliable way to determine whether or not a given diagram is algebraic using the MASG, even without knowing its APD code.

Corollary 6.10 (The MASG of an algebraic tangle diagram)**:**
A tangle diagram T is algebraic if and only if its MASG contains a single subtangle vertex.

Proof:

Let T be a 2-string tangle diagram, let D_{alg} denote the decomposition of T into maximal algebraic subtangles, and let $D_{\max}$ denote the subtangle decomposition of T using a single maximal subtangle. Note that these two subtangle decompositions always exist and satisfy $D_{\text{alg}} \leq D_{\max}$ (Corollary 4.10). By Theorem 4.12, we may classify whether or not T is algebraic by comparing D_{alg} with $D_{\max}$.

The MASG of T is defined by $G(D_{\text{alg}})$. If $G(D_{\text{alg}})$ consists of a single subtangle vertex attached to the special boundary vertex, then $D_{\text{alg}} = D_{\max}$ is a decompositon of T into a single algebraic subtangle, and hence T is algebraic. If $G(D_{\text{alg}})$ contains more than one subtangle vertex, then $D_{\text{alg}} < D_{\max}$, and hence T is non-algebraic. This completes the proof. □

Although we have restricted ourselves to the study of tangle diagrams, it is interesting to imagine how techniques involving subtangle graphs could be extended to tangles more generally. This would require accounting for Reidemeister moves in some way, although Reidemeister I and II moves could be problematic since these indicate the presence of a local knot or trivial tangle, respectively, which we have excluded from our study of subtangle decompositions. While exploring these ideas is beyond the scope of this dissertation, we conclude this section with a conjecture.

Conjecture 6.11 (Reidemeister moves and subtangle graphs)**:**
Reidemeister moves between equivalent 2-string tangle diagrams T_1 and T_2 induce a functorial relationship between $\mathscr{C}_{\text{ASG}}(T_1)$ and $\mathscr{C}_{\text{ASG}}(T_2)$.

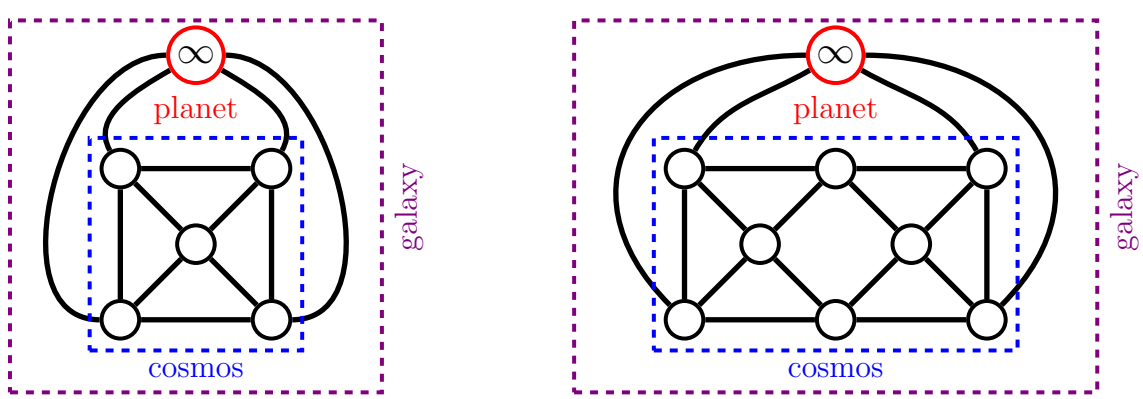
6.2 Constellations and Non-Algebraic Tangle Diagrams

In the same way that we leveraged construction formulas to study algebraic diagrams, maximal algebraic subtangle graphs are a comparable tool for studying non-algebraic diagrams. To classify non-algebraic tangle diagrams, we will start by exploring what types of MASG are possible. In addition to this being a 4-valent planar graph, tangle diagrams which fail to be algebraic impose the additional restriction that there do not exist bigons. In the language of graph theory, this means that the graph contains no double-edges between vertices, which is to say that the MASG is *simple*. This motivates us to define a special type of graph called a *constellation* which formalizes these ideas.

6.2.1 Properties of Constellation Graphs

Definition 6.12 (Constellations)**:**
A $k+1$ *constellation* (C, p) consists of a graph C with k vertices and a special marked $k+1^{st}$ vertex p, where C is connected, planar, 4-valent, and simple (Figure 6.10).

- The unmarked graph C is called the *galaxy* of the constellation.
- The marked vertex p is referred to as the *planet* of the constellation.
- The induced subgraph $C^* = C \setminus \{p\}$ is known as the *cosmos* of the constellation.



(a) A $5+1$ constellation **(b)** An $8+1$ constellation

Figure 6.10: Examples of constellations and their galaxies, planets, and cosmos

Although we usually draw this graph on a flat plane, we can imagine a constellation sitting on the surface of a sphere, with the cosmos in the front and the planet identified with a point at ∞ on the back. The choice of nomenclature reflects the idea that we stand at the planet and gaze up at the cosmos. We prefer to work with planar embeddings where the planet is pushed off to one side and we generally label it ∞ to distinguish it from the other vertices. Many of the results concerning constellations are proven by examining the planar faces in a particular embedding. This includes treating the region outside of the graph as a planar face bounded by the outermost edges.

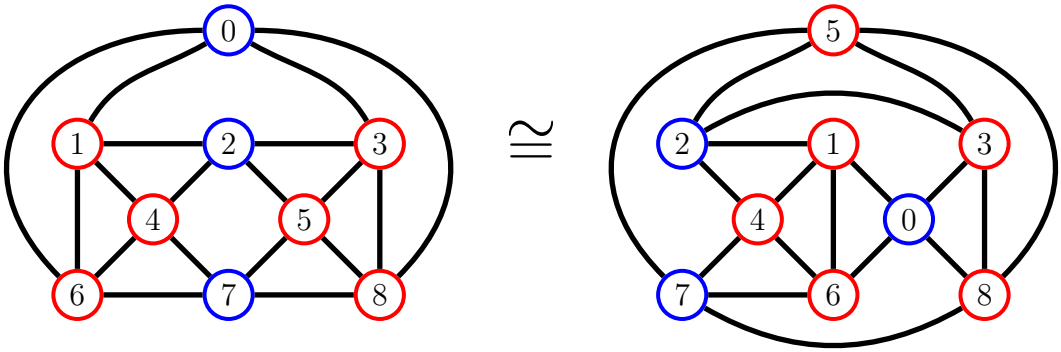
An equivalence of constellations is possible whenever their cosmos represent the same configuration of vertices. Formally, we say that two constellations (C_1, p_1) and (C_2, p_2) are *equivalent*, written $(C_1, p_1) \cong (C_2, p_2)$, if their cosmos are isomorphic as graphs. The next lemma proves that equivalent constellations also have isomorphic galaxies. The reason we define this equivalence using the cosmos C^* rather than the galaxy C is that a different choice of planet in C may yield a non-isomorphic cosmos (Figure 6.11).

Lemma 6.13 (Equivalent constellations have isomorphic galaxies)**:**
If $(C_1, p_1) \cong (C_1, p_2)$ are equivalent constellations, then their galaxies C_1 and C_2 are graph isomorphic. However, the converse does not hold true in general.

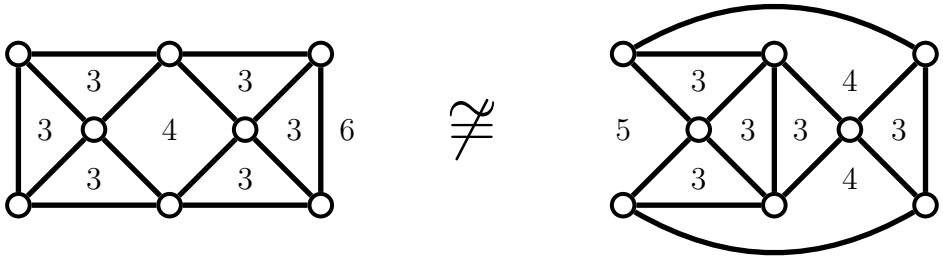
Proof:

Let (C_1, p_1) and (C_2, p_2) denote two equivalent constellations. Hence, these constellations have graph isomorphic cosmos, $C_1^* \cong C_2^*$. Recall that an isomorphism of graphs is a bijection of vertex sets which preserves edges. In other words, there exists a bijection $\phi^* : V(C_1^*) \to V(C_2^*)$ with the property that, if two vertices share an edge in C_1^*, then the two corresponding vertices in C_2^* under ϕ^* also share an edge. The bijection ϕ^* between the cosmos' vertex sets can be extended to a bijection $\phi : V(C_1) \to V(C_2)$ between the galaxies' vertex sets by defining $\phi(p_1) = p_2$. To prove that ϕ defined in this way is a graph isomorphism, it suffices to verify that any vertex adjacent to p_1 in C_1 is mapped to a vertex adjacent to p_2 in C_2.

Although C_1 is a 4-valent graph, the subgraph C_1^* is obtained by deleting the planet in C_1. Hence, the four vertices connected to p_1 in C_1 have degree 3 in C_1^*, but every other vertex has degree 4. We obtain similar results concerning C_2^*. Any two vertices related by ϕ^* must have the same degree in C_1^* and C_2^*, respectively. Hence, ϕ^* maps each of these four degree-3 vertices in C_1^* to one of the four degree-3 vertices in C_2^*. In C_1 and C_2, these are exactly the vertices which are adjacent to the planet. Since $\phi|_{V(C_1^*)} = \phi^*$, it follows that each vertex adjacent to p_1 in C_1 is mapped under ϕ to a vertex adjacent to p_2 in C_2.



(a) Isomorphic galaxies defining non-equivalent $8+1$ constellations



6 triangles 0 pentagons
1 square 1 hexagon

5 triangles 1 pentagon
2 squares 0 hexagons

(b) Non-isomorphic cosmos distinguished by counting planar faces

Figure 6.11: Counterexample showing constellations with a different choice of planet

This proves that ϕ defines a graph isomorphism between C_1 and C_2, establishing the first part of the claim. To prove the second part of the claim, we need only provide a counterexample. We refer to the graphs shown in Figure 6.11 to construct two constellations which have isomorphic galaxies but non-isomorphic cosmos.

Consider the two simple, connected, 4-valent planar graphs shown in Figure 6.11a. Denote these graphs as C_1 and C_2, respectively. These graphs are isomorphic, where the explicit bijection of vertex sets is determined by the labels on the vertices in the figure. One may verify that any pair of adjacent vertices in C_1 corresponds to a pair of adjacent vertices in C_2 with the same labels. These graphs can be used to define a constellation by choosing any vertex as the planet. Let (C_1, p_1) and (C_2, p_2) be the constellations obtained by using the topmost vertex as the planet in each case. By the preceding observations, these two constellations have isomorphic galaxies, $C_1 \cong C_2$.

Figure 6.11b shows the cosmos C_1^* and C_2^* corresponding to these constellations. If these two cosmos were graph isomorphic, then the existence of certain planar faces is one of the features that must be preserved under an isomorphism. We may prove that no such isomorphism exists by comparing the existing planar faces in each graph. As illustrated in the figure, observe that C_1^* contains six triangles, one square, and one hexagon. By contrast, C_2^* contains five triangles, two squares, and one pentagon. Therefore, it is not possible to define a bijection of vertex sets which maps each planar face in C_1^* to a planar face of the same type in C_2^*. This proves that these two cosmos are non-isomorphic even though they are subgraphs of isomorphic galaxies.

□

Any connected, planar, simple, 4-valent graph can be used to define a constellation by choosing a vertex to be the planet. Referring back to Figure 6.11a, one may verify that any constellations obtained by choosing a vertex marked in red as the planet will be equivalent. A similar fact holds using the vertices marked in blue. The counterexample in Lemma 6.13 proves that, in general, a different choice of planet may lead to a non-equivalent constellation. Returning to our cosmic imagery, we see a different shape when viewing the same set of stars from a different location in the galaxy.

Identifying isomorphic graphs is computationally difficult, although small graphs can reasonably be studied with the aid of a computer. The adjacency matrix is a particularly convenient way to study graphs computationally, although these do not carry information about planar embeddings. Two graphs are isomorphic if and only if their adjacency matrices are permutation symmetric. In connection with constellations, the adjacency matrix of the cosmos can be obtained from the matrix of the galaxy by deleting the row and column corresponding to the planet. Although we will not pursue this computational direction, these observations suggest how the task of exhaustively searching for different constellations could be relegated to a computer.

We have seen how choosing a different planet in the same galaxy can lead to a non-equivalent constellation. However, the fact that galaxies are 4-valent makes them easier to study than cosmos in isolation. We may search for different constellations that live in the same galaxy by examining different choices of planet. If two non-equivalent constellations (C_1, p_1) and (C_2, p_2) have graph isomorphic galaxies $C_1 \cong C_2$, then we will say that these constellations are *familiar*, denoted $(C_1, p_1) \sim (C_2, p_2)$. The two graphs from Figure 6.13 are an example of familiar constellations.

The restrictions placed on constellations from Definition 6.12 ensure such a graph must have a minimal number of vertices. In addition to the planet, every constellation must have at least five other vertices. The next result proves that there do not exist any $k+1$ constellations for $k+1 < 6$.

Corollary 6.14 (Minimal number of vertices in a constellation)**:**
If (C, p) is a $k+1$ constellation, then $k+1 \geq 6$.

Proof:

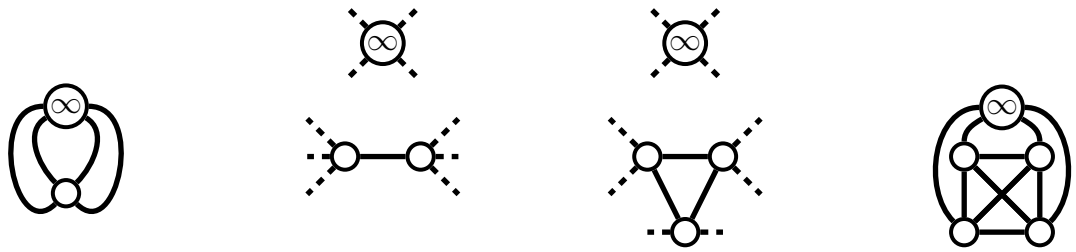
Let (C, p) be a $k+1$ constellation. By Definition 6.12, this graph is connected, simple, planar, and 4-valent. Since each vertex is degree-4, it must be connected to four distinct other vertices; more than a single edge between two vertices would violate simplicity, as would loops. Hence, at minimum, C must contain at least five vertices in total, counting the marked vertex p. If $k+1 = 5$, then all five vertices share exactly one edge with every other vertex. This would mean that C is isomorphic to K_5, the complete graph on five vertices. However, K_5 is known to be non-planar, and hence cannot be isomorphic to a constellation. Therefore, it follows that $k+1 \geq 6$, proving the claim.

□

6.2.2 Constellations and Non-Algebraic MASGs

The definition of constellation introduced in Definition 6.12 is intended to capture the interesting structural features we noticed while studying maximal algebraic subtangle graphs, although there are a few important differences. A subgtangle graph is guaranteed to be connected, planar, and 4-valent, and it has a special marked vertex corresponding to the boundary circle of a tangle diagram collapsed to a point. However, a subtangle graph need not be simple, even when we restrict to the MASG. When we compute the maximal algebraic subtangle graph for a given diagram, we are only allowed to collapse bigons between *subtangle vertices*, which allows for the possibility of bigons involving the boundary vertex.

The MASG will define a constellation if it contains no such bigons, but this is not guaranteed. The first graph in Figure 6.13b shows an exmaple of a possible MASG for which there exists a bigon involving the boundary vertex. However, even if it is not a constellation, the MASG of a non-algebraic diagram must contain a subgraph which isomorphic to the cosmos of some constellation. Before we can prove this relationship, we will need the following preliminary result.



(a) $k = 1$ is algebraic **(b)** Cannot connect all edges if $k = 2$ or $k = 3$ **(c)** $k = 4$ is nonplanar

Figure 6.12: Problems arising for non-algebraic MASGs with $k < 5$ subtangle vertices

Lemma 6.15 (Counting vertices in a non-algebraic MASG)**:**
If T is a non-aglebraic tangle diagram with maximal algebraic subtangle graph G, then G contains no fewer than five subtangle vertices, in addition to the boundary vertex.

Proof:

Let T be a non-algebraic tangle diagram and let G denote the maximal algebraic subtangle graph of T. Since every bigon between subtangle vertices has been collapsed in G, we know that there exists at most one edge between any pair of subtangle vertices. There could exist more than one edge between the boundary vertex and a subtangle vertex, and so the claim does not immediately follow by the argument used in Corollary 6.14, which assumed we worked with a simple graph.

Let k denote the number of subtangle vertices in G. We may prove the claim by showing that every possible placement of edges in G leads to a contradiction if k is 1, 2, 3, or 4. We refer to Figure 6.12 to illustrate the contradictions which arise in each of these cases. In the first case, if $k = 1$, then G contains a single subtangle vertex. Hence, T is an algebraic tangle diagram by Corollary 6.10 (Figure 6.12a). However, this is not possible since it would contradict the assumption that T is non-algebraic.

If $k = 2$ or $k = 3$, then we encounter a problem with placing possible edges (Figure 6.12b). Specifically, since G is 4-valent, each subtangle vertex must have four adjacent edges, but there can exist at most one edge between two subtangle vertices. This means there are at minimum six additional edges adjacent to subtangle vertices that must connect to the boundary vertex. However, the boundary vertex is also degree-4, and so it cannot connect to all of these remaining edges.

If $k = 4$, then the only placement of edges which both satisfies 4-valency and does not introduce a double-edge between subtangle vertices is shown in Figure 6.12c. In this case, all five vertices share exactly one edge with every other vertex, which is the same scenario we encountered in the proof of Corollary 6.14. This means that G is isomorphic to K_5, the complete graph on five vertices. However, K_5 is known to be non-planar, and this violates the fact that the subtangle graph G must be planar.

Therefore, if G is the maximal algebraic subtangle graph for a non-algebraic tangle diagram T, then G must contain at least five subtangle vertices. This completes the proof.

□

Although a non-algebriac MASG may not itself be a constellation, these two types of graph are closely related. If we also collapse bigons which involve the boundary vertex, we will eventually obtain a simple graph satisfying the conditions of Definition 6.12. The cosmos of this constellation exists as a subgraph in the original MASG.

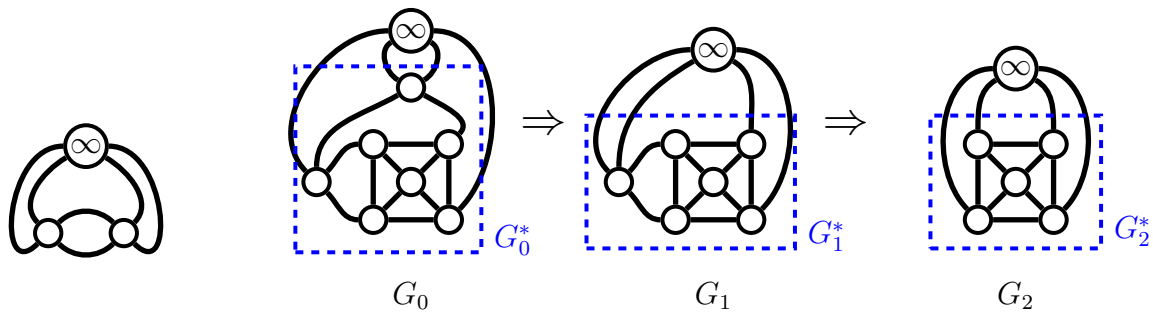
Theorem 6.16 (Subgraph of non-algebraic MASG isomorphic to a cosmos)**:**
If T is a non-algebraic tangle diagram with maximal algebraic subtangle graph G, then either G is a constellation or G contains a subgraph isomorphic to the cosmos of some constellation.

Proof:

Let T be a non-algebraic tangle diagram and let G denote the maximal algebraic subtangle graph of T. As a subtangle graph, G is connected, planar, and 4-valent, and we take the boundary vertex as the special marked vertex. If G is also known to be simple, then G satisfies the conditions of Definition 6.12 and thus defines a constellation. Suppose instead that G is not simple. To prove the claim, we must identify a certain subgraph in G.

Since T is assumed to be non-algebraic, we know by Lemma 6.15 that G contains at least five subtangle vertices. If G is not simple, then it contains one or more bigons. However, since the MASG is obtained by collapsing all bigons between subtangle vertices, any remaining bigons in G must involve the boundary vertex (Figure 6.13b). For any such bigon which does exist, let G_1 be the graph obtained by collapsing this bigon into the boundary vertex. As Figure 6.13b shows, it is possible that such a collapse introduces a new bigon, but this bigon must be also adjacent to the boundary vertex in G_1.

Note that the collapsed bigon corresponds to a multi-edge between the boundary vertex and a subtangle vertex; this collapse is equivalent to absorbing the subtangle vertex into the boundary vertex. Except for the two new edges adjacent to the boundary vertex, note



(a) Contradictory G_{n-2} **(b)** Sequence of intermediate graphs ending at a simple graph G_n

Figure 6.13: Example of intermediate graphs used in the proof of Theorem 6.16

that every other edge between vertices in G_1 corresponds to an edge between vertices in the original graph G. In particular, this means that the graph G_1^* obtained by deleting the boundary vertex is guaranteed to be a subgraph of G. However, G_1^* is not the graph we need if G_1 still contains a bigon, as in the second graph of Figure 6.13b.

By repeating this process as many times as needed, let G_n be the final graph obtained after collapsing all such bigons involving the boundary vertex. Since there are only finitely many vertices, this process must terminate eventually, either with a simple graph or when the boundary vertex is all that remains. The latter case does not have enough vertices to contain the desired subgraph, and so first we must show that this process terminates before all of the subtangle vertices have been removed.

Let $G_0, G_1, \cdots, G_n$ denote the sequence of intermediate graphs used to obtain G_n, where $G_0 = G$ and each other graph G_i is obtained from G_{i-1} by collapsing a bigon which is adjacent to the boundary vertex. Note that number of subtangle vertices decreases by one after each iteration. Since the original graph $G_0 = G$ is connected, planar, and 4-valent, so is each graph G_i in this sequence. Furthermore, as we observed above, each G_i induces a subgraph G_i^* of the original graph G obtained by removing the boundary vertex.

Suppose by way of contradiction that the only remaining vertex in G_n is the boundary vertex. Since $G_0 = G$ is the MASG of a non-algebraic diagram, we know that the original graph had a least five subtangle vertices (Lemma 6.15). Since one of these vertices is collapsed into the boundary vertex after each iteration until none remain in G_n, it follows that $n \geq 5$. In particular, this means that the intermediate graph G_{n-2} contains exactly two subtangle vertices, one of which shares at least two edges with the boundary vertex (Figure 6.13a).

Since loops cannot occur in any of these graphs, the only way that G_{n-2} can be 4-valent is if the edges are connected as shown in Figure 6.13a. However, this graph contains a bigon between the two remaining subtangle vertices. Since the induced graph G^*_{n-2} is a subgraph of G, this means that G also contains a bigon between two subtangle vertices. This contradicts the fact that G is an MASG and hence does not contain any bigons between subtangle vertices.

Thus, we may conclude that the algorithm used to obtain G_n terminates before all subtangle vertices have been removed, as in the third graph of Figure 6.13b. In addition to being connected, planar, and 4-valent, G_n is also simple since it contains no bigons between any vertices. Choosing the boundary vertex as the planet, G_n can be used to define a constellation in accordance with Definition 6.12. Furthermore, the cosmos of this constellation is exactly G^*_n, which we have established is a subgraph of G. This is the subgraph of G asserted in the claim, which completes the proof.

□

Although many non-algebraic tangle diagrams will have an MASG which is a constellation, the intermediate graphs used in the preceding proof show that this will not always happen. However, in these cases, the subgraph which exists by Theorem 6.16 can be used to identify a non-algebraic subtangle. In the same way that we built the MASG using algebraic subtangles that are as large as possible, we can supplement this by identifying non-algebraic subtangles which are as small as possible.

This motivates us to consider the *minimal non-algebraic subtangles* in a non-algebraic tangle diagram, which is to say those non-algebraic subtangles which do not properly contain any other non-algebraic subtangles. If a given non-algebraic diagram does not properly contain a smaller non-algebraic subtangle, then its MASG is a constellation.

Corollary 6.17 (Minimal non-algebraic MASGs)**:**
If T is a non-algebraic tangle diagram which does not properly contain any non-algebraic subtangles, then the maximal algebraic subtangle graph of T is a constellation.

Proof:

Let T be a non-algebraic tangle diagram and let G denote the maximal algebraic subtangle graph of T. Suppose that T does not contain any proper subtangles which are non-algebraic. Suppose by way of contradiction that G is not a constellation. Thus, by Theorem 6.16, G contains a subgraph which is isomorphic to the cosmos of some constellation. Let G^* denote

this subgraph, and note that G^* contains at least 5 subtangle vertices (Corollary 6.14). There must exist at least one other subtangle vertex in G which is not contained in G^*, otherwise G would be a constellation.

Let T' be the subtangle of T which corresponds to the configuration of algebraic subtangles defined by G^*. Hence, T' is chosen so that its MASG is obtained by adding a boundary vertex to G^*. In particular, this means that the MASG of T' contains more than one vertex, and hence T' is non-algebraic by Corollary 6.10. Since there exists at least one other subtangle of T which is disjoint from T', it follows that T' is a proper subtangle of T. However, this implies that T properly contains a non-algebraic subtangle, contradicting our assumption. Therefore, we may conclude that the MASG of T must be a constellation.

□

Note that the converse of Corollary 6.17 is not true in general. A non-algebraic tangle diagram whose MASG is a constellation may contain proper non-algebraic subtangles. This can happen whenever the cosmos of one constellation is a proper subgraph of a larger constellation. For example, two copies of the cosmos of the $5+1$ constellation shown in Figure 6.10a could be combined to create a $10+1$ constellation. A non-algebraic tangle diagram whose MASG matches this constellation could be decomposed into two non-algebraic subtangles. In other words, cosmos of different constellations may be nested inside of each other.

To account for such possibilities, we will say that a constellation is *minimal* if its cosmos does not contain the cosmos of another constellation as a proper subgraph. We can imagine looking for minimal non-algebraic subtangles in a given tangle diagram by searching for the cosmos of minimal constellations in the MASG. If we replace such subgraphs in the MASG with a single vertex representing a minimal non-algebraic subtangle, then we obtain a more refined subtangle graph wherein vertices represent a combination of maximal algebriac subtangles and minimal non-algebraic subtangles. This suggests that constellations could be leveraged to denote non-algebraic diagrams more generally, and we will return to this question in more detail after examining some small constellations.

6.2.3 Counting Small Constellations

For a fixed k, there exist only finitely many $k+1$ constellations, up to isomorphism. For $k < 5$, we have already seen that no such constellations exist (Corollary 6.14). Although the number of possible constellations grows exponentially as k increases, we can reasonably hope to enumerate all constellations up to some sufficiently small fixed value of k. We assert

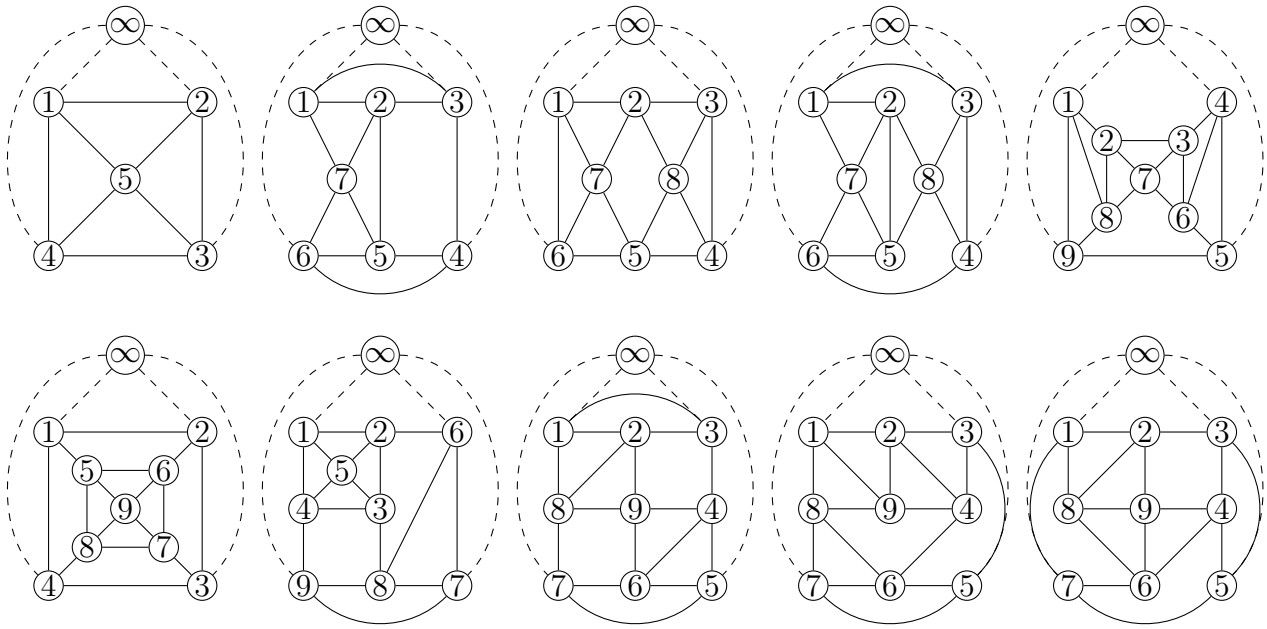


Figure 6.14: List of all possible $k+1$ constellations with $k+1 \leq 10$

that every $k+1$ constellation with $k+1 \leq 10$ is equivalent to one of the ten constellations featured in Figure 6.14, and that each constellation in this list is distinct. Proving these facts is the goal of this subsection.

If we ignore the planet of a constellation, we can start with the easier question of enumerating possible galaxies. Any simple, connected, planar, 4-valent graph can be used as a galaxy, and a constellation is obtained after choosing a planet. Hence, if we first identify all possible galaxies with ten or fewer vertices, then we can analyze possible constellations by examining different choices of planet. Fortunately, this question has already been studied by other mathematicians, and their exist known bounds on number of graphs of this type.

The bounds on the number of possible galaxies in Table 6.1 are based on those determined by Markus Meringer [28], who obtained these results computationally using the computer program GENREG. More expansive tables of planar graphs can also be found on Dr. Merigner's personal website. The number of possible constellations in this table still needs to be proven, but these numbers are bounded below by the corresponding number of galaxies. We can already see that the number of possibilities will rapidly explode past 10 vertices, and hence we will restrict to studying $9+1$ constellations at largest.

$k+1$	Number of Galaxies [28]	Number of Constellations
$\leq 4+1$	0	0
$5+1$	1	1
$6+1$	0	0
$7+1$	1	1
$8+1$	1	2
$9+1$	3	6
$10+1$	3	≥ 13
$11+1$	13	?
$12+1$	21	?
$13+1$	68	?

Table 6.1: Number of distinct $k+1$ galaxies and constellations for small k

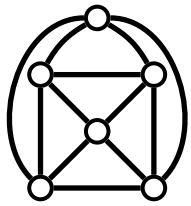
Lemma 6.18 (Enumerating small galaxies [28])**:**
Every *galaxy* (simple, connected, planar, 4-valent graph) of a $k+1$ constellation with at most ten vertices is isomorphic to exactly one of the graphs shown in Figure 6.15.

Proof:

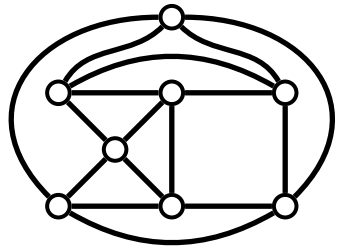
Let G be a galaxy with $n \leq 10$ vertices. By Corollary 6.14, there do not exist any galaxies with fewer than six vertices, and hence $6 \leq n \leq 10$. We can break the claim into several cases depending on the value of n, but most of these will be immediate by the bounds in Table 6.1. There do not exist any galaxies with seven vertices, and there exists exactly one galaxy each with six, eight, and nine vertices, up to isomorphism. Hence, if $n = 6$, 8, or 9, then G must be isomorphic to the galaxy with that many vertices shown in Figure 6.15.

If $n = 10$, then Table 6.1 implies that G is isomorphic to one of three possible galaxies with ten vertices. If we can prove that the three 10-vertex galaxies shown in Figure 6.15d are distinct, then we may conclude that these exhaust the possibilities for G. Since an isomorphism of planar graphs must preserve the faces in a planar embedding, we may prove that these graphs are non-isomorphic by showing they have different combinations of faces.

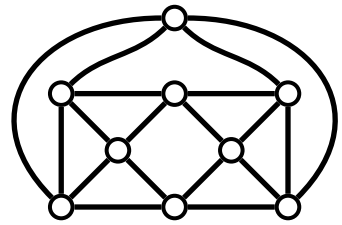
The first graph can immediately be distinguished from the other two by counting the number of triangles, squares, and pentagons in each graph. The remaining two graphs have the same quantities of these planar faces, but each square in the first shares an edge with exactly two other squares, and this feature is not present in the second. Thus, each of these 10-vertex galaxies is distinct, and so these must exhaust the three possible isomorphism classes implied by Table 6.1. Therefore, G must correspond to one of these graphs. □



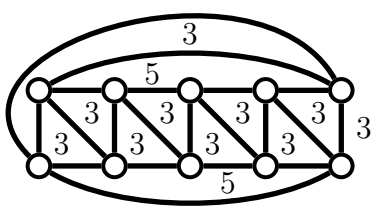
(a) Unique $5+1$ galaxy G_6



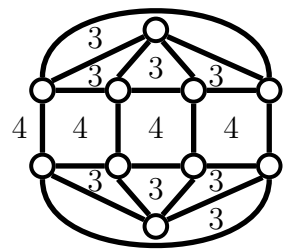
(b) Unique $7+1$ galaxy G_8



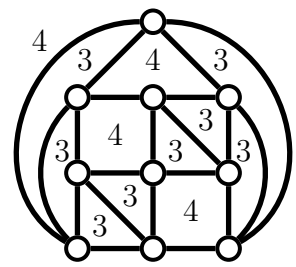
(c) Unique $8+1$ galaxy G_9



10 triangles
0 squares
2 pentagons



8 triangles
4 squares
0 pentagons



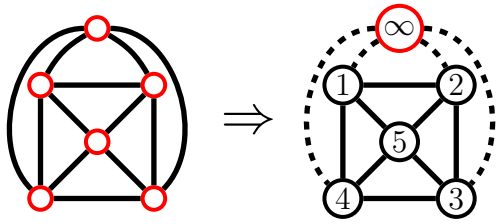
8 triangles
4 squares
0 pentagons

(d) Three possible $9+1$ galaxies G^1_{10}, G^2_{10}, and G^3_{10}, distinguishable by comparing planar faces

Figure 6.15: List of all possible galaxies with ten or fewer vertices, up to isomorphism

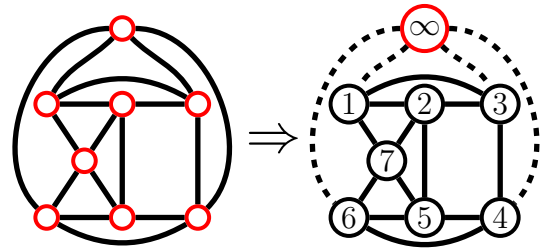
For $k+1 \leq 10$, Lemma 6.18 implies that there are only six possible galaxies. We obtain a constellation from a galaxy by choosing to mark a special vertex as the planet. We defined constellations to be equivalent if they have isomorphic cosmos. Since equivalent constellations must also have isomorphic galaxies (Lemma 6.13), there will be at least six different constellations arising from the different galaxies shown above. However, as the counterexample of Figure 6.11 showed, non-equivalent constellations may arise from the same galaxy by a different choice of planet. Hence, to count possible constellations, we must examine the isomorphism classes of induced subrgraphs obtained by removing a single vertex from any of the galaxies in Figure 6.15.

Theorem 6.19 (Enumerating small constellations)**:**
Up to equivalence, there exist exactly ten distinct $k+1$ constellations with at most 10 vertices, shown in Figure 6.14, each arising from one of the galaxies of Figure 6.15.



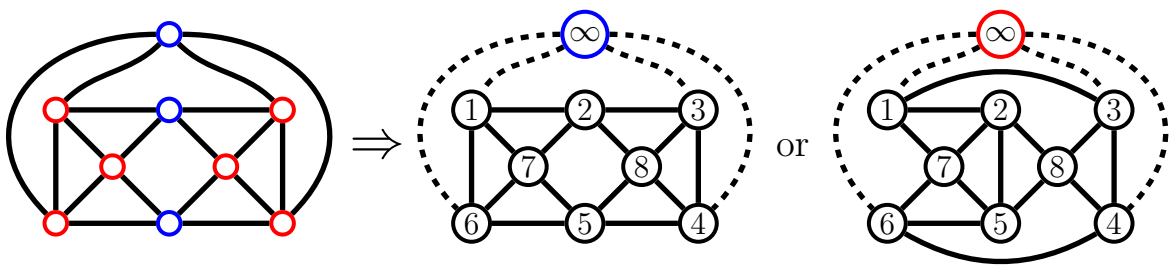
G_6 $C_{5+1} = (G_6, p)$

(a) Unique $5+1$ constellation arising from G_6



G_8 $C_{7+1} = (G_8, p)$

(b) Unique $7+1$ constellation arising from G_8



G_9 $C_{8+1} = (G_9, p)$ $C'_{8+1} = (G_9, p')$

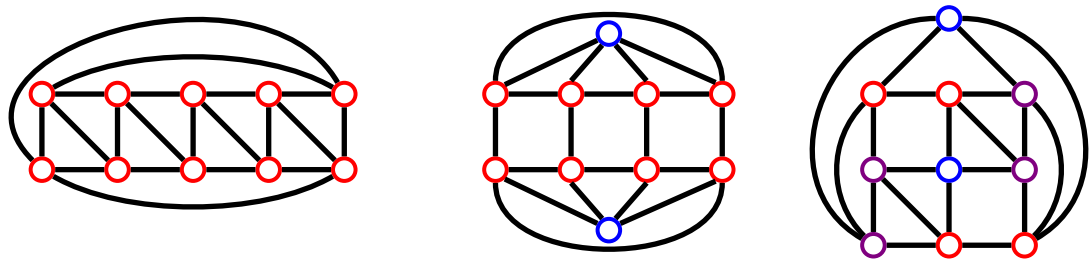
(c) The two possible $8+1$ constellations arising from G_9

Figure 6.16: The first four $k+1$ constellations

Proof:

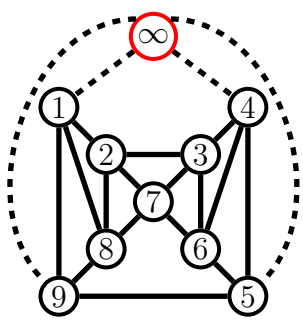
A complete proof of the claim requires that we examine the isomorphism classes of every possible cosmos obtained by choosing a planet in any of the galaxies introduced in Figure 6.15, although we will streamline the more tedious steps. Since equivalent constellations must have isomorphic galaxies (Lemma 6.13), we may consider separately the possible constellations arising from each of the galaxies of Figure 6.15.

Choosing any vertex in a galaxy to serve as a planet also determines the cosmos of the constellation, which is the induced subgraph obtained by removing this vertex. Rather than exhaust all possible choices of planet for each galaxy, we break these into cases which are color coded in Figures 6.16 and 6.17. For a given galaxy, any vertices of the same color will define isomorphic cosmos when chosen as the planet. This is clear by symmetry in some cases, but for these same-color cases more generally, one may obtain an explicit isomorphism of cosmos by constructing an edge-preserving bijection between vertex sets of the induced subgraphs. We will suppress these details here, but these maps can be found by hand.



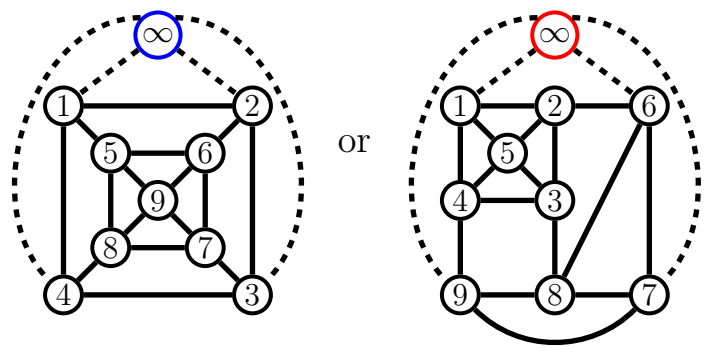
G^1_{10} G^2_{10} G^3_{10}

(a) The three $9+1$ galaxies and the choice of planets which define distinct constellations



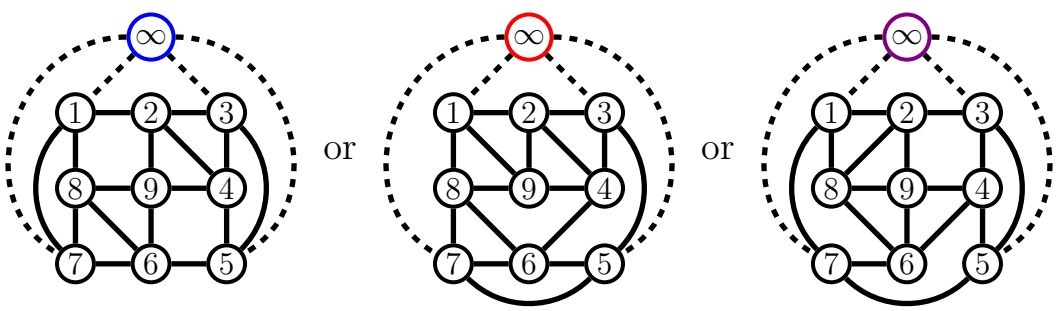
$C^1_{9+1} = (G^1_{10}, p)$

(b) Constellation arising from G^1_{10}



$C^2_{9+1} = (G^2_{10}, p)$ $C^{2'}_{9+1} = (G^2_{10}, p')$

(c) Two constellations arising from G^2_{10}



$C^3_{9+1} = (G^3_{10}, p)$ $C^{3'}_{9+1} = (G^3_{10}, p')$ $C^{3''}_{9+1} = (G^3_{10}, p'')$

(d) Three constellations arising from G^3_{10}

Figure 6.17: The six possible $9+1$ constellations

By the first part of this proof, we obtain the ten possible constellations featured in Figures 6.16 and 6.17, although we have not yet verified that these are distinct. For the second part of the proof, we must show that no two of the proposed constellations are equivalent. Thanks to Lemma 6.13, we may restrict to comparing only familiar constellations, that is, those arising from the same galaxy.

Since every choice of planet in the galaxies G_6, G_8, and G_{10}^1 yields an isomorphic cosmos, it follows immediately that each of these galaxies defines a unique equivalence class of constellations. Hence, the constellations C_{5+1}, C_{7+1}, and C_{9+1}^1 are all distinct from any others (Figures 6.16a, 6.16b, and 6.17b, respectively). Although there is only one $8+1$ galaxy G_9, there are two constellations arising from a different choice of planet (Figure 6.16c). In the counterexample from Figure 6.11, we already showed that these constellations have non-isomorphic cosmos. Hence, we may conclude that C_{8+1} and C'_{8+1} are distinct constellations.

It remains to consider the constellations arising from the $9+1$ galaxies G_{9+1}^2 and G_{9+1}^3 (Figure 6.17a), which we claim will define two and three different constellations, respectively. As in previous proofs, we may verify that the corresponding cosmos are non-isomorphic by comparing their planar faces. For the two familiar constellations C_{9+1}^2 and $C_{9+1}^{2'}$ proposed in Figure 6.17c, observe that the cosmos of the first graph does not contain any pentagons. However, the outermost face of the cosmos of $C_{9+1}^{2'}$ is a pentagon, and hence this constellation must be distinct from C_{9+1}^2.

Finally, consider the three familiar constellations C_{9+1}^3, $C_{9+1}^{3'}$, and $C_{9+1}^{3''}$ introduced in Figure 6.17, each obtained by a choice of planet in G_{10}^3. Observe that the third constellation has a cosmos whose outermost face is a pentagon, but neither of the other two constellations has a cosmos containing any pentagons. Hence, $C_{9+1}^{3''}$ is distinct from the other two. The first two constellations have the same quantities of planar faces of a given type, but their cosmos can be distinguished by examining the squares. Each contains exactly two squares, but these squares share an edge in the cosmos of $C_{9+1}^{3'}$ and they do not in the cosmos of C_{9+1}^3, which proves the two constellations are distinct.

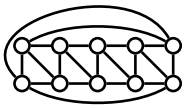
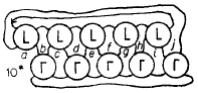
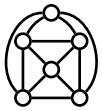
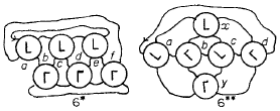
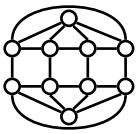
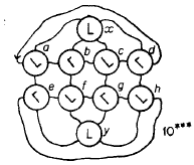
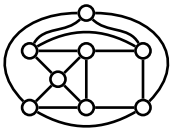
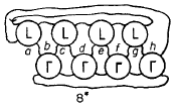
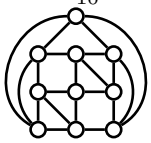
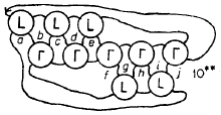
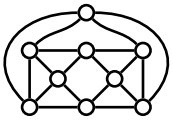
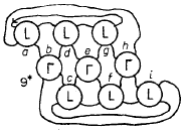
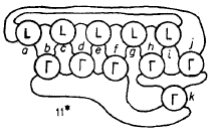
Therefore, no two of the ten constellations we have considered are equivalent. This accounts for every constellation enumerated in Figure 6.14, which completes the proof.

□

We will adopt the notation introduced in Figures 6.16 and 6.17 to reference these constellations. That is, we will refer to a $k+1$ constellation by writing $C_{k+1}^{i'}$, where i is an index used to identify the corresponding galaxy when there exists more than one galaxy contain-

ing $k+1$ vertices. When multiple constellations arise from the same galaxy, we distinguish between these by also including a prime in the superscript, as in C^{3}_{9+1}, $C^{3'}_{9+1}$, and $C^{3''}_{9+1}$. This choice of notation is based in part on how Conway chose to denote his basic polyhedra.

Conway's graphs are closely related to constellations, but they do not use a special marked vertex. However, these graphs are connected, planar, and 4-valent. Except for the Conway graph with a single vertex, they are also simple, which means that Conway's basic polyhedra are equivalent to galaxies. Table 6.2 shows a side-by-side comparison of the galaxies we have considered and the corresponding isomorphic Conway graph. The 1-vertex Conway graph 1^* is not a galaxy, but it is similar to the MASG of an algebraic diagram.

$k+1$	Galaxy	Conway Graph	$k+1$	Galaxy	Conway Graph
1	N/A	1^*	10	G^1_{10} 	10^* 
6	G_6 	$6^* \cong 6^{**}$ 		G^2_{10} 	10^{***} 
8	G_8 	8^* 		G^3_{10} 	10^{**} 
9	G_9 	9^* 	11		11^* 

Conway's graphs reprinted from [7] with permission from Elsevier

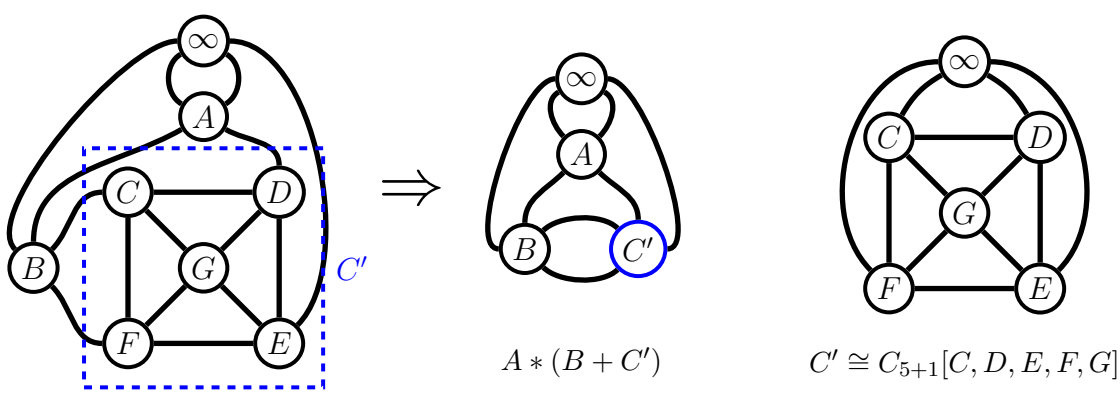
Table 6.2: Comparison of small galaxies with Conway's basic polyhedra

Conway shows two polyhedrons with six vertices, but these are merely different embeddings of the same graph. Since there exists exactly one galaxy with six vertices, we know that $6^* \cong 6^{**} \cong G_6$. Although we have not considered any galaxies with more than ten vertices, note that the 11-vertex Conway graph 11^* would correspond to a galaxy with eleven vertices. However, many more 11-vertex galaxies are possible according to the results stated in Table 6.1.

6.2.4 Constellation Notation

Recall that the basic polyhedra were introduced as a part of Conway's notation for knots, which describes a knot diagram using subtangle decompositions. In the same spirit, we would like to propose another, similar notation for 2-string tangle diagrams. However, unlike with knots, we must also account for a tangle diagram's boundary circle. For this reason, galaxies alone are not sufficient, but an adequate notation may be possible using constellations. When a configuration of subtangles in a decomposition of a tangle diagram matches the cosmos of a constellation, we can identify the boundary with the planet.

Every tangle diagram can be decomposed into algebraic subtangles using the MASG. Although the MASG of a non-algebraic diagram may not be a constellation (Figure 6.18), it must contain a subgraph which is isomorphic to the cosmos of some constellation (Theorem 6.16). This subgraph can be identified with a non-algebraic subtangle, and it may be possible to describe such a diagram using sums and products of both *maximal algebraic subtangles* and *minimal non-algebraic subtangles*. This suggests how we can generalize construction formulas



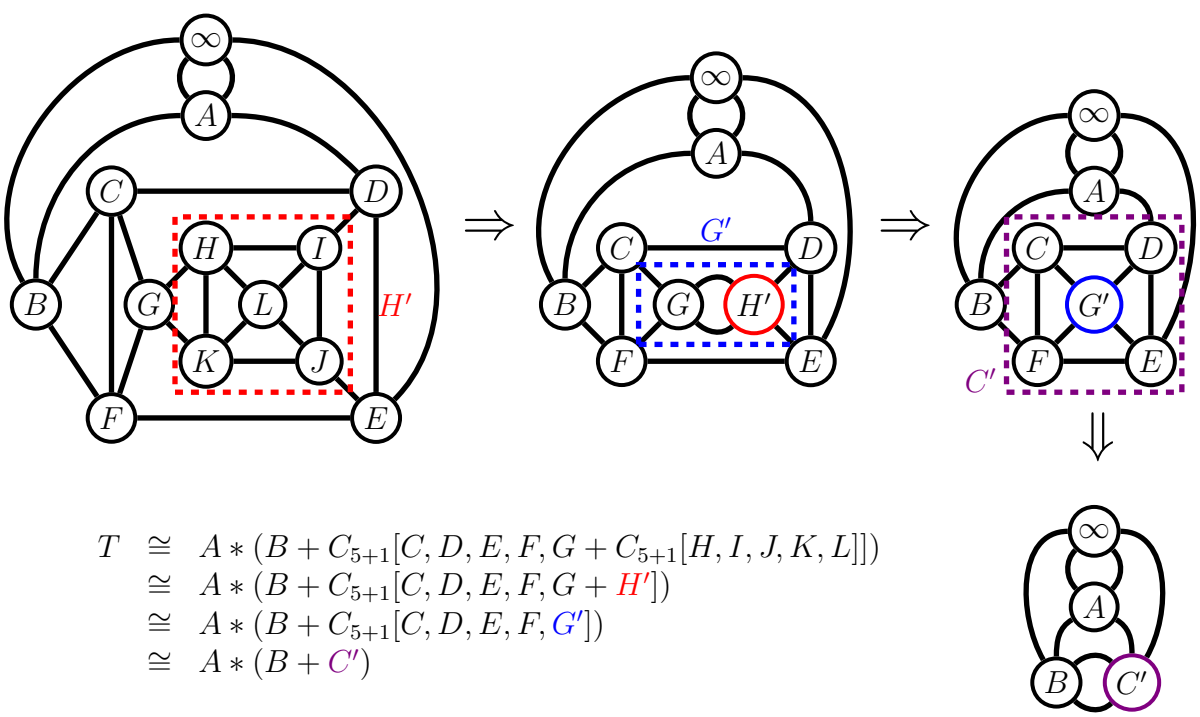
(a) Encoding the MASG using a construction formula **(b)** Minimal non-algebraic subtangle

Figure 6.18: Example of a non-algebraic construction, $T \cong A * (B + C_{5+1}[C, D, E, F, G])$

to non-algebraic diagrams. Just as tangle sum and product represent a combination of two subtangles, we can treat constellations as an operation on several subtangles (Figure 6.18b).

Unfortunately, identifying minimal non-algebraic subtangles may not be enough to describe the remaining subtangles using only tangle algebra. The difficulty is that constellations can be *nested*, by which we mean the cosmos of one constellation may exist as a proper subgraph in a larger constellation. Even among the small constellations we have already examined, we can see that C_{5+1} is nested inside C^2_{9+1} and $C^{2'}_{9+1}$ (Figure 6.17c). In the context of non-algebraic tangle diagrams, this nesting behavior occurs whenever a non-algebraic subtangle properly contains another non-algebraic subtangle.

Similar to how we obtained the MASG through a sequence of collapsing bigons, we may imagine identifying this nesting behavior through a sequence of refinements on the MASG (Figure 6.19). We start by replacing the cosmos of a minimal constellation with a single vertex, keeping track of the configuration of subtangles. This can be thought of as a generalization of a bigon collapse, and we will refer to this operation in the graph as a *constellation replacement*. As in the figure, this replacement could introduce a bigon, which we may then collapse using a tangle sum or product. If collapsing all bigons reduces the graph



$$
\begin{aligned}
T &\cong A * (B + C_{5+1}[C, D, E, F, G + C_{5+1}[H, I, J, K, L]]) \\
&\cong A * (B + C_{5+1}[C, D, E, F, G + H']) \\
&\cong A * (B + C_{5+1}[C, D, E, F, G']) \\
&\cong A * (B + C')
\end{aligned}
$$

Figure 6.19: Example of constellation notation computed from the MASG

to a single subtangle vertex, then we obtain a construction formula. If the configuration is still non-algebraic, such as when there exists a nested constellation, then we may search for another constellation-replacement and repeat.

Each refinement of a non-algebraic MASG results in a smaller subtangle graph representing the same diagram, but now we allow for vertices which represent non-algebraic subtangles. Eventually, this process will lead to a single constellation, or a constellation with some combination of tangle sums and products. As in Figure 6.19, this sequence of refinements can be tracked with a generalized construction formula that allows for non-algebraic subtangles. Building up this formula in stages makes it easy to identify nested constellations. The original diagram can be recovered from this formula by reconstructing the MASG, and then inserting the maximal algebraic subtangles.

We will refer to the construction formula obtained in this way as *constellation notation*, noting that it is consistent with the familiar algebraic construction formulas that we have worked with previously. There are a number of choices involved with denoting the vertices in a constellation, but we pause this question momentarily. If we allow that subtangles arranged in a constellation can be denoted in a reasonable way, then every non-algebraic tangle diagram can be represented with a constellation notation determined from its MASG.

Theorem 6.20 (Constellation notation of a non-algebraic diagram)**:**
If T is a non-algebraic tangle diagram with MASG G, then the constellation notation of T is computed through a sequence of constellation replacements and bigon collapses in G.

Proof:

Let T be a non-algebraic tangle diagram and let G be the corresponding maximal algebraic subtangle graph. By Theorem 6.16, either G is a constellation or else G contains a proper subgraph isomorphic to the cosmos of some constellation. If G is a *minimal* constellation, then this graph determines a constellation notation for T based on the corresponding configuration of maximal algebraic subtangles.

If G is not a constellation or not a minimal constellation, then G contains a proper subgraph corresponding to the cosmos of a smaller constellation. If this constellation is not minimal either, then we may repeat this process until we obtain a subgraph which matches the cosmos of a minimal constellation. Since G only contains finitely many nodes, this search must terminate eventually. Once such a subgraph is identified, we may perform a constellation replacement to obtain a modified subtangle graph G_1, which is identical to G except that we replace the minimal cosmos with a single vertex.

The subgraph replaced in this way corresponds to a minimal non-algebraic subtangle T_1 of T, and it has a constellation notation determined by a corresponding configuration of maximal algebraic subtangles from T. As we continue to refine the subtangle graph for T, we may keep track of the notation for T_1, which can be slotted into a larger construction formula as shown in Figure 6.19. In this way, we may attempt to build up the constellation notation for T in stages.

If G_1 contains any new bigons between subtangle vertices after this constellation replacement, then let G_1' be the graph obtained after performing all possible bigon collapses. In the same way that we originally defined the MASG, this process will eventually terminate with a smaller subtangle graph containing no bigons, except possibly those which involve the boundary vertex. Each bigon collapse corresponds to a sum or product of subtangles in T, and we keep track of each of these while building up a construction formula. If this reduces to a single subtangle vertex, as in Figure 6.18, then this construction formula determines the constellation notation of T.

Otherwise, we may repeat this process starting from the graph G_1', alternating between constellation replacements and sequences of bigon collapses. We allow for possibly empty sequences of bigon collapses, but each constellation replacement reduces the total number of vertices in the subtangle graph. Since there are only finitely many vertices in the graph, this process will eventually terminate when all vertices have been joined together through one of these two operations. Let G_n' be final the final graph obtained in this way:

$$G \xrightarrow{\text{c.r.}} G_1 \xrightarrow{\text{b.c.}} G_1' \xrightarrow{\text{c.r.}} \cdots \xrightarrow{\text{c.r.}} G_n \xrightarrow{\text{b.c.}} G_n'.$$

The final graph G_n' contains a single subtangle vertex and the boundary vertex. Each iteration builds up a constellation notation for a non-algebraic subtangle, until all of these can be combined into a single formula. The final formula obtained in this way is the constellation notation for the original diagram T.

□

The example from Figure 6.19 shows the MASG of a diagram reconstructed from a constellation notation. The sequence of graphs shown in the figure represent the alternating sequence of constellation replacements and bigon collapses. Although the refined graph obtained in this proof only uses minimal constellation replacements, one could use non-minimal constellations in a similar way. However, in the former case, the minimal non-algebraic subtangles are easily identifiable from the constellation notation.

The constellation notation obtained through Theorem 6.20 assumes that the configuration of subtangles identified in each constellation replacement can be encoded in a nice way. In Conway notation, the vertices in the basic polyhedra are ordered and oriented according to a choice made by Conway, and subtangles are slotted into a corresponding formula. We may do something similar using constellations, as the preceding two figures show for the unique $5+1$ constellation C_{5+1}. We will denote such a configuration of subtangles as $C_{k+1}^{i'}[S_1, \cdots, S_k]$, where $C_{k+1}^{i'}$ indicates the constellation and $S_1, \cdots, S_k$ denote the k subtangles that are slotted into the graph. For the small constellations we have studied, we will choose to order and orient these subtangles according the embeddings shown in Figure 6.14.

This is only one possible choice, and it is possible that a different embedding of a constellation may reveal a better choice. We leave open the possibility of modifying the embeddings from Figure 6.14 in favor of something more natural in the future. However, once a choice is a made, we can compute a constellation notation based on the algorithm from Theorem 6.20. This algorithm is independent from a specific embedding, but it assumes that each constellation replacement can be represented with a formula. Hence, this will also apply to larger constellations which we have not yet recorded. Our existing list of ten constellations from Theorem 6.19 will be sufficient for all non-algebraic diagrams with at most 9 crossings.

Corollary 6.21 (Constellations for small non-algebraic diagrams)**:**
If T is a non-algebraic tangle diagram with at most 9 crossings, then its constellation notation can be determined using only the constellations enumerated in Theorem 6.19.

Proof:

Let T be a non-algebraic diagram with 9 or fewer crossings. Since T is non-algebraic, its maximal algebraic subtangle graph contains more than a single vertex (Corollary 6.10). Let G denote the corresponding MASG of this diagram, and let k denote the number of subtangle vertices in G, not counting the boundary vertex. Since every subtangle in the corresponding decomposition of T must contain at least one crossing, the number of subtangle vertices in G cannot exceed the number of crossings in T, which means that $k \leq 9$.

By Theorem 6.16, G contains a subgraph isomorphic to the cosmos of some constellation, but the number of vertices in this subgraph cannot exceed k. Hence, the largest possible constellation that could be used has $k+1 \leq 9+1 = 10$ vertices. Therefore, any constellations with a cosmos matching a subgraph in G must be among those enumerated in Theorem 6.19. Thus, the constellation notation for T computed using the algorithm in Theorem 6.20 will only use these constellations, proving the claim. □

Although the constellations from Figure 6.14 are sufficient for encoding small tangles, this does not account for the fact that the cosmos of a constellation may appear rotated in the MASG of a given diagram. The location of the NW corner in the MASG is inherited from the original tangle diagram. When we make a choice of NW corner for the subtangles in the APD code, we prefer to use the first corner encountered while traveling along the strings with the standard orientation. Hence, the first time we encounter a non-algebraic subtangle corresponding to a subgraph in the MASG, the first vertex we encounter may not be the one in the NW corner of our chosen embedding.

This will not matter for any constellations with rotationally symmetric cosmos, such as C_{5+1} and C^2_{9+1}, but we must account for this behavior more generally. By combining constellations with our notation for tangle rotations, we may use $R_\theta(C^{i'}_{k+1})$ to denote a clockwise rotation of one of these graphs by θ degrees. The embeddings from Figure 6.14 are chosen so that the NW, NE, SE, and SW corners are easily identifiable in the graph; these corners are permuted under such a rotation. In this way, we may account for rotations when computing the constellation notation for small non-algebraic tangle diagrams through a sequence of refinements of the MASG, as we did in Figure 6.19.

Finally, we make a comment on constellation notation in connection with knot diagrams. Although we define this for a tangle diagram, if one knew how to decompose a given knot diagram into maximal algebraic subtangles, it could be converted into a non-algebraic tangle diagram by cutting the strings. If the resulting diagram is algebraic, the knot can be denoted by a standard algebraic construction formula; if it is non-algebraic, it could be encoded with constellation notation. Since subtangle decompositions are also the basis of Conway notation, this suggests how one could convert Conway notation into constellation notation, and vice versa.

Hence, we propose that any knot diagram arising as the closure of a non-algebraic tangle diagram be referred to as an *astroknot*. A link diagram arising in the the same way could be referred as an *astrolink*. Algebraic knots would then be known as *geoknots*. We could then explore many interesting questions inspired by bad puns, such as whether there exists a relationship between astroknots that "live in the same galaxy," meaning defined by MASGs with familiar constellations.

7 Database Development and Future Work

The final chapter of this dissertation is devoted to discussing the next direction of this research, particularly in connection with developing a web-accessible database to disseminate these results. There exist many online resources cataloging knots, including *The Knot Atlas* and *KnotInfo* [2, 23]. We now seek to create a similar resource enumerating 2-string tangles and their structural properties. Towards this, we summarize what has already been accomplished and outline our future goals for this project.

7.1 Database of 2-String Tangles

The creation of such a database involves two parts. First, we must computationally generate and classify a list of tangle diagrams. Second, this information must be organized in an accessible format. Although still under development, significant progress has been made on both aspects of this project. What follows is a brief overview of the computational methods involved and a description of the database prototype.

7.1.1 Tangle Tabulation

One way to enumerate an exhaustive list of tangles is by generating all possible diagram notations. This is the method which Dowker and Thistlewaite used to tabulate knots [11]. This technique can also be applied to tangles, whereby we generate a list of all possible Dowker codes, and then delete those which cannot be realized by a tangle diagram. This method has already been employed in student research projects preceding this dissertation [25, 9], and these have been continued and expanded upon in this project. My contribution to this computational work is a program to calculate the various SPD codes of a diagram [5], starting from the Dowker code.

Since every tangle diagram can be encoded with a Dowker code, every diagram must exist somewhere on this exhaustive list of Dowker codes. Since the number of entries in a Dowker code matches the number of crossings in a diagram, there are only finitely many Dowker codes for a given crossing number. Hence, up to a fixed crossing number, this method will generate an exhaustive list of tangles. However, because a given tangle can be realized by multiple diagrams, this technique will result in many redundancies. Although we do not miss any tangles in this process, we would prefer to exclude repeats. These can be identified by combining the program to tabulate tangles with the program to compute SPD codes.

Tangle classification provides a partial solution to the problem of redundant diagrams, and a complete solution in the cases where a unique canonical diagram is known to exist. After generating a tangle diagram notation, we may examine how it decomposes into maximal subtangles by computing its corresponding subtangle planar diagram codes. Based on these decompositions, the diagram can be classified by Theorem 4.12. In the case of an algebraic diagram, we may use the various SPD codes to obtain an explicit construction formula.

In particular, the 1CPD code of an algebraic diagram can be used to recover an algebraic construction formula using only 1-crossing subtangles. Based on this construction formula, we may identify whether or not the diagram is in semi-canonical form. Since every algebraic tangle diagram is equivalent to a diagram in semi-canonical form (Theorem 5.56), non-semi-canonical diagrams may be safely eliminated as redundant. Furthermore, if an algebraic diagram is integer, rational, Montesinos, or generalized Montesinos, then this can also be inferred from the construction formula. Since the canonical form of these special types of algebraic tangles aligns with the semi-canonical form (Lemma 5.55), this technique will exclude all redundant diagrams for these fully classified tangles.

Algebraic and non-algebraic diagrams can be distinguished by computing the APD code. For algebraic diagrams, the APD code yields an explicit algebraic construction formula using maximal subtangles. For non-algebraic diagrams, the APD code gives the adjacency list for the MASG. Although we do yet have a preferred diagram for non-algebraic tangles, we may restrict to diagrams in which each maximal algebraic subtangle in the MASG is in semi-canonical form, allowing for the elimination of many redundant diagrams. Note that this restriction is relative to the choice of NW corner in these subtangles, which is determined by the first pass of the oriented strings in the APD code.

As crossing number increases, the proportion of diagrams which are non-algebraic grows rapidly, as does the proportion of algebraic diagrams which do not fall into one of the special cases. Hence, tangle tabulation more generally must rely on other methods to identify redundant diagrams. However, for small diagrams, the number of possible redundancies is small enough that it may be possible to search for these on a case-by-case basis. The smallest non-algebraic tangle diagram requires 6 crossings, and the smallest algebraic diagram which is not among the special cases requires 7 crossings. Since the first ten constellations are sufficient for describing non-algebraic diagrams with at most 9 crossings (Corollary 6.21), eliminating all redundancies for tangle diagrams with up to 9 crossings is a realistic starting goal. Table 7.1 shows an exhaustive list of canonical diagrams for tangles with up to 5 crossings, along with the corresponding fraction or construction formula.

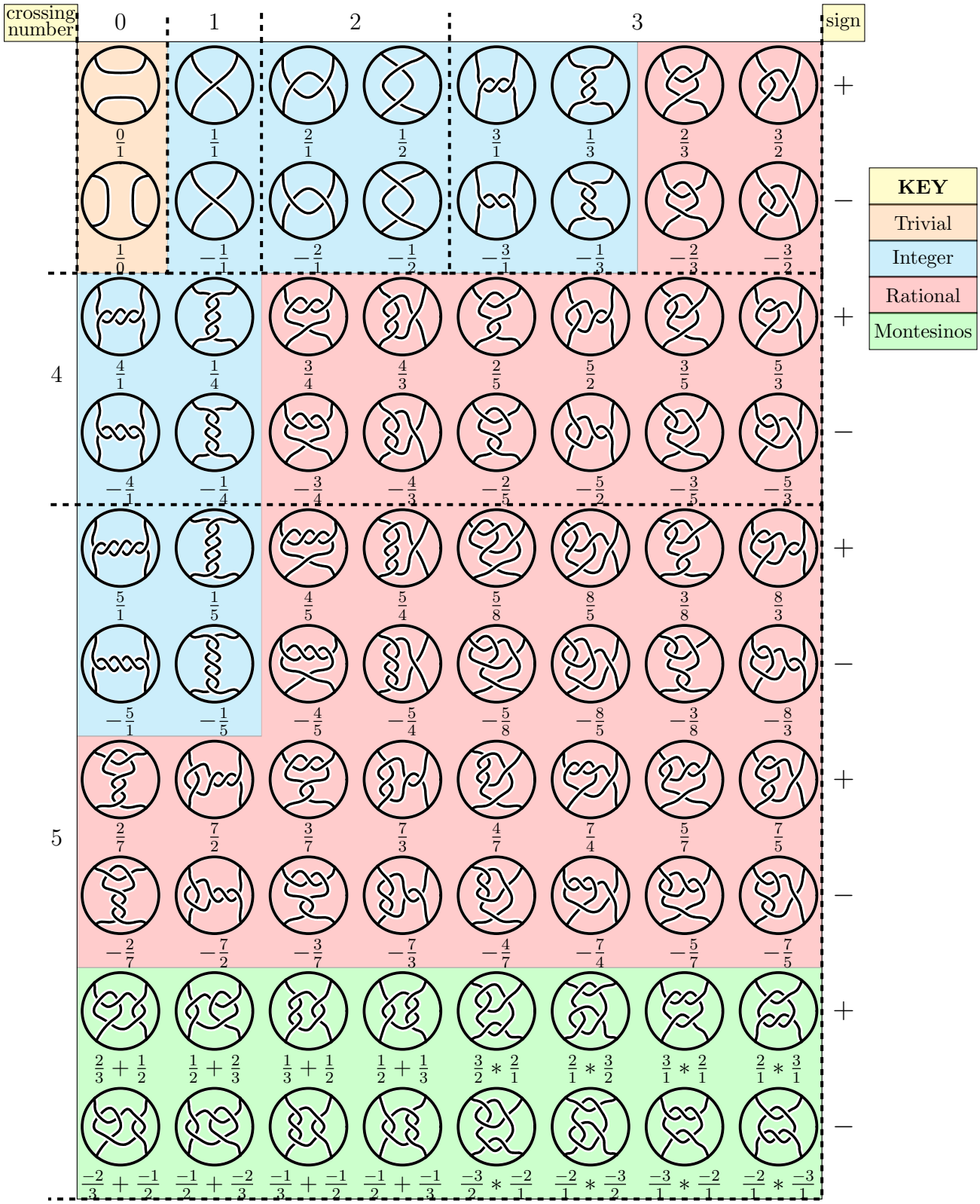


Table 7.1: Table of canonical diagrams for 2-string tangles with 5 or fewer crossings

7.1.2 The Tanglenomicon

That is knot thread
which can eternal tie.

To make the results of the tabulation and classification programs accessible to the broader research community, an online database of 2-string tangles known as *the Tanglenomicon* [6] is currently under development: `http://www.nick-connolly.com/tangles/tangles.php`. This web-accessible database is being developed using SQL, PHP, and HTML, and it is populated with data output by tangle tabulation programs written in C/C++. It is intended to serve as a repository of known tangle diagrams, their notations, classifications, constructions, and their properties.

Although still a prototype at the time of this dissertation, the Tanglenomicon currently includes notations and constructions for the canonical diagrams of all rational tangles with up to 9 crossings, and a subset of representative diagrams for non-rational Montesinos and generalized Montesinos tangles with up to 11 crossings. These existing database entries are a proof of concept and are consistent with existing lists of tangles such as those from [20, 31, 17]. The next step is to update the information in the database with the output from the tabulation and classification programs.

One of the features of the Tanglenomicon is the ability to search for tangles based on a list of criteria. This allows a user to generate a sorted list of tangles which satisfy certain conditions, such as crossing number, parity, or class. Furthermore, each entry in the database includes a page of detailed information unique to that tangle. For example, rational tangles include twist vectors, and Montesinos tangles include the decomposition into maximal rational subtangles. Planned features for the database include incorporating computer generated figures using KnotPlot [35] and adding more diagram notations and tangle invariants.

Since the database will not include entries for tangle diagrams which are known to be redundant, incorporating the diagram classification program into the website is also a planned feature. In this way, a user could input the notation for an unknown tangle diagram, classify it, and then be directed to the page in the database corresponding to the preferred diagram if it exists. This would allow a researcher to utilize the database to study non-canonical diagrams, even if such diagrams are not included directly. This summarizes a few of the features currently under development, but the Tanglenomicon will continue to be expanded as progress is made.

7.2 Future Research Directions

In addition to creating a database, there are a number of promising directions in which this research could be taken. The most obvious next step would be the formalization of constellation notation. Our previous work has shown that the first 10 constellations are sufficient for describing small non-algebraic diagrams, but turning this into a notation will require making a number of choices to account for possible configurations, such as when constellations appear as rotated subgraphs in a non-algebraic MASG. However, using Conway notation as a model, we could choose an order and orientation for the vertices in each constellation, and this would enable us to describe non-algebraic diagrams more generally in a style similar to that of Figure 6.19.

The existence of unique canonical diagrams for generic algebraic tangles is another open question that warrants further study. Just as we have examined subtangle graphs and algebraic tangle trees, probing algebraic tangle diagrams through the lens of another mathematical structure could be a way to approach this. One such possibility is a braid-like representation for algebraic tangle diagrams. Since rational tangles in canonical form are described through alternating horizontal and vertical twists, rational tangles can be represented as the partial plait closure of a 3-string braid. Since algebraic tangle diagrams are constructed from rational subtangles, one can imagine how this braid-like description of rational diagrams could be extended to algebraic diagrams using sums and products.

Based on this idea, I have developed a method in which an algebraic diagram constructed from m rational subtangles could be encoded as the partial plait closure of a $(2m+1)$-string braid. This suggests a way that algebraic diagrams could be represented using braid words, and it may allow for certain results concerning braids to be applied to algebraic tangles. Drawing connections between braids and tangles could lead to some interesting new results that move us closer to classifying generic algebraic tangle diagrams.

In addition to these theoretical questions, there are a number of interesting ways to expand on the computational results of this research. Subtangle decompositions and SPD codes provide a powerful tool for studying knot, link, and tangle diagrams more generally. These could prove especially helpful in computing invariants that rely on recursive methods to examine smaller and smaller diagrams, such as polynomial invariants. Towards this goal, an interesting future direction could involve searching for ways to combine the program to compute the SPD code with other computational algorithms.

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